Solid State Technology Association

COMMITTEE LETTER BALLOT

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COMMITTEE: JC-42.3
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SUBJECT: Proposed DDR6 Standard Draft Rev0.x R0

BACKGROUND: DDR6 Standard Draft Rev0.x R0 with all 31 passed ballots on the voting machine in Q3'26. Ballot details in the revision history at the end.

Material in black is previously passed content, material in red is for ballot. Material in gray is reference content from LPDDR6 or DDR5 and will be replaced by future ballot material.

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PROPOSED

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1. Scope

This standard defines the DDR6 specifications, including features, functionalities, AC and DC characteristics, packages, and ball/signal assignments. The purpose of this standard is to define the minimum set of requirements for JEDEC-compliant DDR6 devices.

This document was created based on the LPDDR6 (JESD209-6) and DDR5 (JESD79-5) standards, and some aspects of the following standards: DDR (JESD79), DDR2 (JESD79-2), DDR3 (JESD79-3), DDR4 (JESD79-4), and LPDDR5 (JESD209-5).

Each aspect of the standard was considered and approved by committee ballot(s). The accumulation of these ballots was then incorporated to prepare the DDR6 standard.

PROPOSED

2. Overview

2.1. Functional Description

DDR6 is a high-speed synchronous DRAM device internally configured with pseudo-split die (PSDs), each containing 3 or 6 double-data-rate (DDR) I/Os (DQs) and a shared differential data strobe (DQS) for both PSDs. The DQs from each PSD contribute to forming pseudo-channels. The bank architecture is two (2) banks in 16 bank groups per PSD. Further description of PSD organization and pseudo-channel configurations is provided in subsequent sections.

The PSD can be accessed pseudo-independently in the 2p3 and 2p6 configurations or combined in the 1p6 and 1p12 configurations. The PSD configurations support operational modes such as 1p6, 1p12, 2p3, and 2p6, where "1p" and "2p" denote the number of pseudo-channels and the second number denotes the number of DQs per PSD. Configuration details are defined in the Addressing Tables section.

The devices use a shared clock (CK) between both PSDs for command/address inputs that operates at the same rate as the shared DQS that is used for data. There are six single-data-rate (SDR) command/address (CA) pins that the memory controller uses to transmit command, address, bank, configuration, and training information to the device. CA signals are latched on the rising CK edges, with the commands starting on an even rising CK edge when indicated by a HIGH signal on the SDR CS pin that is 2N in duration and centered on the even rising CK edge. Commands are 2nCK, 4nCK and 6nCK in duration. See the Command Truth Table for details.

The DDR I/O signaling is designed to operate with on-die termination on both the memory controller and the device.

Data access size is configuration dependent, with 128 data bit prefetch supported. A single BL24 read or write access for a 3 DQ PSD consists of 72-bits (64-bits of data and 8-bits of non-data). A single BL24 read or write access for a 6 DQ PSD consists of 144-bits (128-bits of data and 16-bits of non-data). Non-data refers to bits that may be stored in the array and accessed using address carve-out, or bits that are not stored in the array, such as unstored CRC.

Read and write accesses to the devices are burst oriented. Accesses begin with the registration of an ACTIVATE command, followed by a READ or WRITE command.

The address, bank and bank group bits registered coincident with the ACTIVATE command are used to select the row and the bank to be accessed. The address, bank and bank group bits registered coincident with the READ or WRITE command are used to select the bank and the column location for the burst access.

Prior to normal operation, the device must be initialized. The Power-Up and Initialization section provides detailed information covering device initialization.

PROPOSED

2.1.1. Pad Definition and Description

Table 1 — Pad Definition and Description (Sheet 1 of 2)

Symbol	Type	Description	Note
RESET_n	Input	**RESET:** Reset is active when RESET_n is LOW, and inactive when RESET_n is HIGH. RESET_n must be HIGH during normal operation. RESET_n is an asynchronous signal.	1
CK_t, CK_c	Input	**Clock:** CK_t and CK_c are differential clock inputs. The first crossing point is the rising (falling) edge of CK_t (CK_c) and second crossing point is falling (rising) edge of CK_t (CK_c). Command/Address and CS inputs are sampled on the crossing point that is the rising (falling) edge of CK_t (CK_c). One CK_t and CK_c pair are shared for both pseudo-split die (PSD0, PSD1).	1
CS_0, CS_1	Input	**Chip Select:** CS is part of the command code, the pulse width is 2CK and is sampled on the even rising (falling) edge of CK_t (CK_c) unless the device is in power-down mode where it becomes an asynchronous signal. All commands are masked when CS is registered LOW.	1,3
CA[5:0]_0, CA[5:0]_1	Input	**Command/Address Inputs:** CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus.	1,3,4
DQ[5:0]_0, DQ[5:0]_1	Input / Output	**Data Input/Output:** Bi-directional data bus.	1,3
DQS_t, DQS_c	Input /Output	**Data Strobe:** DQS_t and DQS_c are the differential output clock signals used to strobe data during a READ and differential input clock signals used to latch WRITE data. The differential crossing point is edge-aligned with READ data and centered for WRITE data. One DQS_t and DQS_c pair are shared for both pseudo-split die (PSD0, PSD1).	1
ALERT	Output	**ALERT:** If an error is detected and reported to the fault register, the ALERT encodes the type of error on the pin. See the Alert section for details.	1
ZQ	Reference	**ZQ:** ZQ is used to calibrate the output drive strength and the termination resistance as calibration reference. There is one ZQ pad per die.	2
V_{DD}, V_{DDQ}, V_{PP}	Supply	**Power supplies:** Isolated on the die for improved noise immunity. Includes the main power, DQ power and row activation power.	
V_{SS}	Ground	**Ground Reference:** Power supply ground reference.	
RFU	Input / Output	Reserved for future use .	

PROPOSED

Table 1 — Pad Definition and Description (Continued) (Sheet 2 of 2)

Symbol	Type	Description	Note
NC		**No Connect:** No internal electrical connection is present.	

NOTE 1: This pin is referenced to V_{SS}.
NOTE 2: In a multi-die package stack, one die will arbitrate for all dice.
NOTE 3: Pseudo-Split Die 0 (PSD0) and Pseudo-Split Die 1 (PSD1) have independent pads.
NOTE 4: CA5 (6th CA) is an optional pin on DDR6 that provides Chip ID (CID) address inputs.

PROPOSED

2.1.2. DDR6 SDRAM Ballout

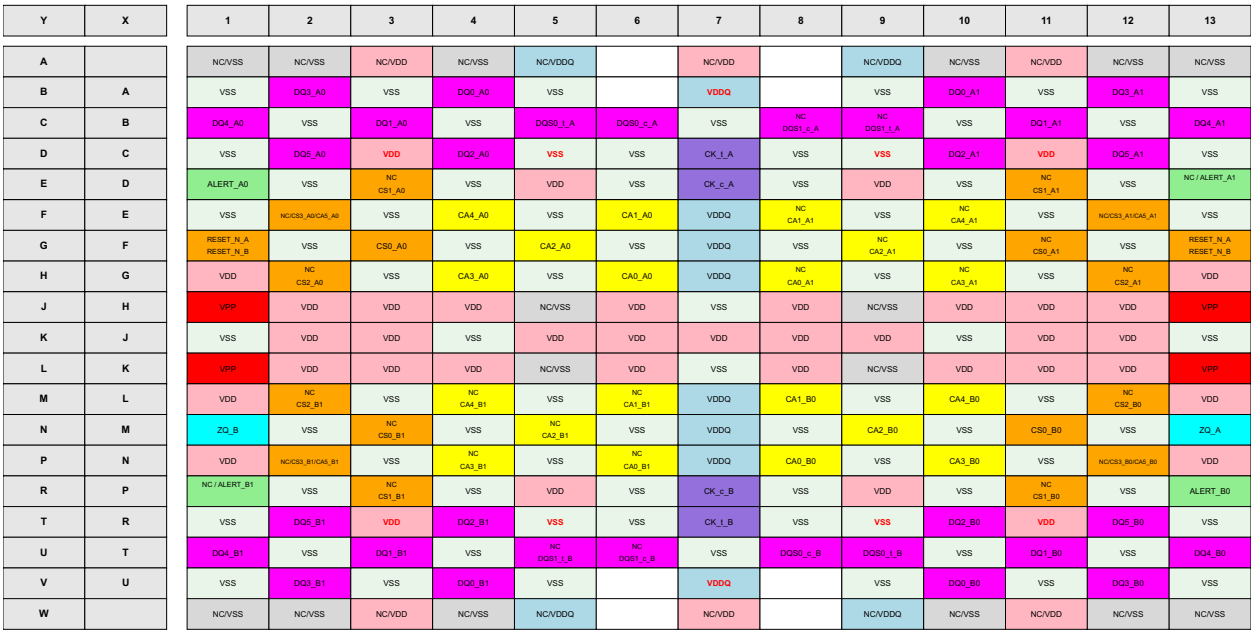
Table 2 — Details

• 2 configurations with 2 Ball Grid Arrays – X12 • X: 13x17 (Rows x Cols) • Y: 13x19 (Rows x Cols)	• X24 – X: 13x17 (Rows x Cols) – Y: 13x19 (Rows x Cols) • Pitch – 0.75(X) x 0.80(Y) mm

Y	X	1	2	3	4	5	6	7	8	9	10	11	12	13
A		NC/VSS	NC/VSS	NC/VDD	NC/VSS	NC/VDDQ		NC/VDD		NC/VDDQ	NC/VSS	NC/VDD	NC/VSS	NC/VSS
B	A	VSS	DQ3_A0	VSS	DQ0_A0	VSS		VDDQ		VSS	DQ0_A1	VSS	DQ3_A1	VSS
C	B	DQ4_A0	VSS	DQ1_A0	VSS	DQS0_t_A	DQS0_c_A	VSS	NC DQS1_c_A	NC DQS1_t_A	VSS	DQ1_A1	VSS	DQ4_A1
D	C	VSS	DQ5_A0	VDD	DQ2_A0	VSS	VSS	CK_t_A	VSS	VSS	DQ2_A1	VDD	DQ5_A1	VSS
E	D	ALERT_A0	VSS	NC CS1_A0	VSS	VDD	VSS	CK_c_A	VSS	VDD	VSS	NC CS1_A1	VSS	NC / ALERT_A1
F	E	VSS	NC/CS3_A0/CA5_A0	VSS	CA4_A0	VSS	CA1_A0	VDDQ	NC CA1_A1	VSS	NC CA4_A1	VSS	NC/CS3_A1/CA5_A1	VSS
G	F	RESET_N_A RESET_N_B	VSS	CS0_A0	VSS	CA2_A0	VSS	VDDQ	VSS	NC CA2_A1	VSS	NC CS0_A1	VSS	RESET_N_A RESET_N_B
H	G	VDD	NC CS2_A0	VSS	CA3_A0	VSS	CA0_A0	VDDQ	NC CA0_A1	VSS	NC CA3_A1	VSS	NC CS2_A1	VDD
J	H	VPP	VDD	VDD	VDD	NC/VSS	VDD	VSS	VDD	NC/VSS	VDD	VDD	VDD	VPP
K	J	VSS	VDD	VDD	VSS	VDD	VDD	VDD	VDD	VDD	VSS	VDD	VDD	VSS
L	K	VPP	VDD	VDD	VDD	NC/VSS	VDD	VSS	VDD	NC/VSS	VDD	VDD	VDD	VPP
M	L	VDD	NC	VSS	NC	VSS	NC	VDDQ	NC	VSS	NC	VSS	NC	VDD
N	M	NC	VSS	NC	VSS	NC	VSS	VDDQ	VSS	NC	VSS	NC	VSS	ZQ_A
P	N	VDD	NC	VSS	NC	VSS	NC	VDDQ	NC	VSS	NC	VSS	NC	VDD
R	P	NC	VSS	NC	VSS	VDD	VSS	NC	VSS	VDD	VSS	NC	VSS	NC
T	R	VSS	NC	VDD	NC	VSS	VSS	NC	VSS	VSS	NC	VDD	NC	VSS
U	T	NC	VSS	NC	VSS	NC	NC	VSS	NC	NC	VSS	NC	VSS	NC
V	U	VSS	NC	VSS	NC	VSS		VDDQ		VSS	NC	VSS	NC	VSS
W		NC/VSS	NC/VSS	NC/VDD	NC/VSS	NC/VDDQ		NC/VDD		NC/VDDQ	NC/VSS	NC/VDD	NC/VSS	NC/VSS

Figure 1 — DDR6 SDRAM X12 ~~DIMM~~ Module Common Ballout

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Y	X	1	2	3	4	5	6	7	8	9	10	11	12	13
A		NC/VSS	NC/VSS	NC/VDD	NC/VSS	NC/VDDQ		NC/VDD		NC/VDDQ	NC/VSS	NC/VDD	NC/VSS	NC/VSS
B	A	VSS	DQ3_A0	VSS	DQ0_A0	VSS		VDDQ		VSS	DQ0_A1	VSS	DQ3_A1	VSS
C	B	DQ4_A0	VSS	DQ1_A0	VSS	DQS0_t_A	DQS0_c_A	VSS	NC DQS1_c_A	NC DQS1_t_A	VSS	DQ1_A1	VSS	DQ4_A1
D	C	VSS	DQ5_A0	VDD	DQ2_A0	VSS	VSS	CK_t_A	VSS	VSS	DQ2_A1	VDD	DQ5_A1	VSS
E	D	ALERT_A0	VSS	NC CS1_A0	VSS	VDD	VSS	CK_c_A	VSS	VDD	VSS	NC CS1_A1	VSS	NC / ALERT_A1
F	E	VSS	NC/CS3_A0/CA5_A0	VSS	CA4_A0	VSS	CA1_A0	VDDQ	NC CA1_A1	VSS	NC CA4_A1	VSS	NC/CS3_A1/CA5_A1	VSS
G	F	RESET_N_A RESET_N_B	VSS	CS0_A0	VSS	CA2_A0	VSS	VDDQ	VSS	NC CA2_A1	VSS	NC CS0_A1	VSS	RESET_N_A RESET_N_B
H	G	VDD	NC CS2_A0	VSS	CA3_A0	VSS	CA0_A0	VDDQ	NC CA0_A1	VSS	NC CA3_A1	VSS	NC CS2_A1	VDD
J	H	VPP	VDD	VDD	VDD	NC/VSS	VDD	VSS	VDD	NC/VSS	VDD	VDD	VDD	VPP
K	J	VSS	VDD	VDD	VSS	VDD	VDD	VDD	VDD	VDD	VSS	VDD	VDD	VSS
L	K	VPP	VDD	VDD	VDD	NC/VSS	VDD	VSS	VDD	NC/VSS	VDD	VDD	VDD	VPP
M	L	VDD	NC CS2_B1	VSS	NC CA4_B1	VSS	NC CA1_B1	VDDQ	CA1_B0	VSS	CA4_B0	VSS	NC CS2_B0	VDD
N	M	ZQ_B	VSS	NC CS0_B1	VSS	NC CA2_B1	VSS	VDDQ	VSS	CA2_B0	VSS	CS0_B0	VSS	ZQ_A
P	N	VDD	NC/CS3_B1/CA5_B1	VSS	NC CA3_B1	VSS	NC CA0_B1	VDDQ	CA0_B0	VSS	CA3_B0	VSS	NC/CS3_B0/CA5_B0	VDD
R	P	NC / ALERT_B1	VSS	NC CS1_B1	VSS	VDD	VSS	CK_c_B	VSS	VDD	VSS	NC CS1_B0	VSS	ALERT_B0
T	R	VSS	DQ5_B1	VDD	DQ2_B1	VSS	VSS	CK_t_B	VSS	VSS	DQ2_B0	VDD	DQ5_B0	VSS
U	T	DQ4_B1	VSS	DQ1_B1	VSS	NC DQS1_t_B	NC DQS1_c_B	VSS	DQS0_c_B	DQS0_t_B	VSS	DQ1_B0	VSS	DQ4_B0
V	U	VSS	DQ3_B1	VSS	DQ0_B1	VSS		VDDQ		VSS	DQ0_B0	VSS	DQ3_B0	VSS
W		NC/VSS	NC/VSS	NC/VDD	NC/VSS	NC/VDDQ		NC/VDD		NC/VDDQ	NC/VSS	NC/VDD	NC/VSS	NC/VSS

Figure 2 — DDR6 SDRAM X24 ~~DIMM~~ Module Common Ballout

2.1.2.1. Package BGA Definition and Description

Table 3 — BGA Definition and Description (Sheet 1 of 3)

Symbol	Type	Description	Note
RESET_N_[A,B]_xy	Input	**RESET:** Reset is active when RESET_n is LOW, and inactive when RESET_n is HIGH. RESET_n must be HIGH during normal operation. RESET_n is an asynchronous signal.	1,4,5,7
CK_t_[A,B]_x, CK_c_[A,B]_x	Input	**Clock:** CK_t and CK_c are differential clock inputs. The first crossing point is the rising (falling) edge of CK_t (CK_c) and second crossing point is falling (rising) edge of CK_t (CK_c). Command/Address and CS inputs are sampled on the crossing point that is the rising (falling) edge of CK_t (CK_c). One CK_t and CK_c pair are shared for both pseudo-split die (PSD0, PSD1).	1,4,7
CS[3:0]_A0_xy, CS[3:0]_A1_xy, CS[3:0]_B0_xy, CS[3:0]_B1_xy	Input	**Chip Select:** CS is part of the command dcode, is two UI the pulse width is 2CK and is sampled on the even rising (falling) edge of CK_t (CK_c) unless the device is in power-down mode where it becomes an asynchronous signal. All commands are masked when CS is registered LOW. Chip Select number signifies the Rank with a min of 1 and a max of 4. Unused/unavailable ranks are NC.	1,3,4,6,7

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Table 3 — BGA Definition and Description (Continued) (Sheet 2 of 3)

Symbol	Type	Description	Note
CA [5:0] _A0_xy, CA[5:0] _A1_xy, CA[5:0]_B0_xy, CA[5:0]_B1_xy	Input	**Command/Address Inputs:** CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. CA5 (6th CA) is an optional pin which provides Chip ID (CID) address inputs. If unused, this is either CS3 or NC depending on package configuration.	1,3,4,6,7
DQ[5:0]_A0_xy, DQ[5:0]_A1_xy, DQ[5:0]_B0_xy, DQ[5:0]_B1_xy	Input / Output	**Data Input/Output:** Bi-directional data bus.	1,3,4,7
DQS0_[t,c]_[A,B]_x, DQS1_[t,c]_[A,B]_x	Input / Output	**Data Strobe:** DQS_t and DQS_c are the differential output clock signals used to strobe data during a READ and differential input clock signals used to latch WRITE data. The differential crossing point is edge-aligned with READ data and centered for WRITE data. One DQS_t and DQS_c pair are shared for both pseudo-split die (PSD0, PSD1). DQS1 is NC for 2p6 and 1p12 package configurations.	1,4,7
ALERT_A0_xy, ALERT_A1_xy, ALERT_B0_xy, ALERT_B1_xy	Output	**ALERT:** If an error is detected and reported to the fault register, the ALERT encodes the type of error on the pin. See the Alert section for details. Each PSD has an Alert. Alert A1 and B1 will be NC for 1p12 configurations.	1,4,7
ZQ_[A,B]_x	Reference	**ZQ:** ZQ is used to calibrate the output drive strength and the termination resistance as calibration reference. There is one ZQ pad per die.	2,4,7
VDD, VDDQ, VPP	Supply	**Power supplies:** Isolated on the die for improved noise immunity. Includes the main power, DQ power and row activation power.	
VSS	Ground	**Ground Reference:** Power supply ground reference.	
RFU	Input / Output	Reserved for future use.	
NC		**No Connect:** No internal electrical connection is present.	
NA		**Not Active:** Internal package connection, not used/needed. Only applicable to block diagram die connection.	

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Table 3 — BGA Definition and Description (Continued) (Sheet 3 of 3)

Symbol	Type	Description	Note
NOTE 1: This pin is referenced to VSS. **NOTE 2:** In a multi-die package stack, one die will arbitrate for all dice. **NOTE 3:** Pseudo-Split Die 0 (PSD0) and Pseudo-Split Die 1 (PSD1) have independent pads. **NOTE 4:** Suffix A [1:0] and B [1:0] signify DRAM package IO Signal Group, X24 has two IO groups, X12 only uses Group A. Each group is split into two sub-groups [1:0]. **NOTE 5:** System designers must connect both RESET_N_[A,B] to a common net on PCB hardware for both X12 and X24 packages. **NOTE 6:** A1 and B1 nets are NC for 1p12 configuration. See package block diagram for details. **NOTE 7:** xy suffix illustrates package BGA to die mapping. x=die/dice and y = PSD within the die/dice. Suffix is not shown in combination BGA map. See block diagram(s) for further details.			

2.1.2.2. Package Connectivity Swizzle Rules

In order to keep package connectivity optimized for signal integrity, there may be cases where die to package connectivity may need to change. The following are acceptable signal swaps:

1) DQ[5:3] within the same PSD.
2) DQ[2:0] within the same PSD.

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3) Full PSD[1:0] which includes all associated signals (i.e. DQ, CA, CS, Alert).

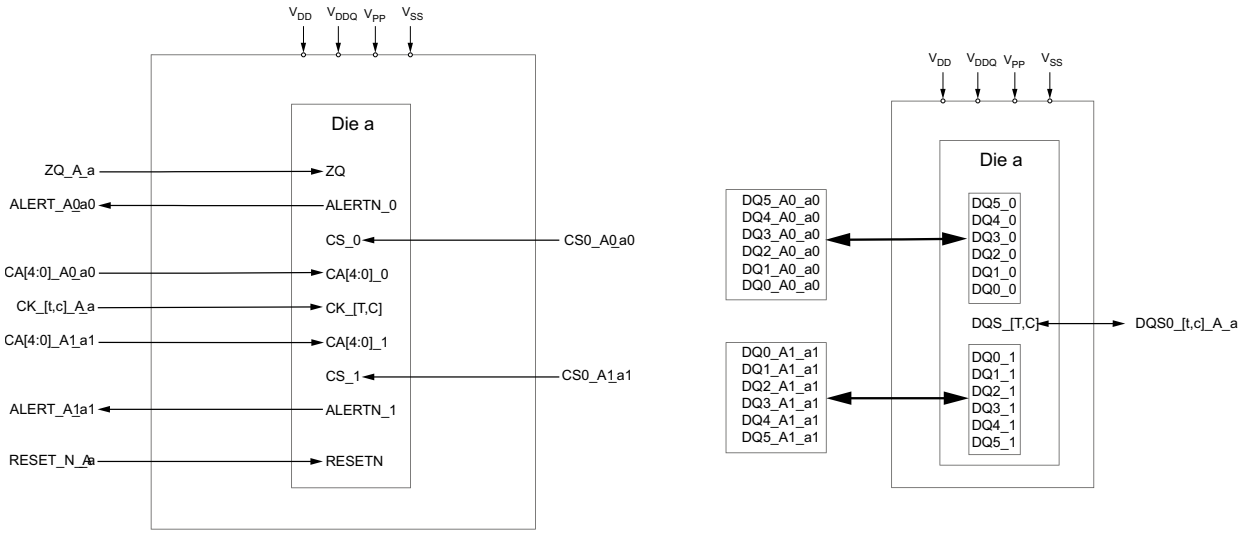
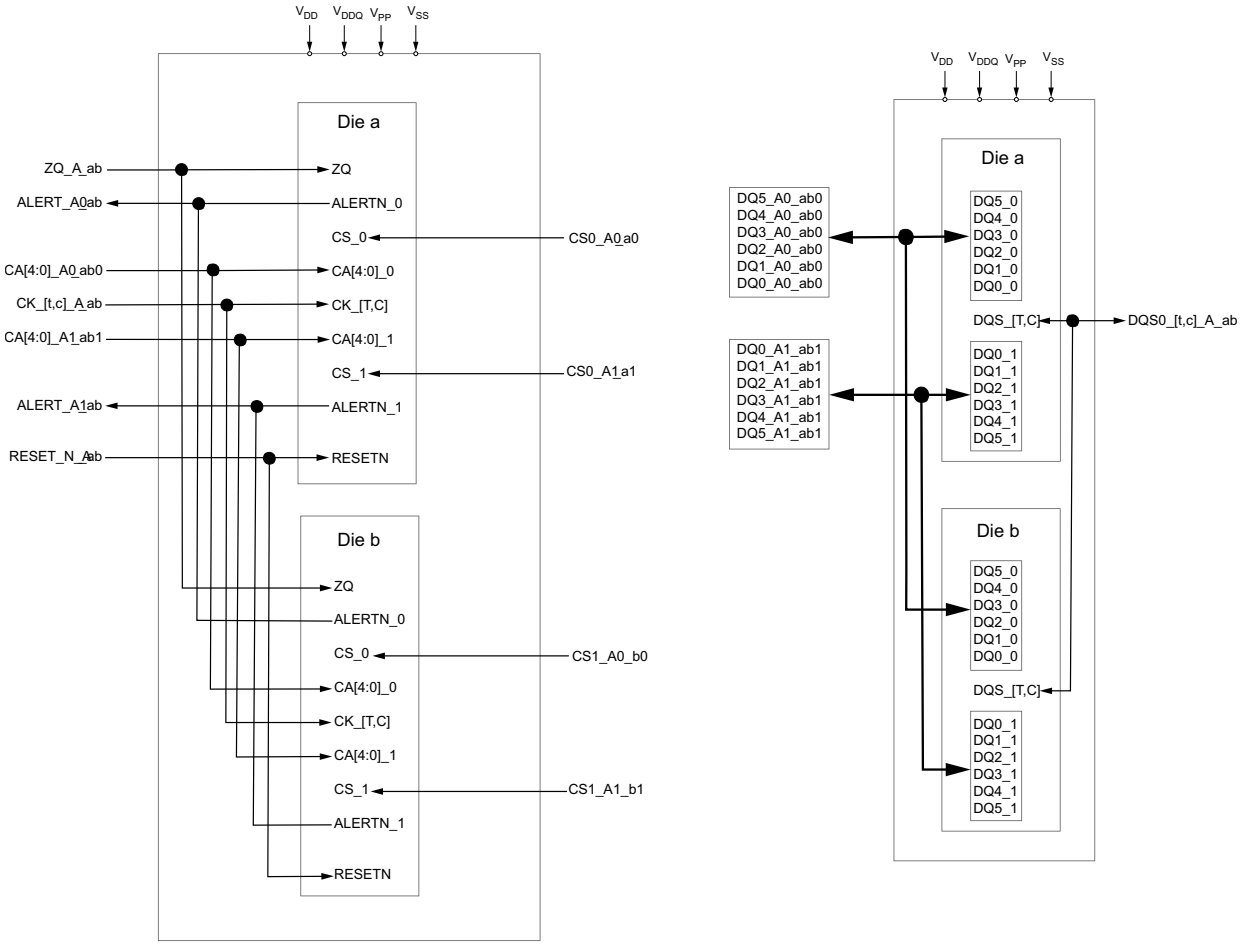
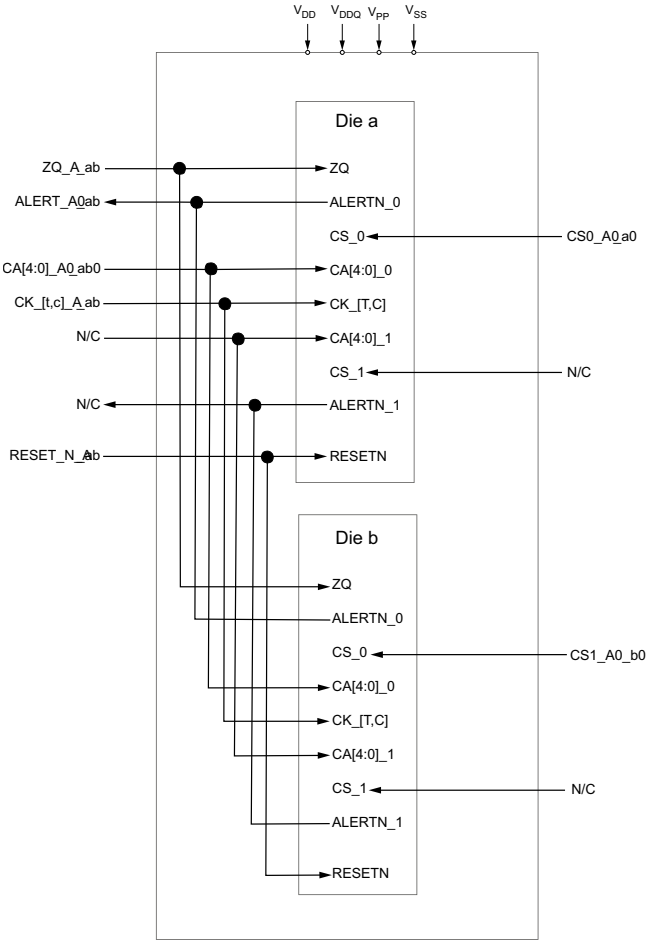


Figure 3 — 1R - 2p6 Block Diagram



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Figure 4 — 2R - 2p6 Block Diagram



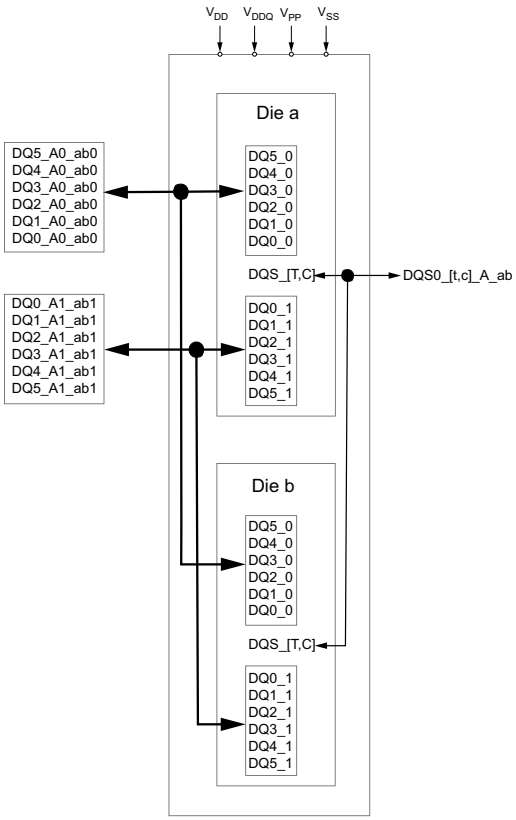


Figure 5 — 2R - 1p12 Block Diagram

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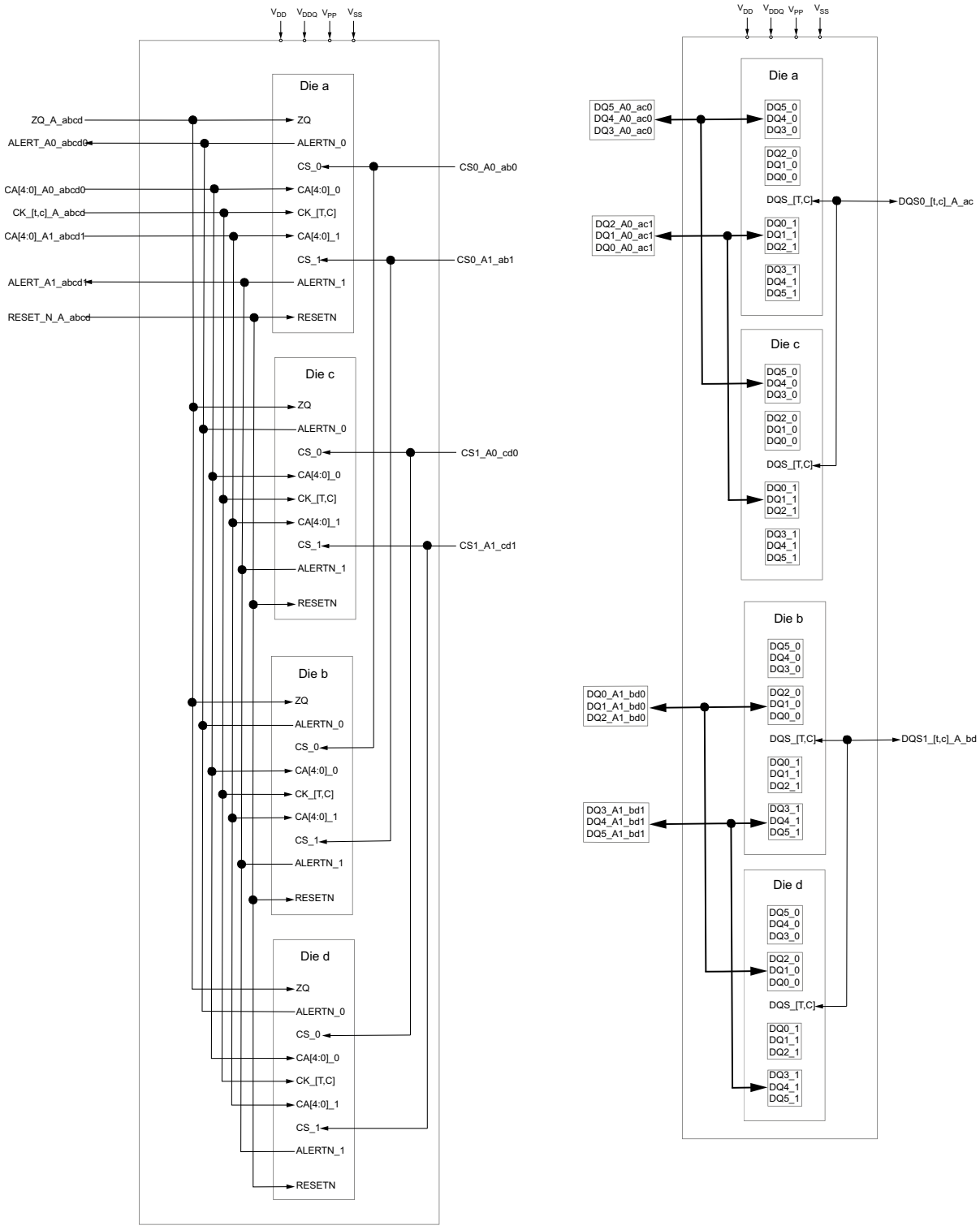


Figure 6 — 2R - 2p3 Block Diagram

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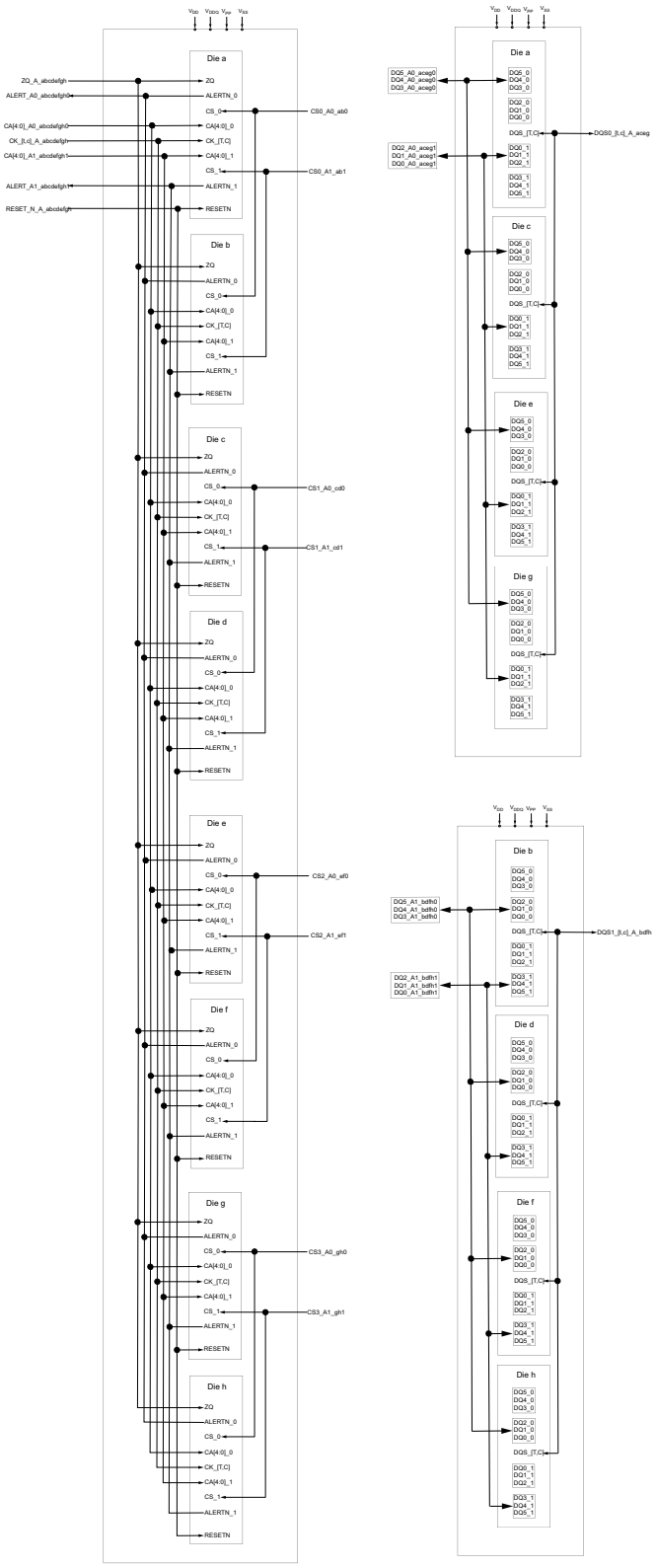


Figure 7 — 4R - 2p3 Block Diagram

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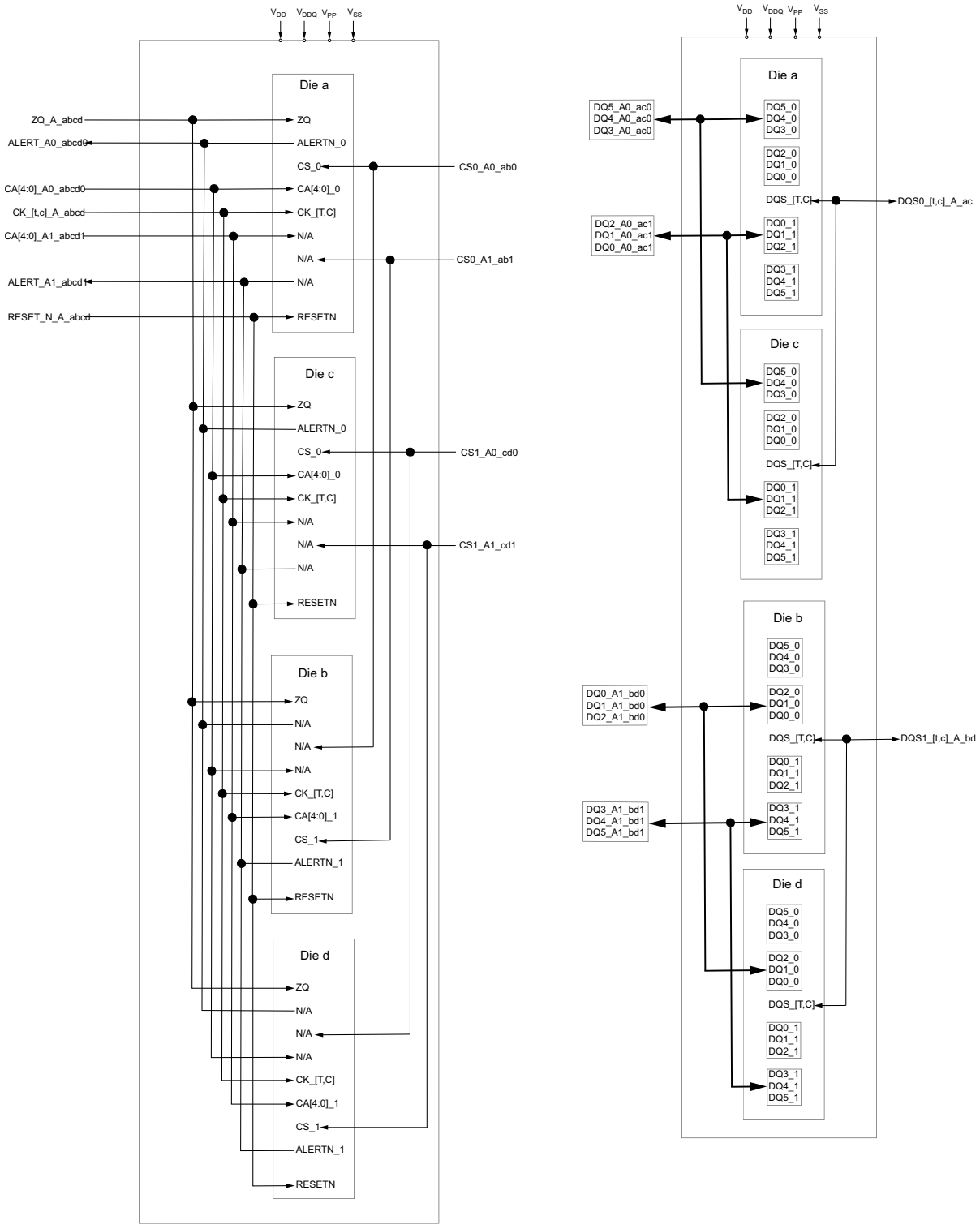


Figure 8 — 2R - 1p6 Block Diagram

2.1.3. Base Die Buffering (BDB) DRAM Package

2.1.3.1. Introduction

This section defines the features and functional operations of Base Die Buffering (BDB) DRAM package. DDR6 can support BDB DRAM which is desinged for high-density multi-die stacked package, facilitating reduced load on CK, CS, CA, and DQ/DQS signals for improved singal integrity as well as lower power relative to unbuffered DRAM package.

In multi-rank package, unbuffered DRAM has multiple indipendent DRAM dies wire bonded together, increasing electrical loading on the system channels. The BDB DRAM technology mitigates this with the “Base DRAM Die” (hereinafter “Base Die”) being a single electrical load, exclusively interfacing the memory controller for the stacked DRAM package no matter if the stack height is 2, 4, or 8. One or more vertically stacked “Core DRAM Dies” (hereinafter “Core Dies”) are directly connected to the Base Die through die-to-die interconnects within the package. This topology isolates external IO between the host and the base die and internal IO between the base die and core dies, improving memory subsystem SI and saving power from Core Dies by turning off redundant IO circuit blocks mainly built for external communications.

MRxx[y]=1 indicates that the DRAM is BDB.

Table 4 — Mode Register Information

Function	Register Type	Operand	Data	Notes
Base Die Buffering DRAM Package	RO	TBD	0: Not BDB DRAM 1: BDB DRAM	

Package implementation details are not in scope of this standard.

2.1.3.2. Fuctional Description

Addressing and Organizations

The BDB DRAM supports an extra CA pin, CA5, for CID address inputs which are used for determining the targeted logical rank that is desired to perform the given commands. Commands that have CID address bits are ACT, RD, WR, PRE, REF and RFM. For other commands that do not have CID address bits, the base die is the targeted logical rank unless otherwise specified.

Targeted logical rank is selected by CID[2:0] and DDR6 can support up to 8 logical ranks. The base die is associated with logical rank 0. The number of available logical ranks in package is provided by MRxx[y:z].

Table 5 — Logical Rank Organizations (Sheet 1 of 2)

CID[2:0]	Logical Rank
000	0
001	1
010	2
011	3
100	4

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Table 5 — Logical Rank Organizations (Continued) (Sheet 2 of 2)

CID[2:0]	Logical Rank
101	5
110	6
111	7

The functional behavior of logical rank(s) shall not deviate from that of physical rank(s) of unbuffered DRAM packages, except when noted in this standard.

Operation

The Base Die serves as the logical host interface for the entire stack. CA inputs are buffered and decoded centrally on the base die, which then redriven to the target core die(s) corresponding to the decoded CID.

Timing Parameters

Because of buffering operation in the base die and in package die-to-die interconnects between the base die and core dies, additional asynchronous delays are needed. TBDns is added to tAA. Different Logical Rank (DLR) timing parameters need to be defined in future ballots.

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2.1.4. Core Architecture

The core architecture will be a Pseudo Split Die (PSD) implementation with separate left and right DRAM arrays (each array is half of the total DRAM density) with a common IO block containing a shared CK and DQS.

For 2p3 and 2p6 configuration, each half array will have an independent Chip Select (CS) and Command/Address (CA) bus. Both half arrays will share a CK and DQS clocking port. The two PSDs cannot transfer data in opposite directions simultaneously as they share strobes; however, one PSD may begin to transfer data before the other. It is the Host controller's responsibility to provide proper turnaround times based on the last PSD transferring in one direction to the first PSD transferring in the other. One PSD may transfer data without the other transferring any data at all.

For the 1p6 configuration, each die will have a single CS, CA and CK/DQS clocking port per device.

For the 1p12 configuration, each die will have a single CS, CA and CK/DQS clocking port per device.

BL24 is the burst length for all configurations.

Example die architectures for each configuration is shown in the sections below.

NOTE: Within the following die diagrams, the placement of the DRAM array BGs/BA and pads are for illustration only.

The meaning of the abbreviations are as follows:

PSD (Pseudo-Split Die)

Portion of a DRAM die with independent chip select(CS), command/address (CA) bus and DQ bus, but a CK port (CK_t/c, DQS_t/c) which is common to both of PSDs

BG

Bank Group

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BA

Bank

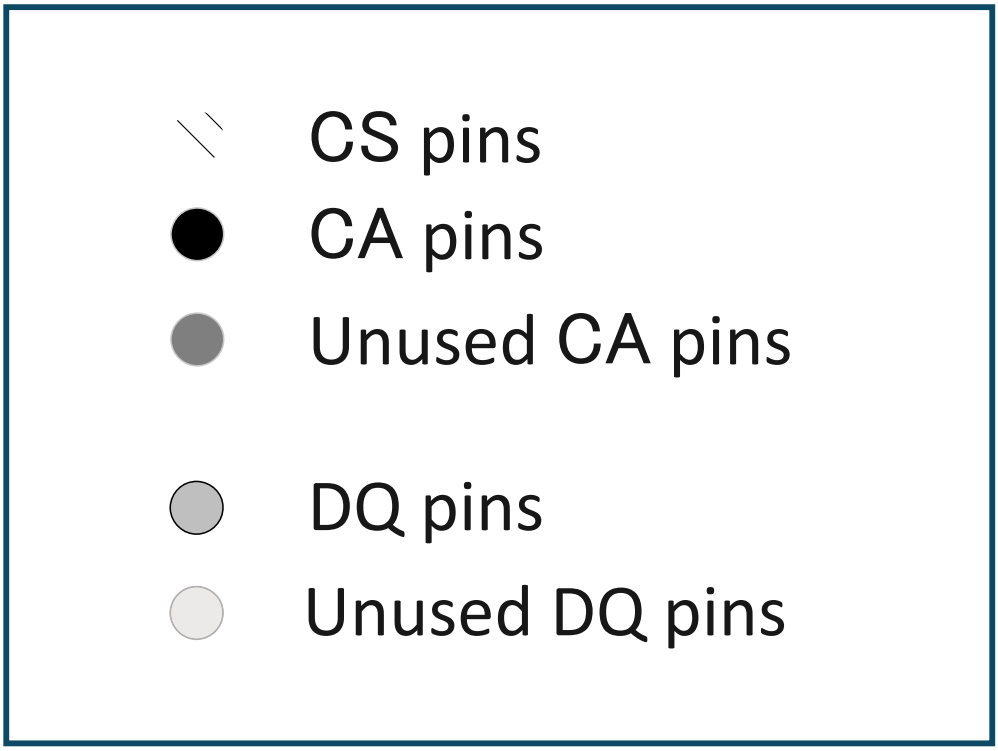


Figure 9 — Figure Descriptor

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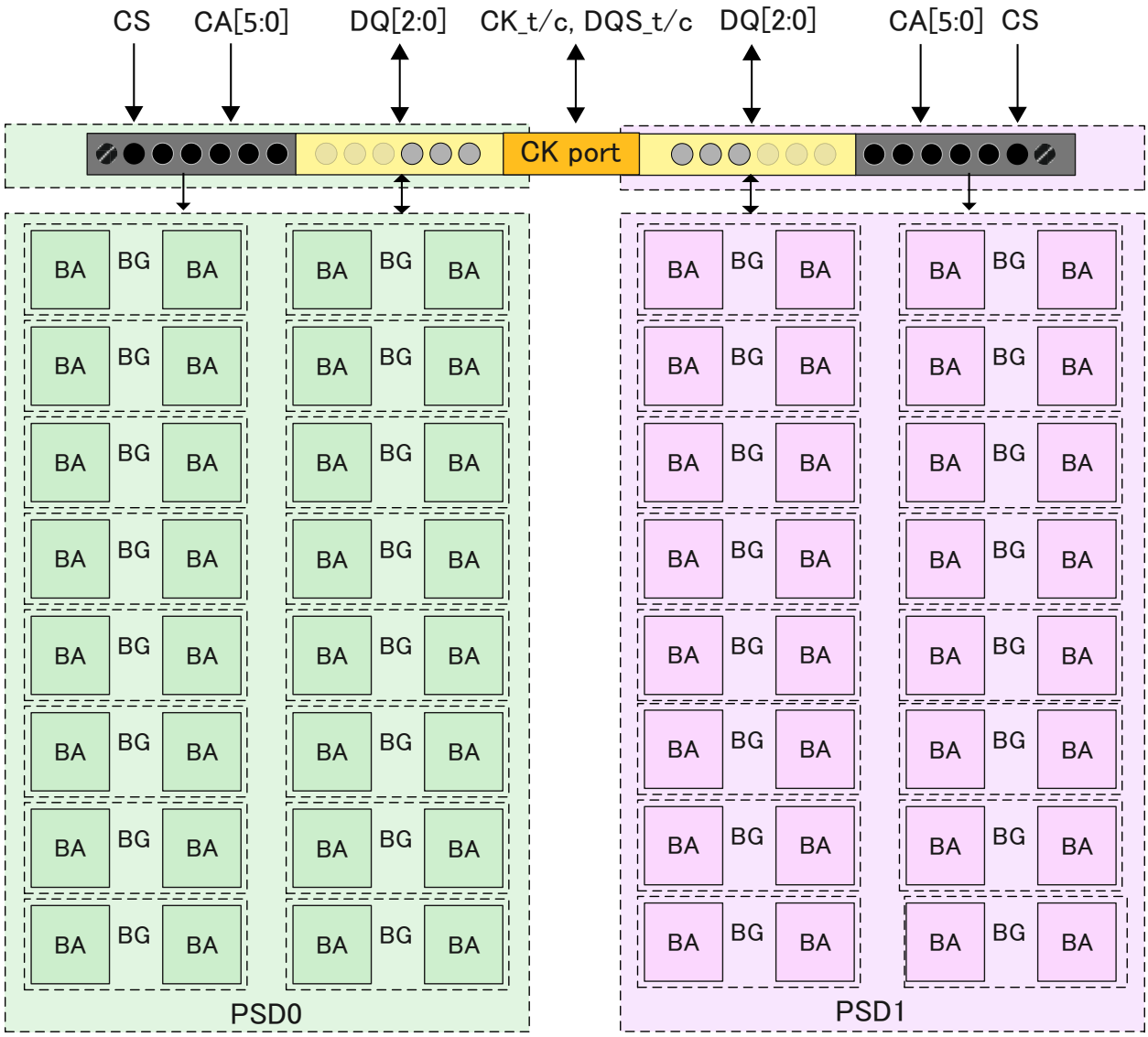


Figure 10 — 2p3 Die Architecture

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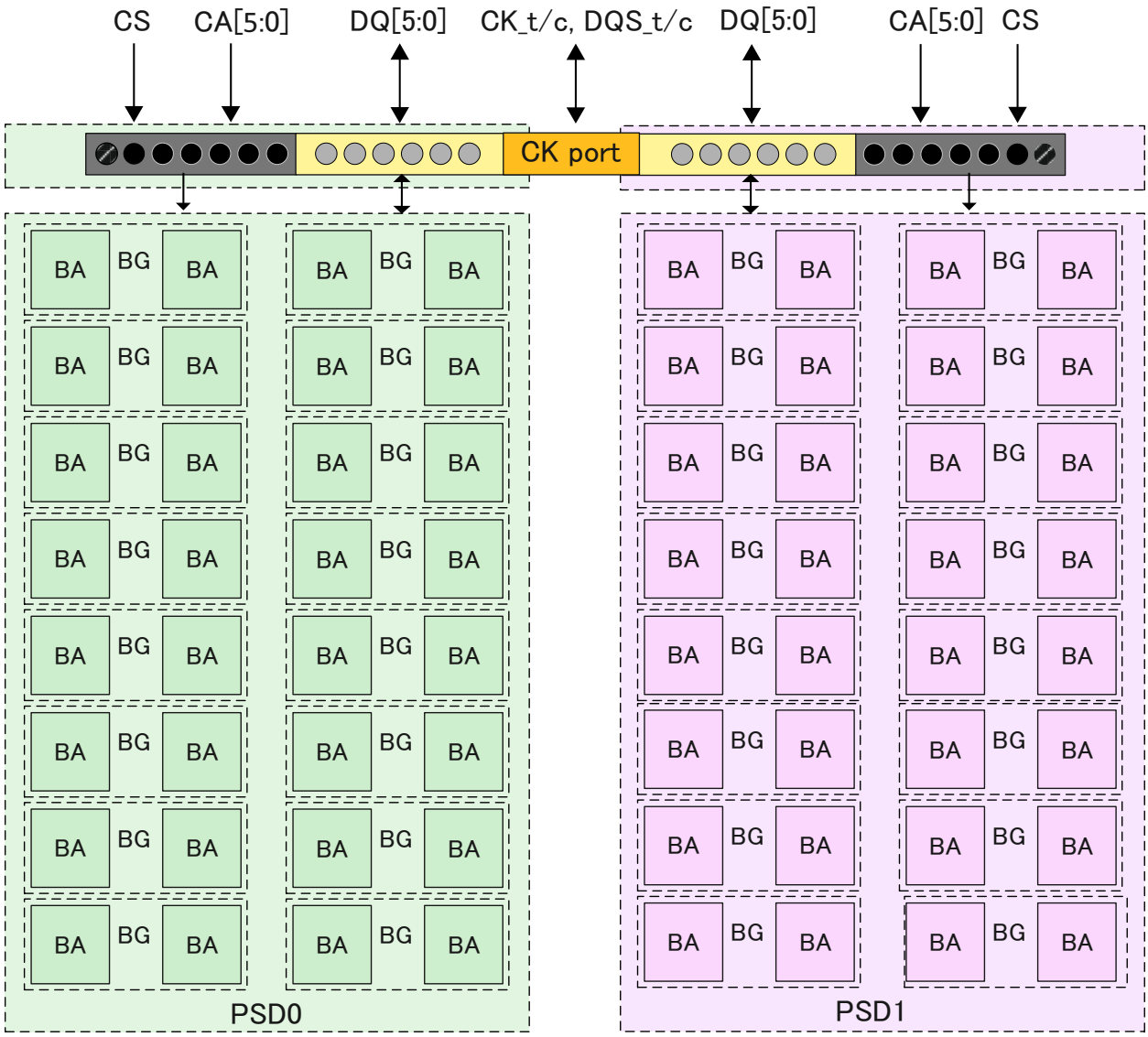


Figure 11 — 2p6 Die Architecture

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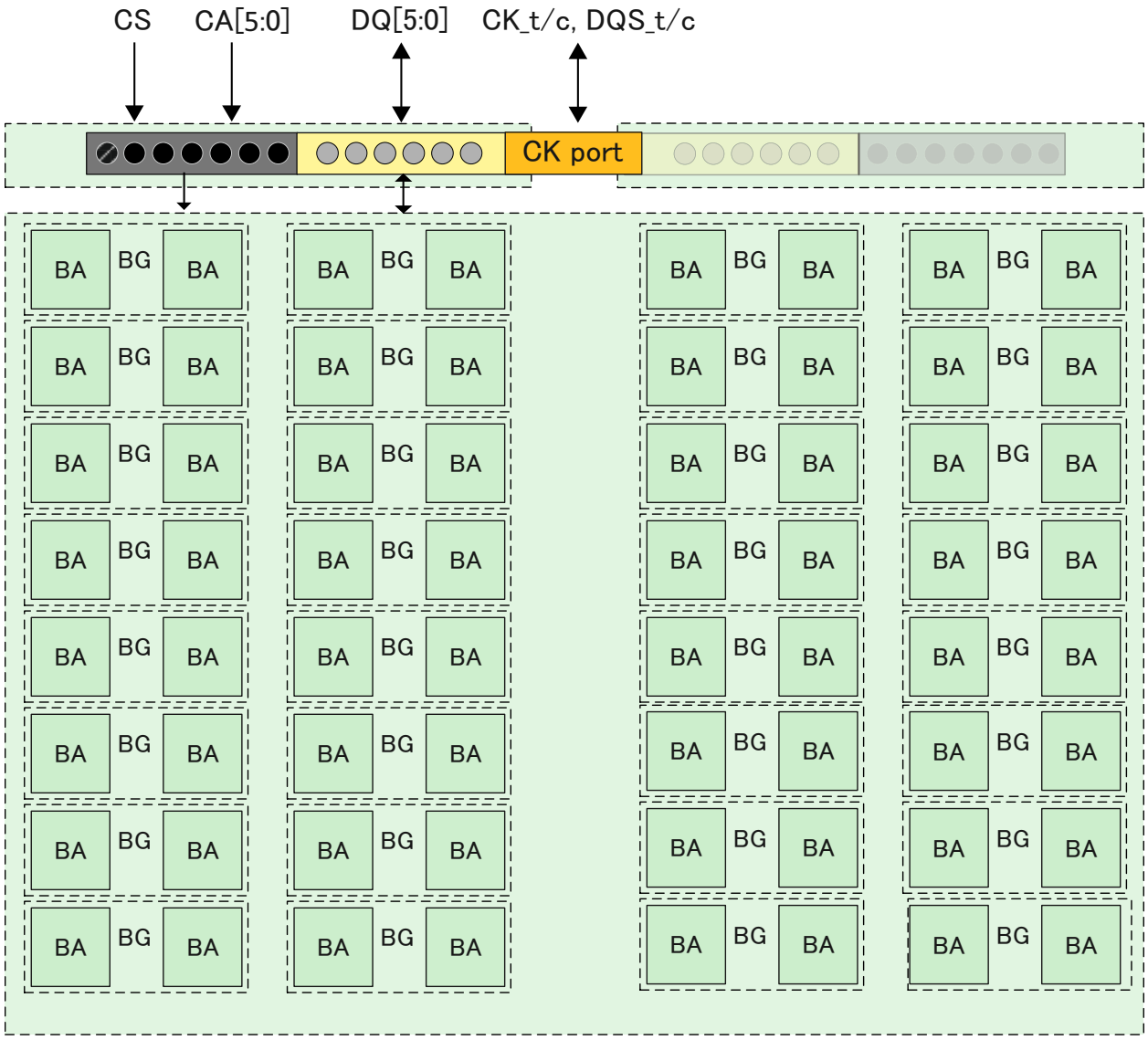


Figure 12 — 1p6 Die Architecture

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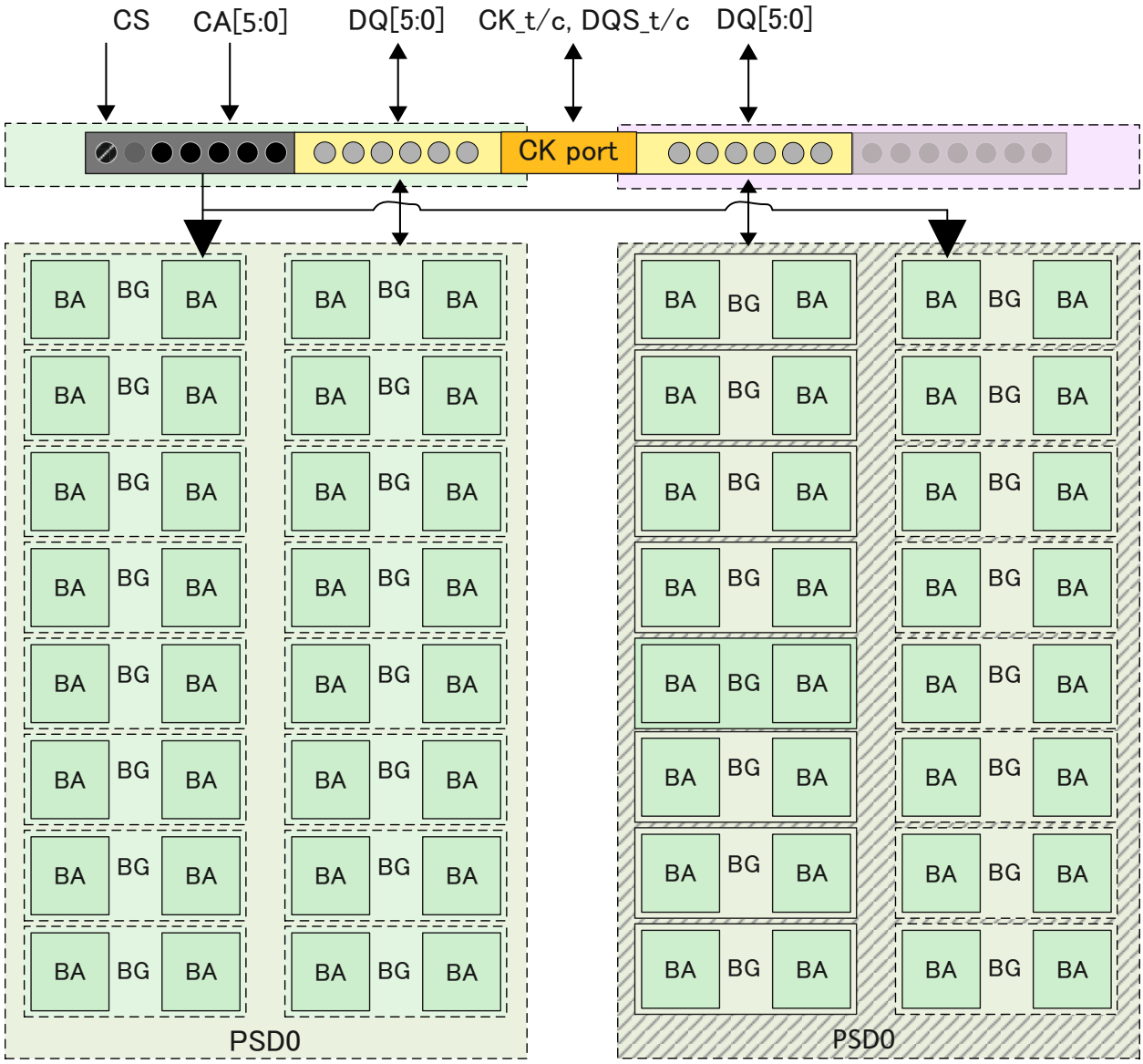


Figure 13 — 1p12 Die Architecture

NOTE: In Figure 1p12, CA[5] is shown grayed out because CA5 is not functionally required for 1p12 devices. The connection is included solely for consistency with combo devices.

2.1.5. Addressing Tables

Table 6 — 2p3 Configuration

Die Density	32Gb	48Gb	64Gb	96Gb	128Gb
Number of PSDs	2	2	2	2	2
Number of banks in BG	2	2	2	2	2
Number of bank groups	16	16	16	16	16
Number of rows	65,536	98,304	131,072	196,608	262,144
Number of columns	128	128	128	128	128
Page size (bytes)	1024	1024	1024	1024	1024
Density per PSD (bits)	17,179,869,184	25,769,803,776	34,359,738,368	51,539,607,552	68,719,476,736
Bank address	BA	BA	BA	BA	BA
Bank group addresses	BG[3:0]	BG[3:0]	BG[3:0]	BG[3:0]	BG[3:0]
Row addresses	R[15:0]	R[16:0] (R15 = 0 when R16 = 1)	R[16:0]	R[17:0] (R16 = 0 when R17 = 1)	R[17:0]
Column addresses	C[6:0]	C[6:0]	C[6:0]	C[6:0]	C[6:0]
Chip ID (CID)	CID[2:0]	CID[2:0]	CID[2:0]	CID[2:0]	CID[2:0]
Native burst length	24	24	24	24	24
NOTE: For non-binary memory densities, only one quarter of the row address space is invalid. When the MSB address bit is HIGH, then the MSB-1 address bit must be LOW.					

Table 7 — 2p6 Configuration (Sheet 1 of 2)

Die Density	32Gb	48Gb	64Gb	96Gb	128Gb
Number of PSDs	2	2	2	2	2
Number of banks in BG	2	2	2	2	2
Number of bank groups	16	16	16	16	16
Number of rows	65,536	98,304	131,072	196,608	262,144
Number of columns	64	64	64	64	64
Page size (bytes)	1024	1024	1024	1024	1024
Density per PSD (bits)	17,179,869,184	25,769,803,776	34,359,738,368	51,539,607,552	68,719,476,736
Bank address	BA	BA	BA	BA	BA
Bank group addresses	BG[3:0]	BG[3:0]	BG[3:0]	BG[3:0]	BG[3:0]
Row addresses	R[15:0]	R[16:0] (R15 = 0 when R16 = 1)	R[16:0]	R[17:0] (R16 = 0 when R17 = 1)	R[17:0]

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Table 7 — 2p6 Configuration (Continued) (Sheet 2 of 2)

Die Density	32Gb	48Gb	64Gb	96Gb	128Gb
Column addresses	C[5:0]	C[5:0]	C[5:0]	C[5:0]	C[5:0]
Chip ID (CID)	CID[2:0]	CID[2:0]	CID[2:0]	CID[2:0]	CID[2:0]
Native burst length	24	24	24	24	24
NOTE: For non-binary memory densities, only one quarter of the row address space is invalid. When the MSB address bit is HIGH, then the MSB-1 address bit must be LOW.					

Table 8 — 1p6 Configuration

Die Density	32Gb	48Gb	64Gb	96Gb	128Gb
Number of banks in BG	2	2	2	2	2
Number of bank groups	32	32	32	32	32
Number of rows	65,536	98,304	131,072	196,608	262,144
Number of columns	64	64	64	64	64
Page size (bytes)	1024	1024	1024	1024	1024
Density (bits)	34,359,738,368	51,539,607,552	68,719,476,736	103,079,215,104	137,438,953,472
Bank address	BA	BA	BA	BA	BA
Bank group addresses	PSD, BG[3:0]	PSD, BG[3:0]	PSD, BG[3:0]	PSD, BG[3:0]	PSD, BG[3:0]
Row addresses	R[15:0]	R[16:0] (R15 = 0 when R16 = 1)	R[16:0]	R[17:0] (R16 = 0 when R17 = 1)	R[17:0]
Column addresses	C[5:0]	C[5:0]	C[5:0]	C[5:0]	C[5:0]
Chip ID (CID)	CID[2:0]	CID[2:0]	CID[2:0]	CID[2:0]	CID[2:0]
Native burst length	24	24	24	24	24
NOTE 1: For the 1p6 Configuration, the CA "PSD" bit becomes a Bank Group and the full device is treated as a single channel. **NOTE 2:** For non-binary memory densities, only one quarter of the row address space is invalid. When the MSB address bit is HIGH, then the MSB-1 address bit must be LOW. **NOTE 3:** BGs with different PSD addresses may have separate command timing constraints between PSDs. See the Command Timing Constraints section for more details.					

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2.2. Speed Grade

2.2.1. Speed Grade

DDR6 SDRAM devices support multiple speed grades defined by their maximum data rate in MT/s (Mega Transfers per second). Speed grades represent the maximum data rate capability of the device. Higher speed grade devices can be operated at lower data rates with appropriate timing adjustments.

Table 9 — DDR6 Speed Grade

Parameter	Speed Grade						Unit
Data Rate[1]	10400	11600	12800	14400	16400	19200	MT/s
tCK(avg)[2]	192.3	172.4	156.3	138.9	122.0	104.2	ps

NOTE 1: 6400, 8000, 9200, and 9600 MT/s are supported as performance down-bins
NOTE 2: tCK(max) shall be less than or equal to 416.7ps
NOTE 3: Higher speed grade devices may be operated at any lower supported data rate with timing parameters adjusted according to Mode Register settings for the target frequency

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2.3. Data Packet Format

The data packet format is based on burst length 24. The entire data packet is divided into 8-bit sub-packets. In the following figures, numbered bit locations are data bits. Shaded bit locations depend on device configuration and Mode Register settings. In the figures, the green and blue bit locations may be Metadata bits or CRC bits, which is selected by Mode Registers upon initialization. (Reference to Metadata section and CRC section.)

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63								
DQ3	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ4	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ5	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127								

Figure 14 — 2p6 and 1p6 Data packet Format

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63								

Figure 15 — 2p3 Data packet Format

2.3.1. Metadata Format

DDR6 supports a Metadata feature for Write and Read operations. In this mode, the data and Metadata bits are written/read in parallel to/from the array. The Metadata feature supports the DRAM storing 0, 8 or 16 bits of Metadata as MD_OFF, MD_8b and MD_16b, respectively, where one of these modes is selected by MRxx:OP[y] at initialization.

For 2p6 and 1p6 DRAM configurations, Metadata sub-packets are 8 bits. When MD_OFF is selected, both sub-packets are non-Metadata (Non MD). Non MD bits are not stored in the DRAM array, but are available for other functions like CRC. MD_8b mode enables blue bit locations for Metadata and green bit locations for Non MD, as shown in the tables below. MD_16b mode enables both blue and green bit locations for Metadata, also as shown in the tables below.

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For the 2p3 DRAM configuration, the sub-packets are half that of 2p6 and 1p6. The tables below show the Metadata sub-packets as 4 bits each, with their respective assignments for MD_OFF, MD_8b and MD_16b.

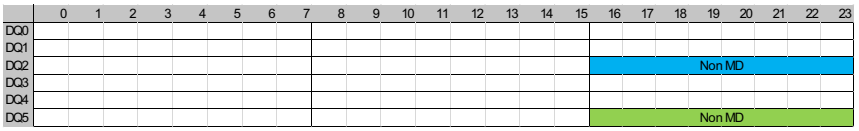


Figure 16 — MD_OFF mode for 2p6 and 1p6

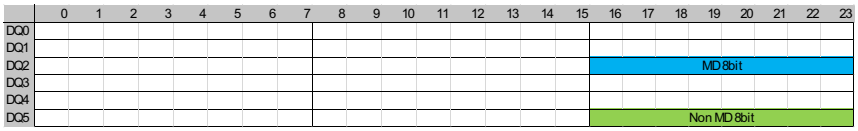


Figure 17 — MD_8b mode for 2p6 and 1p6

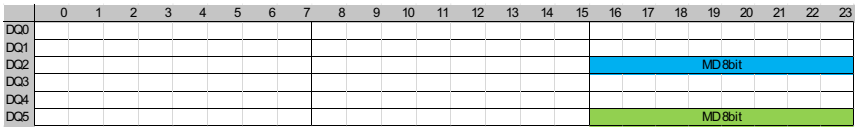


Figure 18 — MD_16b mode for 2p6 and 1p6

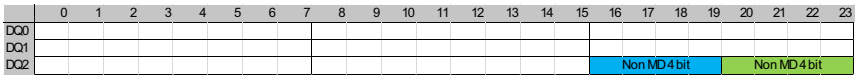


Figure 19 — MD_OFF mode for 2p3

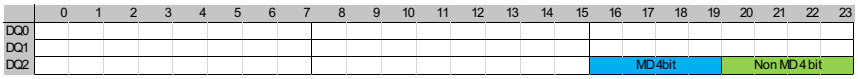


Figure 20 — MD_8b mode for 2p3

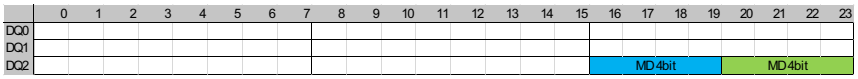


Figure 21 — MD_16b mode for 2p3

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2.3.2. Metadata Contents

TBD

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3. Clocking

DDR6 SDRAM's command and address interface operates from a differential Clock pair (CK_t/CK_c). Commands and Addresses are registered single data rate (SDR) at every rising edge of CK_t and falling edge of CK_c. CS is sampled at every other rising edge of CK_t and the falling edge of CK_c.

DDR6 uses a double data rate (DDR) data interface. The data interface uses two different types of clock pairs for Read (CK_t/CK_c) and Write (DQS_t/DQS_c). CK_t and CK_c are used to drive DQ Read, and they are source synchronous to the DQS during Read operation. The DQS is bi-directional. The DQ Write data is registered by DQS at every rising edge of DQS_t and rising edge of DQS_c for write operation.

DDR6 SDRAM uses an un-matched DQS-DQ path for Write to enable high speed performance and save power in the SDRAM. Therefore, a per-pin training by a host is required to adjust a timing of each DQ signal relative to DQS. This will allow a data eye to arrive at a SDRAM DQ receiver latch being centered on the DQS transition.

CA is matched with CK, and CAs skew training shall be used to compensate the increased CA and CK path delay.

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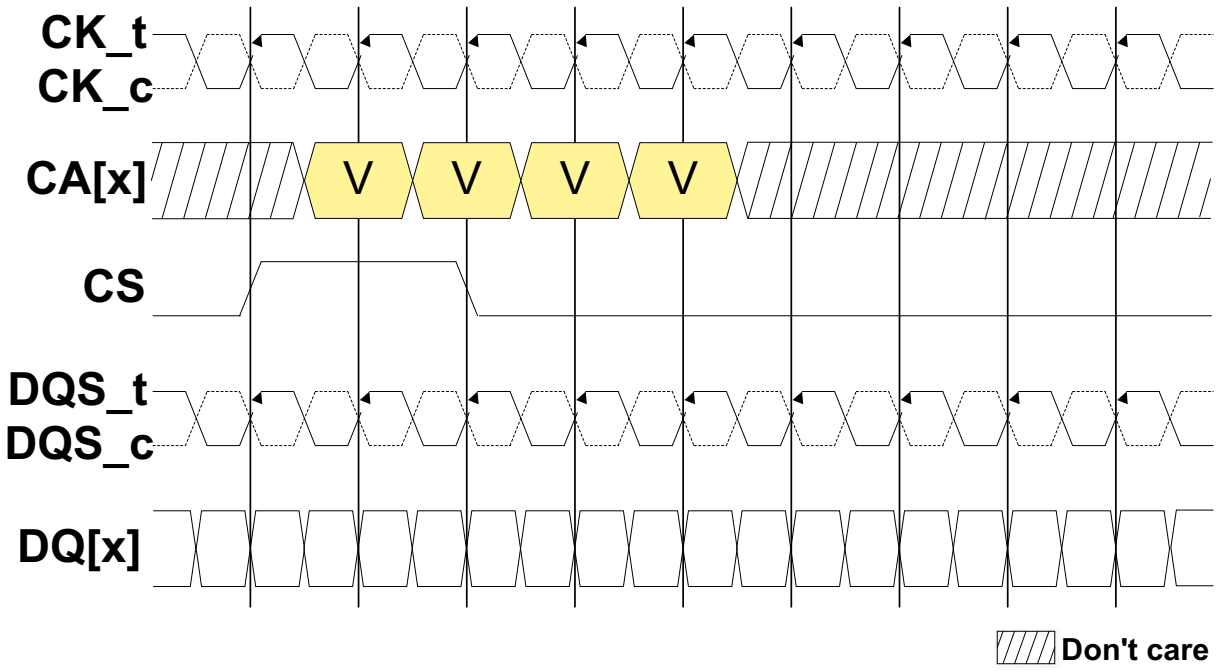


Figure 22 — Clocking and Interface Relationship: Read Operation

NOTE: A nominal example of the signal pin alignments after CSTM and CATM

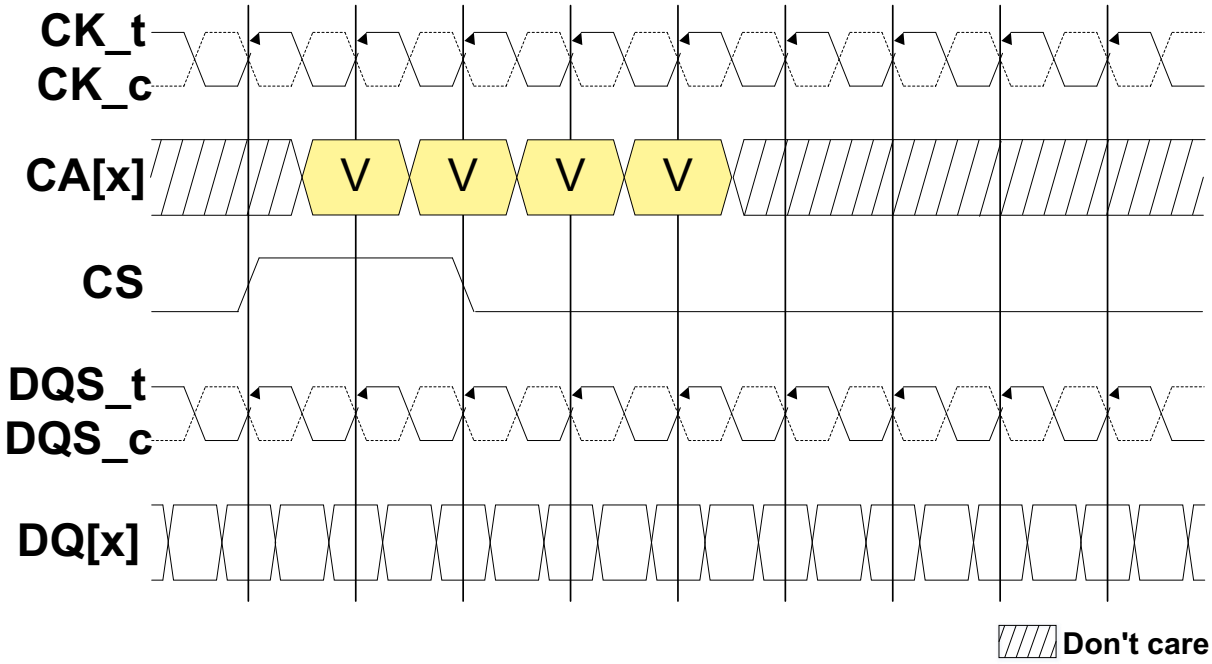


Figure 23 — Clocking and Interface Relationship: Write Operation

NOTE: A nominal example of the signal pin alignments after CSTM and CATM

Table 10 — Example Clock and Interface Signal Frequency Relationship

Pin	Speed	Unit
CK_t, CK_c	5200	MHz
Command/Address	5200	Mbps/pin
Chip Select	2600	Mbps/pin
DQS_t, DQS_c	5200	MHz
DQ	10400	Mbps/pin

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4. Initialization and Training

4.1. Power-up and Initialize sequence

For power-up and reset initialization, in order to prevent DRAM from functioning improperly, default values for the following MR settings need to be defined.

Table 11 — MR default settings

Item	Mode Register	Default Setting	Description
FSP-OP/WR	MR00[3:2]	00_B	FSP-OP/WR[0] is enabled
Frequency	MR09[3:0]	0000_B	4800~6400 Mbps
CAS Latency (CAS Write Latency)	MR08[5:0]	00000_B	68nCK (64nCK)
Parity Latency	MR09[6:4]	000_B	0nCK
nWTP / nRTP	MR0A[3:0]	0000_B	40nCK / 32nCK
CAODT	MRBD[2:0]	CA-ODT is based on CAOTP value[1]	RTT_OFF when CAOTP=L 001_B when CAOTP=H
CSODT	MRBD[6:4]	CS-ODT is based on CSOTP value[1]	RTT_OFF when CSOTP=L 001_B when CSOTP=H
CKODT initialized with CK frequency up toTBD Gbps	MR15[2:0]	CK-ODT is based on CKOTP value[1]	RTT_OFF when CKOTP=L 100_B when CKOTP=H
CKODT initialized with CK frequency higher than TBD Gbps	MR15[2:0]	CK-ODT is based on CKOTP value[1]	RTT_OFF when CKOTP=L TBD_B when CKOTP=H
VREF(CA) Value	MR84[6:0]	1001001_B	VREF(CA):46.5% of VDDQ
VREF(CS) Value	MR83[6:0]	1001001_B	VREF(CS):46.5% of VDDQ
Vref(DQ) Value	MR85[6:0]	1001001_B	VREF(DQ):46.5% of VDDQ
DQS_RTT_PARK	MR18[5:3]	TBD	TBD
RTT_PARK	MR18[2:0]	TBD	TBD
RTT_WR	MR19[2:0]	TBD	TBD
RTT_NOM_WR	MR19[5:3]	TBD	TBD

NOTE 1: CK/CS/CA OTP(One-Time Programmable) registers are set to 0 or 1, by vendor implementation at RESET sequence.

4.1.1. VREFCA, VREFCS and CK/CS/CA ODT control

DDR6 SDRAM supports CKOTP, CSOTP and CAOTP for default operation. Since there are various types of packages and modules, DDR6 internally have OTP registers for CA, CS and CK for initial state. Default RTT value of CA/CS signals, as seen from the package balls, is set to 240Ω by OTP and MR definitions, regardless of the number

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of dies in the package, by enabling CA/CS ODT in a single die. When all dies enable CK ODT in the initial state, the effective RTT becomes the default RTT divided by the number of dies.

Since appropriate VREF level and ODT values are affecting each other, VREF and ODT values need to be changed simultaneously with single command to avoid abnormal operation. DDR6 SDRAM supports Apply function for VREF, and ODT MR for CK, CS, and CA. When MRUPDATE is issued to change VREF or ODT MR, it's stored in shadow-registers. When Apply MRUPDATE is issued, values stored in corresponding shadow-registers are transferred to corresponding normal-registers, and impact on function and MRR operation. Every shadow register's RESET default value is same as normal-registers RESET default value.

Changing CK/CS/CAODT or Apply operations toward these registers are only allowed in MRUPD mode. Other than MRUPD mode, using MRW command, changing VREFCS/CA is allowed, but it directly impact on normal-registers without changing shadow-registers.

In 2PSD configuration, VREFCA, VREFCS, CA ODT, CS ODT normal/shadow registers are PSD Mode Registers, while CK ODT normal/shadow registers are Shared Mode Registers. Apply operation is Functional Mode Registers which is in Shared Registers. So, in 2p3 or 2p6 configurations, VREFCA, VREFCS, CA ODT and CS ODT registers shall be controlled by each of PSD, while CK ODT register can be controlled by either of PSD. Apply operation is shared by MRUPDATE command of either of PSD.

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Table 12 — Apply VREF/ODT definition

Function	Register Type	MA / OP	Data
Apply CK/CS/CA VREF/ODT	R/W	MR05[7:0]=0x80	Apply CK ODT
		MR05[7:0]=0x40	Apply CS ODT Apply VREFCS
		MR05[7:0]=0x20	Apply CA ODT Apply VREFCA

Shadow-registers are activated bypassing normal-registers in train state. When train-state in CSTM is entered, CS VREF and CS ODT MR's shadow-registers are activated without applying operation. When train-state in CATM is entered, CS VREF, CA VREF, CS ODT, CA ODT MR's shadow-registers are activated without applying operation. Bypassing operation is disabled when exit from train-state. Any changes on VREFCS and VREFCA in CSTM and CATM directly stored in shadow-registers and will be applied to normal-registers when Apply MRUPDATE is issued.

DDR6 SDRAM also supports CA-parity recovery mode, defined on MR[y] OP[a]. When this mode and CA parity check modes enabled, VREFCA shadow-register value is automatically applied to VREFCA normal-register when CA parity error occurs.

In physical multi-rank configuration, MRUPDATE command for Apply-VREF/ODT command needs to be issued simultaneously to all ranks. To accomplish successful transition of VREF and ODT, Apply operation is recommended to be issued in MRUPD mode, because broadcasting CS is only allowed in MRUPD mode.

4.1.2. Voltage Ramp and Device Initialization

The following sequence shall be used to power up the DDR6 SDRAM. Unless specified otherwise, these steps are mandatory.

1) While applying power (after Ta), RESET_n is recommended to be LOW (equal or less than 0.2 x VDD) and all other inputs shall be between VSS and VDD. The SDRAM outputs remain at High-Z while RESET_n is held

LOW. Power supply voltage ramp requirements are provided in Table 2. VPP must ramp at the same time or earlier than VDD. VDD must ramp at the same time or earlier than VDDQ.

Table 13 — Voltage Ramp Conditions

After	Applicable Conditions
Ta is reached	VPP must be greater than VDD
	VDD must be greater than VDDQ
NOTE 1: Ta is the point when any power supply first reaches 300mV. **NOTE 2:** Voltage ramp conditions in Table above apply between (Ta) and controlled Power-off. **NOTE 3:** Tb is the point at which all supply voltages are within their defined ranges. **NOTE 4:** Power ramp duration tINIT0(Tb-Ta) must not exceed 20ms.	

2) Following the completion of the voltage ramp (Tb), RESET_n shall be maintained LOW for a minimum of tINIT1, after which RESET_n can be de-asserted to HIGH(Tc). DQ, DQS_t, DQS_c, voltage levels shall be between VSS and VDDQ to avoid latch-up.

3) At least tINIT2 before RESET_n de-assertion, CS level is required to be VSS to prevent malfunction.

4) After RESET_n is de-asserted (Tc), CK/CS/CA/DQ and DQS ODT state should be valid following the default states.

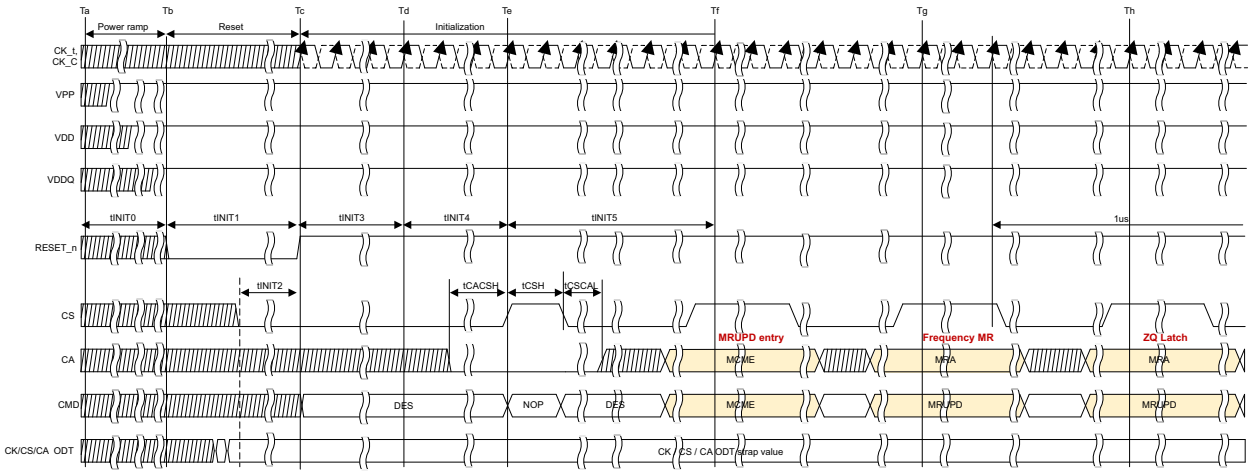


Figure 24 — Power Ramp and Initialization Sequence

NOTE: 1: Training is optional and may be done at the system architect's direction. The training sequence after ZQ Latch in this figure (Th) is simplified recommendation and actual training sequence may vary depending on systems.

NOTE: 2: Initial ZQ Calibration is started automatically by DRAM when RESET_n goes high after tINIT1 and is completed before Td

NOTE: 3: DDR6 SDRAM initialize on IDLE state after CS Pulse. IDLE state shall be maintained for tINIT5 before entering into MRUPD mode

NOTE: 4: In 2PSD configuration, Initialization pulse and MCME shall be issued for both PSDs.

5) CK_t and CK_c are required to be toggled or valid to be complementary level at least tINIT4 before CS receives one toggling (Td).

6) After tINIT4, wait minimum of tINIT5 to issue MCME commands to let DDR6 into MRUPD mode.(Tf)

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7) Between Tg to Th, the following initial configuration modes shall be completed prior to other training modes:
   a) MRUPDATE for setting Frequency MR (MR9) shall be issued. Input clock shall be sustained for tDLYCAL (Tg)
   b) MRUPDATE to execute ZQ Latch shall be issued before entering CSTM. (Th)

8) Host can also change CK/CA/CS ODT and CA/CS VREF values to change SI conditions in MRUPD mode. It is recommended to configure VREF and ODT registers to appropriate level and issue Apply MRUPDATE.

9) After CS/CA training is done, host shall change Mode Registers for RL(WL), nWTP(nRTP) to appropriate values with input clock frequency, before host start the Read and Write related training sequences.

10) Only in MRUPD mode can the host change the clock frequency without Full-PD with Frequency change. Frequency change sequence shall be done accordance with FSP and switching specification.
   a) Host sets FSP-WR to 1.
   b) Host sets Frequency MR that matches the 2nd clock frequency.
   c) If needed, host sets CK/CA/CS ODT and CA/CS VREF values for 2nd clock frequency on shadow registers, and issues CK/CA/CS Apply MRUPDATE to set 2nd FSP registers.
   d) Host issues MRUPDATE to change FSP-OP to 1 and turn off the clock after tCKFC_E. From the rising edge of CK_t (at cycle Tb3) until the rising edge of CK_t at the end of tCKFC_X timing (Tg0), only DES command is allowed and CS should keep low.
   e) If FSP_OP transition includes changes on VREFCA or VREFCS value, following command (Tg0) after the clock frequency change need to be delayed satisfying VREF transition specification.
   f) The period after turning off and before turning on the clock, CK_t and CK_c shall be driven differentially.
   g) The clock with new frequency shall start tCKFC_X before the next MRUPDATE.
   h) The host shall synchronize with DRAM with their internal clock phase by either CPS of CSTM or CK Sync Off functional mode followed by CK sync.

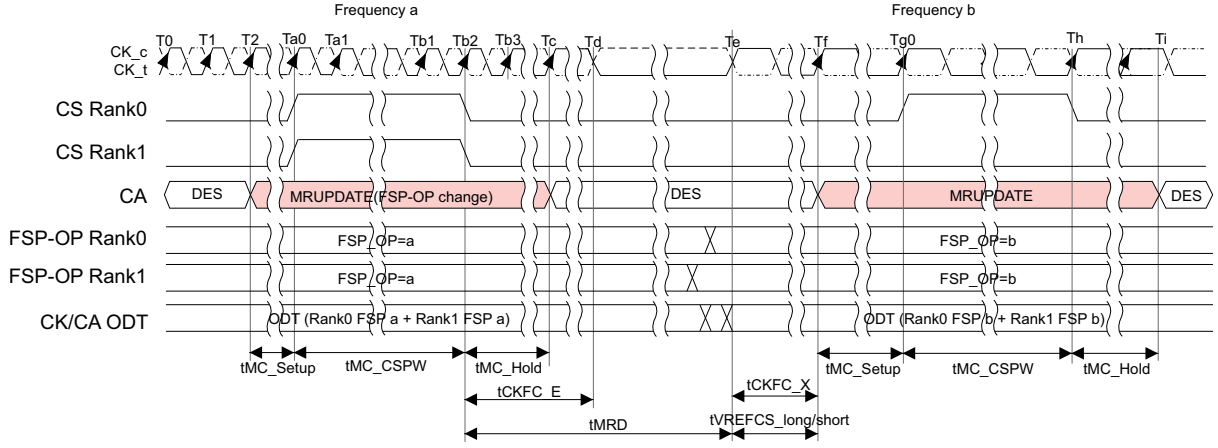


Figure 25 — Frequency change in MRUPD mode in multi-rank configuration

NOTE: 1: In multi-rank configuration, the MRUPDATE command to change FSP-OP shall be issued to all-Ranks simultaneously.
NOTE: 2: In 2p3 or 2p6 configurations, the MRUPDATE command to change FSP-OP shall be issued to either of PSDs.

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Table 14 — Initialization Timing Parameters

Parameter	Value		Unit	Comment
	Min	Max		
tINIT0	-	20	ms	Max voltage-ramp time at power up
tINIT1	200		us	Min RESET_n low time after completion of voltage ramp
tINIT2	10		ns	Min CS low time before RESET_n high
tINIT3	8		ms	Min CS low time after RESET_n high
tINIT4	5ns		ns	Min stable clock before the first CS high
tINIT5	2		us	Min IDLE time before the first MCME command
tZQLAT	Max(30ns,4nCK)		ns	ZQCAL latch quiet time
tDLYCAL	1		us	Valid Clock Requirement after the first Frequency MR set
tCKFC_E	Max(7.5ns, 16nCK)			Valid Clock Requirement after MRUPDATE command before changing Clock Frequency
tCKFC_X	Max(7.5ns, 16nCK)			Valid Clock Requirement after changing Clock Frequency before CA input for MRUPDATE command

4.1.3. Reset Initialization with Stable Power

The following sequence is required for RESET at no power interruption initialization.

1) Assert RESET_n below 0.2 x VDD anytime when reset is needed. RESET_n needs to be maintained for minimum tPW_RESET. CS must be pulled LOW at least tINIT2 before deasserting RESET_n.
2) Repeat steps 4 to 10 in 1.3

Table 15 — Reset Timing Parameters

Parameter	Value		Unit	Comment
Min	Max			
tPW_RESET	100		ns	Min RESET_n low time for Reset initialization with stable power

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4.1.4. DDR6 FSP and switching

The Frequency Set Points (FSP) allow the DDR6 SDRAM CK/CS/CA Bus to be switched between 2 different operating frequencies, with changes in voltage swings and termination values, without ever being in an untrained state which could result in a loss of communication to the SDRAM. This is accomplished by duplicating all CA Bus mode register parameters, as well as other mode register parameters commonly changed with operating frequency.

These doubled registers form 2 sets that use the same mode register addresses, with read/write access controlled by MR bit FSP-WR (Frequency Set-Point Write/Read) and the DRAM operating point controlled by another MR bit FSP-OP (Frequency Set-Point Operation). Changing the FSP-WR bit allows MR parameters to be changed for a selected Frequency Set-Point without affecting the DDR6- SDRAM"s current operation.

Once all necessary parameters have been written to the selected Set-Point, changing the FSP-OP bit will switch operation to use all of the new parameters simultaneously with Full PDE with frequency change, eliminating the possibility of a loss of communication that could be caused by a partial configuration change.

2 mode registers for each of the parameters above can be modified by setting the appropriate FSP-WR value, and how device operation can be switched between operating points by setting the appropriate FSP-OP value. The FSP-WR and FSP-OP functions operate completely independently.

Unlike normal-registers with FSP, shadow-registers are shared within FSPs. When MRUPDATE is issued to change VREF or ODT, it is buffered on shadow-register. When Apply MRUPDATE is issued, FSP-WR value specifies to which FSP the shadow value is applied.

Table 16 — FSP-WR and FSP-OP definition

Function	Register Type	MR address & Operand	Operation	Note
FSP-WR	R/W	MR00[2]	0_B: Data write to Mode Register N for FSP-OP[0] by MRW or MRUPDATE command. Data read from Mode Register N for FSP-OP[0] by MRR command. 1_B: Data write to Mode Register N for FSP-OP[1] by MRW or MRUPDATE command. Data read from Mode Register N for FSP-OP[1] by MRR command.	1
FSP-OP	R/W	MR00[3]	0_B:SDRAM operates with Mode Register N for FSP-OP[0] setting. 1_B: SDRAM operates with Mode Register N for FSP-OP[1] setting.	2
NOTE 1: FSP-WR stands for Frequency Set Point Writ/Read. **NOTE 2:** FSP-OP stands for Frequency Set Point Operating Point.				

4.2. Training

4.2.1. Multi-Cycle CS Mode Enable (MCME) Command

The Multi-Cycle CS Mode Enable (MCME) command is a single-function command to enable MRUPDATE Mode with multi-cycle CS (MCCS).

Prior to successfully completing the CS, VREFCS, CA and VREFCA trainings, or any time the status of the CS, CA and CK alignment is unknown, the CAs are driven with additional setup and hold beyond the CS assertion. For the device to latch the MCME in cases where the alignment between CS, CA and CK may be unknown, the CA inputs must reach the proper command state and provide tMC_Setup of setup prior to CS transitioning from LOW to HIGH, CS must remain HIGH for tMC_CSPW, and the CA inputs must remain in the proper command state and provide tMC_Hold of hold after CS transitions from HIGH to LOW.

While the DRAM may only decode CA[2:0] of the MCME command, all CA inputs are high (H) to ensure proper parity.

Following the issuing of the MCME command, tMCME_Delay needs to be met before the first cycle of an MRUPDATE begins. No commands are allowed during tMCME_Delay.

The MCME command timing diagram and timing parameters are shown in the figure and table below.

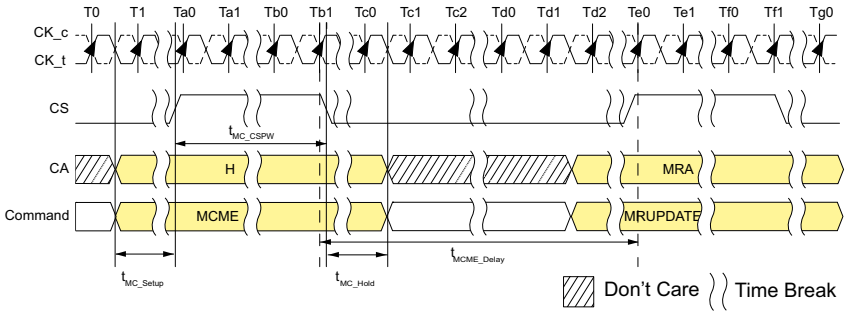


Figure 26 — Multi-Cycle CS Mode Enable (MCME) Command

Table 17 — Multi-Cycle CS Mode Enable (MCME) Command Timing Parameters (Sheet 1 of 2)

Parameter	Symbol	DDR6-6400 to DDR6-14400		>DDR6-14400		Unit	Note
		MIN	MAX	MIN	MAX		
CS Pulse Width to register the state of the CAs	tMC_CSPW	8	-	TBD	-	nCK	
Time between stable CAs and rising CS edge (setup)	tMC_Setup	8	-	TBD	-	nCK	
Time between falling CS edge and stable CAs (hold)	tMC_Hold	8	-	TBD	-	nCK	

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Table 17 — Multi-Cycle CS Mode Enable (MCME) Command Timing Parameters (Continued) (Sheet 2 of 2)

Parameter	Symbol	DDR6-6400 to DDR6-14400		>DDR6-14400		Unit	Note
		MIN	MAX	MIN	MAX		
Time between last CK before the falling CS edge of the MCME command and the first CK after the rising CS edge of the MRUPDATE	tMCME_Delay	tMRD	-	TBD	-	ns	

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4.2.2. Per DRAM Addressability (PDA)

Per DRAM Addressability (PDA) enables the programming of a selected device on a rank. The PDA applies to MRW commands and MRUPDATE Mode, except when entering PDA Enumeration Mode or disabling Mode Register Functional Modes (MR05), and when writing to the PDA mode registers. The Direct Addressable PDA Mode Register (MR02) contains the Enumeration ID which is common to both PSDs, while the Page Space PDA Mode Register MR82 contains the Select ID, which can be unique per PSD.

PDA is a CA interface-only method where a unique Enumeration ID can be programmed in MR02[3:0] and a per PSD unique Select ID is programmed in MR82[3:0]. Programming the unique Enumeration ID requires the use of the DQ signals and the PDA Enumeration Mode. The Enumeration ID is a 4-bit field, and the Select ID is also a 4-bit field. When the Select ID, written by an MRW command or MRUPDATE sequence, matches the Enumeration ID, or when the Select ID is set to the "All DRAM" code of 1111b, an MRW command or the MUPDATE sequence will be executed.

During the RESET procedure, the Enumeration ID and Select ID values are reset to the default 1111b.

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Table 18 — MR02 (PDA Enumeration Mode Register)

MR02				
Bits	**Attribute**	**Default**	**Description**	**Notes**
7:4	RV	0	MR02[7:4] Reserved	
3:0	R	Fh	MR02[3:0] Enumeration ID Read Only, programmed through PDA Enumeration Mode. xxxxb encoding is the Enumeration ID.	

Table 19 — MR82 (PDA Select Mode Register)

MR82				
Bits	**Attribute**	**Default**	**Description**	**Notes**
7:4	RV	0	MR82[7:4] Reserved	
3:0	RW	Fh	MR82[3:0] Select ID The Select ID is set by MRW or MRUPD sequence. Default setting is all DRAMs execute MRW/MRUPD. For all other encodings, the DRAM will execute MRW/MRUPD when Select ID = Enumeration ID, except to this Mode Register which ignores the PDA match and will always be executed.	

4.2.2.1. PDA Enumeration ID Programming

1) PDA Enumeration Mode is a Functional Mode within MR05 that can only be entered by the MRUPDATE sequence with MR05=06h after entering MRUPDATE Mode. PDA Enumeration Mode requires DQS to be low when entering and exiting the mode.
2) Upon entering the PDA Enumeration Mode, the DRAM will internally reset the DQS rising pulse count to 0 to prepare for counting 4 rising pulse edges (DQSR1 - DQSR4)
3) As shown in the timing diagram below, the second DQS rising edge (DQSR2) and the fourth DQS rising edge (DQSR4) are used to sample the data on DQ0 and DQ1

4) The sampled data is then combined to form the Enumeration ID of the device and is stored in the PDA Mode Register, MR02[3:0]

5) PDA Enumeration Mode is exited by latching "All Functions Disabled" in MR05 (MR05=00h).

Table 20 — PDA Enumeration ID Sampled Data

Enumeration ID Bit	DQ Source	DQS Sampling Edge
[3]	DQ[1]	2nd rising edge
[2]	DQ[0]	2nd rising edge
[1]	DQ[1]	4th rising edge
[0]	DQ[0]	4th rising edge

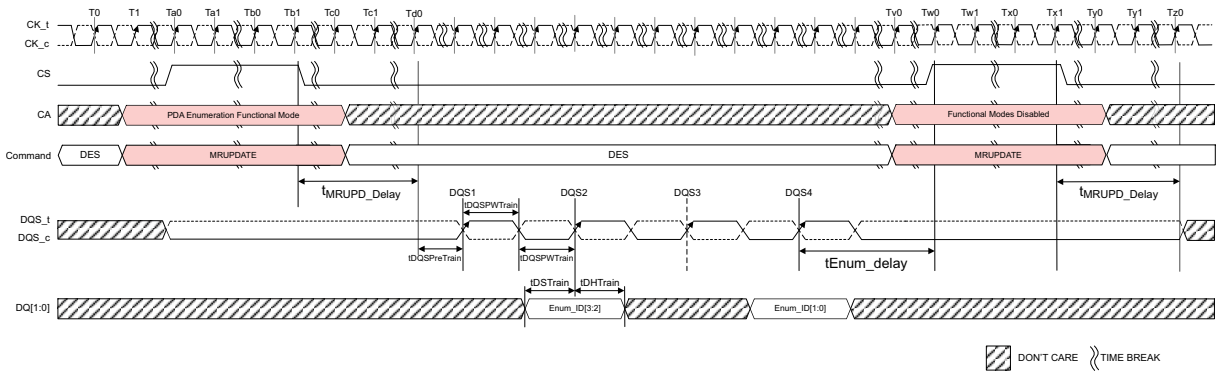


Figure 27 — PDA Enumeration Timing

Table 21 — Timing Parameters for PDA

Item Description	Parameters	Min/ Max	Value	Unit
DQ[1:0] setup	tDSTrain	Min	6	nCK
DQ[1:0] hold	tDHTrain	Min	6	nCK
DQS differentially low prior to 1st DQS rising edge	tDQSPreTrain	Min	6	nCK
DQS differentially low after to 4th DQS falling edge	tDQSPostTrain	Min	6	nCK
DQS pulse width	tDQSPWTrain	Min	6	nCK
Sampled data stored in Enumeration ID MR delay	tEnum_Delay	Min	tMRUPD_DELAY	nS

4.2.2.2. PDA Select ID Operation

Once the PDA Enumeration IDs have been programmed in all devices, the execution of future MRW commands and MRUPDATE sequences depend on the value of the Select ID. If the Select ID is set to 1111b, all DRAM will execute the command/sequence. For all MRW commands and MRUPDATE sequences, except entering PDA Enumeration

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Mode, disabling functional modes, and when writing to the PDA Select Mode Register (MR82), the Select ID will be compared to the Enumeration ID to determine if the command/sequence will be executed. Entering PDA Enumeration Mode, disabling functional modes, and writing to the PDA Select Mode Register will be executed regardless of the Select ID value.

As an example, the following sequence could be used to program unique PDA Select ID fields per device:

1) Send an MRW command with MR82[3:0] and the 'Select ID' opcodes with encoding 0000b
2) Send MRW commands with settings specific to device 0000b. This can be any number of MRWs.
3) Send an MRW command with MR82[3:0] and the 'Select ID' opcodes with encoding 0001b
4) Send MRW commands with settings specific to device 0001b. This can be any number of MRWs.
5) Repeat for any number of devices
6) Send an MRW command with MR82[3:0] and the 'Select ID' opcodes with encoding 1111b to enable execution of all MRW commands and MRUDPATE sequences on all devices

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4.2.3. DDR6 ZQ Calibration

Only Background ZQ Calibration mode is supported in DDR6. In Background Calibration mode, calibration of the output driver and CA/DQ ODT impedance across process, temperature, and voltage occurs in the background of device operation and is designed to eliminate any need for coordination among channels within a single package.

ZQ re-calibration may be required as the DDR6 SDRAM voltage and/or temperature changes due to changes in the system environment. ZQ calibration can only be performed when the VDDQ voltage is ~~at or above its nominal value of 0.6V~~ within the range allowed by the specification, otherwise the memory controller calibration shall halt the ZQ calibration.

Changing CK, CS andCA ODT values and/or DQ, DQS ODT values will not alter the existing recalibration scheme, therefore there is no need for immediate recalibration.

4.2.3.1. Calibration During Powerup and Initialization

ZQ calibration is automatically performed by all DDR6 dies in a package during the initialization/powerup sequence before Td, as shown in Figure xx (Power-up, Initialization and Power-off Procedure). A ZQCal Latch ~~command~~operation shall be initiated ~~issued~~ to all DDR6 dies that share a common clock, on or after Tg regardless of the state of the ZQ Update Flag (ZQUF). See Table 1 for calibration latency and timing.

Asserting ZQ Reset will set the calibration values to their default setting. When ZQ Stop is enabled, the ZQ resource is available for use by other devices.

DDR6 SDRAM package will have one or more ZQ initiator die(s) and ZQ target die(s). ZQ Interval MR setting is only applicable to ZQ Initiator die (MR7 OP[1]). These settings will be ignored by ZQ Target dies.

After the power up initialization and reset sequences have been completed, ZQUF MR7 OP[0] =0B.

DDR6 packages with more than one ZQ pin may include more than one ZQ Initiator die.

4.2.3.2. Background Calibration

In this mode, pull-down/ODT and pull-up calibration will be performed in the background and kept up to date by the DRAM. Re-calibration will be performed by the DDR6 SDRAM within the time interval, tZQINT, specified in MR7 OP[5:4].

Pull-down/ODT calibration is controlled by each DDR6 die using an external ZQ resistor connected between VDDQ and a package ball or pin (ZQ resources). These ZQ resources may be shared among a number of DDR6 dies up to a maximum of NZQ. Calibration will be automatically performed as part of power-up/initialization and after RESET assertion. Subsequent re-calibration will be kept up to date by the DRAM. Self-arbitration by the DRAM insures that up to NZQ dies within a package can share a common external ZQ resistor and avoid conflicts. Noise immunity will not be compromised when sharing the external ZQ calibration resistor.

When automatic pull-down/ODT calibration is complete, pull-up/Voh calibration will start automatically. At the completion of pull-down/ODT and pull-up/Voh calibration, MR7 OP[0] bit (ZQUF) will be set if the new calibration codes do not match the currently latched codes. An MRR of this bit will notify the memory controller that new calibration results are available and that a ZQCal Latch operation (following ZQCal Latch timing constraints) should be issued to ensure accurate calibration of Pull-down, ODT and Voh is consistently maintained. The setting of ZQUF is unique to each die regardless of configuration or sharing of the ZQ pin(s). Alternatively, the memory controller may choose not to monitor ZQUF and periodically initiate a ZQCal Latch operation by setting the appropriate functional mode in the MR6 register.

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Setting the MR7 OP[3] = ZQ Stop will halt all background calibration activity. Re-setting MR7 OP[3] to zero will immediately start a calibration sequence, where all DDR6 dies sharing the ZQ resource will recalibrate in a serial fashion.

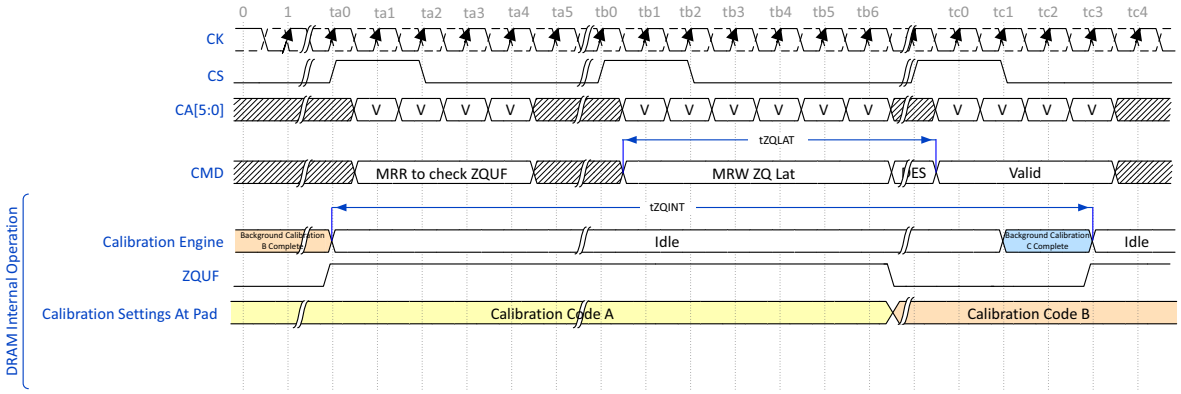


Figure 28 — Background ZQ Cal timing diagram

Latching ZQ Calibration Results in Background Calibration Mode

Latching ZQ calibration results is accomplished with the MRUPDATE sequence for ZQ Latch during initialization and MRW command during runtime as shown in Figure 1. This command loads new calibration results into the pull-down/ODT devices and pull-up drivers.

A ZQCal Latch ~~command~~ operation may be ~~issued~~initiated anytime outside of power-down when there are no DQ operations pending or in progress. The results from the most-recently completed calibration will always be latched with each ZQCal Latch ~~command~~mode register writes through MRUPDATE or MRW. If CA/CS ODT is enabled, the CS shall maintain a deselect state during tZQLAT to allow CA/CSODT calibration settings to be updated. Operations that initiate DQ operations are always dis-allowed during tZQLAT.

The DQ and CA/CSODT calibration values will not be updated until ZQCal Latch ~~command~~operation is performed and tZQLAT has been met, with the following exceptions: ZQ calibration will automatically occur for each die on power up/initialization and after Reset_n pin assertion. A ZQCal Latch ~~command~~ operation may be ~~issued~~initiated to all DDR6 dies with the shared clock, on or after Tg.

When a ZQCal Latch ~~command~~operation is executed the ZQUF bit will be reset to 0 before the expiration of tZQLAT.

Maintaining Accurate Calibration - Background Calibration Mode

To maintain Pull-down/ODT calibration and Voh calibration:

1) Periodically, based on tZQINT, issue an MRR to MR7 to check the ZQUF OP[0] bit status for each DDR6 die.
2) If MR06 OP[0]= 00000001B, issue a ZQCal Latch ~~command~~functional mode register write.

Repeat 1-2 as needed. Alternatively, the memory controller may choose not to monitor ZQUF and periodically issue ZQCal Latch ~~command~~operation. In this case the most-recently-completed calibration results will always be latched and will be no older than tZQINT.

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It may be permissible for the memory controller to ignore ZQUF and not issue ZQCal Latch ~~command~~operation if both of the following are true:

1) CA/CSODT and CK ODT are disabled
2) The DDR6 SDRAM is in an idle state, or in Self Refresh or Power Down mode.

Re-calibration will still occur in the background to ensure accurate driver/ODT settings are available should they be needed. In this case, the memory controller should ensure a ZQCal Latch ~~command~~operation is performed prior to resuming data traffic if ZQUF is set.

When the memory controller de-asserts ZQ Stop it shall reset MR7 OP[3] to zero for all dies sharing the ZQ resource within 100ns as described in 1.2.1.1. The ZQUF will be updated and results from the recalibration can be latched after the appropriate tZQCAL time (tZQCAL4, tZQCAL8 or tZQCALTBD) has been satisfied.

4.2.3.3. ZQ Stop Functionality

ZQ Resistor Sharing in Background Calibration Mode

In Background Calibration mode, a ZQ Stop function is provided to enable other device(s) to share the ZQ resistor. This function is enabled by MR7 OP[3] =ZQ Stop. When another device needs to use the ZQ resistor, MR7 OP[3] shall be set to 1 for all DDR6 dies that share ZQ resources. This will halt background calibration operations within delay time tZQSTOP (see Table 1) for each DDR6 die. Once tZQSTOP has expired for all DDR6 dies that share the ZQ resource, other dies may use the resource. When the ZQ resource is no longer needed by the other die(s) the ZQ Stop MR bit should be reset to 0 to allow background calibrations to continue normally. When the ZQ Stop MR bit is reset to 0, periodic background calibrations will be re-started.

Since ZQ Stop inhibits the DDR6 die from recalibrating, note that changing system conditions while the ZQ Stop MR bit is set may cause the DDR6 ZQ calibration accuracy to deviate from specification. To ensure continued accurate calibration as discussed in section 1.1.2.2, the ZQ Stop MR bit shall not be set for longer than tZQINT in background calibration mode.

Stopping Background Calibration when VDDQ is Powered Off

~~In Background Calibration mode, before entering Power-down mode, the calibration shall may be halted by setting ZQ Stop on initiator die. ZQ Stop may be issued on Target dies asynchronously or simultaneously. When Power-down mode is exited, ZQ Stop should be reset to 0 to re-enable background calibration. Resetting of ZQ Stop on the ZQ Initiator die will start background calibration(s) immediately.~~

In Background Calibration mode, after expiration of the appropriate tZQCAL time (tZQCAL2, tZQCAL4, tZQCAL8 or tZQCALTBD) from resetting of ZQ Stop, the memory controller may check the ZQUF flags or may issue a ZQCal Latch ~~command~~operation to each die sharing the ZQ resource. To guarantee recalibration of all dies sharing the ZQ resource within tZQCAL, the MR7 OP[3] ZQ Stop bit shall be reset for all Target die(s) sharing the ZQ resource either before, or no later than 100ns after, the MR7 OP[3] ZQ Stop bit is reset on the ZQ Initiator die.

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Table 22 — ZQ Calibration Timing Parameters (Sheet 1 of 2)

Parameter	Symbol	Min/Max	Value	Units
From resetting ZQ Stop to Latch Time, NZQ≤2	tZQCAL2	Min	1.5	µs

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Table 22 — ZQ Calibration Timing Parameters (Continued) (Sheet 2 of 2)

Parameter	Symbol	Min/Max	Value	Units
From resetting ZQ Stop to Latch Time, 2<NZQ≤4	tZQCAL4	Min	3	µs
From resetting ZQ Stop to Latch Time,4<NZQ≤8	tZQCAL8	Min	6	µs
From resetting ZQ Stop to Latch Time, NZQ TBD<NZQ≤TBD	tZQCALN	Min	N * 0.75	µs
ZQ Calibration Latch Time	tZQLAT	Min	MAX (30ns, 8nCK)	ns
ZQ Calibration Reset Time	tZQRESET	Min	MAX (50ns, 6nCK)	ns
Delay Time from ZQ Stop Bit Set to ZQ Resistor Available	tZQSTOP	Max	30	ns
Background Calibration Interval	tZQINT	Max	Programmable, 16, 32, 64, 128, or 256	ms
Maximum Number of DDR6 dies Connected to a Single ZQ Resistor	NZQ	Max	8	die(s)
Maximum Capacitive Load on ZQ Network	CZQ_N	Max	TBD	pF

ZQ Reset

Setting the ZQ Reset MR bit resets the output impedance calibration to a default accuracy of ±30% across process, voltage, and temperature. The ZQ Reset ~~command~~ operation is also executed by writing MR7 OP[2] = 1b. The ZQ Reset ~~command~~ operation will also reset the ZQ Stop MR7 OP[3] and ZQUF MR7 OP[0] bits to 0. ZQ Reset will not change the state of the ZQ Interval MR bits MR7 OP[5:4]. If CA/CS ODTis enabled, the CA bus shall maintain a deselect state during tZQRESET to allow CA/CSODT calibration settings to be updated. Operations that initiate DQ operations are always disallowed during tZQRESET. The ZQ Reset MR bit shall be reset to 0b by the DRAM after tZQRESET. To reset the ODT and output impedance in a multi-die package, the ZQ Reset function shall be issued to all dies regardless of ZQ Initiator or ZQ Target designation. The host shall ensure ZQ RESET and ZQ STOP mode register bits are not set up simultaneously.

Multi-die Package Considerations

Up to NZQ DDR6 dies within a single package may connect to the same ZQ resistor. ZQ stop function is executed by writing MR7 OP[2] =1b and resetting MR7 OP[2] =0b is managed by Initiator die and may be issued to other die sharing a ZQ resource asynchronously or simultaneously. ZQCal Latch commands, when required, are necessary for each die. When multiple dies share a ZQ resource and a ZQ Initiator die is managing calibration, the MR7 OP[3] ZQ Stop bit shall be set to 0B for all ZQ Target die sharing the ZQ resource. No other arbitration considerations are required.

Other Considerations in Background Calibration Mode

Each DDR6 die within a package includes a single ZQ pin and associated ZQ calibration circuitry. Calibration value from this circuit will be generated and used according to the following protocol:

- Setting of the ZQ Stop bit always has priority over background calibration processes

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- If the ZQ Stop MR bit is set to 1 while a background ZQ calibration is in progress, the background calibration will be interrupted
- A background calibration will start immediately when the ZQ Stop MR bit is reset to 0
- A background calibration will not start while the ZQ Stop MR bit is set to 1
- The ZQUF bit will not be reset by setting of the ZQ Stop bit to 1
- The ZQUF bit will be reset only by a ZQCal Latch ~~command~~operation or ZQ Reset
- The ZQUF bit will be set if calibration codes do not match the currently latched codes even when the die is in Self-refresh or Power-down mode, providing ZQ Stop is not set.

ZQ External Resistor, Tolerance, and Capacitive Loading

To use the ZQ calibration function, a 120 ohm, ±1% tolerance external resistor shall be connected between the ZQ pin and VDDQ. The total capacitive loading on the ZQ pin shall be limited to CZQ_N.

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Flow Chart Example:

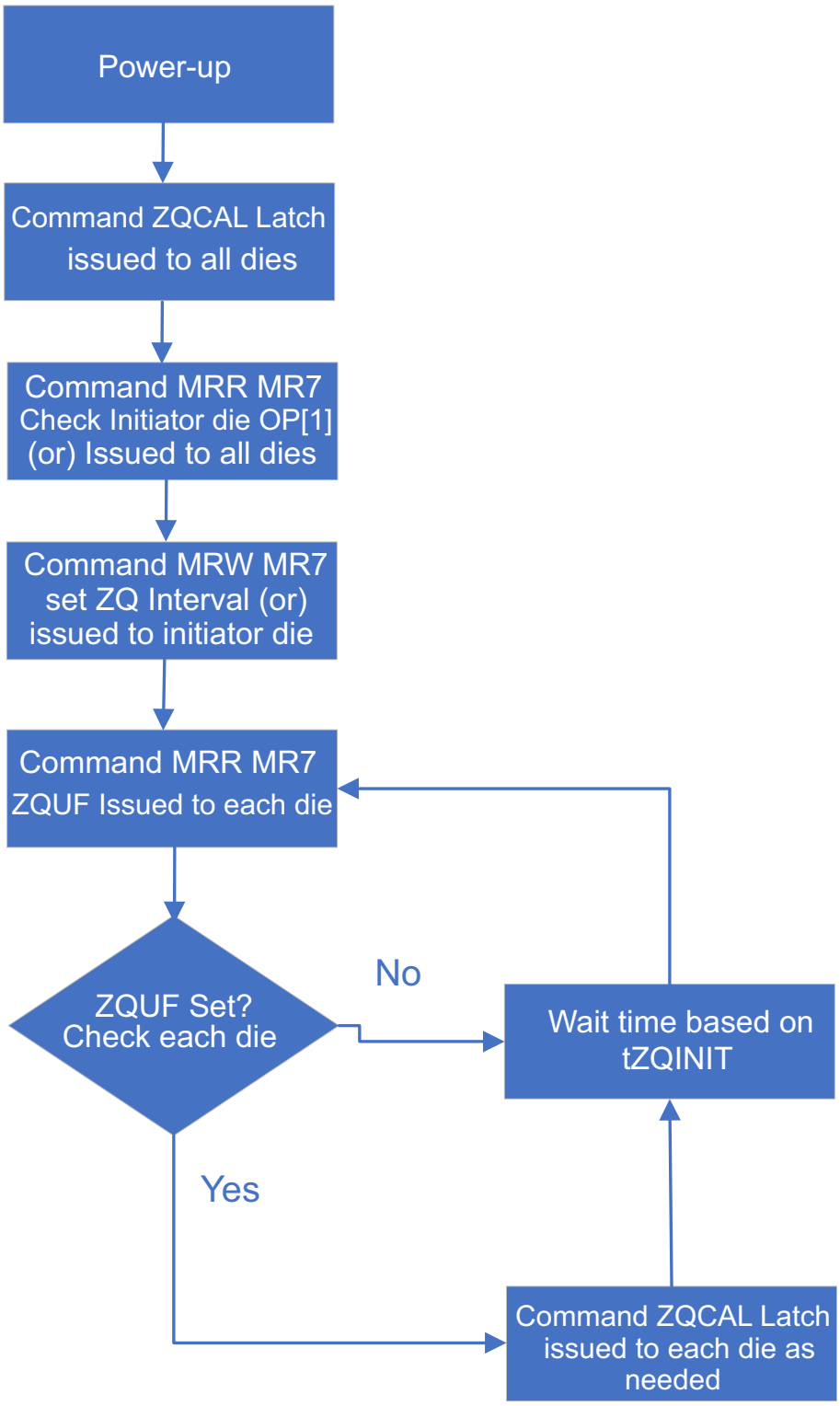


Figure 29 — Example Flow Chart for Background ZQ Calibration

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4.2.4. DQ Swizzling Discovery Mode (DSDM)

During DDR6 DRAM initial training, Data (DQ) bits are used to control the DRAM and receive feedback from it. At this stage, the host has no knowledge of the mapping between host DQ signals and DRAM DQ pins. The following sequence enables the host to discover the DQ mapping before training CA or DQ. The host shall perform mapping discovery for DQ [2:0]. Mapping discovery for the remaining DQ bits at this stage, for different configurations (e.g., 2p6 or 1p6), is optional.

4.2.4.1. DSDM Entry

DQ Swizzle Discovery Mode (DSDM) is entered by issuing an MRUPD sequence that writes OP[x] = 1b to mode register MRxx. Upon entering DSDM, the DRAM shall configure all DQ signals as outputs.

4.2.4.2. DSDM Operation

While in DSDM, the host and DRAM shall follow the procedure below to discover the DQ bit mapping:

1) The host shall issue an MRUPD sequence to program the DQ Mapping Mode register (MRxy) with OP[x] = 1b, as defined in Table 1, to select and enable a single DQ bit.
2) In response, the DRAM shall drive the selected DQ bit to a static High level while driving all other DQ bits to a static Low level. The host shall sample the DQ signals and record the corresponding DQ mapping.
3) Steps 1 and 2 shall be repeated until the host has discovered the mapping for all DQ bits.
4) If the host writes an undefined DQ location (OP[6] and OP[7]) in MRxy, the DRAM shall stop driving all DQ.

Table 23 — Example Mode Registers for DSDM Operation

	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	Notes
MRxx OP[7:0] - Initial Training Control Register	CSTM	CATM	RFU	CKSYN COFF	**DSDM**	RFU	RFU	RFU	
MRxy OP[7:0] - DQ Mapping register 0	DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	RFU	RFU	
MRxz OP[7:0] - DQ Mapping register 1	RFU	RFU	RFU	RFU	RFU	RFU	RFU	RFU	1
NOTE 1: DQ mapping register 1 is required only when 1p12 configuration is enabled.									

4.2.4.3. DSDM Exit

To exit DQ Swizzle Discovery Mode (DSDM), the host shall issue an MRUPD sequence to program MRxx with OP[x] = 0B. After completing the required DQ mapping discovery and exiting DSDM, the host shall proceed to Chip Select Training Mode using the discovered DQ control bit mapping.

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4.2.5. Frequency Set Points (FSP)

4.2.5.1. Input Clock Frequency Change

DDR6 comes with Frequency Set Points (FSP) allowing the DDR6 SDRAM to be switched between 2 different operating frequencies, which can have different termination and Vref values. MRs from these 2 FSPs share the same register addresses. Frequency Set Point Write/Read (FSP-WR) controls which group of the 2 registers are being accessed. Frequency Set Point Operation (FSP-OP) indicates which of the 2 FSP is applied. Prior to an input clock frequency change event, the DRAM is already operating at FSP[x] based on FSP-OP[x], FSP[y] MRs can be updated by setting the FSP-WR[y] ahead of frequency change operation. Setting the FSP-OP[y] bit indicates to the DRAM that new FSP is going to be applied. Power Down Entry or Self Refresh Power Down Entry command is issued to the DRAM. DRAM applies FSP[y] MRs after entering into Power Down or Self Refresh Power Down state. Input clock frequency can be updated while the DRAM is in either Power Down or Self Refresh Power Down state. Once the new input clock frequency is stable, Power Down exit or Self Refresh Power Down Exit command is issued to the DRAM.

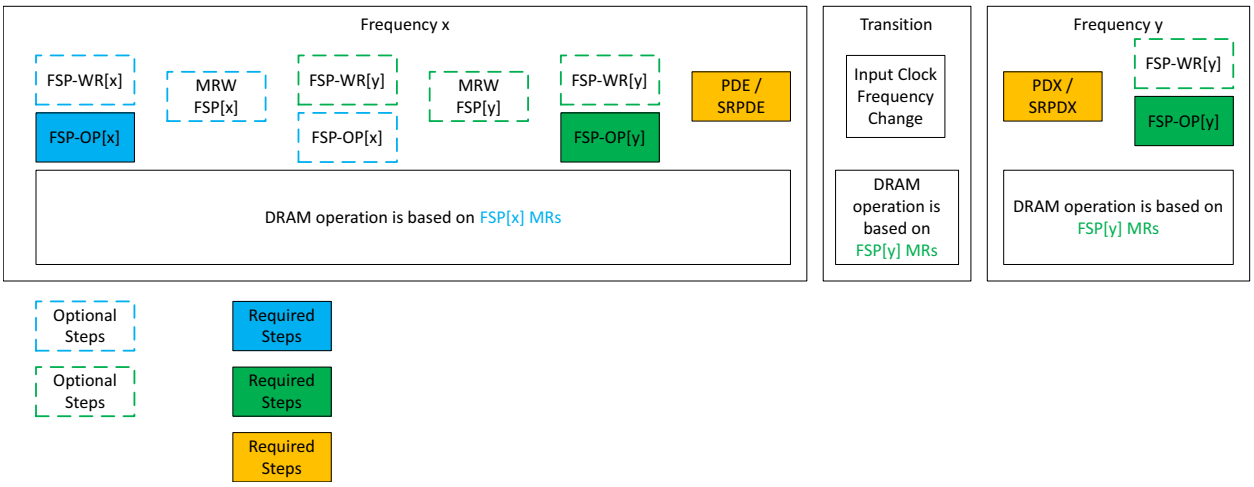


Figure 30 — Frequency Set Points (FSP) State Diagram

DDR6 SDRAM supports input clock frequency change under the following steps:

1) MRs of the FSP[y] used for the next frequency already contain the correct values used for the that frequency
2) FSP-OP[y] used for the next frequency is set indicating that new FSP[y] is going to be applied
3) Power Down Entry or Self Refresh Power Down Entry command is issued to the DRAM
4) DRAM applies FSP[y] MRs after entering to Power Down or Self Refresh Power Down
5) Input clock frequency can be updated

6) Power Down exit or Self Refresh Power Down Exit command is issued to the DRAM

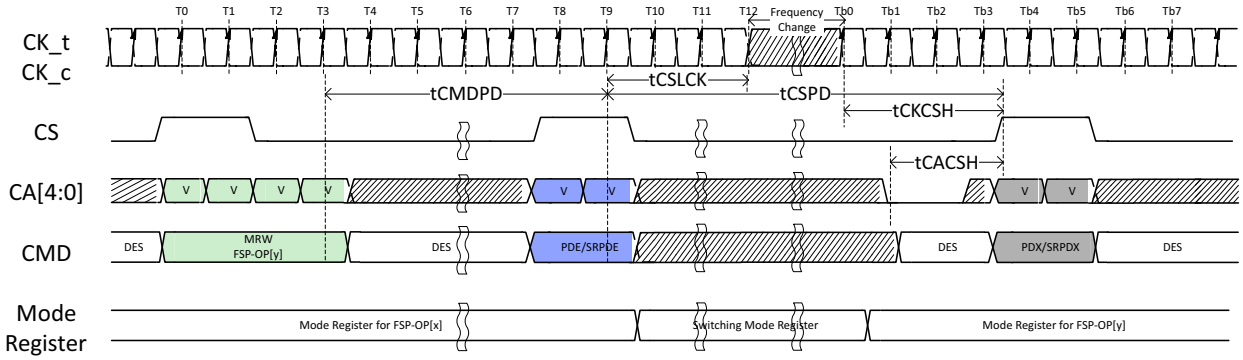


Figure 31 — Input Clock Frequency Change Timing

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4.2.5.2. Frequency MR

Host shall inform the DRAM of operating clock frequency via Frequency MR. At initialize sequence, Frequency MR shall be set by MRUPDATE command to the value matching to the Clock frequency on MR09[3:0]. After the first Frequency MR is set, host shall sustain clock input without changing the frequency at least tDLYCAL.

Only in MRUPD mode can the Frequency MR of the 2nd FSP be set using FSP-WR on MR00[2]. The 2nd FSP Frequency MR can be set without requiring clock frequency matching or a sustained clock signal for tDLYCAL.

Once the registers of Frequency MR are set, and the MRUPD mode is exited, the input clock frequency needs to match the value selected by FSP-OP.Related timing parameters are tCCD_L, tCCD_L_WR, tCCD_L_WR2, tCCD_dlr, tCCD_WR_dlr, nODTLon_WR and nODTLon_WR_NT.

OP[3]	OP[2]	OP[1]	OP[0]
Frequency MR(Data Rate)			

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Function	OP [3:0]	DATA RATE (Mbps)	tCCD_L [nCK]	tCCD_L_WR [nCK]	tCCD_L_WR2 [nCK]	tCCD_dlr[1] [nCK]	tCCD_WR_dlr[2] [nCK]	nODT-Lon_WR /nODT-Lon_WR_NT
Frequency MR (Data Rate)	0000	4800 < Data Rate <= 6400	16	64	32	12	12	WL-12
	0001	6400 < Data Rate <= 8000	20	80	40	12	12	WL-14
	0010	8000 < Data Rate <= 9200	24	92	48	12	12	WL-14
	0011	9200 < Data Rate <= 9600	24	96	48	12	12	WL-14
	0100	9600 < Data Rate <= 10400	28	104	52	12	12	WL-16
	0101	10400 < Data Rate <= 11600	32	116	60	12	12	WL-16
	0110	11600 < Data Rate <= 12800	32	128	64	14	16	WL-16
	0111	12800 < Data Rate <= 14400	36	144	72	16	16	WL-18
	1000	14400 < Data Rate <= 16400	44	164	84	18	20	WL-20
	1001	16400 < Data Rate <= 19200	48	192	96	20	20	WL-22

NOTE 1: tCCD_dlr [nCK] = Max(12nCK, 2x(RU(2ns/(2xtCK))) defines the delay between READ to READ commands to different logical ranks.

Function	OP [3:0]	DATA RATE (Mbps)	tCCD_L [nCK]	tCCD_L_WR [nCK]	tCCD_L_WR2 [nCK]	tCCD_dlr[1] [nCK]	tCCD_WR_dlr[2] [nCK]	nODT-Lon_WR /nODT-Lon_WR_NT
NOTE 2: tCCD_WR_dlr [nCK] = Max(12nCK, 4x(RU(2ns/(4xtCK))) defines the delay between WRITE to WRITE commands to different logical ranks.								

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4.2.6. Chip Select Training Mode (CSTM)

4.2.6.1. Introduction

Chip Select Training Mode (CSTM) is a method to facilitate the loopback of a sampled sequence of the Chip Select (CS) signal. Once this mode is enabled, the device actively samples and drives feedback, where the CS signal is only sampled on the rising edge of even Clock (CK) cycles.

The host follows the CSTM procedure to align all phases of connected devices to achieve the maximum CS eye.

CSTM has two (2) states which can be toggled by DQ[1:0]: Command (CMD) State and Train State. During Command State, only the MRUPDATE sequence is actively processed. The Host drives CA[5:0] low outside CMD State during CSTM. During Train State, every set of 34 CK rising edge samples (68 CK in total) will be included in a logical computation to determine the CSTM sample result that is sent back to the Host on the DQ2.

CSTM includes making changes to VREFCS. VREFCS update and stabilization time depends on the VREF Current Generator (VRCG) for shorter delays. If VRCG is enabled via MRxx[b], tVREFCS_short or tVREFCS_long delay times are used, depending on the number of VREFCS steps being change. However, if VRCG is disabled, all VREFCS changes require the tVREFCS_weak delay.

4.2.6.2. CSTM Entry and Exit

Prior to entering CSTM, MRUPDATE Mode must be enabled by the MCME command. VREFDQ, DQ/DQS ODT and DQ drive strength settings must be adjusted to values that allow slow speed operation.

CSTM is entered and exited through MR05 (Functional Mode Register). Entry or Exit on either PSD (2p3 or 2p6 configuration) will apply the operation to both PSDs. Entry and Exit operations must wait for previous MRUPDATE operations to complete.

4.2.6.3. Entering CSTM

To enter CSTM, the Host issues the MRUPDATE sequence to MR05[7:0]=0~~2~~4h with DQS differentially low prior to the 1st CS pulse of the MRUPDATE sequence. DQS must remain differentially low during CSTM entry. Once tMRUPD_Delay has been satisfied, CSTM has been entered in Command State.

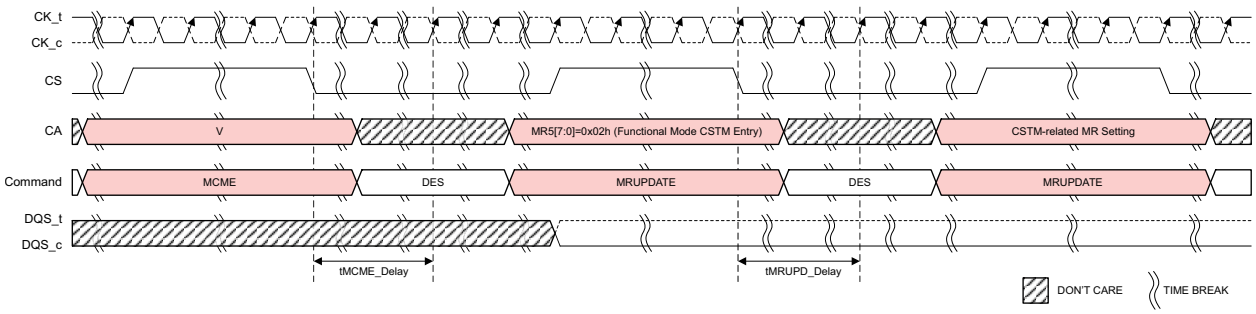


Figure 32 — CSTM Entry Sequence

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4.2.6.4. Exiting CSTM

To exit CSTM, the Host issues the MRUPDATE sequence to MR05[7:0]=00h during Command State with DQS differentially low. During CSTM Exit, DQS must remain differentially low until tMRUPD_Delay is satisfied. After tMRUPD_Delay, CSTM mode is exited.

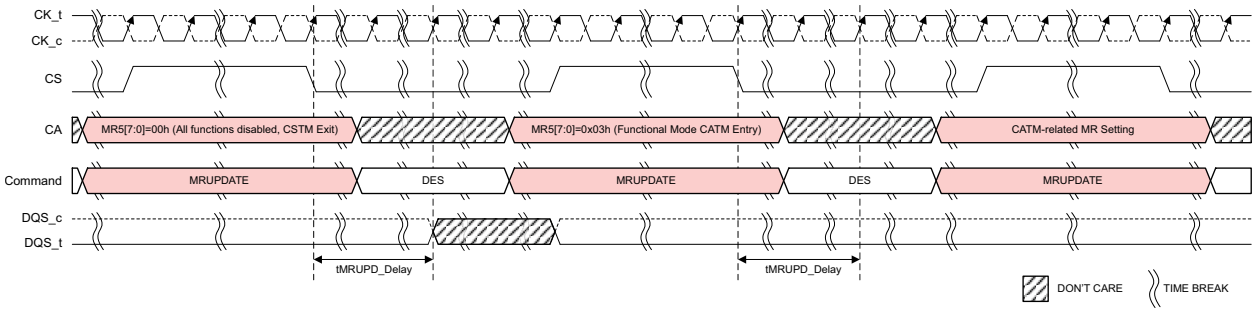


Figure 33 — CSTM Exit Followed by CATM Entry

Issuing an MRUPDATE sequence exit during the CMD state will exit both CSTM and MRUPDATE modes within tMRUPD_Delay.

4.2.6.5. CSTM DQ Function and Mapping

CSTM uses the DQ bus, sampled by DQS, for function control. DQ[1:0] are input pins latched by DQS for CSTM control logic. The host is required to drive identical waveforms on the DQ[1:0] pins of both PSDs.

During CSTM, DQS_t/DQS_c are input only to sample DQ[1:0] as the function code. DQS is required to be differentially low except while sampling DQ[1:0]. CSTM uses a ~~four (4)~~eight (8) bit Function Code (FuncCode[~~3~~7:0]), generated by ~~four~~eight DQS rising pulses to sample DQ[1:0]. Entering CSTM internally resets the DQS rising pulse count to 0 to prepare for counting ~~four~~eight rising pulses. Each Function Code is the combined sample value of the 2nd, 4th, 6th, and 8th DQS differential rising edges (DQSR2, DQSR4, DQSR6, DQSR8, respectively) ~~and the 4th DQS differential rising edge (DQSR4)~~ as shown in the CSTM Function Code Mapping table.

DQ2 is an output pin for the CSTM sample results, per PSD.

All other DQ pins, if present, are unused during CSTM and are disabled.

Table 24 — Data bits (DQ) function during CSTM

DQ	Input/Output	Functions during CSTM
DQ[1:0]	Input	Control logic input
DQ2	Output	Sample results output
DQS_t/DQS_c	Input	DQ[1:0] sample

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Table 25 — CSTM Function Code Mapping

Function Code	DQ Source	DQS Sampling Edge
FuncCode[~~3~~7]	DQ[1]	2nd Rising Edge
FuncCode[~~2~~6]	DQ[0]	2nd Rising Edge
FuncCode[~~1~~5]	DQ[1]	4th Rising Edge
FuncCode[~~0~~4]	DQ[0]	4th Rising Edge
FuncCode[3]	DQ[1]	6th Rising Edge
FuncCode[2]	DQ[0]	6th Rising Edge
FuncCode[1]	DQ[1]	8th Rising Edge
FuncCode[0]	DQ[0]	8th Rising Edge

As shown in the figure below, the leading and trailing DQS differentially low timing (tDQSPreTrain and tDQSPost-Train, respectively) must be met before and after the DQS pulses. tFunc_Delay is the minimum time between Function Codes. Each DQS pulse must meet tDQSPWTrain on both high and low pulse.

DQ[1:0] must meet setup and hold timing (tDSTrain and tDHTrain, respectively). The setup timing tDSTrain is defined as DQ[1:0] leading edge to DQSR2 and the hold timing tDHTrain is defined as DQSR2 to DQ[1:0] trailing edge. The same setup and hold timing requirements apply to DQSR4, DQSR6, and DQSR8.

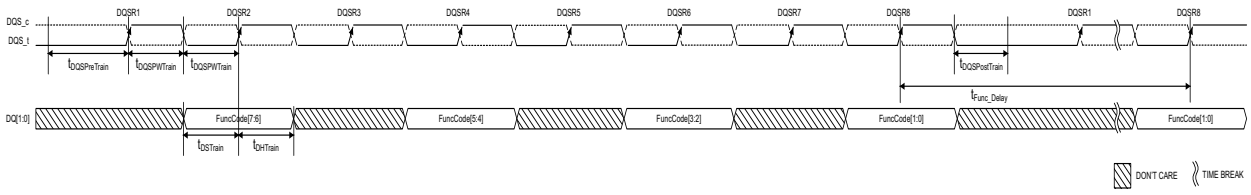


Figure 34 — Example Timing for Functional Codes

NOTE: CK and time breaks are not included.

4.2.6.6. Command and Train States

Command (CMD) State is the default state after CSTM entry. CMD state is MRUPDATE mode enabled with CSTM functions controlled by DQ[1:0] and DQS. Only the MRUPDATE sequence and DES command are valid in the CMD State. During CSTM, the MRUPDATE sequence is only used to trigger functions and to change shadow register bits related to CSTM settings such as VREF CS, CS ODT, CK ODT, DQ/DQS ODT, and DQ driver strength, as needed. Shadow register "Apply" is needed to update any of the normal mode register values for use during CSTM CMD State. The VREFCS generator references the VREF CS Mode Registers settings while in CMD State.

Train State is used to adjust the VREF CS shadow registers setting, perform training sequence sampling, and output training results on DQ2. Train State uses CK ODT and CS ODT shadow register values, as well the VREF CS shadow register value (CA ODT and VREF CA shadow register values are also applied but have not impact since the CA bus is not used). In Train State, the command decoder is disabled and CA is required to be low all the time. VREFCS change functions will only apply to the VREF CS shadow register, which updates the VREFCS generator since it is referencing the shadow register while in Train State. Upon completion of CSTM, FuncCode[~~3~~7:0]=8'b00000000

forces the return from Train State to CMD State, restoring use of the normal VREF CS Mode Register. The Host can apply the VREF CS shadow register value to VREF CS MR (MR~~19~~83[6:0]) per training results during CMD State or after exiting CSTM using the VREFCS and CS ODT Apply Functional Mode (MR0~~5[xx]~~6[7:0]=04h).Shadow register values are not changed by entering or exiting CSTM, by entering CMD State, or by entering Train State.

Issuing the shadow register apply during CMD State or after exiting CSTM, switching from CMD State to Train State, and switching from Train State to CMD State, requires waiting the appropriate VREFCS update and stabilization delay which depends on the status of VRCG and the numbers of VREFCS steps being changed.

DQ[1:0] are input pins during both CMD and Train States. DQ2 is off during CMD State. DQ2 remains off in Train State until the first time it is used as output pin during SMP function. DQS_t/DQS_c are input pins during both states.

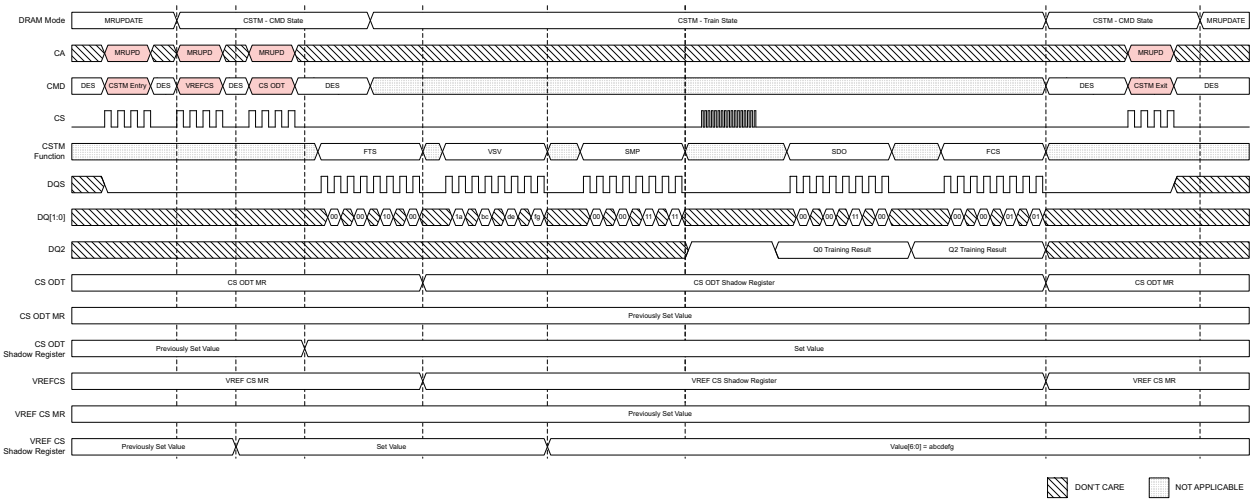


Figure 35 — CSTM Entry/Exit with CMD State and Train State Switch

NOTE: 1: CSTM Functions refer to the next section

NOTE: 2: CK and time break are not included in the figure. The diagram is intended to serve as a reference for behavior rather than for timing purposes.

NOTE: 3: MRUPDATE sequence to set VREFCS and CS ODT during CMD State is only for illustration purposes

NOTE: 4: VREFCS and CS ODT registers are shown as updated to an appropriate previously set value as recommended in section XXX, Voltage Ramp and Device Initialization

4.2.6.7. CSTM Functions

CSTM uses a ~~four (4)~~eight (8) bit function code (FuncCode[~~3~~7:0]) to execute training functions. When a Func-Code[~~3~~7:0] is received, the corresponding functions executes within tFunc_Delay. Some functions can only be used in certain CSTM states. FuncCode[~~3~~7:0] issued outside of an allowed state will be ignored. Any unused Func-Code[~~3~~7:0] is RFU will be ignored.

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Table 26 — CSTM Functions

FuncCode[~~3~~7:0]	Function	Allowed State	Description
00000000	FCS	CMD / Train	Force “CMD State". Use normal VREF and ODT MRs.
00000001	CPS	CMD	CK Phase Shift 90 degrees
00001000	FTS	CMD / Train	Force “Train State". Use VREF and ODT shadow registers.
~~1001~~	~~VSD~~	~~CMD / Train~~	~~Decrease VREF_CS shadow register by 1 (step down)~~
~~1010~~	~~VSU~~	~~CMD / Train~~	~~Increase VREF_CS shadow register by 1 (step up)~~
00001100	SDO	Train	Switch DQ2 output to internal CK phase 180 training results
00001111	SMP	Train	Output CS Pattern Sampling and internal CK phase 0 training results on DQ2
1VVVVVVV	VSV	CMD / Train	Set VREF CS shadow register to FuncCode[6:0]
Others	RFU		

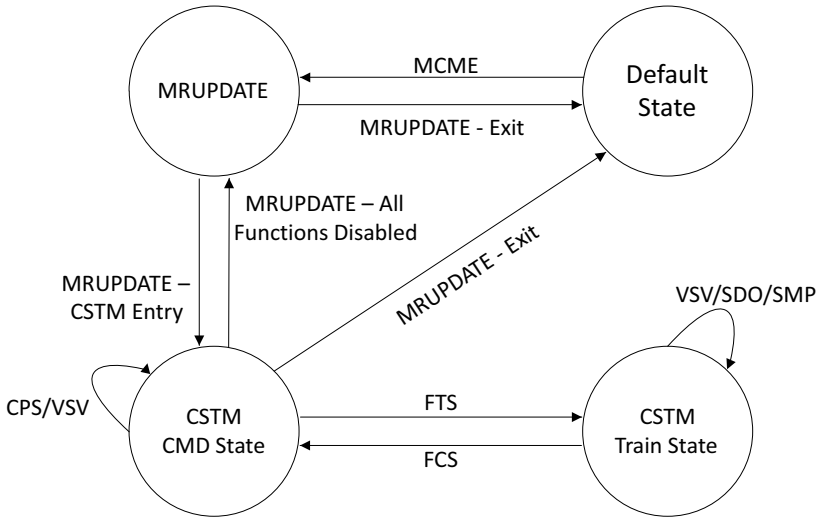


Figure 36 — CSTM Simplified Diagram

NOTE: Default states include Idle and Self-Refresh without Power-Down

FuncCode[~~3~~7:0]=~~4~~8b'00000000: "Force CMD State (FCS)" function. The FCS function may be issued in both the CMD State and the Train State.

FuncCode[37:0]=48b'00000001: "CK Phase Shift (CPS)" function. The CPS function will force an internal CK phase shift 90° (1 CK). This function is only allowed during CMD State. The Host uses the CPS function to align CK phase of all devices that share CK.

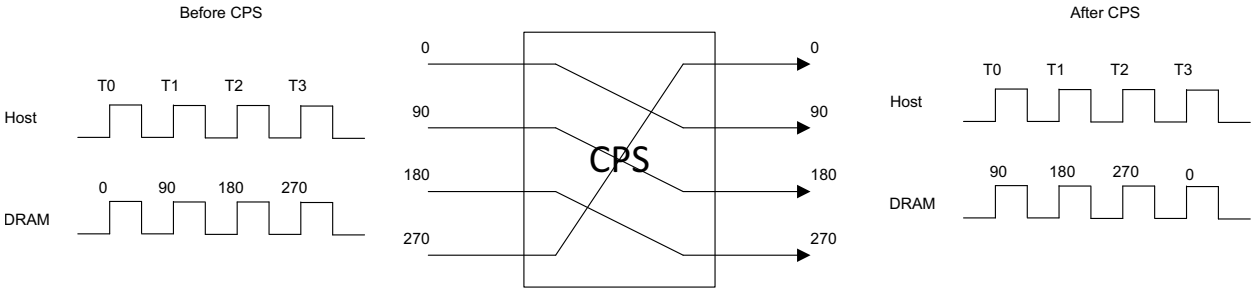


Figure 37 — CPS Function Diagram

NOTE: This diagram is only a high-level function reference for CPS function.
NOTE: Host CK doesn't change before and after CPS, only the internal CK phase is changed by CPS function.

FuncCode[37:0]=48b'00001000: "Force Train State (FTS)" function. The FTS function forces Train State and switches from using normal VREF and ODT MRs to using the VREF and ODT shadow registers. The FTS function can be issued in both the CMD State and Train State. During Train State, VREFCS is controlled by the VREF CS shadow register.

~~FuncCode[3:0]=4b'1001: "VREFCS Step Down (VSD)" function. The VSD function forces the VREF_CS shadow register value to decrease by one step. If the register is at the minimum value, the VREFCS will remain at the minimum value if issued additional VSD functions. The VSD function can be issued in either CMD or Train state.~~

~~FuncCode[3:0]=4b'1010: "VREFCS Step Up (VSU)" function. The VSU function forces the VREF_CS shadow register value to increase by one step. If the register is at the maximum value, the VREFCS will remain at the maximum value if issued additional VSU functions. The VSU function can be issued in either CMD or Train state.~~

FuncCode[7:0]=8b'1VVVVVVV: "VREF Set Value (VSV)" function. The VSV function sets the VREF CS shadow register bits VREFCS[6:0] to FuncCode[6:0].

Note: VREFCS does not use the VREF CS shadow register during the CMD state, so changes will not be applied until Train State is entered. ~~Continuous VSD and VSU operation is possible if tFUNC_Delay between each operation is~~

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~~met. However, t~~The Host must wait the required VREFCS update and settling delay after ~~last VSU or VSD~~ any VSV function.

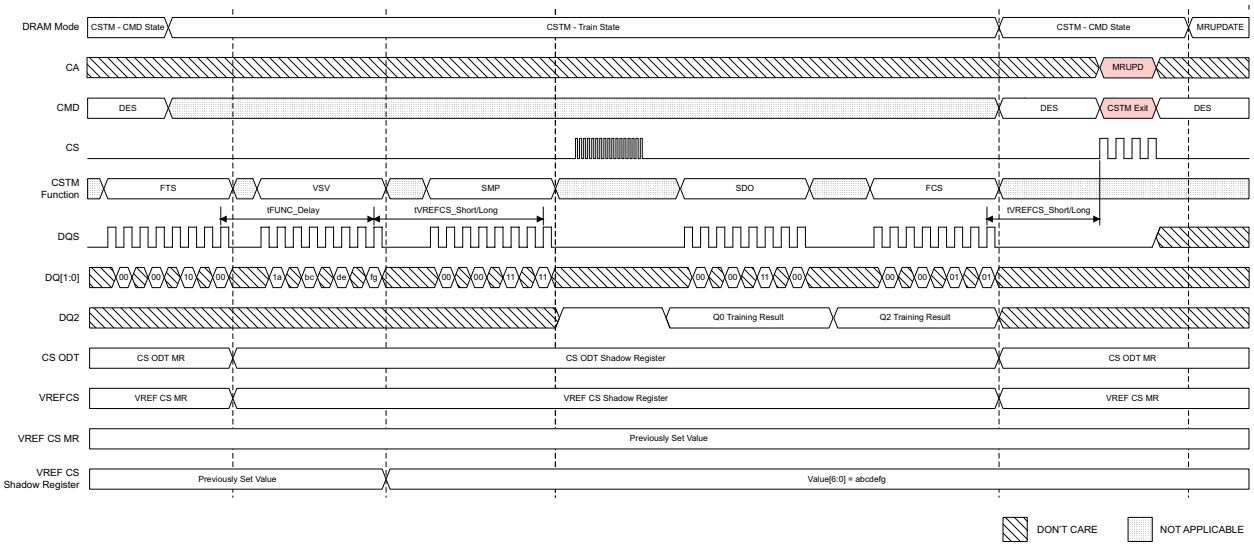


Figure 38 — VSV Function Example and tVREFCS_short/long Requirement

NOTE: 1:CK and time break are not included. The diagram is intended to serve as a reference for consecutive ~~VSU/VSD~~VSV behavior and tVREFCS_short/long timing requirement. Any additional timing requirements not depicted in the figure must also be satisfied.

NOTE: 2: VREF CS registers are shown initially updated to an appropriate previously set value as recommended in section XXX, Voltage Ramp and Device Initialization.

FuncCode[~~3~~7:0]=~~4~~8b'00001100: “Switch DQ2 Output (SDO)" function. The SDO function will switch DQ2 to output training results from the internal CK phase 0 to the internal CK phase 180. After tDQ2Output is met, DQ2 will output the CK phase 180 CS training result. Issuing the SDO function when DQ2 is already outputting CK phase 180 will not have any new effect, however tDQ2Output still needs to be met.

FuncCode[~~3~~7:0]=~~4~~8b'00001111: "Sample (SMP)" function. The SMP function performs the following operations to prepare for CS sampling: reset CS sample counter and logic, reset CK phase 0 and 180 internal training results to fail, and force DQ2 high. After the reset to CSTM delay (tRST2CSTM) is met, the CS pattern can be sampled. The Host uses a “1100" toggle pattern on CS, where each '1' or '0' represents a 2nCK long pulse. The CS is sampled on rising CK edges (internal CK phases 0 and 180). The Host sends 34 CS toggles (17 High and 17 Low pulses, each pulse is 2nCK wide, 68nCK in total). The first CS high sample starts the internal CK counting on the next CS high sample. Once counting begins, the sample and check CS will continue for the even CK edges over the next 64nCK. The CS high pulse width is checked on the internal CK phase 0 (Q0) and the CS low pulse width is checked on internal CK phase 180 (Q2). Passing samples are recorded as '0' and failing samples recorded as '1'. After counting 64nCK cycles, the 16 Q0 results (CS sample on CK phase 0 is expected to be high, when passing is recorded as "0") and the 16 Q2 results (CS sample on CK phase 180 is expected to be low, when passing is recorded as "0") are checked separately with a logical OR. The Q0 training result are output on DQ2, where a “0" indicates a pass and a "1" indicates failure. If a pass is detected, the output remains at "0" until switched to CMD State or next SMP/SDO function. The Host captures the Q0 pass/fail results after tDQ2Output, uses SDO function to switch DQ2 to output Q2 training results, and then captures the Q2 pass/fail results after tDQ2Output. The CS output result are reset by issuing the SMP function before sending out the next CS toggle pattern. Excessive CS pulses will not cause a new CS sampling unless a new SMP function has been issued. DQS is required to be differentially low during SMP function until all CS samples are

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done and tDQ2Output is met. Once the Host issues a new SMP function, DQ2 will be pulled to high and the internal training results will be reset to fail.

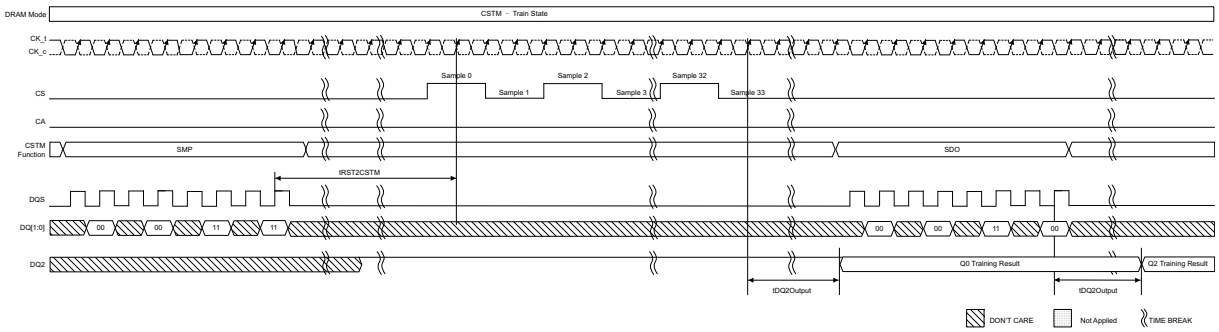


Figure 39 — SMP Function Timing Example

Issuing MRUPDATE sequences or executing a new CSTM function must wait for a previous operation to complete. The only exception is an MRUPDATE sequence for CSTM Exit, which may be used to exit CSTM during CMD State with DQS differentially low at least tFunc_Delay plus the required VREFCS update and settling delay after the last DQS differential rising edge.

The following figure is an example of the sequence to switch to CMD State and exit CSTM when DQS pulse count has been lost. The Host issues at least 816 DQS pulses with DQ[1:0]=00b, where each DQS pulse meets tFunc_Delay between rising edges. CSTM exit command must meet tFunc_Delay the required VREFCS update and settling delay (tVREFCS_long in this example) after the last DQS differential rising edge. CAs remain as "L", as the starting state may be the Train State.

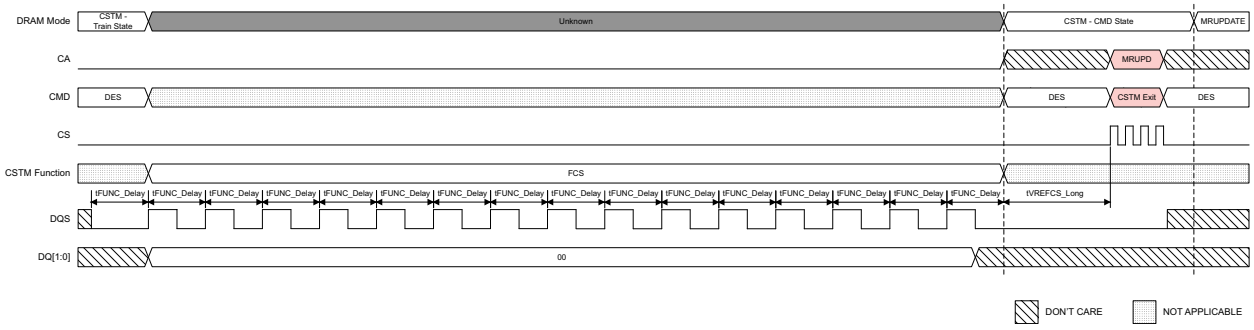


Figure 40 — Example of CSTM Exit in the Middle of 4 DQS Pulses

4.2.6.8. Example CSTM Sequence

An example of the CSTM sequence is described below:

1) If not in MRUPDATE mode, then use MCME to enter MRUPDATE mode.
2) Complete the DQ Swizzling Discovery Mode (DSDM) operation to identify the host to DRAM DQ mapping.
3) Set DQS differentially low. Enter CSTM and wait tMRUPD_Delay.
4) ~~Update the CS_ODT and VREF_CS Mode Register values, if required, and Apply.~~If required, update the CS ODT and VREF CS shadow register values and Apply to normal MR for use in CMD State. The VREF CS normal MR value must be valid in CMD State to allow the MRUPDATE sequence to execute properly.

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5) ~~Update the CS_ODT and VREF_CS shadow register values, if required. The VREFCS value must be valid to allow the MRUPDATE sequence to execute properly, and Train State will use the VREF CS shadow register value.~~If required, update the CS ODT and VREF CS shadow register values for use in Train State.
6) Switch from CMD State to Train State. Wait for tFunc_Delay and VREFCS update and settling delay. The shadow registers are being used.
7) Update the VREFCS setting via the VSU or VSD function. Wait VREFCS update and settling delay after the last VSU/VSD.
8) Issue the SMP function to prepare for sampling. After tFunc_Delay, DRAM is ready for CS sampling can begin and DQ2 output will be “1".
9) Perform CS training (VREFCS, CS timing):
    a) The Host sends the CS toggle pattern 110011001100..., 68nCK in total.
    b) The CS toggle pattern is sampled with the first CS high sampled by an even clock edge. Counting starts on the second CS high sample and counts 32 samples (64nCK) before sending the output results on DQ2. Within the 64nCK sample window, all Q0 CS samples must pass for the DQ2 output to be “0".
    c) The Host reads DQ2 after tDQ2Output and records it, issues the SDO function to output Q2 training results, and again reads DQ2 after tDQ2Output and records it. All Q2 CS samples must pass for the DQ2 output to be “0".
    d) The Host calculates the Q0 and Q2 training results, where both values being “0" is considered a pass. (Both PSDs and both the Q0 and Q2 phases need to be “0" for the training is considered PASS.)
10) Repeat step 7 to 9 as necessary to complete the first round of CS training. Host should analyze training results to identify DRAMs into four CK phase groups based on the center of CS eye. Different groups' CS eye center should be about 1 tCK apart. If all DRAM happens to be in same group then, skip step 11.
11) Use the FCS command to switch to the CMD State, wait tFUNC_Delay and the VREFCS update and settling delay.
12) Choose one of the CK phase group devices as the target group. Issue CPS function to the other CK phase groups to adjust their CS eye mask center to align with target CK phase group. Each CPS function will shift CS eye by 1 nCK (90 degrees). Multi CPS functions may be needed on CK phase groups that are multiple tCK away from target group.
13) Repeat step 6 to 9 for another round of CS training with all devices even CK phase aligned. The Host can analyze the training results and apply the CS delay to the center of common CS eye window to achieve the max CS eye margin.
14) Use the FCS command to switch to the CMD State, wait tFUNC_Delay and the VREFCS update and settling delay.
15) Exit CSTM. After tMRUPD_Delay, CS is “trained".

4.2.6.9. CS-CK Sweeping and CK Phase Alignment

The following explains how to align the internal CK phase of different DRAM.

Fundamentally, the purpose of CSTM is to align the CS eye of each DRAM, which can be described with a 2D shmoo of the CS to CK delay on the X-axis and the VREFCS on the Y-axis. The CS to CK delay (CS rising edge to CK rising edge) is a common setting that impacts all shared signals, while the VREFCS can be used to center each CS eye properly. To maximize the CS margin, all CS eye should be centered and the CS to CK delay is adjusted to compensate for mismatch between the external and internal clocks.

During CSTM, the Host calculates the CS to CK delay referencing a fixed Host clock phase. This chosen phase remains fixed during whole training procedure. The example below chooses the Host clock phase Q2 as the reference clock phase for the CS to CK delay calculation.

The following figure shows four DRAMs (A, B, C, D) to explain the CK phase detection during the CS-CK sweep, with an assumption of a 0.25tCK CS to CK mismatch. The host start CS-CK sweeping with the delay at 0. Due to

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external to internal mismatch, the CS to T0 CK delay is 0.25tCK. DRAM A's Q1 aligns to T0. To have CS centered at Q0, the CS-CK delay will be 1.75tCK which is the center of CS eye mask. DRAM B's Q2 aligns to T0, so the center of CS eye mask will be 0.75tCK. DRAM C's Q3 aligns to T0, so the center of CS eye mask will be -0.25tCK or 3.75tCK. And DRAM D's Q0 aligns to T0, so the center of CS eye mask will be -2.75tCK. In this example, host selects DRAM B as the target for all other DRAMs to align to. The Host issues the CPS function once to DRAM A, twice to DRAM D and three times to DRAM C. After CPS functions, all CS eyes are center aligned to DRAM B, which has internal Q0 aligned to T2 CK. The CS eye center 0.75tCK is the extra delay needed to center CS at T2 CK, which is Q0 of all DRAM. The Host labels the internal phase accordingly.

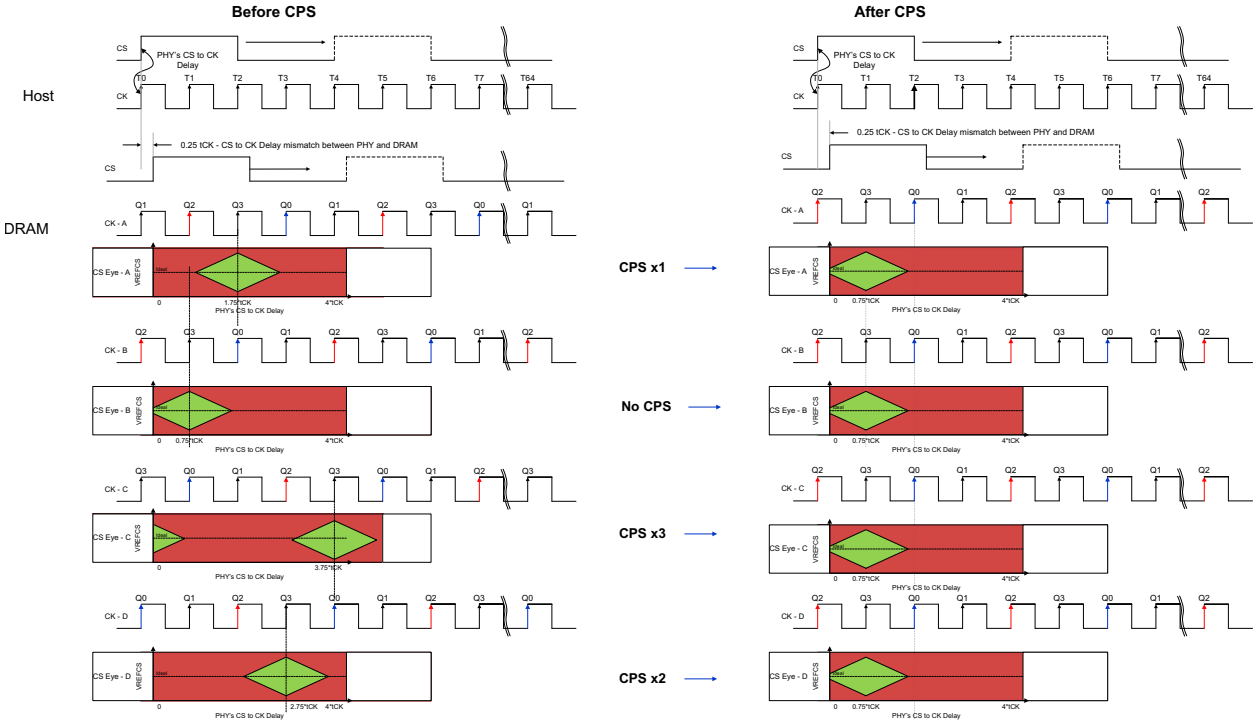


Figure 41 — Example of a CS-CK Sweep with Four DRAMs having Different Internal CK Phases

Table 27 — Timing Parameters for CSTM (Sheet 1 of 2)

Item Description	Parameters	Min/Max	Value	Unit
DQ[1:0] setup	tDSTrain	Min	6	nCK
DQ[1:0] hold	tDHTrain	Min	6	nCK
DQS differentially low prior to 1st DQS rising edge	tDQSPreTrain	Min	6	nCK
DQS differentially low after to ~~4th~~8th DQS falling edge	tDQSPostTrain	Min	6	nCK
DQS pulse width	tDQSPWTrain	Min	6	nCK
Training function to training function delay	tFunc_Delay	Min	tMRUPD_DELAY	nS

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Table 27 — Timing Parameters for CSTM (Continued) (Sheet 2 of 2)

Item Description	Parameters	Min/Max	Value	Unit
SMP function to 1st CS sample CK rising edge	tRST2CTSM	Min	20	nS
VREF Stable - Long with VRCG	tVREFCS_long	Min	350	nS
VREF Stable - Short with VRCG	tVREFCS_short	Min	250	nS
VREF Stable - no VRCG	tVREFCS_weak	Min	1	mS
Training result output on DQ2 delay	tDQ2Output	Min	TBD	nS

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4.2.7. VREFCS

VREFCS is defined as a ratio of the VDDQ voltage. VREFCS is independently controlled per PSD with host MRW to MR86[6:0] in normal operation, or by the VREF_CS shadow register with VREFCS Apply Functional Mode (MR06[2]=1) in MRUPDATE mode.

4.2.7.1. MR83 – VREFCS MR Definition

Table 28 — MR83 Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RV	VREF(CS)						

Table 29 — MR83 Definition

Function	Register Type	Operand	Data	Notes
VREF(CS)	RW	OP[6:0]	0000000 : --Thru -- 1111111 :	1,2,3,4

NOTE 1: A read (MRR) to this register places the contents of OP[x:x] on DQ[y:y]. Any RV bits and unused DQ's shall be set to '0'. See the section on MRR Operation.
NOTE 2: There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Please refer to TBD section for more information on FSP operation.
NOTE 3: Control of VREFCS during CSTM is referred to section 4.2.6 (CSTM)
NOTE 4: The time required for VREFCS to reach the set level depends on the step size from the current level to the new level. See TBD for more information on VREFCS training.

Table 30 — MR VREFCS Setting (Sheet 1 of 3)

Function	Operand	VREF Values (% of VDDQ)								Notes
VREF Setting	OP [6:0]	0000000	10.0[4]	0100000	26.0	1000000	42.0	1100000	58.0	1,2,3,4,5,6
		0000001	10.5[4]	0100001	26.5	1000001	42.5	1100001	58.5	
		0000010	11.0[4]	0100010	27.0	1000010	43.0	1100010	59.0	
		0000011	11.5[4]	0100011	27.5	1000011	43.5	1100011	59.5	
		0000100	12.0[4]	0100100	28.0	1000100	44.0	1100100	60.0	
		0000101	12.5	0100101	28.5	1000101	44.5	1100101	60.5	

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Table 30 — MR VREFCS Setting (Continued) (Sheet 2 of 3)

Function	Operand	VREF Values (% of VDDQ)								Notes
		000010	13.0	010010	29.0	100010	45.0	1100110	61.0	
		000011	13.5	010011	29.5	100011	45.5	1100111	61.5	
		000100	14.0	010100	30.0	100100	46.0	1101000	62.0	
		000101	14.5	010101	30.5	1001001 (default)[7]	46.5	1101001	62.5	
		000110	15.0	010110	31.0	100110	47.0	1101010	63.0	
		000111	15.5	010111	31.5	100111	47.5	1101011	63.5	
		001100	16.0	011100	32.0	100100	48.0	1101100	64.0	
		001101	16.5	011101	32.5	100101	48.5	1101101	64.5	
		0001110	17.0	0101110	33.0	1001110	49.0	1101110	65.0	
		0001111	17.5	0101111	33.5	1001111	49.5	1101111	65.5	
		001000	18.0	011000	34.0	101000	50.0	1110000	66.0	
		001001	18.5	011001	34.5	101001	50.5	1110001	66.5	
		001010	19.0	011010	35.0	101010	51.0	1110010	67.0[5]	
		001011	19.5	011011	35.5	101011	51.5	1110011	67.5[5]	
		001100	20.0	011100	36.0	101100	52.0	1110100	68.0[5]	
		001101	20.5	011101	36.5	101101	52.5	1110101	68.5[5]	
		001110	21.0	0111010	37.0	101110	53.0	1110110	69.0[5]	
		001111	21.5	0111011	37.5	101111	53.5	1110111	69.5[5]	
		001100	22.0	0111000	38.0	101100	54.0	1111000	70.0[5]	

Table 30 — MR VREFCS Setting (Continued) (Sheet 3 of 3)

Function	Operand	VREF Values (% of VDDQ)								Notes
		0011001	22.5	0111001	38.5	1011001	54.5	1111001	70.5[5]	
		0011010	23.0	0111010	39.0	1011010	55.0	1111010	71.0[5]	
		0011011	23.5	0111011	39.5	1011011	55.5	1111011	71.5[5]	
		0011100	24.0	0111100	40.0	1011100	56.0	1111100	72.0[5]	
		0011101	24.5	0111101	40.5	1011101	56.5	1111101	72.5[5]	
		0011110	25.0	0111110	41.0	1011110	57.0	1111110	73.0[5]	
		0011111	25.5	0111111	41.5	1011111	57.5	1111111	73.5[5]	

NOTE 1: These values may be used for MR83[6:0] to set the Vref CS levels in the DDR6 SDRAM.
NOTE 2: Absolute CS VREF low level (%code * VDDQ) must be higher than or equal to 75mV for normal operation. Vref error is not include this calculation.
NOTE 3: Absolute Vref CS high level (%code * VDDQ) must be lower than or equal to VrefCS_Max for nor mal operation. Vref error is not included this calculation. Refer to the CS Rx Mask table for the VrefCS_Max value.
NOTE 4: Vref CS codes from 0000000 to 0000100 are able to be used only for testing purpose not for normal operation. Vref CS accuracy are not guaranteed from 0000000 to 0000100.
NOTE 5: The Vref CS codes from 1110010 to 1111111 are provided to support cases where VDDQ is lower than the typical value, enabling the absolute Vref value to reach 0.4V. When VDDQ is at its typical voltage, codes with an absolute Vref value of higher than 0.4V cannot be used. This restriction also applies during training operations.
NOTE 6: The default VREF CS is defined by CS ODT strap value.

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4.2.8. Command Address Training Mode (CATM)

4.2.8.1. Introduction

The command address bus must be trained before nominal frequency operation. Command Address Training Mode (CATM) is a method to facilitate the loopback of a logical combination of the sampled CA [5:0] signals. In this mode, the CK_t/CK_c differential clock (CK) is running and the Chip Select (CS) qualifies when the CK samples the CA signals. CATM is performed per-CA bit and per even/odd CK phase to adjust the per-bit skew and duty cycle distortion. The Host sends a long CA burst, or multiple CA bursts, with CS high, and the CA pattern is captured on each CS qualified differential CK rising edge.

CATM has two (2) states which can be toggled by DQ[1:0]: Command (CMD) State and Train State. During Command State, only the MRUPDATE sequence is actively processed. During Train State, the burst or bursts of CA data qualified by CS high will be sampled on every differential CK rising edge and compared with a PRBS pattern in a logical computation to determine the CATM output results for feedback to the Host.

The controller and DRAM use matching PRBS generators to compare per even/odd CK phase and per-CA bit in parallel, and feedback is returned on DQ2. The internal VREFCA default level is suitable for MRUPDATE operation during initialization, but the VREFCA must be trained to achieve suitable receiver voltage margin for normal operation. The Host follows the CATM procedure to center the internal VREFCA in the CA data eye and to adjust timing of the CA signals to maximize CA timing margin. While in CATM, no functional commands are executed. The functional command interface is restored only after exiting this mode.

Prior to entering the CATM, the DQ swizzling discovery mode must be completed, and CSTM must have been performed to align the CK phase to achieve the maximum CS eye.

CATM includes making changes to VREFCA. VREFCA update and stabilization time of tVREFCA_short or tVREFCA_long are used, depending on the number of VREFCA steps being change.

~~During CATM, per-pin VREFCA offset is expected to be disabled. If previously enabled, the Host needs to disable any per-pin VREFCA offset before performing CATM.~~

4.2.8.2. CATM Entry and Exit

Prior to entering CATM, MRUPDATE Mode must be enabled by the MCME command. VREFDQ, DQ/DQS ODT and DQ drive strength settings must be adjusted to values that allow slow speed operation.

CATM is entered and exited through MR05 (Functional Mode Register). Issuing the entry or exit on either PSD (2p3 or 2p6 configurations) will apply the operation to both PSDs. Entry and Exit operations must wait for previous MRUPDATE operations to complete.

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4.2.8.3. Entering CATM

To enter CATM, the Host issues the MRUPDATE sequence to MR05[7:0]=03~~3~~5h with DQS differentially low prior to or during the 1st CS pulse of the MRUPDATE sequence. DQS must remain differentially low during CATM entry. Once tMRUPD_Delay has been satisfied, CATM has been entered in the Command State.

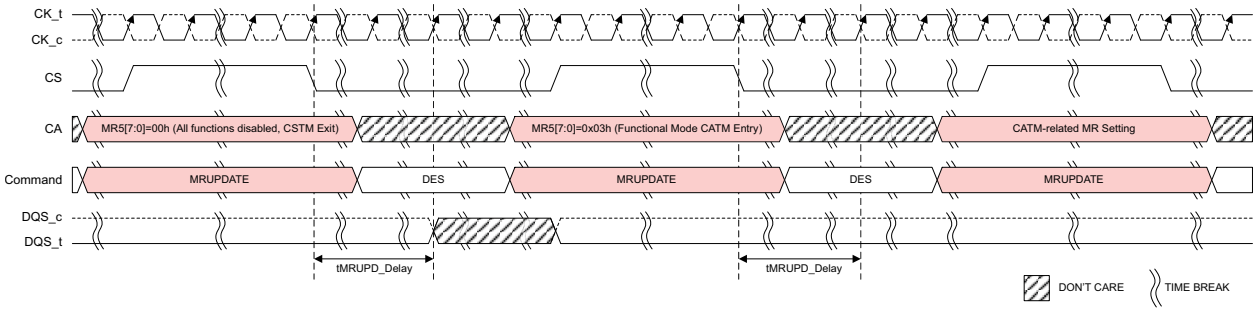


Figure 42 — CATM Entry Sequence Following CSTM Exit

4.2.8.4. Exiting CATM

To exit CATM, the Host issues the MRUPDATE sequence to MR05[7:0] = 00h in the Command State with DQS differentially low. During CATM Exit, DQS must remain differentially low until tMRUPD_Delay is satisfied. After tMRUPD_Delay, CATM mode is exited.

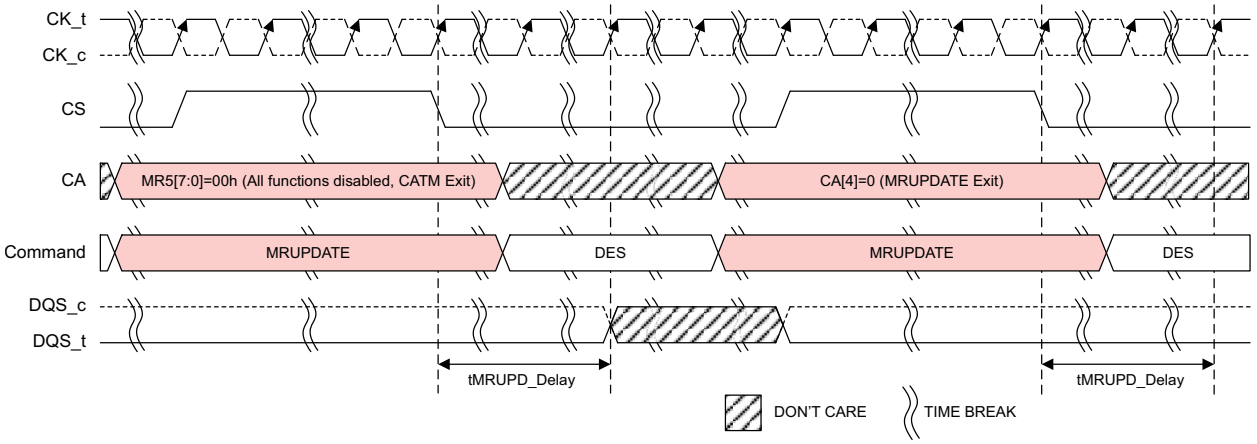


Figure 43 — CATM Exit Followed by MRUPDATE Mode Entry

Issuing an MRUPDATE sequence exit during the CMD state will exit both CATM and MRUPDATE modes within tMRUPD_ Delay.

4.2.8.5. CATM DQ Function and Mapping

CATM uses the DQ bus, sampled by DQS, for function control. DQ[1:0] are input pins latched by DQS for CATM control logic. The Host is required to drive identical waveforms on the DQ[1:0] pins of both PSDs.

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During CATM, DQS_t/DQS_c (DQS) are input only to sample DQ[1:0] as the function code. DQS is required to be differentially low except while sampling DQ[1:0]. CATM uses a ~~four (4)~~eight (8) bit Function Code (FuncCode[~~3~~7:0]), generated by ~~four (4)~~eight (8) DQS rising pulses to sample DQ[1:0]. Entering CATM internally resets the DQS rising pulse count to 0 to prepare for counting the four rising pulses. Each Function Code is the combined sample value of the 2nd, 4th, 6th, and 8th differential DQS rising edges (DQSR2, DQSR4, DQSR6, DQSR8, respectively) ~~and the 4th differential DQS rising edge (DQSR4)~~ as shown the CATM Function Code Mapping table.

DQ2 is an output pin for the CATM sample results, per PSD.

All other DQ pins, if present, are unused during CATM and are disabled.

Table 31 — Data bits (DQ) function during CATM

DQ	Input/Output	Functions during CATM
DQ[1:0]	Input	Control logic input
DQ2	Output	Sample results output
DQS_t/DQS_c	Input	DQ[1:0] sample

Table 32 — CATM Function Code Mapping

Function Code	DQ Source	DQS Sampling Edge
FuncCode[~~3~~7]	DQ[1]	2nd Rising Edge
FuncCode[~~2~~6]	DQ[0]	2nd Rising Edge
FuncCode[~~1~~5]	DQ[1]	4th Rising Edge
FuncCode[~~0~~4]	DQ[0]	4th Rising Edge
FuncCode[3]	DQ[1]	6th Rising Edge
FuncCode[2]	DQ[0]	6th Rising Edge
FuncCode[1]	DQ[1]	8th Rising Edge
FuncCode[0]	DQ[0]	8th Rising Edge

As shown in the figure below, the leading and trailing DQS differentially low timing (tDQSPreTrain and tDQSPostTrain, respectively) must be met before and after the DQS pulses. tFunc_Delay is the minimum time between Function Codes. Each DQS pulse must meet tDQSPWTrain on both high and low pulse.

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DQ[1:0] must meet setup and hold timing (tDSTrain and tDHTrain, respectively). The setup timing tDSTrain is defined as DQ[1:0] leading edge to DQSR2 and the hold timing tDHTrain is defined as DQSR2 to DQ[1:0] trailing edge. The same setup and hold timing requirements apply to DQSR4, DQSR6, and DQSR8.

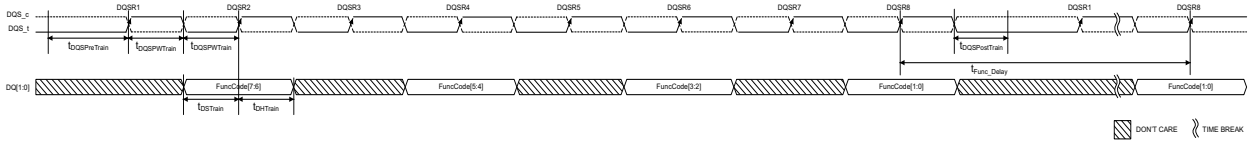


Figure 44 — Example Timing for Functional Codes

NOTE: CK and time breaks are not included.

4.2.8.6. Command and Train States

Command (CMD) State is the default state after CATM entry. CMD state is MRUPDATE mode enabled with CATM functions controlled by DQ[1:0] and DQS. Only the MRUPDATE sequence and DES command are valid in the CMD State. During CATM, the MRUPDATE sequence is only used to trigger functions and to change shadow register bits related to CATM settings such as VREFCA, CA ODT, CK ODT, DQ/DQS ODT, and DQ drive strength, as needed. Shadow register "Apply" is needed to update any of the normal mode register values for use during CATM CMD State. The VREFCA generator references the VREF CA Mode Registers settings while in CMD State.

Train State is used to adjust the VREFCA shadow registers setting, perform training sequence sampling, and output training results on DQ2. Train State uses CK_ODT, CS_ODT, CA_ODT shadow register values, as well the VREF CS and VREF CA shadow register values. In Train State, CA is required to be low except when sending the training sequence. VREFCA change functions will only apply to the VREF CA shadow register, which updates the VREFCA generator since it is referencing the shadow register while in Train State. Upon completion of CATM, Func-Code[~~3~~7:0]=~~4~~8b'00000000b forces the return from Train State to CMD State, restoring use of the normal VREF CA Mode Register. The Host can apply the VREF CA shadow register value to VREF CA MR (MR~~1A~~84[6:0]) per training results during CMD State or after exiting CATM using the VREF CA and CA ODT Apply Functional Mode (MR0~~5[xy]~~6[7:0]=08h).Shadow register values are not changed by entering or exiting CATM, by entering CMD State, or by entering Train State.

Issuing the shadow register Apply during CMD State or after exiting CATM, switching from CMD State to Train State, and switching from Train State to CMD State, requires waiting the appropriate VREFCA update and stabilization delay which depends on the number of VREFCA steps being changed.

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DQ[1:0] are input pins during both CMD and Train States. DQ2 is off during CMD State. DQ2 remains off in Train State until the first time it is used as output pin during SMP function. DQS_t/DQS_c are input pins during both states.

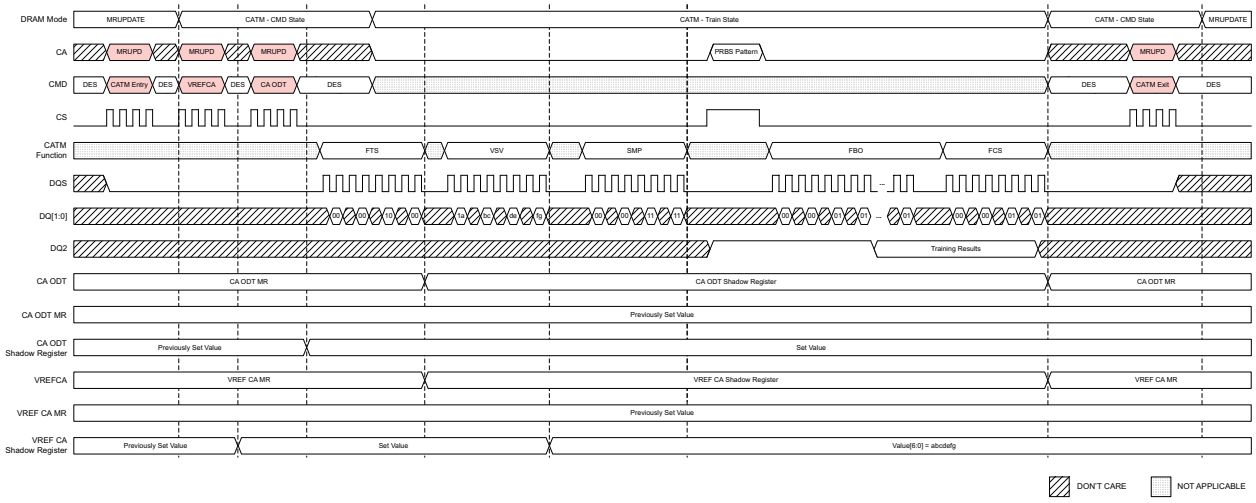


Figure 45 — CATM Entry/Exit with CMD State and Train State Switch

NOTE: 1: CATM Functions refer to the next section

NOTE: 2: CK and time breaks are not included in the figure. The diagram is intended to serve as a reference for behavior rather than for timing purposes.

NOTE: 3: MRUPDATE sequence to set VREFCA and CA ODT during CMD State is only for illustration purposes

NOTE: 4: VREF CA and CA ODT registers are shown as updated to an appropriate previously set value as recommended in section XXX, Voltage Ramp and Device Initialization

4.2.8.7. CATM Functions

CATM uses a ~~four (4)~~eight (8) bit Function Code (FuncCode[~~3~~7:0]) to execute training functions. When a FuncCode[~~3~~7:0] is received, the corresponding functions executes within tFunc_Delay. Some functions can only be used in certain CATM states. FuncCode[~~3~~7:0] issued outside of an allowed state will be ignored. Any unused FuncCode[~~3~~7:0] is RFU and will be ignored.

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Table 33 — CATM Functions

FuncCode[~~3~~7:0]	Function	Allowed State	Description
00000000	FCS	CMD / Train	Force “CMD State”. Use normal VREF and ODT MRs.
00000101	FBO	Train	Output next training results on DQ2
00001000	FTS	CMD / Train	Force “Train State”. Use VREF and ODT shadow registers.
~~1001~~	~~VSD~~	~~CMD / Train~~	~~Decrease VREF_CA shadow register by 1 (step down)~~
~~1010~~	~~VSU~~	~~CMD / Train~~	~~Increase VREF_CA shadow register by 1 (step up)~~
00001111	SMP	Train	Reset LFSR, reset CATM results, enable CS qualified sampling of CA bus
1VVVVVVV	VSV	CMD / Train	Set VREF CA shadow register to FuncCode[6:0]
Others	RFU		

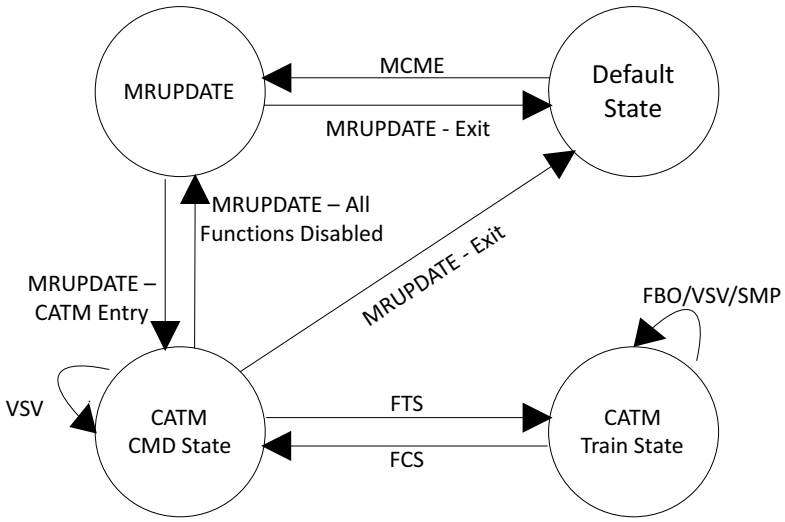


Figure 46 — CATM Simplified Diagram

NOTE: Default states include Idle and Self-Refresh without Power-Down

FuncCode[~~3~~7:0]=~~4~~8b'00000000: "Force CMD State (FCS)" function. The FCS function may be issued in both the CMD State and the Train State.

FuncCode[~~3~~7:0]=~~4~~8b'00001000: "Force Train State (FTS)" function. The FTS function forces Train State and switches from using normal VREF and ODT MRs to using the VREF and ODT shadow registers. The FTS function can be issued in both the CMD State and Train State. During Train State, VREFCA is controlled by the VREF CA shadow register.

~~FuncCode[3:0]=4b'1001: “VREFCA Step Down (VSD)” function. The VSD function forces the VREF_CA shadow register value to decrease by one step. If the register is at the minimum value, the VREFCA will remain at the minimum value if issued additional VSD functions. The VSD function can be issued in either CMD or Train state.~~

~~FuncCode[3:0]=4b'1010: "VREFCA Step Up (VSU)" function. The VSU function forces the VREF_CA shadow register value to increase by one step. If the register is at the maximum value, the VREFCA will remain at the maximum value if issued additional VSU functions. The VSU function can be issued in either CMD or Train state.~~

FuncCode[7:0] = 8b'1VVVVVVV: "VREF Set Value (VSV)" function. The VSV function sets the VREF CA shadow register bits VREFCA[6:0] to FuncCode[6:0].

Note: VREFCA does not use the VREF CA shadow register during the CMD state, so changes will not be applied until Train State is entered. ~~Continuous VSD and VSU operation is possible if tFUNC_Delay between each operation is met. However, t~~The Host must wait the required VREFCA update and settling delay after ~~last VSU or VSD~~ any VSV function.

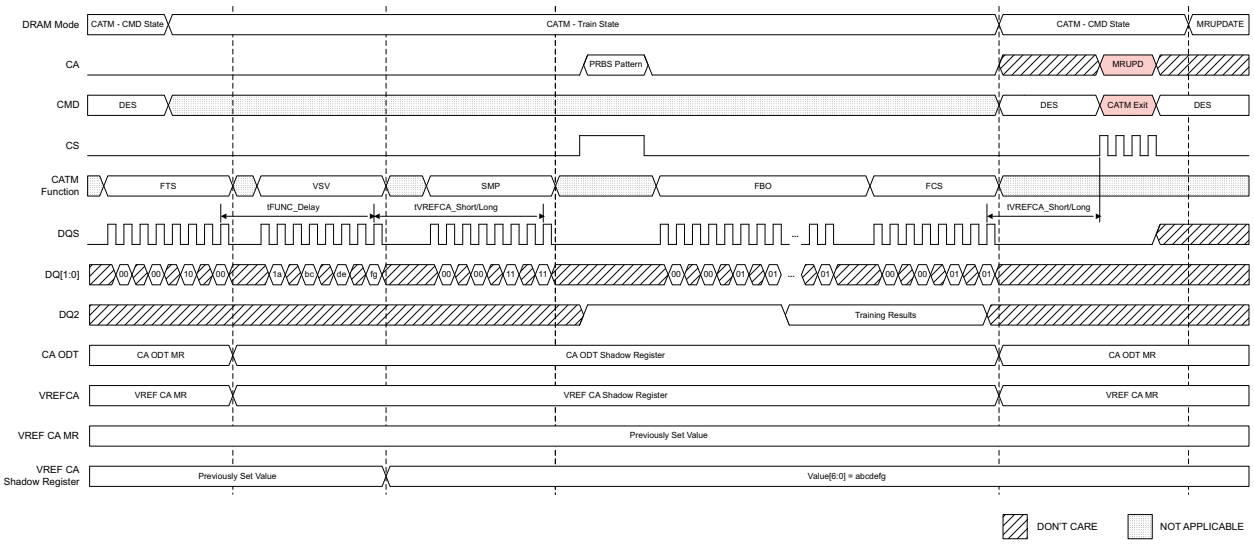


Figure 47 — VSV Function Example and tVREFCA_short/long Requirement

NOTE: 1:CK and time break are not included on purpose. The diagram is intended to serve as a reference for consecutive ~~VSU/VSD~~VSV behavior and tVREFCA_short/long timing requirement. Any additional timing requirements not depicted in the figure must also be satisfied.

NOTE: 2: VREF CA registers are shown initially updated to an appropriate previously set value as recommended in section XXX, Voltage Ramp and Device Initialization.

FuncCode[~~3~~7:0]=~~4~~8b'00001111: "Sample (SMP)" function. The SMP function performs the following operations to prepare for CA sampling: reset the LFSR, reset the CATM internal training results to fail, force DQ2 high, and enable CS high qualified sampling of the CA burst or bursts. After the reset to CATM delay (tRST2CATM) is met, the (first) CA burst can be sampled. The Host sends the correct PRBS pattern on the CA bus, qualified by CS high. The first cycle of the CA burst with CS high must be sent by the Host corresponding to a phase 0° CK rising edge, as previously identified through CSTM. The DRAM will sample the CA burst on this CK differential rising edge and each subsequent differential rising edge of CK until CS is sampled low. The first phase 0° CK rising edge with CS low terminates the burst; only CA samples latched prior to this CK rising edge will be included in the burst pattern comparison. The Host may use any CA burst length that is a non-zero multiple of 4 nCK. The CA sampling may be broken into multiple CS high qualified CA bursts, with each burst separated by "gaps" in which CS is low. In this case, each CA burst length must be a multiple of 4nCK and must be sent corresponding to a phase 0° CK rising edge. The LFSR is not advanced during any CS low "gap" period.

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For each sample of the CA bus, the sampled value is compared against the expected value from the internal PRBS generator. Passing samples are recorded as '0' and failing samples recorded as '1'. These results are accumulated across ~~the full~~ all bursts per CA bit and even/odd phase using a logical 'OR' operation. Note, "even" phase corresponds to both 0° and 180º phases of CK; "odd" phase corresponds to both 90° and 270° phases of CK. This yields 10 separate feedback bits if CA5 is not enabled: 5 CA lanes x 2 even/odd phases. If CA5 is enabled, this increases to 12 separate feedback bits: 6 CA lanes x 2 even/odd phases. See the table below for the feedback sequence.

DQS is required to be differentially low during the CA burst or bursts. After the (last) CA burst is terminated and tCATM2FBO is met, the DRAM is ready to output feedback results. After this point, whenever the DRAM receives FuncCode[~~3~~7:0]=~~4~~8b'00000101 it will execute the Feedback Output (FBO) function. On the first execution of the FBO function, the first feedback bit on DQ2 is CA0_E, with the results being available tADR after the last rising DQ edge of the FBO function. A feedback bit value of '0' indicates the entire burst passed for that CA bit and phase. A feedback bit value of '1' indicates a fail for that CA bit and phase at some point during the burst. On the next execution of the FBO function, DQ2 will output the next feedback bit, CA0_O, and so on for each iteration of the FBO function until the final feedback has been output. If additional FBO functions are executed after the final feedback bit has been output, the final feedback bit will continue to be output. The Host can reset the feedback results and the LFSR for the PRBS pattern by issuing the SMP function again, enabling a new CS qualified sampling of the CA bus.

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Table 34 — CATM Feedback Bits

Feedback Bit	Label	CA Bit and CK Phase Results (Logical OR of all include samples)	Note
0	CA0_E	CA0, even CK differential rising edges	
1	CA0_O	CA0, odd CK differential rising edges	
2	CA1_E	CA1, even CK differential rising edges	
3	CA1_O	CA1, odd CK differential rising edges	
4	CA2_E	CA2, even CK differential rising edges	
5	CA2_O	CA2, odd CK differential rising edges	
6	CA3_E	CA3, even CK differential rising edges	
7	CA3_O	CA3, odd CK differential rising edges	
8	CA4_E	CA4, even CK differential rising edges	
9	CA4_O	CA4, odd CK differential rising edges	1
10	CA5_E	CA5, even CK differential rising edges	2
11	CA5_O	CA5, odd CK differential rising edges	2

NOTE 1: If CA5 is disabled, bit 9 will be the final feedback bit.
NOTE 2: Bits 10 and 11 will only be output if CA5 is enabled. If CA5 is enabled, bit 11 will be the final feedback bit.

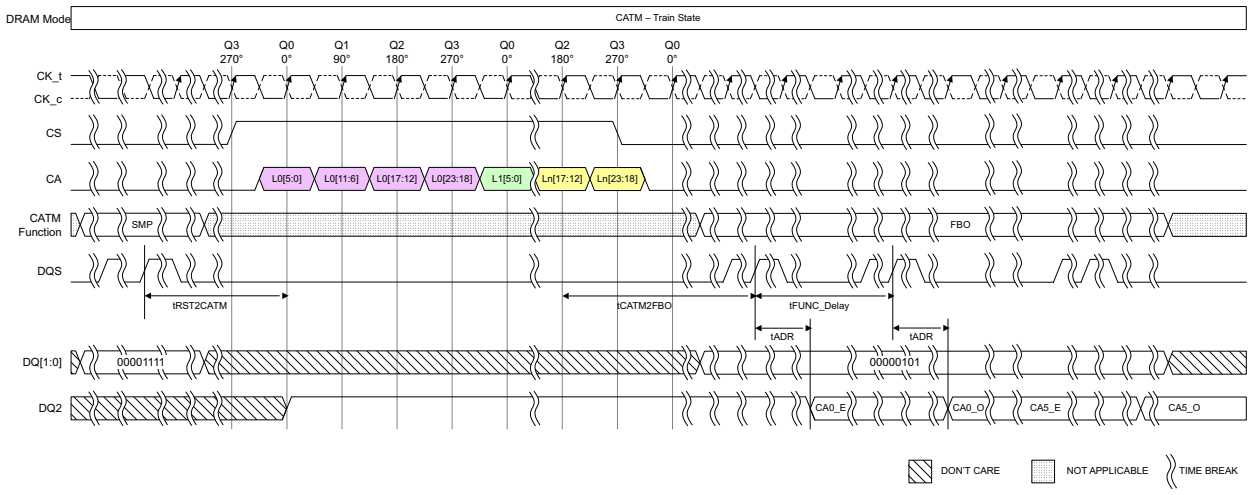


Figure 48 — SMP and FBO Function Timing Example

NOTE: 1: CA burst of PRBS pattern shown for CA5 enabled.
NOTE: 2: SMP and FBO function four cycle sequences shown compressed for illustration purposes. tRST2CATM and tCATM2FBO are measured from the last rising DQS edge of the respective function.
NOTE: 3: Feedback results shown for CA5 enabled (12 values total, CA0_E through CA5_O).

NOTE: 4: SMP and FBO functions are shown as a combined 8-bit FuncCode on DQ[1:0] with only the first and final DQS rising edges shown for readability. Please see the Example Timing for Functional Codes figure above for additional timing details.

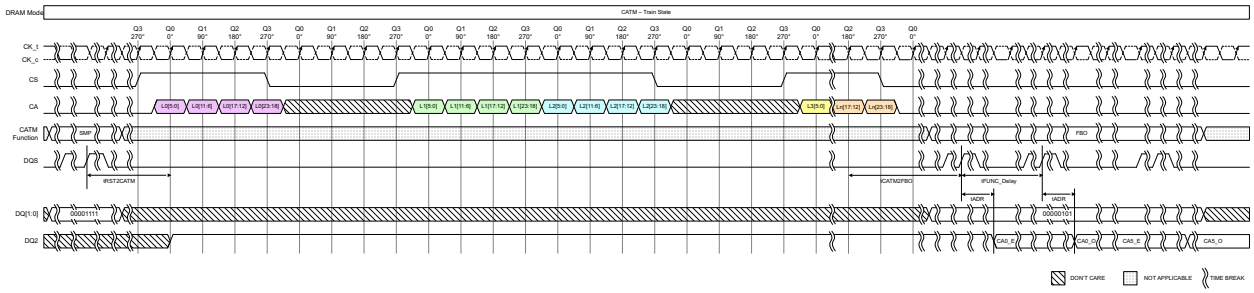


Figure 49 — SMP and FBO Function with Burst Gaps Timing Example

NOTE: 1: CA burst of PRBS pattern shown for CA5 enabled.

NOTE: 2: SMP and FBO function four cycle sequences shown compressed for illustration purposes. tRST2CATM and tCATM2FBO are measured from the last rising DQS edge of the respective function.

NOTE: 3: Feedback results shown for CA5 enabled (12 values total, CA0_E through CA5_O).

NOTE: 4: SMP and FBO functions are shown as a combined 8-bit FuncCode on DQ[1:0] with only the first and final DQS rising edges shown for readability. Please see the Example Timing for Functional Codes figure above for additional timing details.

Issuing MRUPDATE sequences or executing a new CATM function must wait for a previous operation to complete. The only exception is an MRUPDATE sequence for CATM Exit, which may be used to exit CATM during CMD State with DQS differentially low at least tFunc_Delay plus the required VREFCA update and settling delay after the last DQS differential rising edge.

The following figure is an example of the sequence to switch to CMD State and exit CATM when DQS pulse count has been lost. The Host issues at least 816 DQS pulses with DQ[1:0]=00b, where each DQS pulse meets tFunc_Delay between rising edges. CATM exit command must meet tFunc_Delay plus the required VREFCA update and settling delay after the last DQS differential rising edge. CAs remain as "L", as the starting state may be the Train State.

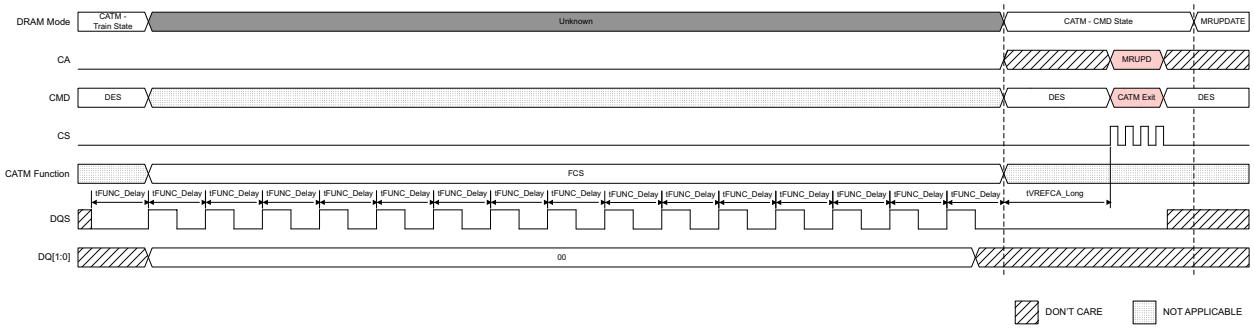


Figure 50 — Example of CATM Exit in the Middle of 4 DQS Pulses

4.2.8.8. CATM PRBS Generator

For CATM, the Host and the DRAM use a Pseudo Random Binary Sequence (PRBS) generated by a 24-bit Galois Linear Feedback Shift Register (LFSR). This sequence generator will be used on the Host to generate the bit stream for the CA burst. The DRAM will use the same sequence generator to generate the expected data and then compare

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the received data with the expected data to determine if the CA bits were transferred successfully from the Host to the DRAM. The feedback results from comparison per CA bit and CK phase will be sent back to the Host.

The polynomial to be used for the 24-bit Galois LFSR is x^24 + x^22 + x^21 + x^19 + x^17 + x^14 + x^10 + x^4 + 1. The figures below show the implementation of the PRBS generator using a 24-bit shift register and XOR gates, as well as the mapping of LFSR bits to the CA bus. Error checking on the CA burst is performed in 4nCK units (corresponding to 20 bits if CA5 is disabled or 24 bits if CA5 is enabled). The LFSR updates to a new 24-bit value every 4nCK. Note that the same bits are mapped to CA[4:0] whether CA5 is enabled or not.

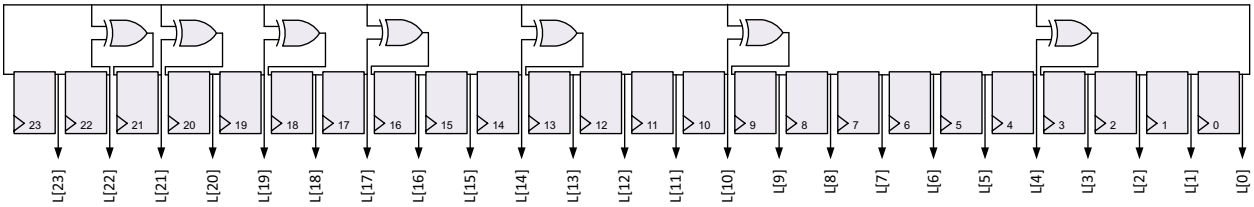


Figure 51 — Galois LFSR 24-bit Circuit

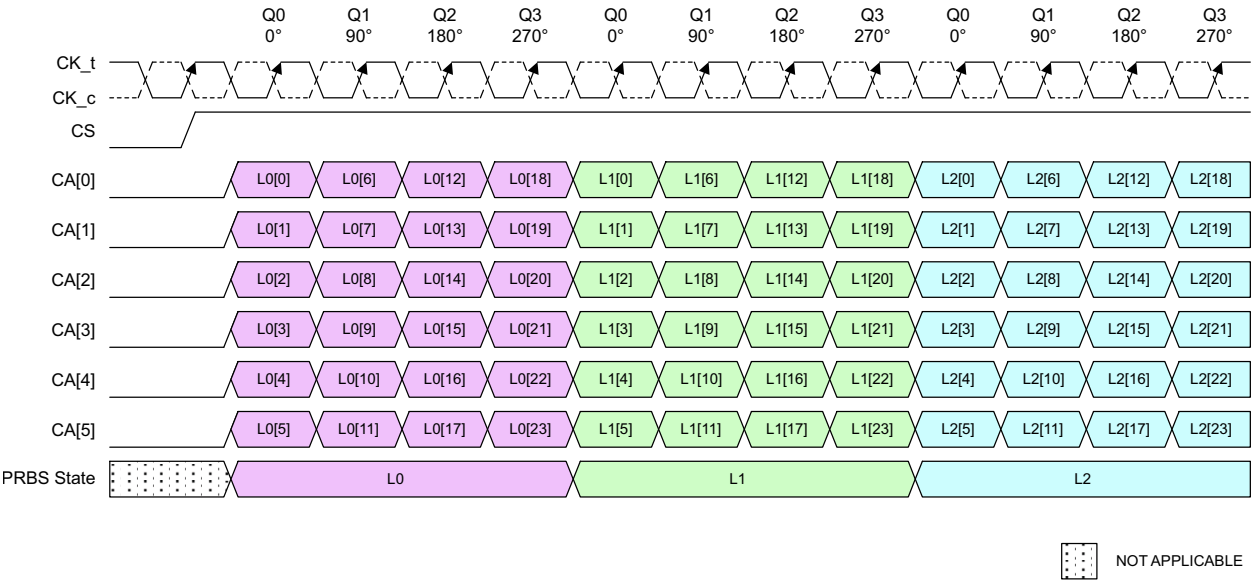


Figure 52 — LFSR 24-bit Mapping to CA Bus

The PRBS generator must be initialized with the same default seed to ensure the same pattern is generated on both the Host and DRAM. This seed is chosen to initialize the PRBS generator such that it generates an alternating 1 and 0 sequence on the CA bus. The initial 8 values seen on the CA bus when the PRBS generator is run with an example default seed of 24'h4894AC are shown in the table below, where L0 is the first data pattern driven from the host and the first PRBS state for comparison.

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Table 35 — LFSR PRBS State

PRBS State	LFSR[23:0]	LFSR Bit																							
		23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Seed	24'h4894AC	0	1	0	0	1	0	0	0	1	0	0	1	0	1	0	0	1	0	1	0	1	1	0	0
L0	24'h244A56	0	0	1	0	0	1	0	0	0	1	0	0	1	0	1	0	0	1	0	1	0	1	1	0
L1	24'h12252B	0	0	0	1	0	0	1	0	0	0	1	0	0	1	0	1	0	0	1	0	1	0	1	1
L2	24'hBC309D	1	0	1	1	1	1	0	0	0	0	1	1	0	0	0	0	1	0	0	1	1	1	0	1
L3	24'hEB3A46	1	1	1	0	1	0	1	1	0	0	1	1	1	0	1	0	0	1	0	0	0	1	1	0
L4	24'h759D23	0	1	1	1	0	1	0	1	1	0	0	1	1	1	0	1	0	0	1	0	0	0	1	1
L5	24'h8FEC99	1	0	0	0	1	1	1	1	1	1	1	0	1	1	0	0	1	0	0	1	1	0	0	1
L6	24'hF2D444	1	1	1	1	0	0	1	0	1	1	0	1	0	1	0	0	0	1	0	0	0	1	0	0
L7	24'h796A22	0	1	1	1	1	0	0	1	0	1	1	0	1	0	1	0	0	0	1	0	0	0	1	0

The PRBS generator will start from the default seed value programmed in ~~MRxa-MRxc~~ MRD8-MRDA whenever the LFSR is reset by the SMP function.

4.2.8.9. Example CATM Sequence

An example of the CATM sequence is described below:

1) If not in MRUPDATE mode, then use MCME to enter MRUPDATE mode.
2) If not previously completed, perform the DQ Swizzling Discovery Mode (DSDM) operation to identify the Host to DRAM DQ mapping.
3) If not previously completed, perform CS Training Mode to align all CK phases to achieve the maximum CS eye.
4) Set DQS differentially low. Enter CATM and wait tMRUPD_Delay.
5) ~~Update the CA_ODT and VREF_CA Mode Register values, if required, and Apply.~~If required, update the CA ODT and VREF CA shadow register values and Apply to normal MR for use in CMD State. The VREF CA normal MR value must be valid in CMD State to allow the MRUPDATE sequence to execute properly.
6) ~~Update the CA_ODT and VREF_CA shadow register values, if required. The VREFCA value must be valid to allow the MRUPDATE sequence to execute properly, and Train State will use the VREF_CA shadow register value.~~If required, update the CA ODT and VREF CA shadow register values for use in Train State.
7) Switch from CMD State to Train State. Wait for tFunc_Delay and VREFCA update and settling delay. The shadow registers are being used.
8) Update the VREFCA setting via the ~~VSU or VSD~~VSV function. Wait VREFCA update and settling delay after ~~the last VSU/VSD~~VSV.
9) Issue the SMP function to prepare for sampling. After tRST2CATM, The DRAM is ready for CA sampling.
10) Perform CA training (VREFCA, CA timing):
    a) The Host sends the CA burst or bursts of the PRBS pattern with CS high.
    b) The CA burst pattern(s) is(are) sampled on each CS high qualified CK differential rising edge, comparing each bit with the expected PRBS pattern and accumulating the feedback results per CA bit and even/odd CK phase using a logical 'OR' operation. A resulting value of '1' indicates a fail while a '0' indicates a

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pass. Note: "even" phase corresponds to both 0° and 180º phases of CK; "odd" phase corresponds to both 90° and 270° phases of CK.

c) After terminating the (last) burst and waiting tCATM2FBO, issue the FBO function repeatedly to output each feedback bit successively on DQ2.

d) Calculate the full training results. When the results for all CA bits and CK phases are '0' for both PSDs, the training is considered a pass.

11) Repeat step 8 to 10 as necessary to complete the first round of CA training.

12) Use the FCS command to switch to the CMD State, wait tFUNC_Delay and the VREFCA update and settling delay.

13) Exit CATM. After tMRUPD_Delay, CA is "trained".

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Table 36 — Timing Parameters for CATM

Item Description	Parameters	Min/ Max	Value	Unit	Notes
Asynchronous Data Read	tADR	Min	TBD	nS	
Last even CK differential rising edge of CA burst to 4th DQS differential rising edge of FBO function	tCATM2FBO	Min	TBD	nS	
DQ[1:0] setup	tDSTrain	Min	6	nCK	
DQ[1:0] hold	tDHTrain	Min	6	nCK	
DQS differentially low prior to 1st DQS rising edge	tDQSPreTrain	Min	6	nCK	
DQS differentially low after to 48th DQS falling edge	tDQSPostTrain	Min	6	nCK	
DQS pulse width	tDQSPWTrain	Min	6	nCK	
Training function to training function delay	tFunc_Delay	Min	tMRUPD_DELAY	nS	
SMP function to 1st CS qualified CA sample	tRST2CATM	Min	20	nS	
VREF Stable - Long	tVREFCA_long	Min	350	nS	1
VREF Stable - Short	tVREFCA_~~long~~short	Min	250	nS	2

NOTE:
1: tVREFCA_long is for a change in VREFCA voltage of at least two steps.

NOTE:
2: tVREFCA_short is for a change in VREFCA voltage of a single step.

4.2.9. Rx CK/DQS/CA/CS CTLE Control

4.2.9.1. Rx CK/DQS/CA/CS CTLE Control

Rx CK/DQS/CA/CS in DDR6 shall have a Continuous-Time Linear Equalizer (CTLE) to compensate channel loss for high frequency operation by providing increased gain at higher frequencies, offering a simple and effective solution for DDR6. Rx CK/DQS/CA/CS CTLE training shall be performed after CSTM and CATM sequence.

4.2.9.2. MR1C– Rx CTLE Control for CK, DQS

Table 37 — MR1C information

Bits	Attribute	Default	Description	Notes
1:0	RW	0h	MR1C[3:2]: RX_CK_CTLE_CNTRL Rx CK CLTE control options 00: No equalization (Default) 01: Weak equalization 10: Middle equalization 11: Strong equalization	1,2,3
3:2	RW	0h	MR1C[3:2]: RX_DQS_CTLE_CNTRL Rx DQS CLTE control options 00: No equalization (Default) 01: Weak equalization 10: Middle equalization 11: Strong equalization	1,2,3
NOTE 1: Rx CTLE is an optional feature in DDR6. MR1D[0] indicates host whether Rx CTLE is supported or not **NOTE 2:** Refer to the vendor data sheets for more information regarding Rx CTLE Control Settings. **NOTE 3:** Since CTLE circuits cannot be typically by-passed, a disable option is not provided. The optimized setting will be given by a vendor, which should be noted that the settings are not specifically linear in relationship to the given setting, so the host can walk through all the provided options and use the setting that works best in their environment.				

4.2.9.3. PG[0]MR09 – Rx CTLE Control for CA, CS

Table 38 — PG[0]MR09 information (Sheet 1 of 2)

Bits	Attribute	Default	Description	Notes
1:0	RW	0h	PG00MR09[1:0]: RX_CA_CTLE_CNTRL Rx CA CTLE control 00: No equalization (Default) 01: Weak equalization 10: Middle equalization 11: Strong equalization	1,2,3

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Table 38 — PG[0]MR09 information (Continued) (Sheet 2 of 2)

Bits	Attribute	Default	Description	Notes
3:2	RW	0h	PG00MR09[3:2]: RX_CS_CTLE_CNTRL Rx CS CTLE control 00: No equalization (Default) 01: Weak equalization 10: Middle equalization 11: Strong equalization	1,2,3
NOTE 1: Rx CTLE is an optional feature in DDR6. MR1D[0] indicates host whether Rx CTLE is supported or not **NOTE 2:** Refer to the vendor data sheets for more information regarding Rx CTLE Control Settings. **NOTE 3:** Since CTLE circuits cannot be typically by-passed, a disable option is not provided. The optimized setting will be given by a vendor, which should be noted that the settings are not specifically linear in relationship to the given setting, so the host can walk through all the provided options and use the setting that works best in their environment.				

4.2.9.4. MR1D – Rx CTLE Support

Table 39 — MR1D information

Bits	Attribute	Default	Description	Notes
0	R	0	MR1D[0]: RX_CTLE_SUPPORT Rx CTLE support indicator 0: Not supported 1: Supported	1
NOTE 1: Rx CTLE is an optional feature in DDR6. It may be needed for DRAMs that operate at >= TBD Mbps. MR1D[0] indicates host whether Rx CTLE is supported or not.				

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4.2.10. Per-pin VREFCA

DDR6 adopts Per-pin VREFCA training to compensate for CA input offset due to SOC transmitter mismatch, DRAM termination mismatch, and DRAM CA path mismatch. Per-pin VREFCA training allows host to program the reference VREF level to an optimum point for each single-ended CA receiver. ~~Per-pin VREFCA training is performed after CATM exit.~~The offset levels among optimized VREFCA levels for each CA pin, which is obtained from VREFCA training in CATM, shall be applied as the per-pin VREFCA offset.

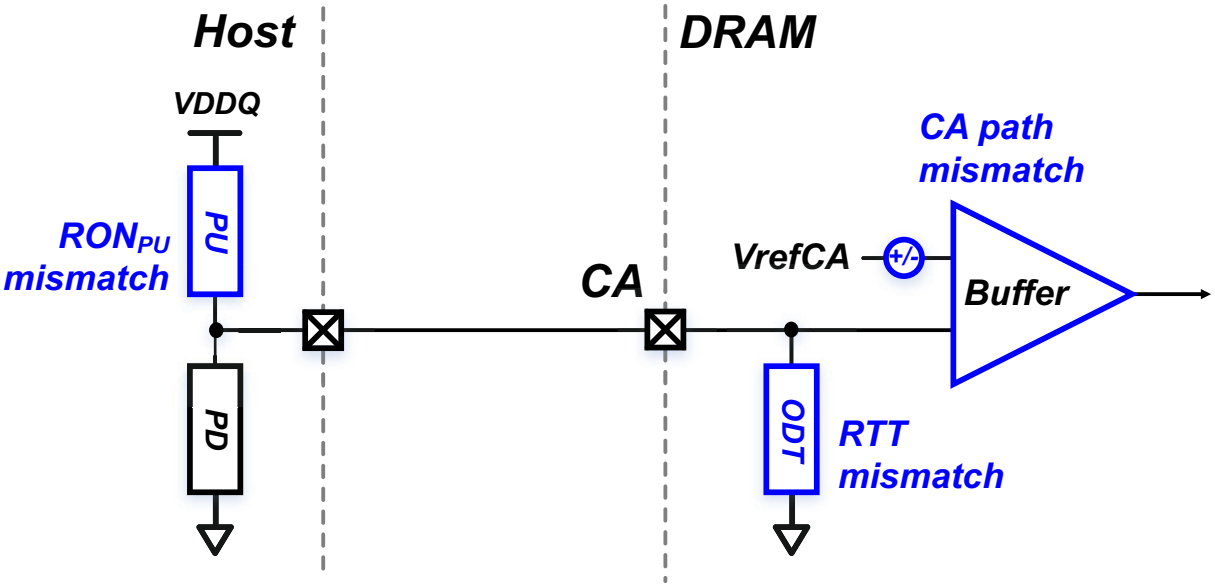


Figure 53 — CA Input offset

4.2.10.1. MR~~1A~~84 – VREFCA MR define

Table 40 — MR~~1A~~84 information

Bits	Attribute	Default	Description	Notes
6:0	RW	TBD	MR84[6:0]: VREFCA VREFCA control options 0000000_B = 10.0% 0000001_B = 10.5% 0000010_B = 11.0% 0000011_B = 11.5% ... 1111110_B = 73.0% 1111111_B = 73.5%	~~1~~,2,3,4,5,6,7

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Table 40 — MR~~1A~~84 information

Bits	Attribute	Default	Description	Notes

NOTE 1: ~~This register controls the VREFCA levels for Frequency Set Point[0]~~
NOTE 2: A read (MRR) to this register places the contents of OP[x:x] on DQ[x:x]. Any RFU bits and unused DQs will be read 0_B. See TBD for more information on MRR Operation.
NOTE 3: There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Please refer to TBD section for more information on FSP operation.
NOTE 4: The time required for VREFCA to reach the set level depends on the step size from the current level to the new level. See TBD for more information on VREFCA training.
NOTE 5: VREFCA can be adjusted by either MRW(MR84[6:0]) in normal mode (not MRUPDATE mode) or VREF_CA shadow register with VREFCA Apply Functional Mode (MR06[3]=1_B) in MRUPDATE mode.
NOTE 6: Control of VREFCA during CATM is referred to section 4.2.8 (CATM)
NOTE 7: Refer to Table 2 for the detailed VREFCA setting values.

Table 41 — MR~~1A~~84 VREFCA Setting

Function	Operand	VREFCA Values (% of VDDQ)								Notes
VREF CA Setting	OP[6:0]	000000_B	10.0*4	010000_B	26.0	100000_B	42.0	1100000_B	58.0	1,2,3,4,5,6
		000000_B	10.5*4	010000_B	26.5	100000_B	42.5	1100001_B	58.5	
		000001_B	11.0*4	010001_B	27.0	100001_B	43.0	1100010_B	59.0	
		000001_B	11.5*4	010001_B	27.5	100001_B	43.5	1100011_B	59.5	
		000010_B	12.0*4	010010_B	28.0	100010_B	44.0	1100100_B	60.0	
		000010_B	12.5	010010_B	28.5	100010_B	44.5	1100101_B	60.5	
		000011_B	13.0	010011_B	29.0	100011_B	45.0	1100110_B	61.0	
		000011_B	13.5	010011_B	29.5	100011_B	45.5	1100111_B	61.5	
		000100_B	14.0	010100_B	30.0	100100_B	46.0	1101000_B	62.0	
		000100_B	14.5	010100_B	30.5	100100_B	46.5	1101001_B	62.5	
		000101_B	15.0	010101_B	31.0	100101_B	47.0	1101010_B	63.0	
		000101_B	15.5	010101_B	31.5	100101_B	47.5	1101011_B	63.5	

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Table 41 — MR~~1A84~~ VREFCA Setting

Function	Operand	VREFCA Values (% of VDDQ)								Notes
		$000110\ 0_B$	16.0	$010110\ 0_B$	32.0	$100110\ 0_B$	48.0	1101100_B	64.0	
		$000110\ 1_B$	16.5	$010110\ 1_B$	32.5	$100110\ 1_B$	48.5	1101101_B	64.5	
		$000111\ 0_B$	17.0	$010111\ 0_B$	33.0	$100111\ 0_B$	49.0	1101110_B	65.0	
		$000111\ 1_B$	17.5	$010111\ 1_B$	33.5	$100111\ 1_B$	49.5	1101111_B	65.5	
		$001000\ 0_B$	18.0	$011000\ 0_B$	34.0	$101000\ 0_B$	50.0	1110000_B	66.0	
		$001000\ 1_B$	18.5	$011000\ 1_B$	34.5	$101000\ 1_B$	50.5	1110001_B	66.5	
		$001001\ 0_B$	19.0	$011001\ 0_B$	35.0	$101001\ 0_B$	51.0	1110010_B	67.0	
		$001001\ 1_B$	19.5	$011001\ 1_B$	35.5	$101001\ 1_B$	51.5	1110011_B	67.5	
		$001010\ 0_B$	20.0	$011010\ 0_B$	36.0	$101010\ 0_B$	52.0	1110100_B	68.0	
		$001010\ 1_B$	20.5	$011010\ 1_B$	36.5	$101010\ 1_B$	52.5	1110101_B	68.5	
		$001011\ 0_B$	21.0	$011011\ 0_B$	37.0	$101011\ 0_B$	53.0	1110110_B	69.0	
		$001011\ 1_B$	21.5	$011011\ 1_B$	37.5	$101011\ 1_B$	53.5	1110111_B	69.5	
		$001100\ 0_B$	22.0	$011100\ 0_B$	38.0	$101100\ 0_B$	54.0	1111000_B	70.0	
		$001100\ 1_B$	22.5	$011100\ 1_B$	38.5	$101100\ 1_B$	54.5	1111001_B	70.5	
		$001101\ 0_B$	23.0	$011101\ 0_B$	39.0	$101101\ 0_B$	55.0	1111010_B *6	71.0	
		$001101\ 1_B$	23.5	$011101\ 1_B$	39.5	$101101\ 1_B$	55.5	1111011_B *6	71.5	
		$001110\ 0_B$	24.0	$011110\ 0_B$	40.0	$101110\ 0_B$	56.0	1111100_B *6	72.0	
		$001110\ 1_B$	24.5	$011110\ 1_B$	40.5	$101110\ 1_B$	56.5	1111101_B *6	72.5	

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Table 41 — MR~~1A84~~ VREFCA Setting

Function	Operand	VREFCA Values (% of VDDQ)								Notes
		$001111 0_B$	25.0	$011111 0_B$	41.0	$101111 0_B$	57.0	1111110_B *6	73.0	
		$001111 1_B$	25.5	$011111 1_B$	41.5	$101111 1_B$	57.5	1111111_B * 6	73.5	

NOTE 1: These values may be used for MR~~1A84~~[6:0] to set the VREFCA levels in the DDR6 SDRAM.
NOTE 2: Absolute VREFCA low level (%code * VDDQ) must be higher than or equal to 75mV for normal operation. VREFCA error is not included this calculation.
NOTE 3: Absolute VREFCA high level (%code * VDDQ) must be lower than or equal to VREFCA max, which refers to values in Table xx (CA Rx mask specification), for normal operation. VREFCA error is not included this calculation.
NOTE 4: VREFCA codes from 0000000_B to 0000100_B are able to be used only for testing purpose not for normal operation. VREFCA accuracy are not guaranteed from 0000000_B to 0000100_B.
NOTE 5: The VREFCA codes from 1110010_B to 1111111_B are provided to support cases where VDDQ is lower than the typical value, enabling the absolute VREFCA value to reach 0.4V.When VDDQ is at its typical voltage, codes with an absolute VREFCA value of higher than 0.4V cannot be used. This restriction also applies during training operations.
NOTE 6: The default VREFCA is defined by CA ODT strap value.

4.2.10.2. MR – VREFCA Offset (CA0~CA5)

~~MR03[3] has 1-bit to support Pseudo-Split Die (PSD) Indicator~~MR80[0] indicates the physical PSD, and Per-pin VREFCA training is performed per each PSD independently.

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Table 42 — ~~PG[0]MR03~~MR86 information (CA0, CA1)

Bits	Attribute	Default	Description	Notes
7:4	RW	0h	~~PG[0] MR03~~ MR86[7:4]: CA1_VREFCA_OFFSET CA1 VREFCA offset 1000_B: Reserved 1001_B: Step -7 1010_B: Step -6 1011_B: Step -5 1100_B: Step -4 1101_B: Step -3 1110_B: Step -2 1111_B: Step -1 0000_B: Disabled (default) 0001_B: Step +1 0010_B: Step +2 0011_B: Step +3 0100_B: Step +4 0101_B: Step +5 0110_B: Step +6 0111_B: Step +7	
3:0	RW	0h	~~PG[0]MR03~~ MR86[3:0]: CA0_VREFCA_OFFSET CA0 VREFCA offset 1000_B: Reserved 1001_B: Step -7 1010_B: Step -6 1011_B: Step -5 1100_B: Step -4 1101_B: Step -3 1110_B: Step -2 1111_B: Step -1 0000_B: Disabled (default) 0001_B: Step +1 0010_B: Step +2 0011_B: Step +3 0100_B: Step +4 0101_B: Step +5 0110_B: Step +6 0111_B: Step +7	

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Table 43 — ~~PG[0]MR04~~MR87 information (CA2, CA3)

Bits	Attribute	Default	Description	Notes
7:4	RW	0h	~~PG[0]MR04~~ MR87[7:4]: CA3_VREFCA_OFFSET CA3 VREFCA offset 1000_B: Reserved 1001_B: Step -7 1010_B: Step -6 1011_B: Step -5 1100_B: Step -4 1101_B: Step -3 1110_B: Step -2 1111_B: Step -1 0000_B: Disabled (default) 0001_B: Step +1 0010_B: Step +2 0011_B: Step +3 0100_B: Step +4 0101_B: Step +5 0110_B: Step +6 0111_B: Step +7	
3:0	RW	0h	~~PG[0]MR04~~ MR87[3:0]: CA2_VREFCA_OFFSET CA2 Vref CA offset 1000_B: Reserved 1001_B: Step -7 1010_B: Step -6 1011_B: Step -5 1100_B: Step -4 1101_B: Step -3 1110_B: Step -2 1111_B: Step -1 0000_B: Disabled (default) 0001_B: Step +1 0010_B: Step +2 0011_B: Step +3 0100_B: Step +4 0101_B: Step +5 0110_B: Step +6 0111_B: Step +7	

Table 44 — ~~PG[0] MR05~~MR88information (CA4, CA5)

Bits	Attribute	Default	Description	Notes
7:4	RW	0h	~~PG[0]MR05~~ MR88[7:4]: CA5_VREFCA_OFFSET CA5 VREFCA offset 1000_B: Reserved 1001_B: Step -7 1010_B: Step -6 1011_B: Step -5 1100_B: Step -4 1101_B: Step -3 1110_B: Step -2 1111_B: Step -1 0000_B: Disabled (default) 0001_B: Step +1 0010_B: Step +2 0011_B: Step +3 0100_B: Step +4 0101_B: Step +5 0110_B: Step +6 0111_B: Step +7	
3:0	RW	0h	~~PG[0]MR05~~ MR88[3:0]: CA4_VREFCA_OFFSET CA4 VREFCA offset 1000_B: Reserved 1001_B: Step -7 1010_B: Step -6 1011_B: Step -5 1100_B: Step -4 1101_B: Step -3 1110_B: Step -2 1111_B: Step -1 0000_B: Disabled (default) 0001_B: Step +1 0010_B: Step +2 0011_B: Step +3 0100_B: Step +4 0101_B: Step +5 0110_B: Step +6 0111_B: Step +7	

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4.2.11. Write Leveling Training Mode

4.2.11.1. Introduction

The DDR6 SDRAM supports a 'write leveling' feature to allow the controller to compensate CK to WRITE DQS timing skew. The DDR6 Write Leveling Training also allows for an unmatched path between CK and DQS within the DRAM, and thus supports an internal Write Leveling Training flow to account for the difference in internal delays.

The DDR6 DRAM also provides a programmable timing in its write logic, controlled by the Write Leveling Internal Cycle Alignment (WICA) mode register (MRy), which provides a means for improved performance of the device's receiver. The proper setting of this register shall be determined by the memory controller, either as described in the sections below, or by some other means. This delay setting is specific to each DRAM, its Write Preamble setting, and the operating frequency being used. PDA select ID is allowed while in WL training mode to program settings specific to each DRAM. Once the proper setting has been determined for a given Write preamble setting, and operating frequency, that setting would be subsequently restored to the DRAM after reset, power-cycle, or return to a previously used operating frequency.

The memory controller can use the 'write leveling' feature and feedback from the DDR6 DRAM to adjust the controller's differential DQS (DQS_t - DQS_c) output delay to align to the phase and cycle that corresponds to the Write Latency (WL) delay after the WRITE command. The memory controller involved in the leveling must have an adjustable delay setting on the DQS pins to align the rising edge of differential DQS with the timing at the receiver that is the pin-level Write Latency (External Write Leveling Training), and to align DQS with the internal DRAM Write Latency timing point (Write Leveling Internal Cycle Alignment). The internal DRAM Write Latency timing point may be skewed from the pin-level Write Latency timing point. The host will minimize this skew (tDQSoffset) through the Write Leveling Training flow.

Since the system and DIMM delays vary, the DRAM will support the ability for the host to align the DQS timings with a pin-level Write Latency differential CK (CK_t-CK_c) edge. This alignment is referred to as External Write Leveling. Once the DQS host timings are aligned at the DRAM Write Latency timing, the internal DRAM timings are optimized for lowest power and internal delay. This is accomplished when the host enables the Internal Write Timing setting in MRx. In order to compensate for the difference in delay, the host will execute an Internal Write Leveling Training sequence, which includes sweeping the Write Leveling Internal Cycle Alignment (WICA) settings in MRy (Write Leveling Internal Cycle Alignment Operation) and then finalizing the differential DQS phase and offset (Write Leveling Internal Phase Alignment and Final Host DQS Timing Operation).

During Write Leveling Training (both External and Internal), the DQS pattern shall include the full programmed write preamble and only the first toggle of the normal data burst sequence. The DRAM will sample the Internal Write Leveling Pulse relative to the first DQS toggle of the data burst sequence (last rising differential DQS edge sent by the host) and asynchronously feed back the result of this sample on the DQ bus, transitioning between tWLOmin and tWLOmax. If the DQ bus output is LOW at tWLOmax, the Internal Write Leveling Pulse signal was sampled while it was deasserted (LOW). Likewise, if the DQ bus output is HIGH at tWLOmax, the Internal Write Leveling Pulse signal was sampled while it was asserted (HIGH). The sampled feedback state will remain on the DQ bus until a subsequent Write Leveling sample changes the state, or until Write Leveling Training is exited.

The Internal Write Leveling Pulse is generated in response to a WRITE command, and held statically low otherwise. To perform the Write Leveling training, the controller repeatedly sends a WRITE command, delays DQS, and monitors the DQ feedback after tWLOmax until a transition from 0 to 1 is detected, indicating alignment with the Internal Write Leveling Pulse.

During Internal Write Leveling Training flows, the host will apply an offset to the starting point or the final setting of the DQS signals. The offsets are referred to as WL_ADJ_start and WL_ADJ_end. This will minimize the tDQSoffset variation across different DRAM devices. The WL_ADJ_start and WL_ADJ_end values depend on the tWPRE setting.

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When External and Internal Write Leveling Training flows are complete and the final WL_ADJ_end offset has been applied to the DQS timings, the DQS is phase aligned and cycle aligned for write operations. During the training sequence, the DRAM in Write Leveling training mode will apply ODT to the strobes in the same P R O P O S E D DDR6 Write Leveling Item # 1897.67 Page 3 of 8 way as for functional operation. All non-target ranks (which will not be in Write Leveling Training Mode) will apply ODT as defined for functional operation. Prior to executing the DDR6 Write Leveling Training Flow, the DRAM tWPRE value must be configured to the functional operation setting.

NOTE: DQS ODT is based on a DQS PARK Mode and is not enabled and disabled with DQ ODT timings.

4.2.11.2. Write Leveling Mode Registers

The MR fields for Write Leveling Training, Internal Write Timing, and Write Leveling Internal Cycle Alignment are part of MRx and MRy. To enter Write Leveling Training Mode, the controller shall program MRx:OP[a]=1. Likewise, to exit Write Leveling Training Mode, the controller will program MRx:OP[a]=0. For pseudo split die, the Write Leveling internal cycle alignment may be performed on either pseudo split die. Once leveling is completed on the selected pseudo split die, the host shall apply the resulting WICA training values into the WICA mode register shared by both pseudo split dies, ensuring consistent alignment within a single die. The Internal Write Timing, once enabled (MRx:OP[b]=1), shall remain enabled through the Internal Write Leveling Training flow and for functional operation. The host is responsible for incrementing the Write Leveling Internal Cycle Alignment setting until the Internal Write Leveling Pulse is pulled early enough to align with the differential DQS signal that was previously aligned with the pin-level Write Latency timing. The Write Leveling Internal Cycle Alignment setting only applies when the Internal Write Timing is enabled.

4.2.11.3. External Write Leveling Training Operation

When Write Leveling Mode is enabled (MRx:OP[a]=1) and the Internal Write Timing is disabled (MRx:OP[b]=0), the DRAM will have an Internal Write Leveling pulse that indicates how the DQS signals shall be aligned to match the pin-level Write Latency timings. The rising edge of the Internal Write Leveling Pulse will align to the rising edge of the differential CK signal that coincides with Write Latency (WL), as shown in the following diagrams.

The differential DQS signals driven by the controller during leveling mode must be terminated by the DRAM based on ranks populated. Similarly, the DQ bus driven by the DRAM must also be terminated at the controller.

All data bits shall carry the leveling feedback to the controller across the DRAM configurations 2p3, 2p6, 1p6 and 1p12.

The following diagram shows the timing sequence to enter Write Leveling Training Mode, operation during Write Leveling Training Mode (with Internal Write Timings disabled), and the timing sequence to exit Write Leveling Training mode. An MRW command is sent to enable Write Leveling Training Mode. Prior to sending the MRW command to enable Write Leveling Training Mode, the host memory controller must drive the DQS signals differentially low. After tWLPEN time, the controller can send a WRITE command, followed by strobe pulses. The DQS signals shall always drive differentially low, except during the WRITE preamble and burst strobe sequence. There is no restriction as to how early the strobe pulses are sent, so long as they are no earlier than WL/2 after the WRITE command. The DRAM will sample the Internal Write Leveling Pulse relative to the first DQS toggle of the data burst sequence and asynchronously feed back the result of this sample on the DQ bus, transitioning between tWLOmin and tWLOmax.

While in Write Leveling Training mode, the host controller may send ACT and PRE commands. The DRAM will ignore these commands. The address associated with the ACT and the WRITE commands may be any value. The following timing diagrams demonstrate the timing requirements associated with Write Leveling Training Mode Entry,

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the Internal Write Leveling Pulse alignment during External Write Leveling Training, and the DQ sample feedback timing.

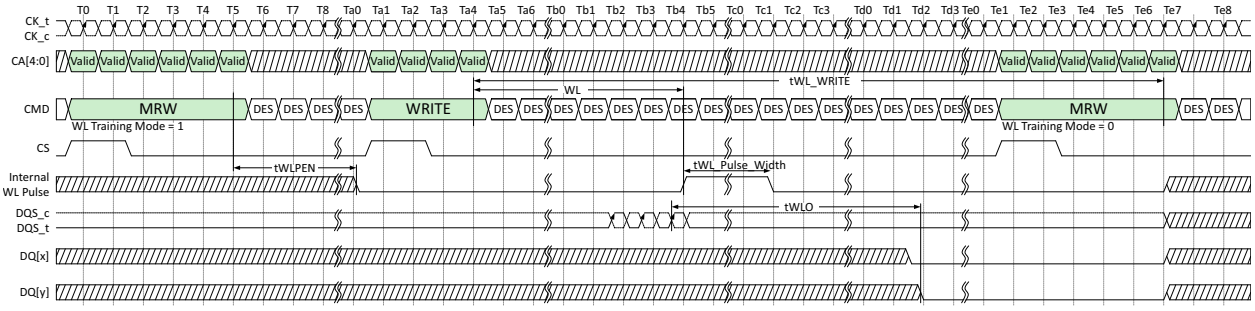


Figure 54 — Write Leveling Training Mode Timing Diagram (External Training, 0 Sample)

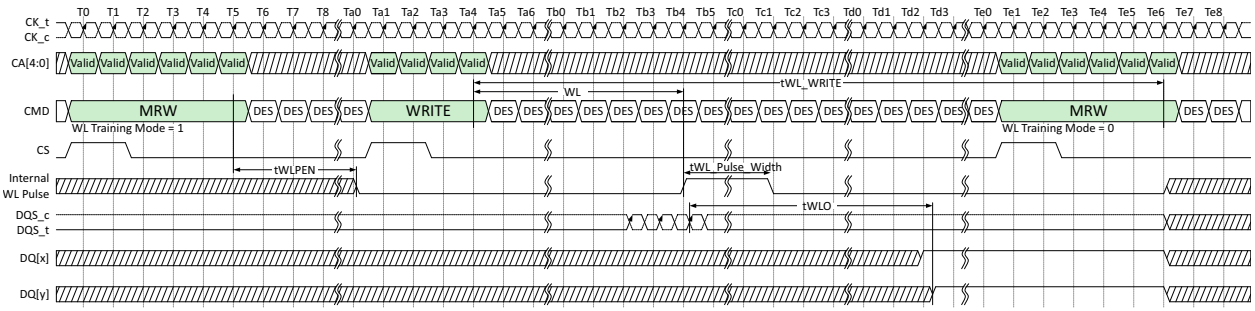


Figure 55 — Write Leveling Training Mode Timing Diagram (External Training, 1 Sample)

The Memory controller initiates Write Leveling mode of all DRAMs in a rank by setting MRx:OP[a]=1. When entering write leveling mode, the DQ pins are in undefined driving mode. Since the controller levels one rank at a time, all non-target ranks will set Write Leveling Mode to disabled. The Controller may assert non-target ODT through the normal WRITE command protocol. DQS_RTT_PARK termination will apply to the DQS_t and DQS_c signals.

The Controller shall drive DQS differentially LOW prior to sending the MRW command to enable Write Leveling Training Mode. The WRITE command must occur after a delay of tWLPEN relative to when the MRW enabled Write Leveling Training Mode.

The DRAM will sample the Internal Write Leveling Pulse and asynchronously feed back the result of this sample on the DQ bus, transitioning between tWLOmin and tWLOmax. The controller can sample the state of any DQ bit after tWLOmax. Based on the sampled state, the controller decides to increment or decrement the DQS delay setting and launches the next WRITE command with associated DQS_t/DQS_c pulse (or pulse sequence) after some time, which is controller dependent. Once a 0 to 1 transition on the DQ bus is detected, the controller locks DQS delay setting and external write leveling is achieved for the device. Write Leveling Training may be executed by issuing consecutive Write commands after entering Write Leveling Mode has been enabled (no requirement to exit and reenter between each Write command). The minimum Write to Write command spacing during Write Leveling Mode is tWL_Write

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and is defined as “max(WL, last DQS differential toggle) + tWLO(max)+ 2nCK”. An example of consecutive Write commands given during Write Leveling Training is shown in the following figure:

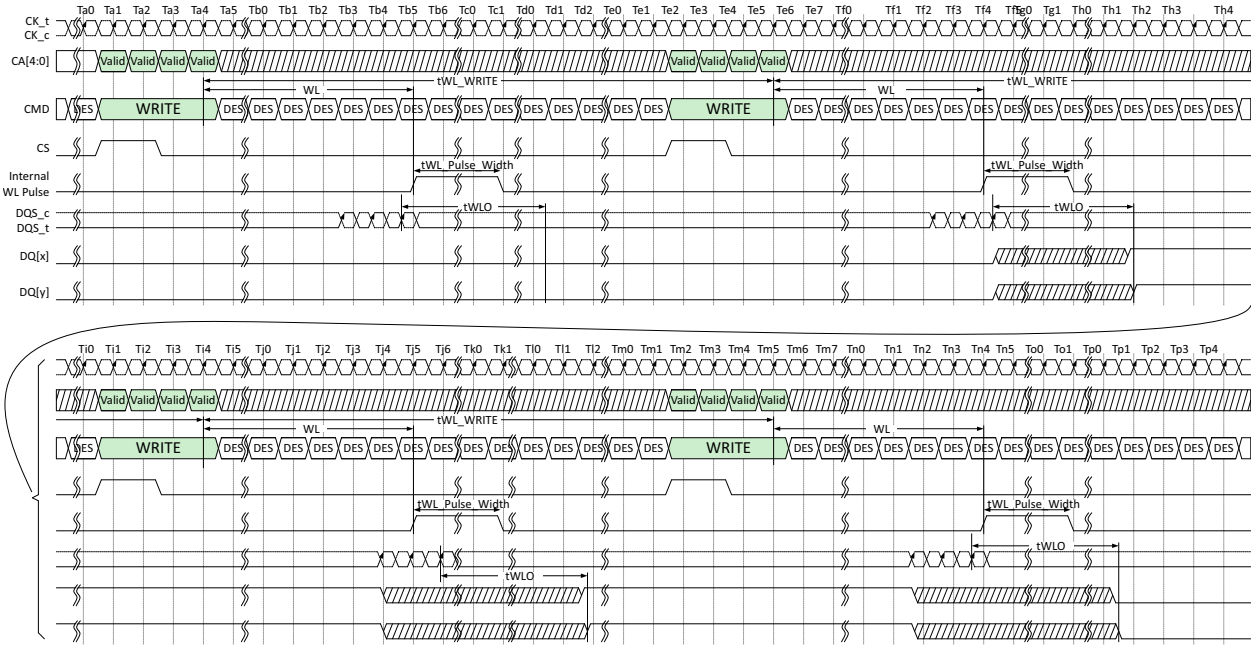


Figure 56 — Consecutive Write Commands during Write Leveling Training Mode Timing Example (External Training, 4 Sample)

4.2.11.4. Write Leveling Internal Cycle Alignment Operation

After the external Write Leveling Training step is completed, and after the host DQS signals have been aligned to the pin-level Write Latency timing, the host will apply a negative offset (WL_ADJ_start) to the DQS timing. The WL_ADJ_start offset moves the DQS timing prior to the Internal Write Leveling Pulse (causing LOW observed on the DQ pins), and it is dependent on the tWPRE setting (shown in table below). This will be a reference point for aligning the Internal Write Leveling Pulse via the WICA settings in MRy.

The Internal Cycle Alignment training begins when an MRW command is issued to enable Internal Write Timing in MRx (keeping OP[a]=1 and setting MRx:OP[b]=1). The host sweeps the Write Leveling Internal Cycle Alignment (WICA) delay settings in MRy, starting with the 0tCK offset default. Increasing the WICA mode register offset setting speeds up the timing of the WRITE command by reducing the WRITE Command to Internal Write Pulse delay until the DQS is aligned with the internal Write Leveling Pulse assertion. Alignment results in a logic HIGH on the DQ pins.

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Table 45 — WL_ADJ_Start and WL_ADJ_End Values per tWPRE Setting

WL_ADJ term	Description	tWPRE = 6 tCK	tWPRE = 8 tCK	tWPRE = TBD	Note
WL_ADJ_start	Offset that the host applies to the DQS_t/DQS_c timing just after external Write Leveling alignment to Write Latency and prior the internal cycle alignment training.	-3.5 tCK	-5.5 tCK	TBD	1
WL_ADJ_end	Offset that the host applies to the DQS_t/DQS_c timing after final phase alignment to the rising edge of the Write Leveling Internal Pulse. This will center the Write Leveling Internal Pulse rising edge within the preamble window.	4.0 tCK	6.0 tCK	TBD	
NOTE 1: tWL_ADJ_start = -(tWL_ADJ_end - 0.5tCK)					

The timing diagrams in the following figure demonstrate Write Leveling Training operation when Internal Write Timings are enabled and the Internal Cycle Alignment is set such that the Internal Write Leveling Pulse has not yet reached the host DQS toggles.

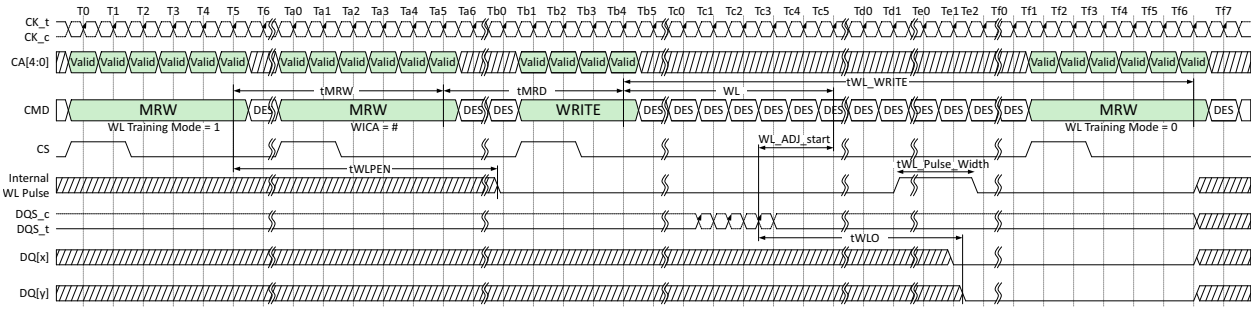


Figure 57 — Write Leveling Training Mode Timing Diagram (Internal Cycle Alignment, 0 Sample)

The timing diagrams in the following figure demonstrate Write Leveling Training operation when Internal Write Timings are enabled and the Internal Cycle Alignment is set such that the Internal Write Leveling Pulse has completed the coarse alignment to the host DQS timing.

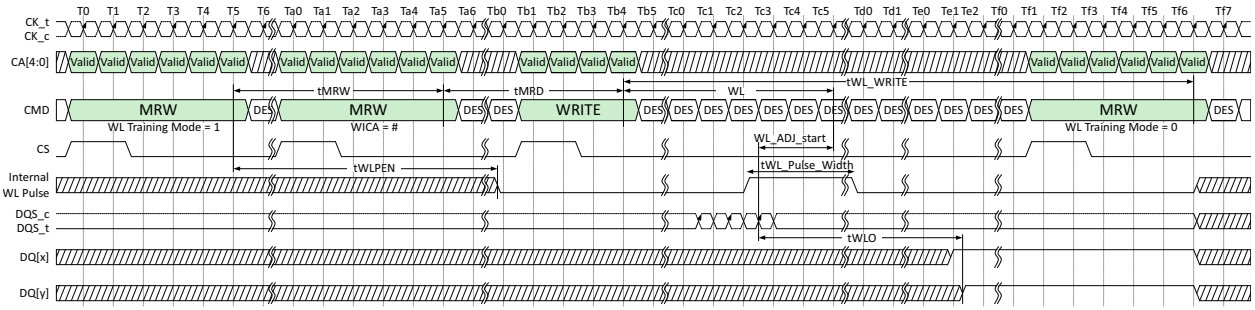


Figure 58 — Write Leveling Training Mode Timing Diagram (Internal Cycle Alignment, 1 Sample)

An example of Write Leveling Training internal Cycle Alignment executed by issuing consecutive Write commands after entering Write Leveling Mode has been enabled (no requirement to exit and reenter between each Write command) is shown in the following figure.

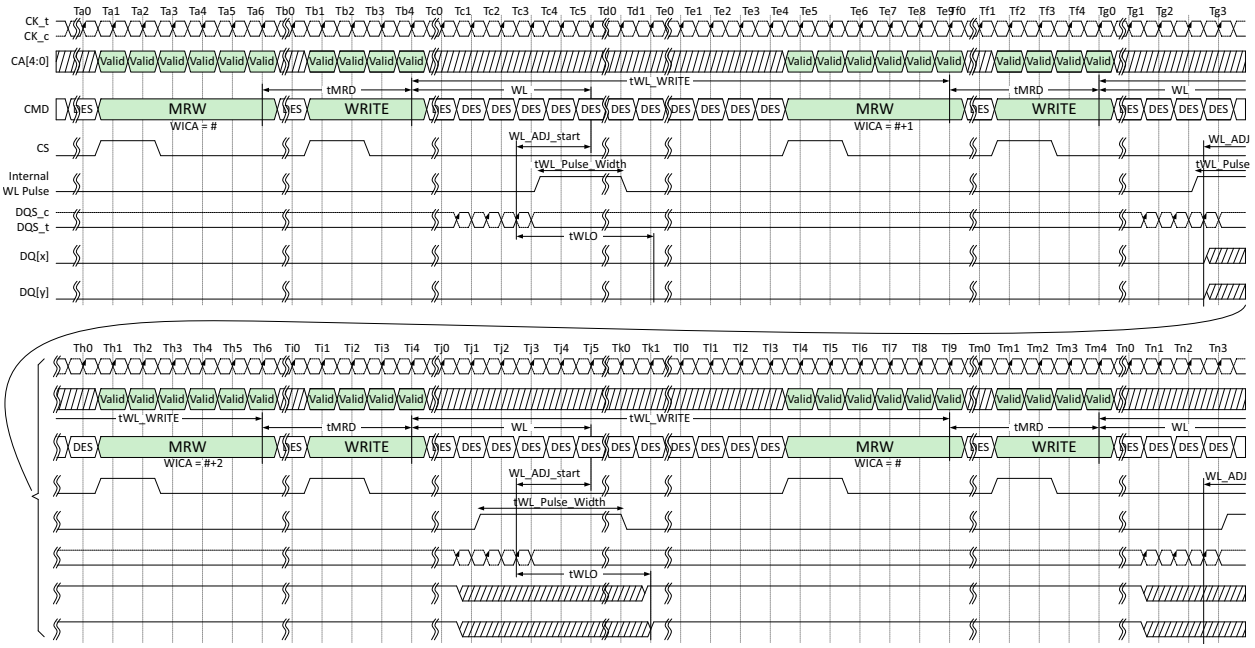


Figure 59 — Consecutive WRITE Commands During Write Leveling Training Mode Timing Example (Internal Training, 4 Sample)

4.2.11.5. Write Leveling Internal Phase Alignment and Final Host DQS Timing Operation

Once the Internal Cycle Alignment is achieved, the host shall perform a final fine sweep of the DQS timings with Internal Write Timing enabled (MRx:OP[b]=1) to establish an even closer phase alignment to the rising edge of the

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internal Write Leveling Pulse. Once this is complete, the host will then add a positive offset of WL_ADJ_end (dependent upon tWPRE) to the DQS timings. This results in a tDQSoffset between -1tCK and +1tCK with measurement adjustments of tWLS/tWLH, placing the rising edge of the Internal Write Leveling Pulse within the preamble. After the WL_ADJ_end offset has been applied, the host will disable Write Leveling Training Mode (MRx:OP[a]=0). The Internal Write Timing will remain enabled and the Internal Cycle Alignment setting will retain the coarse setting that was trained. After every reset, the host must either restore these settings or execute the full Write Leveling Training flow.

The following figure shows the timing relationships for the final placement of the host DQS timings relative to the Internal Write Leveling Pulse. However, it is not necessary to execute this Write Leveling Training Mode Measurement to finalize the setting - this is only for illustration.

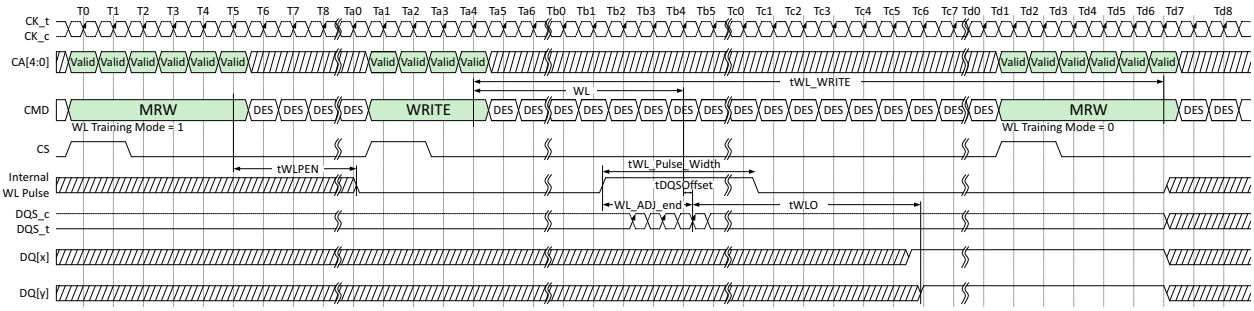


Figure 60 — Timing Diagram for Final Timings after Write Leveling Training is Complete

The following table summarizes the timing parameter ranges associated with Write Leveling Training Mode:

Table 46 — Timing Parameter Ranges Associated with Write Leveling Training Mode (Sheet 1 of 2)

Parameter	Symbol	Min	Max	Units	Notes
Write Leveling Pulse Enable - Time from Write Leveling Training Enable MRW to when Internal Write Leveling Pulse logic level is valid	tWLPEN	-	15	ns	
Write leveling output	tWLO	-	9.5	ns	1
Width of the Write Leveling Internal Pulse	tWL_Pulse_Width	2	-	tCK	

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Table 46 — Timing Parameter Ranges Associated with Write Leveling Training Mode (Continued) (Sheet 2 of 2)

Parameter	Symbol	Min	Max	Units	Notes
Write Leveling Write to subsequent command spacing	tWL_Write	max(WL, last DQS differential toggle) + tWLO(max) + 2nCK	-		

NOTE 1: There is no Max limit for the tWL_Pulse_Width, but the Write Leveling Internal Pulse must begin at zero for each WRITE command.

4.2.11.6. DRAM Termination During Write Leveling

When the DRAM is in Write Leveling Mode, the DQS_c/DQS_t termination (DQS_RTT_PARK) and the Command and Control termination (RTT_CA, RTT_CK, RTT_CS) will be the same as for functional operation. The DQ signals will not be terminated in the DRAM, but instead will be driving values to the controller. The host controller will apply termination for the DQ signals.

Table 47 — Timing Parameter Ranges Associated with Write Leveling Training Mode

ODT Enabled	DQS_t, DQS_c Termination	DQ Termination
RTT_WR	DQS_RTT_PARK	Off
RTT_Park, RTT_WR disabled	DQS_RTT_PARK	Off

NOTE: Termination for TDQS/DM is not included since TDQS and DM are disabled during Write Leveling Training Mode.

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4.2.12. Per-pin VREFDQ

DDR6 supports per-pin VREFDQ offset based on a common VREFDQ setting. Per-pin VREFDQ can adjust input common mode level which is distributed to each pin.

4.2.12.1. MRxx – Common VREFDQ MR Definition

Table 48 — MRxx information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU	VREFDQ						

Function	Register Type	Operand	Data
VREFDQ	R/W	OP[6:0]	0000000B : --Thru -- 1111111B :

Table 49 — MRxx VREFDQ Setting (Sheet 1 of 3)

Function	Operand	VREF Values (% of VDDQ)							
VREF Setting	OP[6:0]	0000000B	10.0[4]	0100000B	26.0	1000000B	42.0	1100000B	58.0
		0000001B	10.5[4]	0100001B	26.5	1000001B	42.5	1100001B	58.5
		0000010B	11.0[4]	0100010B	27.0	1000010B	43.0	1100010B	59.0
		0000011B	11.5[4]	0100011B	27.5	1000011B	43.5	1100011B	59.5
		0000100B	12.0[4]	0100100B	28.0	1000100B	44.0	1100100B	60.0
		0000101B	12.5	0100101B	28.5	1000101B	44.5	1100101B	60.5
		0000110B	13.0	0100110B	29.0	1000110B	45.0	1100110B	61.0
		0000111B	13.5	0100111B	29.5	1000111B	45.5	1100111B	61.5
		0001000B	14.0	0101000B	30.0	1001000B	46.0	1101000B	62.0
		0001001B	14.5	0101001B	30.5	1001001B	46.5	1101001B	62.5
		0001010B	15.0	0101010B	31.0	1001010B	47.0	1101010B	63.0

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Table 49 — MRxx VREFDQ Setting (Continued) (Sheet 2 of 3)

Function	Operand	VREF Values (% of VDDQ)							
		0001011B	15.5	0101011B	31.5	1001011B	47.5	1101011B	63.5
		0001100B	16.0	0101100B	32.0	1001100B	48.0	1101100B	64.0
		0001101B	16.5	0101101B	32.5	1001101B	48.5	1101101B	64.5
		0001110B	17.0	0101110B	33.0	1001110B	49.0	1101110B	65.0
		0001111B	17.5	0101111B	33.5	1001111B	49.5	1101111B	65.5
		0010000B	18.0	0110000B	34.0	1010000B	50.0	1110000B	66.0
		0010001B	18.5	0110001B	34.5	1010001B	50.5	1110001B	66.5
		0010010B	19.0	0110010B	35.0	1010010B	51.0	1110010B	67.0
		0010011B	19.5	0110011B	35.5	1010011B	51.5	1110011B	67.5
		0010100B	20.0	0110100B	36.0	1010100B	52.0	1110100B	68.0
		0010101B	20.5	0110101B	36.5	1010101B	52.5	1110101B	68.5
		0010110B	21.0	0110110B	37.0	1010110B	53.0	1110110B	69.0
		0010111B	21.5	0110111B	37.5	1010111B	53.5	1110111B	69.5
		0011000B	22.0	0111000B	38.0	1011000B	54.0	1111000B	70.0
		0011001B	22.5	0111001B	38.5	1011001B	54.5	1111001B	70.5
		0011010B	23.0	0111010B	39.0	1011010B	55.0	1111010B	71.0
		0011011B	23.5	0111011B	39.5	1011011B	55.5	1111011B	71.5
		0011100B	24.0	0111100B	40.0	1011100B	56.0	1111100B	72.0
		0011101B	24.5	0111101B	40.5	1011101B	56.5	1111101B	72.5

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Table 49 — MRxx VREFDQ Setting (Continued) (Sheet 3 of 3)

Function	Operand	VREF Values (% of VDDQ)							
		0011110B	25.0	0111110B	41.0	1011110B	57.0	1111110B	73.0
		0011111B	25.5	0111111B	41.5	1011111B	57.5	1111111B	73.5

4.2.12.2. VREFDQ MR Offset

Per-pin VREFDQ offset is controlled independently per each Pseudo Split Die (PSD).

Table 50 — MRyy – Per-pin VREFDQ Offset Sign Control

Function	Register Type	Operand	Data
DQ0 VREFDQ Offset Sign	R/W	OP[0]	0B: positive 1B: negative
DQ1 VREFDQ Offset Sign	R/W	OP[1]	0B: positive 1B: negative
DQ2 VREFDQ Offset Sign	R/W	OP[2]	0B: positive 1B: negative
DQ3 VREFDQ Offset Sign	R/W	OP[3]	0B: positive 1B: negative
DQ4 VREFDQ Offset Sign	R/W	OP[4]	0B: positive 1B: negative
DQ5 VREFDQ Offset Sign	R/W	OP[5]	0B: positive 1B: negative

Table 51 — MRzz0 – DQ0 and DQ1 VREF DQ Offset

Function	Register Type	Operand	Data
DQ0 VREFDQ Offset Sign	R/W	OP[3:0]	0000B: disable 0001B: step +/- 1 LSB code 0010B: step +/- 2 LSB code ... 1111B: step +/- 15 LSB code
DQ1 VREFDQ Offset Sign	R/W	OP[7:4]	0000B: disable 0001B: step +/- 1 LSB code 0010B: step +/- 2 LSB code ... 1111B: step +/- 15 LSB code

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Table 52 — MRzz1 – DQ2 and DQ3 VREF DQ Offset

Function	Register Type	Operand	Data
DQ2 VREFDQ Offset Sign	R/W	OP[3:0]	0000B: disable 0001B: step +/- 1 LSB code 0010B: step +/- 2 LSB code ... 1111B: step +/- 15 LSB code
DQ3 VREFDQ Offset Sign	R/W	OP[7:4]	0000B: disable 0001B: step +/- 1 LSB code 0010B: step +/- 2 LSB code ... 1111B: step +/- 15 LSB code

Table 53 — MRzz2 – DQ4 and DQ5 VREF DQ Offset

Function	Register Type	Operand	Data
DQ4 VREFDQ Offset Sign	R/W	OP[3:0]	0000B: disable 0001B: step +/- 1 LSB code 0010B: step +/- 2 LSB code ... 1111B: step +/- 15 LSB code
DQ5 VREFDQ Offset Sign	R/W	OP[7:4]	0000B: disable 0001B: step +/- 1 LSB code 0010B: step +/- 2 LSB code ... 1111B: step +/- 15 LSB code

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4.2.13. Host Receiver Training (HRT) Mode

4.2.13.1. HRT Introduction

DRAM shall support a Host Receiver Training function similar to Write Training Mode, to maintain robust link performance, especially in systems that employ multiple operating modes to enhance I/O scalability and reduce training time. This DRAM feature enables the host to perform receiver data training including internal receiver drift compensation and maintain optimal signal integrity throughout system operation.

4.2.13.2. HRT pattern

HRTM can select the pattern type between user defined pattern vs PRBS with MR setting (MRDD). The PRBS mode provides an independent (different pattern) per DQ lane using a 24-bit Fibonacci LFSR. One 24-bit F-LFSR generator will support x6 (max) DQ lanes. The polynomial for the Fibonacci LFSR is $x^{24} + x^{20} + x^{11} + x^{5} + 1$. For the Host DQ Receiver drift improvements, the DRAM shall drive user defined pattern and/or LFSR pattern during bus idle time. Figure 1 and Figure 2 show the details of left shift Fibonacci LFSR hardware and 24-bit PRBS pattern.

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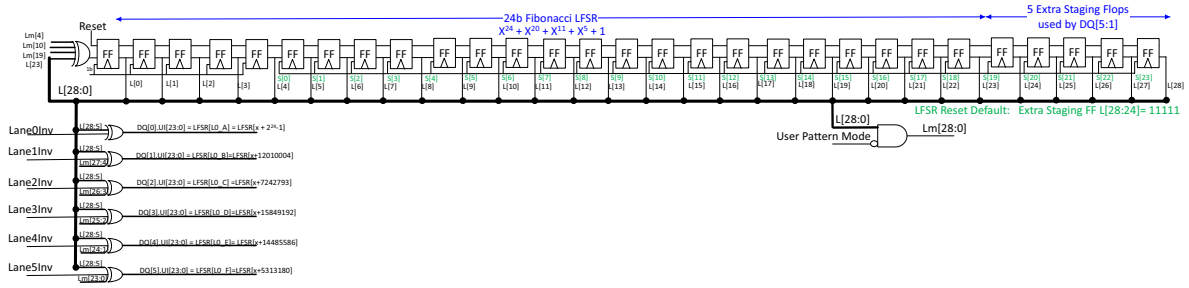


Figure 61 — Fibonacci 24-bit LFSR circuit

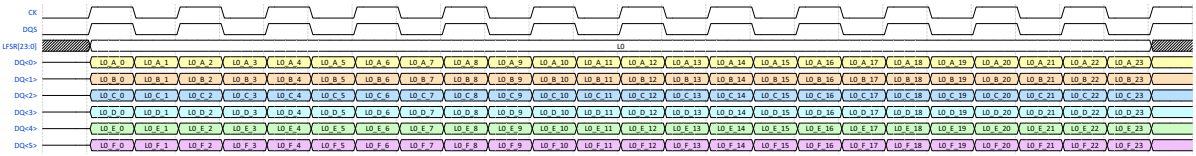


Figure 62 — 24-bit PRBS pattern details

4.2.13.3. Entry and Exit for Host Receiver Training Mode (HRT)

To enter Host Receiver Training Mode (HRT), the host memory controller shall issue a Mode Register Write (MRW) command targeting MR05 OP[4:0] = 10010B, with the HRT Mode entry opcode. When the DRAM is in Receiver Training mode, commands are still actively processed.

To exit HRT Mode, the host memory controller shall issue an MRW command to MR5 OP[7:0] = 00000000B, with the HRT Mode Exit opcode.

4.2.13.4. Host Receiver Training Operation

Host to configure mode register with read pattern or LFSR details and this training is entered by issuing an MRW command. Host to issue a READ command, which causes the DDR6 DRAM to drive the MR content depending on

the mode register settings (LFSR vs Read pattern). The timing of data return and strobe sequence match a regular Read operation which supports back-to-back read the user read pattern / LFSR data. HRT mode is exited by issuing an MRW command. In this training mode, the only valid commands are READ and MRW.

Host Receiver Training Mode Registers:

Table 54 — MRDD OP[7:0]- Host Receiver Training Control Register

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU	RFU	RFU	RFU	Seed Reset	RFU	RFU	Read Training Pattern Selection

Function	Register Type	Operand	Data	Notes
Read Training Pattern Selection	R/W	OP[0]	1b0: User defined read pattern 1b1: LFSR	
RFU	R/W	OP[1]	RFU	
RFU	R/W	OP[2]	RFU	
Seed Reset	R/W	OP[3]	1b0: Disable 1b1: Reset to default both LFSR and 5b-Staging Flip flops (auto clears after reset-ting)	
RFU	R/W	OP[5:4]	RFU	
RFU	R/W	OP[7:6]	RFU	

4.2.13.5. PRBS Pattern Generator for WTM and HRT modes

PRBS generator will start with the seed (Example 24b LFSR Seed: F7410B) and the below LFSR pattern table shows the example of shift mode where each DQ lanes gets different pattern on the lane. The upper 5 extra staging flip-flop bits getting reset to 1, hence, users shall only independently program the LFSR seed [23:5] and need to calculate LFSR seed [4:0] to correctly initialize the LFSR. The LFSR seed S[4:0] is calculated using XOR and the details are given below. This is same as Write Training Mode LFSR pattern. Table 1 illustrates the mapping of the pattern to DQ [5:0].

LFSR Seed Calculation

S4 = XOR(S23,S19,S10,1)

S3 = XOR(S22,S18,S9,1)

S2 = XOR(S21,S17,S8,1)

S1 = XOR(S20,S16,S7,1)

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S0 = XOR(S19,S15,S6,1)

	L0	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12	L13	L14	L15	L16	L17	L18	L19	L20	L21	L22	L23	L24	L25	L26	L27	L28
LFSR Seed	0	1	1	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1	1	1	1
UI	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23					
DQ0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1	1	1	1					
DQ1	0	0	0	1	1	0	0	1	0	1	0	1	0	0	1	1	0	0	0	0	0	0	0	0					
DQ2	1	0	0	1	0	1	0	1	1	1	1	1	1	0	1	0	1	0	0	0	0	0	0	0					
DQ3	1	1	0	1	0	0	1	1	1	0	1	0	1	1	1	0	0	1	0	0	0	0	0	0					
DQ4	1	1	1	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0					
DQ5	0	1	1	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	0					
Shift1	1	0	1	1	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1	1	1
UI	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47					
DQ0	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1	1	1					
DQ1	1	0	0	0	1	1	0	0	1	0	1	0	1	0	0	1	1	0	0	0	0	0	0	0					
DQ2	1	1	0	0	1	0	1	0	1	1	1	1	1	0	1	0	1	0	0	0	0	0	0	0					
DQ3	1	1	1	0	1	0	0	1	1	1	0	1	0	1	1	1	0	0	1	0	0	0	0	0					
DQ4	0	1	1	1	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0					
DQ5	1	0	1	1	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0					
Shift2	1	1	0	1	1	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1	1
UI	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71					
DQ0	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1	1					
DQ1	0	1	0	0	0	1	1	0	0	1	0	1	0	1	0	0	1	1	0	0	0	0	0	0					
DQ2	0	1	1	0	0	1	0	1	0	1	1	1	1	1	1	0	1	0	1	0	0	0	0	0					
DQ3	1	1	1	1	0	1	0	0	1	1	1	0	1	0	1	1	1	0	0	1	0	0	0	0					
DQ4	0	0	1	1	1	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	1	0	0	0					
DQ5	0	1	0	1	1	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0					
Shift3	0	1	1	0	1	1	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1
UI	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95					
DQ0	1	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1					
DQ1	0	0	1	0	0	0	1	1	0	0	1	0	1	0	1	0	0	1	1	0	0	0	0	0					
DQ2	1	0	1	1	0	0	1	0	1	0	1	1	1	1	1	1	0	1	0	1	0	0	0	0					
DQ3	0	1	1	1	1	0	1	0	0	1	1	1	0	1	0	1	1	1	0	0	1	0	0	0					
DQ4	0	0	0	1	1	1	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	1	0	0					
DQ5	1	0	1	0	1	1	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0					
Shift4	1	0	1	1	0	1	1	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1
UI	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119					
DQ0	1	1	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1					
DQ1	1	0	0	1	0	0	0	1	1	0	0	1	0	1	0	1	0	0	1	1	0	0	0	0					
DQ2	0	1	0	1	1	0	0	1	0	1	0	1	1	1	1	1	1	0	1	0	1	0	0	0					
DQ3	0	0	1	1	1	1	0	1	0	0	1	1	1	0	1	0	1	1	1	0	0	1	0	0					
DQ4	1	0	0	0	1	1	1	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	1	0					
DQ5	0	1	0	1	0	1	1	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1					

Figure 63 — LFSR Pattern for Host Receiver Training Mode DQ mapping

DDR6 DRAM devices implement a Host Receiver Training (HRT) function that drives training patterns onto the DQ signal pins to the host and in case of Write Training Mode (WTM), it receives the LFSR pattern from the host on the DQ pins. The output pattern shall be sourced either from a user-defined pattern or from an internally generated Linear Feedback Shift Register (LFSR) sequence.

4.2.13.6. LFSR Registers

An independent 24-bit Fibonacci LFSR generator provides patterns for up to 6 DQ lanes. The source of the DQ output pattern shall be selected via MRDD:OP[0], where this field selects between LFSR output and the user-defined read pattern resource. The MRDD[3] field shall be used to reset the LFSR seed value and associated staging flip-flops. Upon reset, the 5-bit staging flip-flops shall be initialized to their default value of 11111.

If MRDD[0] is set to 1b0, MRD8to MRDAcontains the user-defined pattern. MRD7 is used to assign Pattern Inversion per DQ lane. If MRDD[0] is set to 1b1 ~~and MRxx[2] is set to 1b0~~, then the training shall use LFSR pattern generator and the MRD8 to MRDA registers allow users to select LFSR Seed register UI23-0. These LFSR seed /user pattern, inversion registers are per PSD and common for both Write training mode and Host receiver training modes.

NOTE: For the 2p3 configuration, only DQ[2:0] from the LFSR output shall be used; all remaining DQ bits shall be ignored. For the 1p6 configuration, only DQ[5:0] from PSD0 shall be used. For the 2p6 and 1p12 configurations, DQ[5:0] from PSD0 and DQ[11:6] from PSD1 shall be used.

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Table 55 — MRD8– LFSR Seed / User Pattern Register 0

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
LFSR Seed/User Pattern UI[7:0]							

Function	Register Type	Operand	Data	Notes
LFSR Seed/ Pattern UI0	R/W	OP[0]		
LFSR Seed/ Pattern UI1	R/W	OP[1]		
LFSR Seed/ Pattern UI2	R/W	OP[2]		
LFSR Seed/ Pattern UI3	R/W	OP[3]		
LFSR Seed/ Pattern UI4	R/W	OP[4]		
LFSR Seed/ Pattern UI5	R/W	OP[5]		
LFSR Seed/ Pattern UI6	R/W	OP[6]		
LFSR Seed/ Pattern UI7	R/W	OP[7]		

Table 56 — MRD9 – LFSR Seed / User Pattern Register 1

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
LFSR Seed/User Pattern UI[15:8]							

Function	Register Type	Operand	Data	Notes
LFSR Seed/ Pattern UI8	R/W	OP[0]		
LFSR Seed/ Pattern UI9	R/W	OP[1]		
LFSR Seed/ Pattern UI10	R/W	OP[2]		
LFSR Seed/ Pattern UI11	R/W	OP[3]		
LFSR Seed/ Pattern UI12	R/W	OP[4]		
LFSR Seed/ Pattern UI13	R/W	OP[5]		
LFSR Seed/ Pattern UI14	R/W	OP[6]		
LFSR Seed/ Pattern UI15	R/W	OP[7]		

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Table 57 — MRDA – LFSR Seed / User Pattern Register 2

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
LFSR Seed/User Pattern UI[23:16]							

Function	Register Type	Operand	Data	Notes
LFSR Seed/ Pattern UI16	R/W	OP[0]		
LFSR Seed/ Pattern UI17	R/W	OP[1]		
LFSR Seed/ Pattern UI18	R/W	OP[2]		
LFSR Seed/ Pattern UI19	R/W	OP[3]		
LFSR Seed/ Pattern UI20	R/W	OP[4]		
LFSR Seed/ Pattern UI21	R/W	OP[5]		
LFSR Seed/ Pattern UI22	R/W	OP[6]		
LFSR Seed/ Pattern UI23	R/W	OP[7]		

NOTE: ~~If a per DQ lane based pattern registers are required, then replicate MRxy<2:0> per DQ lane~~

Table 58 — MRD7– Pattern Inversion Control Register 0

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU	RFU	Pattern Inversion DQ5	Pattern Inversion DQ4	Pattern Inversion DQ3	Pattern Inversion DQ2	Pattern Inversion DQ1	Pattern Inversion DQ0

Function	Register Type	Operand	Data	Notes
Pattern Inversion DQ0	R/W	OP[0]	1b0: Normal 1b1: Invert	
Pattern Inversion DQ1	R/W	OP[1]	1b0: Normal 1b1: Invert	
Pattern Inversion DQ2	R/W	OP[2]	1b0: Normal 1b1: Invert	

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Function	Register Type	Operand	Data	Notes
Pattern Inversion DQ3	R/W	OP[3]	1b0: Normal 1b1: Invert	
Pattern Inversion DQ4	R/W	OP[4]	1b0: Normal 1b1: Invert	
Pattern Inversion DQ5	R/W	OP[5]	1b0: Normal 1b1: Invert	
RFU	RFU	OP[6]	RFU	
RFU	RFU	OP[7]	RFU	

Table 59 — ~~MRzz1 – Pattern Inversion Control Register 1~~

~~OP[7]~~	~~OP[6]~~	~~OP[5]~~	~~OP[4]~~	~~OP[3]~~	~~OP[2]~~	~~OP[1]~~	~~OP[0]~~
~~RFU~~	~~RFU~~	~~Pattern Inversion DQ11~~	~~Pattern Inversion DQ10~~	~~Pattern Inversion DQ9~~	~~Pattern Inversion DQ8~~	~~Pattern Inversion DQ7~~	~~Pattern Inversion DQ6~~

~~Function~~	~~Register Type~~	~~Operand~~	~~Data~~	~~Notes~~
~~Pattern Inversion DQ6~~	~~R/W~~	~~OP[0]~~	~~1b0: Normal 1b1: Invert~~	
~~Pattern Inversion DQ7~~	~~R/W~~	~~OP[1]~~	~~1b0: Normal 1b1: Invert~~	
~~Pattern Inversion DQ8~~	~~R/W~~	~~OP[2]~~	~~1b0: Normal 1b1: Invert~~	
~~Pattern Inversion DQ9~~	~~R/W~~	~~OP[3]~~	~~1b0: Normal 1b1: Invert~~	
~~Pattern Inversion DQ10~~	~~R/W~~	~~OP[4]~~	~~1b0: Normal 1b1: Invert~~	
~~Pattern Inversion DQ11~~	~~R/W~~	~~OP[5]~~	~~1b0: Normal 1b1: Invert~~	
~~RFU~~	~~RFU~~	~~OP[6]~~	~~RFU~~	
~~RFU~~	~~RFU~~	~~OP[7]~~	~~RFU~~	

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4.2.14. DQ RX Offset Calibration Training

4.2.14.1. Offset Calibration Training Description

DDR6 SDRAM provides Offset Calibration training to reduce an intrinsic offset of DQ Rx. It is required for the DRAM to meet its receiver eye mask spec and required to operate the training every power-up and initialization training sequence to cope with the change of SDRAM operating condition. Offset calibration is enabled by MR05[4:0]=10100 and this can be issued in either normal mode or MRUPDATE Mode. When the SDRAM starts the training, DQ channel, where all the DRAM dies sharing the DQ, can be floated or can be biased toward VSS. DQ ODTs shall be automatically enabled to values determined by the SDRAM to be suitable for proper operation after MR05[4:0]=10100 is issued for starting an Offset calibration training.

Calibration will be performed by the DDR6 SDRAM automatically within the time interval, tOSCAL, and the calibration results will also be automatically updated within tOSCAL. In any case, DQ operations are disallowed during the DQ RX offset calibration. After calibration is done, offset calibration is disabled by MR05[4:0]=0h as this calibration is not self-clearing.

To reset the offset calibration to a default value, the Rx offset reset MR bit (OCC RESET) needs to be asserted (MR13[7]=1). In any case, operations that initiate DQ operations are always disallowed during tMRD after MR13[7] asserted. The OCC RESET MR bit shall be reset to 0b by the DRAM after tMRD.

4.2.14.2. Offset Calibration Training Sequence

Below is DQ RX Offset Calibration Training sequence.

1) Issue a Rx Offset Cal. Start (MR05[4:0]=10100) for starting a DQ RX Offset Calibration Training
2) Wait tOSCAL until the DDR6 SDRAM completes the Offset Calibration
3) Issue a Rx Offset Cal. Exit (MR05[4:0]=0h) for exiting a DQ RX Offset Calibration Training

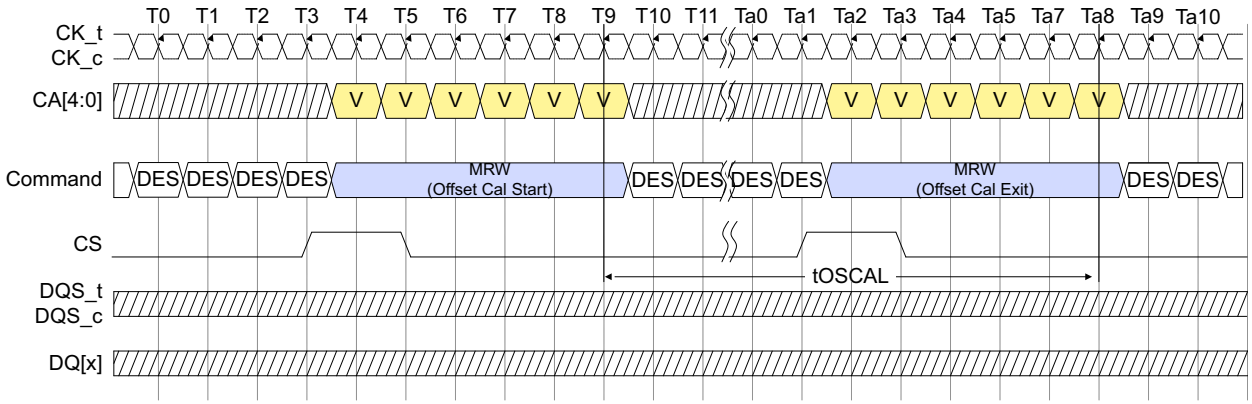


Figure 64 — DQ RX Offset Calibration Training Timing

Table 60 — DQ RX Offset Calibration Training Time Parameter

Item	Symbol	Min/Max	Value	Unit	Note
DQ RX Offset Calibration Training Time	tOSCAL	Max	TBD	µs	-

Table 61 — DQ RX Offset Calibration Training Command Timing Constraint

From	To	Min/Max	Timing Constraint	Unit	Note
MRW Offset Cal. Start	MRW Offset Cal. Exit	Min	TBD	nCK	-

4.2.14.3. MR13 – DQ RX OCC Setting

Table 62 — MR13 Definition

Bits	Attribute	Default	Description	Note
7	RW	0	MR13[7]: RX_OCC_RESET Offset Calibration Control reset 0: Normal Operation 1: RX OCC RESET	-

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4.2.15. Read Preamble Training

Read preamble training supports read leveling of the host receiver timings. This mode supports MRR transactions that access the Read Training Pattern and cannot be used during any other data transactions. Just like Read Training Pattern, Read Preamble Training needs to be entered with CRC disabled. Read preamble training changes the read strobe behavior such that the strobes are always driven by the DRAM, and only toggle during a MR programmed preamble, postamble and the actual burst of the read data. This mode enables the host to detect the timing of when the first data and associated strobe is returned after a read command.

4.2.15.1. Entry and Exit for Preamble Training Mode

The DRAM enters Read Preamble Training Mode by setting MR05:OP[4:0] = 10011B using MRW command. Read Preamble Training Mode is exited by setting MR05:OP[7:0] = 0. Read Preamble Training should not be disturbed by an ACT command until the completion of the training.

4.2.15.2. Preamble Training Mode Operation

Once the DRAM is placed in Read Preamble Training Mode, the only data transactions supported are MRR commands. All non-data commands, such as MRW, are still supported in this mode. Once READ Preamble Training is enabled, the device will drive DQS_t LOW and DQS_c HIGH within tSDOn and remain at these levels until an MRR command is issued.

During read preamble training, a programmed DQS preamble/postamble setting will be used. Once the MRR command is issued to MRyz, the device will drive DQS_t/DQS_c after ~~CL-tRPRE~~ RL-tRPRE-(Read_DQS_Pre-shift),where tRPRE is programmed in MR0C:OP[4:2] register, like a normal READ burst with the Read DQS pre-shift setting MR0C:OP[1:0] programmed in MR0C applied. In response to the MRR to the designated Read Training Pattern Address, the device must also drive the DQ pattern as per the Read Pattern configuration while in this mode. The MRR commands may be sequenced to enable back to back bursts on the DQ bus. Read preamble training mode is exited within tSDOff after setting mode register MR05:OP[7:0] = 0.

Figure 1 illustrates the timing of the strobe being driven differentially low after Read Preamble Training Mode is enabled. It includes the strobe timing sequence following a MRR command issued to access the Read Training Pattern. The sequence shows a programmed preamble/postamble consisting of 2tCK static-low followed by 2tCK toggling (MR0C: OP[4:2] = 001), and a programmed 0.5tCK postamble (MR0C: OP[7:6] = 00). Table 1 specifies the timing requirements for programmed preamble/postamble (2tck static low followed by 2tck toggle preamble and 0.5tck postamble) applicable to Read Preamble Training Mode.

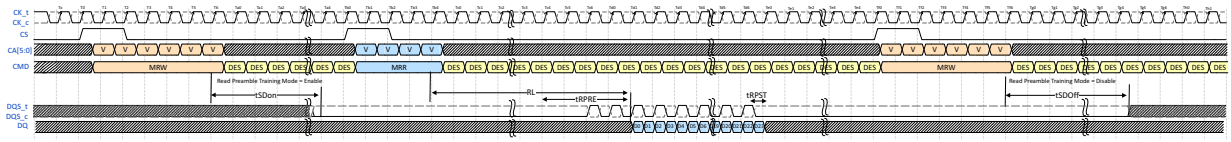


Figure 65 — Timing Diagram for Read Preamble Training Mode Entry, Pattern access and Exit

Table 63 — Timing parameters for Read Preamble Training Mode (Sheet 1 of 2)

Parameter	Symbol	Min	Max
Delay from MRW Command to DQS Driven	tSDOn	-	Max(TBDnCK, TBDns)

Table 63 — Timing parameters for Read Preamble Training Mode (Continued)

Parameter	Symbol	Min	Max
Delay from MRW Command to DQS Disabled	tSDOff	-	Max(TBDnCK, TBDns)

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4.2.16. Read Duty Cycle Adjuster (DCA)

DDR6 SDRAM supports Mode Register-adjustable Read (RD) DCA to allow the memory controller to adjust the DRAM's internally generated read clock tree and DQ duty cycle to compensate for systemic duty cycle errors of all DQS and DQs.

The RD DQS DCA is located in the read clock tree (or equivalent place), following the clock synchronization phase correction scheme.

The controller can adjust the duty cycle through all the DCA mode registers and can determine the optimal Mode Register setting for DCA in multiple different ways.

In case of 4-phase internal clocks, for example, since QCLK(90˚)/IBCLK(180˚)/QBCLK(270˚) are adjusted based on ICLK(0˚), the controller can first confirm that the first BL is synchronized with ICLK(0˚) through clock sync., and then perform the full DCA training operation which needs to have an even number of MRR(or Read) - MRR(or Read) timing to avoid confusion whether the first BL is synchronized with ICLK(0˚) or IBCLK(180˚).

4.2.16.1. Duty Cycle Adjuster Range

The global DCA is applied for all DQS and DQs within the entire PSD of the DRAM die. The step range between the step 0 to 7(-7) is as follows. The difference of actual value between step N and step N+1 is defined to be within a spec range, since the variation of duty cycle by changing DCA code is not linear.

Table 64 — Read Global DCA Step and Range

Parameter	Min/Max	Value		Unit	Notes
		~14.4Gb/s	Data Rate > 14.4Gb/s		
RD DCA Total Range	Min	7	TBD	ps	1, 2
	Max	28	TBD		
RD DCA 1-step Range	Min	1	TBD		1, 3
	Max	4	TBD		

NOTE 1: These values are guaranteed by design
NOTE 2: Total range means the range from step 0 to step +7(-7)
NOTE 3: 1-step range includes non-linearity of each step

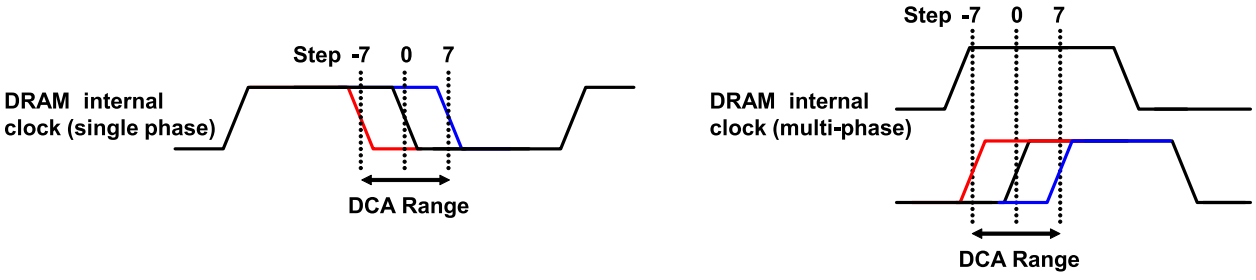


Figure 66 — Read Duty Cycle Adjuster Range

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4.2.16.2. Relationship between DCA Code Change and Single/Two-phase Internal Clock(s)/DQS Timing

In case of the read clock tree used single/two-phase clock(s) scheme, the duty-cycle ratio of the DQS can be adjusted directly according to the internal clock(s) controlled by the DCA code. Note that tDQSCK is not changed by DCA code change.

Using a 2-phase clock scheme, the rising edge of the 0° clock is the reference edge, while the 180° clock is adjusted based on 0° clock. The rising edge of 0° clock is for even burst bit data, and the rising edge of 180° clock is for odd burst bit data.

The global DCA adjustment uses the “RD DCA for single/two-phase clock(s)” mode register bits, MR40:OP[7:4]. A positive DCA adjustment results in a larger duty cycle ratio, while a negative DCA adjustment results in a smaller duty cycle ratio.

In addition to the global DCA adjustment, a RD per-pin DCA adjustment allows an additional step range of -3 to +3, per DQS/DQ. The per-pin DCA adjustment is additive to the global DCA adjustment, as shown in Table 2.

Table 65 — Read DCA Range Examples (Not All Possible Combinations)

Global DCA Adjustment	Per-Pin DCA Adjustment	Total DCA Adjustment at Pin
DCA Step -3	DCA Step -3	DCA Step -6
DCA Step -2	DCA Step +2	DCA Step 0
DCA Step +2	DCA Step -3	DCA Step -1
DCA Step +7	DCA Step +2	DCA Step 9

Like the global DCA adjustment, the actual step size for the per-pin DCA adjustment cannot be defined since the variation of duty cycle by changing DCA code is not linear, however the per-pin DCA adjustment will be approximately the same as the global DCA adjustment.

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Mode register OP bits associated with the IBCLK and QBCLK may not be supported on the DRAM with a 2-phase clock scheme.

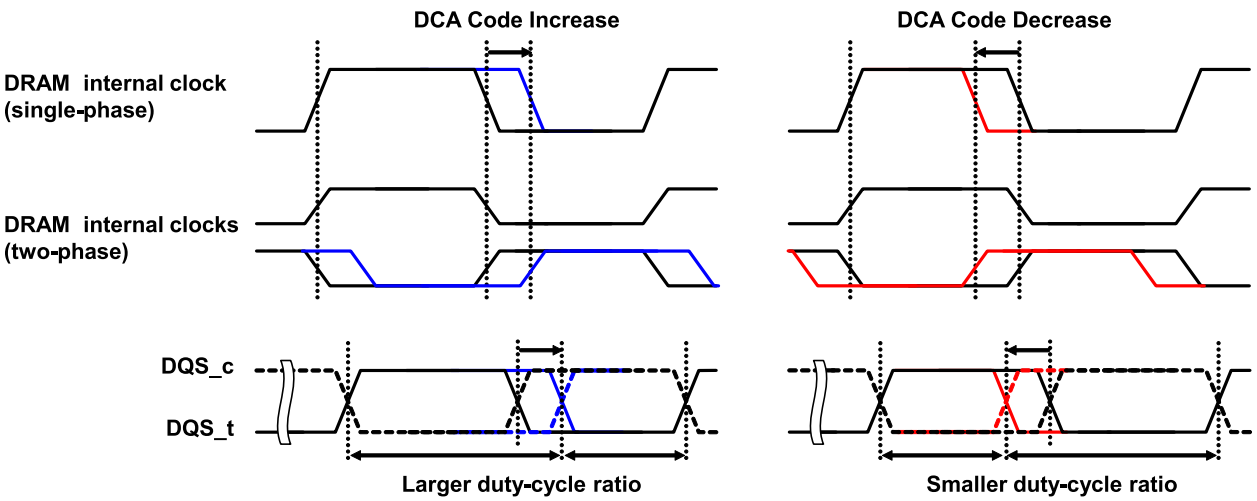


Figure 67 — Relationship between DCA Code Change and the Single/Two-phase

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4.2.16.3. Relationship between DCA Code Change and 4-phase Internal Clock(s)/DQS Timing

In case of the read clock tree used 4-phase clocks scheme, the even and odd duty-cycle ratio of the DQS can be respectively adjusted since the internal 4-phase clocks can be independently controlled by the DCA code.

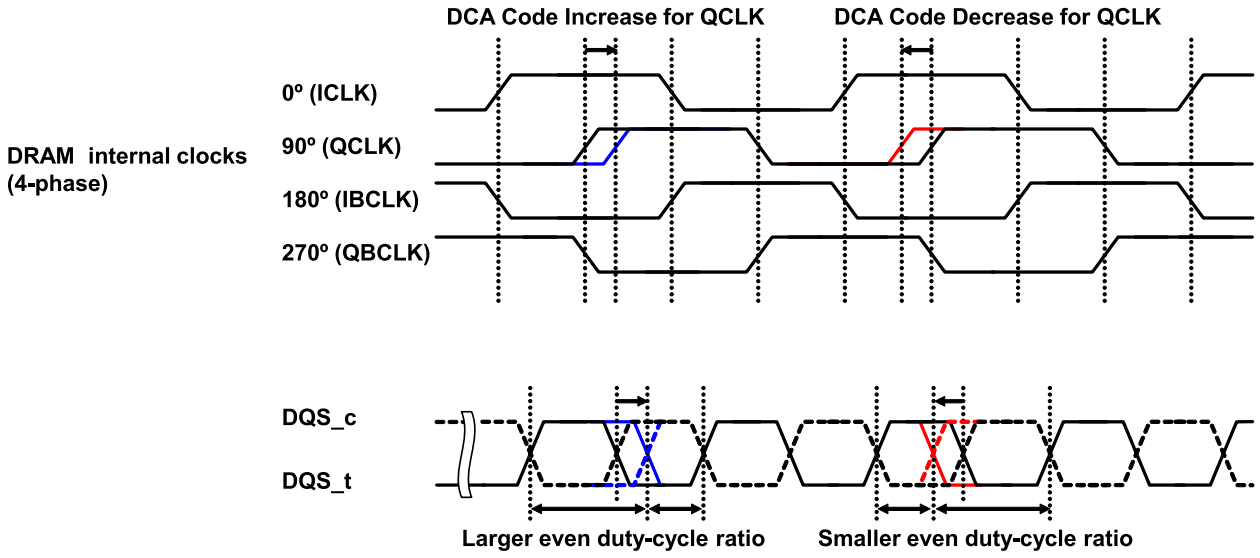


Figure 68 — Relationship between DCA Code Change for QCLK and the 4-phase Internal Clocks/DQS Waveform (Example)

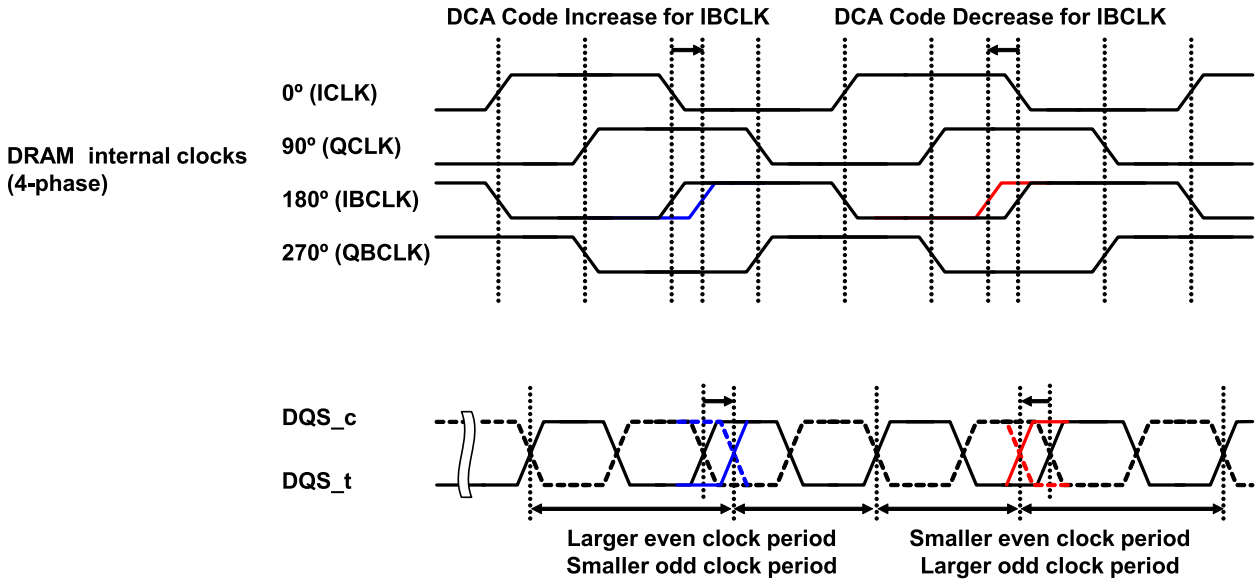


Figure 69 — Relationship between DCA Code Change for IBCLK and the 4-phase Internal Clocks/DQS Waveform (Example)

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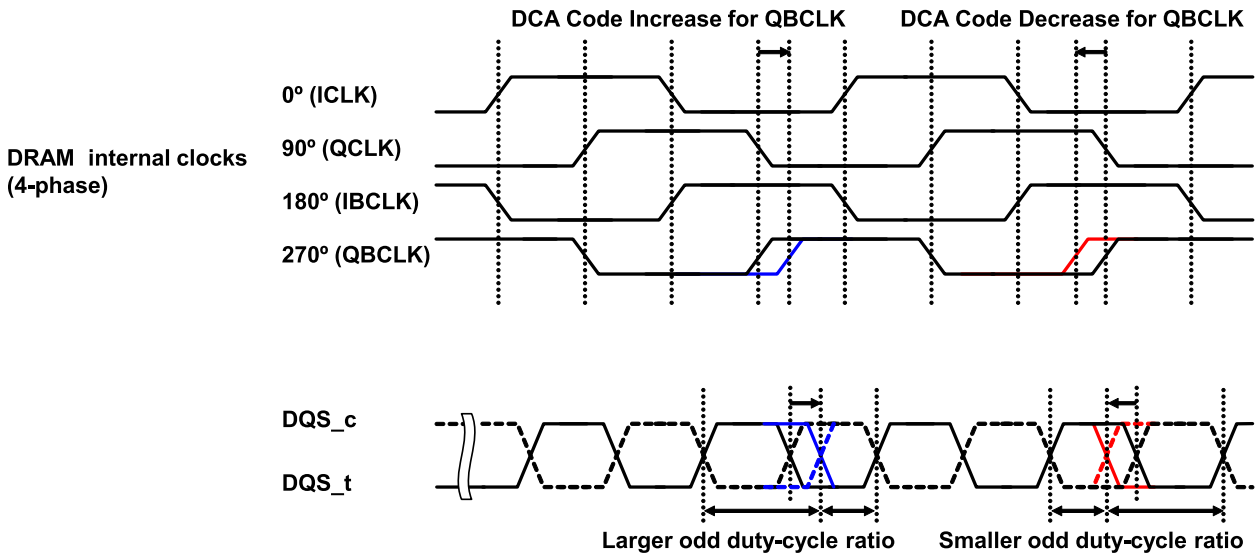


Figure 70 — Relationship between DCA Code Change for QBCLK and the 4-phase Internal Clocks/DQS Waveform (Example)

4.2.16.4. MR40 – RD DCA Types Supported

Table 66 — MR40 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RD DCA for single/two-phase clock(s) or QCLK in 4-phase clocks				RV		RD DCA Types Supported	

Function	Register Type	Operand	Data	Notes
RD DCA for single/two-phase clock(s) or QCLK in 4-phase clocks	RW	OP[7:4]	1000: Reserved 1001: DCA step -7 1010: DCA step -6 1011: DCA step -5 1100: DCA step -4 1101: DCA step -3 1110: DCA step -2 1111: DCA step -1 0000: DCA step +0 (default) 0001: DCA step +1 0010: DCA step +2 0011: DCA step +3 0100: DCA step +4 0101: DCA step +5 0110: DCA step +6 0111: DCA step +7	1
RV	RV	OP[3:2]	RV	

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Function	Register Type	Operand	Data	Notes
RD DCA Types Supported	R	OP[1:0]	00: RD DCA not supported 01: RD DCA supported for single/two-phase internal clock(s) 10: RD DCA supported for 4-phase internal clocks 11: Reserved	
NOTE 1: These settings can only be applied if MR40:OP[1:0]=01 or 10				

4.2.16.5. MR41 – RD DCA Settings

Table 67 — MR41 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RD DCA for QBCLK in 4-phase clocks				RD DCA for IBCLK in 4-phase clocks			

Function	Register Type	Operand	Data	Notes
RD DCA for QBCLK in 4-phase clocks	RW	OP[7:4]	1000: Reserved 1001: DCA step -7 1010: DCA step -6 1011: DCA step -5 1100: DCA step -4 1101: DCA step -3 1110: DCA step -2 1111: DCA step -1 0000: DCA step +0 (default) 0001: DCA step +1 0010: DCA step +2 0011: DCA step +3 0100: DCA step +4 0101: DCA step +5 0110: DCA step +6 0111: DCA step +7	1

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Function	Register Type	Operand	Data	Notes
RD DCA for IBCLK in 4-phase clocks	RW	OP[3:0]	1000: Reserved 1001: DCA step -7 1010: DCA step -6 1011: DCA step -5 1100: DCA step -4 1101: DCA step -3 1110: DCA step -2 1111: DCA step -1 0000: DCA step +0 (default) 0001: DCA step +1 0010: DCA step +2 0011: DCA step +3 0100: DCA step +4 0101: DCA step +5 0110: DCA step +6 0111: DCA step +7	1
NOTE 1: These settings can only be applied if MR40:OP[1:0]=10				

4.2.16.6. MR42 – DQS_t RD per-pin DCA for IBCLK and QCLK

Table 68 — MR42 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQS_t RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks				RV			

Function	Register Type	Operand	Data	Notes
DQS_t RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
RV		OP[3:0]		
NOTE 1: These settings can only be applied if MR40:OP[1:0]=10				

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4.2.16.7. MR43 – DQS_t RD per-pin DCA for QBCLK

Table 69 — MR43 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQS_t RD per-pin DCA for QBCLK				DQS_t RD per-pin DCA for IBCLK			

Function	Register Type	Operand	Data	Notes
DQS_t RD per-pin DCA for QBCLK	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
DQS_t RD per-pin DCA for IBCLK	RW	OP[3:0]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
NOTE 1: These settings can only be applied if MR40:OP[1:0]=10				

4.2.16.8. MR44 – DQS_c RD per-pin DCA for IBCLK and QCLK

Table 70 — MR44 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQS_c RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks				RV			

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Function	Register Type	Operand	Data	Notes
DQS_c RD per-pin DCA for single/two-phase clock(s) or QCLK in 4- phase clocks	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
RV		OP[3:0]		
NOTE 1: These settings can only be applied if MR40:OP[1:0]=01 or 10				

4.2.16.9. MR45 – DQS_c RD per-pin DCA for QBCLK

Table 71 — MR45 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQS_c RD per-pin DCA for QBCLK				DQS_c RD per-pin DCA for IBCLK			

Function	Register Type	Operand	Data	Notes
DQS_c RD per-pin DCA for QBCLK	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
DQS_c RD per-pin DCA for IBCLK	RW	OP[3:0]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
NOTE 1: These settings can only be applied if MR40:OP[1:0]=10				

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4.2.16.10. MRA2 – DQ0 RD per-pin DCA for QCLK

Table 72 — MRA2 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ0 RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks				RV			

Function	Register Type	Operand	Data	Notes
DQ0 RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
RV		OP[0:3]		
NOTE 1: These settings can only be applied if MR40:OP[1:0]=01 or 10				

4.2.16.11. MRA3 – DQ0 RD per-pin DCA for QBCLK and IBCLK

Table 73 — MRA3 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ0 RD per-pin DCA for QBCLK				DQ0 RD per-pin DCA for IBCLK			

Function	Register Type	Operand	Data	Notes
DQ0 RD per-pin DCA for QBCLK	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1

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Function	Register Type	Operand	Data	Notes
DQ0 RD per-pin DCA for IBCLK	RW	OP[3:0]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
NOTE 1: These settings can only be applied if MR40:OP[1:0]=10				

4.2.16.12. MRA4 – DQ1 RD per-pin DCA for QCLK

Table 74 — MRA4 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ1 RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks				RV			

Function	Register Type	Operand	Data	Notes
DQ1 RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
RV		OP[0:3]		
NOTE 1: These settings can only be applied if MR40:OP[1:0]=01 or 10				

4.2.16.13. MRA5 – DQ1 RD per-pin DCA for QBCLK and IBCLK

Table 75 — MRA5 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ1 RD per-pin DCA for QBCLK				DQ1 RD per-pin DCA for IBCLK			

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Function	Register Type	Operand	Data	Notes
DQ1 RD per-pin DCA for QBCLK	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
DQ1 RD per-pin DCA for IBCLK	RW	OP[3:0]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
NOTE 1: These settings can only be applied if MR40:OP[1:0]=10				

4.2.16.14. MRA6 – DQ2 RD per-pin DCA for QCLK

Table 76 — MRA6 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ2 RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks				RV			

Function	Register Type	Operand	Data	Notes
DQ2 RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
RV		OP[0:3]		
NOTE 1: These settings can only be applied if MR40:OP[1:0]=01 or 10				

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4.2.16.15. MRA7 – DQ2 RD per-pin DCA for QBCLK and IBCLK

Table 77 — MRA7 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ2 RD per-pin DCA for QBCLK				DQ2 RD per-pin DCA for IBCLK			

Function	Register Type	Operand	Data	Notes
DQ2 RD per-pin DCA for QBCLK	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
DQ2 RD per-pin DCA for IBCLK	RW	OP[3:0]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
NOTE 1: These settings can only be applied if MR40:OP[1:0]=10				

4.2.16.16. MRA8 – DQ3 RD per-pin DCA for QCLK

Table 78 — MRA8 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ3 RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks				RV			

Function	Register Type	Operand	Data	Notes
DQ3 RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
RV		OP[0:3]		
NOTE 1: These settings can only be applied if MR40:OP[1:0]=01 or 10				

4.2.16.17. MRA9 – DQ3 RD per-pin DCA for QBCLK and IBCLK

Table 79 — MRA9 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ3 RD per-pin DCA for QBCLK				DQ3 RD per-pin DCA for IBCLK			

Function	Register Type	Operand	Data	Notes
DQ3 RD per-pin DCA for QBCLK	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
DQ3 RD per-pin DCA for IBCLK	RW	OP[3:0]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
NOTE 1: These settings can only be applied if MR40:OP[1:0]=10				

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4.2.16.18. MRAA – DQ4 RD per-pin DCA for QCLK

Table 80 — MRAA Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ4 RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks				RV			

Function	Register Type	Operand	Data	Notes
DQ4 RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
RV		OP[0:3]		
NOTE 1: These settings can only be applied if MR40:OP[1:0]=01 or 10				

4.2.16.19. MRAB – DQ4 RD per-pin DCA for QBCLK and IBCLK

Table 81 — MRAB Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ4 RD per-pin DCA for QBCLK				DQ4 RD per-pin DCA for IBCLK			

Function	Register Type	Operand	Data	Notes
DQ4 RD per-pin DCA for QBCLK	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1

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Function	Register Type	Operand	Data	Notes
DQ4 RD per-pin DCA for IBCLK	RW	OP[3:0]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
NOTE 1: These settings can only be applied if MR40:OP[1:0]=10				

4.2.16.20. MRAC – DQ5 RD per-pin DCA for QCLK

Table 82 — MRAC Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ5 RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks				RV			

Function	Register Type	Operand	Data	Notes
DQ5 RD per-pin DCA for single/two-phase clock(s) or QCLK in 4-phase clocks	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
RV		OP[0:3]		
NOTE 1: These settings can only be applied if MR40:OP[1:0]=01 or 10				

4.2.16.21. MRAD – DQ5 RD per-pin DCA for QBCLK and IBCLK

Table 83 — MRAD Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ5 RD per-pin DCA for QBCLK				DQ5 RD per-pin DCA for IBCLK			

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Function	Register Type	Operand	Data	Notes
DQ5 RD per-pin DCA for QBCLK	RW	OP[7:4]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
DQ5 RD per-pin DCA for IBCLK	RW	OP[3:0]	1000-1010: Reserved 1101: step -3 1110: step -2 1111: step -1 0000: Disabled (default) 0001: step +1 0010: step +2 0011: step +3 0100-0111: Reserved	1
NOTE 1: These settings can only be applied if MR40:OP[1:0]=10				

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4.2.17. Write DQ Training Mode (WTM)

4.2.17.1. WTM Introduction

Initial DQ link training is critical with respect to the link performance with multiple modes to enhance IO scaling and training time in system. DDR6 DRAM requires pattern capability, better pattern type that increases robustness for signal integrity interactions. The Write Training Mode provides the ability to select two pattern types ~~with~~ and reports error ~~count~~ on a per-pin for system level optimization. The two pattern types are LFSR independent per-pin and user pattern.

4.2.17.2. WTM pattern

WTM can select the pattern type between ~~clock~~user patternvs LFSR seed with MR setting (MRDE). The LFSR mode provides an independent (different pattern) per lane using a single 24-bit Fibonacci LFSR per Pseudo Split Die (PSD). The polynomial for the Fibonacci LFSR is $x^{24} + x^{20} + x^{11} + x^5 + 1$. The user (Ex: Clock) pattern would operate at the DQ rate for another option for system optimization on drift terms like Rx DQS2DQ. Each mode provides a per pin error ~~count~~ logging. Figure 1 shows the details about left shift Fibonacci LFSR circuit for (X != Y), (Tap[X] XOR

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Tap[Y]) produces Tap[Z] from same polynomial, whereas X and Y are usually < 24, Z is a large integer (1 to 2^{24}-1) and random looking across (X,Y) combos. Figure 2 shows the Fibonacci LFSR-24 pattern details.

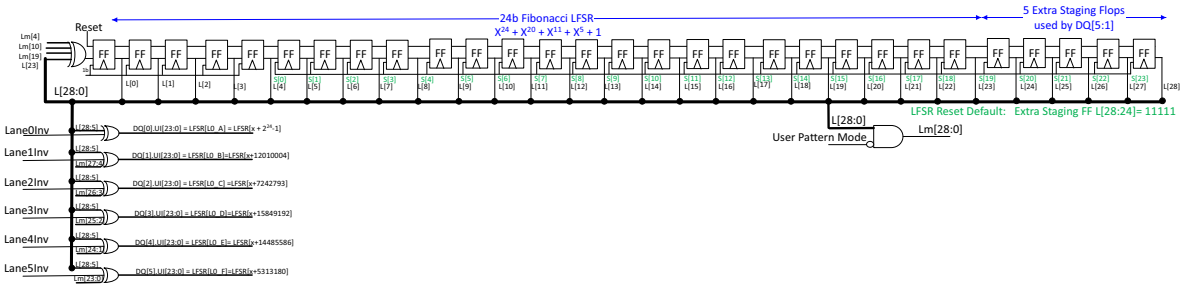


Figure 71 — Fibonacci LFSR 24-bit circuit

	L0	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12	L13	L14	L15	L16	L17	L18	L19	L20	L21	L22	L23	L24	L25	L26	L27	L28
LFSR Seed	0	1	1	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1	1	1	1
UI	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23					
DQ0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1	1	1	1					
DQ1	0	0	0	1	1	0	0	1	0	1	0	1	0	0	1	1	0	0	0	0	0	0	0	0					
DQ2	1	0	0	1	0	1	0	1	1	1	1	1	1	0	1	0	1	0	0	0	0	0	0	0					
DQ3	1	1	0	1	0	0	1	1	1	0	1	0	1	1	1	0	0	1	0	0	0	0	0	0					
DQ4	1	1	1	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0					
DQ5	0	1	1	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	0					
Shift1	1	0	1	1	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1	1	1
UI	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47					
DQ0	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1	1	1					
DQ1	1	0	0	0	1	1	0	0	1	0	1	0	1	0	0	1	1	0	0	0	0	0	0	0					
DQ2	1	1	0	0	1	0	1	0	1	1	1	1	1	1	0	1	0	1	0	0	0	0	0	0					
DQ3	1	1	1	0	1	0	0	1	1	1	0	1	0	1	1	1	0	0	1	0	0	0	0	0					
DQ4	0	1	1	1	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0					
DQ5	1	0	1	1	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0					
Shift2	1	1	0	1	1	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1	1
UI	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71					
DQ0	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1	1					
DQ1	0	1	0	0	0	1	1	0	0	1	0	1	0	1	0	0	1	1	0	0	0	0	0	0					
DQ2	0	1	1	0	0	1	0	1	0	1	1	1	1	1	1	0	1	0	1	0	0	0	0	0					
DQ3	1	1	1	1	0	1	0	0	1	1	1	0	1	0	1	1	1	0	0	1	0	0	0	0					
DQ4	0	0	1	1	1	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	1	0	0	0					
DQ5	1	0	1	1	1	0	0	0	0	1	0	0	0	1	0	0	0	0	1	0	0	0	1	0					
Shift3	0	1	1	0	1	1	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1
UI	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95					
DQ0	1	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1	1					
DQ1	0	0	1	0	0	0	1	1	0	0	1	0	1	0	1	0	0	1	1	0	0	0	0	0					
DQ2	1	0	1	1	0	0	1	0	1	0	1	1	1	1	1	1	0	1	0	1	0	0	0	0					
DQ3	0	1	1	1	1	0	1	0	0	1	1	1	0	1	0	1	1	1	0	0	1	0	0	0					
DQ4	0	0	0	1	1	1	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	1	0	0					
DQ5	1	0	1	0	1	1	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0					
Shift4	1	0	1	1	0	1	1	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1
UI	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119					
DQ0	1	1	1	0	0	0	0	1	0	0	0	1	1	0	0	1	1	1	0	1	1	1	1	1					
DQ1	1	0	0	1	0	0	0	1	1	0	0	1	0	1	0	1	0	0	1	1	0	0	0	0					
DQ2	0	1	0	1	1	0	0	1	0	1	0	1	1	1	1	1	1	0	1	0	1	0	0	0					
DQ3	0	0	1	1	1	1	0	1	0	0	1	1	1	0	1	0	1	1	1	0	0	1	0	0					
DQ4	1	0	0	0	1	1	1	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	1	0					
DQ5	0	1	0	1	0	1	1	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1					

Figure 72 — Fibonacci LFSR 24-bit pattern details

4.2.17.3. PRBS Pattern Generator for WTM

PRBS generator will start with the seed (Example 24b LFSR Seed: F7410B) and the below LFSR pattern table shows the example of shift mode where each DQ lanes gets different pattern on the lane. The upper 5 extra staging bits getting reset to 1, hence, users shall only independently program the LFSR seed [23:5] and need to calculate LFSR seed [4:0]

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to correctly initialize the LFSR. Table X illustrates the mapping of the pattern to DQ [5:0]. Refer to LFSR section x.x.x for more details.

4.2.17.4. Entry and Exit for Write Training Mode (WTM)

To enter Write Training Mode (WTM), the host memory controller shall issue a Mode Register Write (MRW) command targeting MR05 OP[4:0] = 10001B, with the Write Training Mode entry opcode. When the DRAM is in Write Training mode, commands are still actively processed.

To exit Write Training Mode, the host memory controller shall issue an MRW command to MR5 OP[7:0] = 00000000B, with the Write Training Mode Exit opcode.

4.2.17.5. Write Training Mode (WTM) Operation

The host enters WTM using MRW command and follows the sequence below.

1) Perform MRW to enter WTM
2) Host to perform MRW to pick LFSR (24-bit) or User pattern
3) Host to program LFSR seed settings
4) For each unique value of the host parameters,
5) Issue a series of Write bursts (BL 24), with either LFSR or User pattern as shown in Figure 3.
6) DRAM to internally check LFSR pattern per DQ lane
7) DRAM to log error status/error ~~counts~~ on a per lane basis
8) Host to read out the error logs through MRRs
9) MRR to MRR command timing during Write DQ Training Mode shall use tMRR_WTM timing parameter as described in Mode Register Read section x.x.y.z.
10) Clear the error status
11) Repeat the above steps for the next value/parameter.
12) Perform MRW to exit WTM

In this mode the only valid commands are WRITE, MRR and MRW.

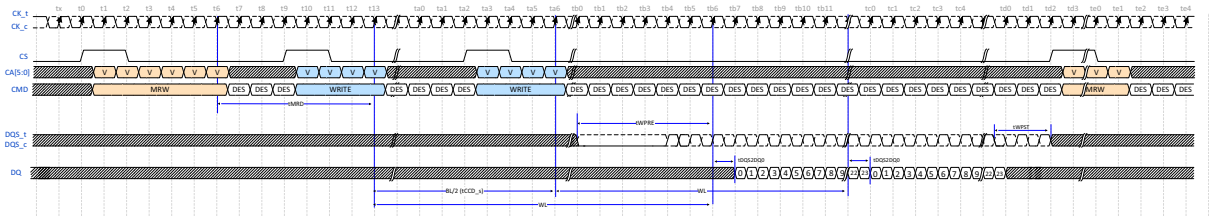


Figure 73 — Write DQ Training Mode Timing

Write Training Mode Registers:

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Table 84 — MRDE OP[7:0]- Write Training Mode Register

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU	RFU	RFU	RFU	LFSR Reset	Clear Errors	Phase Error selection	Write Training Pattern Selection

Function	Register Type	Operand	Data	Notes
Write Training Pattern Selection	R/W	OP[0]	1b0: User defined write pattern 1b1: LFSR	
Phase Error selection	R/W	OP[1]	1b0: Even/Odd 1b1: 4 phase errors	
Clear Errors	R/W	OP[2]	1b0: Do not clear 1b1: Clear phase error registers (auto clears after resetting)	
Seed Reset	R/W	OP[3]	1b0: Disable 1b1: Reset to default both LFSR and 5b-Staging Flip flops (auto clears after resetting)	
RFU	R/W	OP[5:4]	RFU	
RFU	R/W	OP[7:6]	RFU	

An independent 24-bit Fibonacci LFSR generator provides patterns for up to 6 DQ lanes. The source of the DQ output pattern shall be selected via MRDE:OP[0], where this field selects between LFSR output and the user-defined pattern resource. The MRDE[3] field shall be used to reset the LFSR seed value and associated staging flip-flops. Upon reset, the 5-bit staging flip-flops shall be initialized to their default value of 11111.

4.2.17.6. Error Logging

Each DQ lane shall implement phase error status registers to report WTM phase error information. DRAM shall allocate four Mode Registers (MRDF, MRE0, MRE1, and MRE2) for each DQ lane to report either consolidated even/odd phase errors or individual 4-phase errors, as illustrated in Figure 4.

The WTM error reporting mode shall be selected through a Mode Register Write (MRW) to MRDE:OP[1], which determines whether the error status is reported as even/odd phase errors or as individual 4-phase errors.

Even/Odd Error Reporting: MRDE: OP[1] = 0b

The DRAM shall consolidate phase error information as follows:

- Even phase errors: MRDF and MRE0 are combined and reported through MRDF.
- Odd phase errors: MRE1 and MRE2 are combined and reported through MRE1.

Error status shall indicate the presence of any error in the following UIs:

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- Even phase error status reports any error in: UI [0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22]
- Odd phase error status reports any error in: UI [1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23]

4-Phase Error Reporting: MRDE: OP[1] = 1b

The DRAM shall report all four phase error registers (MRDF, MRE0, MRE1, and MRE2) independently.

DRAM shall allocate 4 bits to report the error status for Ph0, Ph1, Ph2, and Ph3, respectively.

Error status shall indicate the presence of any error in the following UIs:

- Ph0 error status reports any error in: UI [0, 4, 8, 12, 16, 20]
- Ph1 error status reports any error in: UI [1, 5, 9, 13, 17, 21]
- Ph2 error status reports any error in: UI [2, 6, 10, 14, 18, 22]
- Ph3 error status reports any error in: UI [3, 7, 11, 15, 19, 23]

Table 85 — MRD8-MRDA WTM Phase Error Register details: (per PSD)

MR#	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
MRDF	RV	DQ2Ph1	DQ1Ph1	DQ0Ph1	RV	DQ2Ph0	DQ1Ph0	DQ0Ph0
MRE0	RV	DQ2Ph3	DQ1Ph3	DQ0Ph3	RV	DQ2Ph2	DQ1Ph2	DQ0Ph2
MRE1	RV	DQ5Ph1	DQ4Ph1	DQ3Ph1	RV	DQ5Ph0	DQ4Ph0	DQ3Ph0
MRE2	RV	DQ5Ph3	DQ4Ph3	DQ3Ph3	RV	DQ5Ph2	DQ4Ph2	DQ3Ph2

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4.2.18. Write Duty Cycle Adjuster (DCA)

DDR6 SDRAM supports a mode-register-adjustable Write(WR) DCA feature that enables the memory controller to adjust the DRAM's internal WR DQS tree duty cycle, enhancing DRAM WR margin. The DQS is shared across different PSDs(Pseudo Split Die). Therefore, the DCA code is determined based on the optimal value for all DQs within the DRAM die.

The memory controller can adjust the DQS duty cycle through the WR DCA mode registers and can determine the optimal Mode Register setting for WR DCA. This function adjusts the SDRAM internal DQS duty cycle (Static).

4.2.18.1. Duty Cycle Adjuster Range

The step range between the step 0 to 7(-7) is as follows. The difference of actual value between step N and step N+1 is defined to be within a spec range, since the variation of duty cycle by changing DCA code is not linear.

Table 86 — Write Duty Cycle Adjuster Step and Range

Parameter	Min/Max	Value		Unit	Notes
		~14.4Gb/s	Data Rate > 14.4Gb/s		
WR DCA Total Range	Min	7	TBD	ps	1, 2
	Max	28	TBD		
WR DCA 1-step Range	Min	1	TBD		1, 3
	Max	4	TBD		

NOTE 1: These values are guaranteed by design
NOTE 2: Total range means the range from step 0 to step +7(-7)
NOTE 3: 1-step range includes non-linearity of each step

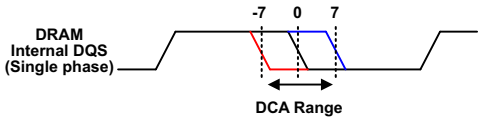


Figure 74 — Write Duty Cycle Adjuster Range

4.2.18.2. The Relationship between DCA Code Change and Single/Two-phase Internal DQS Timing

In case of the DQS tree used single/two-phase DQS scheme, the duty-cycle ratio of all DQS per device can be adjusted directly according to the internal DQS controlled by the DCA code.

Using a 2-phase DQS scheme, the rising edge of the 0° internal DQS is the reference edge, while the 180° internal DQS is adjusted based on 0° internal DQS.

The WR DCA adjustment uses the “DCA for single/two-phase DQS” mode register bits, MR3E:OP[7:4]. A positive DCA adjustment results in a larger duty cycle ratio, while a negative DCA adjustment results in a smaller duty cycle ratio.

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Mode register OP bits associated with the IB-DQS and QB-DQS may not be supported on the DRAM with a 2-phase DQS scheme.

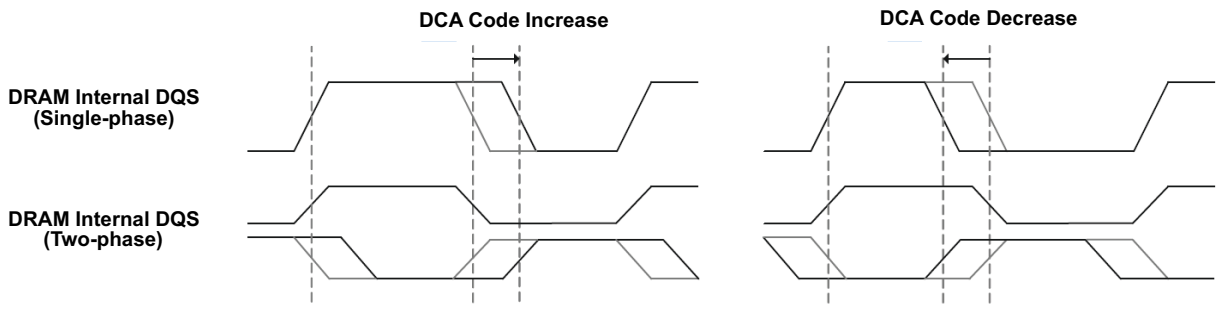


Figure 75 — Relationship between DCA Code Change and the Single/Two-phase Internal DQS Waveform (Example)

4.2.18.3. Relationship between DCA Code Change and 4-phase Internal DQS Timing

In case of the DQS tree used 4-phase DQS scheme, the even and odd duty-cycle ratio of all DQS per device can be respectively adjusted since the internal 4-phase DQS can be independently controlled by the DCA code.

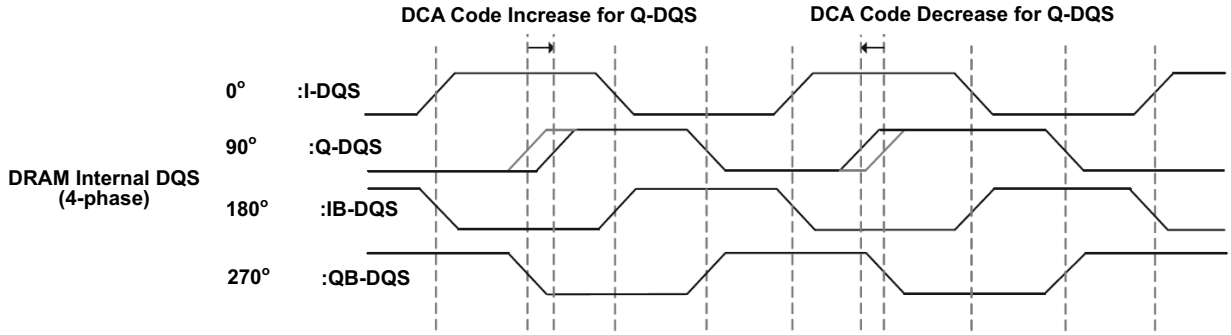


Figure 76 — Relationship between DCA Code Change for Q-DQS and the 4-phase Internal DQS Waveform (Example)

4.2.18.4. MR3E, MR3F – Write DCA Types Supported

Table 87 — MR3E Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
WR DCA for single/2-phase DQS or Q-DQS in 4-phase DQS				RFU		WR DCA Types Supported	

PROPOSED

Function	Register Type	Operand	Data	Notes
WR DCA for single/2-phase DQS or Q-DQS in 4-phase DQS	RW	OP[7:4]	1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	1,2
RV		OP[3:2]		
WR DCA Types Supported	R	OP[1:0]	00: WR DCA not supported 01: WR DCA supported for single/two phase internal DQS 10: WR DCA supported for 4-phase internal DQS 11: Reserved	
NOTE 1: Range: -7~ +7 LSB codes, Step size: 1LSB **NOTE 2:** These settings can only be applied if MR3E:OP[1:0] =01 or 10. (01: WR DCA supported for single/two-phase internal DQS, 10: WR DCA supported for 4-phase internal DQS)				

Table 88 — MR3F Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
WR DCA for QB-DQS in 4-phase DQS				WR DCA for IB-DQS in 4-phase DQS			

PROPOSED

Function	Register Type	Operand	Data	Notes
WR DCA for QB-DQS in 4-phase DQS	RW	OP[7:4]	1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	1,2
WR DCA for IB-DQS in 4-phase DQS	RW	OP[3:0]	1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	1,2

NOTE 1: Range: -7~ +7 LSB codes, Step size: 1LSB
NOTE 2: These settings can only be applied if MR3E:OP[1:0] =10 (10: WR DCA supported for 4-phase internal DQS)

PROPOSED

5. DDR6 Simplified State Diagram

DDR6 SDRAM state diagram provides a simplified illustration of allowed state transitions and the related commands to control them. For a complete definition of the device behavior, the information provided by the state diagram should be integrated with the truth tables and timing specification.

The truth tables provide complementary information to the state diagram; they clarify the device behavior and the applied restrictions when considering the actual state of all banks.

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For the command definition, see section xx Initialization and Training, and section xx Command and Operation.

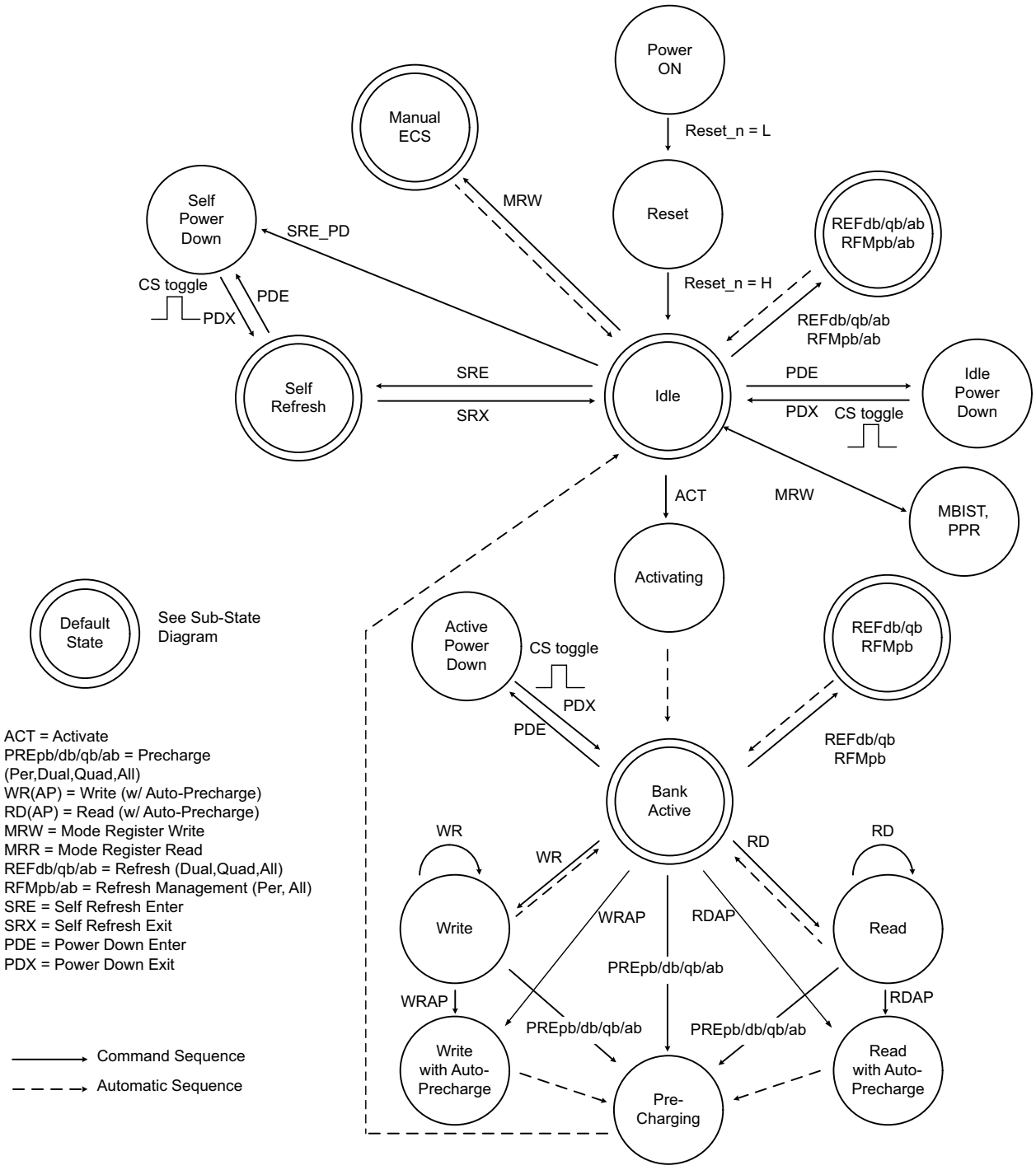


Figure 77 — Simplified State Diagram

PROPOSED

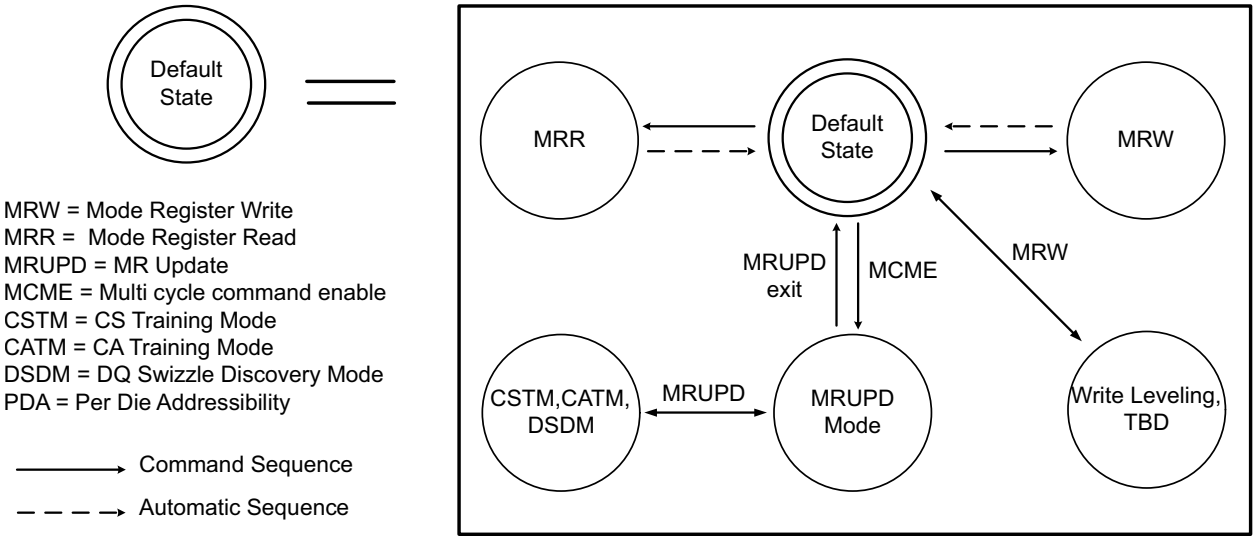


Figure 78 — Sub-State Diagram related to MRR, MRW, MCME, CSTM, CATM, Write Leveling and TBD Commands

NOTE: 1: From the Self-Refresh state the device can enter Power-Down, MRR and MRW states. See TBD, on Self-Refresh, for more information.

NOTE: 2: In IDLE state, all banks are pre-charged.

NOTE: 3: In the case of a MRW command to enter a training mode, the state machine will not automatically return to the IDLE state at the conclusion of training. See TBD, on Mode Register Write (MRW), for more information.

NOTE: 4: This simplified State Diagram is intended to provide an overview of the possible state transitions and the commands to control them. In particular, situations involving more than one bank, the enabling or disabling of on-die termination, and some other events are not captured in full detail.

NOTE: 5: States that have an "automatic return" and can be accessed from more than one prior state (e.g., MRW from either idle or Active states) will return to the state from when they were initiated (e.g., MRW from Idle will return to Idle).

NOTE: 6: The RESET_n pin can be asserted from any state, and will cause the SDRAM to go to the Reset State. The diagram shows RESET state applied from the Power-On as an example, but the Diagram should not be construed as a restriction on RESET_n.

NOTE: 7: "Bank Activate" to "Dual/Quad Bank Refresh" transition only refers to different banks not the same bank

PROPOSED

6. Mode Registers (MRs)

6.1. Mode Register (MR) Naming Convention

Mode Registers (MRs) are arranged as ~~Direct-Mapped~~ Shared Mode Registers and ~~Paged~~ PSD Mode Registers. ~~Direct-Mapped~~ Shared MRs can be accessed (written to and read from) both PSD interfaces. ~~Paged~~ PSD MRs can only be accessed from the respective PSD interface in 2pX configurations. PSD MRs can be accessed using control bits in 1pX configurations, as described in this section, the MRUPDATE section, and the Command Truth Table.

The ~~Direct-Mapped~~ Shared and ~~Paged~~ PSD Mode Register spaces contain up to 128 Mode Registers each, selectable by seven Mode Register Addresses, MRA[6:0]. MRA[7] selects between the ~~Direct-Mapped~~ Shared and ~~Paged~~ PSD spaces, with MRA[7]=0 for ~~Direct-Mapped~~ Shared MRs and MRA[7]=1 for ~~Page~~ PSD MRs. ~~The Paged Mode Register space may contain more than one page, which are selectable by MR00[1:0] if defined. The default page is zero (0) for both PSDs.~~

Each Mode Register contains a byte-wide payload. All unused (RV) bits in a defined Mode Register and all undefined (RV) Mode Registers will be read as a zero (0).

Table 89 — Mode Register Naming Convention

~~Direct-Mapped~~ Shared and PSD Mode Registers	~~Paged Mode Registers~~
MRxx, MR[ww:xx], MR[ww,xx], MRxx[y], MRxx[y:z],MRxx[y,z]	~~PG[a], PG[b:a], PG[b,a], PG[a]MRxx[y]~~
Where: • xx = MR address (using leading zero where applicable) • ww:xx = MR address range • ww,xx = Selected MR addresses • y = Single bit within the MR address • y:z = Bit range within the MR address • y,z = Selected bits within the MR address	~~Where:~~ • ~~a = Page~~ • ~~b:a = Page range~~ • ~~b,a = Selected pages~~ • ~~xx = MR address, address range, or selected addresses within the page~~ • ~~y = Single bit, bit range, or selected bits within the paged MR address~~
Examples: • MR05 (refers to all bits of MR05) • MR[24:20] • MR[60,64,68] • MR0A[0] • MR1F[2:0]	~~Examples:~~ • ~~PG[1:0] (refers to all MR addresses and bits within PG[0] and PG[1]~~ • ~~PG[A,B]MR60~~ • ~~PG[5:4]MR79[7]~~

6.2. Mode Register Attributes

Mode registers can have the following attributes, which are shown in the mode register definition tables.

Table 90 — Mode Register Attributes (Sheet 1 of 2)

Attribute	Abbreviation	Description
Read/Write	RW	This Mode Register can be read or written by software.

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Table 90 — Mode Register Attributes (Continued) (Sheet 2 of 2)

Attribute	Abbreviation	Description
Read Only	R~~O~~	This Mode Register can be read by software. Writes (MRWs) have no effect.
Reserved	RV	These bits are reserved for future expansion. They may not be physically implemented and shall read zero to ensure forward and backward compatibility.
~~Sticky~~	~~S~~	~~This Mode Register is reset to the default only by power cycle, not Reset. This attribute can be combined with RW.~~

Mode Register definition tables must follow these rules to ensure they are machine-readable:

- In the Attribute column, commas must be used to separate attributes.
- In the Default column, “0” and “1” may be used for single-bit values. For all other widths, hex values must be used with an “h” suffix, as in “3h” or “1Ah”.
- In the Description column, names of individual Mode Registers must be in all caps, using underscores in place of spaces, following the address, bit range, and a colon.
- ~~The address in the Description column must not use brackets around the page number, and the page number uses two digits with leading zero. This is the only place where these brackets are not used for the page number.~~

6.3. Mode Register Access

Shared and PSD Mode Registers access differs based on DRAM configuration.

PROPOSED

For 2pX configurations, both PSDs can access the Shared MRs, however each PSD can only access its corresponding PSD MRs.

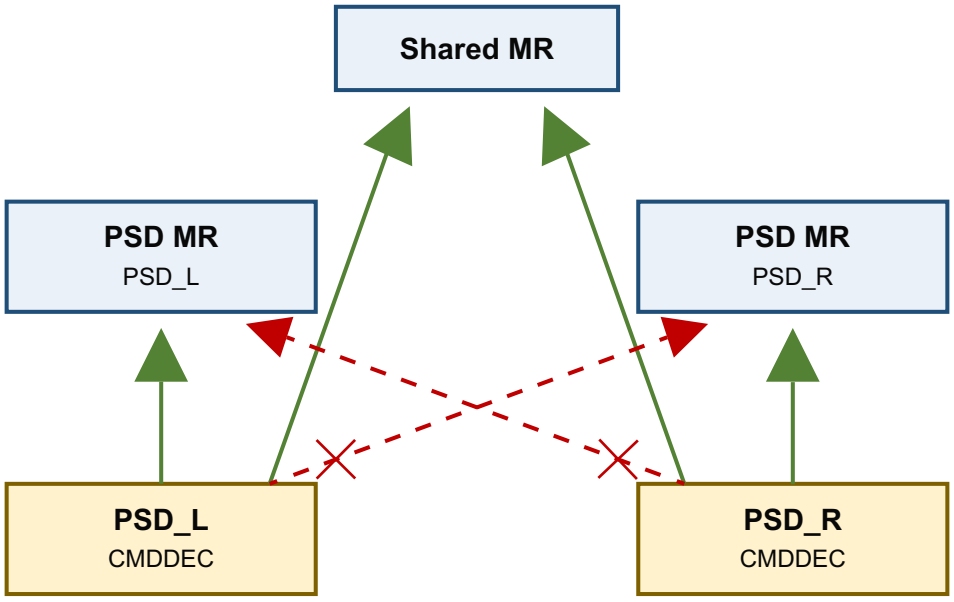


Figure 79 — 2pX Mode Register Access

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For 1pX configurations, the enabled PSD can access the Shared MRs, as well as both physical PSD MRs using CA bit encoding in the MRW and MRR commands and in MRUPATE Mode.

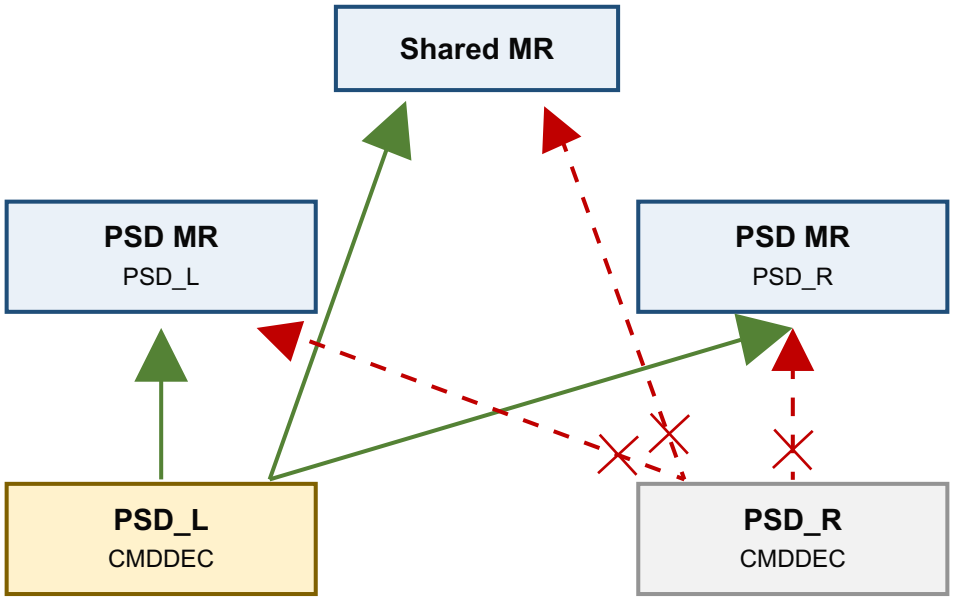
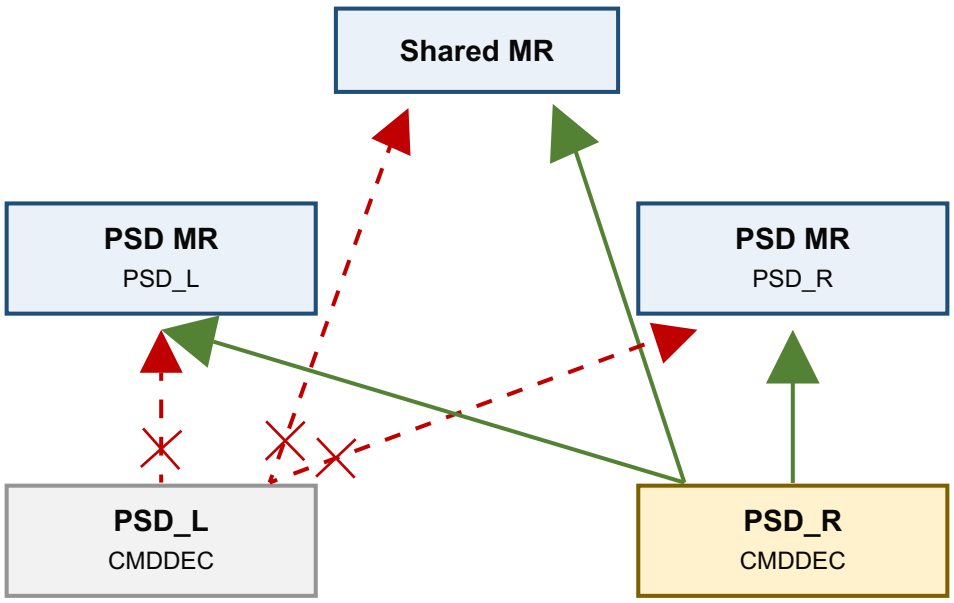


Figure 80 — 1pX PSD Left Connected Mode Register Access



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Figure 81 — 1pX PSD Right Connected Mode Register Access

6.4. Mode Register Functional Modes

Functions or modes (MR Functional Modes) to be executed are defined by ~~a decoded combination of~~ MR05 ~~[7:0]~~ and MR06, as shown in the MR05 and MR06 description s. The target Functional Mode is entered or executed within tMRD after an MRW or the last cycle of an MRUPDATE sequence. The ~~MR05~~ description will also indicate if a Functional Mode can only be enabled while in MRUPDATE Mode.

Functional Modes may be self-clearing or may require to be explicitly disabled, as indicated in the ~~MR05~~ description. MR05 and MR06 ~~is a~~ are RW register s, allowing the state of the function to be read by an MRR command, if the MRR command is permitted while the Functional Mode is executing. If a Functional Mode is self-clearing, the MRR will return to 00h upon completion. Setting MR05[7:0]=00h or MR06[7:0]=00h will disable ~~a~~ the respective Functional Mode, unless it is self-clearing.

Unless explicitly stated that overlap of a self-clearing function by another function is permitted, only one Functional Mode may be entered/executed at a time.

PROPOSED

6.5. Mode Registers (MRs)

6.5.1. Mode Register Maps

The following tables show the mapping of the ~~Direct-Mapped~~ Shared Mode Registers and the ~~Paged~~ PSD Mode Registers.

Table 91 — ~~Direct-Mapped~~ Shared Mode Registers

MR#	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0	FSP
MR00	VRCG	RV			FSP OP	FSP WR	System Metadata Mode		N
MR01	Command Blocking Time		CA Parity Command Blocking	CA Parity Checking	Write CRC Enable	Read CRC Enable	CRC Mode	RV	N
~~MR06~~ MR02	RV				PDA Enumeration ID				N
MR03	# of Ranks in Package			Configuration		Density per Die			N
MR04	Thermal Offset			Temperature Update Flag	Refresh Interval Rate Indicator	Minimum Refresh Rate			N
MR05	RV			Functional Modes					N
MR06	Functional Modes								N
MR07	RV	ZQ Interval			ZQ Stop	ZQ Reset	ZQ Initiator	ZQ Update Flag	N
MR08	RV		CAS Latency						Y
MR09	TM	Parity Latency		Frequency (Data Rate)					Y
MR0A	RV		nWTP/nRTP						Y
MR0B	Write DQS Postamble Length				Write DQS Preamble Length				Y
MR0C	Read DQS Postamble Length				Read DQS Preamble Length				Y
MR0D	RV				Read DQS Preshift				Y
MR0E	RV						Command Ignore Polarity	Command Ignore	N
MR0F	RV		WICA						Y
MR10	Internal Write Leveling								Y
MR11	RV		Shift Register Mode	RV	Seed Reset	RV		Read Training Pattern Selection	N

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Table 91 — ~~Direct Mapped~~ Shared Mode Registers

MR#	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0	FSP
MR12	RV								N
MR13	Rx OCC Reset	RV	PD Output Driver Impedance			PU Output Driver Impedance			Y
MR14	Rx CTLE Support	RV			Rx CTLE Control DQS		Rx CTLE Control CK		Y
MR15	RV								N
~~MR3E~~ MR16	RV				CK ODT Default	CK ODT			Y
MR17	RV								N
~~MR3F~~ MR18	RV		DQS RTT Park			DQ RTT Park			Y
~~MR40~~ MR19	RV		DQ RTT Write Non-Target			DQ RTT Write			Y
~~MR41~~ MR1A	RV		ODTLoff WR Offset			ODTLon WR Offset			Y
~~MR42~~ MR1B	RV		ODTLoff WR NT Offset			ODTLon WR NT Offset			Y
MR1C	RV								N
MR1D	RV						Read DQS Toggle Mode	Enhanced DQS	N
~~MR49~~ MR1E	DQS Interval Oscillator Count LSB								N
~~MR4A~~ MR1F	DQS Interval Oscillator Count MSB								N
~~MR48~~ MR20	DQS Interval Oscillator Run Time Setting								N
MR21	CLK Interval Oscillator Count LSB								N
MR22	CLK Interval Oscillator Count MSB								N
MR23	CLK Interval Oscillator Run Time Setting								N
MR24	RV								N
~~MR31~~ MR25	Serial ID1								N
~~MR32~~ MR26	Serial ID2								N
~~MR33~~ MR27	Serial ID3								N

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Table 91 — ~~Direct Mapped~~ Shared Mode Registers

MR#	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0	FSP
~~MR34~~ MR28	Serial ID4								N
~~MR35~~ MR29	Serial ID5								N
~~MR36~~ MR2A	Serial ID6								N
~~MR37~~ MR2B	Serial ID7								N
~~MR38~~ MR2C	Serial ID8								N
MR2D	RV								N
MR2E	RV								N
MR2F	RV								N
~~MR1E~~ MR30	Guard Key Encoding								N
MR31	RV						mPPR Enable	MBIST Enable	N
~~MR20~~ MR32	RV			sPPR Verification Result	sPPR Verification Enable	sPPR Control		hPPR Enable	N
MR33	RV								N
~~MR4D~~ MR34	RV				Adaptive PRAC		MRFMAACT		N
~~MR4E~~ MR35	RV			Alert Verification	PRAC Testing Init	PRAC Testing Enable	ACI Complete	ACI Enable	N
MR36	RV								N
MR37	RV								N
MR38	Fault Detection Modes								N
MR39	RV					On-Die ECC MBE Fault	Write Link ECC SBE Correction	Write Link ECC/EDC Fault	N
~~MR56~~ MR3A	Temperature Sensor Readout								N
~~MR57~~ MR3B	ECS CE Flag	ECS Writeback Control	ECS Error threshold Count			Row/Codeword Mode	ECS Flag Reset	ECS Reset	N
MR3C	RV								N

Table 91 — ~~Direct Mapped~~ Shared Mode Registers

MR#	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0	FSP
MR3D	RV								N
~~MR68~~ MR3E	Write DCA single/2 phase DQS or 4 phase QDQS				RV		Write DCA Types Supported		Y
~~MR69~~ MR3F	Write DCA 4 phase QBDQS				Write DCA 4 phase IBDQS				Y
~~MR6A~~ MR40	Read DCA single/2 phase CLK or 4 phase QCLK				RV		Read DCA Types Supported		Y
~~MR6B~~ MR41	Read DCA 4 phase QBCLK				Read DCA 4 phase IBCLK				Y
~~MR6D~~ MR42	DQS t Read DCA single/2 phase CLK or 4 phase QCLK				RV				Y
~~MR6E~~ MR43	DQS t Read DCA 4 phase QBCLK				DQS t Read DCA 4 phase IBCLK				Y
~~MR6F~~ MR44	DQS c Read DCA single/2 phase CLK or 4 phase QCLK				RV				Y
~~MR6C~~ MR45	DQS c Read DCA 4 phase QBCLK				DQS c Read DCA 4 phase IBCLK				N
MR46	RV								N
MR47	DRAM Scratch Pad								N
MR48-MR7F	RV								N

Table 92 — ~~Paged~~ PSD Mode Registers

MR#	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0	FSP
MR80	RV						PSD Logical	PSD Physical	N
~~PG[0]MR00~~ MR81	RV		DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	N
~~PG[0]MR02~~ MR82	RV				PDA Select ID				N
~~MR19~~ MR83	RV	VREF CS							Y
~~MR1A~~ MR84	RV	VREF CA							Y
~~MR1B~~ MR85	RV	VREF DQ							Y

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Table 92 — ~~Paged~~ PSD Mode Registers

MR#	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0	FSP
~~PG[0]M R03~~ MR86	CA1 VREFCA Offset				CA0 VREFCA Offset				Y
~~PG[0]M R04~~ MR87	CA3 VREFCA Offset				CA2 VREFCA Offset				Y
~~PG[0]M R05~~ MR88	CA5 VREFCA Offset				CA4 VREFCA Offset				Y
~~PG[0]M R06~~ MR89	DQ1 VREFDQ Offset				DQ0 VREFDQ Offset				Y
~~PG[0]M R07~~ MR8A	DQ3 VREFDQ Offset				DQ2 VREFDQ Offset				Y
~~PG[0]M R08~~ MR8B	DQ5 VREFDQ Offset				DQ4 VREFDQ Offset				Y
MR8C	RV				Rx CTLE Control CS		Rx CTLE Control CA		Y
~~PG[0]M R0A~~ MR8D	RV			Global DFE TAP-4 Enable	Global DFE TAP-3 Enable	Global DFE TAP-2 Enable	Global DFE TAP-1 Enable	Per Pin DFE Control	Y
~~PG[0]M R0B~~ MR8E	RV			DFE TAP-1 DQ0					Y
~~PG[0]M R0C~~ MR8F	RV			DFE TAP-2 DQ0					Y
MR90	DFE TAP-4 DQ0				DFE TAP-3 DQ0				Y
~~PG[0]M R0D~~ MR91	RV			DFE TAP-1 DQ1					Y
~~PG[0]M R0E~~ MR92	RV			DFE TAP-2 DQ1					Y
MR93	DFE TAP-4 DQ1				DFE TAP-3 DQ1				Y
~~PG[0]M R0F~~ MR94	RV			DFE TAP-1 DQ2					Y
~~PG[0]M R10~~ MR95	RV			DFE TAP-2 DQ2					Y

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Table 92 — ~~Paged~~ PSD Mode Registers

MR#	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0	FSP
MR96	DFE TAP-4 DQ2				DFE TAP-3 DQ2				Y
~~PG[0]MR11~~ MR97	RV			DFE TAP-1 DQ3					Y
~~PG[0]MR12~~ MR98	RV			DFE TAP-2 DQ3					Y
MR99	DFE TAP-4 DQ3				DFE TAP-3 DQ3				Y
~~PG[0]MR13~~ MR9A	RV			DFE TAP-1 DQ4					Y
~~PG[0]MR14~~ MR9B	RV			DFE TAP-2 DQ4					Y
MR9C	DFE TAP-4 DQ4				DFE TAP-3 DQ4				Y
~~PG[0]MR15~~ MR9D	RV			DFE TAP-1 DQ5					Y
~~PG[0]MR16~~ MR9E	RV			DFE TAP-2 DQ5					Y
MR9F	DFE TAP-4 DQ5				DFE TAP-3 DQ5				Y
MRA0	RV								N
MRA1	RV								N
~~PG[0]MR1B~~ MRA2	DQ0 Read DCA single/2 phase CLK or 4 phase QCLK				RV				Y
~~PG[0]MR1A~~ MRA3	DQ0 Read DCA 4 phase QBCLK				DQ0 Read DCA 4 phase IBCLK				Y
~~PG[0]MR1D~~ MRA4	DQ1 Read DCA single/2 phase CLK or 4 phase QCLK				RV				Y
~~PG[0]MR1C~~ MRA5	DQ1 Read DCA 4 phase QBCLK				DQ1 Read DCA 4 phase IBCLK				Y
~~PG[0]MR1F~~ MRA6	DQ2 Read DCA single/2 phase CLK or 4 phase QCLK				RV				Y
~~PG[0]MR1E~~ MRA7	DQ2 Read DCA 4 phase QBCLK				DQ2 Read DCA 4 phase IBCLK				Y

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Table 92 — ~~Paged~~ PSD Mode Registers

MR#	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0	FSP
~~PG[0]MR21~~ MRA8	DQ3 Read DCA single/2 phase CLK or 4 phase QCLK				RV				Y
~~PG[0]MR20~~ MRA9	DQ3 Read DCA 4 phase QBCLK				DQ3 Read DCA 4 phase IBCLK				Y
~~PG[0]MR23~~ MRAA	DQ4 Read DCA single/2 phase CLK or 4 phase QCLK				RV				Y
~~PG[0]MR22~~ MRAB	DQ4 Read DCA 4 phase QBCLK				DQ4 Read DCA 4 phase IBCLK				Y
~~PG[0]MR25~~ MRAC	DQ5 Read DCA single/2 phase CLK or 4 phase QCLK				RV				Y
~~PG[0]MR24~~ MRAD	DQ5 Read DCA 4 phase QBCLK				DQ5 Read DCA 4 phase IBCLK				Y
MRAE	RV								N
MRAF	RV								N
MRB0	DQ0 Write DCA single/2 phase CLK or 4 phase QCLK				RV				Y
MRB1	DQ0 Write DCA 4 phase QBCLK				DQ0 Write DCA 4 phase IBCLK				Y
MRB2	DQ1 Write DCA single/2 phase CLK or 4 phase QCLK				RV				Y
MRB3	DQ1 Write DCA 4 phase QBCLK				DQ1 Write DCA 4 phase IBCLK				Y
MRB4	DQ2 Write DCA single/2 phase CLK or 4 phase QCLK				RV				Y
MRB5	DQ2 Write DCA 4 phase QBCLK				DQ2 Write DCA 4 phase IBCLK				Y
MRB6	DQ3 Write DCA single/2 phase CLK or 4 phase QCLK				RV				Y
MRB7	DQ3 Write DCA 4 phase QBCLK				DQ3 Write DCA 4 phase IBCLK				Y
MRB8	DQ4 Write DCA single/2 phase CLK or 4 phase QCLK				RV				Y
MRB9	DQ4 Write DCA 4 phase QBCLK				DQ4 Write DCA 4 phase IBCLK				Y
MRBA	DQ5 Write DCA single/2 phase CLK or 4 phase QCLK				RV				Y
MRBB	DQ5 Write DCA 4 phase QBCLK				DQ5 Write DCA 4 phase IBCLK				Y
MRBC	RV								N

Table 92 — ~~Paged~~ PSD Mode Registers

MR#	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0	FSP
~~MR3D~~ MRBD	CS ODT Default	CS ODT			CA ODT Default	CA ODT			N
MRBE	RV								N
~~MR21~~ MRBF	BG1 BANK1[1:0]		BG1 BANK0[1:0]		BG0 BANK1[1:0]		BG0 BANK0[1:0]		N
~~MR22~~ MRC0	BG3 BANK1[1:0]		BG3 BANK0[1:0]		BG2 BANK1[1:0]		BG2 BANK0[1:0]		N
~~MR23~~ MRC1	BG5 BANK1[1:0]		BG5 BANK0[1:0]		BG4 BANK1[1:0]		BG4 BANK0[1:0]		N
~~MR24~~ MRC2	BG7 BANK1[1:0]		BG7 BANK0[1:0]		BG6 BANK1[1:0]		BG6 BANK0[1:0]		N
~~MR25~~ MRC3	BG9 BANK1[1:0]		BG9 BANK0[1:0]		BG8 BANK1[1:0]		BG8 BANK0[1:0]		N
~~MR2E~~ MRC4	BG11 BANK1[1:0]		BG11 BANK0[1:0]		BG10 BANK1[1:0]		BG10 BANK0[1:0]		N
~~MR27~~ MRC5	BG13 BANK1[1:0]		BG13 BANK0[1:0]		BG12 BANK1[1:0]		BG12 BANK0[1:0]		N
~~MR28~~ MRC6	BG15 BANK1[1:0]		BG15 BANK0[1:0]		BG14 BANK1[1:0]		BG14 BANK0[1:0]		N
~~MR1F~~ MRC7	RV						MBIST Transparency		N
~~MR4F~~ MRC8	RV					Counter Address Report	Counter Error	Alert Back Off	N
~~MR50~~ MRC9	PRAC RAS Event R[7:0]								N
~~MR51~~ MRCA	PRAC RAS Event R[15:8]								N
~~MR52~~ MRCB	PRAC RAS Event BG[3:0]				PRAC RAS Event BA	PRAC RAS Event R[18:16]			N
MRCC	RV							ECS CE Flag	N
~~MR58~~ MRCD	Max CE Row Address R[7:0]								N
~~MR59~~ MRCE	Max CE Row Address R[15:8]								N
~~MR5A~~ MRCF	Max CE BG[3:0]				Max CE BA	Max CE Row Address R[18:16]			N

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Table 92 — ~~Paged~~ PSD Mode Registers

MR#	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0	FSP
~~MR5C~~ MRD0	RV		CA Parity Error Log C0						N
~~MR5D~~ MRD1	RV		CA Parity Error Log C1						N
~~MR5E~~ MRD2	RV		CA Parity Error Log C2						N
~~MR5F~~ MRD3	RV		CA Parity Error Log C3						N
~~MR60~~ MRD4	RV		CA Parity Error Log C4						N
~~MR61~~ MRD5	RV		CA Parity Error Log C5						N
MRD6	RV							Write CRC Error Status	N
~~PG[0]MR4D~~ MRD7	RV		Pattern Inversion DQ5	Pattern Inversion DQ4	Pattern Inversion DQ3	Pattern Inversion DQ2	Pattern Inversion DQ1	Pattern Inversion DQ0	N
~~PG[0]MR46~~ MRD8	LFSR Seed/Pattern UI7	LFSR Seed/Pattern UI6	LFSR Seed/Pattern UI5	LFSR Seed/Pattern UI4	LFSR Seed/Pattern UI3	LFSR Seed/Pattern UI2	LFSR Seed/Pattern UI1	LFSR Seed/Pattern UI0	N
~~PG[0]MR45~~ MRD9	LFSR Seed/Pattern UI15	LFSR Seed/Pattern UI14	LFSR Seed/Pattern UI13	LFSR Seed/Pattern UI12	LFSR Seed/Pattern UI11	LFSR Seed/Pattern UI10	LFSR Seed/Pattern UI9	LFSR Seed/Pattern UI8	N
MRDA	LFSR Seed/Pattern UI23	LFSR Seed/Pattern UI22	LFSR Seed/Pattern UI21	LFSR Seed/Pattern UI20	LFSR Seed/Pattern UI19	LFSR Seed/Pattern UI18	LFSR Seed/Pattern UI17	LFSR Seed/Pattern UI16	N
MRDB	RV								N
MRDC	RV								N
MRDD	RV								N
MRDE	RV		Shift Reg Mode	RV	Seed Reset	Clear Errors	Errors 4Ph, Even/Odd	Write Training Pattern Selection	N
~~PG[0]MR56~~ MRDF	RV	DQ2Ph1	DQ1Ph1	DQ0Ph1	RV	DQ2Ph0	DQ1Ph0	DQ0Ph0	N
~~PG[0]MR57~~ MRE0	RV	DQ2Ph3	DQ1Ph3	DQ0Ph3	RV	DQ2Ph2	DQ1Ph2	DQ0Ph2	N

Table 92 — ~~Paged~~ PSD Mode Registers

MR#	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0	FSP
~~PG[0]MR58~~ MRE1	RV	DQ5Ph1	DQ4Ph1	DQ3Ph1	RV	DQ5Ph0	DQ4Ph0	DQ3Ph0	N
~~PG[0]MR59~~ MRE2	RV	DQ5Ph3	DQ4Ph3	DQ3Ph3	RV	DQ5Ph2	DQ4Ph2	DQ3Ph2	N
MRE3-MRFF	RV								N

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6.6. Mode Register (MR) Definitions

6.6.1. Shared Mode Register Definitions

Note: The Mode Register definitions that follow are intended to reflect the content of balloted sections of the standard. If not yet balloted, these definitions will be updated to reflect the actual standard content.

6.6.1.1. MR00 - Configuration Settings

Bits	Attribute	Default	Description	Note
7:4	RV		MR00[7:4]: RV	
3	RW	0	MR00[3]: FSP_OP Operating Frequency Set Point MR selection 0: FSP-OP[x] 1: FSP-OP[y]	
2	RW	0	MR00[2]: FSP_WR Write/Read Frequency Set Point MR selection 0: FSP-WR[x] 1: FSP-WR[y]	
1:0	RW	0h	MR00[1:0]: SYSTEM_METADATA_MODE System Metadata, to be set during initialization 00: Default 01: MD_8b 10: MD_16b 11: MD_OFF	

6.6.1.2. MR01 - Configuration Settings

Bits	Attribute	Default	Description	Note
7:6	RW	0h	MR01[7:6]: CMD_BLOCKING_TIME Command Blocking time upon CA Parity error (if Command Blocking enabled) 00: TBD 01: TBD 10: TBD 11: TBD	
5	RW	0	MR01[5]: CMD_BLOCKING Command Blocking upon CA Parity error disable / enable 0: Command Blocking disabled (non-blocking) 1: Command Blocking enabled	
4	RW	0	MR01[4]: CA_PARITY_CHECKING CA Parity Checking disable / enable 0: CA Parity Checking disabled 1: CA Parity Checking enabled	

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Bits	Attribute	Default	Description	Note
3	RW	0	MR01[3]: WR_CRC_EN Write CRC disable / enable 0: Write CRC disabled 1: Write CRC enabled	2
2	RW	0	MR01[2]: RD_CRC_EN Read CRC disable / enable 0: Read CRC disabled 1: Read CRC enabled	
1	RW	1	MR01[1]: CRC_CODE_2P3 CRC Code for 2p3 0: CRC-4 1: CRC-8-64	1
0	RV		MR01[0]: RV	

NOTE: 1: : The CRC Code for 2p3 mode register only applies to the 2p3 device configuration. It is ignored for the 1p6/2p6/1p12 device configurations.
NOTE: 2: The host shall disable Write CRC, if it was enabled, prior to entering Write Leveling Training mode.

6.6.1.3. ~~MR06~~ MR02 - Configuration Settings

~~MR06~~ MR02				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RV		MR02[7:4]: RV	
3:0	R	Fh	MR02[3:0]: ENUMERATION_ID Programmed through PDA Enumeration Mode. xxxx encoding is the enumeration ID.	

6.6.1.4. MR03 - Configuration Information

MR03				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	R		MR03[7:5]: #_OF_RANKS_IN_PACKAGE Number of ranks in the DRAM package 000: 1 001: 2 010: 3 011: 4 100: 5 101: 6 110: 7 111: 8	

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MR03				
Bits	**Attribute**	**Default**	**Description**	**Note**
4:3	R		MR03[4:3]: CONFIGURATION Configuration of the DRAM 00: 2p3 01: 1p6 10: 2p6 11: 1p12	
2:0	R		MR03[2:0]: DENSITY_PER_DIE DRAM density per die 000: 32Gb 001: 48Gb 010: 64Gb 011: 96Gb 100: 128Gb 101-111: Reserved	

6.6.1.5. MR04 - Refresh

MR04				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RW	0h	MR04[7:5]: THERMAL_OFFSET Thermal offset applied to the REFRESH_RATE readout 000: No offset 001: 2°C offset 010: 4°C offset 011: 6°C offset 100: 8°C offset 101: 10°C offset 110: 12°C offset 111: 14°C offset	1
4	R	0	MR04[4]: TUF Temperature Update Flag 0: No change in MR04[2:0] since last MR04 read 1: A change occurred in MR04[2:0] since last MR04 read	
3	RW	0	MR04[3]: RIR_INDICATOR Refresh Interval Rate (RIR) disable / enable 0: Disabled 1: Enabled	

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MR04				
Bits	**Attribute**	**Default**	**Description**	**Note**
2:0	R		MR04[2:0]: REFRESH_RATE Average interval between refresh commands sent by the Host to the DRAM 000: Reserved 001: tREFI (1x Refresh Rate), Temperature<75°C nominal 010: tREFI (1x Refresh Rate), 75<Temperature≤80°C nominal 011: tREFI (1x Refresh Rate), 80<Temperature≤85°C nominal 100: tREFI/2 (2x Refresh Rate), 85<Temperature≤90°C nominal 101: tREFI/2 (2x Refresh Rate), 90<Temperature≤95°C nominal 110: tREFI/2 (2x Refresh Rate), Temperature>95°C nominal 111: Reserved	

NOTE: The two PSDs (PSD 0 and PSD 1) on a single die share the same thermal sensor, therefore the thermal offset value is applied to the common thermal sensor.

6.6.1.6. MR05 - Functional Modes

MR05				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR05[7:5]: RV	

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MR05				
Bits	**Attribute**	**Default**	**Description**	**Note**
4:0	RW	0h	MR05[4:0]: FUNCTIONAL MODES DRAM direct execution functional modes 00000: All Functions Disabled MR05[4]=0, Command can only be used in MRUPD mode MR05[3:2]=00: No need operate DQ/DQS • 00001: CK Sync Off • 00010-00011: RV MR05[3:2]=01: Functions on DQ/DQS • 00100: CSTM Entry • 00101: CATM Entry • 00110: PDA Enumeration Mode • 00111: RV MR05[3:2]=10: Output on DQ/DQS • 01000: DSDM Entry • 01001: RV for strap detection • 01010: RV for strap detection • 01011: RV for strap detection MR05[3:2]=11: Reserved • 01100-01111: RV MR05[4]=1, Command can be used in MRUPD and MRW MR05[3:2]=00: Write/Read training • 10000: Write Leveling Training • 10001: Write Training Mode • 10010: Host Receiver Training Mode • 10011: Read Preamble Training MR05[3:2]=01: Other training • 10100: DQ Rx Offset Calibration Training • 10101-10111: RV MR05[3:2]=10: CK/DQS Oscillator • 11000: Start CLK OSC • 11001: Stop CLK OSC • 11010: Start DQS OSC • 11011: Stop DQS OSC MR05[3:2]=11: Reserved • 11100-11111: RV	
NOTE: MR05[4] indicates if the command can be used in MRUPD and MRW.				

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6.6.1.7. MR06 - Functional Modes

MR06				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR06[7:0]: FUNCTIONAL MODES DRAM direct execution functional modes • 00000000: All Functions Disabled MR06[0]: ZQCal • 00000001: ZQCAL Latch MR06[3:1]: Shadow register apply • 00000010: CK ODT Apply • 00000100: CS ODT, VREFCS Apply • 00001000: CA ODT, VREFCA Apply MR06[7:4]: Reserved • 00010000-11111111: RV	
NOTE 1: MR06[3:0] are allowed to be used at the same time. **NOTE 2:** MR06[3:0] are self-clearing.				

6.6.1.8. MR07 - ZQ

MR07				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	RV		MR07[7]: RV	
6:4	RW	0h	MR07[6:4]: ZQ_INTERVAL Period over which the background calibration interval will run 000: Less than 16 001: Less than 32 010: Less than 64 011: Less than 128 100: Less than 256 101-111: Reserved	
3	RW	0	MR07[3]: ZQ_STOP State of ZQ running in the background 0: Normal Operation 1: ZQ Stop	
2	RW	0	MR07[2]: ZQ_RESET Status of ZQ running in the background 0: Normal Operation 1: ZQ Reset	

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MR07				
Bits	**Attribute**	**Default**	**Description**	**Note**
1	RW	0	MR07[1]: ZQ_INITIATOR Indicator is the die is an initiator die for ZQ calibration 0: Not an initiator die 1: Initiator die	
0	RW	0	MR07[0]: ZQUF ZQ Update Flag, indicating if there has been a change in the calibration code since previous ZQ Latch command 0: No change 1: Code has changed	

6.6.1.9. MR08 - Latency Timing

MR08				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MR08[7:6]: RV	
5:0	RW		MR08[5:0]: CL CAS Latency 000000: 68nCK 000001: 72nCK 000010: 76nCK ... 110010: 268nCK 110011: 272nCK 110100: 276nCK 110101-111111: Reserved	

6.6.1.10. MR09 - Frequency

MR09				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	RW	0	MR09[7]: TM Test Mode	

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MR09				
Bits	**Attribute**	**Default**	**Description**	**Note**
6:5	RW	0h	MR09[6:5]: PL Parity Latency 00: 01: 10: 11:	
4:0	RW	00h	MR09[4:0]: DATA_RATE Allowable data rates for MR setting calculations 00000: 4.8Gbps 00001: 6.4Gbps 00010: 8.0Gbps 00011: 9.2Gbps 00100: 9.6Gbps 00101: 10.4Gbps 00110: 11.6Gbps 00111: 12.8Gbps 01000: 14.4Gbps 01001: 16.0Gbps 01010: 18.4Gbps	

6.6.1.11. MR0A - Precharge Timing

MR0A				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MR0A[7:6]: RV	
5:0	RW	0h	MR0A[5:0]: nWTP/nRTP Delay from Write or Read commands to a Precharge command in the same bank 000000: TBD ...	

6.6.1.12. MR0B - Write DQS Preamble/Postamble

MR0B				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR0B[7:4]: WR_PST_LENGTH Write DQS Postamble (PST) length 0000: 2.5*tCK 0001: 4.5*tCK 0010-1111: Reserved	1,2,5,6

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MR0B				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0h	MR0B[3:0]: WR_PRE_LENGTH Write DQS Preamble (PRE) length 0000: 4*tCK static, 2*tCK toggle 0001: 4*tCK static, 4*tCK toggle 0010-1111: Reserved	1,2,3,4
NOTE 1: There are two physical registers assigned to each bit of this MR parameter, designated FSP0 and FSP1. Only the registers for the set point determined by the state of the FSP-WR bit (MR00[2]) will be written with an MRW command to this MR address. **NOTE 2:** There are two physical registers assigned to each bit of this MR parameter, designated FSP0 and FSP1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR00[3]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation. **NOTE 3:** MR0B[3:0]=0001b setting is allowed for lower than TBD Gbps operation. **NOTE 4:** MR0B[3:0]=0000b setting is allowed for lower than TBD Gbps operation. **NOTE 5:** MR0B[7:4]=0001b setting is allowed for lower than TBD Gbps operation. **NOTE 6:** MR0B[7:4]=0000b setting is allowed for lower than TBD Gbps operation.				

6.6.1.13. MR0C - Read DQS Preamble

MR0C				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR0C[7:4]: RD_DQS_PST_LENGTH Read DQS Postamble (PST) Length 0000: 0.5*tCK 0001: 2.5*tCK 0010: 4.5*tCK 0011-1111: Reserved	
3:0	RW	0h	MR0C[3:0]: RD_DQS_PRE_LENGTH Read DQS Preamble (PRE) Length 0000: 4*tCK static, 0*tCK toggle 0001: 2*tCK static, 2*tCK toggle 0010: 4*tCK static, 4*tCK toggle 0011: 2*tCK static, 6*tCK toggle 0100-0111: Reserved	

6.6.1.14. MR0D - Read DQS Preamble

MR0D				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RV		MR0D[7:4]: RV	

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MR0D				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0h	MR0D[3:0]: RD_DQS_PS Read DQS Pre-Shift (PS) 0000: 0*tCK 0001: 2*tCK 0010-0011: Reserved	

6.6.1.15. MR0E - Command Ignore

MR0E				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:2	RV		MR0E[7:2]: RV	
1	RW	1	MR0E[1]: CMD_IGNORE_POLARITY Command Ignore Polarity. This bit is only supported by 2p3 DRAM devices. It is a don't care when the Command Ignore function is disabled. 0: The Ignore bit in C0:CA4 is active low. 1: The Ignore bit in C0:CA4 is active high.	
0	RW	0	MR0E[0]: CMD_IGNORE_EN Command Ignore Enable. This bit is only supported by 2p3 DRAM devices. 0: Disabled 1: Enabled	

6.6.1.16. MR0F - WICA

MR0F				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MR0F[7:6]: RV	
5:0	RW	00h	MR0F[5:0]: WICA Write Leveling internal clock alignment 000000: TBD ...	

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6.6.1.17. MR10 - Write Leveling Training

MR10				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	00h	MR10[7:0]: INTERNAL_WR_LEVELING Internal write leveling settings 00000000: TBD ...	

6.6.1.18. MR11 - Host Receiver Training Control Registers

MR11				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MR11[7:6]: RV	
5	RW	0	MR11[5]: SHIFT_REG_MODE Shift Registers disable / enable 0: Disabled 1: Enabled	
4	RV		MR11[4]: RV	
3	RW	0	MR11[3]: SEED_RESET Seed Reset 0: Disabled 1: Reset to default (auto clears after resetting)	
2:1	RV		MR11[2:1]: RV	
0	RW	0	MR11[0]: RD_TRAINING_PATTERN_SELECTION Read Training Pattern Selection 0: User defined read pattern 1: LFSR	

6.6.1.19. MR12 - Reserved

MR12				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR12[7:0]: RV	

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6.6.1.20. MR13 - IO Settings

MR13				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	RW	0	MR13[7]: RX_OCC_RESET Offset Calibration Control reset 0: Normal Operation 1: RX OCC RESET	
6	RV		MR13[6]: RV	
5:3	RW	0h	MR13[5:3]: PD_OUTPUT_DRIVER_IMPEDANCE Pull-Down Output Driver Impedance 000: TBD ...	
2:0	RW	0h	MR13[2:0]: PU_OUTPUT_DRIVER_IMPEDANCE Pull-Up Output Driver Impedance 000: TBD ...	

6.6.1.21. MR14 - Rx CTLE Control for CK, DQS

MR14				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	RW	0	MR14[7]: RX_CTLE_SUPPORT Rx CTLE support indicator 0: Not supported 1: Supported	1
6:4	RV		MR14[6:4]: RV	
3:2	RW	0h	MR14[3:2]: RX_DQS_CTLE_CNTRL Rx DQS CTLE control options 00: No equalization 01: Weak equalization 10: Middle equalization 11: Strong equalization	2,3,4
1:0	RW	0h	MR14[1:0]: RX_CK_CTLE_CNTRL Rx CK CTLE control options 00: No equalization 01: Weak equalization 10: Middle equalization 11: Strong equalization	2,3,4

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MR14				
Bits	**Attribute**	**Default**	**Description**	**Note**
NOTE: 1: Rx CTLE is an optional feature in DDR6. It may be needed for DRAMs that operate at >= TBD Mbps. MR14[7] indicates host whether Rx CTLE is supported or not. **NOTE:** 2: Rx CTLE is an optional feature on DDR6. MR14[7] indicates whether Rx CTLE is supported or not. **NOTE:** 3: Refer to the vendor data sheets for more information regarding Rx CTLE Control Settings. **NOTE:** 4: Since CTLE circuits cannot be typically by-passed, a disable option is not provided. Instead, a vendor optimized setting is given. It should be noted that the settings are not specifically linear in relationship to the vendor optimized setting, so the host can walk through all the provided options and use the setting that works best in their environment.				

6.6.1.22. MR15 - Reserved

MR15				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR15[7:0]: RV	

6.6.1.23. ~~MR3E~~ MR16 - CK ODT

~~MR3E~~ MR16				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RV		MR16[7:4]: RV	
3	R	TBD	MR16[3]: CK_ODT_DEFAULT CK ODT default 0: Configured to Default On 1: Configured to Default Off	
2:0	RW	TBD	MR16[2:0]: CK_ODT CK ODT settings 000: RTT_OFF - Default Off 001: RZQ/0.5 = 240Ω 010: RZQ/1 = 120Ω 011: RZQ/1.5 = 80Ω 100: RZQ/2 = 60Ω - Default On 101: RZQ/2.5 = 48Ω 110: RZQ/3 = 40Ω 111: RZQ/3.5 = 34Ω	

6.6.1.24. MR17 - Reserved

MR17				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR17[7:0]: RV	

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6.6.1.25. ~~MR3F~~ MR18 - RTT Park

~~MR3F~~ MR18				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MR18[7:6]: RV	
5:3	RW	TBD	MR18[5:3]: DQS_RTT_PARK DQS RTT Park settings 000: TBD ...	
2:0	RW	TBD	MR18[2:0]: DQ_RTT_PARK DQ RTT Park settings 000: TBD ...	

6.6.1.26. ~~MR40~~ MR19 - RTT Write

~~MR40~~ MR19				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MR19[7:6]: RV	
5:3	RW	TBD	MR19[5:3]: DQ_RTT_NOM_WR DQ RTT Write Non-Target setting 000: TBD ...	
2:0	RW	TBD	MR19[2:0]: DQ_RTT_WR DQ RTT Write setting 000: TBD ...	

6.6.1.27. ~~MR41~~ MR1A - ODTL WR

~~MR41~~ MR1A				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MR1A[7:6]: RV	

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~~MR41~~ MR1A				
Bits	**Attribute**	**Default**	**Description**	**Note**
5:3	RW	0h	MR1A[5:3]: ODTLOFF_WR_OFFSET Write ODT off latency offset 000: 0nCK 001: -2nCK 010: -4nCK 011: -6nCK 100: -8nCK 101: -10nCK 110: -12nCK 111: Reserved	
2:0	RW	0h	MR1A[2:0]: ODTLON_WR_OFFSET Write ODT on latency offset 000: 0nCK 001: 2nCK 010: 4nCK 011: 6nCK 100: 8nCK 101: 10nCK 110: 12nCK 111: Reserved	

6.6.1.28. ~~MR42~~ MR1B - ODTL WR NT

~~MR42~~ MR1B				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MR1B[7:6]: RV	
5:3	RW	0h	MR1B[5:3]: ODTLOFF_WR_NT_OFFSET Write non-target ODT off latency offset 000: 0nCK 001: -2nCK 010: -4nCK 011: -6nCK 100: -8nCK 101: -10nCK 110: -12nCK 111: Reserved	

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~~MR42~~ MR1B				
Bits	**Attribute**	**Default**	**Description**	**Note**
2:0	RW	0h	MR1B[2:0]: ODTLON_WR_NT_OFFSET Write non-target ODT on latency offset 000: 0nCK 001: 2nCK 010: 4nCK 011: 6nCK 100: 8nCK 101: 10nCK 110: 12nCK 111: Reserved	

6.6.1.29. MR1C - Reserved

MR1C				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR1C[7:0]: RV	

6.6.1.30. MR1D - Reserved

MR1D				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR1D[7:0]: RV	

6.6.1.31. ~~MR49~~ MR1E - DQS Int Osc

~~MR49~~ MR1E				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR1E[7:0]: DQS_INTERVAL_OSC_COUNT_LSB DQS Interval Oscillator Count - LSB 00000000: TBD ..	

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6.6.1.32. ~~MR4A~~ MR1F - DQS Int Osc

~~MR4A~~ MR1F				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR1F[7:0]: DQS_INTERVAL_OSC_COUNT_MSB DQS Interval Oscillator Count - MSB 00000000: TBD ..	

6.6.1.33. ~~MR48~~ MR20 - DQS Int Osc

~~MR48~~ MR20				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR20[7:0]: DQS_INTERVAL_OSC_RUN_TIME_SET-TING DQS Interval Oscillator run time setting 00000000: TBD ...	

6.6.1.34. MR21 - CK Int Osc

MR21				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR21[7:0]: CK_INTERVAL_OSC_COUNT_LSB CK Interval Oscillator Count - LSB 00000000: TBD ..	

6.6.1.35. MR22 - CK Int Osc

MR22				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR22[7:0]: CK_INTERVAL_OSC_COUNT_MSB CK Interval Oscillator Count - MSB 00000000: TBD ..	

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6.6.1.36. MR23 - CK Int Osc

MR23				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR23[7:0]: CK_INTERVAL_OSC_RUN_TIME_SETTING CK Interval Oscillator run time setting 00000000: TBD ...	

6.6.1.37. MR24 - Reserved

MR24				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR24[7:0]: RV	

6.6.1.38. ~~MR31~~ MR25 - Serial ID

~~MR31~~ MR25				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR25[7:0]: SERIAL_ID1	

6.6.1.39. ~~MR32~~ MR26 - Serial ID

~~MR32~~ MR26				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR26[7:0]: SERIAL_ID2	

6.6.1.40. ~~MR33~~ MR27 - Serial ID

~~MR33~~ MR27				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR27[7:0]: SERIAL_ID3	

PROPOSED

6.6.1.41. ~~MR34~~ MR28 - Serial ID

~~MR34~~ MR28				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR28[7:0]: SERIAL_ID4	

6.6.1.42. ~~MR35~~ MR29 - Serial ID

~~MR35~~ MR29				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR29[7:0]: SERIAL_ID5	

6.6.1.43. ~~MR36~~ MR2A - Serial ID

~~MR36~~ MR2A				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR2A[7:0]: SERIAL_ID6	

6.6.1.44. ~~MR37~~ MR2B - Serial ID

~~MR37~~ MR2B				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR2B[7:0]: SERIAL_ID7	

6.6.1.45. ~~MR38~~ MR2C - Serial ID

~~MR38~~ MR2C				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR2C[7:0]: SERIAL_ID8	

6.6.1.46. MR2D - Reserved

MR2D				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR2D[7:0]: RV	

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6.6.1.47. MR2E - Reserved

MR2E				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR2E[7:0]: RV	

6.6.1.48. MR2F - Reserved

MR2F				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR2F[7:0]: RV	

6.6.1.49. ~~MR1E~~ MR30 - PPR Guard Key

~~MR1E~~ MR30				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR30[7:0]: PPR_KEY_PROTECTION_CODE PPR guard key protection code. Check the PPR section for details	

6.6.1.50. MR31 - MBIST/mPPR

MR31				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:2	RV		MR31[7:2]: RV	
1	RW	0	MR31[1]: MPPR_EN mPPR disable / enable 0: Disabled 1: Enabled	
0	RW	0	MR31[0]: MBIST_EN MBIST disable / enable 0: Disabled 1: Enabled	

PROPOSED

6.6.1.51. ~~MR20~~ MR32 - PPR Control

~~MR20~~ MR32				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR32[7:5]: RV	
4	RW	0	MR32[4]: SPPR_VERIF_RESULT sPPR Verification Result 0: Not successful 1: Successful	
3	RW	0	MR32[3]: SPPR_VERIF_EN sPPR Verification Enable 0: Disabled 1: Enabled	
2:1	RW	0h	MR32[2:1]: SPPR_CNTRL sPPR control options 00: sPPR Disabled 01: sPPR Enable 10: sPPR Undo 11: sPPR Lock	
0	RW	0	MR32[0]: HPPR_EN hPPR disable / enable 0: Disabled 1: Enabled	

6.6.1.52. MR33 - Reserved

MR33				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR33[7:0]: RV	

6.6.1.53. ~~MR4D~~ MR34 - PRAC Config

~~MR4D~~ MR34				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RV		MR34[7:4]: RV	

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~~MR4D~~ MR34				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:2	RW	0h	MR34[3:2]: ADAPTIVE_PRAC Adaptive PRAC 00: Default Mitigation Threshold 01: Mitigation Level A 10: Mitigation Level B 11: Mitigation Level C	
1:0	RW	0h	MR34[1:0]: MRFMAACT MRFMaACT 00: 4 01: 2 10: 1 11: Reserved	

6.6.1.54. ~~MR4E~~ MR35 - PRAC Init & Testing

~~MR4E~~ MR35				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR35[7:5]: RV	
4	RW	0	MR35[4]: ALERT_VERIFICATION Alert Verification 0: ALERT Verification Disabled 1: ALERT Verification Trigger	
3	RW	0	MR35[3]: PRAC_TESTING_INIT PRAC Testing Initialization 0: PRAC Testing Initialization disable 1: PRAC Testing Initialization Enable	
2	RW	0	MR35[2]: PRAC_TESTING_EN PRAC Testing Enable 0: PRAC Testing disabled 1: PRAC Testing enabled	
1	R		MR35[1]: ACI_COMPLETE ACI Complete 0: Initialization not completed 1: Initialization completed	1
0	RW	0	MR35[0]: ACI_EN ACI Enable 0: Normal operating mode 1: Initialization mode	1
NOTE 1: When ACI is enabled (MR35[0]=1), MR35[1] is reset to 0 by the DRAM.				

PROPOSED

6.6.1.55. MR36 - Reserved

MR36				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR36[7:0]: RV	

6.6.1.56. MR37 - Reserved

MR37				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR37[7:0]: RV	

6.6.1.57. MR38 - Reserved

MR38				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR38[7:0]: RV	

6.6.1.58. MR39 - Reserved

MR39				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR39[7:0]: RV	

6.6.1.59. ~~MR56~~ MR3A - Temp Sensor Readout

~~MR56~~ MR3A				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	R	N/A	MR3A[7:0]: TSRO Temp Sensor Readout 00000000: Lower than low temperature 00000001: -40 00000010: -39 00000011: -38 ... 10100110: 125 10100111-11111110: Reserved 11111111: Higher than temperature sensor limit	

PROPOSED

6.6.1.60. ~~MR57~~ MR3B - ECS

~~MR57~~ MR3B				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	R	0	MR3B[7]: ECS_CE_FLAG ECS CE Flag 0: Disabled 1:Enabled	
6	RW	0	MR3B[6]: ECS_WRITEBACK_CNTRL ECS Writeback Control 0: Do not suppress writeback of Data and ECC Check Bits 1: Suppress writeback of Data and ECC Check Bits	
5:3	RW	TBD	MR3B[5:3]: ECS_ERR_THRESHOLD_COUNT ECS Error Threshold Count 000: TBD ...	
2	RW	TBD	MR3B[2]: ROW_CODEWORD_MODE Row Mode/Codeword Mode 0: ECS counts Rows with errors 1: ECS counts Code Words with errors	
1	RW	0	MR3B[1]: ECS_FLAG_RESET ECS Flag Reset 0: Disabled 1: Reset (bit is self-clearing)	
0	RW	0	MR3B[0]: ECS_RESET ECS Reset 0: Disabled 1: Reset (bit is self-clearing)	

6.6.1.61. MR3C - Reserved

MR3C				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR3C[7:0]: RV	

6.6.1.62. MR3D - Reserved

MR3D				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR3D[7:0]: RV	

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6.6.1.63. ~~MR68~~ MR3E - Global Write DCA

~~MR68~~ MR3E				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR3E[7:4]: WR_DCA_QDQS Write DCA for single/2-phase DQS or QDQS in 4-phase DQS 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:2	RV		MR3E[3:2]: RV	
1:0	RW	0h	MR3E[1:0]: WR_DCA_SUPPORT Write DCA types supported 00: DCA not supported 01: DCA supported for single/two-phase internal DQS 10: DCA supported for 4-phase internal DQS 11: Reserved	

6.6.1.64. ~~MR69~~ MR3F - Global Write DCA

~~MR69~~ MR3F				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR3F[7:4]: WR_DCA_QBDQS Write DCA for QBDQS in 4-phase DQS 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	

PROPOSED

~~MR69~~ MR3F				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0h	MR3F[3:0]: WR_DCA_IBDQS Write DCA for IBDQS in 4-phase DQS 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	

6.6.1.65. ~~MR6A~~ MR40 - Global Read DCA

~~MR6A~~ MR40				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR40[7:4]: RD_DCA_QCLK Read DCA for single/two-phase DQS clock(s) or QCLK in 4-phase clocks 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	
3:2	RV		MR40[3:2]: RV	
1:0	RW	0h	MR40[1:0]: RD_DCA_SUPPORT Read DCA type supported 00: Read DCA not supported 01: Read DCA supported for single/two-phase internal clock(s) 10: Read DCA supported for 4-phase internal clock(s) 11: Reserved	

PROPOSED

6.6.1.66. ~~MR6B~~ MR41 - Read Global DCA

~~MR6B~~ MR41				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR41[7:4]: RD_DCA_QBCLK Read DCA for QBCLK in 4-phase clocks 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	
3:0	RW	0h	MR41[3:0]: RD_DCA_IBCLK Read DCA for IBCLK in 4-phase clocks 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	

PROPOSED

6.6.1.67. ~~MR6D~~ MR42 - Read DCA Per-Pin DQS_t for QCLK

~~MR6D~~ MR42				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR42[7:4]: RD_DCA_DQS_T_QCLK Read DCA per-pin DQS_t for single/two-phase DQS clock(s) or QCLK in 4-phase clocks 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RV		MR42[3:0]: RV	

6.6.1.68. ~~MR6E~~ MR43 - Read DCA Per-Pin DQS_t for QBCLK/IBCLK

~~MR6E~~ MR43				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR43[7:4]: RD_DCA_DQS_T_QBCLK Read DCA per-pin DQS_t for QBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RW	0h	MR43[3:0]: RD_DCA_DQS_T_IBCLK Read DCA per-pin DQS_t for IBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	

PROPOSED

6.6.1.69. ~~MR6F~~ MR44 - Read DCA Per-Pin DQS_c for QCLK

~~MR6F~~ MR44				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR44[7:4]: RD_DCA_DQS_C_QCLK Read DCA per-pin DQS_c for single/two-phase DQS clock(s) or QCLK in 4-phase clocks 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RV		MR44[3:0]: RV	

6.6.1.70. ~~MR6C~~ MR45 - Read DCA Per-Pin DQS_c for QBCLK/IBCLK

~~MR6C~~ MR45				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR45[7:4]: RD_DCA_DQS_C_QBCLK Read DCA per-pin DQS_c for QBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RW	0h	MR45[3:0]: RD_DCA_DQS_C_IBCLK Read DCA per-pin DQS_c for IBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	

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6.6.1.71. MR46 - Reserved

MR46				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR46[7:0]: RV	

6.6.1.72. MR47 - DRAM Scratch Pad

MR47				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RW	0h	MR47[7:0]: DRAM_SCRATCH_PAD This DRAM Scratch Pad Mode register is used by the host controller to read back Control Words from the RCD. Control Words are the RCD equivalent of the DRAM Mode registers. The DRAM implements MR47 as a simple read/write register, writable via an MRW to address MR47 and readable via an MRR to address MR47. Any value is valid.	1
NOTE 1: The contents of this register have no function in the DRAM. Details for this function can be found in the DDR5 RCD01 specification.				

6.6.1.73. MR48-MR7F - Reserved

MR[48:7F]				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR[48:7F][7:0]: RV	

PROPOSED

6.7. PSD Mode Register Definitions

6.7.1. MR80 - PSD

MR80				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:2	RV		MR80[7:2]: RV	
1	R		MR80[1]: PSD_LOGICAL Indicator where logical ports are connected in the package 0: PSD 0 1: PSD 1	1
0	R		MR80[0]: PSD_PHYSICAL Indicator how top down physical ports are connected in the package 0: PSD Left 1: PSD Right	1
NOTE 1: PSD indicator bits are to identify logical and physical PSD locations.				

6.7.2. ~~PG[0]MR00~~ MR81 - DQ Mapping

~~PG[0]MR00~~ MR81				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MR81[7:6]: RV	
5	R		MR81[5]: DQ5	
4	R		MR81[4]: DQ4	
3	R		MR81[3]: DQ3	
2	R		MR81[2]: DQ2	
1	R		MR81[1]: DQ1	
0	R		MR81[0]: DQ0	

6.7.3. ~~PG[0]MR02~~ MR82 - PDA Select ID

~~PG[0]MR02~~ MR82				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RV		MR82[7:4]: RV	

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~~PG[0]MR02~~ MR82				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	R	Fh	MR82[3:0]: PDA_SELECT_ID The Select ID is set by MRW or MRUPDATE sequence. Default setting is all DRAMs execute MRW/MRUPD. For all other encodings, the DRAM will execute MRW/MRUP-DATE when Select ID = Enumeration ID, except to this Mode Register which ignores the PDA match and will always be executed.	

6.7.4. ~~MR19~~ MR83 - VREF CS

~~MR19~~ MR83				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	RV		MR83[7]: RV	
6:0	RW	TBD	MR83[6:0]: VREF_CS Vref setting for CS 0000000 = 10.0% 0000001 = 10.5% 0000010 = 11.0% 0000011 = 11.5% ... 1111110 = 73.0% 1111111 = 73.5%	

6.7.5. ~~MR1A~~ MR84 - VREF CA

~~MR1A~~ MR84				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	RV		MR84[7]: RV	
6:0	RW	TBD	MR84[6:0]: VREF_CA Global Vref setting for CA 0000000 = 10.0% 0000001 = 10.5% 0000010 = 11.0% 0000011 = 11.5% ... 1111110 = 73.0% 1111111 = 73.5%	

PROPOSED

6.7.6. ~~MR1B~~ MR85 - VREF DQ

~~MR1B~~ MR85				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	RV		MR85[7]: RV	
6:0	RW	TBD	MR85[6:0]: VREF_DQ Global Vref setting for DQ 0000000 = 10.0% 0000001 = 10.5% 0000010 = 11.0% 0000011 = 11.5% ... 1111110 = 73.0% 1111111 = 73.5%	

6.7.7. ~~PG[0]MR03~~ MR86 - Per Pin VREFCA Offset (CA0, CA1)

~~PG[0]MR03~~ MR86				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0	MR86[7:4]: CA1_VREFCA_OFFSET CA1 VREFCA Offset 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	

PROPOSED

~~PG[0]MR03~~ MR86				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0	MR86[3:0]: CA0_VREFCA_OFFSET CA0 VREFCA Offset 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	

6.7.8. ~~PG[0]MR04~~ MR87 - Per Pin VREFCA Offset (CA2, CA3)

~~PG[0]MR04~~ MR87				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0	MR87[7:4]: CA3_VREFCA_OFFSET CA3 VREFCA Offset 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	

PROPOSED

~~PG[0]MR04~~ MR87				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0	MR87[3:0]: CA2_VREFCA_OFFSET CA2 VREFCA Offset 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	

6.7.9. ~~PG[0]MR05~~ MR88 - Per Pin VREFCA Offset (CA4, CA5)

~~PG[0]MR05~~ MR88				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0	MR88[7:4]: CA5_VREFCA_OFFSET CA5 VREFCA Offset 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	

PROPOSED

~~PG[0]MR05~~ MR88				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0	MR88[3:0]: CA4_VREFCA_OFFSET CA4 VREFCA Offset 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	

6.7.10. ~~PG[0]MR06~~ MR89 - Per Pin VREFDQ Offset (DQ0, DQ1)

~~PG[0]MR06~~ MR89				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0	MR89[7:4]: DQ1_VREFDQ_OFFSET DQ1 VREFDQ Offset 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	

PROPOSED

~~PG[0]MR06~~ MR89				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0	MR89[3:0]: DQ0_VREFDQ_OFFSET DQ0 VREFDQ Offset 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	

6.7.11. ~~PG[0]MR07~~ MR8A - Per Pin VREFDQ Offset (DQ2, DQ3)

~~PG[0]MR07~~ MR8A				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0	MR8A[7:4]: DQ3_VREFDQ_OFFSET DQ3 VREFDQ Offset 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	

PROPOSED

~~PG[0]MR07~~ MR8A				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0	MR8A[3:0]: DQ2_VREFDQ_OFFSET DQ2 VREFDQ Offset 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	

6.7.12. ~~PG[0]MR08~~ MR8B - Per Pin VREFDQ Offset (DQ4, DQ5)

~~PG[0]MR08~~ MR8B				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0	MR8B[7:4]: DQ5_VREFDQ_OFFSET DQ5 VREFDQ Offset 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	

PROPOSED

~~PG[0]MR08~~ MR8B				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0	MR8B[3:0]: DQ4_VREFDQ_OFFSET DQ4 VREFDQ Offset 1000: Reserved 1001: Step -7 1010: Step -6 1011: Step -5 1100: Step -4 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100: Step +4 0101: Step +5 0110: Step +6 0111: Step +7	

6.7.13. MR8C - Rx CTLE Control for CA, CS

MR8C				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RV		MR8C[7:4]: RV	
3:2	RW	0h	MR8C[3:2]: RX_CS_CTLE_CNTRL Rx CS CTLE control 00: No equalization 01: Weak equalization 10: Middle equalization 11: Strong equalization	1,2,3
1:0	RW	0h	MR8C[1:0]: RX_CA_CTLE_CNTRL Rx CA CTLE control 00: No equalization 01: Weak equalization 10: Middle equalization 11: Strong equalization	1,2,3

NOTE: 1: Rx CTLE is an optional feature on DDR6. MR14[7] indicates whether Rx CTLE is supported or not.
NOTE: 2: Refer to the vendor data sheets for more information regarding Rx CTLE Control Settings.
NOTE: 3: Since CTLE circuits cannot be typically by-passed, a disable option is not provided. Instead, a vendor optimized setting is given. It should be noted that the settings are not specifically linear in relationship to the vendor optimized setting, so the host can walk through all the provided options and use the setting that works best in their environment.

PROPOSED

6.7.14. ~~PG[0]MR0A~~ MR8D - DFE Control, Enable

~~PG[0]MR0A~~ MR8D				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR8D[7:5]: RV	
4	RW	0	MR8D[4]: GLOBAL_DFE_TAP-4_EN Global DFE Tap-4 disable / enable 0: Disabled 1: Enabled	3
3	RW	0	MR8D[3]: GLOBAL_DFE_TAP-3_EN Global DFE Tap-3 disable / enable 0: Disabled 1: Enabled	3
2	RW	0	MR8D[2]: GLOBAL_DFE_TAP-2_EN Global DFE Tap-2 enable / disable 0: Enabled 1: Disabled	1,2
1	RW	0	MR8D[1]: GLOBAL_DFE_TAP-1_EN Global DFE Tap-1 enable / disable 0: Enabled 1: Disabled	1,2
0	RW	0	MR8D[0]: PER_PIN_DFE_CNTRL Per-pin DFE enable / disable 0: Enabled 1: Disabled	
NOTE 1: This bit applies to all DQ pins. **NOTE 2:** Setting MR8D[2:1]=11b disables the DFE.				

6.7.15. ~~PG[0]MR0B~~ MR8E - Per-Pin DFE Tap-1 - DQ0

~~PG[0]MR0B~~ MR8E				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR8E[7:5]: RV	

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~~PG[0]MR0B~~ MR8E				
Bits	**Attribute**	**Default**	**Description**	**Note**
4:0	RW	0h	MR8E[4:0]: DFE_TAP-1_DQ0 Per-pin DFE Tap-1 Quantity for DQ0 00000: Step 0 00001: Step 1 00010: Step 2 00011: Step 3 00100: Step 4 00101: Step 5 00110: Step 6 00111: Step 7 01000: Step 8 01001: Step 9 01010: Step 10 01011: Step 11 01100: Step 12 01101: Step 13 01110: Step 14 01111: Step 15 10000-11111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables. **NOTE 3:** If DFE Tap-1 is disabled (MR8D[1]=1), MR8E[4:0] is ignored.				

6.7.16. ~~PG[0]MR0C~~ MR8F - Per-Pin DFE Tap-2 - DQ0

~~PG[0]MR0C~~ MR8F				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR8F[7:5]: RV	

PROPOSED

~~PG[0]MR0C~~ MR8F				
Bits	**Attribute**	**Default**	**Description**	**Note**
4:0	RW	0h	MR8F[4:0]: DFE_TAP-2_DQ0 Per-pin DFE Tap-2 Quantity for DQ0 10000-11000: Reserved 11001: Step -7 11010: Step -6 11011: Step -5 11100: Step -4 11101: Step -3 11110: Step -2 11111: Step -1 00000: Step 0 00001: Step 1 00010: Step 2 00011: Step 3 00100: Step 4 00101: Step 5 00110: Step 6 00111: Step 7 01000-01111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-2 is disabled (MR8D[2]=1), MR8F[4:0] is ignored.				

6.7.17. MR90 - Per-Pin DFE Tap-3, Tap-4 - DQ0

MR90				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR90[7:4]: DFE_TAP-4_DQ0 Per-pin DFE Tap-4 Quantity for DQ0 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Step 0 0001: Step 1 0010: Step 2 0011: Step 3 0100-0111: Reserved	1,2,4

PROPOSED

MR90				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0h	MR90[3:0]: DFE_TAP-3_DQ0 Per-pin DFE Tap-3 Quantity for DQ0 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Step 0 0001: Step 1 0010: Step 2 0011: Step 3 0100-0111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-3 is disabled (MR8D[3]=0), MR90[3:0] is ignored. **NOTE 4:** If DFE Tap-4 is disabled (MR8D[4]=0), MR90[7:4] is ignored.				

6.7.18. ~~PG[0]MR0D~~ MR91 - Per-Pin DFE Tap-1 - DQ1

~~PG[0]MR0D~~ MR91				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR91[7:5]: RV	
4:0	RW	0h	MR91[4:0]: DFE_TAP-1_DQ1 Per-pin DFE Tap-1 Quantity for DQ1 00000: Step 0 00001: Step 1 00010: Step 2 00011: Step 3 00100: Step 4 00101: Step 5 00110: Step 6 00111: Step 7 01000: Step 8 01001: Step 9 01010: Step 10 01011: Step 11 01100: Step 12 01101: Step 13 01110: Step 14 01111: Step 15 10000-11111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-1 is disabled (MR8D[1]=1), MR91[4:0] is ignored.				

PROPOSED

6.7.19. ~~PG[0]MR0E~~ MR92 - Per-Pin DFE Tap-2 - DQ1

~~PG[0]MR0E~~ MR92				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR92[7:5]: RV	
4:0	RW	0h	MR92[4:0]: DFE_TAP-2_DQ1 Per-pin DFE Tap-2 Quantity for DQ1 10000-11000: Reserved 11001: Step -7 11010: Step -6 11011: Step -5 11100: Step -4 11101: Step -3 11110: Step -2 11111: Step -1 00000: Step 0 00001: Step 1 00010: Step 2 00011: Step 3 00100: Step 4 00101: Step 5 00110: Step 6 00111: Step 7 01000-01111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-2 is disabled (MR8D[2]=1), MR92[4:0] is ignored.				

6.7.20. MR93 - Per-Pin DFE Tap-3, Tap-4 - DQ1

MR93				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR93[7:4]: DFE_TAP-4_DQ1 Per-pin DFE Tap-4 Quantity for DQ1 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Step 0 0001: Step 1 0010: Step 2 0011: Step 3 0100-0111: Reserved	1,2,4

PROPOSED

MR93				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0h	MR93[3:0]: DFE_TAP-3_DQ1 Per-pin DFE Tap-3 Quantity for DQ1 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Step 0 0001: Step 1 0010: Step 2 0011: Step 3 0100-0111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-3 is disabled (MR8D[3]=0), MR93[3:0] is ignored. **NOTE 4:** If DFE Tap-4 is disabled (MR8D[4]=0), MR93[7:4] is ignored.				

6.7.21. ~~PG[0]MR0F~~ MR94 - Per-Pin DFE Tap-1 - DQ2

~~PG[0]MR0F~~ MR94				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR94[7:5]: RV	
4:0	RW	0h	MR94[4:0]: DFE_TAP-1_DQ2 Per-pin DFE Tap-1 Quantity for DQ2 00000: Step 0 00001: Step 1 00010: Step 2 00011: Step 3 00100: Step 4 00101: Step 5 00110: Step 6 00111: Step 7 01000: Step 8 01001: Step 9 01010: Step 10 01011: Step 11 01100: Step 12 01101: Step 13 01110: Step 14 01111: Step 15 10000-11111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-1 is disabled (MR8D[1]=1), MR94[4:0] is ignored.				

PROPOSED

6.7.22. ~~PG[0]MR10~~ MR95 - Per-Pin DFE Tap-2 - DQ2

~~PG[0]MR10~~ MR95				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR95[7:5]: RV	
4:0	RW	0h	MR95[4:0]: DFE_TAP-2_DQ2 Per-pin DFE Tap-2 Quantity for DQ2 10000-11000: Reserved 11001: Step -7 11010: Step -6 11011: Step -5 11100: Step -4 11101: Step -3 11110: Step -2 11111: Step -1 00000: Step 0 00001: Step 1 00010: Step 2 00011: Step 3 00100: Step 4 00101: Step 5 00110: Step 6 00111: Step 7 01000-01111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-2 is disabled (MR8D[2]=1), MR95[4:0] is ignored.				

6.7.23. MR96 - Per-Pin DFE Tap-3, Tap-4 - DQ2

MR96				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR96[7:4]: DFE_TAP-4_DQ2 Per-pin DFE Tap-4 Quantity for DQ2 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Step 0 0001: Step 1 0010: Step 2 0011: Step 3 0100-0111: Reserved	1,2,4

PROPOSED

MR96				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0h	MR96[3:0]: DFE_TAP-3_DQ2 Per-pin DFE Tap-3 Quantity for DQ2 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Step 0 0001: Step 1 0010: Step 2 0011: Step 3 0100-0111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-3 is disabled (MR8D[3]=0), MR96[3:0] is ignored. **NOTE 4:** If DFE Tap-4 is disabled (MR8D[4]=0), MR96[7:4] is ignored.				

6.7.24. ~~PG[0]MR11~~ MR97 - Per-Pin DFE Tap-1 - DQ3

~~PG[0]MR11~~ MR97				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR97[7:5]: RV	
4:0	RW	0h	MR97[4:0]: DFE_TAP-1_DQ3 Per-pin DFE Tap-1 Quantity for DQ3 00000: Step 0 00001: Step 1 00010: Step 2 00011: Step 3 00100: Step 4 00101: Step 5 00110: Step 6 00111: Step 7 01000: Step 8 01001: Step 9 01010: Step 10 01011: Step 11 01100: Step 12 01101: Step 13 01110: Step 14 01111: Step 15 10000-11111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-1 is disabled (MR8D[1]=1), MR97[4:0] is ignored.				

PROPOSED

6.7.25. ~~PG[0]MR12~~ MR98 - Per-Pin DFE Tap-2 - DQ3

~~PG[0]MR12~~ MR98				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR98[7:5]: RV	
4:0	RW	0h	MR98[4:0]: DFE_TAP-2_DQ3 Per-pin DFE Tap-2 Quantity for DQ3 10000-11000: Reserved 11001: Step -7 11010: Step -6 11011: Step -5 11100: Step -4 11101: Step -3 11110: Step -2 11111: Step -1 00000: Step 0 00001: Step 1 00010: Step 2 00011: Step 3 00100: Step 4 00101: Step 5 00110: Step 6 00111: Step 7 01000-01111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-2 is disabled (MR8D[2]=1), MR98[4:0] is ignored.				

6.7.26. MR99 - Per-Pin DFE Tap-3, Tap-4 - DQ3

MR99				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR99[7:4]: DFE_TAP-4_DQ3 Per-pin DFE Tap-4 Quantity for DQ3 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Step 0 0001: Step 1 0010: Step 2 0011: Step 3 0100-0111: Reserved	1,2,4

PROPOSED

MR99				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0h	MR99[3:0]: DFE_TAP-3_DQ3 Per-pin DFE Tap-3 Quantity for DQ3 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Step 0 0001: Step 1 0010: Step 2 0011: Step 3 0100-0111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-3 is disabled (MR8D[3]=0), MR99[3:0] is ignored. **NOTE 4:** If DFE Tap-4 is disabled (MR8D[4]=0), MR99[7:4] is ignored.				

6.7.27. ~~PG[0]MR13~~ MR9A - Per-Pin DFE Tap-1 - DQ4

~~PG[0]MR13~~ MR9A				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR9A[7:5]: RV	
4:0	RW	0h	MR9A[4:0]: DFE_TAP-1_DQ4 Per-pin DFE Tap-1 Quantity for DQ4 00000: Step 0 00001: Step 1 00010: Step 2 00011: Step 3 00100: Step 4 00101: Step 5 00110: Step 6 00111: Step 7 01000: Step 8 01001: Step 9 01010: Step 10 01011: Step 11 01100: Step 12 01101: Step 13 01110: Step 14 01111: Step 15 10000-11111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-1 is disabled (MR8D[1]=1), MR9A[4:0] is ignored.				

PROPOSED

6.7.28. ~~PG[0]MR14~~ MR9B - Per-Pin DFE Tap-2 - DQ4

~~PG[0]MR14~~ MR9B				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR9B[7:5]: RV	
4:0	RW	0h	MR9B[4:0]: DFE_TAP-2_DQ4 Per-pin DFE Tap-2 Quantity for DQ4 10000-11000: Reserved 11001: Step -7 11010: Step -6 11011: Step -5 11100: Step -4 11101: Step -3 11110: Step -2 11111: Step -1 00000: Step 0 00001: Step 1 00010: Step 2 00011: Step 3 00100: Step 4 00101: Step 5 00110: Step 6 00111: Step 7 01000-01111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-2 is disabled (MR8D[2]=1), MR9B[4:0] is ignored.				

6.7.29. MR9C - Per-Pin DFE Tap-3, Tap-4 - DQ4

MR9C				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR9C[7:4]: DFE_TAP-4_DQ4 Per-pin DFE Tap-4 Quantity for DQ4 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Step 0 0001: Step 1 0010: Step 2 0011: Step 3 0100-0111: Reserved	1,2,4

PROPOSED

MR9C				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0h	MR9C[3:0]: DFE_TAP-3_DQ4 Per-pin DFE Tap-3 Quantity for DQ4 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Step 0 0001: Step 1 0010: Step 2 0011: Step 3 0100-0111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-3 is disabled (MR8D[3]=0), MR9C[3:0] is ignored. **NOTE 4:** If DFE Tap-4 is disabled (MR8D[4]=0), MR9C[7:4] is ignored.				

6.7.30. ~~PG[0]MR15~~ MR9D - Per-Pin DFE Tap-1 - DQ5

~~PG[0]MR15~~ MR9D				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR9D[7:5]: RV	
4:0	RW	0h	MR9D[4:0]: DFE_TAP-1_DQ5 Per-pin DFE Tap-1 Quantity for DQ5 00000: Step 0 00001: Step 1 00010: Step 2 00011: Step 3 00100: Step 4 00101: Step 5 00110: Step 6 00111: Step 7 01000: Step 8 01001: Step 9 01010: Step 10 01011: Step 11 01100: Step 12 01101: Step 13 01110: Step 14 01111: Step 15 10000-11111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-1 is disabled (MR8D[1]=1), MR9D[4:0] is ignored.				

PROPOSED

6.7.31. ~~PG[0]MR16~~ MR9E - Per-Pin DFE Tap-2 - DQ5

~~PG[0]MR16~~ MR9E				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV		MR9E[7:5]: RV	
4:0	RW	0h	MR9E[4:0]: DFE_TAP-2_DQ5 Per-pin DFE Tap-2 Quantity for DQ5 10000-11000: Reserved 11001: Step -7 11010: Step -6 11011: Step -5 11100: Step -4 11101: Step -3 11110: Step -2 11111: Step -1 00000: Step 0 00001: Step 1 00010: Step 2 00011: Step 3 00100: Step 4 00101: Step 5 00110: Step 6 00111: Step 7 01000-01111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-2 is disabled (MR8D[2]=1), MR9E[4:0] is ignored.				

6.7.32. MR9F - Per-Pin DFE Tap-3, Tap-4 - DQ5

MR9F				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MR9F[7:4]: DFE_TAP-4_DQ5 Per-pin DFE Tap-4 Quantity for DQ5 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Step 0 0001: Step 1 0010: Step 2 0011: Step 3 0100-0111: Reserved	1,2,4

PROPOSED

MR9F				
Bits	**Attribute**	**Default**	**Description**	**Note**
3:0	RW	0h	MR9F[3:0]: DFE_TAP-3_DQ5 Per-pin DFE Tap-3 Quantity for DQ5 1000-1100: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Step 0 0001: Step 1 0010: Step 2 0011: Step 3 0100-0111: Reserved	1,2,3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values. **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap-3 is disabled (MR8D[3]=0), MR9F[3:0] is ignored. **NOTE 4:** If DFE Tap-4 is disabled (MR8D[4]=0), MR9F[7:4] is ignored.				

6.7.33. MRA0 - Reserved

MRA0				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRA0[7:0]: RV	

6.7.34. MRA1 - Reserved

MRA1				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRA1[7:0]: RV	

PROPOSED

6.7.35. ~~PG[0]MR1B~~ MRA2 - Read DCA Per-Pin DQ0 for CLK/QCLK

~~PG[0]MR1B~~ MRA2				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MRA2[7:4]: RD_DCA_DQ0_QCLK Read DCA per-pin DQ0 for single/2 phase CLK or 4 phase QCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RV		MRA2[3:0]: RV	

6.7.36. ~~PG[0]MR1A~~ MRA3 - Read DCA Per-Pin DQ0 for QBCLK and IBCLK

~~PG[0]MR1A~~ MRA3				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MRA3[7:4]: RD_DCA_DQ0_QBCLK Read DCA per-pin DQ0 for 4 phase QBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RW	0h	MRA3[3:0]: RD_DCA_DQ0_IBCLK Read DCA per-pin DQ0 for 4 phase IBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	

PROPOSED

6.7.37. ~~PG[0]MR1D~~ MRA4 - Read DCA Per-Pin DQ1 for CLK/QCLK

~~PG[0]MR1D~~ MRA4				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MRA4[7:4]: RD_DCA_DQ1_QCLK Read DCA per-pin DQ1 for single/2 phase CLK or 4 phase QCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RV		MRA4[3:0]: RV	

6.7.38. ~~PG[0]MR1C~~ MRA5 - Read DCA Per-Pin DQ1 for QBCLK and IBCLK

~~PG[0]MR1C~~ MRA5				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MRA5[7:4]: RD_DCA_DQ1_QBCLK Read DCA per-pin DQ1 for 4 phase QBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RW	0h	MRA5[3:0]: RD_DCA_DQ1_IBCLK Read DCA per-pin DQ1 for 4 phase IBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	

PROPOSED

6.7.39. ~~PG[0]MR1F~~ MRA6 - Read DCA Per-Pin DQ2 for CLK/QCLK

~~PG[0]MR1F~~ MRA6				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MRA6[7:4]: RD_DCA_DQ2_QCLK Read DCA per-pin DQ2 for single/2 phase CLK or 4 phase QCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RV		MRA6[3:0]: RV	

6.7.40. ~~PG[0]MR1E~~ MRA7 - Read DCA Per-Pin DQ2 for QBCLK and IBCLK

~~PG[0]MR1E~~ MRA7				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MRA7[7:4]: RD_DCA_DQ2_QBCLK Read DCA per-pin DQ2 for 4 phase QBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RW	0h	MRA7[3:0]: RD_DCA_DQ2_IBCLK Read DCA per-pin DQ2 for 4 phase IBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	

PROPOSED

6.7.41. ~~PG[0]MR21~~ MRA8 - Read DCA Per-Pin DQ3 for CLK/QCLK

~~PG[0]MR21~~ MRA8				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MRA8[7:4]: RD_DCA_DQ3_QCLK Read DCA per-pin DQ3 for single/2 phase CLK or 4 phase QCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RV		MRA8[3:0]: RV	

6.7.42. ~~PG[0]MR20~~ MRA9 - Read DCA Per-Pin DQ3 for QBCLK and IBCLK

~~PG[0]MR20~~ MRA9				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MRA9[7:4]: RD_DCA_DQ3_QBCLK Read DCA per-pin DQ3 for 4 phase QBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RW	0h	MRA9[3:0]: RD_DCA_DQ3_IBCLK Read DCA per-pin DQ3 for 4 phase IBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	

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6.7.43. ~~PG[0]MR23~~ MRAA - Read DCA Per-Pin DQ4 for CLK/QCLK

~~PG[0]MR23~~ MRAA				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MRAA[7:4]: RD_DCA_DQ4_QCLK Read DCA per-pin DQ4 for single/2 phase CLK or 4 phase QCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RV		MRAA[3:0]: RV	

6.7.44. ~~PG[0]MR22~~ MRAB - Read DCA Per-Pin DQ4 for QBCLK and IBCLK

~~PG[0]MR22~~ MRAB				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MRAB[7:4]: RD_DCA_DQ4_QBCLK Read DCA per-pin DQ4 for 4 phase QBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RW	0h	MRAB[3:0]: RD_DCA_DQ4_IBCLK Read DCA per-pin DQ4 for 4 phase IBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	

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6.7.45. ~~PG[0]MR25~~ MRAC - Read DCA Per-Pin DQ0 for CLK/QCLK

~~PG[0]MR25~~ MRAC				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MRAC[7:4]: RD_DCA_DQ5_QCLK Read DCA per-pin DQ5 for single/2 phase CLK or 4 phase QCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RV		MRAC[3:0]: RV	

6.7.46. ~~PG[0]MR24~~ MRAD - Read DCA Per-Pin DQ5 for QBCLK and IBCLK

~~PG[0]MR24~~ MRAD				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	RW	0h	MRAD[7:4]: RD_DCA_DQ5_QBCLK Read DCA per-pin DQ5 for 4 phase QBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	
3:0	RW	0h	MRAD[3:0]: RD_DCA_DQ5_IBCLK Read DCA per-pin DQ5 for 4 phase IBCLK 1000-1010: Reserved 1101: Step -3 1110: Step -2 1111: Step -1 0000: Disabled 0001: Step +1 0010: Step +2 0011: Step +3 0100-0111: Reserved	

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6.7.47. MRAE - Reserved

MRAE				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRAE[7:0]: RV	

6.7.48. MRAF - Reserved

MRAF				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRAF[7:0]: RV	

6.7.49. MRB0 - Reserved

MRB0				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRB0[7:0]: RV	

6.7.50. MRB1 - Reserved

MRB1				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRB1[7:0]: RV	

6.7.51. MRB2 - Reserved

MRB2				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRB2[7:0]: RV	

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6.7.52. MRB3 - Reserved

MRB3				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRB3[7:0]: RV	

6.7.53. MRB4 - Reserved

MRB4				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRB4[7:0]: RV	

6.7.54. MRB5 - Reserved

MRB5				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRB5[7:0]: RV	

6.7.55. MRB6 - Reserved

MRB6				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRB6[7:0]: RV	

6.7.56. MRB7 - Reserved

MRB7				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRB7[7:0]: RV	

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6.7.57. MRB8 - Reserved

MRB8				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRB8[7:0]: RV	

6.7.58. MRB9 - Reserved

MRB9				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRB9[7:0]: RV	

6.7.59. MRBA - Reserved

MRBA				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRBA[7:0]: RV	

6.7.60. MRBB - Reserved

MRBB				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRBB[7:0]: RV	

6.7.61. MRBC - Reserved

MRBC				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRBC[7:0]: RV	

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6.7.62. ~~MR3D~~ MRBD - CS, CK ODT

~~MR3D~~ MRBD				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	R	TBD	MRBD[7]: CS_ODT_DEFAULT CS ODT default 0: Configured to Default On 1: Configured to Default Off	
6:4	RW	TBD	MRBD[6:4]: CS_ODT CS ODT settings 000: RTT_OFF - Default Off 001: RZQ/0.5 = 240Ω - Default On 010: RZQ/1 = 120Ω 011: RZQ/1.5 = 80Ω 100: RZQ/2 = 60Ω 101: RZQ/2.5 = 48Ω 110: RZQ/3 = 40Ω 111: RZQ/3.5 = 34Ω	
3	R		MRBD[3]: CA_ODT_DEFAULT 0: Configured to Default On 1: Configured to Default Off	
2:0	RW	TBD	MRBD[2:0]: CA_ODT CA ODT settings 000: RTT_OFF - Default Off 001: RZQ/0.5 = 240Ω - Default On 010: RZQ/1 = 120Ω 011: RZQ/1.5 = 80Ω 100: RZQ/2 = 60Ω 101: RZQ/2.5 = 48Ω 110: RZQ/3 = 40Ω 111: RZQ/3.5 = 34Ω	

6.7.63. MRBE - Reserved

MRBE				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRBE[7:0]: RV	

PROPOSED

6.7.64. ~~MR21~~ MRBF - PPR Availability

~~MR21~~ MRBF				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RW	0h	MRBF[7:6]: BG1_BANK1[1:0]	
5:4	RW	0h	MRBF[5:4]: BG1_BANK0[1:0]	
3:2	RW	0h	MRBF[3:2]: BG0_BANK1[1:0]	
1:0	RW	0h	MRBF[1:0]: BG0_BANK0[1:0]	
00: More than one hPPR and one single sPPR resource are available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and more than one hPPR resource remains. 01: One single hPPR and sPPR shared resource is available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and one single hPPR resource remains. 10: hPPR resource is available and sPPR resource is not available. Transition from 00 or 01 after sPPR is executed. 11: hPPR and sPPR resource is not available. Transition from 01 or 10 after hPPR is executed and no hPPR resources remain.				

6.7.65. ~~MR22~~ MRC0 - PPR Availability

~~MR22~~ MRC0				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RW	0h	MRC0[7:6]: BG3_BANK1[1:0]	
5:4	RW	0h	MRC0[5:4]: BG3_BANK0[1:0]	
3:2	RW	0h	MRC0[3:2]: BG2_BANK1[1:0]	
1:0	RW	0h	MRC0[1:0]: BG2_BANK0[1:0]	
00: More than one hPPR and one single sPPR resource are available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and more than one hPPR resource remains. 01: One single hPPR and sPPR shared resource is available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and one single hPPR resource remains. 10: hPPR resource is available and sPPR resource is not available. Transition from 00 or 01 after sPPR is executed. 11: hPPR and sPPR resource is not available. Transition from 01 or 10 after hPPR is executed and no hPPR resources remain.				

6.7.66. ~~MR23~~ MRC1 - PPR Availability

~~MR23~~ MRC1				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RW	0h	MRC1[7:6]: BG5_BANK1[1:0]	
5:4	RW	0h	MRC1[5:4]: BG5_BANK0[1:0]	
3:2	RW	0h	MRC1[3:2]: BG4_BANK1[1:0]	

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~~MR23~~ MRC1				
Bits	**Attribute**	**Default**	**Description**	**Note**
1:0	RW	0h	MRC1[1:0]: BG4_BANK0[1:0]	
00: More than one hPPR and one single sPPR resource are available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and more than one hPPR resource remains. 01: One single hPPR and sPPR shared resource is available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and one single hPPR resource remains. 10: hPPR resource is available and sPPR resource is not available. Transition from 00 or 01 after sPPR is executed. 11: hPPR and sPPR resource is not available. Transition from 01 or 10 after hPPR is executed and no hPPR resources remain.				

6.7.67. ~~MR24~~ MRC2 - PPR Availability

~~MR24~~ MRC2				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RW	0h	MRC2[7:6]: BG7_BANK1[1:0]	
5:4	RW	0h	MRC2[5:4]: BG7_BANK0[1:0]	
3:2	RW	0h	MRC2[3:2]: BG6_BANK1[1:0]	
1:0	RW	0h	MRC2[1:0]: BG6_BANK0[1:0]	
00: More than one hPPR and one single sPPR resource are available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and more than one hPPR resource remains. 01: One single hPPR and sPPR shared resource is available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and one single hPPR resource remains. 10: hPPR resource is available and sPPR resource is not available. Transition from 00 or 01 after sPPR is executed. 11: hPPR and sPPR resource is not available. Transition from 01 or 10 after hPPR is executed and no hPPR resources remain.				

6.7.68. ~~MR25~~ MRC3 - PPR Availability

~~MR25~~ MRC3				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RW	0h	MRC3[7:6]: BG9_BANK1[1:0]	
5:4	RW	0h	MRC3[5:4]: BG9_BANK0[1:0]	
3:2	RW	0h	MRC3[3:2]: BG8_BANK1[1:0]	
1:0	RW	0h	MRC3[1:0]: BG8_BANK0[1:0]	
00: More than one hPPR and one single sPPR resource are available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and more than one hPPR resource remains. 01: One single hPPR and sPPR shared resource is available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and one single hPPR resource remains. 10: hPPR resource is available and sPPR resource is not available. Transition from 00 or 01 after sPPR is executed. 11: hPPR and sPPR resource is not available. Transition from 01 or 10 after hPPR is executed and no hPPR resources remain.				

PROPOSED

6.7.69. ~~MR2E~~ MRC4 - PPR Availability

~~MR2E~~ MRC4				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RW	0h	MRC4[7:6]: BG11_BANK1[1:0]	
5:4	RW	0h	MRC4[5:4]: BG11_BANK0[1:0]	
3:2	RW	0h	MRC4[3:2]: BG10_BANK1[1:0]	
1:0	RW	0h	MRC4[1:0]: BG10_BANK0[1:0]	
00: More than one hPPR and one single sPPR resource are available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and more than one hPPR resource remains. 01: One single hPPR and sPPR shared resource is available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and one single hPPR resource remains. 10: hPPR resource is available and sPPR resource is not available. Transition from 00 or 01 after sPPR is executed. 11: hPPR and sPPR resource is not available. Transition from 01 or 10 after hPPR is executed and no hPPR resources remain.				

6.7.70. ~~MR27~~ MRC5 - PPR Availability

~~MR27~~ MRC5				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RW	0h	MRC5[7:6]: BG13_BANK1[1:0]	
5:4	RW	0h	MRC5[5:4]: BG13_BANK0[1:0]	
3:2	RW	0h	MRC5[3:2]: BG12_BANK1[1:0]	
1:0	RW	0h	MRC5[1:0]: BG12_BANK0[1:0]	
00: More than one hPPR and one single sPPR resource are available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and more than one hPPR resource remains. 01: One single hPPR and sPPR shared resource is available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and one single hPPR resource remains. 10: hPPR resource is available and sPPR resource is not available. Transition from 00 or 01 after sPPR is executed. 11: hPPR and sPPR resource is not available. Transition from 01 or 10 after hPPR is executed and no hPPR resources remain.				

6.7.71. ~~MR28~~ MRC6 - PPR Availability

~~MR28~~ MRC6				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RW	0h	MRC6[7:6]: BG15_BANK1[1:0]	
5:4	RW	0h	MRC6[5:4]: BG15_BANK0[1:0]	
3:2	RW	0h	MRC6[3:2]: BG14_BANK1[1:0]	

PROPOSED

~~MR28~~ MRC6				
Bits	**Attribute**	**Default**	**Description**	**Note**
1:0	RW	0h	MRC6[1:0]: BG14_BANK0[1:0]	
00: More than one hPPR and one single sPPR resource are available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and more than one hPPR resource remains. 01: One single hPPR and sPPR shared resource is available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and one single hPPR resource remains. 10: hPPR resource is available and sPPR resource is not available. Transition from 00 or 01 after sPPR is executed. 11: hPPR and sPPR resource is not available. Transition from 01 or 10 after hPPR is executed and no hPPR resources remain.				

6.7.72. ~~MR1F~~ MRC7 - MBIST

~~MR1F~~ MRC7				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:2	RV		MRC7[7:2]: RV	
1:0	RW	0h	MRC7[1:0]: MBIST_TRANSPARENCY MBIST transparency results 00: No error detected/ MBIST has not run since INIT 01: Error detected: repair resources are available 10: Error detected: no repair resources are available 11: Unsolved operation, abort or retry	

6.7.73. ~~MR4F~~ MRC8 - PRAC Flags

~~MR4F~~ MRC8				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:3	RV		MRC8[7:3]: RV	
2	RW	0	MRC8[2]: COUNTER_ADDRESS_REPORT Counter Address Report 0: Clear 1: Failing row address available in MRC9-MRCB	1
1	RW	0	MRC8[1]: COUNTER_ERR Counter Error 0: Clear 1: Counter Error detected	1
0	R		MRC8[0]: ALERT_BACK_OFF Alert Back-Off 0: Clear 1: Source of ALERT	

PROPOSED

~~MR4F~~ MRC8				
Bits	**Attribute**	**Default**	**Description**	**Note**
NOTE 1: These flags can only be set by the DRAM. MRW of MRC8[1]=0 or MRC8[2]=0 will clear the flag. MRW of MRC8[1]=1 or MRC8[2]=1 by the host will read unchanged from what was previously in the MR (1 if set by the DRAM, otherwise 0).				

6.7.74. ~~MR50~~ MRC9 - PRAC RAS Event

~~MR50~~ MRC9				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	R		MRC9[7:0]: PRAC_RAS_EVENT_R[7:0] Activation Counter Error Row Address R[7:0]	

6.7.75. ~~MR51~~ MRCA - PRAC RAS Event

~~MR51~~ MRCA				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	R		MRCA[7:0]: PRAC_RAS_EVENT_R[15:8] Activation Counter Error Row Address R[15:8]	

6.7.76. ~~MR52~~ MRCB - PRAC RAS Event

~~MR52~~ MRCB				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	R		MRCB[7:4]: PRAC_RAS_EVENT_BG[3:0] Activation Counter Error Bank Group Address BG[3:0]	
3	R		MRCB[3]: PRAC_RAS_EVENT_BA Activation Counter Error Bank BA[0]	
2:0	R		MRCB[2:0]: PRAC_RAS_EVENT_R[18:16] Activation Counter Error Row Address R[18:16]	

6.7.77. MRCC - ECS

MRCC				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:1	RV		MRCC[7:1]:RV	

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MRCC				
Bits	**Attribute**	**Default**	**Description**	**Note**
0	R	0	MRCC[0]: ECS_CE_FLAG ECS CE Flag 0: Disabled 1:Enabled	

6.7.78. ~~MR58~~ MRCD - Max CE Row Address

~~MR58~~ MRCD				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	R		MRCD[7:0]: MAX_CE_ROW_ADDRESS_R[7:0] Max CE Row Address [7:0]	

6.7.79. ~~MR59~~ MRCE - Max CE Row Address

~~MR59~~ MRCE				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	R		MRCE[7:0]: MAX_CE_ROW_ADDRESS_R[15:8] Max CE Row Address [15:8]	

6.7.80. ~~MR5A~~ MRCF - Max CE Row Address

~~MR5A~~ MRCF				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	R		MRCF[7:4]: MAX_CE_BG[3:0] Max CE BG[3:0]	
3	R		MRCF[3]: MAX_CE_BA Max CE BA	
2:0	R		MRCF[2:0]: MAX_CE_ROW_ADDRESS_R[18:16] Max CE Row Address [18:16]	

PROPOSED

6.7.81. ~~MR5C~~ MRD0 - CA Parity

~~MR5C~~ MRD0				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MRD0[7:6]: RV	
5:0	R		MRD0[5:0]: CA_PARITY_REGISTER_CA[5:0]_C0 CA Parity Register for CA[5:0] CK cycle #0	

6.7.82. ~~MR5D~~ MRD1 - CA Parity

~~MR5D~~ MRD1				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MRD1[7:6]: RV	
5:0	R		MRD1[5:0]: CA_PARITY_REGISTER_CA[5:0]_C1 CA Parity Register for CA[5:0] CK cycle #1	

6.7.83. ~~MR5E~~ MRD2 - CA Parity

~~MR5E~~ MRD2				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MRD2[7:6]: RV	
5:0	R		MRD2[5:0]: CA_PARITY_REGISTER_CA[5:0]_C2 CA Parity Register for CA[5:0] CK cycle #2	

6.7.84. ~~MR5F~~ MRD3 - CA Parity

~~MR5F~~ MRD3				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MRD3[7:6]: RV	
5:0	R		MRD3[5:0]: CA_PARITY_REGISTER_CA[5:0]_C3 CA Parity Register for CA[5:0] CK cycle #3	

PROPOSED

6.7.85. ~~MR60~~ MRD4 - CA Parity

~~MR60~~ MRD4				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MRD4[7:6]: RV	
5:0	R		MRD4[5:0]: CA_PARITY_REGISTER_CA[5:0]_C4 CA Parity Register for CA[5:0] CK cycle #4	

6.7.86. ~~MR61~~ MRD5 - CA Parity

~~MR61~~ MRD5				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MRD5[7:6]: RV	
5:0	R		MRD5[5:0]: CA_PARITY_REGISTER_CA[5:0]_C5 CA Parity Register for CA[5:0] CK cycle #5	

6.7.87. MRD6 - CRC

MRD6				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:1	RV		MRD6[7:1]: RV	
0	RW	0	MRD6[0]: WRITE_CRC_ERR_STATUS Write CRC Error Status 0: Clear 1: Error	

6.7.88. ~~PG[0]MR4D~~ MRD7 - Pattern Inversion Control (controls 6DQ lanes)

~~PG[0]MR4D~~ MRD7				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MRD7[7:6]: RV	
5	RW	0	MRD7[5]: PATTERN_INV_DQ5 Pattern Inversion Control DQ5 0: Normal 1: Invert	

PROPOSED

~~PG[0]MR4D~~ MRD7				
Bits	**Attribute**	**Default**	**Description**	**Note**
4	RW	0	MRD7[4]: PATTERN_INV_DQ4 Pattern Inversion Control DQ4 0: Normal 1: Invert	
3	RW	0	MRD7[3]: PATTERN_INV_DQ3 Pattern Inversion Control DQ3 0: Normal 1: Invert	
2	RW	0	MRD7[2]: PATTERN_INV_DQ2 Pattern Inversion Control DQ2 0: Normal 1: Invert	
1	RW	0	MRD7[1]: PATTERN_INV_DQ1 Pattern Inversion Control DQ1 0: Normal 1: Invert	
0	RW	0	MRD7[0]: PATTERN_INV_DQ0 Pattern Inversion Control DQ0 0: Normal 1: Invert	

6.7.89. ~~PG[0]MR46~~ MRD8 - LFSR Seed Register UI7-0

~~PG[0]MR46~~ MRD8				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	RW		MRD8[7]: LFSR_SEED_PATTERN_UI7	
6	RW		MRD8[6]: LFSR_SEED_PATTERN_UI6	
5	RW		MRD8[5]: LFSR_SEED_PATTERN_UI5	
4	RW		MRD8[4]: LFSR_SEED_PATTERN_UI4	
3	RW		MRD8[3]: LFSR_SEED_PATTERN_UI3	
2	RW		MRD8[2]: LFSR_SEED_PATTERN_UI2	
1	RW		MRD8[1]: LFSR_SEED_PATTERN_UI1	
0	RW		MRD8[0]: LFSR_SEED_PATTERN_UI0	

PROPOSED

6.7.90. ~~PG[0]MR45~~ MRD9 - LFSR Seed Register UI15-8

~~PG[0]MR45~~ MRD9				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	RW		MRD9[7]: LFSR_SEED_PATTERN_UI15	
6	RW		MRD9[6]: LFSR_SEED_PATTERN_UI14	
5	RW		MRD9[5]: LFSR_SEED_PATTERN_UI13	
4	RW		MRD9[4]: LFSR_SEED_PATTERN_UI12	
3	RW		MRD9[3]: LFSR_SEED_PATTERN_UI11	
2	RW		MRD9[2]: LFSR_SEED_PATTERN_UI10	
1	RW		MRD9[1]: LFSR_SEED_PATTERN_UI9	
0	RW		MRD9[0]: LFSR_SEED_PATTERN_UI8	

6.7.91. MRDA - LFSR Seed Register UI7-0

MRDA				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	RW		MRDA[7]: LFSR_SEED_PATTERN_UI23	
6	RW		MRDA[6]: LFSR_SEED_PATTERN_UI22	
5	RW		MRDA[5]: LFSR_SEED_PATTERN_UI21	
4	RW		MRDA[4]: LFSR_SEED_PATTERN_UI20	
3	RW		MRDA[3]: LFSR_SEED_PATTERN_UI19	
2	RW		MRDA[2]: LFSR_SEED_PATTERN_UI18	
1	RW		MRDA[1]: LFSR_SEED_PATTERN_UI17	
0	RW		MRDA[0]: LFSR_SEED_PATTERN_UI16	

6.7.92. MRDB - Reserved

MRDB				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRDB[7:0]: RV	

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6.7.93. MRDC - Reserved

MRDC				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRDC[7:0]: RV	

6.7.94. MRDD - Reserved

MRDD				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MRDD[7:0]: RV	

6.7.95. MRDE - Write Training Mode Control Registers

MRDE				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV		MRDE[7:6]: RV	
5	RW	0	MRDE[5]: SHIFT_REG_MODE Shift Registers disable / enable 0: Disabled 1: Enabled	
4	RV		MRDE[4]: RV	
3	RW	0	MRDE[3]: SEED_RESET Seed Reset 0: Disabled 1: Reset to default (auto clears after resetting)	
2	RW	0	MRDE[2]: CLEAR_ERRORS Clear errors 0: Disabled 1: Clear	
1	RW	0	MRDE[1]: ERRORS_CONFIG Configuration setting for 4 phase or even/odd error reporting 0: 4-Phase 1: Even/Odd	
0	RW	0	MRDE[0]: WR_TRAINING_PATTERN_SELECTION Write Training Pattern Selection 0: User defined write pattern 1: LFSR	

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6.7.96. ~~PG[0]MR56~~ MRDF - WTM Error Registers Ph0/1 DQ2:0

~~PG[0]MR56~~ MRDF				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	RV		MRDF[7]: RV	
6	RW	0	MRDF[6]: DQ2_PH1_ERROR DQ2 Phase 1 error in UI [1,5,9,13,17,21] or DQ2 odd phase error 0: No Error 1: Error	
5	RW	0	MRDF[5]: DQ1_PH1_ERROR DQ1 Phase 1 error in UI [1,5,9,13,17,21] or DQ1 odd phase error 0: No Error 1: Error	
4	RW	0	MRDF[4]: DQ0_PH1_ERROR DQ0 Phase 1 error in UI [1,5,9,13,17,21] or DQ0 odd phase error 0: No Error 1: Error	
3	RV		MRDF[3]: RV	
2	RW	0	MRDF[2]: DQ2_PH0_ERROR DQ2 Phase 0 error in UI [0,4,8,12,16,20] or DQ2 even phase error 0: No Error 1: Error	
1	RW	0	MRDF[1]: DQ1_PH0_ERROR DQ1 Phase 0 error in UI [0,4,8,12,16,20] or DQ1 even phase error 0: No Error 1: Error	
0	RW	0	MRDF[0]: DQ0_PH0_ERROR DQ0 Phase 0 error in UI [0,4,8,12,16,20] or DQ0 even phase error 0: No Error 1: Error	

6.7.97. ~~PG[0]MR57~~ MRE0 - WTM Error Registers Ph2/3 DQ2:0

~~PG[0]MR57~~ MRE0				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	RV		MRE0[7]: RV	

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~~PG[0]MR57~~ MRE0				
Bits	**Attribute**	**Default**	**Description**	**Note**
6	RW	0	MRE0[6]: DQ2_PH3_ERROR DQ2 Phase 3 error in UI [3,7,11,15,19,23] 0: No Error 1: Error	
5	RW	0	MRE0[5]: DQ1_PH3_ERROR DQ1 Phase 3 error in UI [3,7,11,15,19,23] 0: No Error 1: Error	
4	RW	0	MRE0[4]: DQ0_PH3_ERROR DQ0 Phase 3 error in UI [3,7,11,15,19,23] 0: No Error 1: Error	
3	RV		MRE0[3]: RV	
2	RW	0	MRE0[2]: DQ2_PH2_ERROR DQ2 Phase 2 error in UI [2,6,10,14,18,22] 0: No Error 1: Error	
1	RW	0	MRE0[1]: DQ1_PH2_ERROR DQ1 Phase 2 error in UI [2,6,10,14,18,22] 0: No Error 1: Error	
0	RW	0	MRE0[0]: DQ0_PH2_ERROR DQ0 Phase 2 error in UI [2,6,10,14,18,22] 0: No Error 1: Error	

6.7.98. ~~PG[0]MR58~~ MRE1 - WTM Error Registers Ph0/1 DQ5:3

~~PG[0]MR58~~ MRE1				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	RV		MRE1[7]: RV	
6	RW	0	MRE1[6]: DQ5_PH1_ERROR DQ5 Phase 1 error in UI [1,5,9,13,17,21] or DQ5 odd phase error 0: No Error 1: Error	
5	RW	0	MRE1[5]: DQ4_PH1_ERROR DQ4 Phase 1 error in UI [1,5,9,13,17,21] or DQ4 odd phase error 0: No Error 1: Error	

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~~PG[0]MR58~~ MRE1				
Bits	**Attribute**	**Default**	**Description**	**Note**
4	RW	0	MRE1[4]: DQ3_PH1_ERROR DQ3 Phase 1 error in UI [1,5,9,13,17,21] or DQ3 odd phase error 0: No Error 1: Error	
3	RV		MRE1[3]: RV	
2	RW	0	MRE1[2]: DQ5_PH0_ERROR DQ5 Phase 0 error in UI [0,4,8,12,16,20] or DQ5 even phase error 0: No Error 1: Error	
1	RW	0	MRE1[1]: DQ4_PH0_ERROR DQ4 Phase 0 error in UI [0,4,8,12,16,20] or DQ4 even phase error 0: No Error 1: Error	
0	RW	0	MRE1[0]: DQ3_PH0_ERROR DQ3 Phase 0 error in UI [0,4,8,12,16,20] or DQ3 even phase error 0: No Error 1: Error	

6.7.99. ~~PG[0]MR59~~ MRE2 - WTM Error Registers Ph2/3 DQ5:3

~~PG[0]MR59~~ MRE2				
Bits	**Attribute**	**Default**	**Description**	**Note**
7	RV		MRE2[7]: RV	
6	RW	0	MRE2[6]: DQ5_PH3_ERROR DQ5 Phase 3 error in UI [3,7,11,15,19,23] 0: No Error 1: Error	
5	RW	0	MRE2[5]: DQ4_PH3_ERROR DQ4 Phase 3 error in UI [3,7,11,15,19,23] 0: No Error 1: Error	
4	RW	0	MRE2[4]: DQ3_PH3_ERROR DQ3 Phase 3 error in UI [3,7,11,15,19,23] 0: No Error 1: Error	
3	RV		MRE2[3]: RV	

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~~PG[0]MR59~~ MRE2				
Bits	**Attribute**	**Default**	**Description**	**Note**
2	RW	0	MRE2[2]: DQ5_PH2_ERROR DQ5 Phase 2 error in UI [2,6,10,14,18,22] 0: No Error 1: Error	
1	RW	0	MRE2[1]: DQ4_PH2_ERROR DQ4 Phase 2 error in UI [2,6,10,14,18,22] 0: No Error 1: Error	
0	RW	0	MRE2[0]: DQ3_PH2_ERROR DQ3 Phase 2 error in UI [2,6,10,14,18,22] 0: No Error 1: Error	

6.7.100. MRE3-MRFF - Reserved

MR[E3:FF]				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	RV		MR[E3:FF][7:0]: RV	

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6.8. DDR6 Scratch Pad Mode Register Information

This DRAM Scratch Pad Mode register is used by the host controller to read back Control Words from the RCD. Control Words are the RCD equivalent of the DRAM Mode registers. The DRAM implements MR47 as a simple read/write register, writable via an MRW to address MR47 and readable via an MRR to address MR47.

MR47: DRAM Scratch Pad Mode register information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DRAM Scratch Pad							

Function	Register type	Operand	Data	Notes
DRAM Scratch Pad	R/W	OP[7:0]	Any value is valid	1
NOTE 1: The contents of this register have no function in the DRAM. Details for this function can be found in the DDR5 RCD01 specification.				

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7. Command and Operation

7.1. Command and Operation

7.1.1. Command Truth Table

Operation or timing that is not specified is illegal, and after such an event and in order to guarantee proper operation, the device must be reset or power-cycled and then restarted through the specified initialization sequence before normal operation can continue.

Table 93 — Command Truth Table (Sheet 1 of 4)

COMMAND	CS	COMMAND PIN						CK_t	Notes
		CA0	CA1	CA2	CA3	CA4	CA5[12]		
DESELECT (DES)	L	V	V	V	V	V	V	C0	1, 2
	L	V	V	V	V	V	V	C1	
NO OPERATION (NOP) ~~POWER DOWN EXIT (PDX)~~	H	L	L	L	V	V	V	C0	1, 2, 6, 9
	X	L	L	L	V	PAR1	V	C1	
SELF REFRESH ENTRY (SRE)	H	L	L	L	PD~~V~~	V	V	C0	1, 2, 9
	X	L	L	H	FC~~PD~~	PAR1	V	C1	
POWER DOWN ENTRY (PDE)	H	L	L	L	V~~ODT~~	ODT~~V~~	V	C0	1, 2,~~6,~~ 9, 10
	X	L	H	L	FC~~V~~	PAR1	V	C1	
SELF REFRESH EXIT (SRX)	H	L	L	L	V	V	V	C0	1, 2, 9
	X	L	H	H	V	PAR1	V	C1	
PRECHARGE (PREpb)	H	L	H	L	BA	PSD/ Ignore	CID0	C0	1, 2, 3, 9
	X	BG0	BG1	BG2	BG3	PAR1	CID1	C1	
PRECHARGE (PREab,PREdb/~~PR E~~qb)	H	L	H	H	BA	~~PSD/~~ Ignore	CID0	C0	1, 2, 3, 4,~~7,~~ 9
	X	BG0	BG1	BG2	AB=H DB/QB=L	PAR1	CID1	C1	

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Table 93 — Command Truth Table (Continued) (Sheet 2 of 4)

COMMAND	CS	COMMAND PIN						CK_t	Notes
		CA0	CA1	CA2	CA3	CA4	CA5[12]		
ACTIVATE (ACT)	H	H	L	L	BA	PSD/ Ignore	CID0	C0	1, 2, 3, 9
	X	BG0	BG1	BG2	BG3	PAR1	CID1	C1	
	L	R14	R15	R16	R17	R18	CID2	C2	
	X	R9	R10	R11	R12	R13	V	C3	
	L	R4	R5	R6	R7	R8	V	C4	
	X	R0	R1	R2	R3	PAR2	V	C5	
WRITE (WR)	H	H	L	H	BA	PSD/ Ignore	CID0	C0	1, 2, 3, 5, 8, 9
	X	BG0	BG1	BG2	BG3	PAR1	CID1	C1	
	L/H	C2	C3	C4	C5	C6	CID2	C2	
	X	C0	C1	V	AP	PAR2	V	C3	
READ (RD)	H	H	H	L	BA	PSD/ Ignore	CID0	C0	1, 2, 3, 5, 9
	X	BG0	BG1	BG2	BG3	PAR1	CID1	C1	
	L	C2	C3	C4	C5	C6	CID2	C2	
	X	C0	C1	V	AP	PAR2	V	C3	
REFRESH (REFab,REFdb/ ~~REF~~qb)	H	H	H	H	BA	V	CID0	C0	1, 2, 3, 7, 9
	X	L	L	L	V	PAR1	CID1	C1	
	L	BG0	BG1	BG2	V	V	CID2	C2	
	X	AB=H DB/QB=L	V	V	V	PAR2	V	C3	
REFRESH MANAGE-MENT (RFMab,RFMdb /qb,RFMpb)	H	H	H	H	BA	PSD	CID0	C0	1, 2, 3, 9, 11
	X	L	L	H	V	PAR1	CID1	C1	
	L	BG0	BG1	BG2	BG3	V	CID2	C2	
	X	AB=H	DB/QB=H PB=L	V	V	PAR2	V	C3	
MODE REGISTER READ (MRR)	H	H	H	H	V	PSD~~V~~	V	C0	1, 2,3,9
	X	L	H	L	CW	PAR1	V	C1	
	L	MA4	MA5	MA6	MA7	V	V	C2	
	X	MA0	MA1	MA2	MA3	PAR2	V	C3	

Table 93 — Command Truth Table (Continued) (Sheet 3 of 4)

COMMAND	CS	COMMAND PIN						CK_t	Notes
		CA0	CA1	CA2	CA3	CA4	CA5[12]		
MODE REGISTER WRITE (MRW)	H	H	H	H	V	PSD~~V~~	V	C0	1, 2,3, 9
	X	L	H	H	CW	PAR1	V	C1	
	L	MA4	MA5	MA6	MA7	V	V	C2	
	X	MA0	MA1	MA2	MA3	V	V	C3	
	L	OP4	OP5	OP6	OP7	V	V	C4	
	X	OP0	OP1	OP2	OP3	PAR2	V	C5	
MULTI-CYCLE MODE ENABLE (MCME)	H	H	H	H	H	H	V	C0	1, 2,13
		H	H	H	H	H	V	C1	
		H	H	H	H	H	V	C2	
		H	H	H	H	H	V	C3	

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Table 93 — Command Truth Table (Continued) (Sheet 4 of 4)

COMMAND	CS	COMMAND PIN						CK_t	Notes
		CA0	CA1	CA2	CA3	CA4	CA5[12]		

NOTE 1: Commands are two (2), four (4) or six (6) clock cycles long in CA 1N mode. They are defined by the states of CS of clocks ~~pairs~~ C0~~/C1~~, C2~~/C3~~, C4~~/C5~~, and by the states of CA[54:0] of clocks C0, C1, C2, C3, C4, C5. CS is 2N pulse width and center aligned to clocks C0, C2 and C4. CS is "Don't Care" at clocks C1, C3, and C5. In CA 2N mode, the CS shall operate identically to CA 1N mode and the command lengths shall be doubled relative to 1N mode, corresponding to four (4), eight (8) or twelve (12) clock cycles long.
NOTE 2: "V" means a defined high "H" or low "L" logic level. "X" means "Don't Care", which includes "H", "L" or floated
NOTE 3: Bank Group addresses BG[3:0] and Bank address BA determine which bank is to be operated upon. Pseudo Split Die "PSD" is for 1pX~~6~~ die configs for PSD ~~die split~~ designation. PSD=L indicates the given command shall be executed at the PSD whose CAs are connected to the host. PSD=H indicates the given command shall be executed at the other PSD whose CAs are not connected to the host. "Ignore" is supported on 2p3 die configuration only for Read-9 and Partial Write operation and used when Ignore feature is enabled (MR0E[0]=1~~MRxx:OP[y]=1B~~). Refer to section 7.9~~XX~~ for more details about the Command Ignore ~~Read-9 and Partial Write operation~~. When not used, the PSD/Ignore bit is "V"
NOTE 4: If C1:CA3 = "H", the command is All Bank PRECHARGE (PREab) and must be applied to all banks with Individual BGs and BA are "V". If CA3:C1 = "L", the command is Dual Bank and Quad Bank PRECHARGE (PREdb/qb) and it is applied to the bank group pair determined by BG[2:0] and BA
NOTE 5: Auto Precharge "AP" = "H" during WRITE or READ commands indicate that the DRAM will precharge the row in the bank associated with the WRITE or READ command automatically at the completion of the respective command
NOTE 6: CS shall be compliant to power down ~~entry/~~exit timing requirements. Refer to Power Down Function and Timing Description
NOTE 7: All Bank REFRESH (REFab) command requires C3:CA[0] = "H" and Dual Bank and Quad Bank REFRESH (REFdb/qb) command requires C3:CA[0] = "L". REFab is applied to all banks with individual BGs and BA are "V". BG[2:0] and BA determine the bank group pair for Dual Bank and Quad Bank REFRESH (REFdb/qb) operation
NOTE 8: When the WRITE command (WR) is issued, CS for the target rank is asserted "H" at C0~~/C1~~ and de-asserted "L" at C2~~/C3~~. The non-target rank CS is asserted "H" during C0~~/C1~~ and C2~~/C3~~ to control the NT-ODT setting and distinguish target and non-target ranks
NOTE 9: When CA Parity Check Mode is enabled (MR01~~xx~~:~~OP~~[4~~y~~]=1~~B~~, the host generates a parity operand PAR1 based on CA[54:0] of C0 and C1 of the command, excluding the PAR1 bit. For commands longer than two cycles, the host generates a additional parity operand PAR2 based on CA[54:0] of C2 and C3 of a four cycle command or C2, C3, C4 and C5 of a six cycle command, excluding the PAR2 bit. When CA Parity Check Mode is enabled, "X" defined in the command truth table shall be replaced by "V", except for DES command, since the defined logic level (H or L) is required in the command input to compute the parity state.
NOTE 10: Setting ODT=H allows On Die Termination (ODT) to persist when the device is in Power Down Mode
NOTE 11: All Bank REFRESH MANAGEMENT (RFMab) command requires C3:CA[1:0] = "VH". Dual Bank and Quad Bank REFRESH MANAGEMENT (RFMdb/qb) command requires C3:CA[1:0] = "HL". Per Bank REFRESH MANAGEMENT (RFMpb) command requires C3:CA[1:0] = "LL". RFMab is applied to all banks with individual BGs and BA are "V". RFMdb is applied to the bank group pair determined by BG[2:0] and BA with BG3 = "V". RFMpb is applied to a single bank as specified by PSD (applicable only for 1pX die configs), BG[3:0] and BA
NOTE 12: CA5 is optionally supported by discretion of DRAM supplier to enable extra CA pin. CA5 transfers CID address inputs to the DRAM for the purpose of capacity expansion. MRxx[y]=1 indicates that CA5 is enabled by the DRAM and shall be properly driven by the host for CID address inputs. When CA Parity Check Mode is enabled (MRxx:OP[y]=1B, the host generates a parity operand PAR1 based on CA[5:0] of C0 and C1 of the command, excluding the PAR1 bit. For commands longer than two cycles, the host generates a additional parity operand PAR2 based on CA[5:0] of C2 and C3 of a four cycle command or C2, C3, C4 and C5 of a six cycle command, excluding the PAR2 bit. When CA Parity Check Mode is enabled, "X" defined in the command truth table shall be replaced by "V", except for DES command, since the defined logic level (H or L) is required in the command input to compute the parity state.
NOTE 13: The MCME command works with additional setup and hold time requirements beyond multi-cycle CS assertion for any time the status of CS, CA and CK alignment may be unknown. It is shown as a 4-cycle command in command truth table but the CA inputs must remain in the proper command state for tMC_Setup + tMC_CSPW

7.1.2. DDR6 CA 2N Mode

2N Mode shall allow the system to provide increased setup and hold margin on the Command/Address (CA) bus. 1N Mode shall be the default operating mode for DDR6 SDRAM devices. MCME command is used to enter into multi-cycle CS MRUPDATE sequence. MRUPDATE sequence shall be used to change mode register MRxx:OP[x] to set 1N vs 2N mode of operation.

In 1N Mode, the DDR6 CA interface shall support command lengths of 6 cycles, 4 cycles, and 2 cycles. In 2N Mode, the duration of each supported command shall be doubled relative to 1N Mode, corresponding to 12 cycles, 8 cycles, and 4 cycles, respectively.

The Chip Select (CS) function shall operate identically in both 1N and 2N modes. The CS# pulse width shall be 2nCK in both modes.

7.1.2.1. Mode Registers

Function	Register Type	Operand	Data	Notes
CA 2N Mode	R/W	OP[1]	0: 1N Mode (Default) 1: 2N Mode	

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7.2. Command Input and Clock Sync

7.2.1. Command Input

Implementation of an internal 4-phase clock requires synchronization (sync) to enable the Host to start commands aligned to an even rising edge of the differential CK_t/CK_c external clock (CK). Internally, the four clock phases, 0°, 90°, 180° and 270°, are represented by Q0, Q1, Q2 and Q3, respectively, where each internal phase period equals four external CK cycles. Commands are only registered on Q0 or Q2, with CS being 2N centered on Q0 or Q2, and the corresponding command/address (CA) bits registered per CK as needed based on the command duration.

The Host is responsible for labeling these quadruples during the CK toggle initiation period of CSTM or CK Sync Operation, prior to issuing commands (exceptions noted in the following section).

4-cycle commands associated with data input or output (i.e., WRITE, READ, and MRR) can only start on Q0.

The MCME command can be started on any internal clock phase.

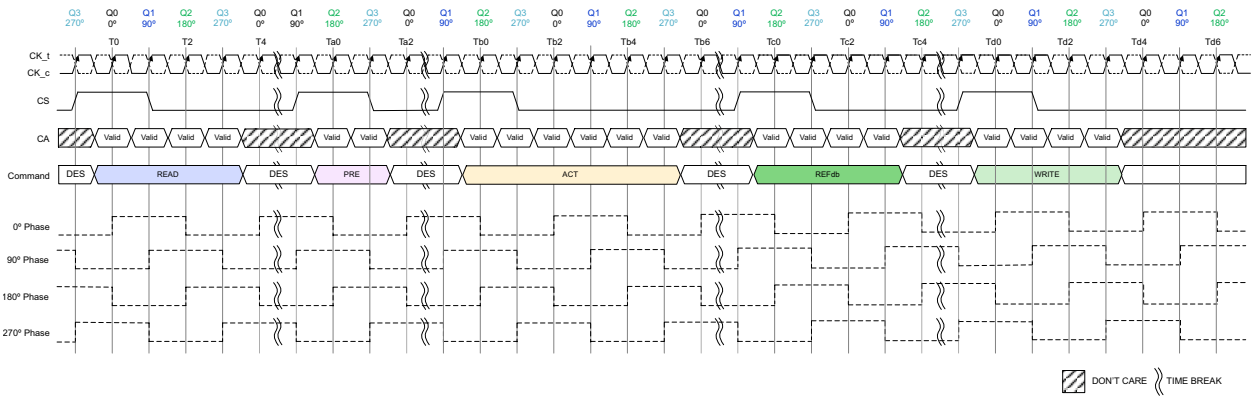


Figure 82 — Command Input

7.2.2. Clock (CK) Operation

During operation, the internal clock phase must be synchronized to the Host clock phase to properly decode commands (this is the "normal" or "CK Sync" state). When the CK Sync state is off, DES and the MCME command are allowed, the MRUPDATE Mode sequence may be executed, and the CK Sync Operation may be performed.

7.2.3. CK Sync Off Operation

The CK Sync state is set to "off' by following conditions:

- Full Power-Down entry, with or without Frequency Change
- CK Sync Off Functional Mode
- Parity Safe-State

Following an MCME command, the CK Sync Off Functional Mode is executed while in MRUPDATE Mode with MR05[7:0]=04h. After the tCKSync_Off delay is met, the CK Sync State will be “Sync Off”. The off state does not

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impact MRUPDATE Mode. CK Sync Off function is a self-clearing function once the CK Sync Off operation has been performed, allowing other MR functional modes to be executed.

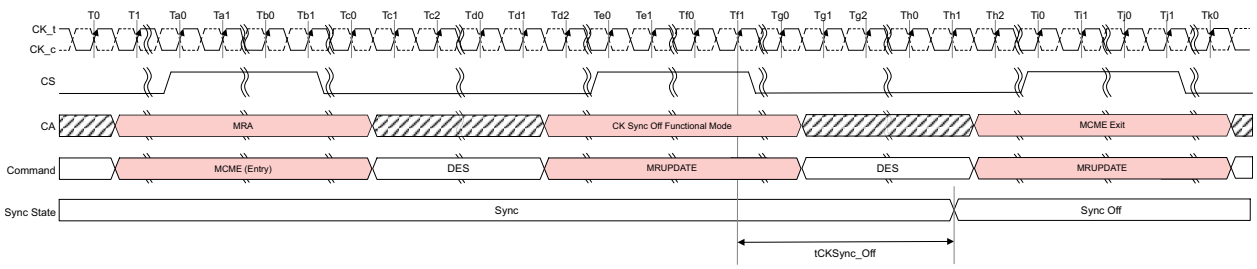


Figure 83 — CK Sync Off Function

The CK Sync Off Functional Mode is executed from one PSD, but the Sync Off affects both PSDs. Once in Sync Off, either PSD can be used to perform the CK Sync Operation, however the "other" PSD must remain unused until tCKSync is satisfied.

7.2.4. CK Sync Operation

Any time the internal clock phase alignment with the Host clock phase is unknown, a clock synchronization operation is required before issuing commands. CSTM can be used to perform this alignment, however a shorter CK Sync Operation is also provided when only the phase alignment is needed, including following MRUPDATE Mode exit when a CK Sync Off was issued.

The CK Sync Operation procedure is as follows:

- If needed, enter MCME on either PSD. Issue CK Sync Off Functional Mode, MR05[7:0]=04h, and wait tCKSync_Off. Exit MUPDATE Mode.
- Issue CA[5:0]=00h with CS low for tCKSync_CAPre to meet the CA bus stable at low time.
- Issue CA[5:0]=00h with CS high for tCKSync_CSPre to ensure CS is latched to release clock gating for the CA buffers.
- Keep CS high for tCKSync_CSPost while issuing a single (1) cycle with CA[5:0]=18h, followed by CA[5:0]=00h. The cycle of CA[5:0]=18h must align with desired Host 0° CK phase (Q0). When CA4 or CA3 is captured as ‘H’, along with CA5 and CA[2:0] as "L", the internal clock is set to 0° to align with Host clock phase.

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- Issue CA[5:0]=00h with CS low for tCKSync_CAPost. After tCKSync is met, the CK Sync operation is done and normal operation is resumed on bth PSDs.

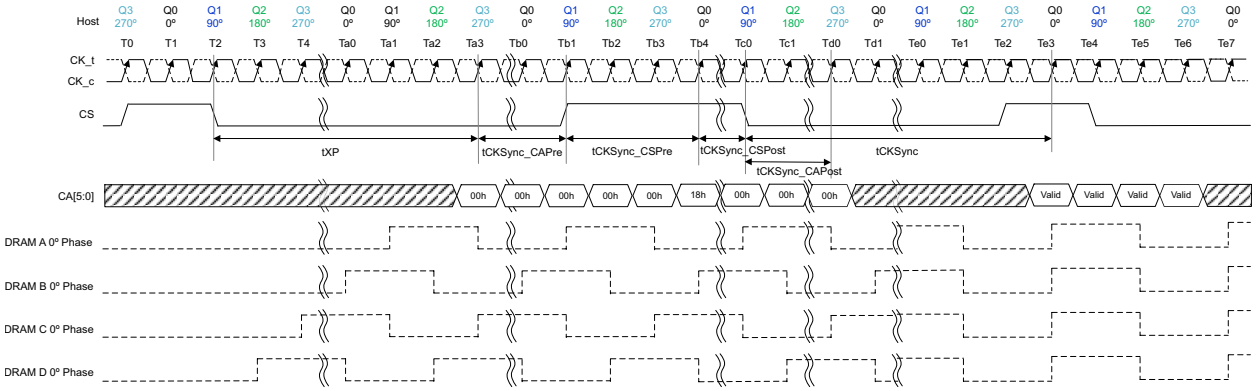


Figure 84 — Clock Sync Timing Operation

Table 94 — CK Sync Timing Parameters

Parameter	Symbol	Min/Max	Value	Unit	Note
CK Sync Off	tCKSync_Off	Min	Max(4ns,8nCK)		
CK Sync Period	tCKSync	Min	4+EVEN(4ns/tCK)	nCK	1
CA Precondition	tCKSync_CAPre	Min	8	nCK	
CA Post-condition	tCKSync_CAPost	Min	4	nCK	
CS Setup	tCKSync_CSPre	Min	3	nCK	
CS Hold	tCKSync_CSPost	Min	3	nCK	
NOTE 1: The EVEN function rounds up to the next even integer. For example, EVEN(4ns/156.25ps) = EVEN(25.6) = 28.					

After the CK Sync Operation, the internal clock phase is align to the Host clock phase, and the command decoder is ready for any valid command.

Since CA Parity is not used during the CK Sync Operation, and if there are any concerns about Host phase tracking, an example method to detect internal clocking phase misalignment utilizes the CA parity check function is provided. In this example, the Host can issue 3 consecutive NOP commands to detect if there is any phase misalignment when CA parity is enabled. An example is shown in the figure below, where the first (1st) NOP has CA3 “L” and CA4 “H” for C0 and C1, the second (2nd) NOP has both CA3 and CA4 “L” for C0 and C1, and the third (3rd) NOP has CA3 “H” and CA4 “L” for C0 and C1. If the internal phase is 90° different from the Host, it will cause a CA parity fail (CA parity check must be enabled).

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If the CK Sync operation wasn't successful, the Host needs to execute the CK Sync Operation again.

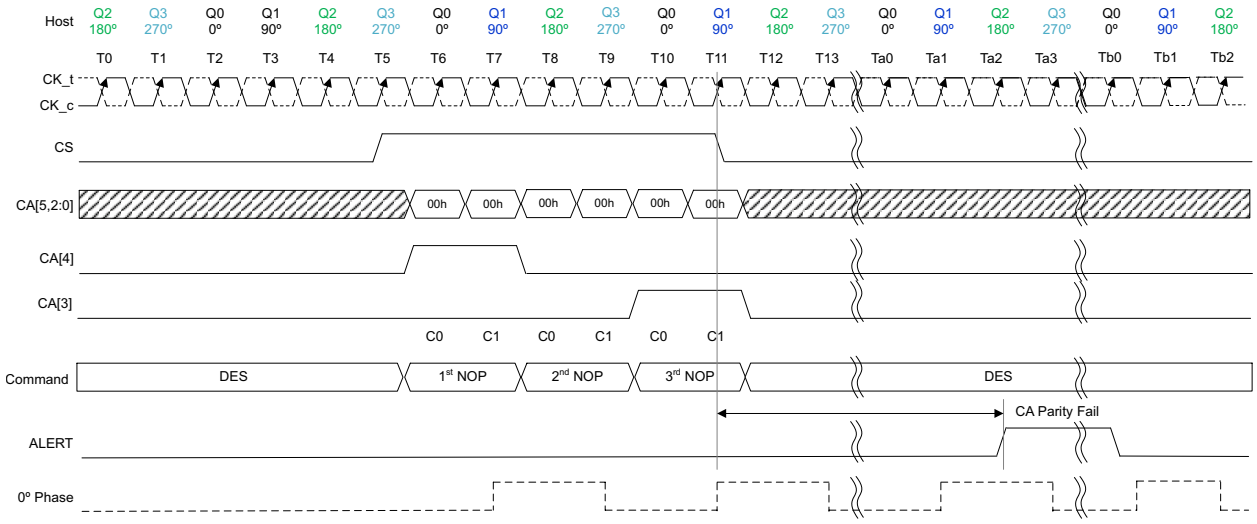


Figure 85 — Phase Misalignment with Parity Check Example

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7.3. Row Operation

7.3.1. ACTIVATE Command

The ACTIVATE command is used to open (or activate) a row in a particular bank for a subsequent access. The value on the BG[3:0] selects the bankgroup; BA inputs selects the bank within the bankgroup, and the address provided on the appropriate CA pins for R[18:0] to select the row (see Command Truth Table). This row remains active (or open) until a precharge command is issued to that bank or a precharge all command is issued. A bank must be precharged before opening a different row in the same bank.

The ACTIVATE command is required to be applied before any READ or WRITE operation can be executed. The SDRAM can accept a READ command at tRCD_R after the ACTIVATE command is issued, and a WRITE command at tRCD_W (or tRCD_W2 for non read-modify-write [1p6, 2p6, 1p12]) after the ACTIVATE command is issued.

The bank active and precharge times are defined as tRAS and tRP respectively. The minimum time interval between ACTIVATE commands to the same bank is determined by the RAS cycle time of the SDRAM (tRC).

Bank-to-bank command timing for ACTIVATE commands uses two different timing parameters, depending on whether the banks are in the same or different bank group. tRRD_S (short) is used for timing between banks located in different bank groups. tRRD_L (long) is used for timing between banks located in the same bank group. Consecutive ACTIVATE commands, allowed to be issued at tRRD_S, are restricted to a maximum of four within the time period tFAW (four activate window).

NOTE: – The following diagrams are just examples and the intention is to show the CK clock to CA is center-aligned. Expectation is the DRAM will have a delay from CK input to CA sampling that will need to be aligned to the center of the CA bit.

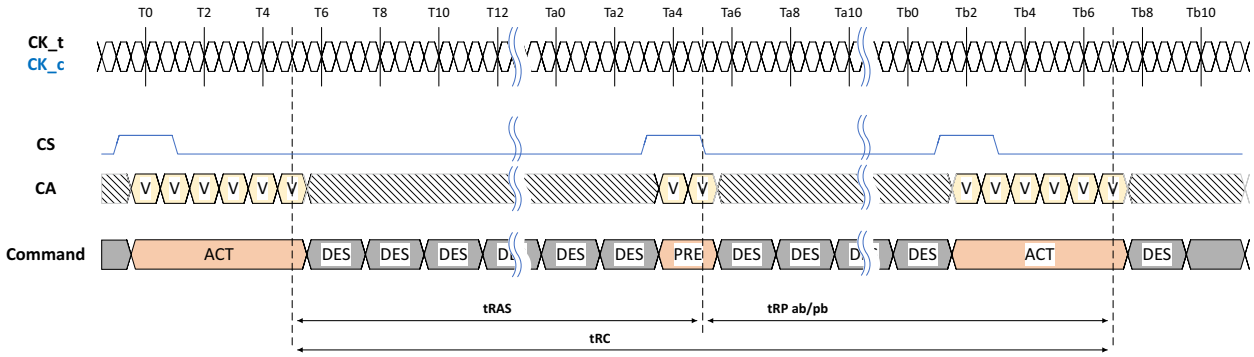


Figure 86 — tRAS, tRP and tRC Timing

PROPOSED

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.

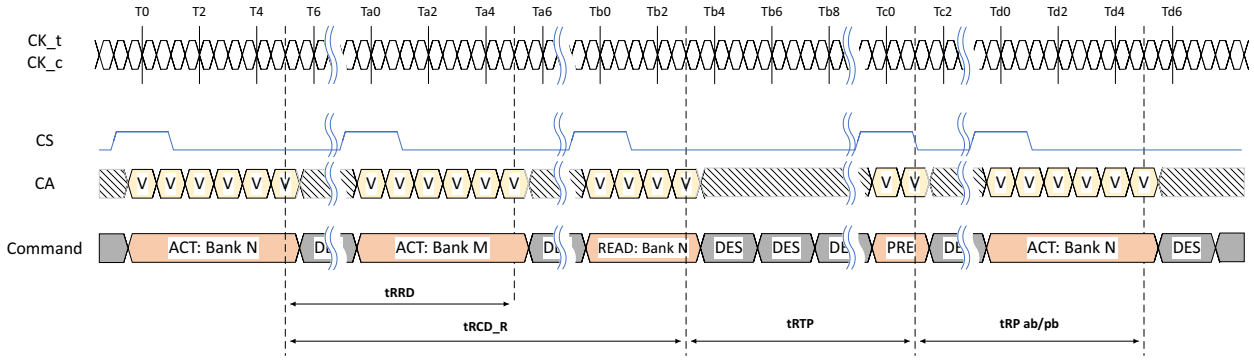


Figure 87 — Activate Command Followed By a Read Command

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.

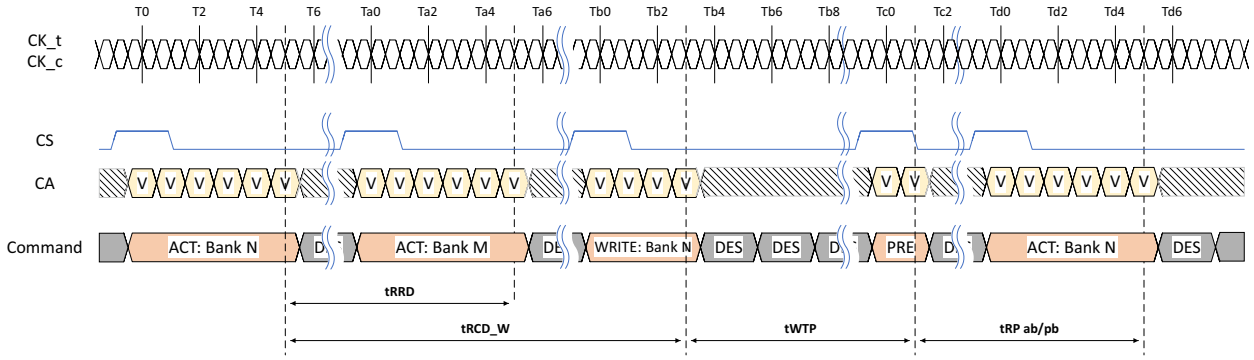


Figure 88 — Activate Command Followed By a Write Command

PROPOSED

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.

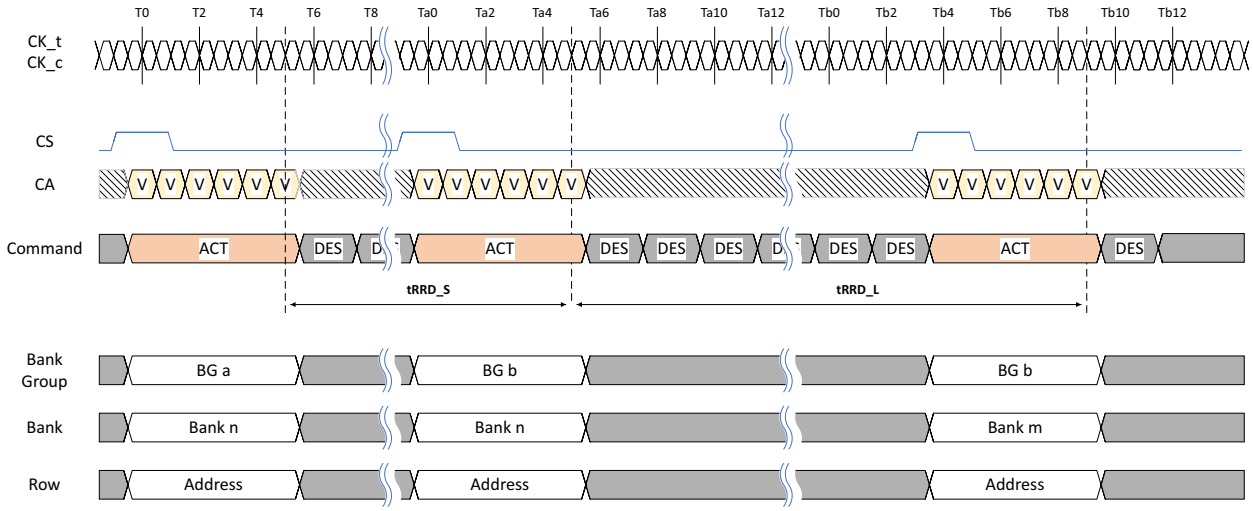


Figure 89 — Activate to Activate Command Period: tRRD_x

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.
NOTE: 2: tRRD_S: ACTIVATE to ACTIVATE Command period: Applies to consecutive ACTIVATE Commands to different Bank Groups
NOTE: 3: tRRD_L: ACTIVATE to ACTIVATE Command period: Applies to consecutive ACTIVATE Commands to different Banks of the same Bank Group.

7.3.1.1. Sequential Bank Activation Restriction

Sequential ACTIVATE commands are restricted to a maximum of four (4) within a rolling window determined by the tFAW timing parameter.

A REFdb command counts toward the four activate count within the tFAW window. ACT-ACT-ACT-ACT or ACT-ACT-ACT-REFdb is allowed within tFAW, but ACT-ACT-ACT-ACT-REFdb is not allowed.

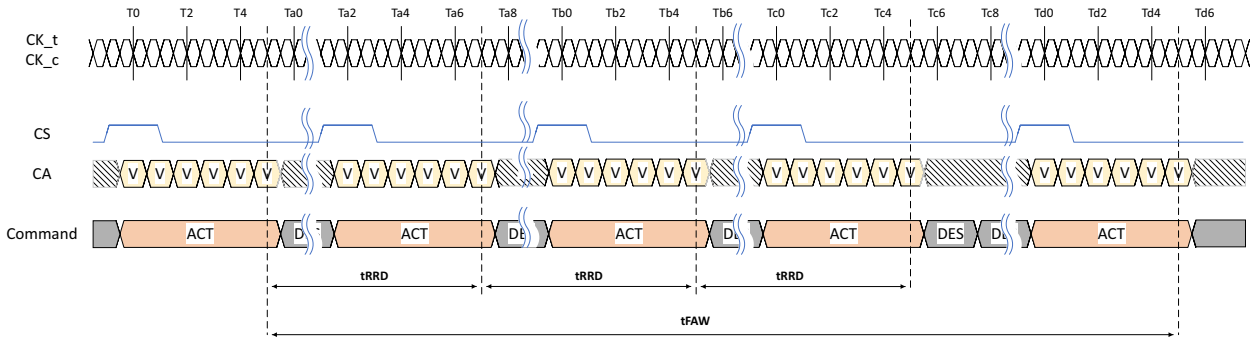


Figure 90 — tFAW Timing

PROPOSED

7.3.2. Precharge Operation

7.3.2.1. PRECHARGE Command

The PRECHARGE command is used to deactivate the open row in a particular bank or the open row in all banks. The bank(s) shall be available for subsequent row activation a specified time (tRP) after the PRECHARGE command is issued. Once a bank has been precharged, it is in the idle state and must be activated prior to any READ or WRITE commands being issued to that bank. Issuing a PRECHARGE command to a bank in idle state is allowed—the classic case is some bank being in idle state when an All-Bank PRECHARGE command is issued. tRP period needs to be satisfied before issuing next commands such as Activate, Self-refresh Entry, and Refresh, even if the target bank of the PRECHARGE command is already in the idle state. The tRP period shall be determined by the last PRECHARGE command issued to the bank.

The precharge-to-precharge delay is defined by tPPD in the core timing tables. tPPD applies to any combination of PRECHARGE commands (PREpb, PREdb, PREqb, PREab). To ensure that DDR6 devices can meet the instantaneous current demands, the row-Precharge time for an all-bank Precharge (tRPab) is longer than the per-bank Precharge time (tRPpb).

7.3.2.2. PRECHARGE Command Modes

DDR6 will support three different PRECHARGE commands: Per-Bank PRECHARGE (PREpb), Dual-Bank(/Quad-Bank) PRECHARGE (PREdb/PREqb) and All-Bank PRECHARGE (PREab). See the Command Truth Table for Encodings.

The Per-Bank PRECHARGE command (PREpb) applies precharge to a specific bank defined by BA in a specific bank group defined by BG[3:0], while an All-Bank PRECHARGE applies precharge to all banks in all bank groups.

The Dual-Bank PRECHARGE command (PREdb) applies precharge to specific bank pairs, as defined in Table 1. The Quad-Bank PRECHARGE command (PREqb) is an extension of PREdb, applying precharge to two sets of bank pairs simultaneously. PREqb uses the same bank pair mappings as the REFdb command in Table 1, but it operates across both PSDs at once. This command is unique to 1p6 devices and replaces PREdb. Note: PREdb and PREqb follow the same bank pair mappings as the REFRESH commands.

Table 95 — Precharge Bank Selection 2 Bank/ 16 Bank Group (Sheet 1 of 2)

Command	BG2	BG1	BG0	BA	PreCharge Bank Pair
PREdb/(PREqb)	0	0	0	0	0/16
PREdb/(PREqb)	0	0	0	1	1/17
PREdb/(PREqb)	0	0	1	0	2/18
PREdb/(PREqb)	0	0	1	1	3/19
PREdb/(PREqb)	0	1	0	0	4/20
PREdb/(PREqb)	0	1	0	1	5/21
PREdb/(PREqb)	0	1	1	0	6/22
PREdb/(PREqb)	0	1	1	1	7/23
PREdb/(PREqb)	1	0	0	0	8/24
PREdb/(PREqb)	1	0	0	1	9/25

PROPOSED

Table 95 — Precharge Bank Selection 2 Bank/ 16 Bank Group (Continued) (Sheet 2

Command	BG2	BG1	BG0	BA	PreCharge Bank Pair
PREdb/(PREqb)	1	0	1	0	10/26
PREdb/(PREqb)	1	0	1	1	11/27
PREdb/(PREqb)	1	1	0	0	12/28
PREdb/(PREqb)	1	1	0	1	13/29
PREdb/(PREqb)	1	1	1	0	14/30
PREdb/(PREqb)	1	1	1	1	15/31

A PRECHARGE command following a READ command to the same bank can be issued after tRTP has been satisfied. However, PRECHARGE to the same bank cannot be issued until after tRAS is also satisfied. A new bank ACTIVATE command can be issued to the same bank after the row precharge time (tRP) has elapsed, see the below figure.

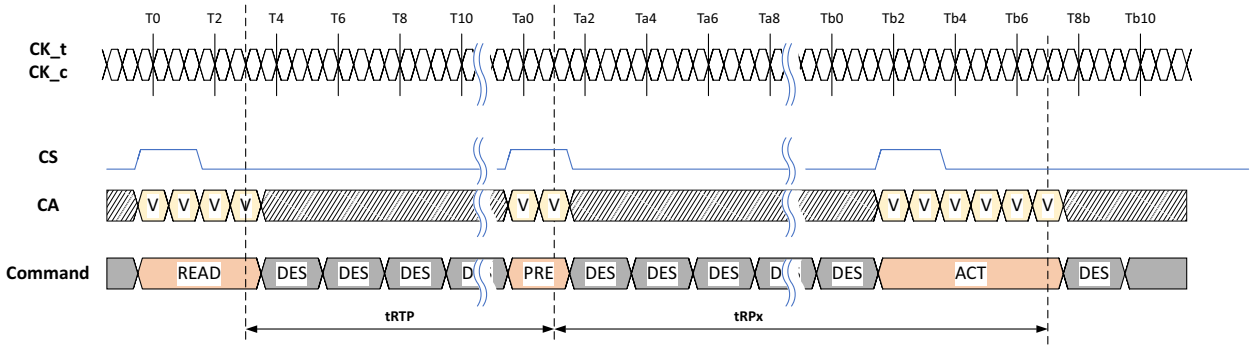


Figure 91 — READ followed by PRECHARGE

NOTE: DES commands are shown for ease of illustration; other commands may be valid at these times.

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DDR6 SDRAM devices write data to the memory array in prefetch multiples. The minimum Write-to-Precharge time for commands to the same bank is tWTP clock cycles.

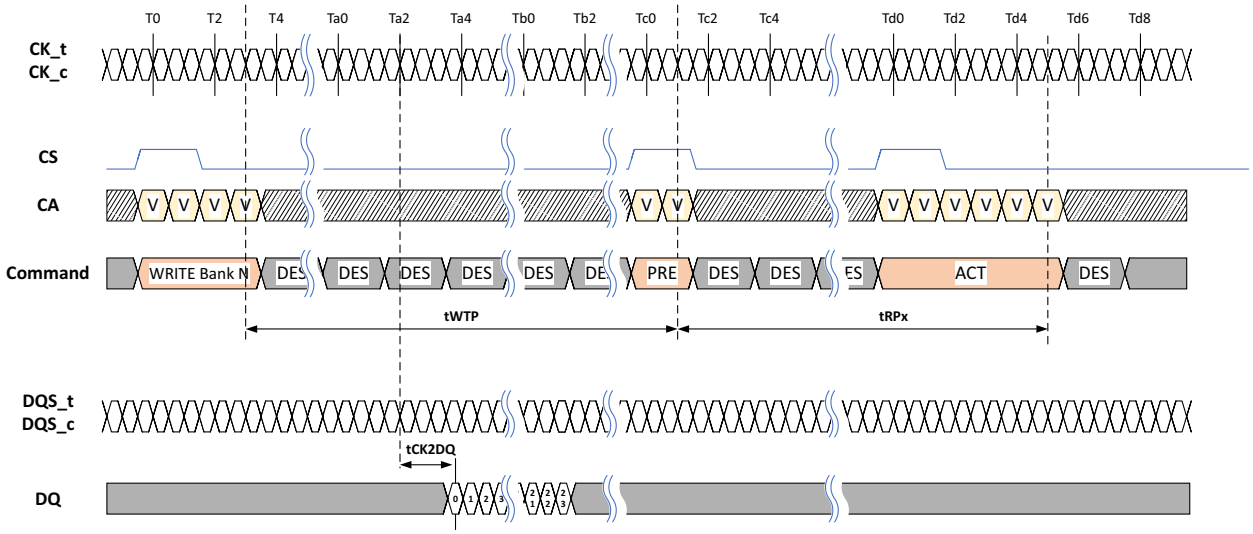


Figure 92 — WRITE Followed by PRECHARGE

NOTE: DES commands are shown for ease of illustration; other commands may be valid at these times.

PROPOSED

7.3.2.3. Auto-Precharge Operation

When a READ or WRITE command is issued to the DDR6 SDRAM, the AP bit (CA3/C3) can be set to enable the activated bank to automatically begin the precharge operation at the earliest possible moment after the read or write operation.

If AP is LOW when the READ or WRITE command is issued, then the normal read or write operation is executed and the bank remains activated at the completion of the operation.

If AP is HIGH when the READ or WRITE command is issued, the Auto-Precharge function is engaged. This feature enables the precharge operation to be partially or completely hidden when performing a read or write operation (dependent upon read or write latency), thus improving system performance for random data access. READ with Auto-Precharge or WRITE with Auto-Precharge commands may be issued after tRCD_R or tRCD_W, respectively, has been satisfied. The DRAM bases the internal tRAS on tWTP or tRTP in MRxx to schedule the internal Precharge. tRC needs to be satisfied prior to issuing subsequent ACTIVATE commands to the same bank. The following figure shows an example of Meeting internal tRAS timings.

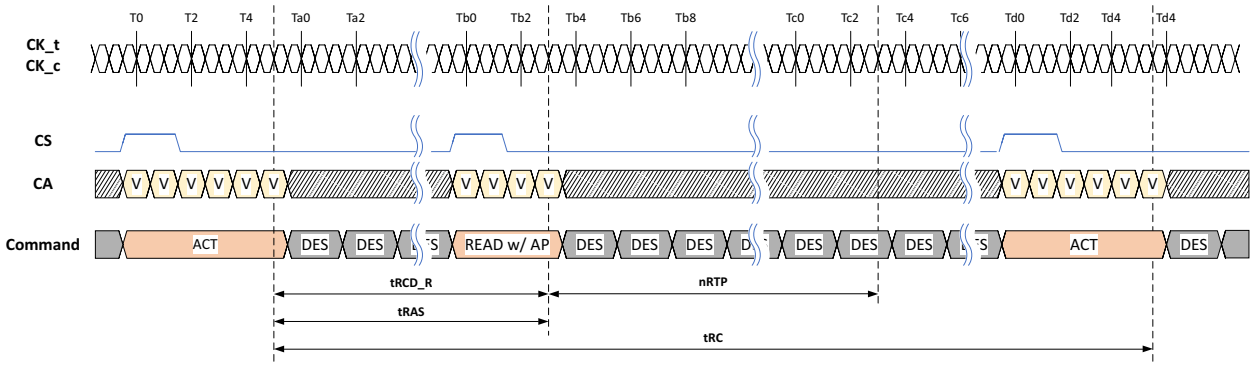


Figure 93 — Example of Meeting Internal tRAS Timing Constraints

NOTE: DES commands are shown for ease of illustration; other commands may be valid at these times.

Delay time from WRITE to READ with Auto-Precharge

In the case of a WRITE command followed by READ with Auto-Precharge, the delay time from WRITE or READ command to ACTIVATE command to the same bank is required to satisfy timings shown in following two figures. Note that the diagrams in the Precharge Operation sections are just examples and the intention is to show CK aligned

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to the center of the CA bits. The expectation is that the DRAM will have a delay from CK input to CA sampling that will need to be aligned to the center of the CA bit.

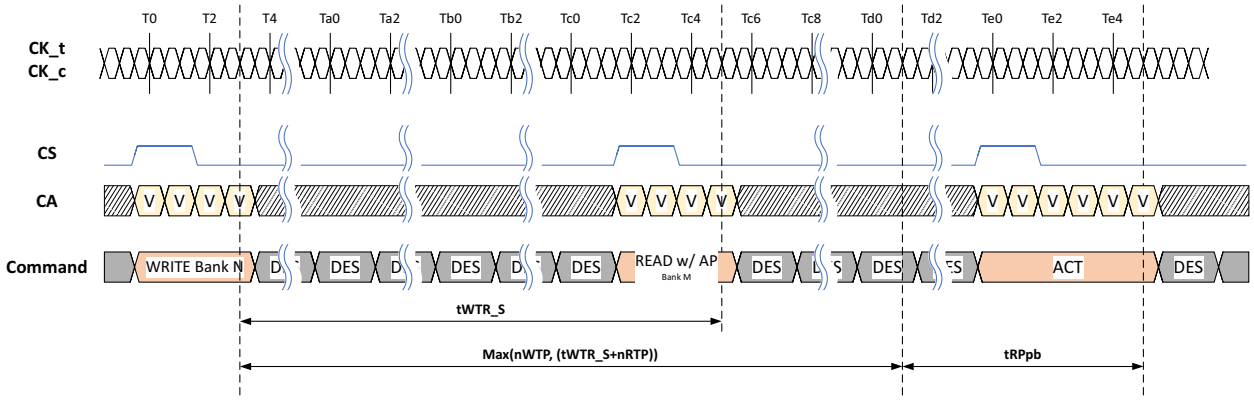


Figure 94 — Delay time from WRITE to Different Bank READ with Auto-Precharge

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.
NOTE: 2: tWTR_S starts at the Seventh rising edge of CK_t after WL + BL/2 from Write command.

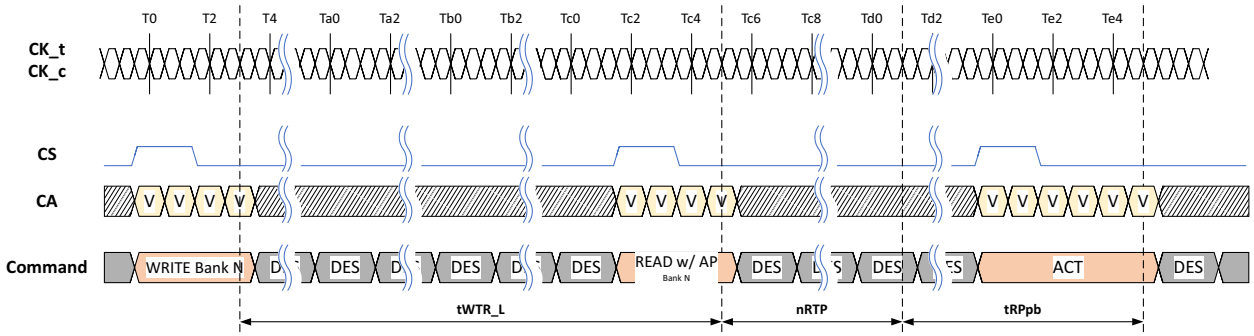


Figure 95 — Delay time from WRITE to Same Bank READ with Auto-Precharge

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.
NOTE: 2: tWTR_L starts at the Seventh rising edge of CK_t after WL + BL/2 from Write command.

READ with Auto-Precharge

If AP is HIGH when a READ command is issued, the READ with Auto-Precharge function is engaged. After the read operation, an internal precharge operation starts with a time delay.

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After a READ with Auto-Precharge command, an ACTIVATE command can be issued to the same bank if the following two conditions are both satisfied:

1) The RAS Precharge time (tRP) has been satisfied from the clock at which the internal precharge operation began, which is estimated as nRTP after the READ with Auto-Precharge command.
2) The RAS cycle time (tRC) from the previous bank activation has been satisfied.

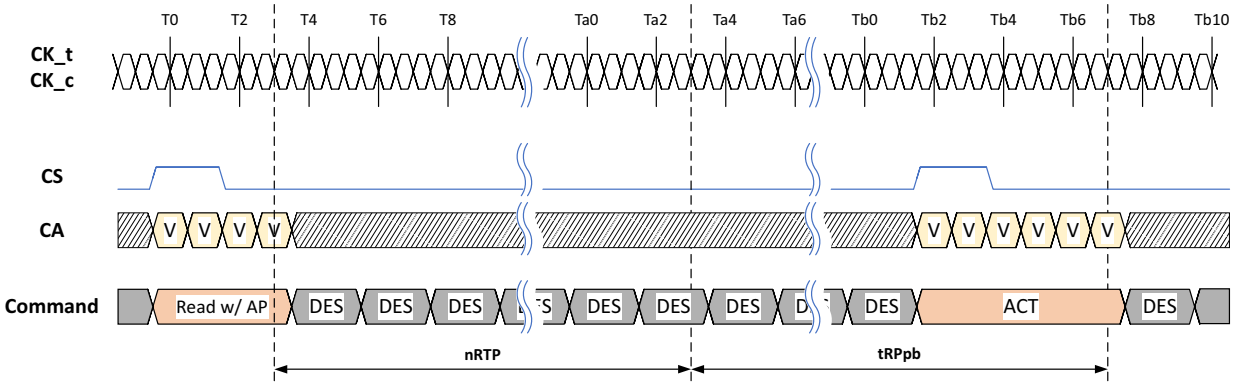


Figure 96 — READ with Auto-Precharge

NOTE: DES commands are shown for ease of illustration; other commands may be valid at these times.

WRITE with Auto-Precharge

If AP is HIGH when a WRITE command is issued, the WRITE with Auto-Precharge function is engaged. The device starts an Auto-Precharge on the rising edge nWTP cycles after the completion of the write operation.

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WRITE with Auto-Precharge, an ACTIVATE command can be issued to the same bank if the following conditions are met:

1) The RAS Precharge time (tRP) has been satisfied from the clock at which the internal precharge operation began, which is estimated as nWTP + after the WRITE with Auto-Precharge command.
2) The RAS cycle time (tRC) from the previous bank activation has been satisfied.

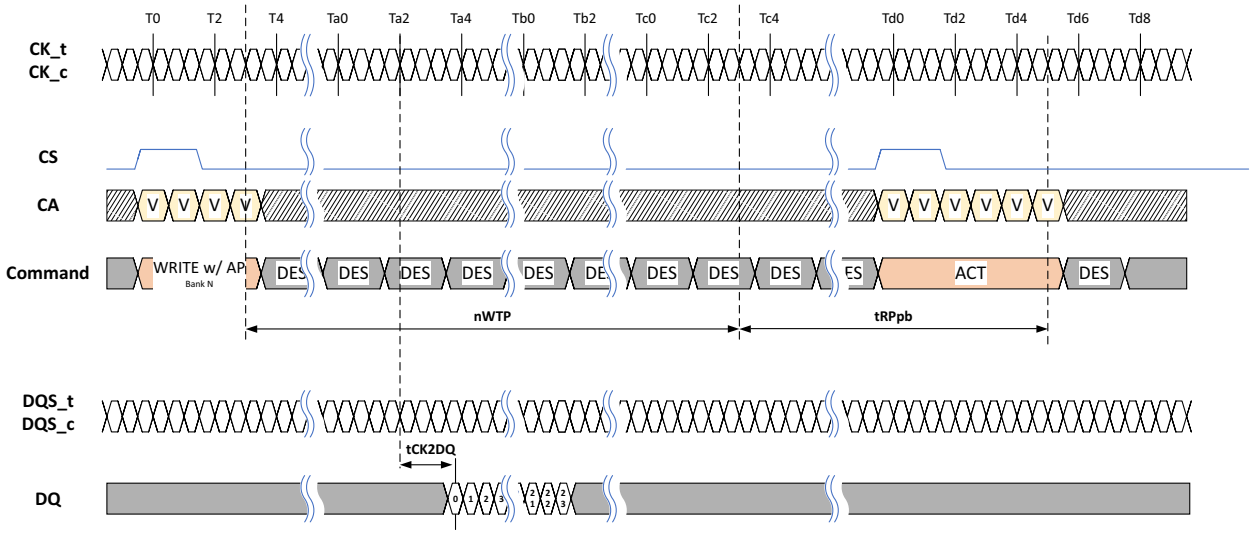


Figure 97 — WRITE with Auto-Precharge

NOTE: DES commands are shown for ease of illustration; other commands may be valid at these times.

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7.4. Read/Write Operations

7.4.1. READ Command

After a bank has been activated, the Read Operation causes DDR6 SDRAM to retrieve and output data stored in its array. The Read Operation is initiated by the Read Command during which the beginning column address and Bank/Bank group address for the data to be retrieved from the array is provided.

The data is driven by the SDRAM on its DQ pins CAS Read Latency (RL) cycles after the Read Command along with the proper waveform on the Read DQS (RDQS) outputs. Read Latency is defined and measured from final cycle of the Read command to the first effective rising DQS (excluding read preamble). DDR6 SDRAM requires a DQS post-amble after completing all read DQ bursts. The length of DQS post-amble can be set by mode register.

7.4.1.1. READ Timing

A read command is initiated with CS, and CA[4:0] asserted to the proper state at the rising and falling edges of CK, as defined by the Command Truth Table, Table XXX.

The read latency (RL) is defined from the 4th rising edge of the CK_t of the command that starts a read command to the rising edge of the CK_t from which (tDQS2offset + tDQS2DQo) is measured.

The first valid data is available of RL * tCK + (tDQss (Max) + tDQS2DQo) or RL * tCK + (tDQS2ss (Min) + tDQS2DQO) after the CK_t rising edge of a read command.

The DQ-data is valid for tQW (DQ output window) and the controller is required to periodically train its internal capture clock to stay centered in the tQW window to compensate for timing changes due to temperature and voltage variation. Burst data is captured by the controller on successive edges of DQS until the 24 bit data burst is complete. The DQS is required to remain active (toggling) for tRPST (DQS postamble) after the completion of the burst read.

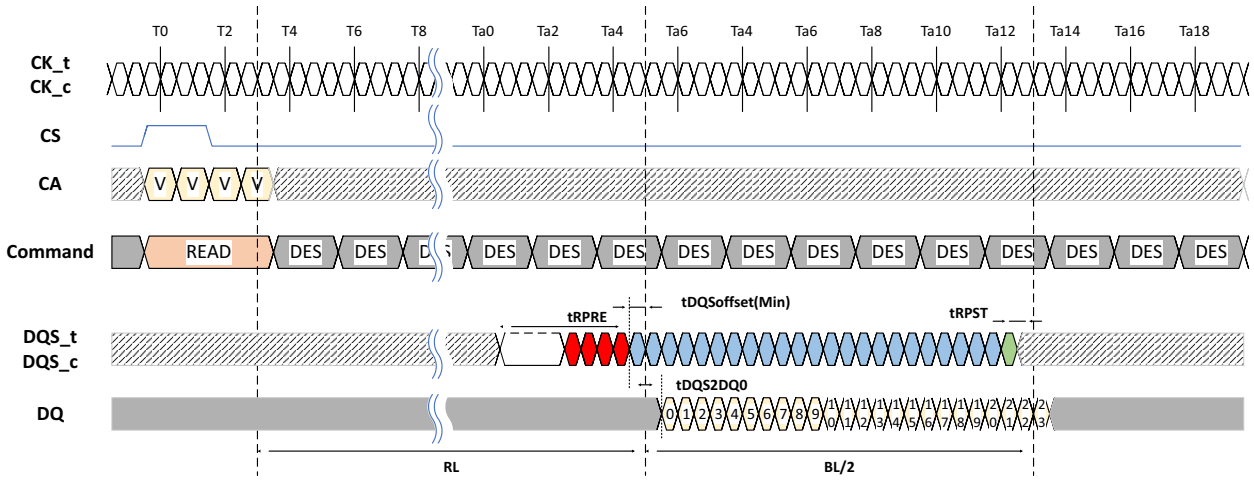


Figure 98 — Burst Read Operation

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.
NOTE: 2: BL=24, tRPRE=4nCK, tRPST=0.5nCK(Toggle)

NOTE: 3: The DQ signal is shown as "Don't Care" before the first Read data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be low for a minimum of 8UI prior to the first Read data bit for proper DFE synchronization. If inter-amble is only 2tCK, the DRAM will drive the DQ signal low for 4UI prior to the first Read data bit for proper DFE.

7.4.1.2. Back to Back Read Operation within tCCD_S

The following figures show timing diagram of back-to-back read operations where the 2nd Read command issued at tCCD_S.

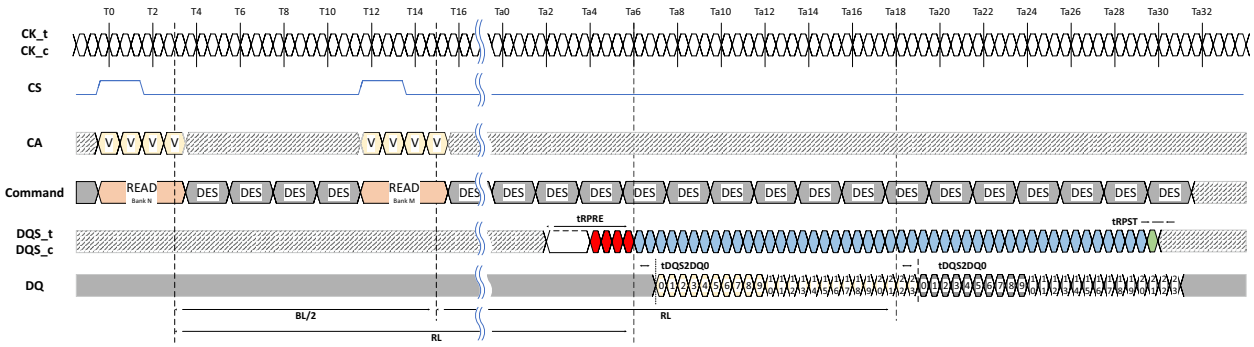


Figure 99 — tCCD_S Back to Back Read Operation (within PSD)

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.
NOTE: 2: tDQSoffset = 0ps for illustration
NOTE: 3: BL=24, tRPRE=4nCK, tRPST=0.5nCK(Toggle)
NOTE: 4: The DQ signal is shown as "Don't Care" before the first Read data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be low for a minimum of 8UI prior to the first Read data bit for proper DFE synchronization. If inter-amble is only 2tCK, the DRAM will drive the DQ signal low for 4UI prior to the first Read data bit for proper DFE.

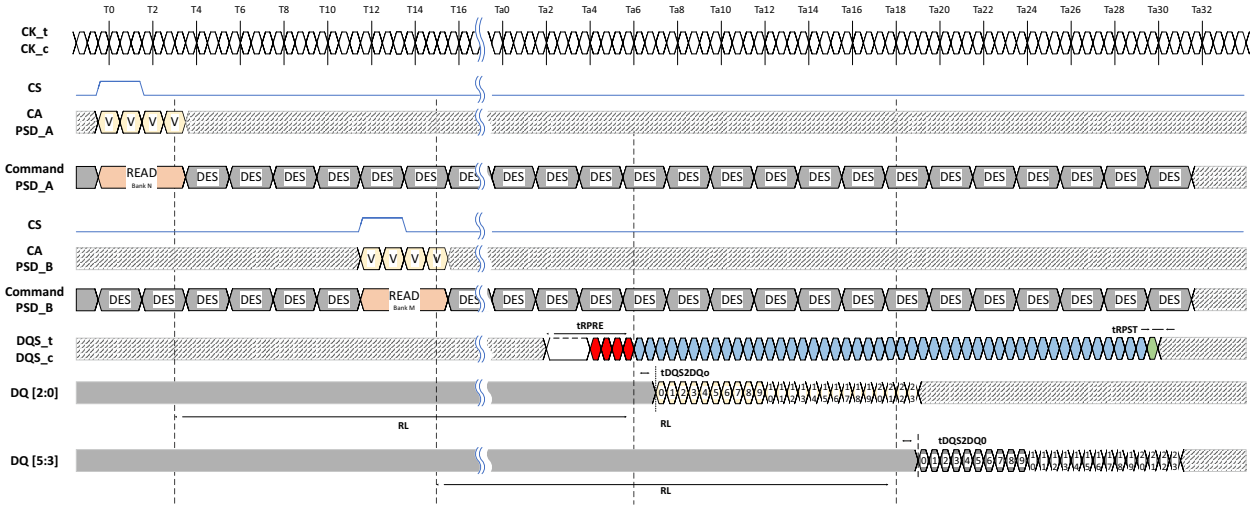


Figure 100 — Back to Back Read Operation (PSD to PSD)

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.
NOTE: 2: tDQSoffset = 0ps for illustration

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NOTE: 3: BL=24, tRPRE=4nCK, tRPST=0.5nCK(Toggle)
NOTE: 4: The DQ signal is shown as "Don't Care" before the first Read data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be low for a minimum of 8UI prior to the first Read data bit for proper DFE synchronization. If inter-amble is only 2tCK, the DRAM will drive the DQ signal low for 4UI prior to the first Read data bit for proper DFE.

7.4.1.3. Example Read to Read Operation with the 2nd Read is 2tCK or later

When the delay between two Read commands is slightly larger than tCCD_S, the second Read command must be delayed to the Modulus 2tCK command slot. In the case of only a 2tCK delay, the SoC sends 2tCK worth of DQS toggles regardless of the PRE and POST options. When the delay between the two Reads is 4tCK or larger, the behavior depends on the selected PRE and POST options.

In Figure 4, tRPRE = 4tCK and tRPST = 0.5tCK fit within the 4tCK gap on the Read channel. If tRPRE = 6tCK and tRPST = 0.5tCK (not shown), the DRAM would toggle for the full 4tCK gap. Figure 4 illustrates a within-PSD example, but the same concept can be extended to PSD-to-PSD scenarios as well.

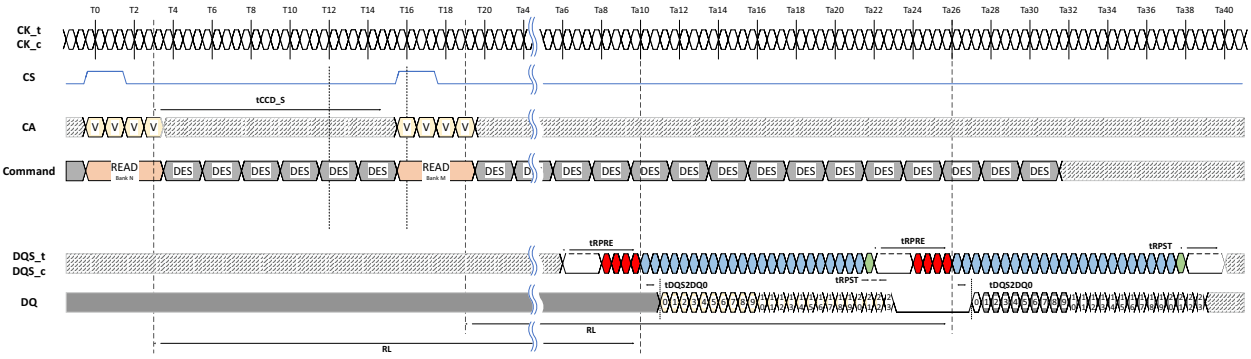


Figure 101 — Back to Back Read Operation with 2nd Read 4tCK later (Within PSD)

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.
NOTE: 2: tDQSoffset = 0ps for illustration
NOTE: 3: BL=24, tRPRE=4nCK, tRPST=2.5nCK(Toggle)
NOTE: 4: The DQ signal is shown as "Don't Care" before the first Read data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be low for a minimum of 8UI prior to the first Read data bit for proper DFE synchronization. If inter-amble is only 2tCK, the DRAM will drive the DQ signal low for 4UI prior to the first Read data bit for proper DFE.

7.4.1.4. Read Operation followed by Write Operation

The timing that a Write command following a Read command is issued on WCK Always on mode is below. Refer to clause 8.3.1:Read to Write Timing (tRTW) for detail on Read to Write timing.

PROPOSED

When a write command follows a read command, the minimum gap between two commands is defined as tRTW. tRTW is defined from the 4th rising edge of the CK_t that starts a read command to the 4th rising edge of the CK_t that starts the write command.

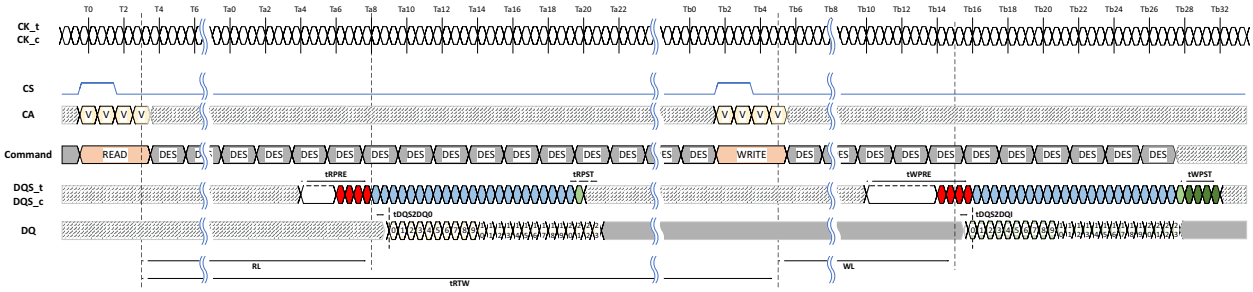


Figure 102 — Read Operation followed by Write Operation: Read and Write to Same BG

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.

NOTE: 2: tDQSoffset = 0ps for illustration

NOTE: 3: BL=24, tRPRE=4nCK, tRPST=2.5nCK(Toggle), tWPRE=6nCK, tWPST=2.5nCK

NOTE: 4: The DQ signal is shown as "Don't Care" before the first Read data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be low for a minimum of 8UI prior to the first Read data bit for proper DFE synchronization. If inter-amble is only 2tCK, the DRAM will drive the DQ signal low for 4UI prior to the first Read data bit for proper DFE.

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7.4.1.5. Read DQS Preamble Postamble Pre-Shift

Table 96 — MR0C Definition

Bits	Attribute	Default	Description	Notes
7:6	RW	0h	MR0C[7:6]:RD_DQS_P-ST_LENGTH Read DQS Postamble (PST) Length 00: 0.5*tCK (default) 01: 2.5*tCK 10: 4.5*tCK 11: Reserved	1
5	RV			
4:2	RW	0h	MR0C[4:2]:RD_D-QS_PRE_LENGTH Read DQS Preamble (PRE) Length 000: 2*tCK static, 2*tCK toggle (default) 001: 2*tCK static, 4*tCK toggle 010: 4*tCK static, 4*tCK toggle 011: 2*tCK static, 6*tCK toggle 100-111: Reserved	1,2,3
1:0	RW	0h	MR0C[2:0]:RD_DQS_PS Read DQS Pre-Shift (PS) 00: 0*tCK (default) 01: 2*tCK 10: 4*tCK 11: Reserved	1,4,5

NOTE 1: There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Please refer to TBD section for more information on FSP operation.
NOTE 2: MR0C[1:0] = 00 supported over all frequency ranges.
NOTE 3: MR0C[1:0] = 01/10 setting is allowed for greater than TBD Mbps operation.
NOTE 4: MR0C[4:2] = 0000 supported over all frequency ranges.
NOTE 5: MR0C[4:2] = 0001/0010/0011 setting is allowed for greater than TBD Mbps operation.

Read Preamble and Postamble

The Read DQS strobe for the SDRAM requires a preamble prior to the first latching edge (the rising edge of Read DQS_t with data "valid"), and it requires a postamble after the last latching edge. The preamble and postamble lengths are set via mode register writes (MRW).

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Preamble/postamble behavior such as static (no-toggle), toggling or combination of static and toggle, and its length can be set by mode register.

Read Pre Shift

DDR6 SDRAM supports programmable option to drive Read DQS early by nCK for low power host architecture to compensate extra delay on the path of Read DQS propagation path compared to DQ.

Read DQS Pre-Shift: MR0C[1:0]= 00B: 0*tCK

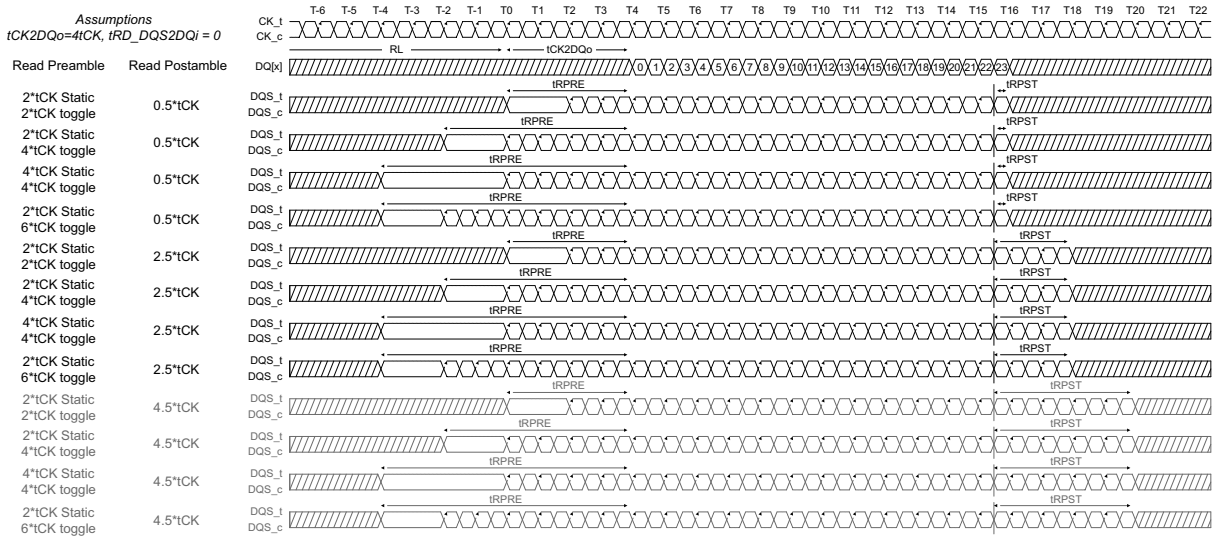


Figure 103 — Read DQS Read Preamble and Toggle Postamble with Pre-Shift = 0*tCK

NOTE: Read DQS Pre-Shift: MR0C[1:0]= 00B: 0*tCK

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Read DQS Pre-Shift: MR0C[1:0]=01B: 2*tCK

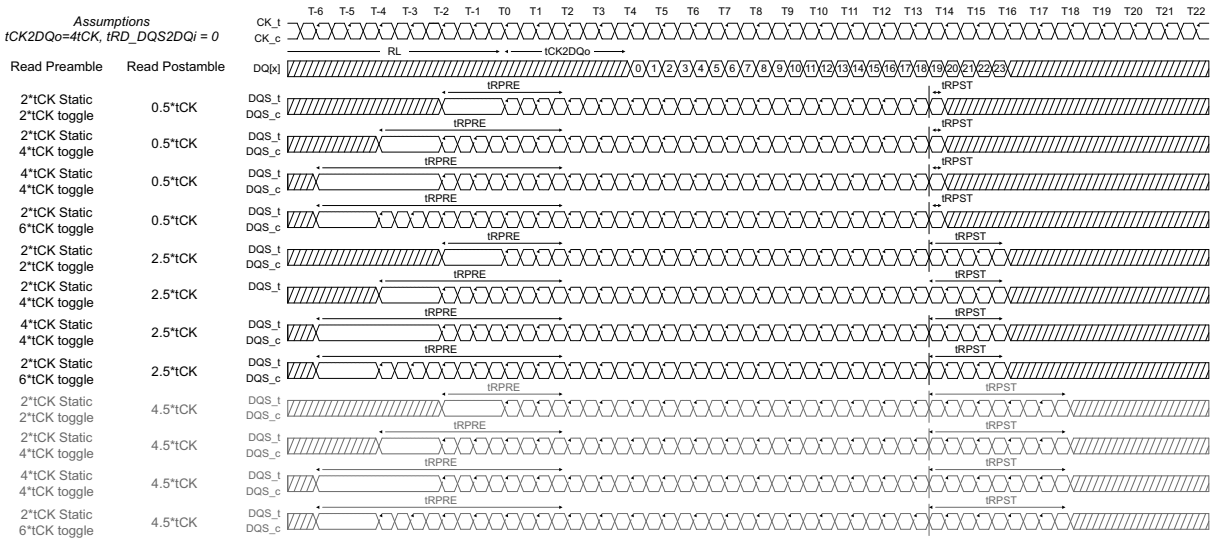


Figure 104 — Read DQS Read Preamble and Toggle Postamble with Pre-Shift = 2*tCK

NOTE: Read DQS Pre-Shift: MR0C[1:0]=01B: 2*tCK

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7.4.1.6. DQ Preamble

The DQ Preamble is the defined state of the DQ bus right before the normal data packets are transfered in order to keep the bus stable for the DFE. In DDR6, this is defined as the amount time in clocks that the DQ bus is kept low. As like regular preambles, the DQ Preamble will not force Read or Write gaps when doing back to back transactions. DQ bursts will always take priority over DQ Preambles.

Table 97 — MRxx Register Information (MA[7:0]=xxH)

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU				RFU		Read DQ Preamble	

Table 98 — MRxx Definition

Function	Register Type	Operand	Data	Notes
Read DQ Preamble	R/W	OP[1:0]	00B: 8UI / 4tCK DQ Preamble (Default) 01B: RFU 10B: RFU 11B: RFU	
RFU	RFU	OP[3:2]	RFU	
RFU	RFU	OP[6:4]	RFU	

Read DQ Preamble

The number of cycles for Read DQ Preamble is configured via MRxx.



Figure 105 — Example: Consecutive READ Operation: tCCD=MIN+8 (8 Clock Gap and Postambles=Toggle w/8UI Read DQ Preamble)

NOTE: Figure not available from ballot sponsor.

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Write DQ Preamble

The number of cycles for Write DQ Preamble is configured via the SPD. The Write DQ Preamble is currently limited to 8UI/4tCK but additional ranges may be defined for future speed bins.



Figure 106 — Example: Consecutive WRITE Operation: tCCD=MIN+8 (8 Clock Gap w/8UI Write DQ Preamble)

NOTE: Figure not available from ballot sponsor.

PROPOSED

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7.4.2. WRITE Command

After a bank is activated, the Write operation stores input data into the DDR6 SDRAM array. It begins with a Write command, which specifies the starting column address and the Bank/Bank Group address.

Data is driven onto the DQ inputs after CAS Write Latency (CWL) cycles from the Write command, accompanied by the appropriate DQS waveform. WL is measured from the last cycle of the Write command to the first valid rising edge of DQS (excluding the write preamble).

After all write bursts, DDR6 SDRAM requires a DQS postamble, whose length is programmable via a mode register.

7.4.2.1. WRITE Timing

A WRITE command is initiated with CS, and CA[4:0] asserted to the proper state at the rising and falling edges of CK, as defined by the Command Truth Table, Table XXXX. The write latency (WL) is defined from the 4th rising edge of the CK_t that starts a write command to the rising edge of the CK_t from which (tDQS2offset+ tDQS2D) is measured. The first valid "latching" edge of DQS is required to be driven WL * tCK + tDQss (Max) or WL * tCK -tDQss(Min) after the rising edge of CK_t that completes a write command.

The DDR6 SDRAM uses an un-matched DQS-DQ path for lower power, so the DQS is required to arrive at the SDRAM ball prior to the DQ signal by the amount of tDQS2DQi. The DQS is required to be trained to arrive at the DQ pad center-aligned with the DQ-data. The DQ-data is required to be held for tDIVW (data input valid window) and the DQS is required to be periodically trained to stay centered in the tDIVW window to compensate for timing changes due to temperature and voltage variation. Burst data is captured by the SDRAM on successive edges of DQS until the 24 data burst is complete. DQS is required to remain active for tWPST (DQS postamble) after the completion of the burst WRITE. Pin input timings are measured relative to the crosspoint of CK_t and CK_c.

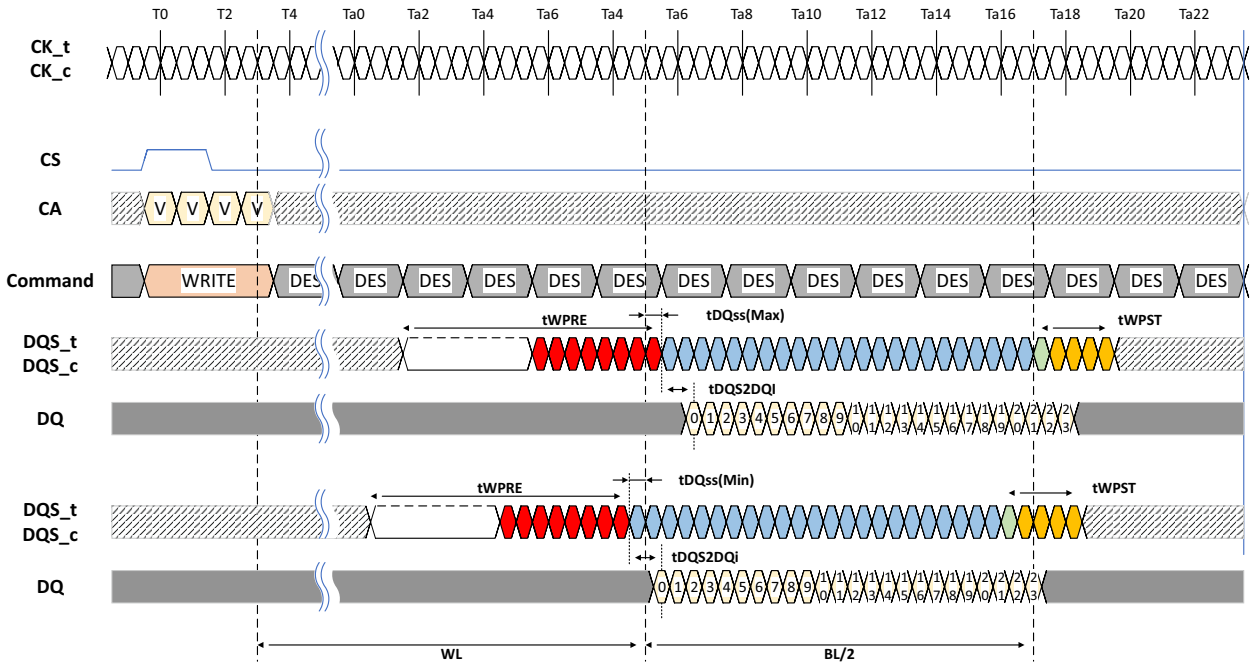


Figure 107 — Burst Write Operation

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.

NOTE: 2: tDQSoffset = 0ps for illustration
NOTE: 3: BL=24, tWPRE=8nCK, tWPST=2.5nCK(Toggle).
NOTE: 4: The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be low for a minimum of 8UI prior to the first Write data bit for proper DFE synchronization.

7.4.2.2. Back to Back Write Operation within tCCD_S

Figure 2A and Figure 2B shows timing diagram of back to back write operations where the 2nd Write command issued at tCCD_S for the within PSD and PSD to PSD cases.

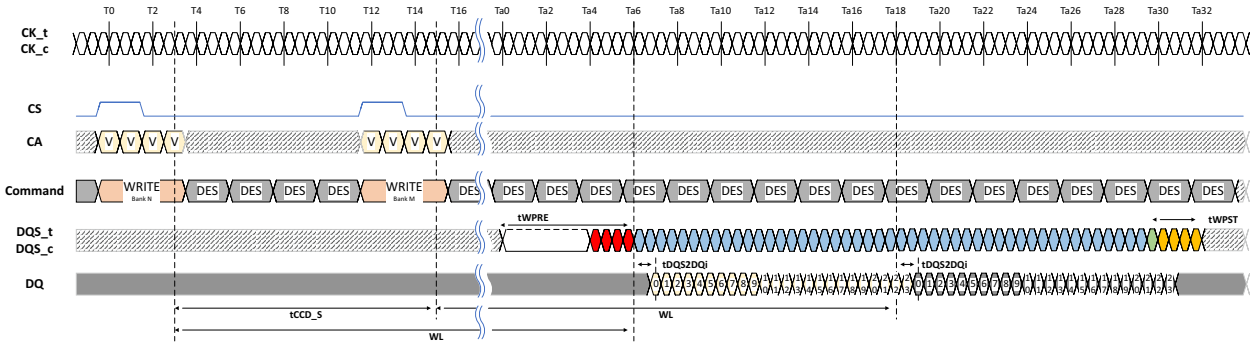


Figure 108 — tCCD_S Back to Back Write Operation (within PSD)

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.
NOTE: 2: tDQSoffset = 0ps for illustration
NOTE: 3: BL=24, tWPRE=6nCK, tWPST=2.5nCK(Toggle)
NOTE: 4: The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be low for a minimum of 8UI prior to the first Write data bit for proper DFE synchronization.

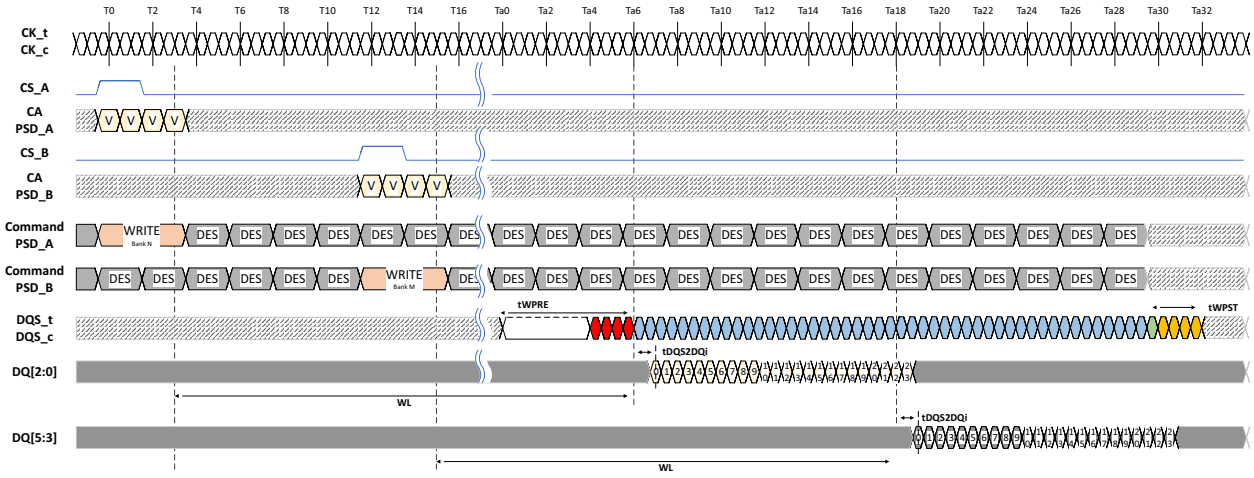


Figure 109 — tCCD_S Back to Back Write Operation (PSD to PSD)

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.
NOTE: 2: tDQSoffset = 0ps for illustration
NOTE: 3: BL=24, tWPRE=6nCK, tWPST=2.5nCK(Toggle)

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NOTE: 4: The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be low for a minimum of 8UI prior to the first Write data bit for proper DFE synchronization.

7.4.2.3. Write to Write Operation with an 4tCK Command Slot Gap Example

When the delay between two Write commands is slightly larger than tCCD_S, the 2nd Write Command needs to move to the next 4tCK Write command slot. In Figure 3, the inter-amble is smaller than a full tWPST and tWPRE, so the DQS is expected to toggle 4tCK before the 2nd Write command data is sampled on DQ.

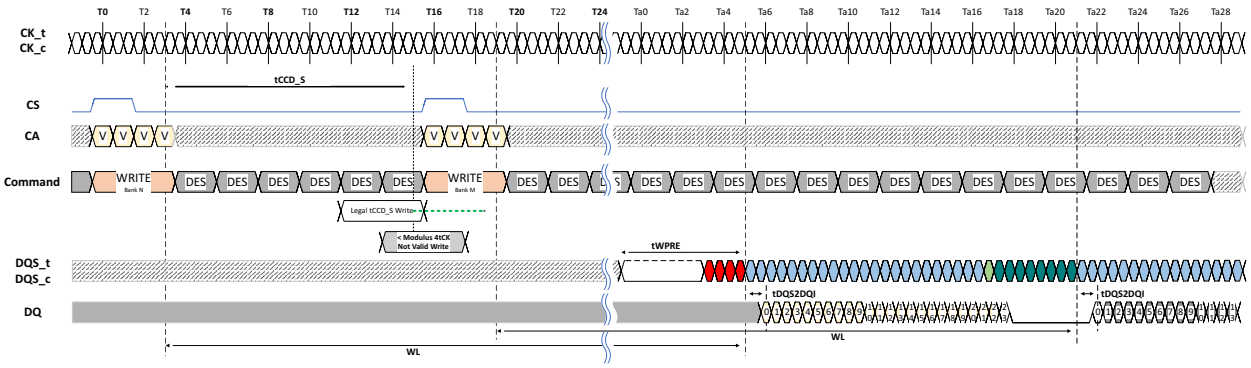


Figure 110 — Back to Back Write Operation with 4tCK Inter-amble (Within PSD)

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.
NOTE: 2: tDQS2offset = 0ps for illustration
NOTE: 3: BL=24, tWPRE=6nCK, tWPST=2.5nCK(Toggle)
NOTE: 4: The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be low for a minimum of 8UI prior to the first Write data bit for proper DFE synchronization.

7.4.2.4. Write to Write Operation with an 8tCK Command Slot Gap Example

When the delay between two Write commands is greater than tCCD_S, the second Write command must move to the next 4tCK modulus Write command slot. Figure 4 shows an example in which the PSD-to-PSD inter-amble gap is

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8tCK, and the tPRE option used in this example is 8nCK. As a result, the combined tWPST and tWPRE is still larger than the inter-amble gap, so DQS will toggle for 8tCK before the data of the second Write command.

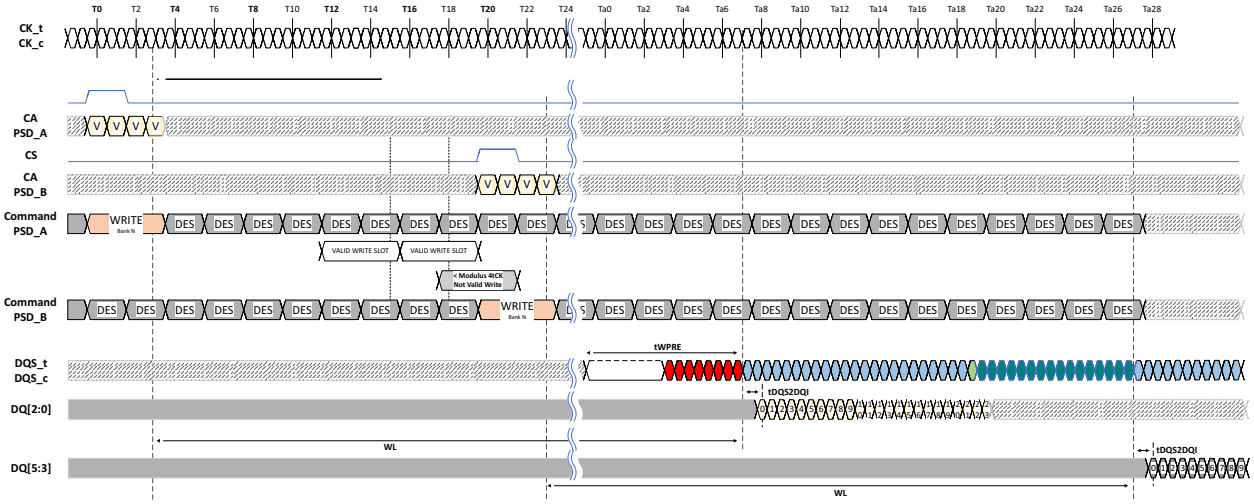


Figure 111 — Back to Back Write Operation with 8tCK Inter-amble (PSD-PSD)

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.
NOTE: 2: tDQS2offset = 0ps for illustration
NOTE: 3: BL=24, tWPRE=8nCK, tWwPST=2.5nCK(Toggle)
NOTE: 4: The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be low for a minimum of 8UI prior to the first Write data bit for proper DFE synchronization.

7.4.2.5. Write Operation followed by Read Operation

When a read command follows a write command, the minimum gap between two commands is defined as min(tWTR_S/L). tWTR_L (Same BG) / tWTR_S (Different BG) is measured from the 4th rising edge of the CK_t command that satisfies tWTR_L (Same BG) / tWTR_S (Different BG) respectively after a write command to the following read command.

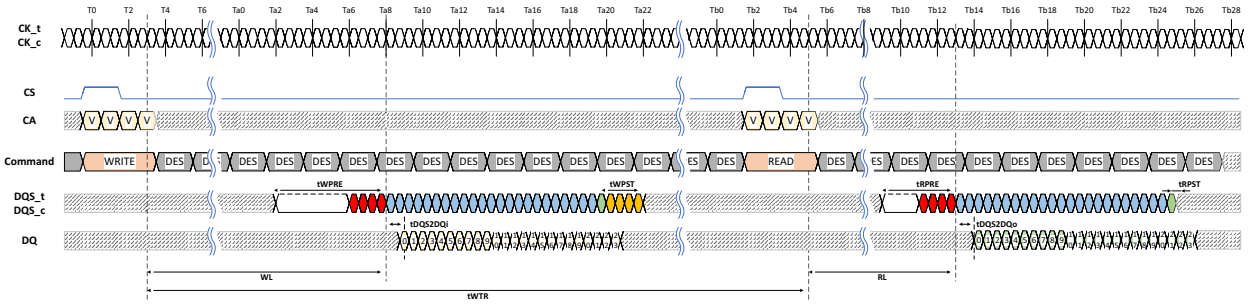


Figure 112 — Write Operation followed by Read Operation: Read and Write to Same BG

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.

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NOTE: 2: tDQS2offset = 0ps for illustration

NOTE: 3: BL=24, tWPRE=6nCK, tWPST=2.5nCK(Toggle), tRPRE=4nCK, tRPST=0.5nCK

NOTE: 4: The DQ signal is shown as “Don’t Care” before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be low for a minimum of 8UI prior to the first Write data bit for proper DFE synchronization.

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7.4.2.6. DDR6 Write Preambles, Postambles

Table 99 — MRxx Register Information (MA [3:0] = 0AH)

OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Write DQS PST Length		Write DQS PRE Length		

Table 100 — MRxx Definition

Function	Register Type	Operand	Data	Notes
Write DQS PRE Length	R/W	OP[2:0]	000_B: 4*tCK static, 2*tCK toggle (default) 001_B: 4*tCK static, 4*tCK toggle All others are RFU	1,2,3,4
Write DQS PST Length		OP[4:3]	00_B: 2.5*tCK toggle (default) 01_B: 4.5*tCK toggle 10_B: Reserved 11_B: Reserved	1,2,5,6

NOTE 1: There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MRyy OP[a]) will be written with an MRW command to this MR address.
NOTE 2: There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MRyy OP[b]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.
NOTE 3: MRxx OP[2:0] = 001B setting is allowed for lower than TBD Mbps operation.
NOTE 4: MRxx OP[2:0] = 000B setting is allowed for lower than TBD Mbps operation.
NOTE 5: MRxx OP[4:3] = 01B setting is allowed for lower than TBD Mbps operation.
NOTE 6: MRxx OP[4:3] = 00B setting is allowed for lower than TBD Mbps operation.

Write Preamble and Postamble

DDR6 SDRAM supports a programmable write pre-amble and post-amble. The Write DQS strobe for the SDRAM requires a pre-amble prior to the first latching edge (the rising edge of Write DQS_t with data "valid"), and it requires a post-amble after the last latching edge. The pre-amble and post-amble lengths are set via mode register writes (MRW).

Preamble/postamble behavior such as static (no-toggle), toggling or combination of static and toggle, and its length can be set by mode register.

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Write pre-amble is configured as 4*tCK static, 2*tCK toggle or 4*tCK static, 4*tCK toggle via MRxx:OP[1:0]. Write post-amble is configured as 2.5*tCK or 4.5*tCK via MRxx:OP[3:2].

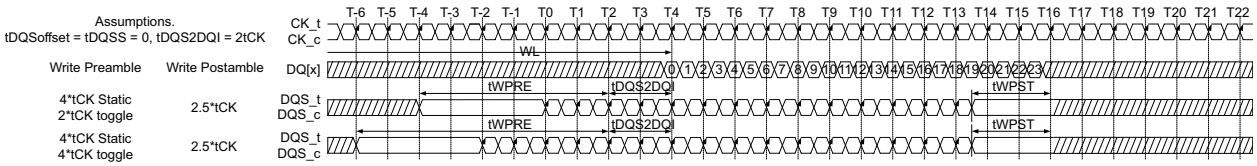


Figure 113 — Write DQS Preamble Modes (Default) with 2.5tCK Postamble

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7.4.3. Interamble

The DQS strobe requires a preamble prior to the first latching edge (the rising edge of DQS_t with data valid) and a postamble after the last latching edge. The preamble and postamble options are set via MRW commands. Additionally, the postamble- and preamble-configured size does not force the host to add command gaps in the command interval just to satisfy postamble or preamble settings (that is, preamble = 4 tCK + postamble = 2.5tCK does not force tCCD +6).

Regardless of READ or WRITE interambles, any time the Postambles of one transaction would overlap the beginning of the next transactions preamble, the interamble will be composed of only full toggles.

7.4.3.1. Read Interamble Timing Diagrams

In READ-to-READ operations with tCCD = BL/2, postamble for the first command and preamble for the second command disappear to create consecutive DQS latching edges for seamless burst operations. In the case of READ-to-READ operations with a command interval of tCCD +1, tCCD +2, etc., the toggles will always continue if the postambles and preambles overlap.

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Static preambles will only be used if there is a gap that exceeds the sum of both selected preamble & postamble.

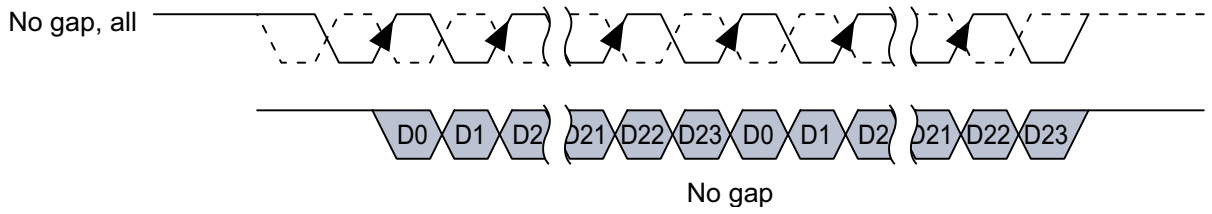


Figure 114 — Example: Seamless READ Operation: tCCD=MIN

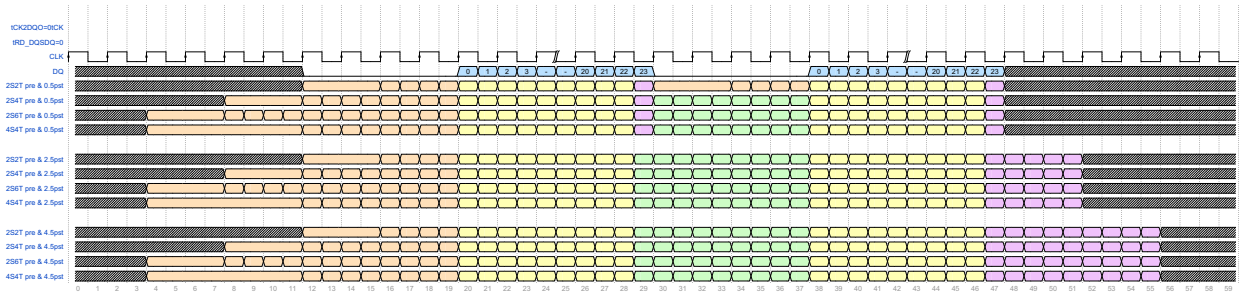


Figure 115 — Example: Consecutive READ Operation: CCD=MIN+4 (4 Clock Gap and Postambles=Toggle)

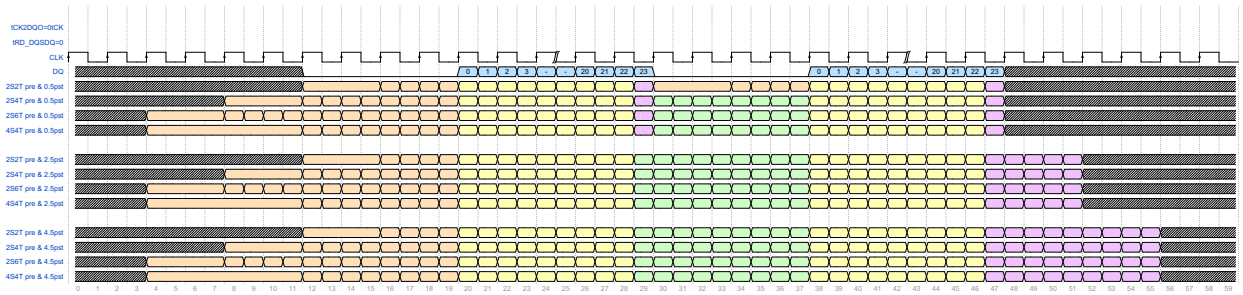


Figure 116 — Example: Consecutive READ Operation: CCD=MIN+4 (4 Clock Gap and Postambles=Static)

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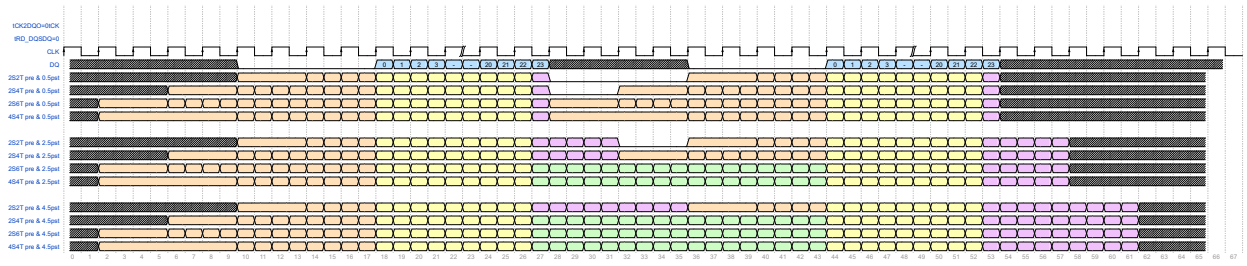


Figure 117 — Example: Consecutive READ Operation: tCCD=MIN+8 (8 Clock Gap and Postambles=Toggle)

Placeholder

Figure 118 — Example: Consecutive READ Operation: tCCD=MIN+8 (8 Clock Gap and Postambles=Static)

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NOTE: Figure not available from ballot sponsor.

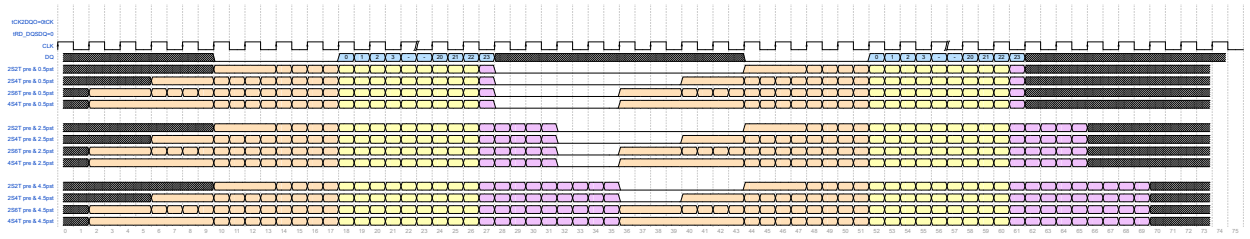


Figure 119 — Example: Consecutive READ Operation: tCCD=MIN+12 (12 Clock Gap and Postambles=Toggle)



Figure 120 — Example: Consecutive READ Operation: tCCD=MIN+12 (12 Clock Gap and Postambles=Static)

NOTE: Figure not available from ballot sponsor.

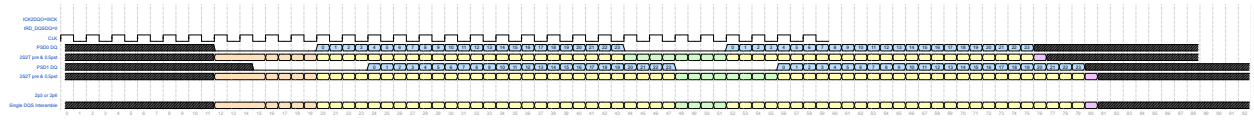


Figure 121 — Example: Consecutive Read Operations: tCCD=MIN+4 but across two different PSD, with PSD1 following PSD0 by 2tCK. This example just shows how the individual PSDs would normally have

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preambles/intermables/postambles but how the two PSDs share a single DQS.



Figure 122 — Example: Consecutive Read Operations: tCCD=MIN+4 but across two different PSD, with PSD1 following PSD0 by a burst + 2tCK. This example just shows how the individual PSDs would normally have preambles/intermables/postambles but how the two PSDs share a single DQS.

NOTE: Figure not available from ballot sponsor.

7.4.3.2. Write Interamble Timing Diagrams

In WRITE-to-WRITE operations with tCCD=BL/2, postamble for the first command and preamble for the second command disappear to create consecutive DQS latching edges for seamless burst operations. In the case of WRITE-to-WRITE operations with a command interval of tCCD+1, tCCD+2, etc., the toggles will always continue if

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the postambles and preambles overlap. Static preambles will only be used if there is a gap that exceeds the sum of both selected preamble and postamble.

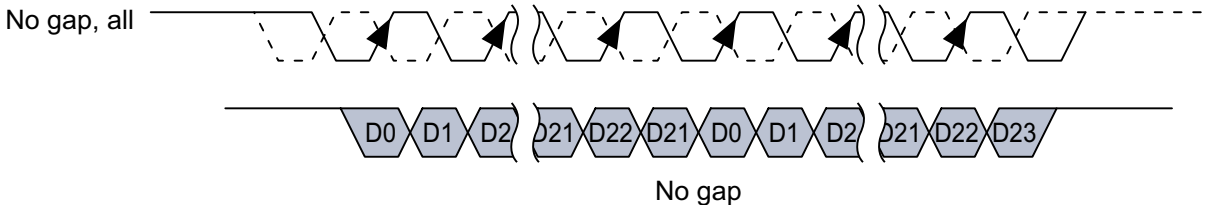


Figure 123 — Example: Seamless WRITE Operation: tCCD=MIN

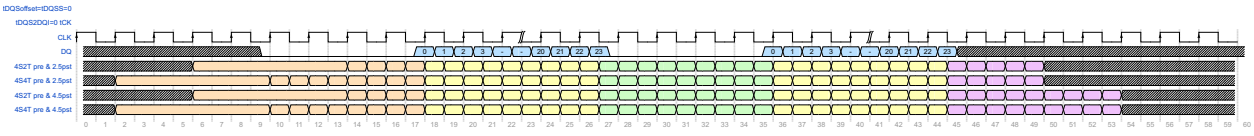


Figure 124 — Example: Consecutive WRITE Operation: tCCD=MIN+4 (4 Clock Gap)

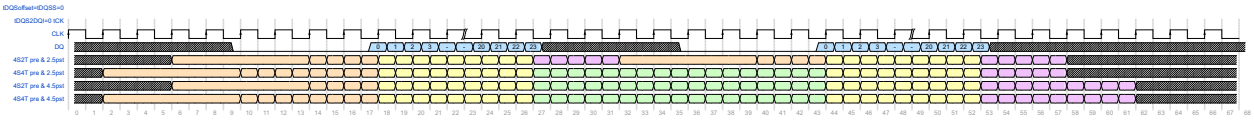


Figure 125 — Example: Consecutive WRITE Operation: tCCD=MIN+8 (8 Clock Gap)

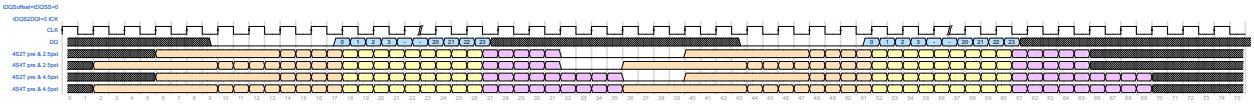


Figure 126 — Example: Consecutive WRITE Operation: tCCD=MIN+12 (12 Clock Gap)

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7.4.4. Read/Write Latencies

Latencies related to Read timing are measured from the 4th rising edge of CK_t of the Read command. The Read Latencies depend on the settings of 1p6/1p12 device, Read Link protection and Command blocking Disable/Enable. WL is calculated to be RL-4tCK.

For each MR08:OP[5:0] setting, only the data rate ranges shown are supported.

Table 101 — Parity Latency

Data Rate [Mbps]		Parity Latency [nCK]
Lower Limit [> Mbps]	Upper Limit [<= Mbps]	
4800	6400	12
6400	8000	
8000	9200	
9200	9600	
9600	10400	
10400	11600	16
11600	12800	
12800	14400	
14400	16400	20
16400	19200	24
NOTE 1: Parity latency (PL) is calculated per each data rate in the case of CMD blocking enabled by CA Parity.		

Table 102 — Read/Write Latencies - Command Blocking Disabled (Sheet 1 of 2)

Data Rate [Mbps]		tCK [ps][4]	2pX Device				1pX Device[6]			
			Read Link CRC Disabled		Read Link CRC Enabled[7]		Read Link CRC Disabled		Read Link CRC Enabled[7]	
Lower Limit [> Mbps]	Upper Limit [<= Mbps]		RL [nCK]	WL [nCK]	RL [nCK]	WL [nCK]	RL [nCK]	WL [nCK]	RL [nCK]	WL [nCK]
4800	6400	312.5	68	64	76	72	76	72	84	80
6400	8000	250.0	84	80	96	92	92	88	104	100
8000	9200	217.4	100	96	112	108	108	104	120	116
9200	9600	208.3	104	100	116	112	112	108	124	120
9600	10400	192.3	112	108	124	120	120	116	136	132

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Table 102 — Read/Write Latencies - Command Blocking Disabled (Continued)

Data Rate [Mbps]		tCK [ps][4]	2pX Device				1pX Device[6]			
			Read Link CRC Disabled		Read Link CRC Enabled[7]		Read Link CRC Disabled		Read Link CRC Enabled[7]	
Lower Limit [> Mbps]	Upper Limit [<= Mbps]		RL [nCK]	WL [nCK]	RL [nCK]	WL [nCK]	RL [nCK]	WL [nCK]	RL [nCK]	WL [nCK]
10400	11600	172.4	124	120	140	136	136	132	148	144
11600	12800	156.3	136	132	152	148	148	144	164	160
12800	14400	138.9	152	148	172	168	168	164	184	180
14400	16400	122.0	176	172	196	192	192	188	212	208
16400	19200	104.2	204	200	228	224	224	220	248	244

NOTE 1: DDR6 SDRAM device should not be operated at a frequency above the upper frequency limit, or below the lower frequency limit, shown for each RL or WL value.
NOTE 2: RL [nCK] = 4 x RU(RL_{min} / 4tCK)
NOTE 3: WL [nCK] = RL [nCK] – 4
NOTE 4: tCK = 2 / Data Rate (max)
NOTE 5: RL_{min} = 21ns
NOTE 6: 1pX Device adder for RL_{min} = 2ns
NOTE 7: Read Link CRC adder for RL_{min} = 2.5ns

Table 103 — Read/Write Latencies - Command Blocking Enabled (Sheet 1 of 2)

Data Rate [Mbps]		tCK [ps][4]	2pX Device				1pX Device[6]			
			Read Link CRC Disabled		Read Link CRC Enabled[7]		Read Link CRC Disabled		Read Link CRC Enabled[7]	
Lower Limit [> Mbps]	Upper Limit [<= Mbps]		RL [nCK]	WL [nCK]	RL [nCK]	WL [nCK]	RL [nCK]	WL [nCK]	RL [nCK]	WL [nCK]
4800	6400	312.5	80	76	88	84	88	84	96	92
6400	8000	250.0	96	92	108	104	104	100	116	112
8000	9200	217.4	112	108	124	120	120	116	132	128
9200	9600	208.3	116	112	128	124	124	120	136	132
9600	10400	192.3	124	120	136	132	132	128	148	144
10400	11600	172.4	140	136	156	152	152	148	164	160
11600	12800	156.3	152	148	168	164	164	160	180	176

Table 103 — Read/Write Latencies - Command Blocking Enabled (Continued)

Data Rate [Mbps]		tCK [ps][4]	2pX Device				1pX Device[6]			
			Read Link CRC Disabled		Read Link CRC Enabled[7]		Read Link CRC Disabled		Read Link CRC Enabled[7]	
Lower Limit [> Mbps]	Upper Limit [<= Mbps]		RL [nCK]	WL [nCK]	RL [nCK]	WL [nCK]	RL [nCK]	WL [nCK]	RL [nCK]	WL [nCK]
12800	14400	138.9	168	164	188	184	184	180	200	196
14400	16400	122.0	196	192	216	212	212	208	232	228
16400	19200	104.2	228	224	252	248	248	244	272	268

NOTE 1: DDR6 SDRAM device should not be operated at a frequency above the upper frequency limit, or below the lower frequency limit, shown for each RL or WL value.
NOTE 2: RL [nCK] = 4 x RU(RL_{min} / 4tCK) + PL [nCK]
NOTE 3: WL [nCK] = RL [nCK] – 4
NOTE 4: tCK = 2 / Data Rate (max)
NOTE 5: RL_{min} = 21ns
NOTE 6: 1pX Device adder for RL_{min} = 2ns
NOTE 7: Read Link CRC adder for RL_{min} = 2.5ns
NOTE 8: Please refer to the Table 1 that describes Parity Latency in the case of Command blocking enabled.

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7.4.5. Metadata

7.4.5.1. Introduction

DDR6 SDRAM supports a Metadata feature for Write and Read operations. In this mode, the data and Metadata are written/read in parallel to/from the array. This feature supports the MD_OFF, MD_8b and MD_16b modes. The modes can be chosen by MRxx OP[y] at initialization. MD_16b refers to the provision of 16 bits of metadata for every 128b of data. In a 2p3 device, MD_8b refers to the provision of 4 bits of metadata for every 64b of data and MD_16b refers to the provision of 8 bits of metadata for every 64b of data. The metadata bits will be placed in the existing location of metadata as defined in the data packet for both the Read and Write operations

Table 104 — MRxx Decode of Metadata Modes

Mode	MRxx:OP[y]	MRxx:OP[z]	Notes
Default	0	0	1
MD_8b	0	1	
MD_16b	1	0	
MD_OFF	1	1	
NOTE: DDR6 SDRAM powers up in this mode			

7.4.5.2. Description

The Metadata feature is implemented through carveout, where columns in the array are reserved for storing the meta-data bits. The number of columns reserved will depend on the metadata mode being selected. The DDR6 SDRAM shall powerup in the Default Mode (MRxx OP[y:z] =00) mode. In the modes where metadata is enabled, the metadata bits are part of the codeword of the on-die ECC engine.

Table 105 — Data Address Range by Mode

Mode	Data Address Range	Notes
MD_OFF	0x00 - 0x3F	
MD_8b	0x00 - 0x3B	1
MD_16b	0x00 - 0x37	1
NOTE: Access to the remaining column address space is prohibited		

MD_OFF

In this mode, the entire addressable column space is available for storing data. This mode is selected by setting MRxx OP[y:z]=11.

MD_8b

In the MD_8b mode, the DDR6 SDRAM provides 8 bits of metadata for every 128 bits of data. This mode is selected by setting MRxx OP[y:z]=01.In this mode Column addresses C [5:0] =0x00-0x3B is available for data and access to the remaining column space is prohibited.

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MD_16b

In the MD_16b mode, the DDR6 SDRAM provides 16 bits of metadata for every 128 bits of data. This mode is selected by setting MRxx OP[y:z]=10.In this mode Column addresses C [5:0] =0x00-0x37 is available for data and access to the remaining column space is prohibited.

7.4.5.3. Metadata Mode Selection

The Metadata mode can be selected by setting the MRxx OP[y:z] to the required state at initialization.

Timing

When the metadata mode is toggled, a delay of tMDMode_Select_Delay is required, before any command is issued to the SDRAM.

Table 106 — Metadata Mode Timing Parameters

Parameter	Symbol	Value	Unit
Delay between Metadata Mode change and next command	tMDMode_Select_Delay	tINIT3	ms

Miscellaneous

Any background training shall be restarted. DDR6 SDRAM shall be in idle state when the Metadata mode is selected. The CA bus shall maintain a DESELECT state during tMDMode_Select_Delay

Upon the selection of the Metadata mode, the DDR6 SDRAM shall reset ACI Complete Flag (MRxx OP[y]=0) until a full array Activation Counter Initialization (ACI) is completed by the host to set the Activation counter bits to a known state.

7.4.5.4. Metadata Mode Enable Sequence

The DDR6 SDRAM shall powerup in a Default Mode as shown in Table 1.In the Default mode Activate commands shall not be issued to the DRAM and any issued Activates are illegal.Illegal Activates shall be blocked by the DRAM. Once a Metadata mode is selected by the user the Mode Register bits used in selecting the Metadata mode are locked and cannot be rewritten unless a RESET is issued. Activate Commands are enabled once tMDMode_Select Delay has elapsed.

Initialization Sequence Pertaining to Metadata

The following sequence pertaining to Metadata shall be used.

1) DDR6 SDRAM powers up into the Default Mode.Activate Commands shall not be issued. Any issued Activates shall be blocked by the DDR6 SDRAM
2) The Metadata Mode is selected and MRxx:OP[y:z] set to something other than Default Mode.This action locks the MRxx:OP[y:z] and prevents the MRxxOP[y:z] bits from being rewritten. The CA bus shall maintain a DESELECT state during the tMDMode_Select_Delay time
3) Activate commands can be issued to the DRAM once tMDMode_Select_Delay has elapsed.
4) ACI operation needs to be performed after the Metadata Mode selection

7.5. Refresh Operation

7.5.1. Refresh Command

Two refresh commands are supported: All-bank REFRESH (REFab) and Dual-bank REFRESH (REFdb). These commands are available for 2p3 and 2p6 configuration devices. For 1p6 configuration devices, the Dual-bank REFRESH command becomes a Quad-bank REFRESH (REFqb) command.

An All-bank REFRESH command (REFab) refreshes one or more rows in all banks as indicated by the internal row refresh counter. A "refresh" to a row means the row is activated, all bits on that row are sensed and restored (refreshed) to the sensed state, and the row is precharged.

A Dual-bank REFRESH command (REFdb) refreshes one or more rows in two banks as indicated by bank group addresses BG[2:0] and the bank address BA.

A Dual-bank REFRESH command (REFdb) can be issued in any BG[2:0] and BA order, with the internal bank counter being incremented with each REFdb command. The order of REFdb commands may be sequential round robin: (0,16)-(1,17)-(2,18)-(3,19)-(4,20)-(5,21)-(6,22)-(7,23)-(8,24)-(9,25)-(10,26)-(11,27)-(12,28)-(13,29)-(14,30)-(15,31) . The order of REFdb commands may be random, for example, the REFdb commands may be issued in the following order:
(0,16)-(9,25)-(8,24)-(2,18)-(1,17)-(10,26)-(3,19)-(12,28)-(4,20)-(11,27)-(13,29)-(6,22)-(5,21)-(15,31)-(7,23)-(14,30) . After all banks have been refreshed using REFdb commands, the internal bank counter is reset or rolls over to 0, and the internal row refresh counter will be incremented. The controller can send another set of REFdb commands in the same order or a different order. It is illegal to send a REFdb command to the same bank unless all other banks have been refreshed using the REFdb command. The bank counter starts with the first REFdb command after a synchronization event.

A Quad-bank REFRESH command (REFqb) has the same behavior as the REFdb command, but is performed on the same two banks in both PSDs for a single REFqb command.

The bank counter is synchronized between the controller and DRAM by resetting it to 0. Resetting occurs upon issuing a RESET_n or at every exit from Self-Refresh. The REFab command also is a counter synchronization event and resets the bank counter to 0. Entering Self-Refresh or issuing a REFab command can be done at any valid time without cycling through all banks using the REFdb (or REFqb) command. After the bank counter is synchronized, the controller can issue REFdb (or REFqb) commands in any order as described in the previous paragraphs.

A REFab command issued when the bank counter is not zero will reset the bank counter to 0 and refreshes are performed to all banks as needed and as indicated by the internal row refresh counter. If another refresh command (REFab or REFdb or REFqb) is issued after the REFab command, then it uses an incremented value of the internal row refresh counter. The table below shows examples of both bank and refresh counter increment behavior.

Table 107 — Bank and Refresh Counter Increment Behavior (Sheet 1 of 3)

#	Sub #	Command	BG2	BG1	BG0	BA	Refresh Bank #	Bank Counter #	Ref. Counter # (Row Address #)
0	0	**RESET_n, SRX, or REFab**						To 0	

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Table 107 — Bank and Refresh Counter Increment Behavior (Continued) (Sheet 2

#	Sub #	Command	BG2	BG1	BG0	BA	Refresh Bank #	Bank Counter #	Ref. Counter # (Row Address #)
1	1	REFdb / REFqb	0	0	0	0	0/16	0 to 1	
2	2	REFdb / REFqb	0	0	0	1	1/17	1 to 2	
3	3	REFdb/ REFqb	0	0	1	0	2/18	2 to 3	
4	4	REFpb/ REFqb	0	0	1	1	3/19	3 to 4	
5	5	REFdb/ REFqb	0	1	0	0	4/20	4 to 5	
6	6	REFdb/ REFqb	0	1	0	1	5/21	5 to 6	
7	7	REFdb/ REFqb	0	1	1	0	6/22	6 to 7	
8	8	REFdb/ REFqb	0	1	1	1	7/23	7 to 8	
9	9	REFdb/ REFqb	1	0	0	0	8/24	8 to 9	n
10	10	REFdb/ REFqb	1	0	0	1	9/25	9 to 10	
11	11	REFdb/ REFqb	1	0	1	0	10/26	10 to 11	
12	12	REFdb/ REFqb	1	0	1	1	11/27	11 to 12	
13	13	REFdb/ REFqb	1	1	0	0	12/28	12 to 13	
14	14	REFdb/ REFqb	1	1	0	1	13/29	13 to 14	
15	15	REFdb/ REFqb	1	1	1	0	14/30	14 to 15	
16	16	REFdb/ REFqb	1	1	1	1	15/31	15 to 0	

Table 107 — Bank and Refresh Counter Increment Behavior (Continued) (Sheet 3

#	Sub #	Command	BG2	BG1	BG0	BA	Refresh Bank #	Bank Counter #	Ref. Counter # (Row Address #)
17	1	REFdb/ REFqb	0	0	0	0	0/16	0 to 1	n + 1
18	2	REFdb/ REFqb	1	0	0	1	9/25	1 to 2	
				–					
				–					
				–					
31	15	REFdb/ REFqb	0	0	1	1	3/19	14 to 15	
32	16	REFdb/ REFqb	1	0	1	1	11/27	15 to 0	
33	1	REFdb/ REFqb	0	0	0	0	0/16	0 to 1	n + 2
34	2	REFdb/ REFqb	0	0	0	1	1/17	1 to 2	
35	3	REFdb/ REFqb	0	0	1	0	2/18	2 to 3	
36	**0**	**REFab**	**V**	**V**	**V**	**V**	**0~31**	**To 0**	**n + 2**
37	1	REFdb/ REFqb	0	0	1	0	2/18	0 to 1	n + 3
38	2	REFdb/ REFqb	0	1	0	1	5/21	1 to 2	
Snip									

NOTE 1: REFdb and REFqb cannot be mixed since they are supported by different device configurations.
NOTE 2: Refresh Bank # is per PSD.

A bank shall be idle before it can be refreshed. The controller must track the bank pair being refreshed by the REFdb (or REFqb) command.

The REFdb (or REFqb) command must not be issued to the device until the following conditions are met:

- tRFCab has been satisfied after the prior REFab command.
- tdbR2dbR has been satisfied after the prior REFdb command (or tqbR2qbR has been satisfied after the prior REFqb command).
- tRP has been satisfied after the prior PRECHARGE command to that bank.
- tRRD has been satisfied after the prior ACTIVATE command (if applicable, for example after activating a row in a different bank than the one affected by the REFdb or REFqb command).

The target bank pair is inaccessible during the REFdb (or REFqb) command cycle time tRFCdb (or tRFCqb), however, other banks not being refreshed within the device are still accessible. During the REFdb (or REFqb) operation, any of

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the non-refreshing banks can receive an ACTIVATE or PRECHARGE command, be maintained in an active state, or be accessed by a READ or a WRITE command. When the REFdb (or REFqb) command cycle has completed, the affected bank pair will be in the idle state.

After issuing REFdb (or REFqb), these conditions must be met:

- tRFCdb (or tRFCqb) must be satisfied before issuing a REFab command.
- tRFCdb (or tRFCqb) must be satisfied before issuing an ACTIVATE command to the same bank or bank pair.
- tdbR2act (or tqbR2act) must be satisfied before issuing an ACTIVATE command to a different non-refreshing bank.
- tdbR2dbR (or tqbR2qbR) must be satisfied before issuing another REFdb (or REFqb) command.

A REFab command issues a REFRESH command to all banks. All banks shall be idle when REFab is issued (for instance, by issuing a PRECHARGE-ALL command prior to issuing the REFab command). REFab also synchronizes the bank counter to 0. The REFab command must not be issued to the device until the following conditions have been met:

- tRFCab has been satisfied following the prior REFab command.
- tRFCdb (or tRFCqb) has been satisfied following the prior REFdb (or REFqb) command.
- tRP has been satisfied following the prior PRECHARGE command.

When an all-bank refresh cycle has completed, all banks will be idle. After issuing REFab:

- tRFCab must be satisfied before issuing an ACTIVATE command.
- tRFCab must be satisfied before issuing a REFab or REFdb or REFqb command.

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Table 108 — REFRESH Command Scheduling Separation Requirements

Symbol	Minimum Delay From	To	Notes
tRFCab	REFab	REFab	
		Activate command to any bank	
		REFdb (or REFqb)	
tRFCdb (or tRFCqb)	REFdb (or REFqb)	REFab	
		Activate command to same bank as REFdb (or REFqb)	
tdbR2act (or tqbR2act)	REFdb (or REFqb)	Activate command to different bank than REFdb (or REFqb)	
tRRD_L	Activate	REFdb (or REFqb)	1
tdbR2dbR (or tqbR2qbR)	REFdb (or REFqb)	REFdb (or REFqb)	

NOTE 1: A bank shall be in the idle state before it is refreshed, so following an ACTIVATE command REFab is prohibited; REFdb (or REFqb) is supported only if it affects a bank that is in the idle state.
NOTE 2: "Same bank" and "different bank" here indicate "same bank pair" and “different bank pair", respectively. Please refer to "Refresh Bank#" row in previous Table - Bank and Refresh counter increment behavior for further information.

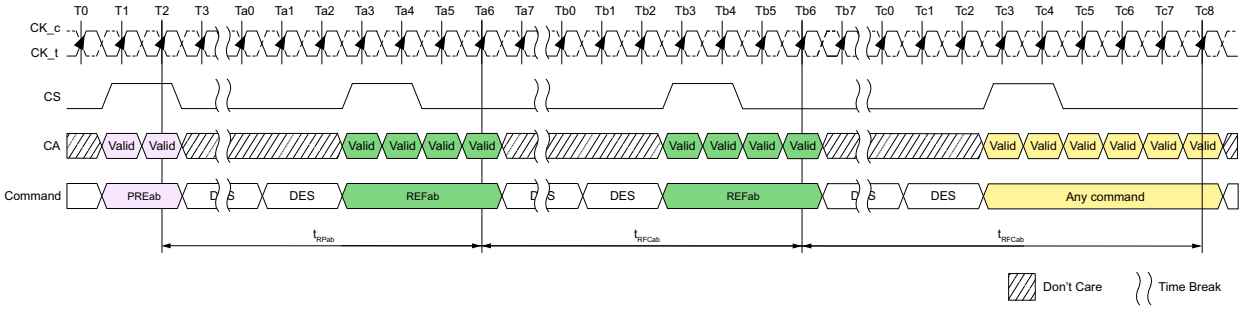


Figure 127 — All-Bank Refresh Operation

NOTE: 1: PREab is only shown for illustration. The requirement to issue the REFab command is that all banks are in an idle state (precharged and tRP satisfied).
NOTE: 2: Only DES, NOP, PDE, MRR and MRW commands are allowed after the REFab command is registered until tRFCab(min) expires.

NOTE: 3: If "Any command" is an ACT command, the end point of tRFCab is last edge (C5) of CK_t. Refer to below figure.

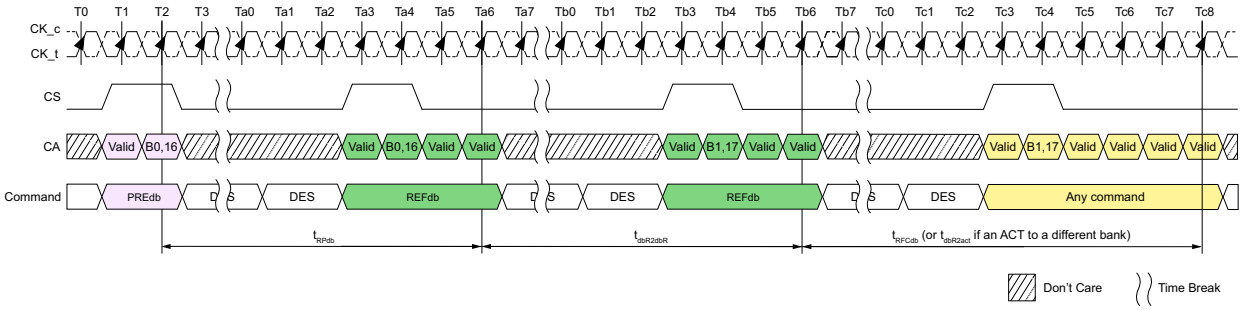


Figure 128 — Dual-Bank Refresh Operation

NOTE: 1: PREdb is only shown for illustration. The requirement to issue the REFdb command is that the banks receiving the REFdb command are in an idle state (precharged and tRP satisfied).
NOTE: 2: Operation to banks other than the banks being refreshed are supported during tRFCdb period.

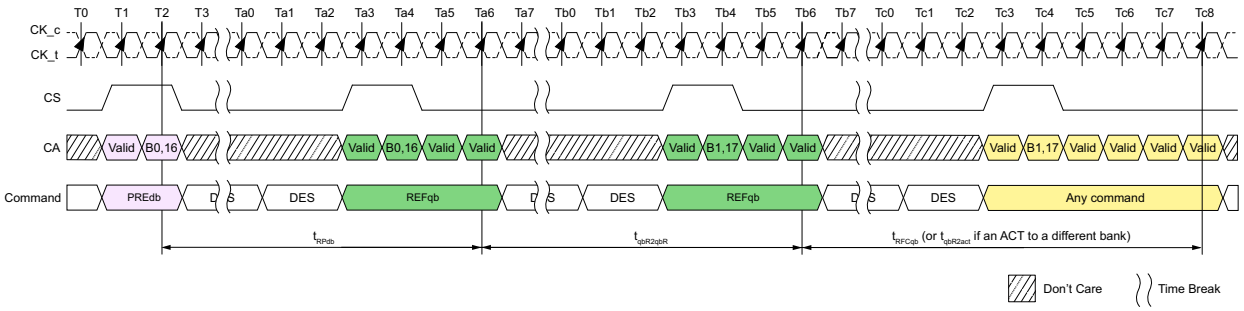


Figure 129 — Quad-Bank Refresh Operation

NOTE: 1: PREdb is only shown for illustration. The requirement to issue the REFqb command is that the banks receiving the REFqb command are in an idle state (precharged and tRP satisfied).
NOTE: 2: Operation to banks other than the banks being refreshed are supported during tRFCqb period.

The maximum average refresh interval (tREFI) requirement depends on the device's junction temperature (Tj), as indicated by bits MR4:OP[4:0]. When REFab commands are issued and the temperature is Tj ≤ 85°C, the maximum average refresh interval is tREFI. When REFab commands are issued and the temperature is 85°C < Tj ≤ 95°C, the maximum average refresh interval is tREFI/2.

The device includes a method for the host to indicate when REFRESH commands are being issued at the tREFI/2 refresh interval rate. The Refresh Interval Rate (RIR) indicator bit, MR4:OP[5], can be set if the host supports the refresh interval rate indication as part of the REFRESH command using C2:CA4. When enabled (MR4:OP[5]=1), the host issues REFab commands at tREFI with C2:CA4=L when Tj ≤ 85°C, and the host issues REFab commands at tREFI/2 with C2:CA4=H (Tj any allowable temperature).

When issuing Dual-bank REFRESH (REFdb) or Quad-bank REFRESH (REFqb) commands, the maximum average refresh interval is divided by 16 (tREFI or tREFI/2), depending on the MR4 temperature requirement.

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tREFI is based on the 8192 REFRESH commands per bank that need to be issued within the baseline tREFW=32ms refresh period on the device.

In general, REFRESH commands need to be issued every tREFI interval. To allow for improved efficiency in scheduling and switching between tasks, some flexibility in the absolute refresh interval is provided. A maximum of two (2) REFRESH commands can be postponed during operation.

In the case that 2 REFRESH commands are postponed in a row, the resulting maximum interval between the surrounding REFRESH commands is limited to 3 x tREFI. At any given time, a maximum of 3 REFab commands can be issued within 1x tREFI. Self-Refresh may be entered with a maximum of 2 REFRESH commands being postponed. After exiting Self-Refresh with one or more REFRESH commands postponed, additional REFRESH commands may be postponed to the extent that the total number of postponed REFRESH commands (before and after the Self-Refresh) will never exceed 2. During Self-Refresh, the number of postponed REFRESH commands does not change.

The following figure describes postponed All-bank REFRESH commands.

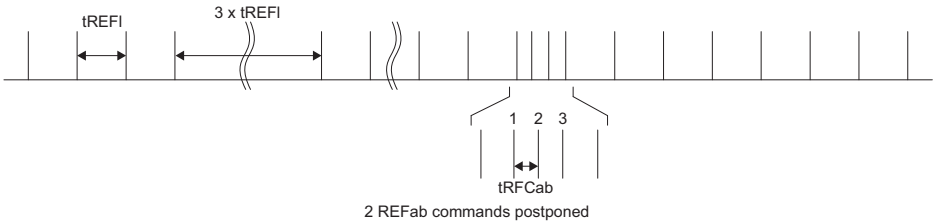


Figure 130 — Postponing Refresh Commands (Example)

7.5.1.1. Refresh Requirements

Table 109 — Refresh Requirement Parameters (Sheet 1 of 2)

Refresh Requirements		Symbol	Density per Die					Units	Notes
			32Gb	48Gb	64Gb	96Gb	128Gb		
Refresh window (tREFW) (1x Refresh)		tREFW	32					ms	2
Average refresh interval (1x refresh)	REFab	tREFI	3.906					µs	2
	REFdb(or REFqb)	tREFIdb(or tREFIqb)	244					ns	2
Refresh Cycle Time (All-bank)		tRFCab	505	715	715	TBD	TBD	ns	
Refresh Cycle Time (Dual-bank or Quad-bank)		tRFCdb (or tRFCqb)	160	210	210	TBD	TBD	ns	

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Table 109 — Refresh Requirement Parameters (Continued) (Sheet 2 of 2)

Refresh Requirements	Symbol	Density per Die						
		32Gb	48Gb	64Gb	96Gb	128Gb	Units	Notes
REFdb to REFdb or REFqb to REFqb different bank delay	tdbR2dbR (or tqbR2qbR)	90	90	90	TBD	TBD	ns	
REFdb or REFqb to ACT different bank	tdbR2act (or tqbR2act)	20	20	20	TBD	TBD	ns	

NOTE 1: Refresh for each pseudo-split die is independent of the other pseudo-split die. Power delivery in the user's system should be verified to make sure the DC operating conditions are maintained when multiple pseudo-split die are refreshed simultaneously.
NOTE 2: 1x refresh rate (tREFW = 32ms) is available for junction temperatures at or below 85°C, based on MR4. Refer to MR4:OP[4:0] for refresh rate details.

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7.5.2. Self-Refresh Operation

The Self-Refresh (SREF) ~~command~~ operation retains data in the device without external Refresh commands. While in SREF with Full Power-Down, the device retains data without external clocking~~DDR6 SDRAM even when the rest of the system is powered down and external clocking is disabled~~. The device has a built-in timer to accommodate the SREF operation. While in SREF, the device adjusts its internal average periodic refresh interval as needed, based on the on-die temperature sensor. The internal periodic refresh interval adjustment (increasing, decreasing or staying constant) does not require any external control.

The device enters Self-Refresh through the Self-Refresh Entry (SRE) command. SRE ~~is issued by asserting CS for two consecutive clock cycles~~ is a 2-cycle command with CA[2:0] = 000_b during the first cycle and CA[2:0] = 100_b during the second cycle. If the SRE command is issued with CA[3] HIGH during the second cycle, Self-Refresh is initiated with Full Power-Down. For 2p3 and 2p6 devices, a SRE command without Full Power-Down may be issued to each PSD independently. Before issuing the SRE command, the target PSD shall be idle with all banks precharged.

DDR6 SDRAM can operate in Self-Refresh across standard and elevated temperature ranges, managing its own power consumption as the operating temperature changes. For proper Self-Refresh operation, power supply pins (V_{DD}, V_{DDQ}, and V_{PP}) shall remain at valid operating levels.

During Self-Refresh without Full Power-Down, all input pins of the PSD are activated and ODT states are maintained according to the state of the mode registers, as well as NT ODT. If not in Full Power-Down, the device may accept the following commands during Self-Refresh after t_{SRED} is met: PDE, PDX, SRX, MRR, MRW, DES, NOP, and MCME (including MRUPDATE mode). All defined training modes in Chapter 4 are supported during Self-Refresh. Training modes must be properly exited and all required exit timing delays must be satisfied before issuing SRX. The minimum time that the device is required to remain in Self-Refresh is t_{SR}.

The device exits Self-Refresh through the Self-Refresh Exit (SRX) command. SRX ~~is issued by asserting CS for two consecutive clock cycles~~ is a 2-cycle command with CA[2:0] = 000_b during the first cycle and CA[2:0] = 110_b during the second cycle. If SRX is issued while the device is not in Self-Refresh, the command will not execute. For 2p3 and 2p6 devices, SRX shall be issued to each PSD independently. Once SRX is registered, only MRR, MRW, and MCME (including MRUPDATE mode) commands are allowed until t_{XSR} is satisfied.

The use of Self-Refresh introduces the possibility that an internally timed Refresh event can be missed when Self-Refresh Exit is registered because entering and exiting Self-Refresh will reset the internal bank counter. Upon exit from Self-Refresh, at least one REFab or sixteen REFdb/REFqb commands shall be issued before entry into a subsequent Self-Refresh. Refer to the Refresh section when postponing or pulling in Refresh commands. ~~This Refresh command does not count toward the t_{REFI}-based periodic refresh requirement and does not modify the postponed or~~

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~~pulled-in refresh counts; the Refresh command does count toward the number of maximum refreshes permitted within $2 \times t_{REFI}$.~~

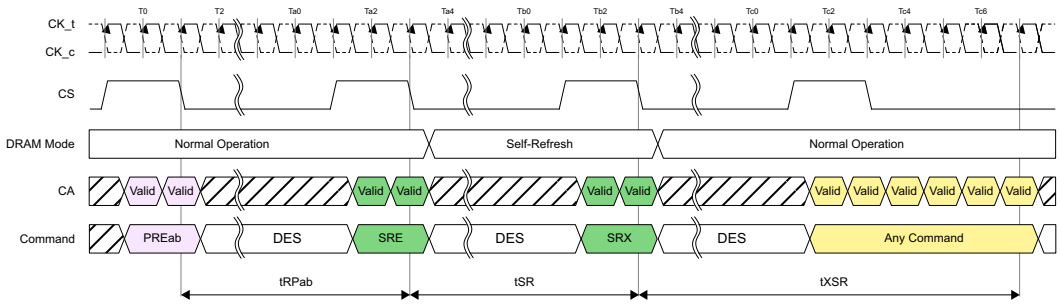


Figure 131 — Self-Refresh Entry/Exit Timing

NOTE: 1: Only PDE, PDX, SRX, MRR, MRW, DES, NOP, and MCME (including MRUPDATE mode) commands are allowed during Self-Refresh.

NOTE: 2: DES commands are shown for ease of illustration; other commands may be valid at these times where the CA input is indicated as "Don't Care".

7.5.2.1. Power-Down Entry and Exit during Self-Refresh

Entry to and exit from Full Power-Down mode is permitted while the device is in Self-Refresh. There are two ways to enter Full Power-Down during Self-Refresh: issue the SRE command with CA[3] HIGH during the second clock cycle or issue the Full Power-Down Entry (PDE) command after waiting t_{SRED} following the SRE command. ~~During~~For Full Power-Down in Self-Refresh on 2p3 and 2p6 devices, the internal clock is stopped and Full Power-Down is entered after both PSDs have received the Full PDE command. Before Self-Refresh can be exited on the target PSD, the Power-Down Exit (PDX) command shall be issued and CK Sync shall be completed. The timing parameters and operation of Full Power-Down are defined in the Power-Down section.

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The related timing parameters between Self-Refresh entry/exit and power-down entry/exit are shown in the following figures.

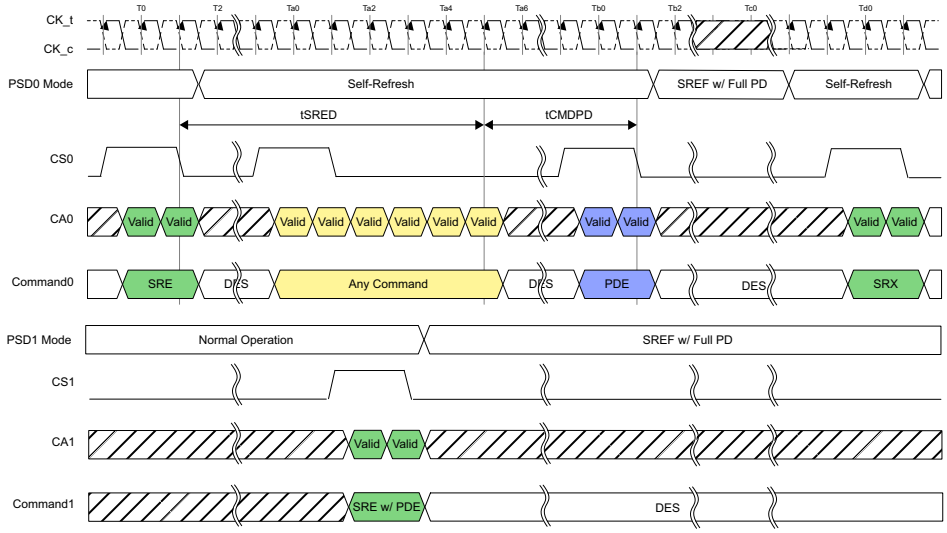


Figure 132 — Self-Refresh Entry/Exit Timing with Power-Down Entry/Exit in 2PSD Operation

NOTE: 1: SRX command on PSD0 after Full Power-Down is shown for ease of illustration; PDE, PDX, MRR, MRW, DES, NOP and MCME commands are valid where SRX is shown.

NOTE: ~~2: PDX and CK Sync are not shown on PSD0 between PDE and SRX for ease of illustration. Full Power-Down and CK Sync operation requirements are defined in their respective sections of the standard.~~

7.5.2.2. Self-Refresh Timing Parameters

Table 110 — Self-Refresh Timing Parameters

Parameters	Symbol	Min/Max	Value	Unit
Minimum Self-Refresh time (SRE to SRX)	tSR	Min	Max (15ns, 8nCK)	ns,nCK
Delay from SRE to first valid command	tSRED	Min	Max (15ns, 8nCK)	ns,nCK
Exit Self-Refresh to valid commands	tXSR	Min	tRFCab + Max(7.5ns, 8nCK)	ns,nCK

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7.5.3. Per Row Activation Counting (PRAC)

7.5.3.1. Introduction

Per Row Activation Counting (PRAC) monitors for row activity that may adversely affect the data stored in cells on neighboring rows within the DRAM array.

The intended implementation for PRAC is to add Activation Counter bits corresponding to every row in the DRAM. These bits store a count associated with the number of activations and the duration of the activations received by a row since the last time it was refreshed. Activations include any commands that result in an activation and precharge of a row.

7.5.3.2. Activation Counter Initialization (ACI)

Upon power-up or any time that device refresh requirements are violated, the Activation Counter bits may be in an unknown state, requiring initialization to put the bits into a known state.

Prior to use of the Activation Counter bits, a full array Activation Counter Initialization (ACI) must be performed. The DRAM will not track activation counts, nor issue an Alert Back-Off (ABO), until the Activation Counter bits are initialized.

To initialize the activation counter bits, the host sets MR~~4E~~35[0] = 1b to indicate to the device that a full array refresh will take place. Only REF, MRR, NOP and NT_ODT commands are permitted during initialization. Upon completion of a full ~~refresh~~ ACI cycle, the DRAM indicates completion by setting MR~~4E~~35[1] = 1b, which the host then follows by setting MR~~4E~~35[0] = 0b. The DRAM starts counting activations from that point forward and issues the ABO as necessary.

~~During ACI, the DRAM maintains refresh in the main array.~~ If the host re-enters ACI by setting MR~~4E~~35[0] = 1b, the DRAM will reset MR~~4E~~35[1] = 0b until the initialization has been completed.

During ACI, the DRAM will perform refresh commands in the main array based on the ACI Main Array Refresh Control bit, MR34[4]. Performing an ACI with MR34[4]=0 (default) does not require the DRAM to refresh the main array, and any data previously written may be corrupted. Performing an ACI with MR34[4]=1 requires the DRAM to refresh the main array.

Like normal DRAM cells, the activation counter bits require refresh to maintain the stored values. Anytime refresh is violated during ACI or after ACI, the counter bits need set to a known state by repeating the ACI. ~~A full array refresh will set counter bits to a known state, but ABOs may be generated during the refresh.~~

During hPPR and MBIST/mPPR, the system does not maintain refresh to the DRAM. Therefore, the host must perform a full array ACI before rewriting data following the hPPR and MBIST/mPPR operations. During sPPR, the system is required to maintain refresh to DRAM. However, before the sPPR operation, the DRAM is not required to maintain data integrity (refresh) for the row element designated to be used for the repair. tPGM_Exit timing provides sufficient time to the DRAM to update the activation count of the new row as well as proactively perform mitigation for neighboring rows of the new row before completion of sPPR operation. Likewise for sPPR_Undo, the DRAM may update the activation count of the old row as well as proactively perform mitigation for neighboring rows of the old row before completion of sPPR operation.

7.5.3.3. Per Row Activation Counting Core Timing Parameters

PRAC requires additional counter cells per row in the DRAM to store Activation (ACT) command counts. Updating the Activation Counter cells requires a read-modify-write to the Activation Counter cells.

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An example of the timings affected by the updating of the activation counter bits is shown below. The "i" references DRAM internal commands, not external commands issued by the Host. In this example, the PRE command initiates an internal read (iRD) followed by an internal write (iWR) to increment the Activation Counting bits for all open rows affected by the PRE command. This is the Activation Counter Update (ACU), with the associated tACU timing.

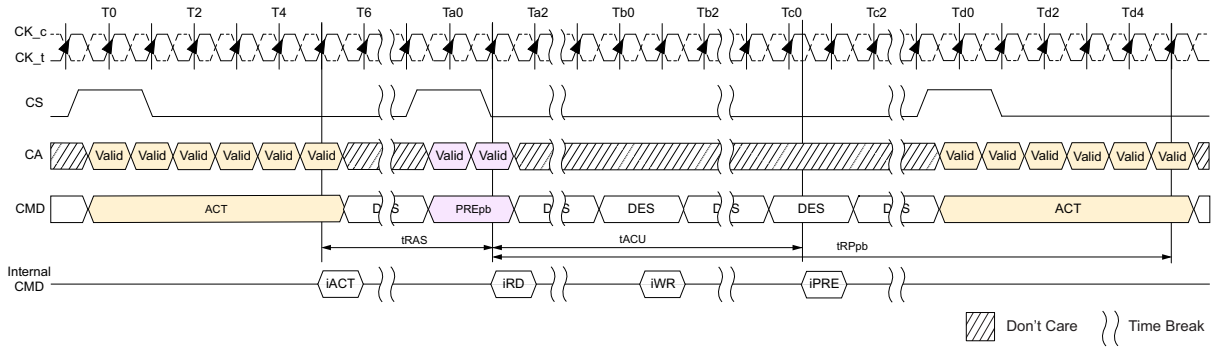


Figure 133 — Example ACT-PRE with ACU

7.5.3.4. Per Row Activation Counting Response

Since the DRAM is updating row activation counts based on row activity, back-off action is required upon one or more rows reaching a counter threshold level or levels, or when registers holding row addresses for targeted refreshing reaches or passes a critical level. This back-off action is a request by the DRAM for the host to pause normal activity for a recovery period so that refresh management (RFM) commands or other functions can be performed. The back-off protocol is described in the next section.

The counter and/or register threshold level(s) may be changed (reduced) by setting the adaptive activation counting levels in MR~~4D~~34[3:2] to something other than the default MR~~4D~~34[3:2] = 00b. These adaptive levels are based on the default starting value. The threshold level(s) reductions are approximately level A = default - 10%, level B = default - 20%, and level C = default - 30%. Changing levels requires an ACI ~~or~~with full array refresh to set/reset all counters to a known state below critical threshold levels and to empty the registers.

In addition to the back-off which is required to prevent activation counts from exceeding the threshold level, the device should proactively perform targeted refreshes during Refresh Management (RFM) and/or Refresh (REF) commands to reduce and/or prevent the occurrence of the alert back-off being triggered. RFM commands must be issued either periodically or based on a bank activation count, as described below.

The host may implement either Periodic RFM (PRFM) or Bank Activation Threshold (BAT), but is not required to support both modes simultaneously. When BAT is used to ensure proper refresh management, Periodic RFM commands are not required. Likewise when BAT is not active, PRFM shall be used to ensure proper refresh management.

7.5.3.5. Periodic Refresh Management (PRFM)

Periodic Refresh Management (PRFM) requires the host to send an RFM command to each bank within every tPRFMi window, with the maximum gap of 1.5*tPRFMi between each RFM command. RFMab, RFMdb and RFMpb commands are applicable, with RFMdb and RFMpb requiring the same time gap stated above on a per equivalent bank basis.

When using PRFM, time is partitioned into consecutive tPRFMi windows. The host shall issue at least one RFM command to each bank during each window. To allow scheduling flexibility across window boundaries, the maximum

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allowable gap between RFM commands to the same bank is 1.5 × tPRFMi. To avoid a strictly deterministic refresh pattern, the host is encouraged to vary the timing of RFM commands within and across tPRFMi windows, such that RFM activity remains periodic, not fixed to a rigid schedule.

Since tPRFMi is a window, no more than the equivalent of 1 RFM command may be postponed.

All references to dual bank, REFdb or RFMdb apply to 2p3 and 2p6 configuration devices. For 1p6 configuration devices, all references to “dual bank” become “quad bank”, references to REFdb become REFqb, and references to “RFMdb” become “RFMqb”, with equivalent command separation and timings.

7.5.3.6. Bank Activation Threshold (BAT)

Bank Activation Threshold (BAT) requires that the host track the level of activity that occurs on a per-bank basis. When the bank activation threshold (BAT) is reached (shown below), a RFMab, RFMdb or RFMpb command is issued to the bank or banks at the BAT. Upon issuing an RFM command, any bank or banks receiving the command can have its BAT count decremented by the BAT threshold, with the result not less than zero (0).

Bank activation counting is done for every ACT/PRE combination, and additional counts are added by dividing the activation time by tRC or longer to account for long tRAS.

The duration of the RFM command (tRFM) is sufficient for the row refresh span of ±1 and ±2 physically adjacent rows and the target row to be refreshed. Each row refresh span is determined as tRRFab, tRRFdb and tRRFpb.

Table 111 — Refresh Management Interval Timing Parameters

Parameter	Symbol	Value	Unit
Average Periodic RFM Interval	tPRFMi(avg)	32	uS
Bank Activation Threshold	BAT(max)	256	Activations

Table 112 — Row Refresh Timing Parameters

Parameter	Symbol	Value	Unit
All bank row refresh	tRRFab(min)	80	ns
Dual bank row refresh	tRRFdb(min)	75	ns
Per bank row refresh	tRRFpb(min)	70	ns

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Table 113 — Refresh Management Command Timing

Parameter	Symbol	Value	Unit
All bank RFM (RFMab)	tRFMab(min)	(4+1) * tRRFab	ns
Dual bank RFM (RFMdb)	tRFMdb(min)	(4+1) * tRRFdb	ns
Per bank RFM (RFMpb)	tRFMpb(min)	(4+1) * tRRFpb	ns
NOTE 1: PRAC-related MR changes are prohibited during the tRFMab(min), tRFMdb(min) and tRFMpb(min) periods.			

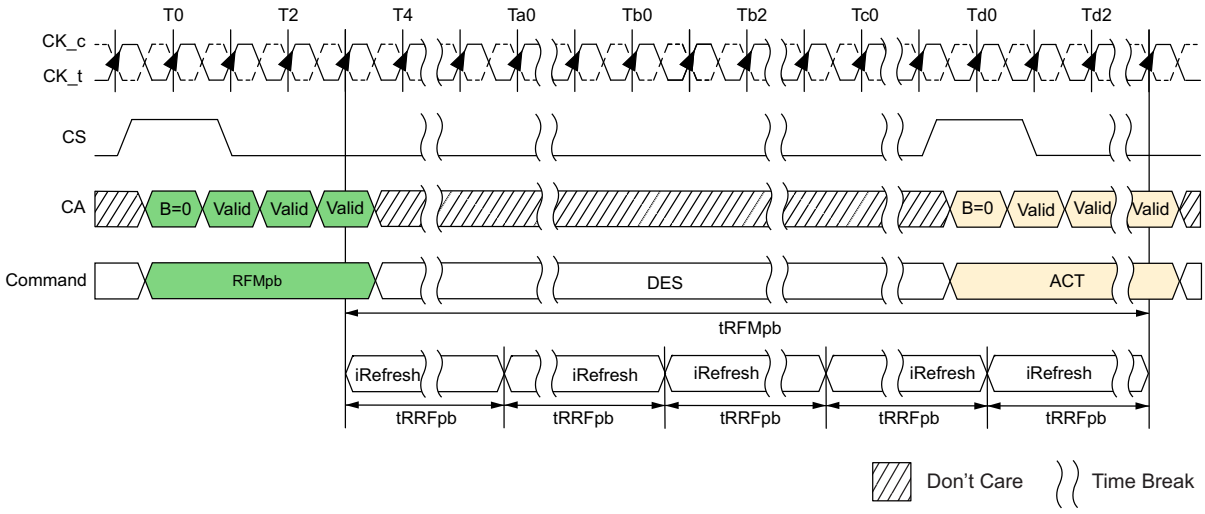


Figure 134 — tRRF Timing Definition

Similar to Refresh commands, a bank shall be idle before it can receive the RFM command.

The bank(s) receiving the RFM command will be inaccessible for the cycle time tRFM, however, other banks not being refreshed within the device during RFMpb or RFMdb commands are still accessible. During the RFMpb or RFMdb operation, any of the non-refreshing banks can receive an ACTIVATE or PRECHARGE command, be maintained in an active state, or be accessed by a READ or a WRITE command. When the RFM command has completed, the affected bank(s) will be in the idle state.

The RFM Command Scheduling Separation Requirements must be met when issuing RFM commands.

Table 114 — RFM Command Scheduling Separation Requirements (Sheet 1 of 2)

Symbol	Minimum Delay From	To	Notes
tRFMab	RFMab	RFMab, REFab	
		Activate command to any bank	
		RFMpb, RFMdb, REFdb	

Table 114 — RFM Command Scheduling Separation Requirements (Continued)

Symbol	Minimum Delay From	To	Notes
tRFMdb	RFMdb	RFMab, REFab	
		Activate command to same bank	2
		RFMpb, RFMdb, REFdb to same bank	2
tRFMpb	RFMpb	RFMab, REFab	
		Activate command to same bank	2
		RFMpb, RFMdb, REFdb to same bank	2
tbRFM2bRFM	RFMpb or RFMdb	RFMpb, RFMdb, REFdb to different bank	2
tbRFM2act	RFMpb or RFMdb	Activate command to different bank	2
NOTE 1: A bank shall be in the idle state before it is refreshed, so following an ACTIVATE command, RFMab is prohibited; RFMpb, RFMdb is permitted only if it affects a bank that is in the idle state. NOTE 2: "Same bank" and "different bank" indicate "either bank within the Refresh bank pair" and "a bank not within the Refresh bank pair", respectively. Please refer to "Refresh Bank#" column in the Refresh Operation section table "Bank and Refresh Counter Increment Behavior" for further information.			

Table 115 — Command Separation Timing Parameters

Requirement	Symbol	Max/Min/Typ	Value	Unit
RFMpb or RFMdb to REFdb, RFMdb or RFMpb different bank	tbRFM2bRFM	Min	90	nS
RFMpb or RFMdb to ACT different bank	tbRFM2act	Min	20	nS

7.5.3.7. Back-Off Protocol

To maintain data integrity, the DRAM may request the host to pause normal activity for a maintenance period where refresh management (RFM) commands or other functions may be performed. This is called a "back-off."

7.5.3.8. Alert Back-Off Protocol

The back-off protocol uses the ALERT signal and mode register settings to provide a handshake between the DRAM and the host. This is the alert back-off (ABO).

7.5.3.9. PRAC-Triggered Alert Back-Off

Upon reaching critical threshold levels by one or more rows, or when registers holding addresses for targeted refreshing reaches or passes a critical level, the DRAM asserts ALERT and sets the ABO flag at MR~~4E~~C8[0] = 1b, indicating that ALERT was asserted due to additional refresh management (RFM) being required. All commands received by the device are executed while ALERT is asserted to maintain synchronization.

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Commands that may trigger the ABO include PREab/PREdb/PREpb, WRA, RDA, REFab/REFdb, and RFMab/RFMdb/RFMpb. ALERT will be asserted after issuing the trigger command but before ~~the max(5ns,10nCK)~~ a maximum of 8ns following the completion of the respective command duration (e.g. PREab/PREdb/PREpb = tRP, REFab/REFdb = tRFC, RFMab/RFMdb/RFMpb = tRFM).

Since the PREpb required during hPPR/sPPR and the PREab during mPPR have different functions than closing a row, no counter bits will be incremented associated with the repaired row, therefore no ABO will be triggered in this instance.

7.5.3.10. Alert Back-Off

To instantiate the ABO, ALERT is asserted with a fixed duration of clock cycles as defined by tALERT_PW.

Table 116 — Alert Back-Off Timing Parameters

Parameter	Symbol	Max/Min/Typ	Value	Unit
tALERT_PW	Alert back-off pulse width	Typ	TBD	nCK
tABO_ACT	Normal traffic to the device allowed	Max	180	ns

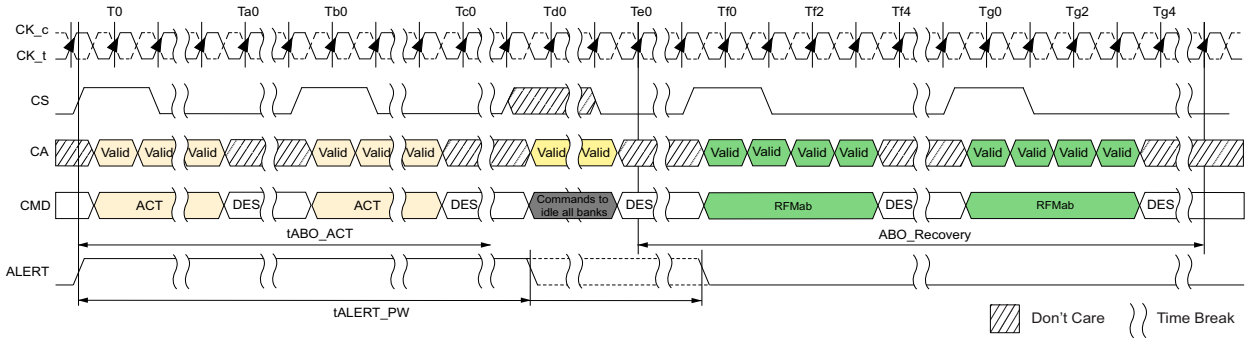


Figure 135 — ABO: MRFMaACT = 2 Case (MR~~4D~~34[1:0] = 01b)

After the ALERT is asserted, a bounded number of ACTIVATE (ACT) commands are permitted before the host starts the RFM-based recovery period. Issuing ACT commands along with normal traffic is allowed within the tABO_ACT window, however no additional ACT commands are allowed at the end of the tABO_ACT window until ABO_Recovery is complete. The host shall complete idling all banks within tREFI after the ALERT is asserted, which is followed by the recovery period (ABO_Recovery).

The recovery period begins when all banks have been idled in preparation for RFMab commands. The DRAM self-clears the ABO flag by setting MR~~4FC8~~[0]=0b by the end of the ABO_Recovery period. ABO_Recovery ends tRFMab after the last RFMab command or after tALERT_PW, whichever takes longer.

The only RFM command allowed during the ABO_Recovery period is RFMab. The duration of the RFMab command is described in the previous section (Per Row Activation Counting Response). The number of RFMab commands required to be issued per ABO maintenance period is defined by mode register ABO_RFM bits, MR~~4D~~34[1:0]. Systems that have stricter requirements for isochronous bandwidth may limit the number of RFMab commands by setting MR~~4D~~34[1:0] to a lower value.

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If additional RFM commands are needed, the ABO protocol is restarted following the ABO delay, as described below.

During the ABO_Recovery period, a host can issue only RFMab, REFab/REFdb, PDE, PDX, SRE, PREab/PREdb/PREpb, NOP, NT_ODT, MRW (PRAC-related MR changes not allowed during ABO_Recovery) and MRR commands.

Refresh commands are issued to meet device refresh requirements, but they do not perform any of the required refresh management.

Power Down Entry and Exit (PDE, PDX) and Self Refresh Entry and Exit (SRE, SRX) are allowed during recovery period.

In Self Refresh, the device de-asserts ALERT if triggered by an ABO. Upon exiting Self Refresh, the device re-evaluates and issues a new ALERT if ABO is still needed. In power-down, the device keeps the current ALERT state.

7.5.3.11. Alert Back-Off Delays

ABO_Delay defines an interval in which the device services a minimum number of commands that may trigger an ABO before it can reassert ALERT. The ABO_Delay count is specified by MR34[1:0]. Adherence to the ABO_Delay count is set by the ABO_Delay Control bit, MR34[5], where the default MR34[5]=0 requires that ALERT will not be reasserted until the ABO_Delay count is satisfied, and MR34[5]=1 allows the ALERT to be reasserted as needed to maintain data integrity.

~~ABO_Delay is specified by the minimum number of commands in MR4D[1:0].~~

Commands that may trigger ALERT include PREab/PREdb/PREpb, WRA, RDA, REFab/REFdb~~,~~ and RFMab/RFMdb/RFMpb ~~and Manual ECS~~. Any of these commands that occur are included in the total ABO_Delay count allowed.

Following ABO_Delay, any of the commands included in the PRAC-Triggered Alert Back-Off section above may trigger the ABO~~, as well as an ACT command~~.

ALERT may be shared between sub-channels in a channel, but parameters associated with ABO (tABO_ACT, tALERT_PW, ABO_Recovery, ABO_Delay) are independent per sub-channel.

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Table 117 — Alert Back-Off Delay Parameter

Parameter	Symbol	Value	Unit
ABO_Delay	Minimum number of ~~ACTIVATION commands~~ activations allowed ~~per bank~~ prior to reasserting ALERT	MR~~4D~~34[1:0]	Activations

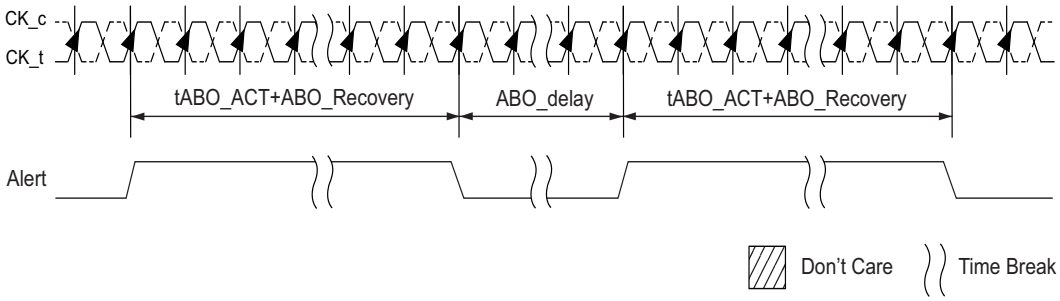


Figure 136 — ~~Bank-to-Bank~~ Alert Back-Off Delay

7.5.3.12. ALERT Priorities

At times, Write CRC, Alert Back-Off, and CA parity errors may occur simultaneously ~~Alert Back-Off may occur simultaneously with other functions that also use ALERT~~, resulting in an overlap of ALERT assertions or the occurrence of ~~the two~~ events back-to-back (non-overlapped) may result in a short "glitch" on ALERT. For all events, the DRAM shall treat the ALERT as an "OR" function where ALERT is issued whenever an event occurs, and the DRAM sets the corresponding Write CRC ~~or~~, Alert Back-Off (MR~~4F~~C8[0]), or CA parity error flag indicating which type of event, or possibly ~~both~~ multiple, occurred.

7.5.3.13. Activation Counter Errors

The DRAM has the capability to detect errors that occur within the counter bits, and the following protocol defines how the device reports the associated failing row address to the host.

If ~~a counter bit error~~ an Activation Counter Error (ACE) is detected, the device sets the Counter Error Flag, MR~~4F~~C8[1] = 1b, if not previously set, and. ~~While the Counter Error Flag is set,~~ the counter function for the specific row with the error is disabled. The DRAM does not assert the ABO to request RFM commands for errors associated with that row. All other rows without counter errors continue to count activity and assert ALERT, if necessary.

During ECS, if the Counter Error Flag is set and the DRAM logs a failing row address, the DRAM will update MR~~50-MR53~~C9-MRCB with the failing row address and set the Counter Address Report Flag, MR~~4F~~C8[2]=1b, when the ECS reaches the failing row. Only the first failing row will be reported. Additional rows will be reported in a subsequent scrub if they continue to fail after the Counter Address Report Flag is cleared by the host.

A counter RAS error that occurs during the ECS may still trigger the counter error flag, but may not set the counter address report flag and report the failing row address.

The host clears the counter error flag and counter address report flag, once the necessary information about the failing row has been collected and mitigative actions have been taken.

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7.5.3.14. Per Row Activation Counting Testing

Per Row Activation Counting (PRAC) testing is activated by setting MR4E35[2] = 1b. Once enabled, the counter for a target row to be tested is initialized by the host setting MR4E35[3] = 1b, followed by an ACT/PRE to the target row. The host then sets MR4E35[3] = 0b and issues 32 activations or fewer to the target row or rows to trigger the ABO (multiple increments to the counter bits may occur depending on tRAS duration, mimicking normal operations). ABO is required as described previously before testing subsequent rows.

Depending on counter bit starting values, all DRAMs in the package or on a DIMM devices may not trigger the ABO at the same time (detected by MR4FC8[0]); however, all devices will trigger within 32 activations.

After PRAC testing is complete, the test mode is cleared by the Host setting MR4E35[2] = 0b. Since the counter cells will be in an unknown state following testing, a full ACI is required before resuming normal operations.

Refresh commands are not allowed while PRAC Testing is enabled. Data in normal cells are not maintained during in PRAC testing.

7.5.3.15. Alert Verification

ALERT verification is triggered by setting MR4E35[4] = 1b. The DRAM asserts ALERT and sets the ABO flag at MR4FC8[0] = 1b within tMRD after the MRW and will keep ALERT asserted for tALERT_PW. The host should follow all normal ABO procedures for clearing the ALERT. MR4E35[4] and MR4FC8[0] will be self-cleared by the device prior to the end of the ABO_Recovery period.

Data in normal cells are maintained during ALERT verification, as long as refresh is maintained.

7.5.3.16. Mode Registers

NOTE: Self-clearing mode register bits (MR35[4], MRC8[0]) are automatically reset by the DRAM and do not require host intervention to clear.

7.5.3.17. MR4D34 - PRAC Config

MR#	OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
4D	RV		ABO_Delay Control	ACI Main Array Refresh Control	Adaptive PRAC		MRFMaACT	

MR4D34[7:0]				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:6	RV			
5	RW	0	MR34[5]: ABO_DELAY_CTRL ABO_Delay Control 0: Enabled 1: Disabled	

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MR~~4D~~34[7:0]				
Bits	**Attribute**	**Default**	**Description**	**Note**
4	RW	0	MR34[4]: ACI_REF_CTRL ACI Main Array Refresh Control 0: No main array refresh requirement 1: Main array refresh required	
3:2	RW	0h	MR~~4E~~34[3:2]: ADAPTIVE_PRAC Adaptive PRAC 00: Default Mitigation Threshold 01: Mitigation Level A 10: Mitigation Level B 11: Mitigation Level C	
1:0	RW	0h	MR~~4E~~34[1:0]: MRFMAACT ~~MRFMaACT~~Minimum RFMab Commands during ABO_Recovery 00: 4 01: 2 10: 1 11: Reserved	

7.5.3.18. MR~~4E~~35 - PRAC Init & Testing

MR#	OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
~~4E~~35	RV			Alert Verification	PRAC Testing Initialization	PRAC Testing Enable	ACI Complete	ACI Enable

MR~~4E~~35[7:0]				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:5	RV			
4	RW	0	MR~~4E~~35[4]: ALERT_VERIF Alert Verification 0: ALERT Verification Disabled 1: ALERT Verification Trigger	
3	RW	0	MR~~4E~~35[3]: PRAC_TESTING_INIT PRAC Testing Initialization 0: PRAC Testing Initialization disable 1: PRAC Testing Initialization enable	

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MR4E35[7:0]				
Bits	**Attribute**	**Default**	**Description**	**Note**
2	RW	0	MR4E35[2]: PRAC_TESTING_EN PRAC Testing Enable 0: PRAC Testing disabled 1: PRAC Testing enabled	
1	R		MR4E35[1]: ACI_COMPLETE ACI Complete 0: Initialization not completed 1: Initialization completed	1
0	RW	0	MR4E35[0]: ACI_EN ACI Enable 0: Normal operating mode 1: Initialization mode	1
NOTE 1: When ACI is enabled (MR4E35[0]=1), MR4E35[1] is reset to 0 by the DRAM.				

7.5.3.19. MR4FC8 - PRAC Flags

MR#	OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
4FC8	RFU					Counter Address Report	Counter Error	Alert Back-Off

MR4FC8[7:0]				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:3	RV			
2	RW	0	MR4FC8[2]: COUNTER_ERROR_REPORT Counter Address Report 0: Clear 1: Failing row address available in MR50-MR53C9-MRCC	1
1	RW	0	MR4FC8[1]: COUNTER_ERR Counter Error 0: Clear 1: Counter Error detected	1
0	R		MR4FC8[0]: ALERT_BACK_OFF Alert Back-Off 0: Clear 1: Source of ALERT	
NOTE 1: These flags can only be set by the DRAM. MRW of MR4FC8[1]=0 or MR4FC8[2]=0 will clear the flag. MRW of MR4FC8[1]=1 or MR4FC8[2]=1 by the host will be ignored(1 if set by the DRAM, otherwise 0).				

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7.5.3.20. MR~~50~~C9 - PRAC RAS Event

MR#	OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
~~50~~C9	PRAC RAS Event Rows[7:0]							

MR~~50~~C9[7:0]				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	R		MR~~50~~C9[7:0]: PRAC_RAS_EVENT_R[7:0] Activation Counter Error Row Address R[7:0]	

7.5.3.21. MR~~51~~CA - PRAC RAS Event

MR#	OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
~~51~~CA	PRAC RAS Event Rows[15:8]							

MR~~51~~CA[7:0]				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:0	R		MR~~51~~CA[7:0]: PRAC_RAS_EVENT_R[15:8] Activation Counter Error Row Address R[15:8]	

7.5.3.22. MR~~52~~CB - PRAC RAS Event

MR#	OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
~~52~~CB	PRAC RAS Event BG[3:0]				PRAC RAS Event BA	PRAC RAS Event Rows[18:16]		

MR~~52~~CB[7:0]				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:4	R		MR~~52~~CB[7:4]: PRAC_RAS_EVENT_BG[3:0] Activation Counter Error Bank Group Address BG[3:0]	
3	R		MR~~52~~CB[3]: PRAC_RAS_EVENT_BA Activation Counter Error Bank BA[0]	
2:0	R		MR~~52~~CB[2:0]: PRAC_RAS_EVENT_R[18:16] Activation Counter Error Row Address R[18:16]	

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7.5.3.23. MR~~53~~CC - PRAC RAS Event

MR#	OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
~~53~~CC	RV							~~PRAC_RAS_EVENT_PSD~~RV

MR~~53~~CC[7:0]				
Bits	**Attribute**	**Default**	**Description**	**Note**
7:~~1~~0	RV			
~~0~~	~~R~~		~~MR53[7:0]: PRAC_RAS_EVENT_PSD~~ ~~PRAC RAS Event PSD~~ ~~0: Activation Counter Error PSD0~~ ~~1: Activation Counter Error PSD1~~	

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7.6. Power-down mode

7.6.1. Types of Power-Down modes

DDR6 SDRAM supports three power-down modes: Full Power-Down (Full-PD), Full Power-Down with Frequency Change (Full-PD FC), and ODT Power-Down (ODT-PD).

- Full-PD : In this mode, power consumption is minimized by disabling both the CK and CA input buffers.
- ODT-PD : In this mode, only the CA input buffers are disabled, while the CK buffer remains active. This configuration allows the DDR6 to support the Write NT-ODT command.

7.6.2. Power-Down Entry and Exit

Full-PDis entered when the PDE command is registered with C0:CA4=L. Full-PD with Frequency change is entered when the PDE command is registered with C0:CA4=Land C1:CA3=H. ODT-PDis entered when the PDE command is registered with C0:CA4=H.

Full-PDalso can be entered with Self-Refresh Entry (SRE) command with C0:CA3=H. Likewise, Full-PD with Frequency change also can be entered with C1:CA3=H.

Table 118 — . Power-down mode command definition

Command	CS	CA0	CA1	CA2	CA3	CA4	CA5	CK_t
Full PDE w/o Freq. Change	H	L	L	L	V	ODT=L	V	C0
	X	L	H	L	FC=L	PAR1	V	C1
Full PDE w/ Freq. Change	H	L	L	L	V	ODT=L	V	C0
	X	L	H	L	FC=H	PAR1	V	C1
ODT PDE	H	L	L	L	V	ODT=H	V	C0
	X	L	H	L	V	PAR1	V	C1
SRE w/ Full PDE w/o Freq. Change	H	L	L	L	PD=H	V	V	C0
	X	L	L	H	FC=L	PAR1	V	C1
SRE w/ Full PDE w/ Freq. Change	H	L	L	L	PD=H	V	V	C0
	X	L	L	H	FC=H	PAR1	V	C1

The PDE command must not be issued while the following operations are in progress on the same PSD:

- Mode Register Read
- Mode Register Write
- Read
- Write

Also, DDR6 SDRAM cannot be placed in Power Down state during following state and until those modes have completely terminated. (SRE with Power Down Entry command is contained in Power Down Entry command. See Command Truth Table for detail).

- In MRUPD Mode

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- When any Functional Mode was enabled via MRW: MR5=2'h00 ~ 2'hxy
- When MBIST enabled via MRW: MR31[0]=1B
- When mPPR enabled via MRW: MR31[1]=1B
- When hPPR enabled via MRW: MR32[0]=1B
- When sPPR enabled via MRW: MR32[1]=1B
- When ACI enabled via MRW: MR35[0]=1B
- VREF(CS) Value setting via MRW: MR83[6:0]
- VREF(CA) Value setting via MRW: MR84[6:0]
- VREF(DQ) Value setting via MRW: MR85[6:0]

7.6.3. Full-PD Entry and Exit

Entering Full-PDis permitted when the PSD is in IDLE or Self-Refresh. In IDLE PD, all banks are closed and all other commands to the PSD needs to be completed. The PSD will be in an Idle state and ready to enter a Power-Down mode.

Entering Full-PDdeactivates the input and output buffers, excluding CS and RESET_n which remain enabled throughout the duration of this mode. CS assertion is not permitted until the Power-Down Exit (PDX).

To exit Full-PD, the host asserts CS high with the tCSH period to wake up the deactivated input buffers. After tXP_-FULL, CK Sync must be initiated.

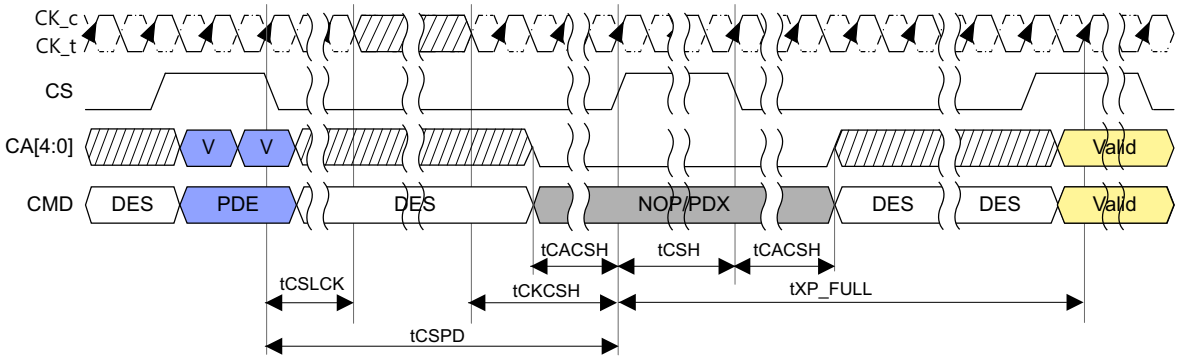


Figure 137 — Full Power-down entry and exit

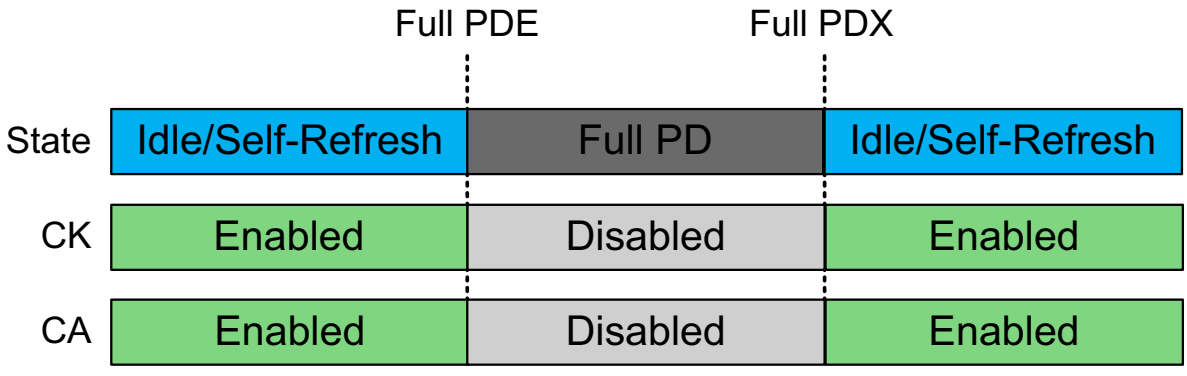


Figure 138 — Buffer-control on Full Power-down entry and exit

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In the Full-PD with Frequency change, the DRAM flips the FSP-OP value at PDX sequence and begins referencing the FSP-related Mode Registers (MRs). Frequency change sequence is finally determined by the CS pulse-width of PDX sequence. The host shall assert CS for duration tCSH during a frequency change, and for duration tCSH_FC_-Cancel when canceling the change.

Since the frequency change operation may involve VREF and ODT values, the changes will be performed after the CS is de-asserted. After the FSP MR transition, deactivated input buffers are enabled. After tXP_FC, CK sync must be initiated.

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Since each rank could have different ODT values in a multiple physical-rank package/module, the FSP-OP transition needs to be changed concurrently. The host initiates the FSP update sequence concurrently on all ranks by asserting CS on same clock edge.

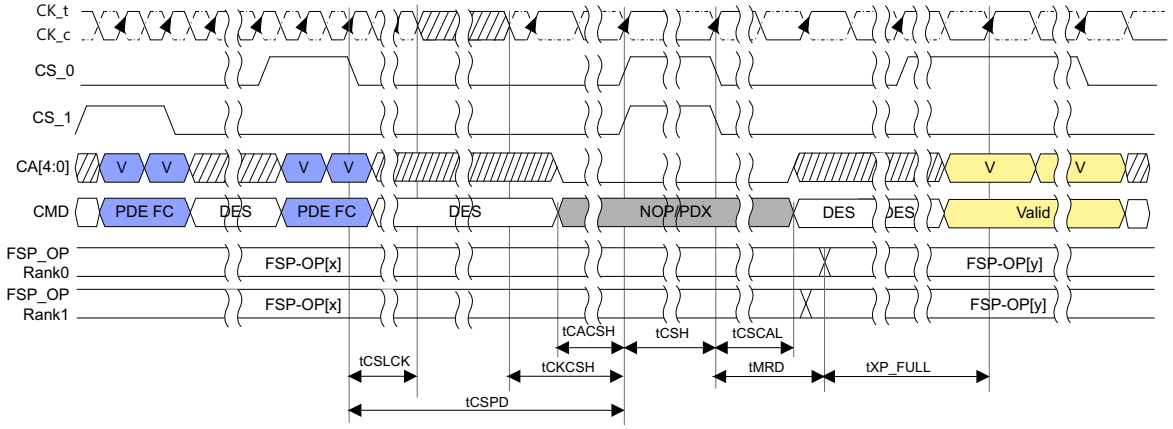


Figure 139 — PDE/PDX with Frequency Change without VREFCA/CS change in Multi-rank

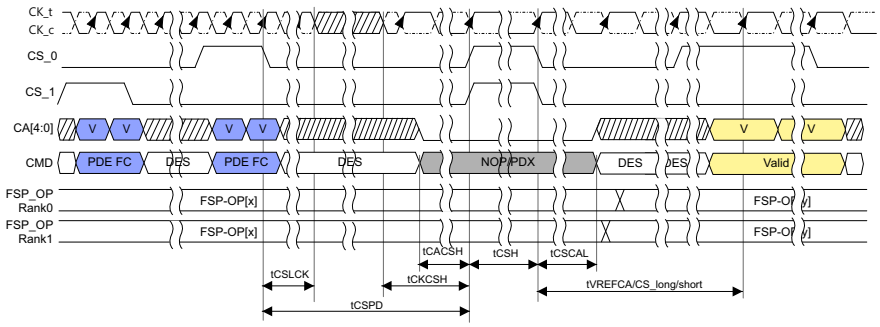


Figure 140 — PDE/PDX with Frequency Change with VREFCA/CS change in Multi-rank

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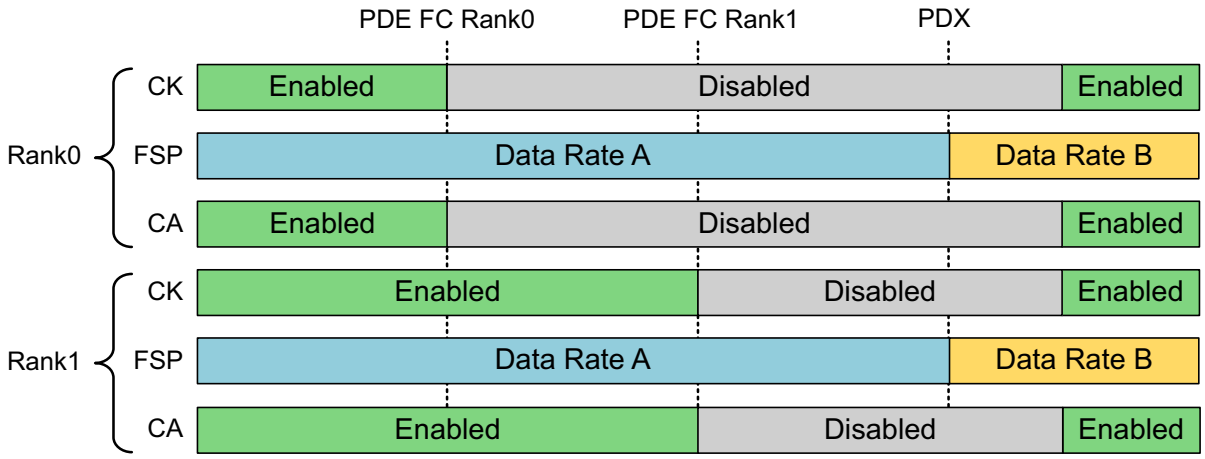


Figure 141 — Buffer control on PDE/PDX with frequency change in multiple physical-ranks

7.6.4. ODT-PD Entry and Exit

Entering ODT-PD is permitted when the PSD is in IDLE. In ODT-PD, all banks are closed and all other commands to the PSD needs to be completed. The PSD will be in an Idle state and ready to enter a Power-Down mode.

Entering ODT PD deactivates the input and output buffers, excluding CS, CK_t/c, and RESET_n. While in ODT PD Mode, only Write NT ODT commands and PDX ODT commands are allowed tCPDED after ODT-PDE command. Clock input buffers CK_t and CK_c remain enabled, allowing the appropriate Write NT ODT command to be passed through and decoded while ODT PD Mode (i.e., the DRAM monitors CS and will not exit ODT PD as a result of a valid Write NT ODT command being registered). Table 1 shows the command truth table for ODT PD Mode.

If CA-parity check is enabled upon entering the ODT-PD, the DRAM disables CA parity checking and operates based on CS only. However, the WR NT-ODT or PDX_ODT operation remains delayed by tCA_PAR_Delay in the same

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manner as the command issued before entering ODT-PD mode. Upon issuance of PDX-ODT, the DRAM resumes checking CA-parity after the tXP_ODT period.

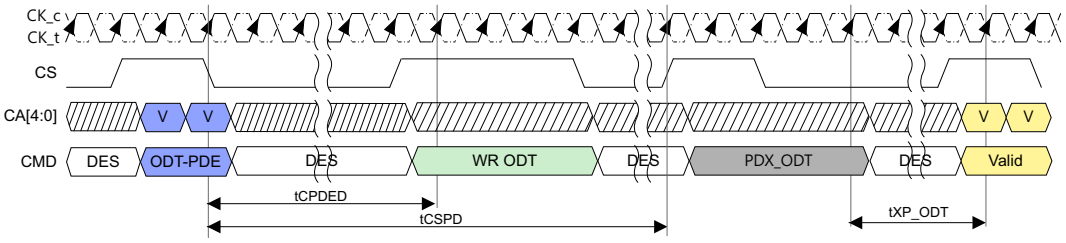


Figure 142 — ODT Power-Down mode entry and exit

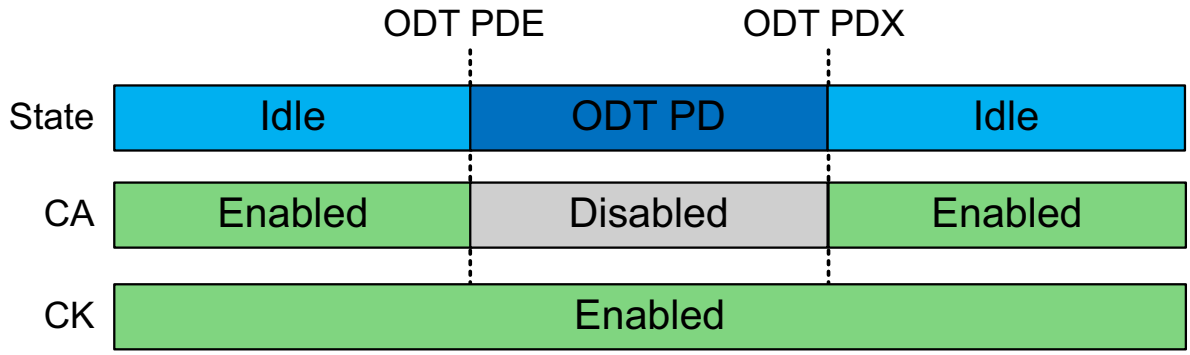


Figure 143 — Buffer control on ODT Power-Down mode entry and exit

Table 119 — . Command Truth Table in ODT Power-down mode

Command	CS	CK_t edge
PDX_ODT	H	C0
		C1
	L	C2
		C3
Write NT ODT	H	C0
		C1
		C2
		C3

When ODT PD is entered, the DRAM will be switching from decoding all CA bus command bits to only decoding CS during the tCPDED period. During tCPDED, all CA command bits must be valid when CS is asserted for a Write NT ODT command. Likewise, following the PDX_ODT command, all CA bits must be valid for Write NT ODT command also, since the DRAM will be transitioning to decoding all bits. All CA bits must be valid when CS is asserted between

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the PDE command and the completion of tXP_ODT upon PDX_ODT.The host shall issue Write NT ODT command tCPDED after ODT-PDE and tXP_ODT after PDX_ODT.

Table 120 — . Timing parameter related to power-down

Parameter	Symbol	Min	Max	Note
Command path disable delay	tCPDED	Max(5ns,16nCK) +tCA_PAR_Delay	-	1
Delay time from PDE to CS goes High for PDX	tCSPD	Max(10ns,16nCK) +tCA_PAR_Delay	3*tREFI	1
Valid Clock Requirement after PDE	tCSLCK	Max(5ns,16nCK)+PL		2
Valid Low Requirement for CA before CS goes High	tCACSH	3ns		
Minimum CA Low Duration time @ PDX	tCSCAL	3ns		
Valid Clock Requirement before CS goes High	tCKCSH	5ns		
Full Power-Down Exit command to Valid command	tXP_FULL	Max(10ns, 16nCK)		
ODT Power-Down Exit command to valid command	tXP_ODT	Max(7.5ns, 16nCK)		
CS High Pulse-width at Full-PD exit	tCSH	5ns	7.5ns	
CS High Pulse-width at Full-PD exit with Frequency Change Cancel	tCSH_FC_Cancel	50ns	-	
MRR command to Full PDE command	tMRRPDEN	CL+RU(tCK2DQO/tCK) +BL/2+tRPST		
MRW command to Full PDE command	tMRWPDEN	tMRD		
Read w/Auto Precharge command to Full PDE command	tRDAPDEN	Max(tMRRPDEN, nRTP)		
Write w/Auto Precharge command to Full PDE command	tWRAPDEN	nWTP		

NOTE 1: tCA_PAR_Dealy is considered as Parity Latency (MR09[6:4]) when both CA parity Check and CA Parity Command Blocking(MR01[4:5]) are enabled and considered as 0 when any of them are disabled.
NOTE 2: PL is considered as Parity Latency (MR09[6:4]) when CA parity Check (MR01[5]) is enabled and 0 when disabled.

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7.6.5. PSD operations in Power-down

For 2p3 or 2p6 device configurations, Full PDE and PDX can be issued to each PSD independently. CK buffer is disabled only when both PSDs enter Full PD.

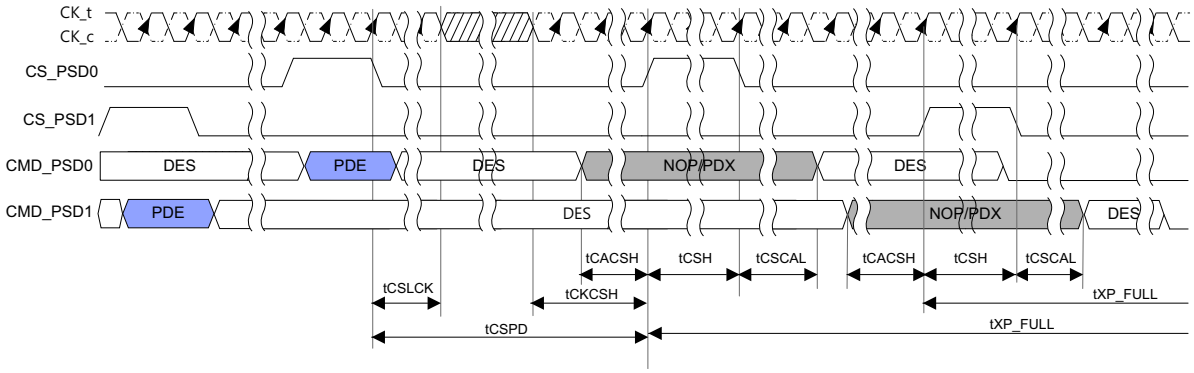


Figure 144 — Full Power-down entry and exit at 2p3 or 2p6

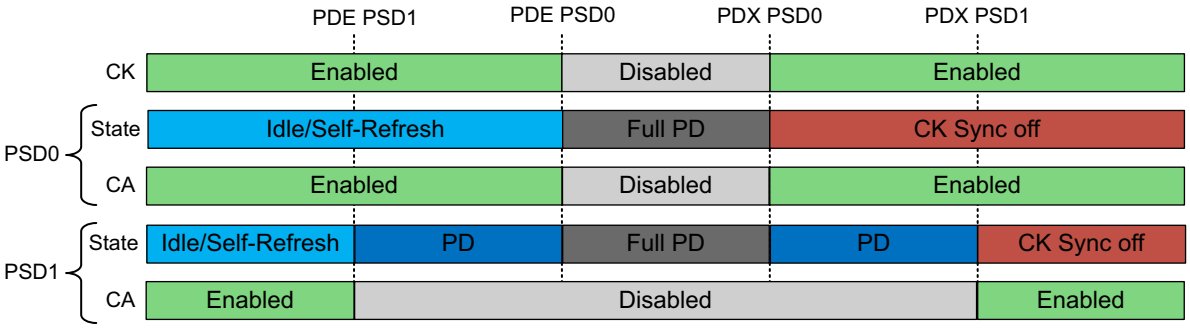


Figure 145 — Buffer control on Full Power-down entry and exit at 2p3 or 2p6

For 1p6 or 1p2 configurations, Full Power-down mode entry and exit is controlled with corresponding CS and CA BUS.

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ODT PD mode supports independent operation for each PSD. When one PSD enters ODT PD Mode, the PSD not in ODT PD Mode supports normal commands, except for the MCME and MRW commands.

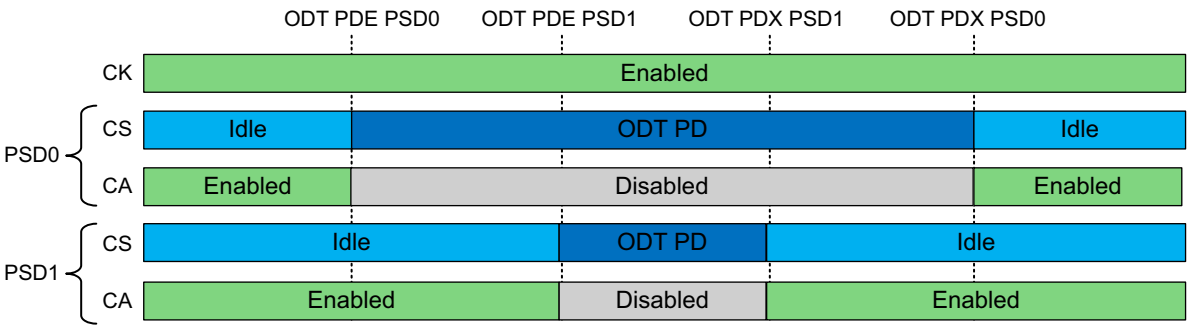


Figure 146 — Buffer control on ODT Power-down entry and exit at 2p3 or 2p6

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7.7. Wake-up CS (WUCS)

7.7.1. Introduction

DDR6 supports Wake-up CS (WUCS) functionality. The feature allows the DRAM to enter a clock-gating state on the CMD/ADDR interface during periods of inactivity and to exit that state in a controlled, deterministic manner ahead of the first command of a new access sequence. When Wake-up CS is enabled, all four internal clock phases (CK/4_0°, CK/4_90°, CK/4_180°, and CK/4_270°) may be gated during idle periods, and every CS assertion acts as a wake-up event.

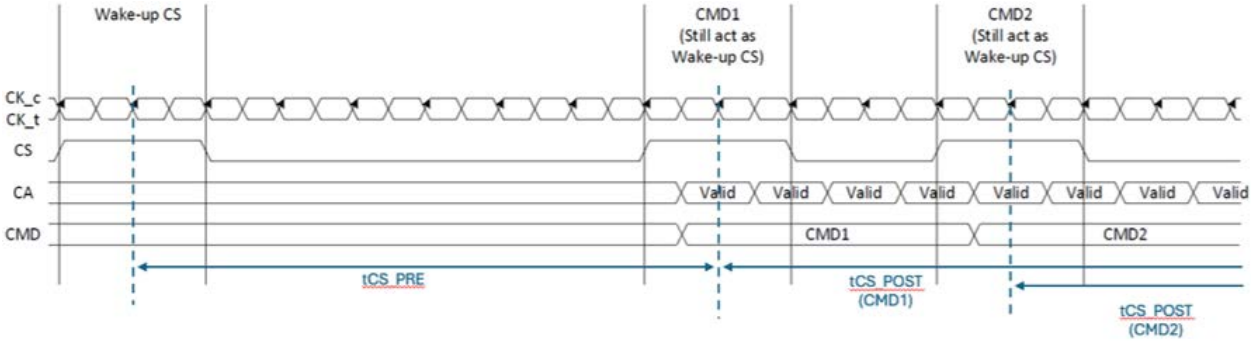


Figure 147 — Wake-up CS Timing Diagram Example

7.7.2. tCS_PRE - Pre-command Wake-up Period

tCS_PRE is the minimum time between the assertion of the Wake-up CS and the first valid command. tCS_PRE is an analog timing parameter. During tCS_PRE, the CMD/ADDR receivers are enabled and the device prepares to sample incoming commands. The minimum value of tCS_PRE(min) is 1.5ns. Once tCS_PRE has elapsed from the Wake-up CS assertion, the CMD/ADDR interface remains ready to sample commands indefinitely, with no clock gating applied while in the ready state.

7.7.3. tCS_POST - Post-command Inactivity Timeout

tCS_POST is a digital, programmable parameter that defines the number of clock cycles after the last command (or last CS pulse assertion) before the CMD/ADDR interface automatically returns to the power-save (clock-gating) state, provided no new command has been issued in the interim. tCS_POST is set via MRW as shown in Table 1. The host may extend the active window deterministically by issuing periodic NOP commands before tCS_POST expires.

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Table 121 — MRW Settings for Wake-up CS

Function	MR Access	Operand	Data	Notes
Wake-up CS post-command timeout (tCS_POST)	Write	OP[1:0]	00b: Disabled (Default) 01b: tCS_POST = 64 tCK 10b: tCS_POST = 128 tCK 11b: tCS_POST = 256 tCK	1
NOTE: 1: The tCS_POST values of 64/128/256 tCK are an initial proposal. Alternative durations (e.g., 32/64/128 tCK) may be considered.				

7.7.4. Wake-up CS in MRUPDATE and CSTM Modes

Wake-up CS settings shall only be modified while the device is in MRUPDATE mode. Upon issuing the MCME command, Wake-up CS functionality is automatically disabled after tMCME_Delay. Wake-up CS is re-enabled following MRUPDATE mode exit, using either the previous settings (if unchanged) or the newly programmed settings (if reprogrammed during MRUPDATE).

7.7.5. Wake-up CS Sequence

The Wake-up CS sequence operates as follows:

1) After power-up or reset, Wake-up CS functionality is disabled.
2) The host shall enable Wake-up CS operation by programming the associated mode register field to a non-default value. The default MR setting 00b disables the feature.
3) When Wake-up CS is enabled, the DRAM may place the CMD/ADDR receive path into a power-saving state after tCS_POST of command inactivity.
4) Prior to issuing a command while the CMD/ADDR receive path is in the power-saving state, the host shall issue a Wake-up CS pulse.
5) The first valid command shall not be issued until at least tCS_PRE(min) after the rising edge of the Wake-up CS pulse.
6) Following a Wake-up CS pulse, the CMD/ADDR receive path shall remain in the active-ready state and capable of accepting commands.
7) Any command CS assertion occurring while the receive path is already active will also reset the tCS_POST activity timer.
8) If no command activity occurs for tCS_POST after the previous command, the CMD/ADDR receive path may automatically return to the power-saving state.

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7.8. Mode Register Operations

7.8.1. Mode Register Read

The Mode Register Read (MRR) command is used to read configuration and status data from the DDR6 SDRAM registers. The MRR command is initiated with CS and CA [5:0] in the proper state as defined by the Command Truth Table. The mode register address operands (MA [7:0]) allow the user to select one of the available Mode Registers ~~(MRs)~~. MRR has a command burst length (BL) 24. The mode register contents are available at the start of 8th and ends at 23rd UI data bits of DQ [2:0] after RL + tCK2DQO following the MRR command. Every Opcode, OP [7:0], is available for 2 UI lengths as shown in Table 1-6. The minimum MRR to MRR command period is defined by tMRR. The minimum MRR command to Any command period is defined by tMRD as shown in Table 7. The truth table for MRR is defined in Table 8.

To avoid a potentially worst-case pattern, DQ1 bits shall have their contents inverted relative to DQ0 and DQ2 (represented with "!"). The data in the burst (BL0-7) shall be either "0" or "1", with "1" indicating that the content of the later UIs (BL8-23) is inverted. Subsequent data bits DQ [5:3] for 2p6, 1p6 and DQ [11:6] for 1p12 configuration contain valid but undefined content. DQS is toggled for the duration of the Mode Register Read burst.

7.8.1.1. MRR Timing Diagram

The timing diagram below depicts the tMRR and tMRD timing for MRR command to MRR command and MRR to 6-cycle MRW command to PSD specific MR ~~or any other valid~~.

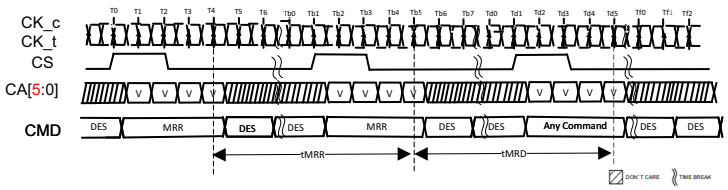


Figure 148 — Mode Register Read Timing Diagram

NOTE: 1: Only DES command is allowed during tMRR and tMRD periods.

7.8.1.2. MRR behavior for Shared MR and PSD MR

For MRR, 2pX and 1pX operate differently for Shared MR and PSD MR accesses.

2pX Mode

- Each PSD reads its own PSD MR.
- The corresponding OP values are output through the DQ pins associated with that PSD.

1pX Mode

- Shared MR Read: The OP values are output through all DQs, with the data replicated in 3-DQ groups.
- PSD MR Read: Each PSD's OP values are output through the DQ pins associated with that PSD.

Note: For detailed information regarding Shared and PSD MR access mechanisms, please refer to Mode Register Access in Section x.x.y

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Table 122 — DQ Output Mapping for 2p3 Configuration

BL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0/3	0								OP0		OP1		OP2		OP3		OP4		OP5		OP6		OP7	
DQ1/4	1								!OP0		!OP1		!OP2		!OP3		!OP4		!OP5		!OP6		!OP7	
DQ2/5	0								OP0		OP1		OP2		OP3		OP4		OP5		OP6		OP7	

Table 123 — DQ Output Mapping for 2p6~~/1p6~~ Configuration

BL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	0								OP0		OP1		OP2		OP3		OP4		OP5		OP6		OP7	
DQ1	1								!OP0		!OP1		!OP2		!OP3		!OP4		!OP5		!OP6		!OP7	
DQ2	0								OP0		OP1		OP2		OP3		OP4		OP5		OP6		OP7	
DQ3	~~V~~0								~~V~~OP0		~~V~~OP1		~~V~~OP2		~~V~~OP3		~~V~~OP4		~~V~~OP5		~~V~~OP6		~~V~~OP7	
DQ4	~~V~~1								~~V~~!OP0		~~V~~!OP1		~~V~~!OP2		~~V~~!OP3		~~V~~!OP4		~~V~~!OP5		~~V~~!OP6		~~V~~!OP7	
DQ5	~~V~~0								~~V~~OP0		~~V~~OP1		~~V~~OP2		~~V~~OP3		~~V~~OP4		~~V~~OP5		~~V~~OP6		~~V~~OP7	

~~NOTE: Option is to repeat the pattern on DQ3-5, V = Valid~~

Table 124 — DQ Output Mapping for 1p6 Configuration – Shared MR Access

BL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0/3	0								OP0		OP1		OP2		OP3		OP4		OP5		OP6		OP7	
DQ1/4	1								!OP0		!OP1		!OP2		!OP3		!OP4		!OP5		!OP6		!OP7	
DQ2/5	0								OP0		OP1		OP2		OP3		OP4		OP5		OP6		OP7	

Table 125 — DQ Output Mapping for 1p6 Configuration – PSD MR Access (Sheet 1 of 2)

BL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0/3_L	0								OP0_L		OP1_L		OP2_L		OP3_L		OP4_L		OP5_L		OP6_L		OP7_L	

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Table 125 — DQ Output Mapping for 1p6 Configuration – PSD MR Access (Continued) (Sheet 2 of 2)

BL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ1/4_L	1								!OP0_L		!OP1_L		!OP2_L		!OP3_L		!OP4_L		!OP5_L		!OP6_L		!OP7_L	
DQ2/5_L	0								OP0_L		OP1_L		OP2_L		OP3_L		OP4_L		OP5_L		OP6_L		OP7_L	
DQ3/0_R	0								OP0_R		OP1_R		OP2_R		OP3_R		OP4_R		OP5_R		OP6_R		OP7_R	
DQ4/1_R	1								!OP0_R		!OP1_R		!OP2_R		!OP3_R		!OP4_R		!OP5_R		!OP6_R		!OP7_R	
DQ5/2_R	0								OP0_R		OP1_R		OP2_R		OP3_R		OP4_R		OP5_R		OP6_R		OP7_R	

Table 126 — DQ Output Mapping for 1p12 Configuration– Shared MR Access (Sheet 1 of 2)

BL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0_L	0								OP0		OP1		OP2		OP3		OP4		OP5		OP6		OP7	
DQ1_L	1								!OP0		!OP1		!OP2		!OP3		!OP4		!OP5		!OP6		!OP7	
DQ2_L	0								OP0		OP1		OP2		OP3		OP4		OP5		OP6		OP7	
DQ3_L	~~V~~0								~~V~~OP0		~~V~~OP1		~~V~~OP2		~~V~~OP3		~~V~~OP4		~~V~~OP5		~~V~~OP6		~~V~~OP7	
DQ4_L	~~V~~1								~~V~~!OP0		~~V~~!OP1		~~V~~!OP2		~~V~~!OP3		~~V~~!OP4		~~V~~!OP5		~~V~~!OP6		~~V~~!OP7	
DQ5_L	~~V~~0								~~V~~OP0		~~V~~OP1		~~V~~OP2		~~V~~OP3		~~V~~OP4		~~V~~OP5		~~V~~OP6		~~V~~OP7	
~~DQ6~~ DQ0_R	~~V~~0								~~V~~OP0		~~V~~OP1		~~V~~OP2		~~V~~OP3		~~V~~OP4		~~V~~OP5		~~V~~OP6		~~V~~OP7	
~~DQ7~~ DQ1_R	~~V~~1								~~V~~!OP0		~~V~~!OP1		~~V~~!OP2		~~V~~!OP3		~~V~~!OP4		~~V~~!OP5		~~V~~!OP6		~~V~~!OP7	
~~DQ8~~ DQ2_R	~~V~~0								~~V~~OP0		~~V~~OP1		~~V~~OP2		~~V~~OP3		~~V~~OP4		~~V~~OP5		~~V~~OP6		~~V~~OP7	
~~DQ9~~ DQ3_R	~~V~~0								~~V~~OP0		~~V~~OP1		~~V~~OP2		~~V~~OP3		~~V~~OP4		~~V~~OP5		~~V~~OP6		~~V~~OP7	

Table 126 — DQ Output Mapping for 1p12 Configuration– Shared MR Access (Continued) (Sheet 2 of 2)

BL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
~~DQ10~~ DQ4_R	~~V~~1								~~V~~!OP0		~~V~~!OP1		~~V~~!OP2		~~V~~!OP3		~~V~~!OP4		~~V~~!OP5		~~V~~!OP6		~~V~~!OP7	
~~DQ11~~ DQ5_R	~~V~~0								~~V~~OP0		~~V~~OP1		~~V~~OP2		~~V~~OP3		~~V~~OP4		~~V~~OP5		~~V~~OP6		~~V~~OP7	

~~NOTE: Option is to repeat the pattern on DQ3-11, V = Valid~~

Table 127 — DQ Output Mapping for 1p12 Configuration – PSD MR Access

BL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0_L	0								OP0_L		OP1_L		OP2_L		OP3_L		OP4_L		OP5_L		OP6_L		OP7_L	
DQ1_L	1								!OP0_L		!OP1_L		!OP2_L		!OP3_L		!OP4_L		!OP5_L		!OP6_L		!OP7_L	
DQ2_L	0								OP0_L		OP1_L		OP2_L		OP3_L		OP4_L		OP5_L		OP6_L		OP7_L	
DQ3_L	0								OP0_L		OP1_L		OP2_L		OP3_L		OP4_L		OP5_L		OP6_L		OP7_L	
DQ4_L	1								!OP0_L		!OP1_L		!OP2_L		!OP3_L		!OP4_L		!OP5_L		!OP6_L		!OP7_L	
DQ5_L	0								OP0_L		OP1_L		OP2_L		OP3_L		OP4_L		OP5_L		OP6_L		OP7_L	
DQ0_R	0								OP0_R		OP1_R		OP2_R		OP3_R		OP4_R		OP5_R		OP6_R		OP7_R	
DQ1_R	1								!OP0_R		!OP1_R		!OP2_R		!OP3_R		!OP4_R		!OP5_R		!OP6_R		!OP7_R	
DQ2_R	0								OP0_R		OP1_R		OP2_R		OP3_R		OP4_R		OP5_R		OP6_R		OP7_R	
DQ3_R	0								OP0_R		OP1_R		OP2_R		OP3_R		OP4_R		OP5_R		OP6_R		OP7_R	
DQ4_R	1								!OP0_R		!OP1_R		!OP2_R		!OP3_R		!OP4_R		!OP5_R		!OP6_R		!OP7_R	
DQ5_R	0								OP0_R		OP1_R		OP2_R		OP3_R		OP4_R		OP5_R		OP6_R		OP7_R	

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Table 128 — Mode Register Read AC Timing

Parameter	Symbol	Min/Max	Value	Unit	Note
MODE REGISTER READ command period	tMRR	Min	Max(10ns,~~16nCK~~16nCK)	ns,nCK	
Mode REGISTER SET command delay	tMRD	Min	Max(14ns,~~40nCK~~40nCK)	ns,nCK	
MRR command to Read command	tMRR_RD	Min	Max(10ns,~~16nCK~~16nCK)	ns,nCK	
Read command to MRR command	tRD_MRR	Min	RL	ns,nCK	
Write command to MRR command	tWR_MRR	Min	tWTR_L	ns,nCK	
MRR command to MRR command in WTM	tMRR_WTM	Min	Max(5ns,~~16nCK~~16nCK)	ns,nCK	1
PSD MRR to PSD MRR command delay for shared mode register access	tMRR	Min	Max(10ns, 16nCK)	ns,nCK	
PSD MRR to PSD MRW command delay for shared mode register access	tMRR	Min	Max(10ns, 16nCK)	ns,nCK	
NOTE1: MRR to MRR during WTM is specific to MR~~(xx-xy)~~DF-MRE2 mode registers.					

Table 129 — Truth Table for Mode Register Read (MRR)

Current State	Command	Intermediate State	Next State
DDR6 DRAM		DDR6 DRAM	DDR6 DRAM
All banks Idle	MRR	Mode Register Reading (All Banks Idle)	All Banks Idle
Bank(s) Active	MRR	Mode Register Reading	Bank(s) Active

7.8.2. Mode Register Write

The Mode Register Write (MRW) command is used to write configuration data to the mode registers. The MRW command is initiated by setting CS and CA [5:0] to valid levels as defined in Command Truth Table. The mode register address operands (MA [7:0]) allow the user to select one of available Mode Registers ~~(MRs)~~. The Mode Register Write command is an MRW 6-cycle long command. The minimum MRW command period is defined by tMRW. The minimum MRW command to any command period is defined by tMRD as shown in Table 1. The truth table for mode register write (MRW) is defined in Table 2. The Mode Register Write command to read-only registers has no impact on the functionality of the device.

The timing diagram below depicts the timing for MRW command to MRW command and MRW to 4-cycle or any other valid command.

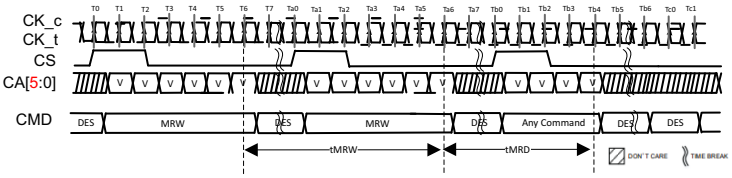


Figure 149 — Mode Register Write Timing

NOTE: Only DES command is allowed during tMRW and tMRD periods.

Table 130 — Mode Register Write AC Timing

Parameter	Symbol	Min/Max	Value	Unit	Note
Mode Register Write Timing					
MODE REGISTER WRITE command period	tMRW	Min	Max(10ns, 40nCK)	ns, nCK	
Mode REGISTER SET command delay	tMRD	Min	Max(14ns, 40nCK)	ns, nCK	
PSD MRW to PSD MRW command delay for shared mode register space	tMRD	Min	Max(14ns, 40nCK)	ns, nCK	
PSD MRW to PSD MRR command delay for shared mode register space	tMRD	Min	Max(14ns, 40nCK)	ns, nCK	

Table 131 — Truth table for Mode Register Write (MRW) (Sheet 1 of 2)

Current State	Command	Intermediate State	Next State
DDR6 DRAM		DDR6 DRAM	DDR6 DRAM
All banks Idle	MRW	Mode Register Writing (All Banks Idle)	All Banks Idle
Bank(s) Active[1]	MRW	Mode Register Writing	Bank(s) Active

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Table 131 — Truth table for Mode Register Write (MRW) (Continued) (Sheet 2 of 2)

Current State	Command	Intermediate State	Next State
DDR6 DRAM		DDR6 DRAM	DDR6 DRAM
NOTE 1: The MRW command may only be issued after all read and write bursts, including postambles and termination timing have fully completed, and all applicable timing requirements (such as tRP, tACI, etc.) have been satisfied.			

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7.8.3. Mode Register Update (MRUPDATE) Mode

The Mode Register Update (MRUPDATE) Mode performs the equivalent function as the Mode Register Write (MRW) command, but without the command bits, saving two Command/Address (CA) cycles for every Mode Register write operation. MRUPDATE is multi-cycle CS per C/A cycle as shown in the first figure below, as compared to the single-cycle CS MRW command.

7.8.3.1. MRUPDATE Mode Entry

MRUPDATE Mode is entered by issuing an MCME command, which also enables multi-cycle CS.

Entering MRUPDATE Mode resets the starting reference point for the DRAM to know that all subsequent commands received will be a function of four (4) CA cycles. The 4 CA cycles are shown in the table below.

Table 132 — MRUPDATE Mode Multi-Cycle Command/Address

FUNCTION	COMMAND PIN					Notes
	CA0	CA1	CA2	CA3	CA4	
MRUPDATE	MA4	MA5	MA6	MA7	H=ExecuteL =Disable	1
	MA0	MA1	MA2	MA3	H=ExecuteL =Disable	1
	OP4	OP5	OP6	OP7	H=ExecuteL =Disable	1
	OP0	OP1	OP2	OP3	H=ExecuteL =Disable	1

NOTE 1: CA4 is the control bit for the MRUPDATE. CA4 must be HIGH for all four MRUPDATE cycles for the MRUPDATE to be executed. When CA4 of all four MRUPDATE cycles is LOW, MRUPDATE mode is disabled/exited. The MRUPDATE will not be performed with any CA4 mixed combination of HIGH and LOW.
NOTE 2: CS behavior is shown in the MRUPDATE Timing diagram.

7.8.3.2. MRUPDATE Mode Execution

For the device to latch the MRUPATE in cases where the alignment between CS, CA and CK may be unknown, each CA cycle must meet the required timings. The CA inputs must reach the proper command state and provide tMC_Setup of setup prior to CS transitioning from LOW to HIGH, CS must remain HIGH for tMC_CSPW, and the CA inputs must remain in the proper command state and provide tMC_Hold of hold after CS transitions from HIGH to LOW.

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Following the fourth (4th) CA cycle, tMRUPD_Delay needs to be met to latch the data into the Mode Register prior to starting the next MRUPATE.

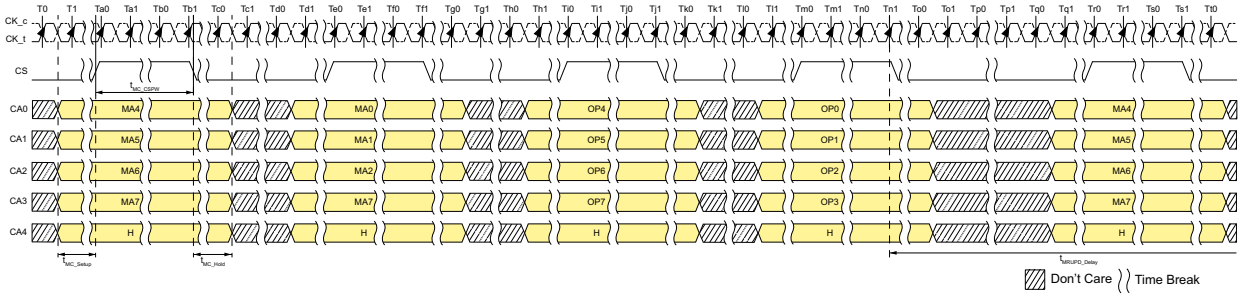


Figure 150 — MRUPDATE Timing

NOTE: CA4=HIGH to execute the MRUPDATE. CA4=LOW to disable/exit MRUPDATE mode.

Table 133 — Mode Register Update (MRUPDATE) Timing Parameters

Parameter	Symbol	DDR6-6400 to DDR6-12800		>DDR6-12800		Unit	Note
		MIN	MAX	MIN	MAX		
CS Pulse Width to register the state of CS	tMC_CSPW	8	-	TBD	-	nCK	
Time between stable CAs and rising CS edge (setup)	tMC_Setup	8	-	TBD	-	nCK	
Time between falling CS edge and stable CAs (hold)	tMC_Hold	8	-	TBD	-	nCK	
Time between last CK prior to the 4th falling CS edge of an MRUPDATE and the first CK after the 4th rising CS edge of the next MRUPDATE	tMRUPD_Delay	tMRD	-	TBD	-	nS	

7.8.3.3. MRUPDATE Mode Exit

MRUPDATE Mode is disabled by executing four cycles of MRUPDATE with CA4=LOW. The MRA and OP bits states may be any combination of HIGH and LOW. Disabling MRUPDATE will also disable multi-cycle CS.

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7.9. Other Operations

7.9.1. On-Die Termination for DQ, DQS

On-Die Termination (ODT) is a feature of the DDR6 SDRAM that allows the DRAM to change termination resistance for each DQ and DQS. The ODT feature is designed to improve signal integrity of the memory channel by allowing the DRAM controller to independently change termination resistance for any or all DRAM devices. In addition to the control capability of the DQ ODT, the DQS ODT can be independently programmed via MR63:OP[5:3]. All ODT control is command & mode register based. ODT is enabled via Write and non-target Write commands, disabled via Read and Mode Register Read commands, or set to the default "PARK" value through settings in MR63-MR66.

A simple functional representation of the DRAM ODT feature is shown in Figure 1.

DQ, DQS

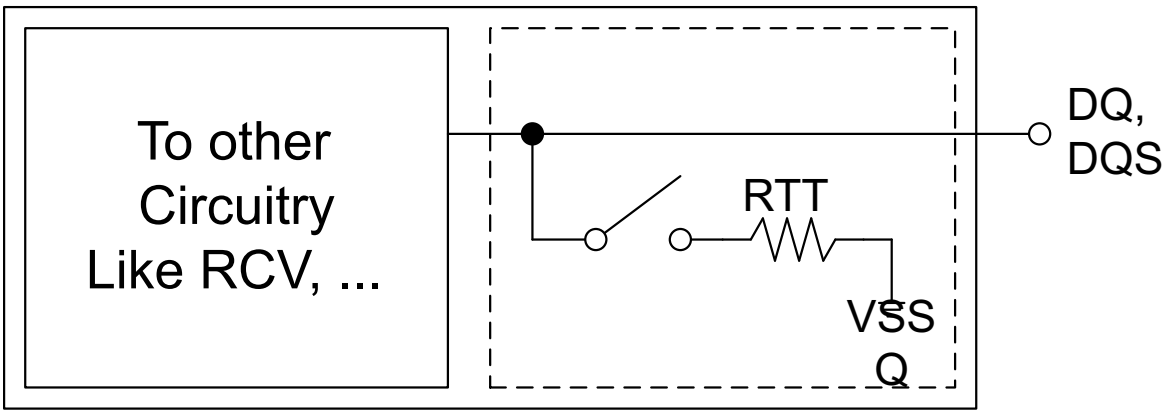


Figure 151 — Functional Representation of ODT

The switch is enabled by the internal ODT control logic, which uses command decode, Mode Register Settings and other control information, see below. The value of RTT is determined by the settings of Mode Register bits.

7.9.1.1. ODT Modes and State Table

The ODT Mode of DDR6 SDRAM has four states: Data/Strobe Termination Disable, RTT_WR, RTT_NT_WR and RTT_PARK/DQS_RTT_PARK. The ODT Mode is enabled based on Mode Registers for each RTT listed below. In this case, the value of RTT is determined by the settings of these mode register bits.

Application: Controller can control each RTT condition with Write (WR), Read (RD) or Mode Register Read (MRR) commands and use of ODT Offset Control Mode Registers.

- Data Termination Disable: DRAM driving data upon receiving Read command disables the termination after RL-2-8UI, and re-enables the termination at RL + BL/2 (RL and BL/2 defined in clock cycles).
- Strobe Termination Disable: DRAM driving strobe upon receiving Read command disables the termination after RL-2-tRPRE-(Read_DQS_Pre-shift) and re-enables the termination at RL+BL/2- 0.5+tRPST-(Read_DQS_Pre-shift) (RL and BL/2 defined in clock cycles).
- RTT_WR: For the rank that is being written to provide termination, it adjusts timing based on ODT Control Mode Register settings.
- RTT_NT_WR: DRAM applies RTT_NT_WR if it sees a CS asserted during the third and fourth pulse of the Write command, and it adjusts timing based on ODT Control Mode Register settings.

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- RTT_PARK: Default parked value set via MR63:OP[2:0]; enabled when a Read or Write is not active.
- DQS_RTT_PARK: Default parked value for DQS set via MR63:OP[5:3]; enabled when a Read is not active.

Those RTT values have priority as follows:

1) Data Termination Disable and Strobe Termination Disable
2) RTT_WR
3) RTT_NT_WR
4) RTT_PARK and DQS_RTT_PARK

This means if there is a Write command, then the DRAM turns on RTT_WR, not RTT_NT_WR, and also if there is a Read command, then the DRAM disables data termination and goes into driving mode.

If CS is driven high during the third and fourth cycle of a Write command, then RTT_NT_WR is enabled for the non-target rank. This provides additional and potentially different termination options for the other ranks on the channel.

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Table 134 — Termination State Table for DQ

Command	Mode Register Config. Settings			Results		Note
	RTT_PARK	RTT_WR	RTT_NT_WR	Target DRAM Term	Non-Target DRAM Term	
ANY	Disabled			Off	Off	
Any Non-Term CMD	Enabled	Don't Care		RTT_PARK	RTT_PARK	3
WR	Disabled		Enabled	Off	RTT_NT_WR	
	Disabled	Enabled	Disabled	RTT_WR	Off	2
	Don't Care	Enabled		RTT_WR	RTT_NT_WR	2
	Enabled		Disabled	RTT_WR	Off	2, 4
	Enabled	Disabled	Enabled	RTT_PARK	RTT_NT_WR	
	Enabled	Disabled		RTT_PARK	Off	4
RD/MRR	Enabled	Don't Care		Off	RTT_PARK	1

NOTE 1: When a Read command is executed, the DRAM termination state of the target rank will be Hi-Z for a defined period independent of MR setting of RTT_PARK/RTT_WR/RTT_NT_WR.
NOTE 2: If RTT_WR is enabled, RTT_WR will be activated by a Write command for a defined period of time independent of the MR setting of RTT_PARK/RTT_WR/RTT_NT_WR.
NOTE 3: If RTT_PARK is enabled, DRAM RTT_PARK termination will be enabled while WR/RD/MRR are not being executed.
NOTE 4: When a Non-Target ODT command is executed and the RTT_NT_WR is disabled, the DRAM termination state of the non-target rank will be Hi-Z for a defined period, independent of the MR setting of RTT_PARK.

Table 135 — Termination State Table for DQS

Command	Mode Register Config. Settings	Results		Note
	RTT_DQS_PARK	Target DRAM Term	Non-Target DRAM Term	
ANY	Disabled	Off	Off	
Any Non-Term CMD	Enabled	DQS_RTT_PARK	DQS_RTT_PARK	1
WR	Enabled	DQS_RTT_PARK	DQS_RTT_PARK	1
RD/MRR	Enabled	Off	DQS_RTT_PARK	

NOTE 1: If RTT_DQS_PARK is enabled, DRAM RTT_DQS_PARK termination will be enabled while RD/MRR are not being executed.

7.9.1.2. Dynamic ODT

Dynamic ODT is the feature that allows the DDR6 SDRAM to switch between pre-programmed ODT termination states in response to command activity without requiring MRW during normal traffic.

ODT Functional Description

The function is described as follows:

- Four RTT values are available: RTT_NT_WR, RTT_WR, RTT_PARK and DQS_RTT_PARK.
  - The value for RTT_NT_WR is preselected via MR64:OP[5:3].
  - The value for RTT_WR is preselected via MR64:OP[2:0].
  - The value for RTT_PARK is preselected via MR63:OP[2:0].
  - The value for DQS_RTT_PARK is preselected via MR63:OP[5:3].
- During operation without RD, MRR, WR, Non-target WR commands, the termination is controlled as follows:
  - RTT_PARK is selected for DQ
  - DQS_RTT_PARK is selected for DQS.
- When a Write command (WR) is registered, the termination is controlled as follows:
  - A latency ODTLon_WR after the Write command, termination strength RTT_WR is selected.
  - A latency ODTLoff_WR after the Write command, termination strength RTT_WR is de-selected.
- The termination, RTT_NT_WR, for the non-target Write command is selected and de-selected by latencies ODTLon_WR_NT and ODTLoff_WR_NT, respectively.
- When a Read command (RD) is registered, the termination is controlled as follows:
  - A latency ODTLoff_RD after the Read command, data termination is disabled. Then, ODTLon_RD after the Read command, data termination is enabled.
  - A latency ODTLoff_RD_DQS after the Read command, strobe termination is disabled. ODTLon_RD_DQS after the Read command, strobe termination is enabled.

To achieve the minimum write burst duration, ODTLoff_X and ODTLon_X latencies contain independent programmable mode register offsets:

- The values for the Write command ODT control offsets are preselected via MR65.

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  - MR65:OP[2:0] preselects ODTLon_WR_Offset
  - MR65:OP[5:3] preselects ODTLoff_WR_Offset

- The values for the non-target Write command ODT control offsets are preselected via MR66.
  - MR66:OP[2:0] preselects ODTLon_WR_NT_Offset
  - MR66:OP[5:3] preselects ODTLoff_WR_NT_Offset

The duration of a Write or Read command is a full burst cycle, BL/2. The termination select ("ODTLon_...") and de-select ("ODTLoff_...") latency settings shall not result in an ODT pulse width which violates a burst cycle (BL/2) minimum duration. The equation "ODTLoff_X - ODTLon_X >= BL/2" must be met, where X is the termination latency setting associated with a particular command type (WR, WR_NT).

See Table 3 for ODT latency and timing parameter details.

Table 136 — Latencies and Timing Parameters for Dynamic ODT (Sheet 1 of 2)

Name and Description	Abbreviation	Defined from	Define to	Value	Unit	Note
ODT Latency On from Write command to RTT enable	ODTLon_WR	Registering external Write command	Change RTT strength from Previous State to RTT_WR	WL + ODTLon_WR_Offset - EVEN(RU((tODTon,max-tDQS2DQi,min)/tCK) + 8.5UI - tDQSoffset,min - tDQSS_min)	nCK	1, 2 ,3
ODT Latency On from NT Write command to RTT Enable	ODTLon_WR_NT	Registering external Write command	Change RTT strength from Previous State to RTT_NT_WR	WL + ODTLon_WR_NT_Offset - EVEN(RU((tODTon,max - tDQS2DQi,min)/tCK) + 8.5UI - tDQSoffset,min - tDQSS_min)	nCK	1, 2 ,3
ODT Latency Off from Write command to RTT Disable	ODTLoff_WR	Registering external Write command	Change RTT strength from RTT_WR to Next state	WL + ODTL-off_WR_Offset + EVEN(RU((tDQS2DQI,max – tODTon,min)/tCK) + 23.5UI + tDQSoffset,max + tDQSS,max)	nCK	1, 2 ,3

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Table 136 — Latencies and Timing Parameters for Dynamic ODT (Continued)

Name and Description	Abbreviation	Defined from	Define to	Value	Unit	Note
ODT Latency Off from NT Write command to RTT Disable	ODTLoff_WR_NT	Registering external Write command	Change RTT strength from RTT_NT_WR to Next state	WL + ODTL-off_WR_NT_Offset + EVEN(RU((tDQS2DQI,max – tODTon,min)/tCK) + 23.5UI + tDQSoffset,max + tDQSS,max)	nCK	1, 2 ,3
Data Termination Disable	ODTLoff_RD	Registering external Read command	Disables the termination upon driving data	RL-2-8UI	nCK	4
Data Termination Enable	ODTLon_RD	Registering external Read command	Re-enables the termination after driving data	RL+BL/2	nCK	4
Strobe Termination Disable	ODTLoff _RD_DQS	Registering external Read command	Disables the termination upon driving strobe	RL-2-tRPRE-(Read DQS Pre-shift)	nCK	4
Strobe Termination Enable	ODTLon _RD_DQS	Registering external Read command	Re-enables the termination after driving strobe	RL+BL/2-0.5+tRPST-(Read DQS Pre-shift)	nCK	4

NOTE 1: All "_Offset" parameters refer to the ODT Configuration Mode Registers settings in MR65 to MR66.
NOTE 2: See Table 4, 5
NOTE 3: The EVEN function rounds numbers up to the next even integer. For example, EVEN(3ns/1.25ns)=EVEN(2.4)=4.
NOTE 4: For simplicity, Read commands are assigned the same type of timing parameter; however, unlike others, it is a fixed timing and does not have an offset mode register to control it. To indicate this, it was named Data (or Strobe)

Table 137 — ODTLon_WR, ODTLoff_WR Latency values when ODTLon_WR_Offset = 0 and ODTLoff_WR_Offset = 0 (Sheet 1 of 2)

Data Rate		Clock Frequency		ODTLon_WR [nCK]	ODTLoff_WR [nCK]	Note
Lower Limit [>Mbps]	Upper Limit [<=Mbps]	Lower Limit [>MHz]	Upper Limit [<=MHz]			
4800	6400	2400	3200	WL-12	WL+16	1, 2, 3
6400	8000	3200	4000	WL-14	WL+16	1, 2, 3

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Table 137 — ODTLon_WR, ODTLoff_WR Latency values when ODTLon_WR_Offset = 0 and ODTLoff_WR_Offset = 0 (Continued) (Sheet 2 of 2)

Data Rate		Clock Frequency		ODTLon_WR [nCK]	ODTLoff_WR [nCK]	Note
Lower Limit [>Mbps]	Upper Limit [<=Mbps]	Lower Limit [>MHz]	Upper Limit [<=MHz]			
8000	9200	4000	4600	WL-14	WL+16	1, 2, 3
9200	9600	4600	4800	WL-14	WL+16	1, 2, 3
9600	10400	4800	5200	WL-16	WL+16	1, 2, 3
10400	11600	5200	5800	WL-16	WL+16	1, 2, 3
11600	12800	5800	6400	WL-16	WL+16	1, 2, 3
12800	14400	6400	7200	WL-18	WL+16	1, 2, 3
14400	16400	7200	8200	WL-20	WL+16	1, 2, 3
16400	19200	8200	9600	WL-22	WL+16	1, 2, 3

NOTE 1: ODTLon_WR = WL + ODTLon_WR_Offset - EVEN(RU((tODTon,max-tDQS2DQi,min)/tCK) + 8.5UI - tDQSoffset,min - tDQSS_min)
NOTE 2: WL + ODTLoff_WR_Offset + EVEN(RU((tDQS2DQI,max – tODTon,min)/tCK) + 23.5UI + tDQSoffset,max + tDQSS,max)

Table 138 — ODTLon_WR_NT, ODTLoff_WR_NT Latency values when ODTLon_WR_NT_Offset = 0 and ODTLoff_NT_WR_Offset = 0 (Sheet 1 of 2)

Data Rate		Clock Frequency		ODTLon_WR _NT [nCK]	ODTLoff_WR _NT [nCK]	Note
Lower Limit [>Mbps]	Upper Limit [<=Mbps]	Lower Limit [>MHz]	Upper Limit [<=MHz]			
4800	6400	2400	3200	WL-12	WL+16	1, 2, 3
6400	8000	3200	4000	WL-14	WL+16	1, 2, 3
8000	9200	4000	4600	WL-14	WL+16	1, 2, 3
9200	9600	4600	4800	WL-14	WL+16	1, 2, 3
9600	10400	4800	5200	WL-16	WL+16	1, 2, 3
10400	11600	5200	5800	WL-16	WL+16	1, 2, 3
11600	12800	5800	6400	WL-16	WL+16	1, 2, 3
12800	14400	6400	7200	WL-18	WL+16	1, 2, 3
14400	16400	7200	8200	WL-20	WL+16	1, 2, 3
16400	19200	8200	9600	WL-22	WL+16	1, 2, 3

Table 138 — ODTLon_WR_NT, ODTLoff_WR_NT Latency values when ODTLon_WR_NT_Offset = 0 and ODTLoff_NT_WR_Offset = 0 (Continued) (Sheet 2

Data Rate		Clock Frequency		ODTLon_WR_NT [nCK]	ODTLoff_WR_NT [nCK]	Note
Lower Limit [>Mbps]	Upper Limit [<=Mbps]	Lower Limit [>MHz]	Upper Limit [<=MHz]			

NOTE 1: ODTLon_WR_NT = WL + ODTLon_WR_NT_Offset - EVEN(RU((tODTon,max - tDQS2DQi,min)/tCK) + 8.5UI - tDQSoffset,min - tDQSS_min)
NOTE 2: ODTLoff_WR_NT = WL + ODTLoff_WR_NT_Offset + EVEN(RU((tDQS2DQI,max – tODTon,min)/tCK) + 23.5UI + tDQSoffset,max + tDQSS,max)
NOTE 3: ODTLon_WR_NT_offset, ODTLoff_WR_NT_offset = 0 (default value) in this Table.

tODTQoff, tODTQon During Read Operations

At the start of a read burst, the ODT is disabled and transitions to the DRAM drive state tODTQoff before the start of the DQ, DQS burst. tODTQoff(min) and tODTQoff(max) are minimum and maximum RTT change timing skews between different RTT states with respect to the DQ, DQS toggle start point. At the end of a read burst, the ODT is enabled tODTQon after the end of the DQ, DQS burst. tODTQon(min) and tODTQon(max) are minimum and maximum RTT change timing skews between different RTT states with respect to the DQ, DQS toggle end point. Unless the RTT is specifically disabled, no High-Z state shall be allowed during RTT transition. During DRAM Drive state, the DRAM RON shall keep the DQ signal low prior to the first DQ transition.

Figure 2 is an example showing the tODTQoff/on(min), tODTQoff/on(avg) and tODTQoff/on(max) with respect to the RTT status and effects on the DQ lines prior to the burst.

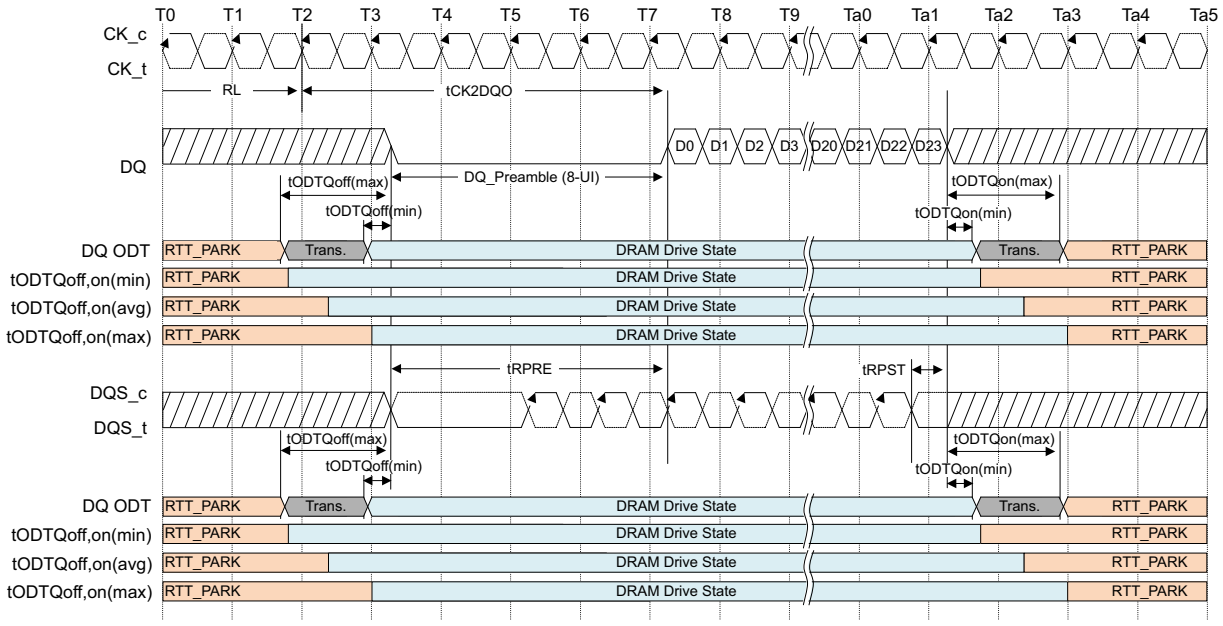


Figure 152 — tODTQoff,on Clarification – DQ, DQS RTT Park to Read

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NOTE: 1: The follow diagram shows a transition from RTT_PARK to Read DRAM Drive state. During transitions from RTT to Read Drive state, the DRAM RON from the driver will keep the DQ signal low prior to the data driven in D0. No High-Z time during RTT transition is allowed in this example.

NOTE: 2: The follow diagram shows a transition from DQS_RTT_PARK to Read DRAM Drive state. During transitions from RTT to Read Drive state, the DRAM RON from the driver will keep the DQS signal low prior to the DQS preamble start. No High-Z time during RTT transition is allowed in this example.

Table 139 — tODTQoff, tODTQon Timing

Parameter	Value	Unit
tODTQoff,min	0.03	ns
tODTQoff,max	2tCK-0.03	ns
tODTQon,min	0.03	ns
tODTQon,max	2tCK-0.03	ns

tODTon, tODToff During Write and Non-target Write Operations

tODTon is defined as the time required for the DRAM to transition from the RTT_PARK state to the write- termination state (RTT_WR for Write, RTT_NT_WR for Non-Target Write). tODTon is measured after the corresponding ODT turn-on latency (ODTLon_WR or ODTLon_NT_WR) from the Write command has been satisfied. tODToff is defined as the time required for the DRAM to transition from the write-termination state back to the RTT_PARK state, and tODToff is measured after the corresponding ODT turn-off latency (ODTLoff_WR or ODTLoff_NT_WR) from the Write command has been satisfied. Unless the RTT is specifically disabled, no High-Z state shall be allowed during RTT transition.

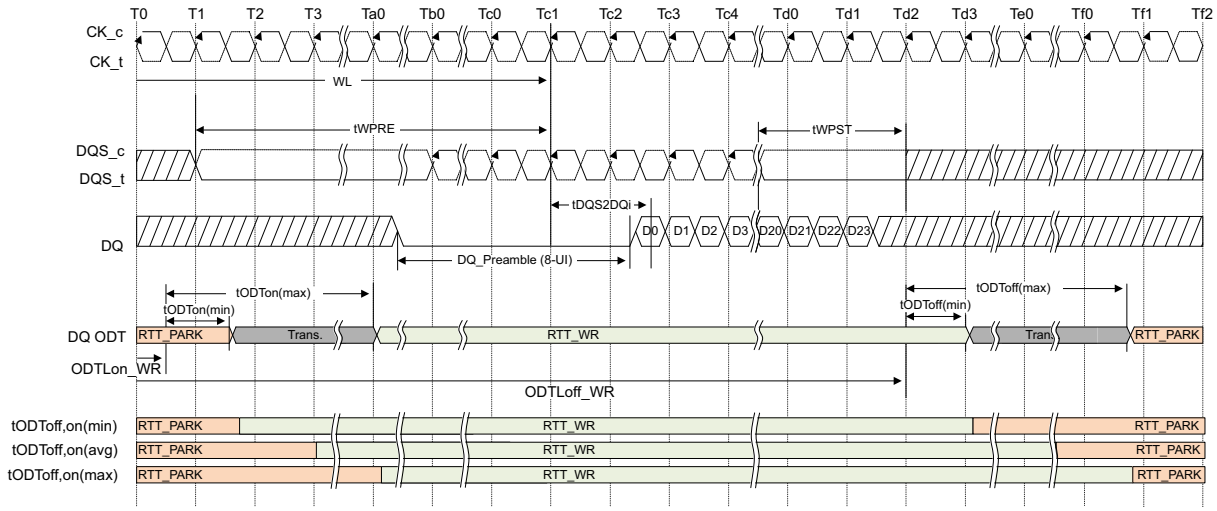


Figure 153 — tODTon,off Clarification – DQ RTT Park to Write

NOTE: 1: The host will keep the DQ signal LOW 8UI prior to the data driven in D0.

NOTE: 2: The DFE should assume that 8UI prior to D0 the signal is LOW.

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Table 140 — tODTon, tODToff Timing

Parameter	Value	Unit
tODTon,min	0.45	ns
tODTon,max	1.5	ns
tODToff,min	0.45	ns
tODToff,max	1.5	ns

7.9.1.3. Mode Register Definition

Table 141 — MR65 (Direct Addressable Mode Registers) Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU		ODTLoff_WR_Offset			ODTLon_WR_Offset		

Function	Register Type	Operand	Data	Notes
ODTLon_WR_Offset	R/W	OP[2:0]	000B : 0nCK (Default) 001B : 2nCK 010B : 4nCK 011B : 6nCK 100B : 8nCK 101B : 10nCK 110B : 12nCK 111B : RFU	
ODTLoff_WR_Offset	R/W	OP[5:3]	000B : 0nCK (Default) 001B : -2nCK 010B : -4nCK 011B : -6nCK 100B : -8nCK 101B : -10nCK 110B : -12nCK 111B : RFU	
RFU	RFU	OP[7:6]	RFU	

Table 142 — MR66 (Direct Addressable Mode Registers) Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU		ODTLoff_WR_NT_Offset			ODTLon_WR_NT_Offset		

Function	Register Type	Operand	Data	Notes
ODTLon_WR_NT_Offset	R/W	OP[2:0]	000B : 0nCK (default) 001B : 2nCK 010B : 4nCK 011B : 6nCK 100B : 8nCK 101B : 10nCK 110B : 12nCK 111B : RFU	
ODTLoff_WR_NT_Offset	R/W	OP[5:3]	000B : 0nCK (Default) 001B : -2nCK 010B : -4nCK 011B : -6nCK 100B : -8nCK 101B : -10nCK 110B : -12nCK 111B : RFU	
RFU	RFU	OP[7:6]	RFU	

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7.9.1.4. ODT Electrical Characteristic

ZQ=120ohm, entire operation temperature range, after proper ZQ calibration.

Table 143 — ODT Electrical Characteristic (Sheet 1 of 2)

RTT	Vout	Min	Nom	Max	Unit	Note
240ohm	VOLdc=0.2*VDDQ	0.8	1	1.1	RZQ/0.5	1,2,3
	VOMdc=0.5*VDDQ	0.9	1	1.1	RZQ/0.5	1,2,3
	VOHdc=0.75*VDDQ	0.9	1	1.3	RZQ/0.5	1,2,3
120ohm	VOLdc=0.2*VDDQ	0.8	1	1.1	RZQ	1,2,3
	VOMdc=0.5*VDDQ	0.9	1	1.1	RZQ	1,2,3
	VOHdc=0.75*VDDQ	0.9	1	1.3	RZQ	1,2,3
80ohm	VOLdc=0.2*VDDQ	0.8	1	1.1	RZQ/1.5	1,2,3
	VOMdc=0.5*VDDQ	0.9	1	1.1	RZQ/1.5	1,2,3
	VOHdc=0.75*VDDQ	0.9	1	1.3	RZQ/1.5	1,2,3
60ohm	VOLdc=0.2*VDDQ	0.8	1	1.1	RZQ/2	1,2,3
	VOMdc=0.5*VDDQ	0.9	1	1.1	RZQ/2	1,2,3
	VOHdc=0.75*VDDQ	0.9	1	1.3	RZQ/2	1,2,3
48ohm	VOLdc=0.2*VDDQ	0.8	1	1.1	RZQ/2.5	1,2,3
	VOMdc=0.5*VDDQ	0.9	1	1.1	RZQ/2.5	1,2,3
	VOHdc=0.75*VDDQ	0.9	1	1.3	RZQ/2.5	1,2,3
40ohm	VOLdc=0.2*VDDQ	0.8	1	1.1	RZQ/3	1,2,3
	VOMdc=0.5*VDDQ	0.9	1	1.1	RZQ/3	1,2,3
	VOHdc=0.75*VDDQ	0.9	1	1.3	RZQ/3	1,2,3
34ohm	VOLdc=0.2*VDDQ	0.8	1	1.1	RZQ/3.5	1,2,3
	VOMdc=0.5*VDDQ	0.9	1	1.1	RZQ/3.5	1,2,3
	VOHdc=0.75*VDDQ	0.9	1	1.3	RZQ/3.5	1,2,3
DQ-DQ Mismatch within Byte	VOMdc=0.5*VDDQ	-		10	%	1,2,3,4,5
CA-CA Mismatch within Device	VOMdc=0.5*VDDQ	-		10	%	1,2,3,6

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Table 143 — ODT Electrical Characteristic (Continued) (Sheet 2 of 2)

RTT	Vout	Min	Nom	Max	Unit	Note

NOTE 1: The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see "ZQ Calibration Commands" section

NOTE 2: Pull-down ODT resistors are required to be calibrated at 0.5*VDDQ

NOTE 3: The tolerance limits are specified under the typ voltage condition that VDDQ=0.6v, VDD=1.025v

NOTE 4: DQ to DQ mismatch within byte variation for a given component including DQS_T and DQS_C (characterized)

NOTE 5: RTT variance range ratio to RTT Nominal value in a given component, including DQS_t and DQS_c

$$\text{DQ-DQ (mismatch)} = \frac{R_{TT}(\text{MAX}) - R_{TT}(\text{MIN})}{R_{TT}(\text{NOM})} \times 100$$

NOTE 6: CA to CA mismatch within device variation for a given component including CS, CK_t and CK_c (characterized)

$$\text{CA-CA (mismatch)} = \frac{R_{TT}(\text{MAX}) - R_{TT}(\text{MIN})}{R_{TT}(\text{NOM})} \times 100$$

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7.9.2. Temperature Sensor and Temperature Gradient

DDR6 devices feature a temperature sensor whose status (MR04:OP[2:0]) can be read to determine the minimum required refresh rate. Either the temperature sensor readout or the device TOPER may be used by the system to determine whether the refresh rate and operating temperature requirements are being met. Additionally, the device's temperature status (MR04:OP[2:0]) can be read by the system to determine the current approximate temperature range sensed by the device.

DDR6 devices shall monitor device temperature and update MR4 according to tTSI. Upon completion of device initialization, the device temperature status bits shall be no older than tTSI. MR4 will be updated even when device is in Self Refresh state.

When using the temperature sensor, the actual device case temperature may be higher than the TOPER specification that applies for the standard or elevated temperature ranges. For example, TCASE may be above 85°C when MR04:OP[2:0]=100_B. DDR6 devices shall allow for 2°C temperature margin between the point at which the device updates the MR04 value and the point at which the controller reconfigures the system accordingly.

Each MR04[2:0] setting specifies a nominal temperature range, and the six ranges defined by MR04[2:0] are determined by five temperature thresholds which will be nominally 75°C, 80°C, 85°C, 90°C, and 95°C. The 3rd threshold (nominally 85°C) is used by the system to switch to 2x refresh. The 5th threshold (nominally 95°C) is used by the system to throttle activity to keep the DRAM at a safe operating temperature. The 2nd threshold (nominally 80°C) allows the system to take actions which delay reaching the 3rd threshold. The 4rd threshold (nominally 90 °C) allows the system to take actions which delay reaching the 5th threshold. The 1st threshold (nominally 75°C) has no minimum value specified and the 5th threshold (nominally 95°C) has no maximum temperature value specified.

To ensure proper operation using the temperature sensor, applications should consider the following factors:

- TempGradient is the maximum temperature gradient experienced by the memory device at the temperature of interest over a range of 2°C, measured in the range of interest 75-95°C.
- ReadInterval is the maximum time period between Temperature Sensor mode register bits being read by the system.
- TempSensorInterval (tTSI) is the maximum delay between internal updates of the Temperature Sensor mode register bits.
- SysRespDelay is the maximum time between a read of the Temperature Sensor mode register bits and the response by the system.

The required frequency of polling the Temperature Sensor mode register bits by the system uses the maximum TempGradient and the maximum response time of the system in the following equation:

TempGradient x (ReadInterval + tTSI + SysRespDelay) ≤ TempMargin

Table 144 — Temperature Sensor Parameters

Parameters	Symbol	Min/Max	Value	Unit	Notes
System Temperature Gradient	TempGradient	Max	System Dependent	°C/s	
Read Interval	ReadInterval	Max	System Dependent	ms	
Temperature Sensor Interval	tTSI	Max	32	ms	
System Response Delay	SysRespDelay	Max	System Dependent	ms	
Device Temperature Margin	TempMargin	Max	2	°C	1

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For example, if TempGradient is 10°C/s and the SysRespDelay is 100 ms:

(10 °C/s) x (ReadInterval + 32 ms + 100 ms) ≤ 2 °C

In this case, ReadInterval shall be no greater than 68 ms.

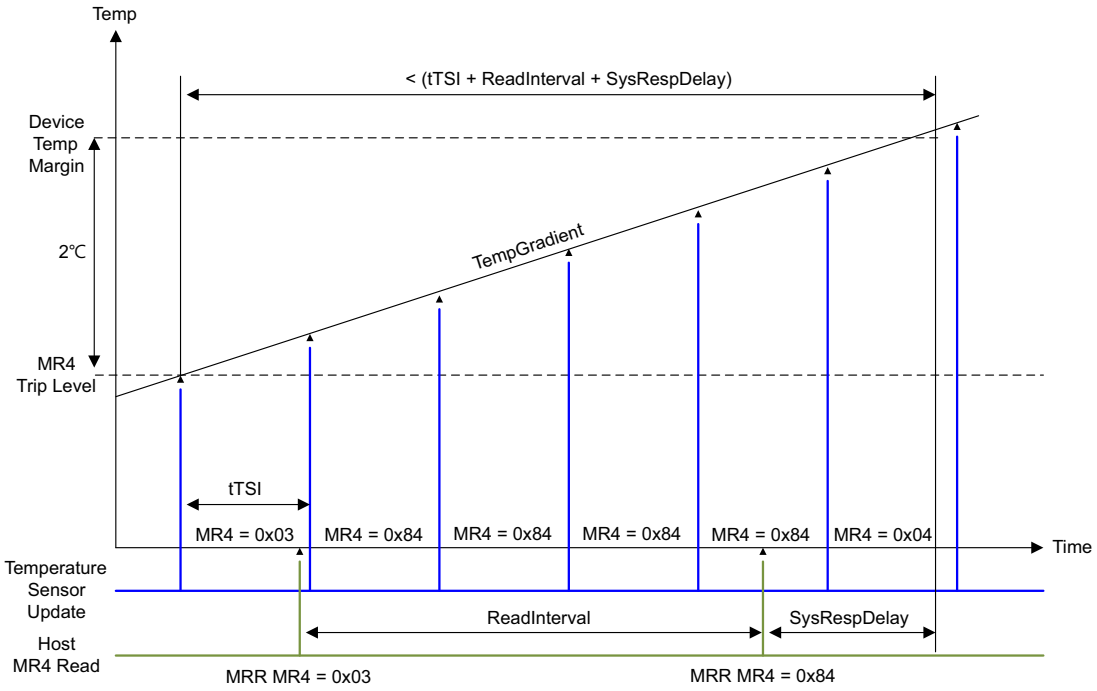


Figure 154 — Temp Sensor Timing

7.9.2.1. Temperature Sensor Read Out

DDR6 provides temperature sensor readout information as reference. Temperature is not located in the same area as memory array. Only limited position temperature is measured.

DDR temperature sensor cannot provide accurate enough temperature due to circuits and placement. Host can use this information as temperature gradient information. Temperature accuracy is not guaranteed by design values.

MR56 is assigned for temperature sensor report. Please refer MR56 definition for more detailed information.

Mode register read out shows on-die temperature sensor value.

7.9.2.2. Thermal Offset

The host may provide a temperature-offset to the DDR6 SDRAM, allowing the memory's self-refresh timing or the 2x-refresh decision point to be adjusted for reliable operation. This offset is programmed via the appropriate MR (e.g., MR04:OP[7:5]). Once applied, the DRAM's Temperature-Compensated Self-Refresh (TCSR) circuit uses the adjusted temperature reference to modify the refresh interval as required, and the update is reflected in the refresh-control field (e.g. MR04:OP[2:0]) within approximately 200 µs.

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7.9.2.3. Mode Register Definition

Table 145 — MR56 (TSRO Mode Register)

Bits	Attribute	Default	Description	Notes
7:0	R		MR56[7:0]: TSRO Temperature Sensor Readout 00000000: Lower than low temperature sensor limit	1
			00000001: -40°C 00000010: -39°C 00000011: -38°C ... 01000000: 23°C 01000001: 24°C	1, 2
			01000010: 25°C 01000011: 26°C ... 10010001: 104°C 10010010: 105°C	1, 3
			10010011: 106°C 10010100: 107°C ... 10100101: 124°C 10100110: 125°C	1, 2, 3
			10100111: RV 10101000: RV ... 11111101: RV 11111110: RV	3
			11111111: Higher than high temperature sensor limit	1

NOTE 1: DDR temperature sensor cannot provide accurate enough temperature due to circuits and placement. Host can use this information as temperature gradient information. Temperature accuracy is not guaranteed by design values.
NOTE 2: These temperature ranges are optional. Please consult with vendor.
NOTE 3: When operating above 85°C, the host shall provide appropriate Refresh mode controls associated with the increased temperature range.

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Table 146 — MR04 (Refresh Mode Register) (Sheet 1 of 2)

Bits	Attribute	Default	Description	Notes
7:5	RW	0h	MR04[7:5]: THERMAL_OFFSET Thermal offset applied to the REFRESH_RATE 000: No offset (default) 001: 2°C offset 010: 4°C offset 011: 6°C offset 100: 8°C offset 101: 10°C offset 110: 12°C offset 111: 14°C offset	8
4	R	0	MR04[4]: TUF Temperature Update Flag 0: No change in MR04[2:0] since last MR04 read (default 1: A change occurred in MR04[2:0] since last MR04 read	4,5,6,7
3	RW	0	MR04[3]: RIR_INDICATOR Refresh Interval Rate (RIR)disable/enable 0: Disabled (Default) 1: Enabled	
2:0	R		MR04[2:0]:REFRESH_RATE Refresh command interval sent by the Host to the DRAM 000: Reserved 001: tREFI (1x Refresh Rate), <= 75°C nominal 010: tREFI (1x Refresh Rate), 75 <Temperature <= 80°C nominal 011: tREFI (1x Refresh Rate), 80 < Temperature <= 85°C nominal 100: tREFI/2 (2x Refresh Rate), 85 < Temperature <= 90°C nominal 101: tREFI/2 (2x Refresh Rate), 90 < Temperature <= 95°C nominal 110: tREFI/2 (2x Refresh Rate), > 95°C nominal 111: Reserved	1,2,3,4,5,6,7

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Table 146 — MR04 (Refresh Mode Register) (Continued) (Sheet 2 of 2)

Bits	Attribute	Default	Description	Notes
NOTE 1: DRAM vendors must report all of the possible settings over the operating temperature range of the device. Each vendor guarantees that their device will work at any temperature within the range using the refresh interval requested by their device. **NOTE 2:** The 2x Refresh Rate must be provided by the system before the DRAM Tj has gone up by more than 2°C (Temperature Margin) since the first report out of MR04[2:0]=100. This condition is reset when MR04[2:0] is equal to 011. **NOTE 3:** The device may not operate properly when MR04[2:0]=110 and the DRAM Tj has gone up by more than 2 °C (Temperature Margin) since the first report out of MR04[2:0]=110. This condition is reset when MR04[2:0] is equal to 101. MR04[2:0]=110 must be a temporary condition of the DRAM, to be addressed by immediately reducing the Tj of the DRAM by throttling its power, and/or the power of nearby devices. **NOTE 4:** MR04[4] = 0 at power-up. MR04[2:0] bits are valid after initialization sequence (Te). **NOTE 5:** When MR04[4]=1, the refresh rate reported in MR04[2:0] has changed since the last MR04 read. A mode register read from MR04 will reset MR04[4] to '0'. **NOTE 6:** MR04[2:0] bits indicate the latest Refresh Rate whenever MR04[4] = 1 **NOTE 7:** See the section on “Temperature Sensor” for information on the recommended frequency of reading MR04 **NOTE 8:** The two PSDs (PSD0 and PSD1) on a single die share the same thermal sensor, therefore the thermal offset value in MR04[7:5] is applied to the common thermal sensor				

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7.9.3. DQS and CLK Interval Oscillator

As voltage and temperature change on the SDRAM die, the DQS and CLK clock tree delay will shift and may require re-training. The DDR6-SDRAM includes an internal DQS and CLK clock tree oscillator to measure the amount of delay over a given time interval (determined by the controller), allowing the controller to compare the trained delay value to the delay value seen at a later time. The DQS and CLK interval oscillator will provide the controller with important information regarding the need to re-train, and the magnitude of potential error.

The DQS and CLK interval oscillator is started by issuing an MRW MR05 [Start DQS/CLK Interval Osc] command as described in the Mode Register Functional Modes section, which will start an internal ring oscillator that counts the number of times a signal propagates through a replica of the DQS or CLK clock tree.

The DQS and CLK interval oscillator may be stopped by issuing an MRW MR05 [Stop DQS/CLK Interval Osc] command as described in the MRW Functional Mode section, or the controller may instruct the SDRAM to count for a specific number of clocks and then stop automatically ~~(See MR45xx,46xx, and 47xx for more information).~~If MR20~~xx~~ is set to automatically stop the DQS ~~and CLK~~ interval oscillator, then the MRW MR05 [Stop DQS~~/CLK~~ Interval Osc] command shall not be used. Similarly, if MR23 is set to automatically stop the CLK interval oscillator, then the MRW MR05 [Stop CLK Interval Osc] command shall not be used.

When the DQS and CLK interval oscillator is stopped by either method, the result of the oscillator counter is automatically stored in MR1E~~xx~~/MR1F~~xx~~ and MR21~~xx~~/MR22~~xx~~

MRW commands to MR20~~xx~~ during an ongoing DQS~~/CLK~~ interval oscillator operation are not permitted. MRW commands to MR23 during an ongoing CLK interval oscillator operation are not permitted.MR20~~xx~~and MR23may be reprogrammed after the tMRD delay following the automatic oscillator stop.

DQS interval oscillator and CLK interval oscillator cannot be operated simultaneously. After completing DQS (or CLK) interval oscillator operation, MRW MR05 [Start CLK (or DQS) Interval Osc] command can be issued.

The DRAM shall respond in one of two ways if an MRW MR05 [Start DQS/CLK Interval Osc] is issued during an ongoing DQS/CLK interval oscillator operation (automatic or manual). One option is the DRAM will ignore the concurrent start command, in which case the DQS/CLK interval oscillator operation will proceed as normal with all timing constraints referencing the original start command. The other option is for the DRAM to restart the DQS interval oscillator operation, in which case all timing constraints will reference the most recent (subsequent) start command. The host must account for the worst-case option in the event that a concurrent start command is issued. If issuing a concurrent MRW MR05 [Start DQS/CLK Interval Osc] results in a loss of run time tracking, the results stored in MR1E~~xx~~/MR1F and MR21~~xx~~/MR22 should be ignored and the full operation restarted.

Entering Full Power-down during an ongoing DQS/CLK interval oscillator operation is permitted. Upon exiting Full Power-down, an operation started in automatic mode shall be allowed to complete naturally based upon the specified number of clocks (cumulative before and after Full Power-down). If a manual operation started prior to entering Full Power-down is not stopped prior as well, the operation shall be manually stopped upon Full Power-down exit. The results stored in MR1E~~xx~~/MR1F~~xx~~ and MR21~~xx~~/MR22~~xx~~ for a DQS and CLK interval oscillator operation that spans Full Power-down entry/exit should be ignored and the full operation restarted.

The controller may adjust the accuracy of the result by running the DQS/CLK interval oscillator for shorter (less accurate) or longer (more accurate) duration. The accuracy of the result for a given temperature and voltage is determined by the following equation:

$$DQS\ Oscillator\ Granularity\ Error = \frac{2 * (DQS\ delay)}{Run\ time}$$

$$CLK\ Oscillator\ Granularity\ Error = \frac{2 * (CLK\ delay)}{Run\ time}$$

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Where:

Run time

total time between start and stop commands

DQS delay

the value of the DQS clock tree delay (~~tRX_DQS2DQ~~tDQS2DQi min/max)

CLK delay

the value of the CLK clock tree delay (~~tTX_CLK2DQ~~tCK2DQo min/max)

Additional matching error must be included, which is the difference between DQS/CLK replica circuit and the actual DQS/CLK clock tree across voltage and temperature. The matching error is vendor specific.

Therefore, the total accuracy of the DQS or CLK interval oscillator counter is given by:

- *DQS(or CLK) Interval Oscillator Accuracy = 1 - Granularity Error - Matching Error*

Example: If the total time between start and stop commands is 100 ns, and the maximum DQS clock tree delay is TBD ps (~~tRX_DQS2DQ~~tDQS2DQi max), then the DQS interval oscillator granularity error is:

$$\text{DQS Interval Oscillator Granularity Error} = \frac{2 * (TBD \text{ ns})}{100 \text{ ns}} = TBD\%$$

This equates to a granularity timing error of TBD ps.

Assuming a circuit Matching Error of 5 ps across voltage and temperature, then the accuracy is:

$$\text{DQS Interval Oscillator Accuracy} = 1 - \frac{TBD + 5}{TBD} = TBD\ \%$$

The result of the DQS/CLK interval oscillator is defined as the number of DQS/CLK clock tree delays that can be counted within the "run time," determined by the controller. The result for DQS~~CLK~~ is stored in MR1E~~xx~~ and MR1F~~xx~~, and that for CLK is also stored in MR21~~xx~~ and MR22~~xx~~. MR1E~~xx~~ and MR21~~xx~~ contain the least significant bits (LSB) of the result for DQS and CLK, respectively. MR1F~~xx~~ and MR22~~xx~~ contain the most significant bits (MSB) of the result for DQS and CLK, respectively. MR1E~~xx~~ and MR1F~~xx~~ are overwritten by the SDRAM when an MRW MR05 [Stop DQS Interval Osc] command is received. Similarly, for a CLK interval oscillator stop command, MR21~~xx~~ and MR22~~xx~~ are also overwritten when SDRAM receives MRW MR05 [Stop CLK Interval Osc] command.

The SDRAM counter will count to its maximum value (=2^16) and stop. If the maximum value is read from the mode registers, then the memory controller must assume that the counter overflowed the register and discard the result. The longest "run time" for the oscillator that will not overflow the counter registers can be calculated as follows:

- Longest ~~tRX_DQS2DQ~~tDQS2DQi Run Time Interval = 2^{16}* ~~tRX_DQS2DQ~~tDQS2DQi (min) =2^{16} * TBD = TBD
- Longest ~~tTX_CK2DQS~~tCK2DQo Run Time Interval = 2^{16}* ~~tTX_CK2DQS~~tCK2DQo (min) =2^{16} * TBD = TBD

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The interval oscillator matching error is defined as the difference between the DQS/CLK replica circuit (interval oscillator) and the actual DQS and CLK clock tree across voltage and temperature.

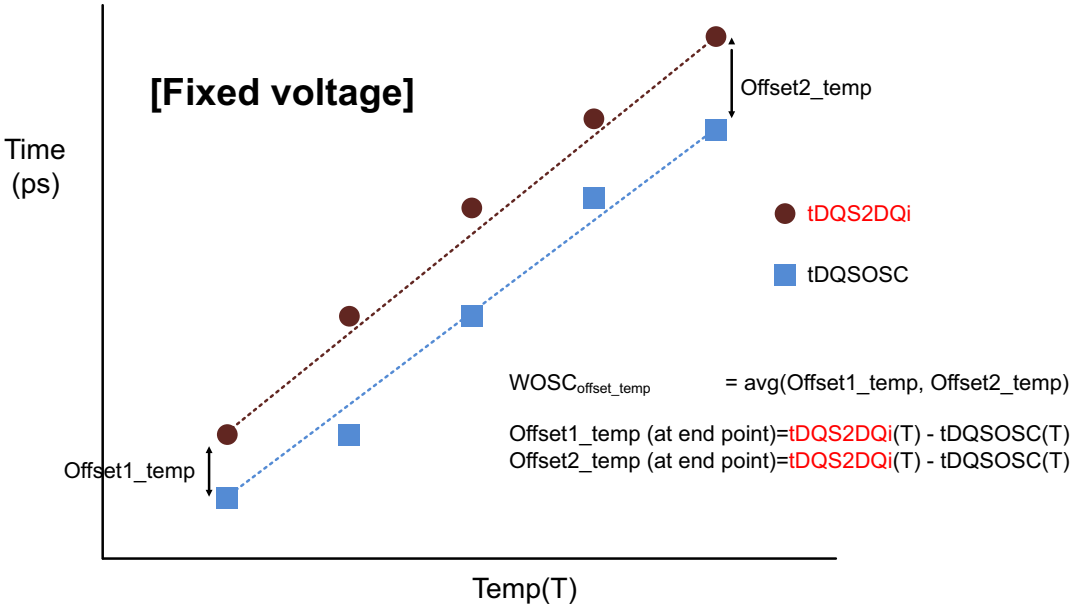


Figure 155 — Interval Oscillator WOSCOffset_temp

- $WOSC_{Match_temp}$: $WOSC_{Match_temp} = [\sout{tRX_DQS2DQ}tDQS2DQi(T) - tDQSOSC(T) - WOSC_{offset_temp}]$

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- tDQSOSC(T): *tDQSOSC(T) = Run time / 2*Count*

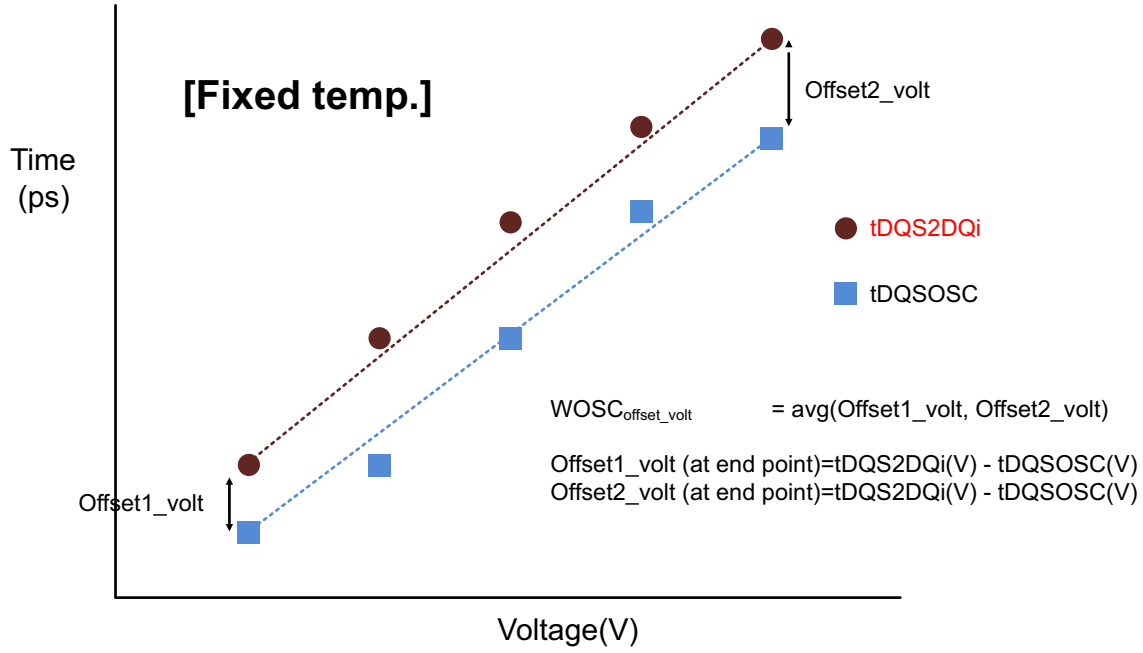


Figure 156 — Interval Oscillator WOSCOffset_volt

- WOSC$_{Match_volt}$: *WOSC$_{Match_volt}$ = [~~tRX_DQS2DQ~~tDQS2DQi(V) - tDQSOSC(V) - WOSC$_{offset_volt}$]*
- tDQSOSC(V): *tDQSOSC(V) = Run time / 2*Count*

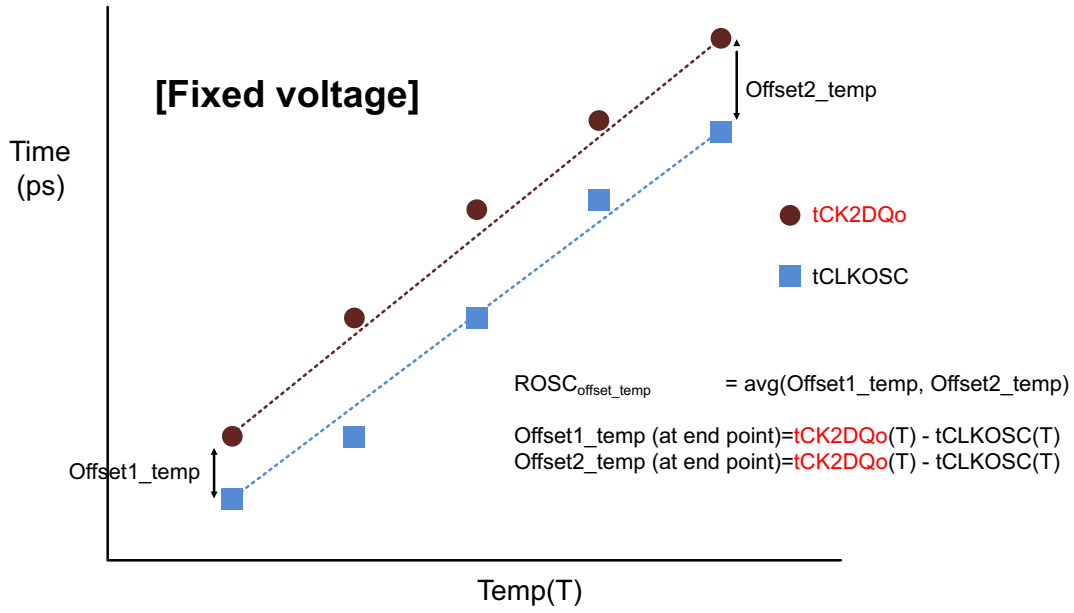


Figure 157 — Interval Oscillator ROSCOffset_temp

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- ROSC$_{Match_temp}$: *ROSC$_{Match_temp}$ = [~~tTX_CLK2DQ~~tCK2DQo(T) - tCLKOSC(T) - ROSC$_{offset_temp}$]*
- tCLKOSC(T): *tCLKOSC(T) = Run time / 2*Count*

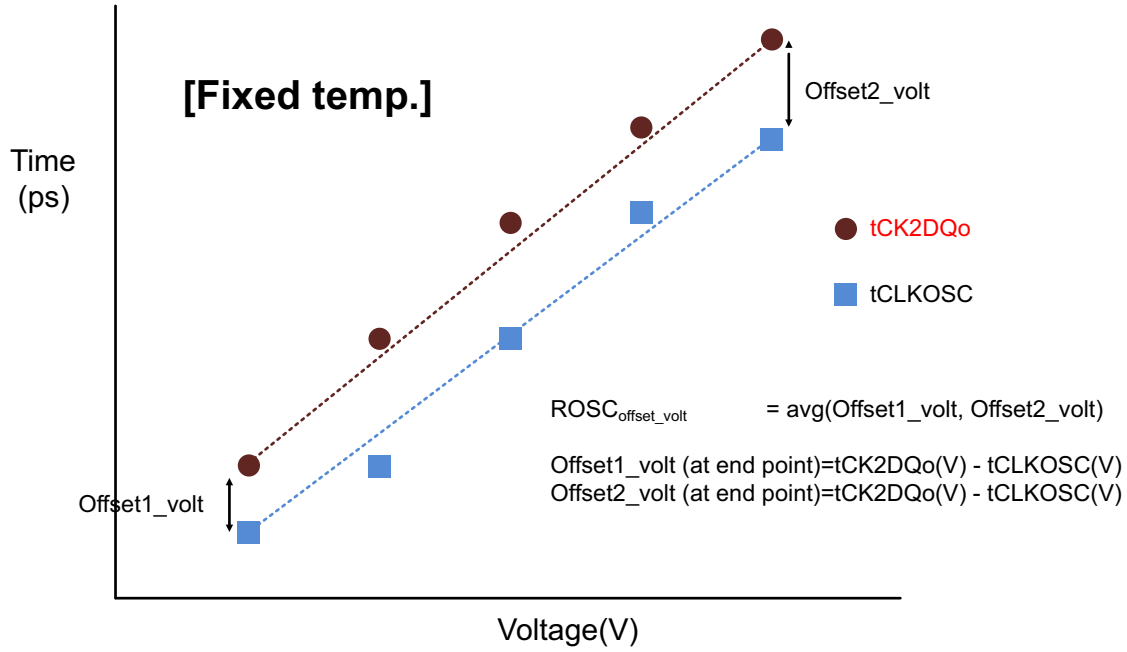


Figure 158 — Interval Oscillator ROSCOffset_volt

- ROSC$_{Match_volt}$: *ROSC$_{Match_volt}$ = [~~tTX_CLK2DQ~~tCK2DQo(V) - tCLKOSC(V) - ROSC$_{offset_volt}$]*
- tCLKOSC(V): *tCLKOSC(V) = Run time / 2*Count*

Table 147 — DQS/CLK Interval Oscillator Matching Error Specification (Sheet 1 of 2)

Parameter	Symbol	Min	Max	Units	Note	
DQS Interval Oscillator Matching Error: Voltage variation	WOSC$_{Match_volt}$	-7.5	+7.5	ps	1,2,3,4,5,6,7,8	
DQS Interval Oscillator Matching Error: Temperature variation	WOSC$_{Match_temp}$	-6.0	+6.0	ps	1,2,3,4,5,6,7,8	
DQS Interval Oscillator Offset for Voltage variation	WOSC$_{Offset_volt}$	-100	+100	ps	2,4,6,8	
DQS Interval Oscillator Offset for Temperature variation	WOSC$_{Offset_temp}$	-100	+100	ps	2,4,6,8	
CLK Interval Oscillator Matching Error: Voltage variation	ROSC$_{Match_volt}$	-20	+20	ps	1,2,3,4,5,6,7,9	

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Table 147 — DQS/CLK Interval Oscillator Matching Error Specification (Continued) (Sheet 2 of 2)

Parameter	Symbol	Min	Max	Units	Note	
CLK Interval Oscillator Matching Error: Temperature variation	$ROSC_{Match_temp}$	-20	+20	ps	1,2,3,4,5,6,7,9	
CLK Interval Oscillator Offset for Voltage variation	$ROSC_{Offset_volt}$	-200	+200	ps	2,4,6,9	
CLK Interval Oscillator Offset for Temperature variation	$ROSC_{Offset_temp}$	-200	+200	ps	2,4,6,9	

NOTE 1: The $WOSC_{Match}$ is the matching error between the actual DQS and DQS interval oscillator over voltage and temp. The $ROSC_{Match}$ is the matching error between the actual CLK and CLK interval oscillator over voltage and temp.
NOTE 2: This parameter will be characterized or guaranteed by design.
NOTE 3: The $WOSC_{Match}$ and $ROSC_{Match}$ are defined as the following: $WOSC_{Match}$ = *[~~tRX_DQS2DQtDQS2DQi~~(V, T) - tDQSOSC(V, T) - $WOSC_{offset}$]* Where ~~tRX_DQS2DQ~~tDQS2DQi(V,T), tDQSOSC(V,T) and $WOSC_{Offset}$(V,T) are determined over the same voltage and temp conditions. $ROSC_{Match}$ = *[~~tTX_CLK2DQ~~tCK2DQo(V, T) - tCLKOSC(V, T) - $ROSC_{Offset}$]* Where ~~tTX_CLK2DQ~~tCK2DQo(V,T), tCLKOSC(V,T) and $ROSC_{Offset}$(V,T) are determined over the same voltage and temp conditions.
NOTE 4: The runtime of the oscillator must be at least 200ns for determining tDQSOSC(V,T) and tCLKOSC(V,T).
NOTE 5: The input stimulus for ~~tRX_DQS2DQ~~tDQS2DQi and ~~tTX_CLK2DQ~~tCK2DQo will be consistent over voltage and temp conditions.
NOTE 6: The $WOSC_{Offset}$ is the average difference of the endpoints across voltage and temp.
NOTE 7: ~~tRX_DQS2DQ~~tDQS2DQi(V,T) delay will be the average of DQS to DQ delay over the runtime period. ~~tTX_CLK2DQ~~tCK2DQo(V,T) delay will be the average of CLK to DQ delay over the runtime period.
NOTE 8: The matching error and offset of WOSC came from DQS interval oscillator.
NOTE 9: The matching error and offset of ROSC came from CLK interval oscillator.

Figure 5 and figure 6 illustrate the mode register readout timing of the count value after stop DQS interval oscillator command.

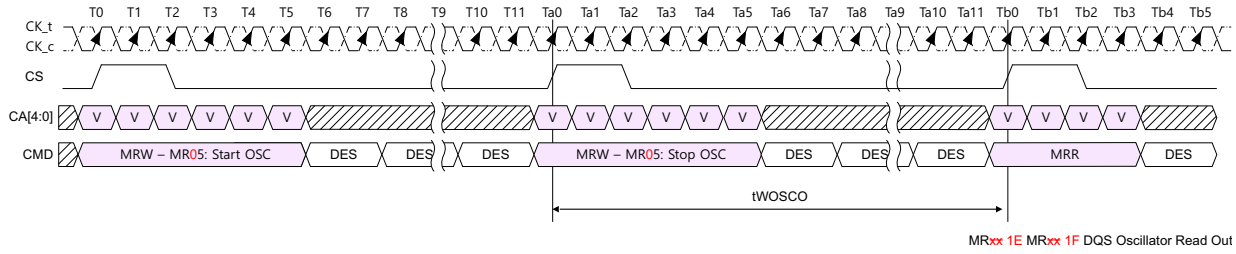


Figure 159 — DQS Interval Oscillator Manual Mode Timing Diagram

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.

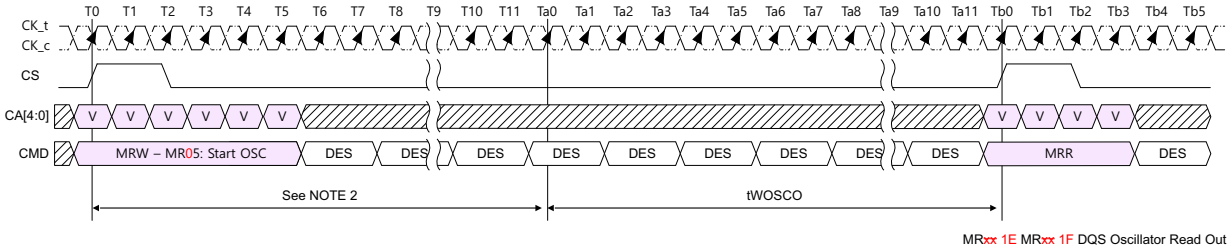


Figure 160 — DQS Interval Oscillator Automatic Mode Timing Diagram

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.
NOTE: 2: DQS interval run time : MR20~~xx~~ OP [7:0].

Figure 7 and figure 8 illustrate the mode register readout timing of the count value after stop CLK interval oscillator command.

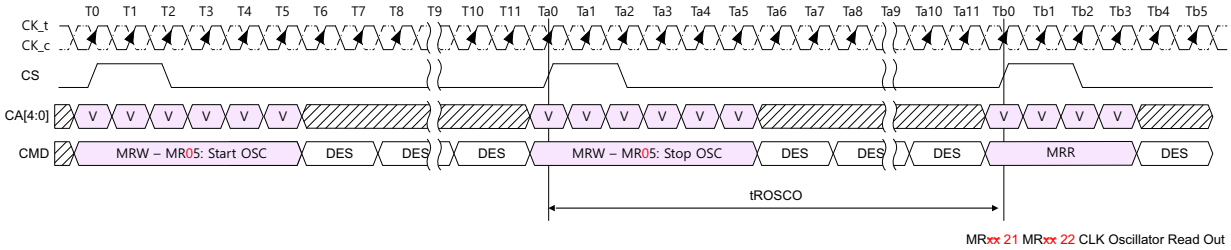


Figure 161 — CLK Interval Oscillator Manual Mode Timing Diagram

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.

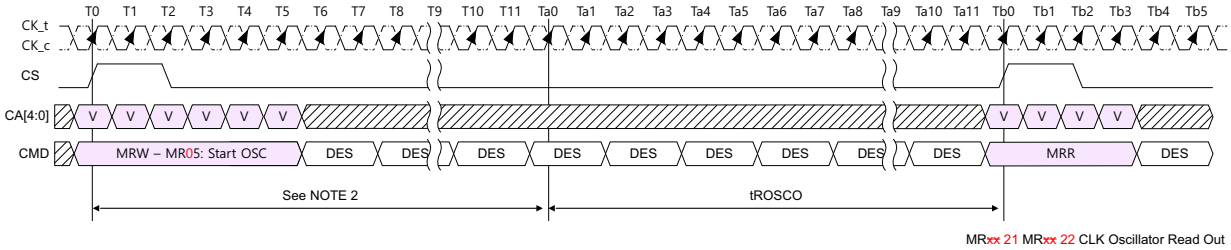


Figure 162 — CLK Interval Oscillator Automatic Mode Timing Diagram

NOTE: 1: DES commands are shown for ease of illustration; other commands may be valid at these times.
NOTE: 2: CLK interval run time: MR23~~xx~~ OP [7:0].

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Start/stop DQS and CLK interval oscillator command constraints is shown in Figure 9.

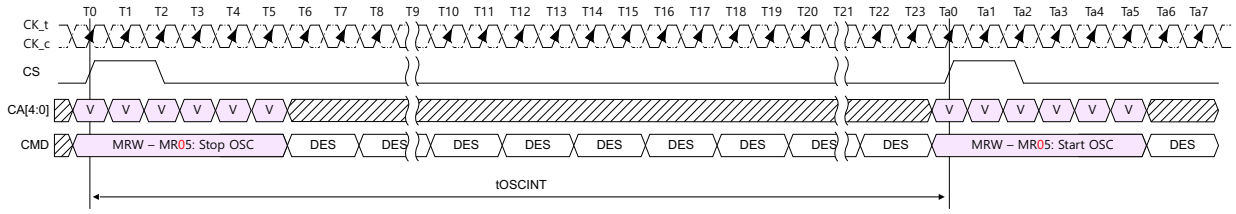


Figure 163 — Start/Stop DQS and CLK Interval Oscillator Command Constraints Timing

NOTE: 1: This parameter applies to following combination.

- MRW MR05 [Stop DQS Interval Osc] to MRW MR05 [Start CLK Interval Osc].
- MRW MR05 [Stop CLK Interval Osc] to MRW MR05 [Start DQS Interval Osc].
- MRW MR05 [Stop DQS Interval Osc] to MRW MR05 [Start DQS Interval Osc].
- MRW MR05 [Stop CLK Interval Osc] to MRW MR05 [Start CLK Interval Osc].

Table 148 — DQS/CLK Interval Oscillator AC Timing

Parameter	Symbol	Min/Max	Value	Units	Notes
Delay time from Stop DQS interval oscillator command to Mode Register Readout from MR1E~~xx~~/MR1F~~xx~~	tWOSCO	Min	Max(40ns, 16nCK)	nCK	1
Delay time from Stop CLK interval oscillator command to Mode Register Readout from MR21~~xx~~/MR22~~xx~~	tROSCO	Min	Max(40ns, 16nCK)	nCK	2
Delay time from Stop DQS/CLK interval oscillator command to Start DQS/CLK interval oscillator command	tOSCINT	Min	Max(40ns, 16nCK)	nCK	
NOTE 1: Issuing all Mode Register Read (MRR) command except for MR1E~~xx~~/MR1F~~xx~~ is allowed in tWOSCO period. **NOTE 2:** Issuing all Mode Register Read (MRR) command except for MR21~~xx~~/MR22~~xx~~ is allowed in tROSCO period.					

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Table 149 — MR05 Functional Modes

MR05[7:0]				
Bits	Attribute	Default	Description	Note
7:0	RW	0h	MR05[7:0]: FUNCTIONAL MODES DRAM direct execution functional modes 00000000: All Functions Disabled 00000001: ZQCal Latch 00000010: CSTM Entry 00000011: CATM Entry 00000100: CK Sync Off 00000101: DSDM Entry 00000110: PDA Enumeration Mode 00000111: Write Leveling Training 00001000: Write Training Mode 00001001: Host Receiver Training Mode 00001010: Read Preamble Training 10000001: Start DQS Interval Oscillator 10000010: Stop DQS Interval Oscillator 10000011: Start CLK Interval Oscillator 10000100: Stop CLK Interval Oscillator xxxxxxxx: Apply Shadow MRs to Normal MRs 00001011-11111111: Reserved (No operation)	

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7.9.3.1. Mode Register Definition

Table 150 — MR1Exx Register Information~~(MA[7:0] = xxH)~~

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQS Interval Oscillator Count - LSB							

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Table 151 — MR1Exx Definition

Function	Register Type	Operand	Data	Notes
DQS Interval Oscillator – LSB	Read-only	OP[7:0]	0-255 LSB DRAM DQS Interval Oscillator Count	1,2,3

NOTE 1: MR1Exx reports the LSB bits of the DRAM DQS interval oscillator count. The DRAM DQS interval oscillator count value is used to train DQS to the DQ data valid window. The value reported by the DRAM in this mode register can be used by the memory controller to periodically adjust the phase of DQS relative to DQ.
NOTE 2: Both MR1Exx and MR1Fxx must be read (MRR) and combined to get the value of the DQS interval oscillator count.
NOTE 3: A MRW MR05 [Start DQS Interval Osc] could be issued to reset the contents of MR1Exx/MR1Fxx.

Table 152 — MR1Fxx Register Information (MA[7:0] = xxH)

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQS Interval Oscillator Count - MSB							

Table 153 — MR1Fxx Definition

Function	Register Type	Operand	Data	Notes
DQS Interval Oscillator – MSB	Read-only	OP[7:0]	0-255 MSB DRAM DQS Interval Oscillator Count	1,2,3

NOTE 1: MR1Fxx reports the MSB bits of the DRAM DQS interval oscillator count. The DRAM DQS interval oscillator count value is used to train DQS to the DQ data valid window. The value reported by the DRAM in this mode register can be used by the memory controller to periodically adjust the phase of DQS relative to DQ.
NOTE 2: Both MR1Exx and MR1Fxx must be read (MRR) and combined to get the value of the DQS interval oscillator count.
NOTE 3: A new MRW MR05 [Start DQS Interval Osc] could be issued to reset the contents of MR1Exx/MR1Fxx.

Table 154 — MR20xx Register Information (MA[7:0] = xxH)

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQS interval timer run time setting							

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Table 155 — MR20~~xx~~ Definition

Function	Register Type	Operand	Data	Notes
DQS interval timer run time	Write/Read	OP[7:0]	00000000B: DQS interval timer stop via MR Functional Modes Command (Default) 00000001B: DQS timer stops automatically at 16th clock after timer start 00000010B: DQS timer stops automatically at 32nd clock after timer start 00000011B: DQS timer stops automatically at 48th clock after timer start 00000100B: DQS timer stops automatically at 64th clock after timer start ---------- Thru ---------- 00111111B: DQS timer stops automatically at (63X16)th clock after timer start 01XXXXXXB: DQS timer stops automatically at 2048th clock after timer start 10XXXXXXB: DQS timer stops automatically at 4096th clock after timer start 11XXXXXXB: DQS timer stops automatically at 8192nd clock after timer start	1,2

NOTE 1: MRW MR05 command with OP[76:0]= 00011011_B~~xxxxxxxxB~~ (Stop DQS interval oscillator) stops DQS interval timer in case of MR20~~xx~~ OP[7:0] = 00000000_B~~xxxxxxxxB~~.
NOTE 2: MRW MR05 command with OP[76:0]= 00011011_B~~xxxxxxxxB~~ (Stop DQS interval oscillator) is illegal with non-zero values in MR20~~xx~~ OP[7:0].

Table 156 — MR21~~xx~~ Register Information ~~(MA[7:0] = xxH)~~

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
CLK Interval Oscillator Count - LSB							

Table 157 — MR21~~xx~~ Definition

Function	Register Type	Operand	Data	Notes
CLK Interval Oscillator – LSB	Read-only	OP[7:0]	0-255 LSB DRAM CLK Interval Oscillator Count	1,2,3

Table 157 — MR21~~xx~~ Definition

Function	Register Type	Operand	Data	Notes
NOTE 1: MR21~~xx~~ reports the LSB bits of the DRAM CLK interval oscillator count. The DRAM CLK interval oscillator count value is used to train CLK to the DQ data valid window. The value reported by the DRAM in this mode register can be used by the memory controller to periodically adjust the phase of CLK relative to DQ. **NOTE 2:** Both MR21~~xx~~ and MR22~~xx~~ must be read (MRR) and combined to get the value of the CLK interval oscillator count. **NOTE 3:** A MRW MR05 [Start CLK Interval Osc] could be issued to reset the contents of MR21~~xx~~/MR22~~xx~~.				

Table 158 — MR22~~xx~~ Register Information ~~(MA[7:0] = xxH)~~

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
CLK Interval Oscillator Count - MSB							

Table 159 — MR22~~xx~~ Definition

Function	Register Type	Operand	Data	Notes
CLK Interval Oscillator – MSB	Read-only	OP[7:0]	0-255 MSB DRAM CLK Interval Oscillator Count	1,2,3
NOTE 1: MR22~~xx~~ reports the MSB bits of the DRAM CLK interval oscillator count. The DRAM CLK interval oscillator count value is used to train CLK to the DQ data valid window. The value reported by the DRAM in this mode register can be used by the memory controller to periodically adjust the phase of CLK relative to DQ. **NOTE 2:** Both MR21~~xx~~ and MR22~~xx~~ must be read (MRR) and combined to get the value of the CLK interval oscillator count. **NOTE 3:** A new MRW MR05 [Start CLK Interval Osc] could be issued to reset the contents of MR21~~xx~~/MR22~~xx~~.				

Table 160 — MR23~~xx~~ Register Information ~~(MA[7:0] = xxH)~~

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
CLK interval timer run time setting							

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Table 161 — MR23~~xx~~ Definition

Function	Register Type	Operand	Data	Notes
CLK interval timer run time	Write/Read	OP[7:0]	00000000B: CLK interval timer stop via MR Functional Modes Command (Default) 00000001B: CLK timer stops automatically at 16th clock after timer start 00000010B: CLK timer stops automatically at 32nd clock after timer start 00000011B: CLK timer stops automatically at 48th clock after timer start 00000100B: CLK timer stops automatically at 64th clock after timer start ---------- Thru ---------- 00111111B: CLK timer stops automatically at (63X16)th clock after timer start 01XXXXXXB: CLK timer stops automatically at 2048th clock after timer start 10XXXXXXB: CLK timer stops automatically at 4096th clock after timer start 11XXXXXXB: CLK timer stops automatically at 8192nd clock after timer start	1,2

NOTE 1: MRW MR05 command with OP[7~~6~~:0]= 00011001_B~~xxxxxxxxB~~ (Stop CLK interval oscillator) stops CLK interval timer in case of MR23~~xx~~ OP[7:0] = 00000000_B~~xxxxxxxxB~~.

NOTE 2: MRW MR05 command with OP[7~~6~~:0]= 00011001_B~~xxxxxxxxB~~ (Stop CLK interval oscillator) is illegal with non-zero values in MR23~~xx~~ OP[7:0].

7.9.4. Post Package Repair (PPR)

DDR6 supports three functions of Post Package Repair (PPR) for failed row addresses: Hard Post Package Repair (hPPR) and MBIST PPR (mPPR) for permanent row repair, and Soft Post Package Repair (sPPR) for temporary row repair. mPPR is to be used in conjunction with the MBIST function to repair failing addresses identified during MBIST.

7.9.4.1. Guard Key Protection

Access to hPPR, sPPR, MBIST, and mPPR modes is protected through a sequential MRW guard key ('PPR guard key'), which prevents unintended PPR programming. The sequential MRW guard key is the same for hPPR, sPPR, MBIST and mPPR.

The PPR guard key requires a sequence of four MRW commands to be issued immediately after issuing the MRW to enter PPR or MBIST. The guard key sequence is entered in the specified order as stated below and shown in the guard key encoding for the MR30 table. Interruptions to the guard key sequence by other MRW/MRR commands or non-MR commands (e.g., ACT, WR, RD, REFab, REFdb, REFqb) are prohibited. Although interruption of the guard key entry is not allowed, if the guard key is not entered in the required order or is interrupted by other commands, hPPR/sPPR/MBIST/mPPR entry will not execute and the offending command that interrupted the PPR guard key may or may not execute correctly. However, the offending command will not cause the device to lock up and subsequent ACT and WR commands will be conducted as normal DRAM commands. The DRAM does not provide an error indication if an incorrect PPR guard key sequence is entered.

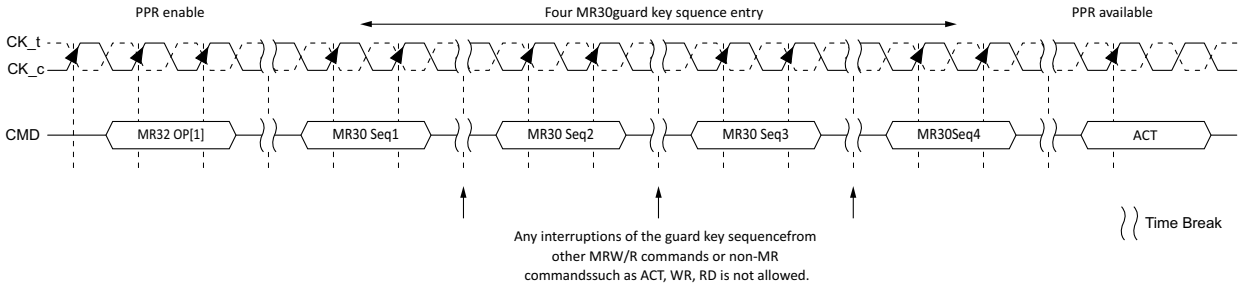


Figure 164 — Guard Key Timing Diagram

NOTE: Only DES commands are allowed during t_{MRD}.

Table 162 — Guard Key Encoding for MR30

Command	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
MR30 Seq1	1	1	0	0	1	1	1	1
MR30 Seq2	0	1	1	1	0	0	1	1
MR30 Seq3	1	0	1	1	1	0	1	1
MR30 Seq4	0	0	1	1	1	0	1	1

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7.9.4.2. Hard Post Package Repair (hPPR)

hPPR resources are allocated per bank in the PSD Mode Registers. The hPPR/sPPR resource designation (MRBF-MRC6) will indicate the hPPR resource availability and should be read/checked prior to attempting a repair. It is important to note that hPPR repairs are permanent.

Table 163 — MRBF-MRC6 OP[7:0] hPPR/sPPR Register Information

MR #	Name	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
MRBF	PPR Availability	BG1, Bank 1[1:0]		BG1, Bank 0[1:0]		BG0, Bank 1[1:0]		BG0, Bank 0[1:0]	
MRC 0		BG3, Bank 1[1:0]		BG3, Bank 0[1:0]		BG2, Bank 1[1:0]		BG2, Bank 0[1:0]	
MRC 1		BG5, Bank 1[1:0]		BG5, Bank 0[1:0]		BG4, Bank 1[1:0]		BG4, Bank 0[1:0]	
MRC 2		BG7, Bank 1[1:0]		BG7, Bank 0[1:0]		BG6, Bank 1[1:0]		BG6, Bank 0[1:0]	
MRC 3		BG9, Bank 1[1:0]		BG9, Bank 0[1:0]		BG8, Bank 1[1:0]		BG8, Bank 0[1:0]	
MRC 4		BG11, Bank 1[1:0]		BG11, Bank 0[1:0]		BG10, Bank 1[1:0]		BG10, Bank 0[1:0]	
MRC 5		BG13, Bank 1[1:0]		BG13, Bank 0[1:0]		BG12, Bank 1[1:0]		BG12, Bank 0[1:0]	
MRC 6		BG15, Bank 1[1:0]		BG15, Bank 0[1:0]		BG14, Bank 1[1:0]		BG14, Bank 0[1:0]	

00: More than one hPPR and one single sPPR resource are available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and more than one hPPR resource remains.
01: One single hPPR and sPPR shared resource is available. Transition from 00 after hPPR is executed, or from 10 after sPPR Undo is executed, and one single hPPR resource remains.
10: hPPR resource is available and sPPR resource is not available. Transition from 00 or 01 after sPPR is executed.
11: hPPR and sPPR resource is not available. Transition from 01 or 10 after hPPR is executed and no hPPR resources remain.

7.9.4.3. hPPR Fail Row Address Repair

Entry into hPPR is through a mode register enable and an ACT command is used to transmit the bank and row address of the row to be repaired. For DRAM configured as 2p3 or 2p6, an hPPR shall only be done on one PSD at a time and the other PSD shall remain idle. After tRCD time, a WRA command is used to select the individual DRAM through the DQ bits and to transfer the repair address to the DRAM. Following the program time and a PREpb command, exit hPPR through the mode register disable and resume normal operation after tPGMPST_hPPR.

hPPR Fail Row Address Repair Procedure:

1) Verify hPPR resource availability by issuing an MRR to read the hPPR/sPPR resource designation registers in the PSDMode Registers (MRBF-MRC6). If no repair resource is available (11, hPPR and sPPR resource is not available), the host controller should not initiate hPPR mode.
2) Before entering hPPR mode, precharge all banks and ensure they are idle; disable CRC mode.

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3) Enable hPPR by setting MR32[0] = 1 and wait t_{MRD}.
4) Issue the guard key as four consecutive MR30[7:0] MRW commands as described above.
5) Issue an ACT command with the BG, Bank and Row fail address; wait t_{RCD}.
6) Issue a WRA command with a valid column address; wait WL. The DRAM will ignore the column address.
7) After WL, drive DQ[2:0] HIGH for $12t_{CK}$ on the target DRAM. For non-target DRAMs, drive DQ[2:0] LOW for $12t_{CK}$. If DQ[2:0] data bits are neither all LOW nor all HIGH for $12t_{CK}$, then hPPR mode execution is unknown. For 2p6, 1p6 and 1p12 devices, data bits other than DQ[2:0] are don't cares.
8) Wait t_{PGM_hPPRa} or t_{PGM_hPPRb} for internal repair, then issue PREpb.
9) Wait $t_{PGM_EXIT_hPPR}$ after PREpb command to enable recognition of the repaired row address.
10) Exit hPPR by setting MR32[0] = 0 and wait t_{PGMPST_hPPR}.
11) Resume normal operation.
12) When repairing more than one failed address in a bank with more than one hPPR resource available, repeat steps 3–10. Otherwise, repeat steps 1-10 for repairs in other banks.

During hPPR mode, REFab/REFdb commands are allowed, but array contents are not guaranteed. Upon receiving a REFab or REFdb command in hPPR mode, the DRAM may ignore the Refresh operation but will not disrupt the repair operation. Other commands except REFab/REFdb during t_{PGM_hPPR} can cause incomplete repair so no other

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command except REF is allowed during t_{PGM_hPPR}. After exiting hPPR mode (MR32:OP[0] = 0 after t_{PGMPST_hPPR}), the host can verify the repaired row by writing data to the target row and subsequently reading it back.

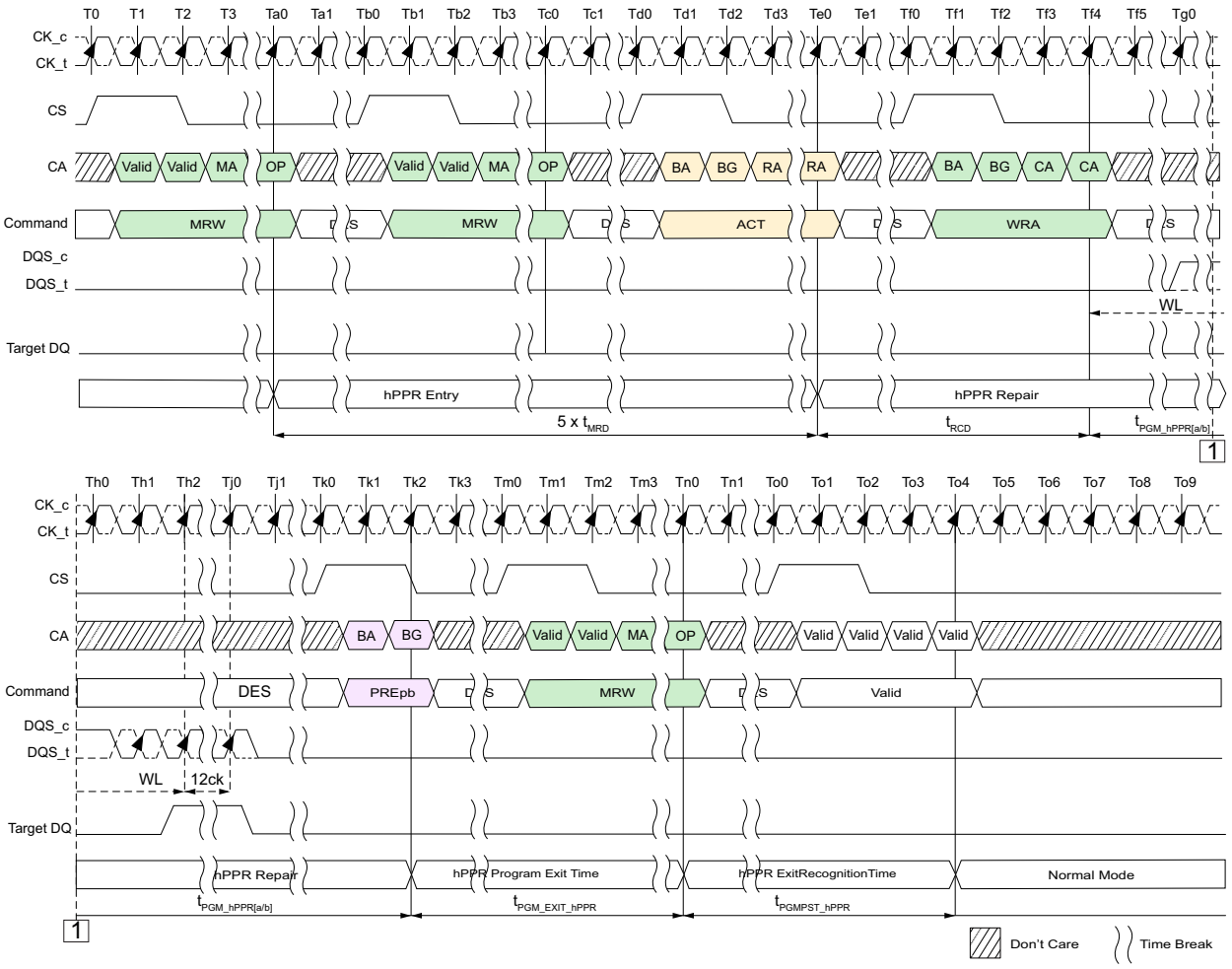


Figure 165 — hPPR Timing

NOTE: 1: With one hPPR command, only one row can be repaired at one time.
NOTE: 2: Allow REFab/REFdb/REFqb from $WL+12t_{CK}+t_{WR}+t_{RP}$ after WR, but does not guarantee array contents are refreshed during hPPR.
NOTE: 3: Timing diagram shows possible commands but not all shown can be issued at the same time. All regular timings must still be satisfied.

Table 164 — hPPR Timing Parameters (Sheet 1 of 2)

Parameter	Symbol	Value	Min/Max	Unit
hPPR Programming Time: 2p3, 2p6, 1p6	t_{PGM_hPPRa}	1000	Min	ms
hPPR Programming Time: 1p12	t_{PGM_hPPRb}	2000	Min	ms
hPPR Program Exit Time	$t_{PGM_EXIT_hPPR}$	t_{RPab}	Min	ns

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Table 164 — hPPR Timing Parameters (Continued) (Sheet 2 of 2)

Parameter	Symbol	Value	Min/Max	Unit
hPPR Exit Recognition Time	t_{PGMPST_hPPR}	500	Min	µs

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7.9.5. Soft Post Package Repair (sPPR)

Soft Post Package Repair (sPPR) enables rapid, temporary repair of a single row address per bank on DDR6 DRAM devices. In contrast, Hard Post Package Repair (hPPR) provides permanent row repair with a longer execution time. sPPR and hPPR differ in their operational limitations and intended use cases.

The host controller must ensure that refresh requirements for all other rows are maintained during sPPR operations. sPPR information is retained as long as VDD remains within the operating range. If power is removed or the device is RESET, the sPPR will revert to the unrepaired state. If hPPR or mPPR modes are entered while sPPR is enabled, the DRAM will disable sPPR in the target bank and clear the current address prior to executing. MBIST and sPPR may not be enabled at the same time. sPPR must have been disabled, cleared and unlocked prior to entering MBIST.

Each bank has one sPPR resource shared with hPPR; when the hPPR resources for a bank are used up, the bank has no more available resources for sPPR. To preserve sPPR availability, this shared resource will be used for hPPR by the DRAM only after all other hPPR resources are exhausted. The host is responsible for reading the hPPR/sPPR resource registers prior to issuing sPPR to ensure the resource is available in the targeted bank. The sPPR Verification mode can be used to validate if the targeted row was repaired correctly. The use of the sPPR resource will be reflected in the hPPR/sPPR resource registers as 10_B.

DDR6 shall support the sPPR Undo and sPPR Lock operations. sPPR Undo will undo a previous sPPR and cause the sPPR resource to be taken back to its unused condition. The original row will appear back in the memory map at its original location following an sPPR Undo and will be reflected in the hPPR/sPPR resource register state. sPPR Lock will lock the specific sPPR resource at its current location and not allow another sPPR or an sPPR Undo to be performed to that resource.

Rows that have been replaced by spare rows do not require refresh by the DRAM. When a spare row is moved in or out of the memory map, the host controller is responsible for issuing sufficient ACT commands to any unmapped row to maintain required data retention. The host controller is also responsible for performing all necessary data copy operations between the original row and the spare row.

7.9.5.1. sPPR Fail Row Address Repair

Entry into sPPR is through a mode register enable and an ACT command is used to transmit the bank and row address of the row to be repaired. For DRAM configured as 2p3 or 2p6, an sPPR shall only be done on one PSD at a time and the other PSD shall remain idle. After t_{RCD} time, a WR command is used to select the individual DRAM through the DQ bits and to transfer the repair address to the DRAM. Following the program time and a PREpb command, exit sPPR through the mode register disable and resume normal operation after t_{PGMPST_sPPR}.

sPPR Fail Row Address Repair Procedure:

1) Verify sPPR resource availability by issuing an MRR to read the hPPR/sPPR resource designation registers in the PSD Mode Registers (~~MR33-48~~ MRBF-MRC6). If no resources are available (10_B, hPPR resource is available and sPPR resource is not available, 11_B, hPPR and sPPR resource is not available), do not initiate sPPR mode.
2) Back up the data of the target row address in the bank prior to executing sPPR. The backup data should be one row per bank.
3) Precharge all banks and ensure they are idle; disable CRC mode.
4) Enable sPPR by setting MR32:OP[2:1] = 01_B and wait t_{MRD}.
5) Issue the guard key as four consecutive MR30:OP[7:0] MRW commands as described in PPR Guard Key Protection.
6) Issue an ACT command with the BG, Bank, and Row fail address; wait t_{RCD}.
7) Issue a WR command with a valid column address; wait WL. The DRAM will ignore the column address.

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8) After WL, drive DQ[2:0] HIGH for $12t_{CK}$ on the target DRAM. For non-target DRAMs, drive DQ[2:0] LOW for $12t_{CK}$. If all the DQ[2:0] data bits are neither all LOW nor all HIGH for $12t_{CK}$, then sPPR mode execution is unknown. For 2p6, 1p6 and 1p12 devices, data bits other than DQ[2:0] are don't cares.

9) Wait t_{PGM_sPPR} for the internal repair register to be written, then issue PREpb to the bank.

10) Wait $t_{PGM_EXIT_sPPR}$ after PREpb to enable recognition of the repaired row address.

11) Exit sPPR by setting MR32:OP[2:1] = 00_B and wait t_{PGMPST_sPPR}.

12) Restore the backed-up data to the repaired array

13) sPPR can be performed without affecting the hPPR previously performed, provided a row is available in that repair region. In the case of conducting a subsequent sPPR in the same bank, a sPPR Undo must be performed before initiating sPPR. In the case of conducting sPPR in a different bank, repeat steps 1–12. During sPPR, a REFRESH command is not allowed between sPPR MRW entry and exit.

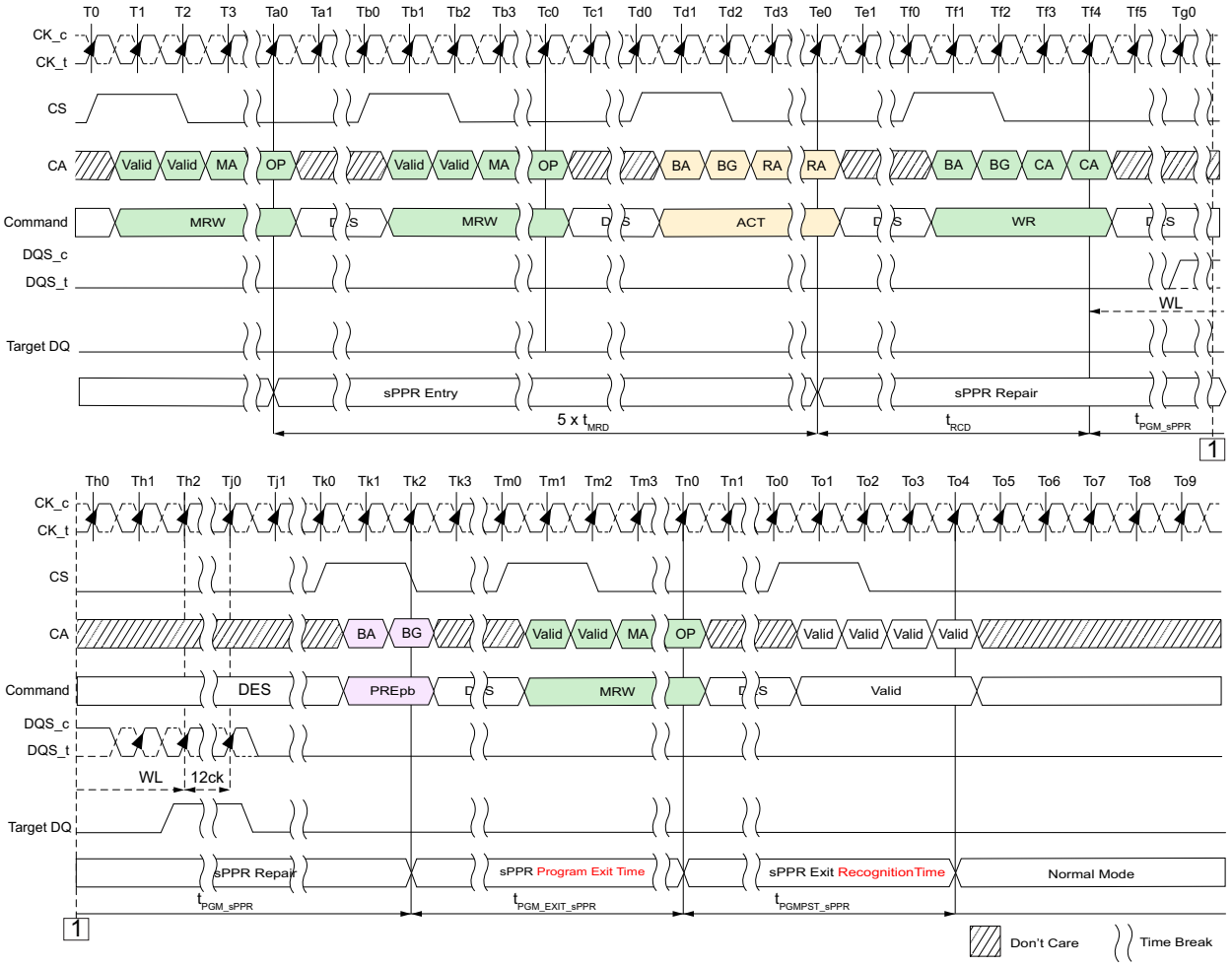


Figure 166 — sPPR Timing

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Table 165 — sPPR Timing Parameters

Parameter	Symbol	Value	Min/Max	Unit
sPPR Programming Time	t_{PGM_sPPR}	$CL+12*t_{CK}+t_{WTP}$	Min	ns
sPPR Program Exit Time	$t_{PGM_EXIT_sPPR}$	$t_{RPab}+t_{RFMpb}$	Min	ns
sPPR Exit Recognition Time	t_{PGMPST_sPPR}	t_{MRD}	Min	ns

7.9.5.2. sPPR Undo

sPPR Undo restores an sPPR resource to its unused state, equivalent to its condition following reset, and will be reflected in the hPPR/sPPR resource register state. ~~This operation uses~~ The sPPR Undo procedure follows the same protocol as the sPPR sequence below~~, except MR32:OP[2:1] is set to 10_B~~.

1) Precharge all banks and ensure they are idle; disable CRC mode.
2) Enable sPPR Undo by setting MR32:OP[2:1] = 10_B and wait t_{MRD}.
3) Issue the guard key as four consecutive MR30:OP[7:0] MRW commands as described in PPR Guard Key Protection.
4) Issue an ACT command with the BG, Bank, and Row fail address; wait t_{RCD}.
5) Issue a WR command with a valid column address; wait WL. The DRAM will ignore the column address.
6) After WL, drive DQ[2:0] HIGH for $12t_{CK}$ on the target DRAM. For non-target DRAMs, drive DQ[2:0] LOW for $12t_{CK}$. If all the DQ[2:0] data bits are neither all LOW nor all HIGH for $12t_{CK}$, then sPPR mode execution is unknown. For 2p6, 1p6 and 1p12 devices, data bits other than DQ[2:0] are don't cares.
7) Wait t_{PGM_sPPR} for the internal repair register to be un-written, then issue PREpb to the bank.
8) Wait $t_{PGM_EXIT_sPPR}$ after PREpb to enable recognition of the repaired row address.
9) Exit sPPR Undo by setting MR32:OP[2:1] = 00_B and wait t_{PGMPST_sPPR}.

The host controller must issue an ACT command with the same BG bits, Bank bits, and row address used in the most recent sPPR for that resource. If the row address does not match the most recent sPPR, the DRAM may or may not perform the undo. All required data copy operations are the responsibility of the host controller.

After an sPPR Undo, the resource can be reassigned to a new or previous location. Data within the sPPR resource is retained but does not require refresh by the DRAM. If data validity must be maintained, the host controller shall issue sufficient ACT commands while the row is mapped in.

7.9.5.3. sPPR Lock

sPPR Lock secures an sPPR resource in its current location, preventing any subsequent sPPR or sPPR Undo operations on that resource. Locks are applied individually to each sPPR resource. To remove a lock, a hardware reset or power cycle must be performed. A hardware reset or power cycle is also required before executing any hPPR operation if any sPPR resources remain locked.

The sPPR Lock procedure follows the same protocol as the sPPR sequence below~~, except MR32:OP[2:1] is set to 11_B~~.

1) Precharge all banks and ensure they are idle; disable CRC mode.
2) Enable sPPR Lock by setting MR32:OP[2:1] = 11_B and wait t_{MRD}.

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3) Issue the guard key as four consecutive MR30:OP[7:0] MRW commands as described in PPR Guard Key Protection.
4) Issue an ACT command with the BG, Bank, and Row fail address; wait t_{RCD}.
5) Issue a WR command with a valid column address; wait WL. The DRAM will ignore the column address.
6) After WL, drive DQ[2:0] HIGH for $12t_{CK}$ on the target DRAM. For non-target DRAMs, drive DQ[2:0] LOW for $12t_{CK}$. If all the DQ[2:0] data bits are neither all LOW nor all HIGH for $12t_{CK}$, then sPPR mode execution is unknown. For 2p6, 1p6 and 1p12 devices, data bits other than DQ[2:0] are don't cares.
7) Wait t_{PGM_sPPR} for the internal repair register to be locked, then issue PREpb to the bank.
8) Wait $t_{PGM_EXIT_sPPR}$ after PREpb to enable recognition of the repaired row address.
9) Exit sPPR Lock by setting MR32:OP[2:1] = 00_B and wait t_{PGMPST_sPPR}.

The ACT command shall include BG bits, Bank bits, and the row address associated with the most recent sPPR for that resource. If the row address does not match the most recent sPPR, the device may or may not perform the lock.

7.9.5.4. sPPR Verification

sPPR Verification is a DDR6 feature that validates whether a specified row address matches the most recent sPPR assignment for a given bank. This operation follows the same protocol as the sPPR sequence, with MR32:OP[3] set to 1. The verification result is stored in MR32:OP[4], where a value of 1 indicates success and 0 indicates failure. MR32:OP[4] is shared across all banks within each PSD and is reset to 0 after every MRR to the verification register. Multiple row addresses from different banks can be verified; however, the host controller must read MR32:OP[4] after each ACT command to prevent overwriting the previous verification status.

The following is the procedure for sPPR Verification:

1) Precharge all banks and ensure they are idle; disable CRC mode.
2) Enable sPPR Verification by setting MR32:OP[3] = 1 and wait t_{MRD}.
3) Issue the guard key as four consecutive MR30:OP[7:0] MRW commands as described in PPR Guard Key Protection.
4) Issue an ACT command with the BG, Bank and Row fail address and wait t_{RAS}.
5) Issue a PREpb command to the bank and wait t_{RP}.
6) Issue an MRR command to MR32:OP[4] to read the verification status and wait t_{MRR}.
7) Exit sPPR Verification by setting MR32:OP[3] = 0 and wait t_{MRD}.
8) In the case of verifying a subsequent Row address from another bank, repeat steps 4–6.

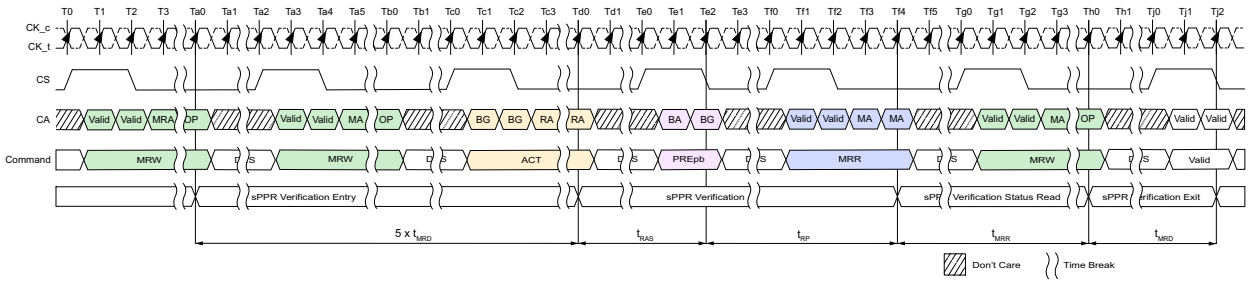


Figure 167 — sPPR Verification Timing

7.9.6. MBIST / mPPR

DDR6 devices support Memory Built-In Self-Test (MBIST) and Memory Built-In Self-Test Post Package Repair (mPPR) functions to address hard failures, such as single-bit or multi-bit errors within a device. These functions enable detection and repair of weak cells.

- **MBIST:** The DRAM executes vendor-specific test patterns to scan the memory array and identify hard failures.
- **mPPR:** The DRAM repairs failing addresses identified during MBIST.

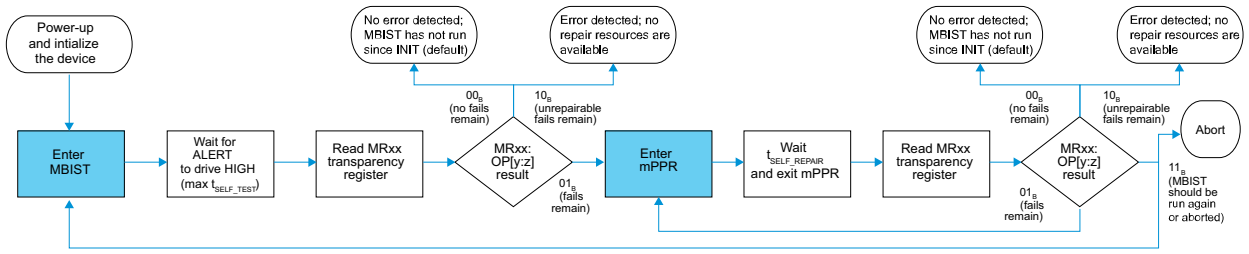


Figure 168 — MBIST Flow Diagram

7.9.6.1. MBIST

MBIST shall only be executed when all banks are in the idle state after device initialization but must be completed prior to mPPR. To enable MBIST, the host shall issue an MRW command with MR31[0] = 1. During MBIST, the DRAM applies vendor-specific test patterns to the memory array to detect hard failures. The DRAM asserts ALERT for the duration of MBIST and de-asserts ALERT upon completion. For 2p3 or 2p6 devices, initiating MBIST through either PSD causes the DRAM to execute MBIST on both PSDs.

After MBIST concludes and ALERT is LOW, the DRAM updates the MBIST transparency status in MRC7[1:0], which the host shall read. If the transparency status indicates remaining failures and the controller elects to perform mPPR, mPPR shall be executed immediately after reading the transparency status. Detailed transparency status definitions are provided in MRC7.

Table 166 — MBIST Mode Registers

MR#	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0	FSP
MR31	RV						mPPR Enable	MBIST Enable	N

Table 167 — MBIST Transparency Register

MR#	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0	FSP
MRC7	RV						MBIST Transparency		N

PROPOSED

The following steps shall be performed to execute MBIST. For 2p3 or 2p6 devices, steps 2-4 performed through one PSD executes MBIST on both PSDs.

1) Precharge all banks and ensure they are idle; disable CRC mode.
2) Enable MBIST by setting MR31[0] = 1 and wait t_{MRD}.
3) Issue the guard key as four consecutive MR30[7:0] MRW commands as described in PPR Guard Key Protection. Upon completion of the guard key sequence, the DRAM enters MBIST operation and asserts ALERT HIGH for t_{SELF_TEST}.
   a) While ALERT is HIGH, only DES commands are permitted.
   b) When MBIST concludes, the DRAM de-asserts ALERT LOW to indicate completion. MBIST transparency status is updated in MRC7.
4) Exit MBIST by setting MR31[0] = 0 and wait t_{MRD}.
5) Issue an MRR to MRC7[1:0] and wait t_{MRR}. If transparency indicates that failures remain and repair resources are available, proceed to the mPPR procedure.

The following are the MBIST transparency outcomes of MRC7[1:0]:

- **00:** No error detected / MBIST has not run since INIT (default)
- **01:** Error detected; repair resources are available
- **10:** Error detected; no repair resources are available
- **11:** Unresolved operation, abort or retry

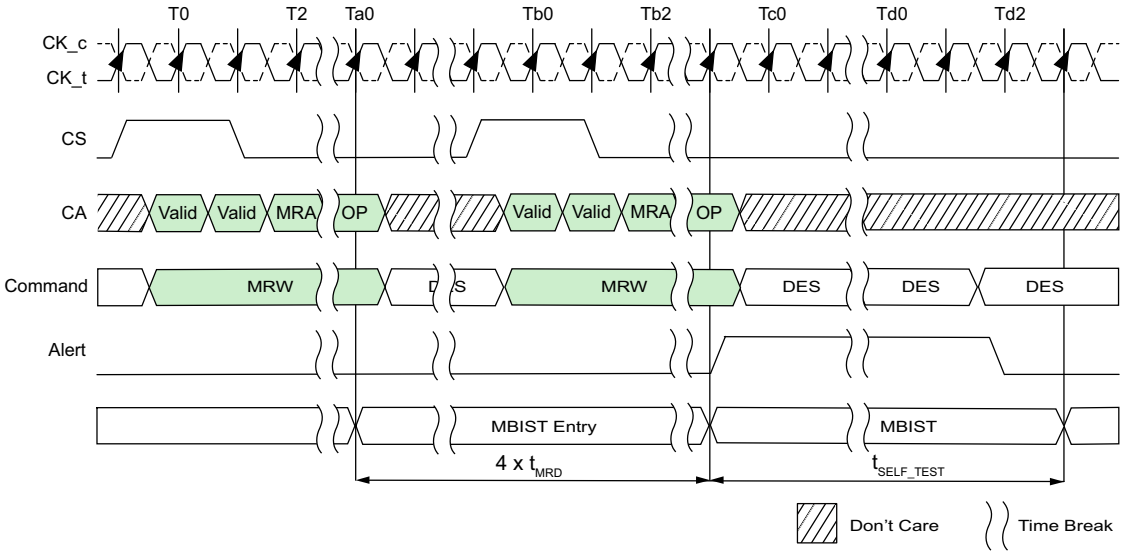


Figure 169 — MBIST Entry

7.9.6.2. mPPR

If the host issues an MRR to MRC7[1:0] and obtains outcome 01 (error detected; repair resources are available) then mPPR shall be performed by issuing an MRW with MR31[1] = 1. The mPPR procedure will repair a single row identified during MBIST. Attempting mPPR without MBIST may result in an unexpected repair. mPPR resources are separated from normal hPPR/sPPR resources.

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The following steps shall be performed to execute mPPR. For 2p3 or 2p6 devices, all steps shall be performed through each PSD.

1) Precharge all banks and ensure they are idle; disable CRC mode.
2) Enable mPPR by setting MR31[1] = 1 and wait t_{MRD}.
3) Issue the guard key as four consecutive MR30[7:0] MRW commands as described in PPR Guard Key Protection.
4) Issue an ACT command with a valid BG, Bank, and Row address; wait t_{RCD}. The DRAM will ignore the address.
5) Issue a WRA command with a valid column address; wait WL. The DRAM will ignore the column address.
6) After WL, drive DQ[2:0] HIGH for $12t_{CK}$ on the target DRAM. Only DRAMs which most recently have had an MBIST transparency result of 01 may be considered “target DRAMs”. For non-target DRAMs, drive DQ[2:0] LOW for $12t_{CK}$. If DQ[2:0] data bits are neither all LOW nor all HIGH for $12t_{CK}$, then mPPR mode execution is unknown. For 2p6, 1p6 and 1p12 devices, data bits other than DQ[2:0] are don't cares.
7) Wait t_{PGM_mPPR} for internal row repair, then issue a PREab command.
8) Wait $t_{PGM_EXIT_mPPR}$ after PREab command to enable the DRAM to update MBIST transparency.
9) Exit mPPR by setting MR31[1] = 0 and wait t_{PGMPST_mPPR}.

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10) Issue an MRR to MRC7[1:0] and wait t_{MRR}. If MBIST transparency result is 01 (error detected; repair resources are available), repeat steps 2-10 to perform additional mPPR. Otherwise, the DRAM may continue to the next planned operation.

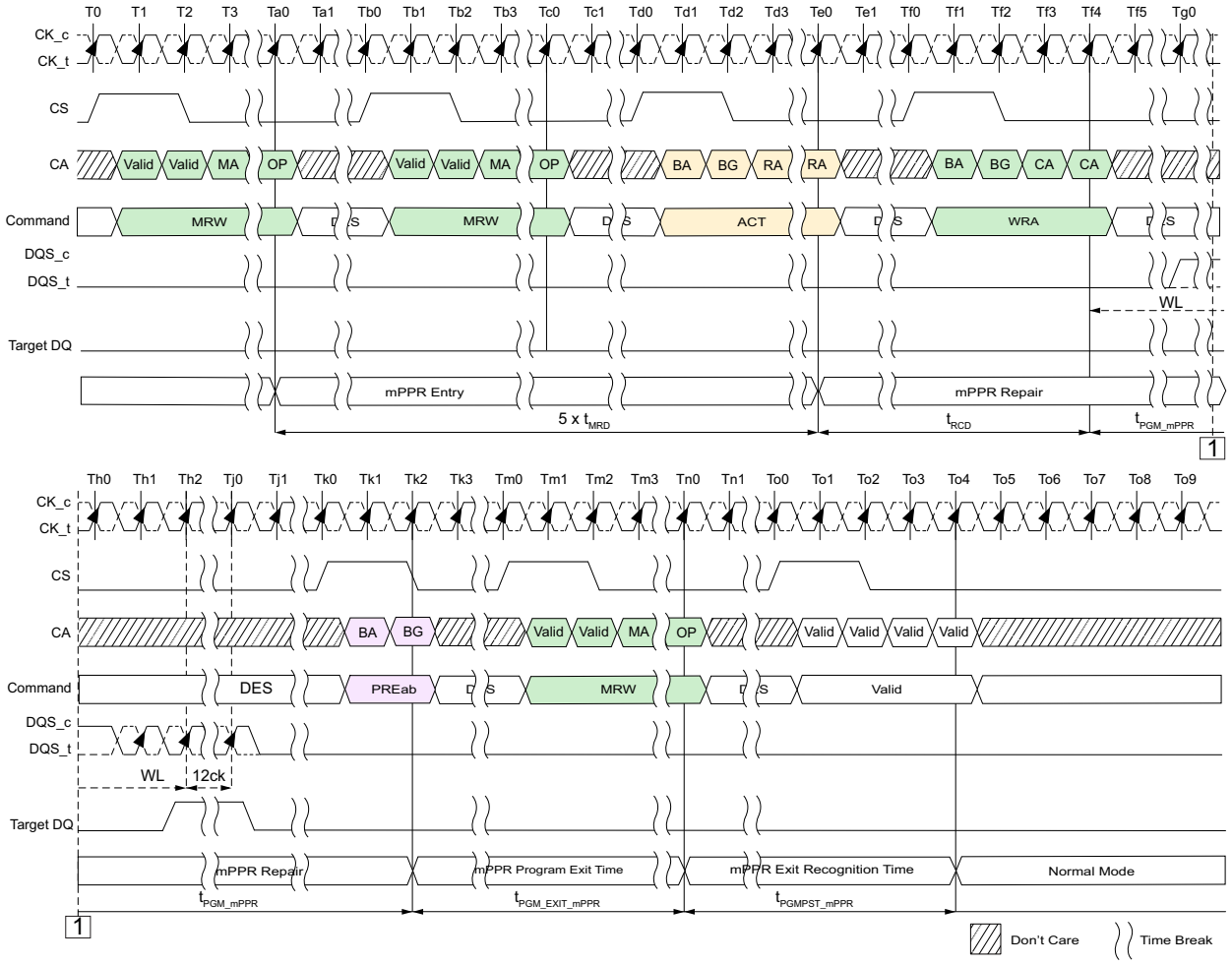


Figure 170 — mPPR Entry

Table 168 — MBIST Timing Parameters Test Times

Paramter	Symbol	Density			Min/ Max	Unit
		32Gb	48Gb	64Gb		
MBIST Time	t_{SELF_TEST}	28.7	42	55.4	Max	s
mPPR Programming Time	t_{PGM_mPPR}	5.6			Max	s
mPPR Program Exit Time	$t_{PGM_EXIT_mPPR}$	t_{RPab}			Min	ns
mPPR Exit Recognition Time	t_{PGMPST_mPPR}	500			Min	µs

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7.9.7. DQ RX Decision Feedback Equalization (DFE)

The DDR6 supports high data rates. Operating at elevated frequencies within the memory channel can result in significant inter-symbol interference (ISI) due to insertion loss and reflections. To compensate channel characteristics and enhance signal integrity, equalization techniques are required on the DRAM receiver (RX) side.

DDR6 provides per-pin controlled Decision Feedback Equalization (DFE) for DQ RX. A 2-tap DFE is selected to equalize the received DQ signals, as it effectively reduces the impact of insertion loss and reflections without amplifying the noise.

To eliminate ISI, the per-pin DFE Tap-1 and Tap-2 bits should be enabled and appropriately configured for the data rates. When the DFE is enabled, the DQs shall be driven low for a minimum of 8UI prior to the first Write data bit to ensure proper DFE synchronization. In case of back to back write, there is no need to add pre-drive.

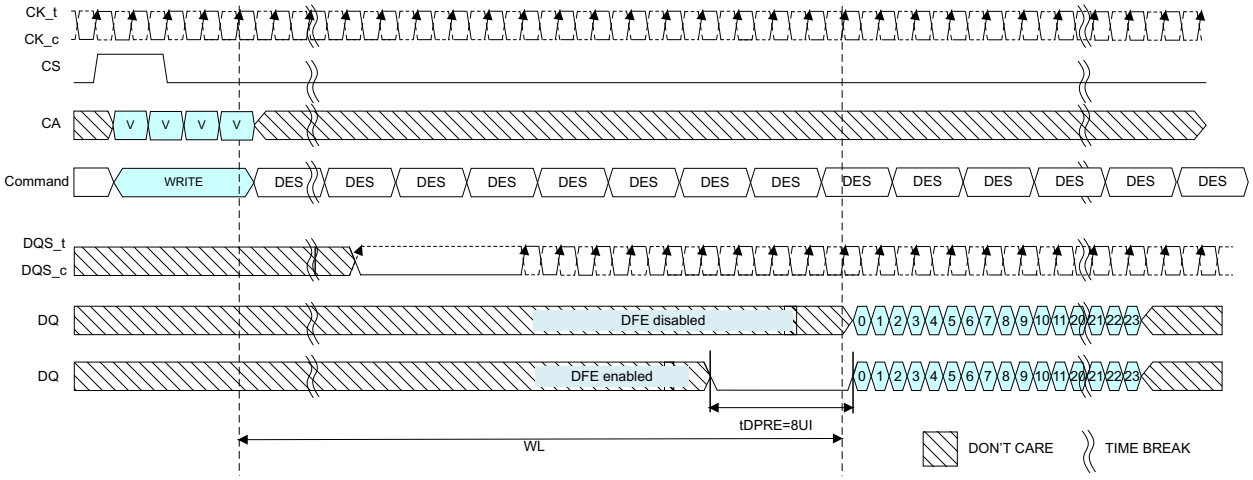


Figure 171 — Per-pin DFE Pre-Drive Requirement

7.9.7.1. DFE Tap Coefficients

The DFE tap coefficients shown in Table 1 are used by the memory controller to adjust the strengths of Tap-1 and Tap-2, to adapt ISI cancellation based on channel performance. Tables 2 and 3 list the DFE Tap-1 and Tap-2 coefficient values, respectively, according to their step size, minimum, and maximum values. Table 4 is DFE Tap-2 Coefficient Step Table for polarity inversion.

Table 169 — Min/Max Ranges and Steps for DFE Tap Coefficients (Sheet 1 of 2)

Parameter	Min/Max/Typ	Value		Unit	Note
		~12.8Gb/s	Data Rate > 12.8Gb/s		
DFE Tap-1 coefficient	Min	-75	TBD	mV	1, 2
	Max	0	TBD	mV	1, 2
DFE Tap-1 coefficient step size	Typ	5	TBD	mV	1, 2

Table 169 — Min/Max Ranges and Steps for DFE Tap Coefficients (Continued)

Parameter	Min/Max/Typ	Value		Unit	Note
		~12.8Gb/s	Data Rate > 12.8Gb/s		
DFE Tap-2 coefficient	Min	-35	TBD	mV	1, 2
	Max	35	TBD	mV	1, 2
DFE Tap-2 coefficient step size	Typ	5	TBD	mV	1, 2

NOTE 1: DFE Quantity setting is specified across DC voltage and temperature. NOTE 2 These numbers are guaranteed by design.

Table 170 — DFE Tap-1 Coefficient Step Table

Parameter	Value	Unit	Notes
DFE Tap-1 step 0	0	mV	
DFE Tap-1 step 1	-5	mV	
DFE Tap-1 step 2	-10	mV	
DFE Tap-1 step 3	-15	mV	
DFE Tap-1 step 4	-20	mV	
DFE Tap-1 step 5	-25	mV	
DFE Tap-1 step 6	-30	mV	
DFE Tap-1 step 7	-35	mV	
DFE Tap-1 step 8	-40	mV	
DFE Tap-1 step 9	-45	mV	
DFE Tap-1 step 10	-50	mV	
DFE Tap-1 step 11	-55	mV	
DFE Tap-1 step 12	-60	mV	
DFE Tap-1 step 13	-65	mV	
DFE Tap-1 step 14	-70	mV	
DFE Tap-1 step 15	-75	mV	

Table 171 — DFE Tap-2 Coefficient Step Table (Sheet 1 of 2)

Parameter	Value	Unit	Notes
DFE Tap-2 step 0	0	mV	
DFE Tap-2 step 1	-5	mV	

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Table 171 — DFE Tap-2 Coefficient Step Table (Continued) (Sheet 2 of 2)

Parameter	Value	Unit	Notes
DFE Tap-2 step 2	-10	mV	
DFE Tap-2 step 3	-15	mV	
DFE Tap-2 step 4	-20	mV	
DFE Tap-2 step 5	-25	mV	
DFE Tap-2 step 6	-30	mV	
DFE Tap-2 step 7	-35	mV	

Table 172 — DFE Tap-2 Coefficient Step Table for Polarity Inversion

Parameter	Value	Unit	Notes
DFE Tap-2 step 0	0	mV	
DFE Tap-2 step 1	5	mV	
DFE Tap-2 step 2	10	mV	
DFE Tap-2 step 3	15	mV	
DFE Tap-2 step 4	20	mV	
DFE Tap-2 step 5	25	mV	
DFE Tap-2 step 6	30	mV	
DFE Tap-2 step 7	35	mV	

7.9.7.2. DFE Mode Register Definition

Table 173 — DFE Mode Registers (Sheet 1 of 2)

MR #	Name	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
MR8D	DFE control/ DFE enable	RV					DFE Tap-2 Enable	DFE Tap-1 Enable	Per-pin DFE Control
MR8E	DFE DQ0	RV			Per-pin DFE Tap-1 Quantity for DQ0				
MR8F		RV			Per-pin DFE Tap-2 Quantity for DQ0				
MR91	DFE DQ1	RV			Per-pin DFE Tap-1 Quantity for DQ1				
MR92		RV			Per-pin DFE Tap-2 Quantity for DQ1				

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Table 173 — DFE Mode Registers (Continued) (Sheet 2 of 2)

MR #	Name	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
MR94	DFE DQ2	RV			Per-pin DFE Tap-1 Quantity for DQ2				
MR95		RV			Per-pin DFE Tap-2 Quantity for DQ2				
MR97	DFE DQ3	RV			Per-pin DFE Tap-1 Quantity for DQ3				
MR98		RV			Per-pin DFE Tap-2 Quantity for DQ3				
MR9A	DFE DQ4	RV			Per-pin DFE Tap-1 Quantity for DQ4				
MR9B		RV			Per-pin DFE Tap-2 Quantity for DQ4				
MR9E	DFE DQ5	RV			Per-pin DFE Tap-1 Quantity for DQ5				
MR9F		RV			Per-pin DFE Tap-2 Quantity for DQ5				

Table 174 — MR8D Definition: DFE Control and Enable

Function	Register Type	Operand	Data	Notes
Per-pin DFE Control	RW	OP[0]	0: Per-pin DFE enable (Default) 1: Per-pin DFE disable & Broadcast DFE DQ0 to all DQs	
DFE Tap-1 Enable	RW	OP[1]	0: DFE Tap-1 enable (Default) 1: DFE Tap-1 disable	1, 2
DFE Tap-2 Enable	RW	OP[2]	0: DFE Tap-2 enable (Default) 1: DFE Tap-2 disable	1, 2

NOTE 1: This bit applies to all DQ pins
NOTE 2: Setting MR8D[2:1] = 11 disables the DFE

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Table 175 — MR8E Definition: DFE Tap-1 DQ0

Function	Register Type	Operand	Data	Notes
Per-pin DFE Tap-1 Quantity for DQ0 (DFE TAP-1 DQ0)	RW	OP[4:0]	00000: DFE Tap-1 quantity step 0 (Default) 00001: DFE Tap-1 quantity step 1 00010: DFE Tap-1 quantity step 2 . . 01111: DFE Tap-1 quantity step 15 10000: Reserved . . 11111: Reserved	1, 2, 3

NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values
NOTE 2: The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables
NOTE 3: If DFE Tap1 is disabled (MR8D[1]=1), MR8E[4:0] is ignored

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Table 176 — MR8F Definition: DFE Tap-2 DQ0

Function	Register Type	Operand	Data	Notes
Per-pin DFE Tap-2 Quantity for DQ0 (DFE TAP-2 DQ0)	RW	OP[4:0]	11001: DFE Tap-2 quantity step -7 11010: DFE Tap-2 quantity step -6 11011: DFE Tap-2 quantity step -5 11100: DFE Tap-2 quantity step -4 11101: DFE Tap-2 quantity step -3 11110: DFE Tap-2 quantity step -2 11111: DFE Tap-2 quantity step -1 00000: DFE Tap-2 quantity step 0 (Default) 00001: DFE Tap-2 quantity step +1 00010: DFE Tap-2 quantity step +2 00011: DFE Tap-2 quantity step +3 00100: DFE Tap-2 quantity step +4 00101: DFE Tap-2 quantity step +5 00110: DFE Tap-2 quantity step +6 00111: DFE Tap-2 quantity step +7 01000: Reserved . . 11000: Reserved	1, 2, 3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient tables **NOTE 3:** If DFE Tap2 is disabled (MR8D[2]=1), MR8F[4:0] is ignored				

Table 177 — MR91 Definition: DFE Tap-1 DQ1

Function	Register Type	Operand	Data	Notes
Per-pin DFE Tap-1 Quantity for DQ1 (DFE TAP-1 DQ1)	RW	OP[4:0]	00000: DFE Tap-1 quantity step 0 (Default) 00001: DFE Tap-1 quantity step 1 00010: DFE Tap-1 quantity step 2 . . 01111: DFE Tap-1 quantity step 15 10000: Reserved . . 11111: Reserved	1, 2, 3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap1 is disabled (MR8D[1]=1), MR91[4:0] is ignored				

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Table 178 — MR92 Definition: DFE Tap-2 DQ1

Function	Register Type	Operand	Data	Notes
Per-pin DFE Tap-2 Quantity for DQ1 (DFE TAP-2 DQ1)	RW	OP[4:0]	11001: DFE Tap-2 quantity step -7 11010: DFE Tap-2 quantity step -6 11011: DFE Tap-2 quantity step -5 11100: DFE Tap-2 quantity step -4 11101: DFE Tap-2 quantity step -3 11110: DFE Tap-2 quantity step -2 11111: DFE Tap-2 quantity step -1 00000: DFE Tap-2 quantity step 0 (Default) 00001: DFE Tap-2 quantity step +1 00010: DFE Tap-2 quantity step +2 00011: DFE Tap-2 quantity step +3 00100: DFE Tap-2 quantity step +4 00101: DFE Tap-2 quantity step +5 00110: DFE Tap-2 quantity step +6 00111: DFE Tap-2 quantity step +7 01000: Reserved . . 11000: Reserved	1, 2, 3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient tables **NOTE 3:** If DFE Tap2 is disabled (MR8D[2]=1), MR92[4:0] is ignored				

Table 179 — MR94 Definition: DFE Tap-1 DQ2

Function	Register Type	Operand	Data	Notes
Per-pin DFE Tap-1 Quantity for DQ2 (DFE TAP-1 DQ2)	RW	OP[4:0]	00000: DFE Tap-1 quantity step 0 (Default) 00001: DFE Tap-1 quantity step 1 00010: DFE Tap-1 quantity step 2 . . 01111: DFE Tap-1 quantity step 15 10000: Reserved . . 11111: Reserved	1, 2, 3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap1 is disabled (MR8D[1]=1), MR94[4:0] is ignored				

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Table 180 — MR95 Definition: DFE Tap-2 DQ2

Function	Register Type	Operand	Data	Notes
Per-pin DFE Tap-2 Quantity for DQ2 (DFE TAP-2 DQ2)	RW	OP[4:0]	11001: DFE Tap-2 quantity step -7 11010: DFE Tap-2 quantity step -6 11011: DFE Tap-2 quantity step -5 11100: DFE Tap-2 quantity step -4 11101: DFE Tap-2 quantity step -3 11110: DFE Tap-2 quantity step -2 11111: DFE Tap-2 quantity step -1 00000: DFE Tap-2 quantity step 0 (Default) 00001: DFE Tap-2 quantity step +1 00010: DFE Tap-2 quantity step +2 00011: DFE Tap-2 quantity step +3 00100: DFE Tap-2 quantity step +4 00101: DFE Tap-2 quantity step +5 00110: DFE Tap-2 quantity step +6 00111: DFE Tap-2 quantity step +7 01000: Reserved . . 11000: Reserved	1, 2, 3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient tables **NOTE 3:** If DFE Tap2 is disabled (MR8D[2]=1), MR95[4:0] is ignored				

Table 181 — MR97 Definition: DFE Tap-1 DQ3

Function	Register Type	Operand	Data	Notes
Per-pin DFE Tap-1 Quantity for DQ3 (DFE TAP-1 DQ3)	RW	OP[4:0]	00000: DFE Tap-1 quantity step 0 (Default) 00001: DFE Tap-1 quantity step 1 00010: DFE Tap-1 quantity step 2 . . 01111: DFE Tap-1 quantity step 15 10000: Reserved . . 11111: Reserved	1, 2, 3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap1 is disabled (MR8D[1]=1), MR97[4:0] is ignored				

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Table 182 — MR98 Definition: DFE Tap-2 DQ3

Function	Register Type	Operand	Data	Notes
Per-pin DFE Tap-2 Quantity for DQ3 (DFE TAP-2 DQ3)	RW	OP[4:0]	11001: DFE Tap-2 quantity step -7 11010: DFE Tap-2 quantity step -6 11011: DFE Tap-2 quantity step -5 11100: DFE Tap-2 quantity step -4 11101: DFE Tap-2 quantity step -3 11110: DFE Tap-2 quantity step -2 11111: DFE Tap-2 quantity step -1 00000: DFE Tap-2 quantity step 0 (Default) 00001: DFE Tap-2 quantity step +1 00010: DFE Tap-2 quantity step +2 00011: DFE Tap-2 quantity step +3 00100: DFE Tap-2 quantity step +4 00101: DFE Tap-2 quantity step +5 00110: DFE Tap-2 quantity step +6 00111: DFE Tap-2 quantity step +7 01000: Reserved . . 11000: Reserved	1, 2, 3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient tables **NOTE 3:** If DFE Tap2 is disabled (MR8D[2]=1), MR98[4:0] is ignored				

Table 183 — MR9A Definition: DFE Tap-1 DQ4

Function	Register Type	Operand	Data	Notes
Per-pin DFE Tap-1 Quantity for DQ4 (DFE TAP-1 DQ4)	RW	OP[4:0]	00000: DFE Tap-1 quantity step 0 (Default) 00001: DFE Tap-1 quantity step 1 00010: DFE Tap-1 quantity step 2 . . 01111: DFE Tap-1 quantity step 15 10000: Reserved . . 11111: Reserved	1, 2, 3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap1 is disabled (MR8D[1]=1), MR9A[4:0] is ignored				

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Table 184 — MR9B Definition: DFE Tap-2 DQ4

Function	Register Type	Operand	Data	Notes
Per-pin DFE Tap-2 Quantity for DQ4 (DFE TAP-2 DQ4)	RW	OP[4:0]	11001: DFE Tap-2 quantity step -7 11010: DFE Tap-2 quantity step -6 11011: DFE Tap-2 quantity step -5 11100: DFE Tap-2 quantity step -4 11101: DFE Tap-2 quantity step -3 11110: DFE Tap-2 quantity step -2 11111: DFE Tap-2 quantity step -1 00000: DFE Tap-2 quantity step 0 (Default) 00001: DFE Tap-2 quantity step +1 00010: DFE Tap-2 quantity step +2 00011: DFE Tap-2 quantity step +3 00100: DFE Tap-2 quantity step +4 00101: DFE Tap-2 quantity step +5 00110: DFE Tap-2 quantity step +6 00111: DFE Tap-2 quantity step +7 01000: Reserved . . 11000: Reserved	1, 2, 3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient tables **NOTE 3:** If DFE Tap2 is disabled (MR8D[2]=1), MR9B[4:0] is ignored				

Table 185 — MR9E Definition: DFE Tap-1 DQ5

Function	Register Type	Operand	Data	Notes
Per-pin DFE Tap-1 Quantity for DQ5 (DFE TAP-1 DQ5)	RW	OP[4:0]	00000: DFE Tap-1 quantity step 0 (Default) 00001: DFE Tap-1 quantity step 1 00010: DFE Tap-1 quantity step 2 . . 01111: DFE Tap-1 quantity step 15 10000: Reserved . . 11111: Reserved	1, 2, 3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient Tables **NOTE 3:** If DFE Tap1 is disabled (MR8D[1]=1), MR9E[4:0] is ignored				

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Table 186 — MR9F Definition: DFE Tap-2 DQ5

Function	Register Type	Operand	Data	Notes
Per-pin DFE Tap-2 Quantity for DQ5 (DFE TAP-2 DQ5)	RW	OP[4:0]	11001: DFE Tap-2 quantity step -7 11010: DFE Tap-2 quantity step -6 11011: DFE Tap-2 quantity step -5 11100: DFE Tap-2 quantity step -4 11101: DFE Tap-2 quantity step -3 11110: DFE Tap-2 quantity step -2 11111: DFE Tap-2 quantity step -1 00000: DFE Tap-2 quantity step 0 (Default) 00001: DFE Tap-2 quantity step +1 00010: DFE Tap-2 quantity step +2 00011: DFE Tap-2 quantity step +3 00100: DFE Tap-2 quantity step +4 00101: DFE Tap-2 quantity step +5 00110: DFE Tap-2 quantity step +6 00111: DFE Tap-2 quantity step +7 01000: Reserved . . 11000: Reserved	1, 2, 3
NOTE 1: Refer to the DDR6 DFE specification for information on step size and range values **NOTE 2:** The step size and step values are related to the min/max ranges specified in the DFE Tap Coefficient tables **NOTE 3:** If DFE Tap2 is disabled (MR8D[2]=1), MR9F[4:0] is ignored				

7.9.7.3. DFE Mode Register Write Update Timing

The following parameters are applied through the Mode Register for DFE update setting.

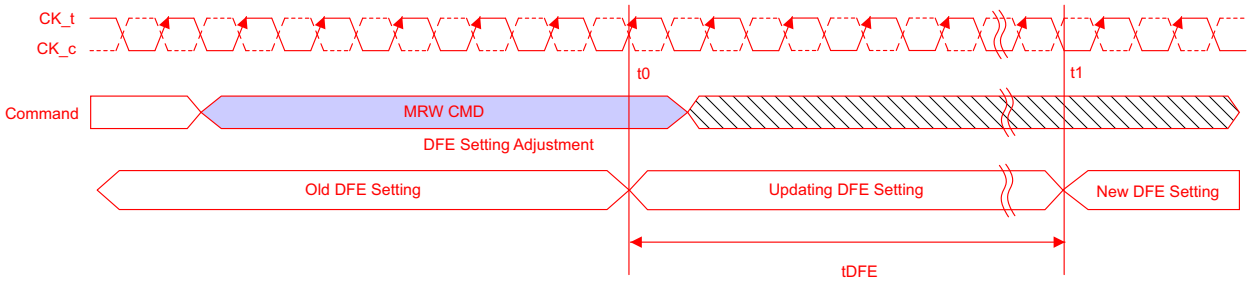


Figure 172 — DFE Update Setting

Table 187 — DFE Mode Register AC Timing

Parameter	Min/Max/Typ	Value (~12.8Gb/s)	Unit	Note
DFE Mode Register Write Update Delay Time (tDFE)	Min	80	ns	1
NOTE 1: This parameter applies only to MRW's to DFE registers and is defined as the settling time before a new DFE setting is active				

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7.9.8. CRC

DDR6 supports CRC for write and read operations. Write and read CRC can be enabled by separate Write CRC Enable and Read CRC Enable mode register bits in MR01~~OP~~[3] and MR01~~OP~~[2], respectively.

7.9.8.1. Write CRC

The controller generates the CRC checksum and forms the write data packet as shown in Section 1.1.4.3.

The DRAM checks for an error in a received write data packet by comparing the received checksum

against the computed checksum and reports errors using the ALERT signal if there is a mismatch.

DRAM can write data to the DRAM core without waiting for CRC check for full writes. If bad data is written to the DRAM core then controller will retry the transaction and overwrite the bad data. Controller is responsible for data coherency.

There is no write latency adder when Write CRC is enabled.

Write CRC Error Handling

When DRAM detects CRC error on a received write data packet, then it drives ALERT signal to '1' for tCRC_ALERT_PW.

The latency to ALERT signal is defined as tCRC_ALERT in Figure 1.

DRAM ~~will~~shall set the Write CRC Error Status bit in MRxx~~OP~~[0] to '1' upon detecting a CRC error within (tCRC_ALERT__ON + tMRD). The Write CRC Error Status bit remains '1' until the host clears it explicitly using an MRW command or MRUPDATE Mode.

The controller upon seeing an error as a pulse width ~~will~~may retry the write transactions. The controller understands the worst-case delay for ALERT (during initialization) and can back up the transactions accordingly or the controller can be made more intelligent and try to correlate the write CRC error to a specific rank or a transaction. The controller is also responsible for opening any pages and ensuring that retrying of writes is done in a coherent fashion.

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~~The pulse width may be seen longer than CRC_ALERT_PW at the controller if there are multiple CRC errors as the ALERT is a daisy chain bus or a many-to-one bus.~~

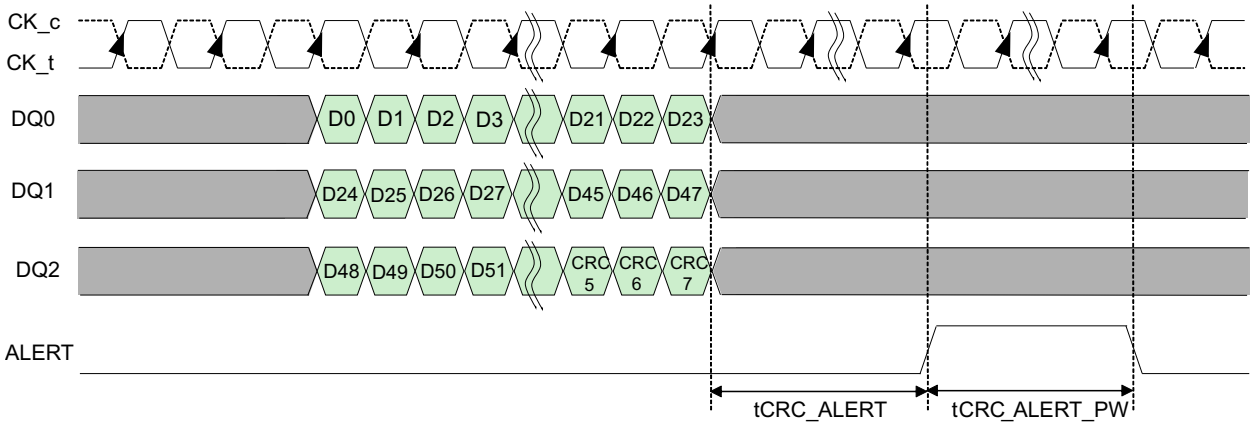


Figure 173 — CRC Error Reporting Timing Diagram for CRC-8-D64 in 2p3 with MD_OFF

NOTE: ~~1: CRC_ALERT_PW is specified from the point where the DRAM starts to drive the signal high to the point where the driver releases and the controller starts to pull the signal down.~~
NOTE: ~~2: Timing diagram applies to 2p3 devices in MD_OFF Metadata Mode with CRC-4-SC-8 CRC-8-D64.~~

7.9.8.2. Read CRC

The DDR6 SDRAM generates the CRC checksum and forms the read data packet as shown in Section 1.1.4.4. The controller can check for an error in a received data packet by comparing the received checksum against the computed checksum and if there is a mismatch then controller may retry the transaction.

Read latency adder exists when read CRC is enabled ~~depends on data rate~~.

7.9.8.3. CRC Codes

The CRC-4 and CRC-8polynomials used by DDR6 are $X^4 + X^3 + X^2 + 1$ (0x1d)and $X^8 + X^4 + X^3 + X^2 + 1$ (0x11d), respectively, which are different from DDR5. ~~The CRC-4-SC-8 is a subcode derived from row-splitting the CRC-4 code to get 8 checksum bits.~~The CRC-4 code includes 64 data bits, 4 metadata bits, and 4 CRC checksum bits. The CRC-8-D64 code includes 64 data bits and 8 CRC checksum bits. The CRC-8-D136 code includes 128 data bits, 8 metadata or zero pad bits, and 8 CRC checksum bits. The CRC code for the 2p3 device configuration is configured as CRC-4 or CRC-8-D64 using the CRC Code for 2p3 mode register bit in MR01[1]. The CRC code for 1p6/2p6/1p12 device configurations is CRC-8-D136. The selected CRC code will be instantiated once for the 2p3/1p6/2p6 device configurations or twice for the 1p12 device configuration.

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CRC-4

Figure 2 shows the CRC-4 H-matrix, and the logic equations are shown afterwards in Figure 3.

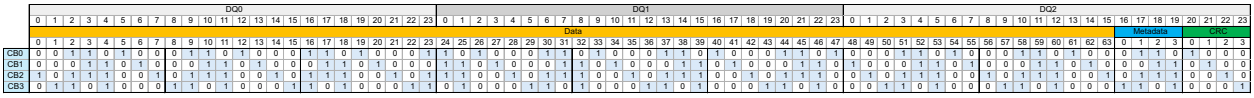


Figure 174 — CRC-4 H-matrix

CRC0=D2^D3^D5^D9^D10^D12^D16^D17^D19^D23^D24^D26^D30^D31^D33^D37^D38^D40^D44^D45^D47^D51^D52^D54^D58^D59^D61^M1^M2

CRC1=D3^D4^D6^D10^D11^D13^D17^D18^D20^D24^D25^D27^D31^D32^D34^D38^D39^D41^D45^D46^D48^D52^D53^D55^D59^D60^D62^M2^M3

CRC2=D0^D2^D3^D4^D7^D9^D10^D11^D14^D16^D17^D18^D21^D23^D24^D25^D28^D30^D31^D32^D35^D37^D38^D39^D42^D44^D45^D46^D49^D51^D52^D53^D56^D58^D59^D60^D63^M1^M2^M3

CRC3=D1^D2^D4^D8^D9^D11^D15^D16^D18^D22^D23^D25^D29^D30^D32^D36^D37^D39^D43^D44^D46^D50^D51^D53^D57^D58^D60^M0^M1^M3

~~**CRC-4-SC-8CRC-8-D64**~~

Figure 4 shows the ~~CRC-4-SC-8~~CRC-8 H-matrix, and the logic equations are shown afterwards in Figure 5.

~~Figure 4 – CRC-4-SC-8 H-matrix~~

~~**CRC0**=D2^D3^D9^D10^D16^D17^D19^D30^D31^D37^D38^D44^D45^D47^D51^D58^D59~~

~~**CRC1**=D3^D10^D11^D17^D18^D31^D38^D39^D45^D46^D48^D59~~

~~**CRC2**=D0^D2^D3^D9^D10^D11^D16^D17^D18^D28^D30^D31^D37^D38^D39^D44^D45^D46^D49^D51^D56^D58^D59~~

~~**CRC3**=D1^D2^D8^D9^D11^D16^D18^D29^D30^D36^D37^D39^D44^D46^D50^D51^D57^D58~~

~~**CRC4**=D5^D12^D23^D24^D26^D33^D40^D52^D54^D61~~

~~**CRC5**=D4^D6^D13^D20^D24^D25^D27^D32^D34^D41^D52^D53^D55^D60^D62~~

~~**CRC6**=D4^D7^D14^D21^D23^D24^D25^D32^D35^D42^D52^D53^D60^D63~~

~~**CRC7**=D4^D15^D22^D23^D25^D32^D43^D53^D60~~

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~~Figure 5 – CRC-4-SC-8 Logic Equations~~

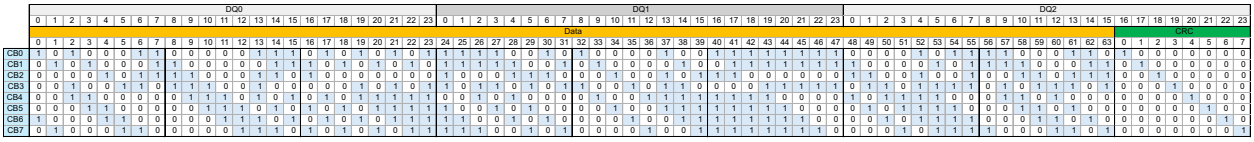


Figure 175 — CRC-8-D64 H-matrix

CRC0=D0^D2^D6^D7^D13^D14^D15^D17^D19^D21^D23^D24^D25^D26^D27^D30^D32^D37^D40^D41^D42^D43^D44^D45^D46^D47^D52^D54^D55^D56^D57^D61^D62

CRC1=D1^D3^D7^D8^D14^D15^D16^D18^D20^D22^D24^D25^D26^D27^D28^D31^D33^D38^D41^D42^D43^D44^D45^D46^D47^D48^D53^D55^D56^D57^D58^D62^D63

CRC2=D4^D6^D7^D8^D9^D13^D14^D16^D24^D28^D29^D30^D34^D37^D39^D40^D41^D48^D49^D52^D55^D58^D59^D61^D62^D63

CRC3=D2^D5^D6^D8^D9^D10^D13^D19^D21^D23^D24^D26^D27^D29^D31^D32^D35^D37^D38^D43^D44^D45^D46^D47^D49^D50^D52^D53^D54^D55^D57^D59^D60^D61^D63

CRC4=D2^D3^D9^D10^D11^D13^D15^D17^D19^D20^D21^D22^D23^D26^D28^D33^D36^D37^D38^D39^D40^D41^D42^D43^D48^D50^D51^D52^D53^D57^D58^D60

CRC5=D3^D4^D10^D11^D12^D14^D16^D18^D20^D21^D22^D23^D24^D27^D29^D34^D37^D38^D39^D40^D41^D42^D43^D44^D49^D51^D52^D53^D54^D58^D59^D61

CRC6=D0^D4^D5^D11^D12^D13^D15^D17^D19^D21^D22^D23^D24^D25^D28^D30^D35^D38^D39^D40^D41^D42^D43^D44^D45^D50^D52^D53^D54^D55^D59^D60^D62

CRC7=D1^D5^D6^D12^D13^D14^D16^D18^D20^D22^D23^D24^D25^D26^D29^D31^D36^D39^D40^D41^D42^D43^D44^D45^D46^D51^D53^D54^D55^D56^D60^D61^D63

CRC-8-D136

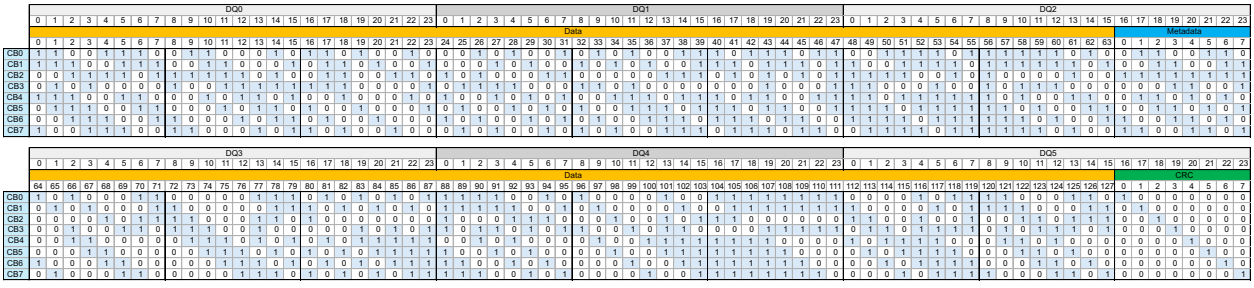


Figure 176 — CRC-8-D136 H-matrix

CRC0=D0^D1^D4^D5^D6^D9^D10^D14^D16^D17^D19^D22^D26^D28^D31^D33^D35^D38^D39^D40^D42^D43^D44^D46^D47^D50^D51^D52^D53^D55^D56^D57^D58^D59^D60^D62^M1^M2^M5^M6^D64^D66^D70

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^D71^D77^D78^D79^D81^D83^D85^D87^D88^D89^D90^D91^D94^D96^D101^D104^D105^D106^D107^D108^D109^D110^D111^D116^D118^D119^D120^D121^D125^D126

CRC1=D0^D1^D2^D5^D6^D7^D10^D11^D15^D17^D18^D20^D23^D27^D29^D32^D34^D36^D39^D40^D41^D43^D44^D45^D47^D48^D51^D52^D53^D54^D56^D57^D58^D59^D60^D61^D63^M2^M3^M6^M7^D65^D67^D71^D72^D78^D79^D80^D82^D84^D86^D88^D89^D90^D91^D92^D95^D97^D102^D105^D106^D107^D108^D109^D110^D111^D112^D117^D119^D120^D121^D122^D126^D127

CRC2=D2^D3^D4^D5^D7^D8^D9^D10^D11^D12^D14^D17^D18^D21^D22^D24^D26^D30^D31^D37^D38^D39^D41^D43^D45^D47^D48^D49^D50^D51^D54^D56^D61^M0^M1^M2^M3^M4^M5^M6^M7^D68^D70^D71^D72^D73^D77^D78^D80^D88^D92^D93^D94^D98^D101^D103^D104^D105^D112^D113^D116^D119^D122^D123^D125^D126^D127

CRC3=D1^D3^D8^D11^D12^D13^D14^D15^D16^D17^D18^D23^D25^D26^D27^D28^D32^D33^D35^D43^D47^D48^D49^D53^D56^D58^D59^D60^M3^M4^M7^D66^D69^D70^D72^D73^D74^D77^D83^D85^D87^D88^D90^D91^D93^D95^D96^D99^D101^D102^D107^D108^D109^D110^D111^D113^D114^D116^D117^D118^D119^D121^D123^D124^D125^D127

CRC4=D0^D1^D2^D5^D6^D10^D12^D13^D15^D18^D22^D24^D27^D29^D31^D34^D35^D36^D38^D39^D40^D42^D43^D46^D47^D48^D49^D51^D52^D53^D54^D55^D56^D58^D61^D62^M1^M2^M4^M6^D66^D67^D73^D74^D75^D77^D79^D81^D83^D84^D85^D86^D87^D90^D92^D97^D100^D101^D102^D103^D104^D105^D106^D107^D112^D114^D115^D116^D117^D121^D122^D124

CRC5=D1^D2^D3^D6^D7^D11^D13^D14^D16^D19^D23^D25^D28^D30^D32^D35^D36^D37^D39^D40^D41^D43^D44^D47^D48^D49^D50^D52^D53^D54^D55^D56^D57^D59^D62^D63^M2^M3^M5^M7^D67^D68^D74^D75^D76^D78^D80^D82^D84^D85^D86^D87^D88^D91^D93^D98^D101^D102^D103^D104^D105^D106^D107^D108^D113^D115^D116^D117^D118^D122^D123^D125

CRC6=D2^D3^D4^D7^D8^D12^D14^D15^D17^D20^D24^D26^D29^D31^D33^D36^D37^D38^D40^D41^D42^D44^D45^D48^D49^D50^D51^D53^D54^D55^D56^D57^D58^D60^D63^M0^M3^M4^M6^D64^D68^D69^D75^D76^D77^D79^D81^D83^D85^D86^D87^D88^D89^D92^D94^D99^D102^D103^D104^D105^D106^D107^D108^D109^D114^D116^D117^D118^D119^D123^D124^D126

CRC7=D0^D3^D4^D5^D8^D9^D13^D15^D16^D18^D21^D25^D27^D30^D32^D34^D37^D38^D39^D41^D42^D43^D45^D46^D49^D50^D51^D52^D54^D55^D56^D57^D58^D59^D61^M0^M1^M4^M5^M7^D65^D69^D70^D76^D77^D78^D80^D82^D84^D86^D87^D88^D89^D90^D93^D95^D100^D103^D104^D105^D106^D107^D108^D109^D110^D115^D117^D118^D119^D120^D124^D125^D127

Error Detection Coverage

Table 1 shows error detection coverage of DDR6 CRC codes.

Table 188 — Error Detection Details (Sheet 1 of 2)

Error Type	Detection Capability		
	CRC-4	~~CRC-4-SC-8~~CRC-8-D64	CRC-8-D136
Random Single Bit Error	100%	100%	100%
Random Double Bit Error	≥ 86%	~~≥ 92%~~100%	100%
Random Triple Bit Error	100%	≥ 99.5%	≥ 99.6%
Random Odd Count Error	100%	~~100%~~≥ 99.6%	≥ 99.6%

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Table 188 — Error Detection Details (Continued) (Sheet 2 of 2)

Error Type	Detection Capability		
	CRC-4	~~CRC-4-SC-8~~CRC-8-D64	CRC-8-D136
Random Multi-Bit Error	≥ 93%	≥ 99.6%	≥ 99.6%

7.9.8.4. CRC Configurations with Metadata Modes

CRC is supported for metadata modes MD_OFF, MD_8b_A, MD_8b_B, and MD_16b.~~, and the CRC mode configured separately as CRC-4 or CRC-4-SC-8 using the CRC Mode mode register bit in MRxx OP[y]. The device configuration as 2p3/1p6/2p6/1p12 will determine how many times the CRC code is instantiated in the data packet.~~ Table 2 shows which CRC codes are supported for various device configurations and metadata modes.

Table 189 — Supported CRC Codes for Metadata Modes

Device Configuration	Metadata Mode	Supported CRC Codes					
		1 x CRC-4	~~2 x CRC-4~~	~~4 x CRC-4~~	1 x CRC-8-D64~~4-SC-8~~	~~2~~1 x CRC-8-D1~~364-SC-8~~	~~4~~2 x CRC-8-D1~~364-SC-8~~
2p3	MD_OFF	No	~~No~~	~~No~~	Yes	No	No
	MD_8b_A	Yes	~~No~~	~~No~~	No	No	No
	MD_8b_B	Yes	~~No~~	~~No~~	No	No	No
	MD_16b	Yes	~~No~~	~~No~~	Yes	No	No
1p6, 2p6	MD_OFF	No	~~No~~	~~No~~	No	Yes	No
	MD_8b_A	No	~~Yes~~	~~No~~	No	No	No
	MD_8b_B	No	~~Yes~~	~~No~~	No	No	No
	MD_16b	No	~~Yes~~	~~No~~	No	Yes	No
1p12	MD_OFF	No	~~No~~	~~No~~	No	No	Yes
	MD_8b_A	No	~~No~~	~~Yes~~	No	No	Yes
	MD_8b_B	No	~~No~~	~~Yes~~	No	No	Yes
	MD_16b	No	~~No~~	~~Yes~~	No	No	Yes

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Location of CRC Checksum Bits in the Data Packet

CRC-4 checksum bits ~~always~~occupy the last 4 bits of DQ2~~, and in addition, the last 4 bits of DQ5 for 1p6/2p6/1p12, and in addition, the last 4 bits of DQ8 and DQ11 for 1p12~~. ~~CRC-4-SC-8~~CRC-8-D64checksum bits ~~always~~occupy the last 8 bits of DQ2~~, and in addition, the last 8 bits of DQ5 for 1p6/2p6/1p12, and in addition, the last 8 bits of DQ8 and DQ11 for 1p12~~. CRC-8-D136 checksum bits occupy the last 8 bits of DQ5 for 1p6/2p6/1p12, and in addition, the last 8 bits of DQ11 for 1p12.

Stored CRC

~~In certain combinations of metadata modes and CRC codes, the CRC parity bits are a subset of the metadata bits. During writes, the CRC parity bits will be checked and also stored as metadata bits.~~Stored CRC is configured by using MR01~~OP~~[1:~~0~~] to select a CRC code, whose checksum bits are located within a subset of the metadata bits associated with the metadata mode. The read CRC function may then be disabled by setting MR01~~OP~~[2] to 0, as reading the metadata bits will produce the CRC parity bits. This concept, known as stored CRC, avoids additional read latency normally associated with read CRC. If the CRC parity bits are a subset of the metadata bits but the read CRC function is enabled by setting MR01~~OP~~[2] to 1, then the CRC parity bits shall be regenerated during a read and output on the DQ bus.

Zero Pad

When non-data bit locations in the data packet are unused, for example, by metadata or CRC checksum bits, the unused bits in the data packet shall be set to 0s.

Mapping Data Packet Bits to CRC Codes

Mapping 2p3 Data Packet Bits to CRC Codes

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Figures 8-13 show detailed bit mapping for the CRC codewords for the 2p3 device configuration for various CRC and metadata modes.

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	0	0	0	0	0	0	0	0

Figure 177 — Bit Mapping for CRC Disabled in 2p3 with MD_OFF

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	CRC0	CRC1	CRC2	CRC3	CRC4	CRC5	CRC6	CRC7

Figure 178 — CRC Bit Mapping for ~~CRC-4-SC-8~~ CRC-8-D64 in 2p3 with MD_OFF

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	M0	M1	M2	M3	CRC0	CRC1	CRC2	CRC3

Figure 179 — CRC Bit Mapping for CRC-4 in 2p3 with MD_8b_A

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	0	0	0	0	M0 CRC0	M1 CRC1	M2 CRC2	M3 CRC3

Figure 180 — CRC Bit Mapping for CRC-4 in 2p3 with MD_8b_B and Stored CRC

PROPOSED

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	M0	M1	M2	M3	M4 CRC0	M5 CRC1	M6 CRC2	M7 CRC3

Figure 181 — CRC Bit Mapping for CRC-4 in 2p3 with MD_16b and Stored CRC

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	M0 CRC0	M1 CRC1	M2 CRC2	M3 CRC3	M4 CRC4	M5 CRC5	M6 CRC6	M7 CRC7

Figure 182 — CRC Bit Mapping for ~~CRC-4-SC-8~~ CRC-8-D64 in 2p3 with MD_16b and Stored CRC

Mapping 1p6/2p6 Data Packet Bits to CRC Codes

PROPOSED

Figures 14-18 show detailed bit mapping for the CRC codewords for the 1p6/2p6 device configurations for various CRC and metadata modes.

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	0	0	0	0	0	0	0	0
DQ3	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ4	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ5	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	0	0	0	0	0	0	0	0

Figure 183 — Bit Mapping for CRC Disabled in 1p6/2p6 with MD_OFF

PROPOSED

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	0	0	0	0	0	0	0	0
DQ3	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ4	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ5	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	CRC0	CRC1	CRC2	CRC3	CRC4	CRC5	CRC6	CRC7

Figure 184 — CRC Bit Mapping for CRC 1 x CRC-8-D136 in 1p6/2p6 with MD_OFF

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	M0	M1	M2	M3	M4	M5	M6	M7
DQ3	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ4	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ5	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	CRC0	CRC1	CRC2	CRC3	CRC4	CRC5	CRC6	CRC7

Figure 185 — CRC Bit Mapping for 1 x CRC-8-D136 in 1p6/2p6 with MD_8b_A

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	0	0	0	0	0	0	0	0
DQ3	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ4	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ5	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	M0 CRC0	M1 CRC1	M2 CRC2	M3 CRC3	M4 CRC4	M5 CRC5	M6 CRC6	M7 CRC7

Figure 186 — CRC Bit Mapping for 1 x CRC-8-D136 in 1p6/2p6 with MD_8b_B and Stored CRC

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	M0	M1	M2	M3	M4	M5	M6	M7
DQ3	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ4	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ5	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	M8 CRC0	M9 CRC1	M10 CRC2	M11 CRC3	M12 CRC4	M13 CRC5	M14 CRC6	M15 CRC7

Figure 187 — CRC Bit Mapping for 1 x CRC-8-D136 in 1p6/2p6 with MD_16b and Stored CRC

Mapping 1p12 Data Packet Bits to CRC Codes

PROPOSED

Figures 19-23 show detailed bit mapping for the CRC codewords for the 1p12 device configuration for various CRC and metadata modes.

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	0	0	0	0	0	0	0	0
DQ3	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ4	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ5	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	0	0	0	0	0	0	0	0
DQ6	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ7	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ8	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	0	0	0	0	0	0	0	0
DQ9	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ10	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ11	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	0	0	0	0	0	0	0	0

Figure 188 — Bit Mapping for CRC Disabled in 1p12 with MD_OFF

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	0	0	0	0	0	0	0	0
DQ3	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ4	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ5	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	CRC0	CRC1	CRC2	CRC3	CRC4	CRC5	CRC6	CRC7
DQ6	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ7	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ8	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	0	0	0	0	0	0	0	0
DQ9	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ10	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ11	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	CRC0	CRC1	CRC2	CRC3	CRC4	CRC5	CRC6	CRC7

Figure 189 — CRC Bit Mapping for 2 x CRC-8-D136 in 1p12 with MD_OFF

PROPOSED

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	M0	M1	M2	M3	M4	M5	M6	M7
DQ3	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ4	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ5	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	CRC0	CRC1	CRC2	CRC3	CRC4	CRC5	CRC6	CRC7
DQ6	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ7	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ8	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	M0	M1	M2	M3	M4	M5	M6	M7
DQ9	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ10	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ11	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	CRC0	CRC1	CRC2	CRC3	CRC4	CRC5	CRC6	CRC7

Figure 190 — CRC Bit Mapping for 2 x CRC-8-D136 in 1p12 with MD_8b_A

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	0	0	0	0	0	0	0	0
DQ3	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ4	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ5	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	M0 CRC0	M1 CRC1	M2 CRC2	M3 CRC3	M4 CRC4	M5 CRC5	M6 CRC6	M7 CRC7
DQ6	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ7	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ8	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	0	0	0	0	0	0	0	0
DQ9	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ10	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ11	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	M0 CRC0	M1 CRC1	M2 CRC2	M3 CRC3	M4 CRC4	M5 CRC5	M6 CRC6	M7 CRC7

Figure 191 — CRC Bit Mapping for 2 x CRC-8-D136 in 1p12 with MD_8b_B and Stored CRC

PROPOSED

	Transfer																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DQ0	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ1	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ2	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	M0	M1	M2	M3	M4	M5	M6	M7
DQ3	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ4	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ5	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	M8 CRC0	M9 CRC1	M10 CRC2	M11 CRC3	M12 CRC4	M13 CRC5	M14 CRC6	M15 CRC7
DQ6	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
DQ7	D24	D25	D26	D27	D28	D29	D30	D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D45	D46	D47
DQ8	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57	D58	D59	D60	D61	D62	D63	M0	M1	M2	M3	M4	M5	M6	M7
DQ9	D64	D65	D66	D67	D68	D69	D70	D71	D72	D73	D74	D75	D76	D77	D78	D79	D80	D81	D82	D83	D84	D85	D86	D87
DQ10	D88	D89	D90	D91	D92	D93	D94	D95	D96	D97	D98	D99	D100	D101	D102	D103	D104	D105	D106	D107	D108	D109	D110	D111
DQ11	D112	D113	D114	D115	D116	D117	D118	D119	D120	D121	D122	D123	D124	D125	D126	D127	M8 CRC0	M9 CRC1	M10 CRC2	M11 CRC3	M12 CRC4	M13 CRC5	M14 CRC6	M15 CRC7

Figure 192 — CRC Bit Mapping for 2 x CRC-8-D136 in 1p12 with MD_16b and Stored CRC

7.9.8.5. Mode Registers

Function	Register Type	Mode Register	Operand	Data	Notes
CRC ~~Mode~~Code for 2p3	R~~/~~W	MR01	~~OP~~[1~~:0~~]	~~00~~$_B$ ~~0h~~ – CRC-4 ~~01~~$_B$ ~~1h~~ – ~~CRC-4-SC-8~~ CRC-8-D64 (Default) ~~10~~$_B$ ~~2h – CRC-8-D136~~	1
Read CRC Enable	R~~/~~W	MR01	~~OP~~[2]	0_B – Disable (Default) 1_B – Enable	
Write CRC Enable	R~~/~~W	MR01	~~OP~~[3]	0_B – Disable (Default) 1_B – Enable	
Write CRC Error Status	R~~/~~W	MRxx (PSD)	~~OP~~[0]	0_B – Clear 1_B – Error	~~1~~2

NOTE 1: The CRC Code for 2p3 mode register only applies to the 2p3 device configuration. It is ignored for the 1p6/2p6/1p12 device configurations.
NOTE 2: The host shall disable Write CRC, if it was enabled, prior to entering Write Leveling Training mode.

PROPOSED

7.9.8.6. Timing

Table 190 — CRC Error Handling Timing Parameters

Symbol	Description	Min	Max	Unit
tCRC_ALERT	CRC Alert Delay Time	~~3~~ -	13	ns
~~tCRC_ALERT_PW~~	~~CRC Alert Pulse Width~~	~~512~~	~~512~~	~~nCK~~

PROPOSED

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7.9.9. On-Die ECC

[DIRECTIONAL] The DDR6 SDRAM device shall have the On-Die ECC and timings aligned to correction capability.

[DIRECTIONAL] DDR6 SDRAM device shall also provide additional transparency (i.e., Uncorrectable Error information) for the use of System RAS solution.

The device supports on-die ECC covering the array bits. The on-die ECC may be whatever type of ECC the device manufacturer determines is to best support its design objectives. However, the multi-bit error (MBE) diagnostic coverage (DC) shall be available to host.

7.9.9.1. Error Definitions

SBE = Single bit error: one single bit error per ECC code word; sometimes referred to as payload or mission data.

MBE = Multi-bit error: two or more single bit errors per ECC code word; sometimes referred to as payload or mission data.

7.9.9.2. ECC Diagnostic Coverage

The on-die ECC corrects one bit error (SBE) detected in the ECC code word. More than one bit error detected in the ECC code word is classifieds as an uncorrectable MBE.

The device's on-die ECC has the ability to meet or exceed the following diagnostics:

100% SBE correction

- On-die ECC corrects one bit error (SBE) detected in the ECC code word

≥

to 95% MBE detection

- On-die ECC classifies more than one bit error detected in the ECC code word as an uncorrectable MBE and the detected MBE should be registered

Table 191 — On-Die ECC Diagnostic Coverage

SBE Correction	MBE Detection
100%	≥99.5%

7.9.9.3. MBE Registration

The on-die ECC MBE detection should be registered in MR97 OP[3].

7.9.10. Bounded Fault

[DIRECTIONAL]

7.9.10.1. 2p6/2p4

A bounded fault shall not impact more than 32 bits in a 128-bit data fetch.

The data bits impacted by a bounded fault will be confined to at most 4 sub-packets of the data packet.

The impacted data sub-packets can be present anywhere in the data packet.

7.9.10.2. 2p3/2p2

A bounded fault shall not impact more than 24 bits in a 64-bit data fetch.

The data bits impacted by a bounded fault will be confined to at most 3 sub-packets of the data packet.

The impacted data sub-packets can be present anywhere in the data packet.

PROPOSED

7.9.11. Error Check and Scrub (ECS)

To be balloted in Q4'26.

PROPOSED

7.9.12. Alert Output

Unballoted.

PROPOSED

PROPOSED

7.9.13. Fault Registers

To be balloted in Q4'26.

7.9.13.1. Fault Registers

Two fault registers, MR97 and MR98, register up to 15 functions when a fault(s) is detected by a covered function. Fault registers MR97 and MR98 are default LOW and remain LOW unless a covered function has detected a fault and sets the associated register bit HIGH. The affected fault registers' bit(s) retain their fault status until either (1) the affected register is read from; (2) a RESET command is applied; or (3) power is removed.

The supported functions, when enabled via MR99 and MR100, control the alert output when a fault occurs. FMR3 and FMR4 may be written to when the device is in idle state and no data is driven. The alert output is driven active, asynchronously, upon fault occurrence and remains active until either (1) the affected MR97 and/or MR98 is read from, with one exception: MR97 OP[0] clearingw is controlled by PRAC operation and is cleared as defined by PRAC alert back-off protocol (ABO); (2) a RESET command is applied; or (3) power is removed. The logical diagram below depicts the functionality.

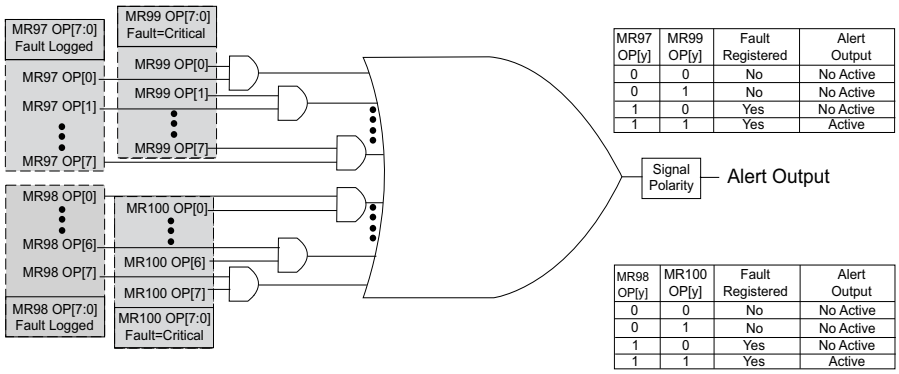


Figure 193 — Fault Register Logical Diagram

Fault register MR97 OP[7] is derived by ORing MR98 OP[7:0], logical diagram below depicts this functionality. When the host interrogates MR97 and if MR97 OP[7] is LOW, then the host does not need to read MR98.

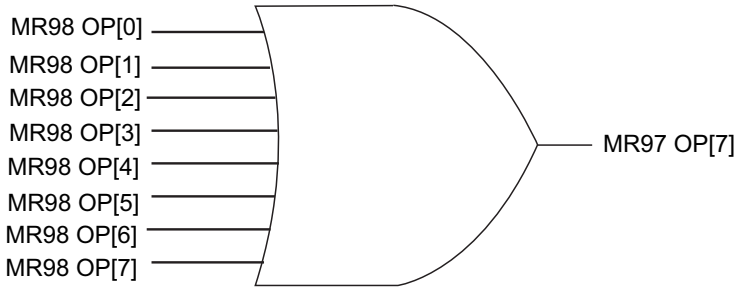


Figure 194 — Fault Register MR97 OP[7]

7.9.13.2. Fault Register Definitions

Fault registers MR97 and MR98 define which functions register fault occurrence. These functions are not required to be supported by the device; however, if a function is supported, it should use the defined register bit. MR97 and MR98 are default LOW and remain LOW unless a covered function has a detected fault and sets the associated register bit HIGH. The affected fault registers' bit(s) retain their fault status until either (1) register is read from (Note: MR97 OP[7] is not required to be reset by MRR command; MR97 OP[7] may be cleared by clearing MR98); (2) a RESET command applied, or (3) power is removed.

Table 192 — MR97 (FMR1) Mode Register Definition

NOTE: FMR1 OP[7] is the sum of FMR2 OP[7:0]

Function	Register Type	Operand	Data	Notes
PRAC limit exceeded	Read	OP[0]	0b: Fault not detected 1b: Fault detected	
Write link ECC MBE fault /write link EDC SBE/MBE fault	Read	OP[1]	0b: Fault not detected 1b: Fault detected	
Write link ECC SBE correction attempted	Read	OP[2]	0b: Fault not detected 1b: Fault detected	
On-die ECC MBE fault	Read	OP[3]	0b: Fault not detected 1b: Fault detected	
TBD	Read	OP[4]	0b: Fault not detected 1b: Fault detected	
CA parity fault	Read	OP[5]	0b: Fault not detected 1b: Fault detected	
TBD	Read	OP[6]	0b: Fault not detected 1b: Fault detected	
Sum of MR98 OP[7:0]	Read	OP[7]	0b: Fault not detected 1b: Fault detected	1

Table 193 — MR98 (FMR2) Mode Register Definition (Sheet 1 of 2)

Function	Register Type	Operand	Data
RFU	Read	OP[0]	0b: Fault not detected 1b: Fault detected

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Table 193 — MR98 (FMR2) Mode Register Definition (Continued) (Sheet 2 of 2)

Function	Register Type	Operand	Data
RFU	Read	OP[1]	0b: Fault not detected 1b: Fault detected
RFU	Read	OP[2]	0b: Fault not detected 1b: Fault detected
RFU	Read	OP[3]	0b: Fault not detected 1b: Fault detected
RFU	Read	OP[4]	0b: Fault not detected 1b: Fault detected
RFU	Read	OP[5]	0b: Fault not detected 1b: Fault detected
RFU	Read	OP[6]	0b: Fault not detected 1b: Fault detected
RFU	Read	OP[7]	0b: Fault not detected 1b: Fault detected

7.9.13.3. Fault Alert Control

The two registers, MR99 and MR100, control which functions with registered faults affect the alert output status. MR99 and MR100 are default LOW and remain LOW until written to; these two MR registers may be written to when the device is in idle state and no data is driven. These two registers retain set values until either (1) rewritten to different state (2) a RESET command is applied, or (3) power is removed.

Table 194 — MR99 (FMR3) Mode Register Definition

NOTE: MR99 OP[0] controls whether PRAC ABO protocol drives ALERT active.

Function	Register Type	Operand	Data	Notes
PRAC limit exceeded	Write	OP[0]	0b: Fault does not affect alert output 1b: Fault affects alert output	1
Write link ECC MBE fault /write link EDC SBE/MBE fault	Write	OP[1]	0b: Fault does not affect alert output 1b: Fault affects alert output	
Write link ECC SBE correction attempted	Write	OP[2]	0b: Fault does not affect alert output 1b: Fault affects alert output	

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Function	Register Type	Operand	Data	Notes
On-die ECC MBE fault	Write	OP[3]	0b: Fault does not affect alert output 1b: Fault affects alert output	
TBD	Write	OP[4]	0b: Fault does not affect alert output 1b: Fault affects alert output	
CA parity fault	Write	OP[5]	0b: Fault does not affect alert output 1b: Fault affects alert output	
TBD	Write	OP[6]	0b: Fault does not affect alert output 1b: Fault affects alert output	
Sum of FMR2 OP[7:0]	Write	OP[7]	0b: Fault does not affect alert output 1b: Fault affects alert output	

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Table 195 — MR100 (FMR4) Mode Register Definition

NOTE: MR99 OP[7] set to 0b overrides MR100 OP[7:0] and prevents MR100 OP[7:0] from affecting alert output.

Function	Register Type	Operand	Data	Notes
RFU	Write	OP[0]	0b: Fault does not affect alert output 1b: Fault affects alert output	1
RFU	Write	OP[1]	0b: Fault does not affect alert output 1b: Fault affects alert output	1
RFU	Write	OP[2]	0b: Fault does not affect alert output 1b: Fault affects alert output	1
RFU	Write	OP[3]	0b: Fault does not affect alert output 1b: Fault affects alert output	1
RFU	Write	OP[4]	0b: Fault does not affect alert output 1b: Fault affects alert output	1
RFU	Write	OP[5]	0b: Fault does not affect alert output 1b: Fault affects alert output	1
RFU	Write	OP[6]	0b: Fault does not affect alert output 1b: Fault affects alert output	1
RFU	Write	OP[7]	0b: Fault does not affect alert output 1b: Fault affects alert output	1

7.9.13.4. Fault to Alert Timing

The timing for fault occurrence to detection and notification is defined in table below.

Table 196 — Fault Timing

NOTE: Triggering: Several events can trigger PRAC limit exceeded, large variation required since relationship is asynchronous.
NOTE: The second CK rising edge is timing reference; that is, CK rising (R2) of error command.

Function	Operand	Detection to Valid MRR	Valid MRR to Alert Output	Notes
PRAC limit exceeded	MR97 OP[0]	t ≤ Max (85*n*CK or 10*n*CK+500ns)		1
Write link ECC MBE fault	MR97 OP[1]	≤WL+BL/n+2*n*CK+RU(13ns/^{t}CK)	≤9ns/^{t}CK	2
Write link EDC SBE/MBE fault	MR97 OP[1]	≤WL+BL/n+2*n*CK+RU(13ns/^{t}CK)	≤9ns/^{t}CK	2
Write link ECC SBE correction attempted	MR97 OP[2]	≤WL+BL/n+2*n*CK+RU(13ns/^{t}CK)	≤9ns/^{t}CK	2

Function	Operand	Detection to Valid MRR	Valid MRR to Alert Output	Notes
On-die ECC MBE fault	MR97 OP[3]	≤RL+RU(4ns/^{t}CK)	≤9ns/^{t}CK	2
TBD	MR97 OP[4]	TBD	TBD	
CA parity fault	MR97 OP[5]	≤Max (2*n*CK or 8ns/tCK)	≤9ns/^{t}CK	2
TBD	MR97 OP[6]	TBD	TBD	
Sum of FMR2 MR98 OP[7:0]	MR97 OP[7]	NA	≤9ns/^{t}CK	
RFU	MR98 OP[0]	TBD	NA	
RFU	MR98 OP[1]	TBD	NA	
RFU	MR98 OP[2]	TBD	NA	
RFU	MR98 OP[3]	TBD	NA	
RFU	MR98 OP[4]	TBD	NA	
RFU	MR98 OP[5]	TBD	NA	
RFU	MR98 OP[6]	TBD	NA	
RFU	MR98 OP[7]	TBD	NA	

The time for clearing fault from alert output is as follows:

- If MR97 OP[7] is 0, then the time required is **TBD**.
- If MR97 OP[7] is 1, then time required is **TBD**.

7.9.13.5. Error Definitions

SBE: Single bit error: one single bit error per ECC code word; sometimes referred to as payload or mission data.

DBE: Double bit error: two single bit errors per ECC code word; sometimes referred to as payload or mission data.

MBE: Multi-bit error: two or more single bit errors per ECC code word; sometimes referred to as payload or mission data.

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7.9.14. Command Address Parity (CA Parity)

7.9.14.1. CA Parity Checking

CA parity checking detects bit errors in the command address bus. The CA Parity Checking mode register, MR01~~OP~~[4], enables or disables CA parity checking. The default is ~~disen~~abled. The DRAM must be in idle state when CA Parity checking mode register is changed. When CA parity checking is disabled, the DRAM does not need to check CA parity bit(s) in each command. When CA parity checking is enabled, the CA parity shall be checked and ALERT shall be asserted if a CA parity error is detected.

The CA Parity Command Blocking mode register, MR01~~OP~~[5], selects blocking or nonblocking mode when a CA parity error is detected. The default is ~~non~~blocking.

When CA parity checking is enabled in nonblocking mode, an erroneous command will be executed, but the DRAM may execute an unknown command. There is no additional latency increase compared to when CA parity checking is disabled, but DRAM will be in unknown state when a CA parity error is detected, so it is highly recommended that the host reset the DRAM.

When CA parity checking is enabled in blocking mode, an erroneous command that fails the CA parity check shall not be executed. An additional delay (~~PL : Parity Latency~~tPAR_CA_Delay) is applied to all commands. The command is held for the time of the Parity Latency (PL) before it is executed inside the device. This means that issuing timing of the internal command is determined with PL. Since some commands on one or more DRAM devices will be blocked after the first CA parity error, while these commands may be executed on one or more other DRAM devices that did not have a CA parity error, the host should replay commands starting from the first command that might have had a CA parity error. The earliest command that could possibly have had a detected CA parity error can be determined by working backwards from the time the host detects the ALERT, the propagation delay of ALERT from the DRAM to the host, and tPAR_ALERT_ON. The exact command that had a CA parity error can be determined by reading the CA Parity Log C0 to C5 mode registers.

7.9.14.2. Parity Calculation

Most commands in the Command Truth Table have a PAR1 bit in the 2nd cycle in CA4. 4-cycle and 6-cycle commands in the Command Truth Table have a PAR2 bit in the 4th or 6th cycle, respectively, in CA4. The convention of parity is even parity, meaning that valid parity is defined as an even number of ones across the inputs used for parity computation combined with the parity signal. In other words, the parity bit is chosen so that the total number of 1's in the transmitted signal, including the parity bit is even. The parity check passes if the PAR1 is valid in 2-cycle commands and passes if both PAR1 and PAR2 are valid in 4- and 6-cycle commands.

To generate valid parity bits, the host shall calculate PAR1 and PAR2 as follows. PAR1 and PAR2 satisfy even parity. The host shall calculate PAR1 as the XOR of all of the bits except itself in CA0, CA1, CA2, CA3, CA4, and CA5 (if supported) in the 1st and 2nd cycles of the command, excluding itself. In 4-cycle commands, the host shall calculate PAR2 as the XOR of all of the bits except itself in CA0, CA1, CA2, CA3, CA4, and CA5 (if supported) in the 3rd and 4th cycles of the command. In 6-cycle commands, the host shall calculate PAR2 as the XOR of all of the bits except itself in CA0, CA1, CA2, CA3, CA4, and CA5 (if supported) in the 3rd, 4th, 5th, and 6th cycles of the command. Only DES is allowed between valid commands to prevent DRAM from any malfunctioning.

7.9.14.3. CA Parity Error Detection and Actions

When CA parity checking is enabled, and the DRAM detects a CA parity error in any command as qualified by CS, then it must perform the following steps:

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Table 197 — CA Parity Error DRAM Handling Actions (Sheet 1 of 2)

Applies to Non-blocking Mode	Applies to Blocking Mode	Description
No	Yes	In blocking mode only, the DRAM shall delay the erroneous command by PL and then ignore it.
Yes	Yes	In both blocking and nonblocking modes, log the error by storing the erroneous command and address bits in the CA Parity Error Log C0-C5 mode registers for the PSD with the error. The DRAM maintains the CA Parity Error Log C0-C5 mode registers for the first erroneous command until the CA Parity Error Status mode register bit is cleared to 0.
Yes	Yes	Set the CA Parity Error Status bit in the mode register to '1'. ~~The CA Parity Error Status bit must be setbefore tCMD_BLOCKING finishes~~ within (tPAR_ALERT_ON + tMRD). The CA Parity Error Status bit remains 1 until the host clears it explicitly using an MRW command or MRUPDATE Mode.
Yes	Yes	All in-progress commands shall continue without interruption.
No	Yes	In blocking modeonly, the DRAM shall ignore subsequent commands immediately after the erroneous command on the same PSD's CA bus and shall continue to ignore commands on the same PSD's CA bus until after tCMD_BLOCKING.
Yes	Yes	~~If the Alert State Enable on CA Parity Error mode register bit is 1,~~Assert the ALERT signal to the host within tPAR_ALERT_ON time for tPAR_ALERT_PW. ~~After tPAR_ALERT_PW has been satisfied, the DRAM shall de-assert ALERT.~~
~~Yes~~	~~No~~	~~In nonblocking mode only, disable DQS outputs starting between tPAR_ALERT_ON and (RL - tRPRE) after the erroneous command and keep the DQS outputs disabled until after tCMD_BLOCKING; and disable DQ outputs starting between tPAR_ALERT_ON and RL after the erroneous command and keep DQ outputs disabled until after tCMD_BLOCKING. This means that READ and MRR commands that were issued prior to the CA parity error may have all, some, or none of the typical DQS or DQ activity after tPAR_ALERT_ON. It is highly recommended that the host ignore read data on DQ after ALERT is asserted for a CA parity error in nonblocking mode. It is highly recommended that the host assert RESET_n when a CA parity error is detected in nonblocking mode.~~

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Table 197 — CA Parity Error DRAM Handling Actions (Continued) (Sheet 2 of 2)

Applies to Non-blocking Mode	Applies to Blocking Mode	Description
No	Yes	In blocking mode only, if the PSD is not already in self-refresh mode, the DRAM shall internally issue PREab ~~and SREREFab~~after waiting PL + tRFCab, followed by SRE at least Max(tSR, tSRED) before tCMD_BLOCKING finishes. Similarly, the host may stop normal command sequences to all DRAMs that share the same ALERT pin, wait for all previous commands to be fully executed, and then issue PREab~~, REFab,~~and SRE commands to put the devices in a safe state after ALERT is detected and before ~~(~~tCMD_BLOCKING ~~× 80%)~~elapses.
No	Yes	In blocking mode only, if a CA parity error is detected and the CA Parity Recovery Mode mode register bit is 1~~b~~, then the DRAM shall copy the VREFCA shadow mode register value to the VREFCA normal mode register before tCMD_BLOCKING elapses.
No	Yes	In blocking mode only, after tCMD_BLOCKING, the DRAM shall automatically stop blocking commands from the host.
No	Yes	In blocking mode only, the host shall stop sending commands before ~~(~~tCMD_BLOCKING ~~× 80%)~~ elapses. After ~~(~~tCMD_BLOCKING ~~× 120%)~~, the DRAM is ready to accept commands for normal operation. Parity latency will be in effect and parity checking will resume immediately.
Yes	Yes	It is possible that the DRAM might have ignored a refresh command during the tCMD_BLOCKING window or the refresh command is the first erroneous frame, so it is recommended that the controller issues extra refresh cycles as needed.
~~Yes~~	~~Yes~~	~~The CA Parity Error Status mode register bit may be read any time by the host after(tCMD_BLOCKING× 120%), to determine which DRAM had the error. The DRAM maintains the CA Parity Error Log C0-C5 mode registers for the first erroneous command until the CA Parity Error Status mode register bit is reset to '0'.~~

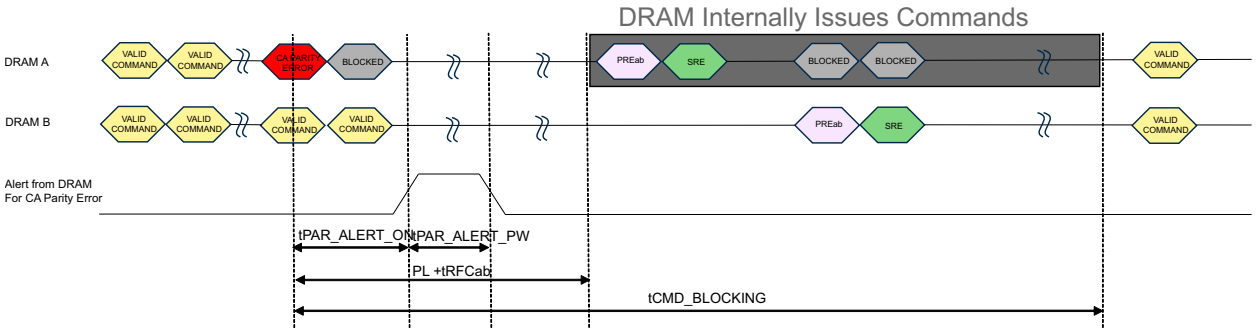


Figure 195 — CA Parity Checking in Nonblocking Mode

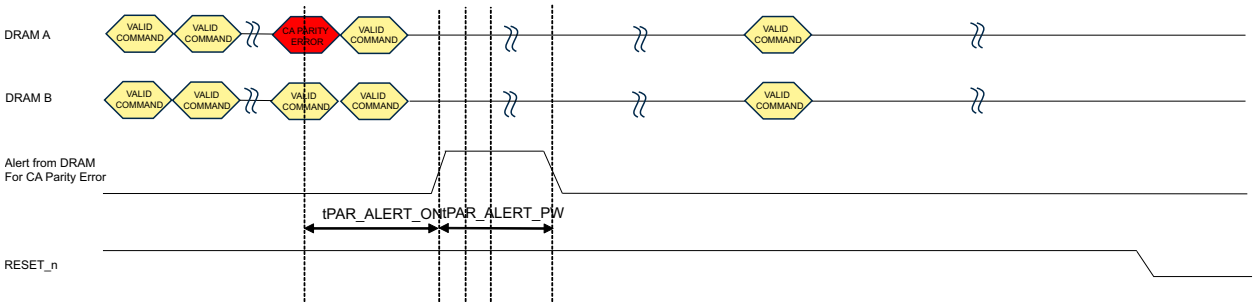


Figure 196 — CA Parity Checking in Blocking Mode

7.9.14.4. 4- and 6-Cycle Command Cancel

DDR6 DRAM commands ACT, RD, REFab, REFdb, REFqb, RFMab, RFMdb, RFMqb, RFMpb, MRR, MRW are 4- and 6-cycle commands without associated ODT control requirements. The DRAM will not execute these 4- and 6-cycle commands if the CS is HIGH on the last two cycles of the command (command cancel).

For example, if the RCD or IMB detects a parity error on the 1st and/or 2nd cycles of a 2-, 4-, or 6-cycle command using the PAR1 bit, the CS will remain LOW for the duration of the 2, 4, or 6 cycles, respectively.

For example, if the RCD or IMB detects a parity error on the 3rd and/or 4th cycles of a 4-cycle command using the PAR2 bit, the CS will remain HIGH for the 1st, 2nd, 3rd, and 4th cycles of the command. If the command is a Write, then it will be converted to non-target termination command in the DRAM. If the command is either RD, REFab, REFdb, REFqb, RFMab, RFMdb, RFMqb, RFMpb, or MRR, then the erroneous command will be canceled in the DRAM.

For example, if the RCD or IMB detects a parity error on the 3rd, 4th, 5th, and/or 6th cycles of a 6-cycle command using the PAR2 bit, the CS will remain HIGH for the 1st and 2nd cycles, LOW for the 3rd, and 4th cycles, and HIGH for 5th, and 6th cycles of the command. If the command is either ACT or MRW, then the erroneous command will be canceled in the DRAM.

Command cancel is not intended by the host rather it is a result of a CA parity error detected by the RCD or IMB. So the relationship between the canceled command and the next valid command shall not be illegal.

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If CA parity checking is enabled, the check also applies to cancelled commands. For example, if the RCD or IMB detects a parity error using the PAR2 bit, the RCD or IMB may recalculate the PAR2 bit, so the DRAM should still expect the PAR2 check to pass.

After a command is cancelled, the CS will remain LOW for a period of time. The last cycle of the next valid command must be at least tCMD_cancel after the last cycle of the cancelled command.

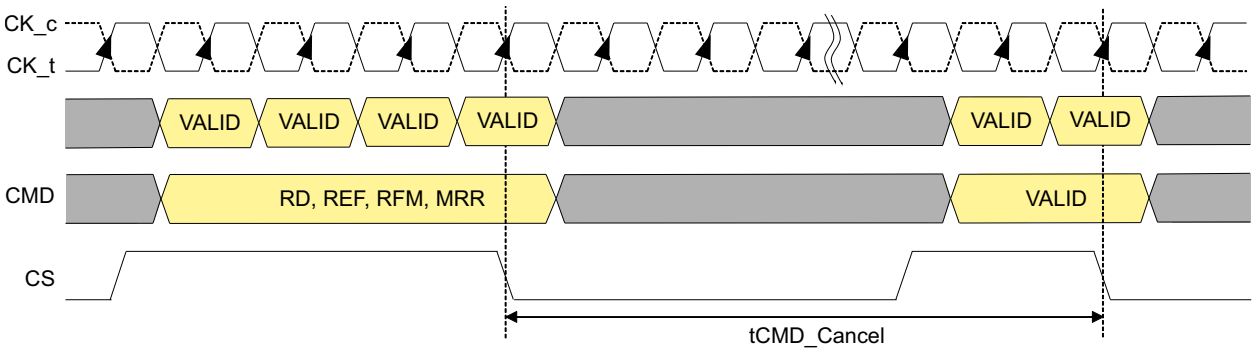


Figure 197 — 4-Cycle Command Cancel Timing

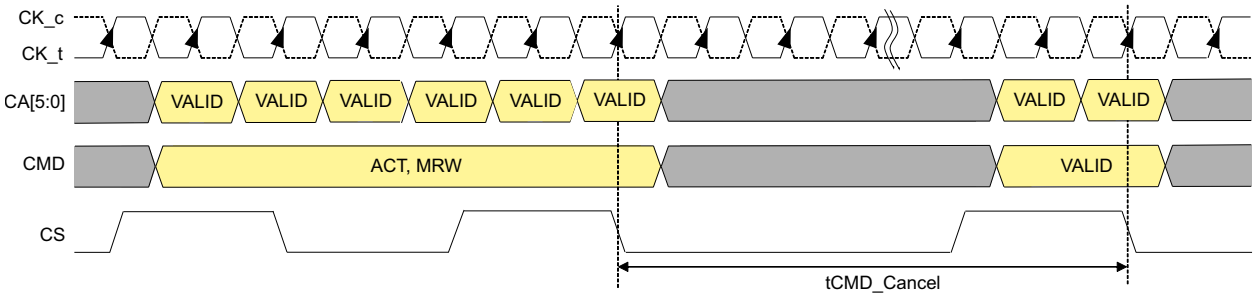


Figure 198 — 6-Cycle Command Cancel Timing

7.9.14.5. Mode Registers

Function	Register Type	Mode Register	Operand	Data	Notes
CA Parity Checking	R~~/~~W	MR01	[4]	0_B – ~~Disable~~ Enable (Default) 1_B – ~~Enable~~ Disable	
CA Parity Command Blocking	R~~/~~W	MR01	[5]	0_B – ~~Nonblocking~~ Blocking (Default) 1_B – ~~Blocking~~ Nonblocking	

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Function	Register Type	Mode Register	Operand	Data	Notes
Parity Latency	R~~/~~W	MR09	[5:4]	Parity Latency (PL) ~~00~~$_B$ 0h – 12 nCK (Default) ~~01~~$_B$ 1h – 16 nCK 2h – 20 nCK 3h – 24 nCK	1~~,2~~
Command Blocking Time	R~~/~~W	MR01	[7:6]	tCMD_BLOCKING ~~00~~$_B$ 0h = ~~1024~~ 768 ns (Default) ~~01~~$_B$ 1h = ~~1536~~ 1024 ns ~~10~~$_B$ 2h = ~~1536~~ 1536 ns ~~11~~$_B$ 3h = ~~RV~~ 2048 ns	~~32~~
CA Parity Error Status	R~~/~~W	MRxx (PSD)	[y]	0_B – Clear 1_B – Error	
CA Parity Error Log C0	R	MRD0	[5:0]	CA[5:0] contents in C0	3
CA Parity Error Log C1	R	MRD1	[5:0]	CA[5:0] contents in C1	3
CA Parity Error Log C2	R	MRD2	[5:0]	CA[5:0] contents in C2	3
CA Parity Error Log C3	R	MRD3	[5:0]	CA[5:0] contents in C3	3
CA Parity Error Log C4	R	MRD4	[5:0]	CA[5:0] contents in C4	3
CA Parity Error Log C5	R	MRD5	[5:0]	CA[5:0] contents in C5	3
~~Alert State Enable on CA Parity Error~~	~~R/W~~	~~MRxx (Shared)~~	~~[y]~~	~~0_B – Disable Alert not active for CA parity error (Default)~~ ~~1_B – Enable Alert active for CA parity error~~	
CA Parity Recovery Mode Enable	R~~/~~W	MRxx (Shared)	[y]	0_B – Do not use the VREFCA shadow register (Default) 1_B – Copy VREFCA shadow mode register value to the VREFCA normal mode register when a CA Parity Error is detected in blocking mode.	

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Function	Register Type	Mode Register	Operand	Data	Notes

~~NOTE 1: Parity Latency is applied to all commands.~~
NOTE 1: Parity Latency can be changed only when CA Parity Checking is disabled.
NOTE 2: +/- 20% variation of delay expected in DRAM
NOTE 3: CA Parity Error Log C0-C5 only stores the first erroneous command, until CA Parity Error Status is cleared.

7.9.14.6. Timing

Table 198 — CA Parity Timing Parameters

Parameter	Symbol	DDR6 TBD ~ TBD		Unit	Notes
		Min	Max		
Delay from erroneous command to ALERT assertion	tPAR_ALERT_ON	-	PL + 5 ns		
~~Pulse width of ALERT signal when asserted~~	~~tPAR_ALERT_PW~~	~~512~~	~~768~~	~~nCK~~	~~1~~
Additional latency for all commands for parity checking	tPAR_CA_DELAY	0	PL	nCK	1, 2

~~NOTE 1: +/- 20% variation of delay expected in DRAM~~
NOTE 1: tPAR_CA_DELAY is applied to all commands.
NOTE 2: If CA parity check is enabled and CA parity command blocking mode is enabled, then tPAR_-CA_DELAY = PL; else tPAR_CA_DELAY = 0.

Table 199 — Command Cancel Timing

Parameter	Symbol	DDR6 TBD ~ TBD		Unit
		Min	Max	
Command cancel timing for ACT, RD, REFab, REFdb, REFqb, RFMab, RFMdb, RFMqb, RFMpb, MRR, MRW when CS is high on the last two cycles	tCMD_cancel	8	-	nCK

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7.10. Command Ignore

The host can dynamically choose which Activate, Read, Write, and/or Precharge commands to ignore or execute for selected DRAM die(s) by configuring mode register bits and by setting an Ignore bit in C0:CA4 in the command to low (active low) or high (active high). Ignored commands (including Write-NTODT) shall be treated like NOP commands. ~~Write-NTODT commands will also be treated like NOP commands.CA parity checks, when enabled, will also apply to ignored commands.~~When CA parity checking is enabled, responses to CA parity errors will be the same whether a command is ignored or not. The Command Ignore function is only supported by 2p3 DRAM devices.

All command timing constraints need to be satisfied as though the command is being executed, even if the command is ignored. If the current or next command is ignored, assume different bank groups when applying command timing constraints.

The Command Ignore Enable/Disable bit, MR0E~~OP~~[0], may be set to 0~~b~~ (active low) or 1~~b~~ (active high) to dynamically ignore Activate, Read, Write, or Precharge commands for a DRAM die when the Ignore bit in the Activate, Read, Write, or Precharge command is low (active low) or high (active high) as defined in the Command Ignore Polarity bit, MR0E~~OP~~[1].

When the Command Ignore function is disabled (i.e., MR0E~~OP~~[0] = 0~~b~~), then the Ignore bit in C0:CA4 becomes don't care. When the Command Ignore function is enabled (i.e., MR0E OP[0] = 1~~b~~) and the Command Ignore Polarity is 'low' (i.e., MR0E~~OP~~[1] = 0~~b~~) then the DRAM die shall dynamically ignore Activate, Read, Write, or Precharge commands when the Ignore bit in C0:CA4 is low. When the Command Ignore function is enabled (i.e., MR0E~~OP~~[0] = 1~~b~~) and the Command Ignore Polarity is 'high' (i.e., MR0E~~OP~~[1] = 1~~b~~) then the DRAM die shall dynamically ignore Activate, Read, Write, or Precharge commands when the Ignore bit in C0:CA4 is high.

A Mode Register Write command with Per-DRAM Addressability (PDA) may be used to set the Command Ignore Enable/Disable bit or the Command Ignore Polarity bit.

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7.10.1. Mode Registers

Function	Register Type	Mode Register	Operand	Data	Notes
Command Ignore Enable/Disable	R/W	MR0E	[0]	0_B – The Command Ignore function is disabled. The Ignore bit in C0:CA4 doesn't take any effect. (Default) 1_B – The Command Ignore function is enabled. The DRAM die will dynamically ignore Activate, Read, Write, or Precharge commands depending on the Ignore bit in C0:CA4 and the 'Command Ignore Polarity' as set in the MR0E ~~OP~~[1]. This bit is only supported by 2p3 DRAM devices.	
Command Ignore Polarity	R/W	MR0E	[1]	This bit is a don't care when the Command Ignore function is disabled. 0_B – The Ignore bit in C0:CA4 is active low. 1_B – The Ignore bit in C0:CA4 is active high. (Default) This bit is only supported by 2p3 DRAM devices.	

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8. Command Constraints and AC Timing

8.1. DDR6 Command Timing Constraints

Table 200 — Command Timing Constraints for Same Bank

Current CMD	Next CMD				
	ACTIVATE	**READ**	**WRITE**	**PRECHARGE Per-Bank / Dual-Bank / Quad-Bank**	PRECHARGE All-Bank
ACTIVATE	N/A	RU(tRCD_R/tCK)	RU(tRCD_W/tCK) or RU(tRCD_W2/tCK)[3]	RU(tRAS/tCK)	RU(tRAS/tCK)
READ	N/A	RU(tCCD_L/tCK)[5]	tRTW[2]	RU(tRTP/tCK)	RU(tRTP/tCK)
WRITE	N/A	tWTR_L	RU(tCCD_L_WR/tCK) or RU(tCCD_L_WR2/tCK)[4]	tWTP	tWTP
PRECHARGE Per-Bank / Dual-Bank / Quad-Bank	RU(tRPpb/tCK)	N/A	N/A	tPPD	tPPD
PRECHARGE All-Bank	RU(tRPab/tCK	N/A	N/A	tPPD	tPPD

NOTE 1: Even nCK command gap is required to comply with DDR6 every other nCK command protocol.
NOTE 2: Read to Write Timing (tRTW) varies depending on DQ On-Die-Termination (ODT), Non-Target DQ ODT, Non-target DQ ODT, Read DQS. Refers to the Read to Write Timing (tRTW) section.
NOTE 3: tRCD_W for read-modify-write (2p3), and tRCD_W2 for non read-modify-write (other configurations).
NOTE 4: tCCD_L_WR for read-modify-write (2p3), and tCCD_L_WR2 for non read-modify-write (other configurations).
NOTE 5: If DDR6 operates below this minimum data rate, this constraint shall be tCCD_S rather than RU(tCCD_L/tCK).
NOTE 6: Unit of all parameters is nCK.

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Table 201 — Command Timing Constraints for Different Banks in Same Bank Group

Current CMD	Next CMD				
	ACTIVATE	READ	WRITE	PRECHARGE Per-Bank / Dual-Bank / Quad-Bank	PRECHARGE All-Bank
ACTIVATE	RU(tRRD_L/tCK)[5]	4	4	2 or 4[4]	RU(tRAS/tCK)
READ	6	RU(tCCD_L/tCK)[5]	tRTW[2)]	2 or 4[4)]	RU(tRTP/tCK)
WRITE	6	tWTR_L	RU(tCCD_L_WR/tCK) or RU(tCCD_L_WR2/tCK)[3]	2 or 4[4]	tWTP
PRECHARGE Per-Bank / Dual-Bank / Quad-Bank	6	4	4	tPPD	tPPD
PRECHARGE All-Bank	6	N/A	N/A	tPPD	tPPD

NOTE 1: Even nCK command gap is required to comply with DDR6 every other nCK command protocol.
NOTE 2: Read to Write Timing (tRTW) varies depending on DQ On-Die-Termination (ODT), Non-Target DQ ODT, Non-target DQ ODT, Read DQS. Refers to the Read to Write Timing (tRTW) section.
NOTE 3: tCCD_L_WR for read-modify-write (2p3), and tCCD_L_WR2 for non read-modify-write (other configurations).
NOTE 4: 2nCK for 2-cycle PRECHARGE command, and 4nCK for 4-cycle PRECHARGE command.
NOTE 5: If DDR6 operates below this minimum data rate, this constraint shall be tCCD_S rather than RU(tCCD_L/tCK).
NOTE 6: Unit of all parameters is nCK.

Table 202 — Command Timing Constraints for Different Bank Groups in 2pX/1pX Device and Different PSDs in 1pX Device (Sheet 1 of 2)

Current CMD	Next CMD				
	ACTIVATE	READ	WRITE	PRECHARGE Per-Bank / Dual-Bank / Quad-Bank	PRECHARGE All-Bank
ACTIVATE	tRRD_S	4	4	2 or 4[3]	RU(tRAS/tCK)
READ	6	tCCD_S	tRTW[2]	2 or 4[3]	RU(tRTP/tCK)
WRITE	6	tWTR_S	tCCD_S	2 or 4[3]	tWTP

Table 202 — Command Timing Constraints for Different Bank Groups in 2pX/1pX Device and Different PSDs in 1pX Device (Continued) (Sheet 2 of 2)

Current CMD	Next CMD				
	ACTIVATE	READ	WRITE	PRECHARGE Per-Bank / Dual-Bank / Quad-Bank	PRECHARGE All-Bank
PRECHARGE Per-Bank / Dual-Bank / Quad-Bank	6	4	4	tPPD	tPPD
PRECHARGE All-Bank	6	N/A	N/A	tPPD	tPPD

NOTE 1: Even nCK command gap is required to comply with DDR6 every other nCK command protocol.
NOTE 2: Read to Write Timing (tRTW) varies depending on DQ On-Die-Termination (ODT), Non-Target DQ ODT, Non-target DQ ODT, Read DQS. Refers to the Read to Write Timing (tRTW) section.
NOTE 3: 2nCK for 2-cycle PRECHARGE command, and 4nCK for 4-cycle PRECHARGE command.
NOTE 4: Unit of all parameters is nCK.

Table 203 — Command Timing Constraints between Different PSDs in 2pXDevice

Current CMD (PSD 0/1)	Next CMD (PSD 1/0)				
	ACTIVATE	READ	WRITE	PRECHARGE Per-Bank / Dual-Bank / Quad-Bank	PRECHARGE All-Bank
ACTIVATE	-	-	-	-	-
READ	-	-	tRTW[2]	-	-
WRITE	-	tWTR_S	-	-	-
PRECHARGE Per-Bank / Dual-Bank / Quad-Bank	-	-	-	-	-
PRECHARGE All-Bank	-	-	-	-	-

NOTE 1: Even nCK command gap is required to comply with DDR6 every other nCK command protocol.
NOTE 2: Read to Write Timing (tRTW) varies depending on DQ On-Die-Termination (ODT), Non-Target DQ ODT, Non-target DQ ODT, Read DQS. Refers to the Read to Write Timing (tRTW) section.
NOTE 3: Unit of all parameters is nCK.
NOTE 4: '-' means no timing constraints required between current CMD and next CMD.

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Table 204 — ACT-to-ACT(Long)/PRE/READ/WRITE Timing Constraints

Data Rate [Mbps]		tCK [ps][2]	tRAS [nCK]	tRCD_R [nCK]	tRCD_W [nCK]	tRCD_W2 [nCK]	tRRD_L [nCK]
Lower Limit [> Mbps]	Upper Limit [<= Mbps]						
4800	6400	312.5	64	58	58	26	16
6400	8000	250.0	80	72	72	32	20
8000	9200	217.4	92	84	84	38	24
9200	9600	208.3	96	88	88	40	24
9600	10400	192.3	104	94	94	42	28
10400	11600	172.4	116	106	106	48	32
11600	12800	156.3	128	116	116	52	32
12800	14400	138.9	144	130	130	58	36
14400	16400	122.0	164	148	148	66	44
16400	19200	104.2	192	174	174	78	48

NOTE 1: DDR6 SDRAM device should not be operated at a frequency above the upper frequency limit, or below the lower frequency limit
NOTE 2: tCK = 2 / Data Rate (max)

Table 205 — READ-to-READ(Long)/PRE and WRITE-to-WRITE(Long) Timing Constraints (Sheet 1 of 2)

Data Rate [Mbps]		tCK [ps][2]	tRTP [nCK]	tCCD_L [nCK]	tCCL_L_WR [nCK]	tCCD_L_WR2 [nCK]
Lower Limit [> Mbps]	Upper Limit [<= Mbps]					
4800	6400	312.5	32	16	64	32
6400	8000	250.0	40	20	80	40
8000	9200	217.4	46	24	92	48
9200	9600	208.3	48	24	96	48
9600	10400	192.3	52	28	104	52
10400	11600	172.4	58	32	116	60
11600	12800	156.3	64	32	128	64
12800	14400	138.9	72	36	144	72
14400	16400	122.0	82	44	164	84
16400	19200	104.2	96	48	192	96

Table 205 — READ-to-READ(Long)/PRE and WRITE-to-WRITE(Long) Timing Constraints (Continued) (Sheet 2 of 2)

Data Rate [Mbps]		tCK [ps][2]	tRTP [nCK]	tCCD_L [nCK]	tCCL_L_WR [nCK]	tCCD_L_WR2 [nCK]
Lower Limit [> Mbps]	Upper Limit [<= Mbps]					
NOTE 1: DDR6 SDRAM device should not be operated at a frequency above the upper frequency limit, or below the lower frequency limit **NOTE 2:** tCK = 2 / Data Rate (max)						

8.1.1. READ-to-WRITE Command Timing Constraints (tRTW)

The following tables show the command constraint from Read to Write timing when DDR6 SDRAM can control NT-ODT turn on/off timing for DQ and Read DQS separately.

Each of tRTW (DQ) and tRTW (DQS) is described in Table 7/8, depending on ODT enable/disable options, where tRTW timing is defined Max (tRTW (DQ), tRTW (DQS)).

Table 206 — READ to WRITE Command Timing Constraints (DQ-Bound) (Sheet 1 of 2)

RTT_PARK	RTT_DQS_PARK	RTT_WR	tRTW Timing Decomposition				Unit	Note
			RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- ODTLon_WR + ODTLon_WR_Offset (min) + tODTon (min)	- WL + 8.5 UI + RU(tDQS2DQi(min)/tCK) + tDQSoffset (min) + tDQSS (min)	tODTQon (max)		
Disable	Disable	Disable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	-	- WL + 8.5 UI + RU(tDQS2DQi(min)/tCK) + tDQSoffset (min) + tDQSS (min)	-	nCK	1, 2, 3
Disable	Disable	Enable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- ODTLon_WR + ODTLon_WR_Offset (min) + tODTon (min)	-	-	nCK	1, 2, 3
Disable	Enable	Disable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	-	- WL + 8.5 UI + RU(tDQS2DQi(min)/tCK) + tDQSoffset (min) + tDQSS (min)	-	nCK	1, 2, 3

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Table 206 — READ to WRITE Command Timing Constraints (DQ-Bound) (Continued) (Sheet 2 of 2)

RTT_PARK	RTT_DQS_PARK	RTT_WR	tRTW Timing Decomposition				Unit	Note
			RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- ODTLon_WR + ODTLon_WR_Offset (min) + tODTon (min)	- WL + 8.5 UI + RU(tDQS2DQi(min)/tCK) + tDQSoffset (min) + tDQSS (min)	tODTQon (max)		
Disable	Enable	Enable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- ODTLon_WR + ODTLon_WR_Offset (min) + tODTon (min)	-	-	nCK	1, 2, 3
Enable	Disable	Disable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	-	- WL + 8.5 UI + RU(tDQS2DQi(min)/tCK) + tDQSoffset (min) + tDQSS (min)	tODTQon (max)	nCK	1, 2, 3
Enable	Disable	Enable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- ODTLon_WR + ODTLon_WR_Offset (min) + tODTon (min)	-	tODTQon (max)	nCK	1, 2, 3
Enable	Enable	Disable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	-	- WL + 8.5 UI + RU(tDQS2DQi(min)/tCK) + tDQSoffset (min) + tDQSS (min)	tODTQon (max)	nCK	1, 2, 3
Enable	Enable	Enable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- ODTLon_WR + ODTLon_WR_Offset (min) + tODTon (min)	-	tODTQon (max)	nCK	1, 2, 3

NOTE 1: Even nCK command gap is required to comply with DDR6 every other nCK command protocol.
NOTE 2: Please refer to the Read/Write Latencies section for more details on RL/WL latencies.
NOTE 3: Write command includes non-target Write command.

Table 207 — READ to WRITE Command Timing Constraints (DQS-Bound) (Sheet 1 of 2)

RTT_PARK	RTT_DQS_PARK	RTT_WR	tRTW Timing Decomposition				Unit	Note
			RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- Read DQS Pre-Shift + tRPST – 0.5tCK	tODTQon (max)	- WL + tWPRE - tDQSoffset (min) - tDQSS (min)		
Disable	Disable	Disable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- Read DQS Pre-Shift + tRPST – 0.5tCK	-	- WL + tWPRE - tDQSoffset (min) - tDQSS (min)	nCK	1, 2, 3
Disable	Disable	Enable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- Read DQS Pre-Shift + tRPST – 0.5tCK	-	- WL + tWPRE - tDQSoffset (min) - tDQSS (min)	nCK	1, 2, 3
Disable	Enable	Disable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- Read DQS Pre-Shift + tRPST – 0.5tCK	tODTQon (max)	- WL + tWPRE - tDQSoffset (min) - tDQSS (min)	nCK	1, 2, 3
Disable	Enable	Enable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- Read DQS Pre-Shift + tRPST – 0.5tCK	tODTQon (max)	- WL + tWPRE - tDQSoffset (min) - tDQSS (min)	nCK	1, 2, 3
Enable	Disable	Disable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- Read DQS Pre-Shift + tRPST – 0.5tCK	-	- WL + tWPRE - tDQSoffset (min) - tDQSS (min)	nCK	1, 2, 3
Enable	Disable	Enable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- Read DQS Pre-Shift + tRPST – 0.5tCK	-	- WL + tWPRE - tDQSoffset (min) - tDQSS (min)	nCK	1, 2, 3
Enable	Enable	Disable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- Read DQS Pre-Shift + tRPST – 0.5tCK	tODTQon (max)	- WL + tWPRE - tDQSoffset (min) - tDQSS (min)	nCK	1, 2, 3

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Table 207 — READ to WRITE Command Timing Constraints (DQS-Bound) (Continued) (Sheet 2 of 2)

RTT_PARK	RTT_DQS_PARK	RTT_WR	tRTW Timing Decomposition				Unit	Note
			RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- Read DQS Pre-Shift + tRPST – 0.5tCK	tODTQon (max)	- WL + tWPRE - tDQSoffset (min) - tDQSS (min)		
Enable	Enable	Enable	RL + BL24 + RU(tCK2DQo(max)/tCK) + 4	- Read DQS Pre-Shift + tRPST – 0.5tCK	tODTQon (max)	- WL + tWPRE - tDQSoffset (min) - tDQSS (min)	nCK	1, 2, 3

NOTE 1: Even nCK command gap is required to comply with DDR6 every other nCK command protocol.
NOTE 2: Please refer to the Read/Write Latencies section for more details on RL/WL latencies.
NOTE 3: Write command includes non-target Write command.

8.1.2. DDR6 WRITE-to-READ (tWTR_S/L) and WRITE-to-Precharge (tWTP) Command Timing Constraints

Table 208 — WRITE-to-READ Short Timing (tWTR_S) (Sheet 1 of 2)

Data Rate [Mbps]		tCK [ps][5]	tWTR_S [nCK]			
			2pX Device[2]		1pX Device[2]	
Lower Limit [> Mbps]	Upper Limit [<= Mbps]		Read Link CRC Disabled	Read Link CRC Enabled	Read Link CRC Disabled	Read Link CRC Enabled
4800	6400	312.5	96	104	112	120
6400	8000	250.0	118	130	134	146
8000	9200	217.4	138	150	154	166
9200	9600	208.3	142	154	160	172
9600	10400	192.3	154	166	172	188
10400	11600	172.4	170	186	192	204
11600	12800	156.3	184	200	210	226
12800	14400	138.9	206	226	236	252
14400	16400	122.0	236	256	268	288
16400	19200	104.2	272	296	312	336

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Table 208 — WRITE-to-READ Short Timing (tWTR_S) (Continued) (Sheet 2 of 2)

Data Rate [Mbps]		tCK [ps][5]	tWTR_S [nCK]			
			2pX Device[2]		1pX Device[2]	
Lower Limit [> Mbps]	Upper Limit [<= Mbps]		Read Link CRC Disabled	Read Link CRC Enabled	Read Link CRC Disabled	Read Link CRC Enabled

NOTE 1: The DDR6 SDRAM device should not be operated at a frequency above the upper frequency limit, or below the lower frequency limit, shown for each RL or WL value.
NOTE 2: tWTR_S [nCK] = (WL-PL) [nCK] + tCCD_S [nCK] + 2x(RU(K / 2tCK)) where K=6.25ns for 2pX Device and K=8.25ns for 1pX Device
NOTE 3: When Command Blocking is disabled, PL = 0 [nCK]. Refers "Parity Latency" table for details.
NOTE 4: tCCD_S = 12nCK
NOTE 5: tCK = 2 / Data Rate (max)

Table 209 — WRITE-to-READ Long Timing (tWTR_L) & WRITE-to-Precharge Timing (tWTP)

Data Rate [Mbps]		tCK [ps][5]	tWTR_L & tWTP [nCK]			
			2pX Device[2]		1pX Device[2]	
Lower Limit [> Mbps]	Upper Limit [<= Mbps]		Read Link CRC Disabled	Read Link CRC Enabled	Read Link CRC Disabled	Read Link CRC Enabled
4800	6400	312.5	120	128	134	142
6400	8000	250.0	148	160	164	176
8000	9200	217.4	176	188	192	204
9200	9600	208.3	182	194	200	212
9600	10400	192.3	198	210	216	232
10400	11600	172.4	220	236	244	256
11600	12800	156.3	242	258	266	282
12800	14400	138.9	272	292	302	318
14400	16400	122.0	312	332	344	364
16400	19200	104.2	364	388	404	428

NOTE 1: The DDR6 SDRAM device should not be operated at a frequency above the upper frequency limit, or below the lower frequency limit, shown for each RL or WL value.
NOTE 2: tWTR_L and tWTP [nCK] = (WL - PL) [nCK] + 2x(RU(tCCD_L [ns] + K) / 2tCK) where K=12ns for 2pX Device and K=14ns for 1pX Device
NOTE 3: When Command Blocking is disabled, PL = 0 [nCK]. Refers "Parity Latency" table for details.
NOTE 4: tCCD_L = 5ns
NOTE 5: tCK = 2 / Data Rate (max)
NOTE 6: In this case, Write recovery time is 12ns + tACU (2pX Device) or 14ns+tACU (1pX Device).

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8.1.3. DDR6 Auto Precharge Command Timing Constraints

Table 210 — Auto Precharge Command Timing Constraints for Same Bank

Current CMD	Next CMD				
	ACTIVATE	READ or READ with AP	WRITE or WRITE with AP	PRECHARGE Per-Bank / Dual-Bank / Quad-Bank	PRECHARGE All-Bank
READ with AP	RU(tRTP/tCK) + RU(tRPpb/tCK)	N/A	N/A	RU(tRTP/tCK)	RU(tRTP/tCK)
WRITE with AP	tWTP + RU(tRPpb/Tck)	N/A	N/A	tWTP	tWTP

NOTE 1: Even nCK command gap is required to comply with DDR6 every other nCK command protocol.

Table 211 — Auto Precharge Command Timing Constraints for Different Banks in Same Bank Group

Current CMD	Next CMD				
	ACTIVATE	READ or READ with AP	WRITE or WRITE with AP	PRECHARGE Per-Bank / Dual-Bank / Quad-Bank	PRECHARGE All-Bank
READ with AP	6	RU(tCCD_L/tCK)[5]	tRTW[2)]	2 or 4[4]	RU(tRTP/tCK)
WRITE with AP	6	tWTR_L	RU(tCCD_L_WR /tCK) or RU(tCCD_L_WR 2/tCK)[3]	2 or 4[4]	tWTP

NOTE 1: Even nCK command gap is required to comply with DDR6 every other nCK command protocol.
NOTE 2: Read to Write Timing (tRTW) varies depending on DQ On-Die-Termination (ODT), Non-Target DQ ODT, Non-target DQ ODT, Read DQS. Refers to the Read to Write Timing (tRTW) section.
NOTE 3: tCCD_L_WR for read-modify-write (2p3), and tCCD_L_WR2 for non read-modify-write (other configurations).
NOTE 4: 2nCK for 2-cycle PRECHARGE command, and 4nCK for 4-cycle PRECHARGE command.
NOTE 5: If DDR6 operates below this minimum data rate, this constraint shall be tCCD_S rather than RU(tCCD_L/tCK).
NOTE 6: Unit of all parameters is nCK.

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Table 212 — Auto Precharge Command Timing Constraints for Different Bank Groups

Current CMD	Next CMD				
	ACTIVATE	READ or READ with AP	WRITE or WRITE with AP	PRECHARGE Per-Bank / Dual-Bank / Quad-Bank	PRECHARGE All-Bank
READ with AP	6	tCCD_S	tRTW[2]	2 or 4[3]	RU(tRTP/tCK)
WRITE with AP	6	tWTR_S	tCCD_S	2 or 4[3]	tWTP

NOTE 1: Even nCK command gap is required to comply with DDR6 every other nCK command protocol.
NOTE 2: Read to Write Timing (tRTW) varies depending on DQ On-Die-Termination (ODT), Non-Target DQ ODT, Non-target DQ ODT, Read DQS. Refers to the Read to Write Timing (tRTW) section.
NOTE 3: 2nCK for 2-cycle PRECHARGE command, and 4nCK for 4-cycle PRECHARGE command.
NOTE 4: Different PSDs in 1p6 device comply with command timing constraints for different bank groups.
NOTE 5: Unit of all parameters is nCK.

Table 213 — Auto Precharge Command Timing Constraints for Different PSDs in 2p3 and 2p6 Device

Current CMD (PSD 0/1)	Next CMD (PSD 1/0)				
	ACTIVATE	READ or READ with AP	WRITE or WRITE with AP	PRECHARGE Per-Bank / Dual-Bank / Quad-Bank	PRECHARGE All-Bank
READ with AP	-	-	tRTW[2]	-	-
WRITE with AP	-	tWTR_S	-	-	-

NOTE 1: Even nCK command gap is required to comply with DDR6 every other nCK command protocol.
NOTE 2: Read to Write Timing (tRTW) varies depending on DQ On-Die-Termination (ODT), Non-Target DQ ODT, Non-target DQ ODT, Read DQS. Refers to the Read to Write Timing (tRTW) section.
NOTE 3: Unit of all parameters is nCK.

8.1.4. DDR6 MRR/MRW Timing Constraints

Table 214 — MRR/MRW Timing Constraints (Sheet 1 of 2)

Current Command	Next Command	Min Delay	Unit	Note
MODE REGISTER READ	MODE REGISTER READ	tMRR	nCK	
	READ / READ with AP	tMRR_RD	nCK	

Table 214 — MRR/MRW Timing Constraints (Continued) (Sheet 2 of 2)

Current Command	Next Command	Min Delay	Unit	Note
	WRITE / WRITE with AP	tRTW	nCK	1,2
	MODE REGISTER WRITE	RL+tCCD_S+RU(tCK2DQo(max)/tck)	nCK	3
READ	MODE REGISTER READ	RL	nCK	
READ with AP		RL	nCK	
WRITE		tWTR_L	nCK	
WRITE with AP		tWTR_L	nCK	
MODE REGISTER WRITE		tMRD	ns	
MODE REGISTER WRITE	READ / READ with AP	tMRD	ns	
	WRITE / WRITE with AP	tMRD	ns	
	MODE REGISTER WRITE	tMRW	ns	
READ Training		RL+tCCD_S+RU(tCK2DQo(max)/tck)	nCK	
READ with AP		RL+tCCD_S+RU(tCK2DQo(max)/tck)	nCK	
WRITE Training		WL+RU(tCCD_L/tCK)+RU(7.5 ns/tCK)	nCK	
WRITE with AP		WL+RU(tCCD_L/tck)+RU(7.5ns/tck)	nCK	

NOTE 1: Even nCK command gap is required to comply with DDR6 every other nCK command protocol.
NOTE 2: Mode Register Read to Mode Register Write Timing varies depending on DQ On-Die-Termination (ODT), Non-Target DQ ODT, Non-target DQ ODT, Read DQS. Refers to the Read to Write Timing (tRTW) section.
NOTE 3: If the ensuing MRW changes the settings of a Mode Register which affects data output condition and/or timing, the delay time from MRR to MRW must be as follows when NT-ODT is enabled.

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8.2. MRR/MRW Timing Constraints

To be balloted in Q4'26.

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8.3. DQS2DQ and CK2DQ AC Parameters

tDQS2DQi defines the timing offset from DQS input to DQ input for write operation and it reflects an internal delay of DRAM from a DQS receiver to a DQ latch. tCK2DQo defines the timing offset from CK input to DQ output for read operation and it reflects an internal delay of DRAM from a CK receiver to an DQ output driver. These parameters apply to write/read/ODT command timing equations for command scheduling.

Table 215 — 6400 – 14400Mbps tDQS2DQi AC Parameters

Parameter	Symbol	6400 – 12800 [Mbps]		14400 [Mbps]		Unit	Notes
		Min	Max	Min	Max		
DQS to DQ input offset	tDQS2DQi	100	550	50	500	ps	1
DQS to DQ input offset voltage variation	tDQS2DQi_volt	-	20	-	20	ps/50mV	
DQS to DQ input offset temperature variation	tDQS2DQi_temp	-	0.5	-	0.5	ps/°C	
NOTE 1: The range includes variations across process, voltage, and temperature							

Table 216 — 6400 – 14400Mbps tCK2DQo AC Parameters

Parameter	Symbol	6400 – 12800 [Mbps]		14400 [Mbps]		Unit	Notes
		Min	Max	Min	Max		
CK to DQ output offset	tCK2DQo	200	1500	100	1400	ps	1
CK to DQ output offset voltage variation	tCK2DQo_volt	-	2.5	-	2.5	ps/mV	
CK to DQ output offset temperature variation	tCK2DQo_temp	-	1.4	-	1.4	ps/°C	
NOTE 1: The range includes variations across process, voltage, and temperature							

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8.4. tCCD

Unballoted.

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8.5. Training-Related Timing Constraints

Unballoted.

The device can enter a read or write training state by issuing commands such as READ DQ CALIBRATION. For training commands that read or write the device, READ latency (RL) and WRITE latency (WL) are counted from the rising CK edge of each training command with the same timing relationship as any typical READ or WRITE command.

Table 217 — Training-Related Timing Constraints

NOTE: ^{t}WRWTR = WL + BL/n_max + max[RU(7.5ns/^{t}CK), 6*n*CK]

NOTE: ^{t}RTRRD = RL + BL/n_max + max[RU(7.5ns/^{t}CK), 6*n*CK].

NOTE: ^{t}RTW is TBD and has a different value based on ODT state.

NOTE: The value of ^{t}RTW that is applied for different banks in different bank groups should be used for this constraint.

Previous Command	Next Command	Min Delay	Unit	Note
WRITE (WR_S/WR_L)	READ DQ CALIBRATION	WL+BL/n_max +RU(^{t}WTR/^{t}CK)	*n*CK	
READ (RD_S/RD_L)/ MRR	READ DQ CALIBRATION	^{t}RTRRD	*n*CK	3
WRITE FIFO	WRITE (WR_S/WR_L)	Not allowed	–	2
	READ (RD_S/RD_L)/ MRR	Not allowed	–	2
	READ DQ CALIBRATION	Not allowed	–	2
READ FIFO	WRITE (WR_S/WR_L)	^{t}RTRRD	*n*CK	3
	READ (RD_S/RD_L)/ MRR	^{t}RTRRD	*n*CK	3
	READ DQ CALIBRATION	^{t}RTRRD	*n*CK	3
READ DQ Calibration	WRITE (WR_S/WR_L)	^{t}RTRRD	*n*CK	3
	READ (RD_S/RD_L)/ MRR	^{t}RTRRD	*n*CK	3
	READ DQ CALIBRATION	6	*n*CK	

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8.6. Logical Rank-to-Rank Timings

Unballoted

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8.7. 1p6 Command Timing Constraints

Unballoted.

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9. Core Timings

9.1. DDR6 Core AC Timing Parameters

Table 218 — Core AC Timing Parameters for 1p6/1p12 mode disabled, Read Link CRC Disabled and Command blocking Disabled

Parameter	Symbol	Min/Max	Value	Unit	Notes
ACTIVATE to PRECHARGE delay time	tRAS	Min	20	ns	
		Max	300	ns	
Activation Counter Update time	tACU	Min	22	ns	4
Row Precharge time (all Banks)	tRPab	Min	nACU + 21ns	-	
Row Precharge time (Single/Dual)	tRPpb/db	Min	nACU + 18ns	-	
ACTIVATE to ACTIVATE delay time for same bank	tRC	Min	tRAS+tRPab with all-bank precharge tRAS+tRPpb with per-bank precharge	-	
ACTIVATE to READ delay time	tRCD_R	Min	18	ns	
ACTIVATE to WRITE delay time	tRCD_W	Min	18	ns	
ACTIVATE to WRITE delay time WRITE not RMW	tRCD_W2	Min	8	ns	
Read Latency	RL	Min	21	ns	1
Write Latency	WL	-	RL-4	nCK	
WRITE to PRECHARGE delay time	tWTP	Min	WL +RU(tCCD_L/tCK) + RU(12ns/tCK)	nCk	2, 4
READ to PRECHARGE delay time	tRTP	Min	10	ns	

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Table 218 — Core AC Timing Parameters for 1p6/1p12 mode disabled, Read Link CRC Disabled and Command blocking Disabled

Parameter	Symbol	Min/Max	Value	Unit	Notes
ACTIVATE to ACTIVATE delay time for same bank group	tRRD_L	Min	5	ns	3
ACTIVATE to ACTIVATE delay time for different bank group	tRRD_S	Min	12	nCK	
Four bank ACTIVATE window	tFAW	Min	Max (4 *tRRD_S, 7.5ns)	ns, nCK	
WRITE to READ delay time for same bank group	tWTR_L	Min	WL + RU(tCCD_L/tCK) +RU(12ns/tCK)	nCK	2
WRITE to READ delay time for different bank group	tWTR_S	Min	WL + tCCD_S + RU(6.25ns/tCK)	nCK	2
PRECHARGE to PRECHARGE delay time	tPPD	Min	8	nCK	
READ to READ or WRITE to WRITE delay time for different bank group	tCCD_S	Min	12	nCK	
READ to READ delay time for same bank group	tCCD_L	Min	5	ns	3
WRITE to WRITE delay time for same bank group	tCCD_L_WR	Min	20	ns	
WRITE to WRITE delay time for same bank group, second WRITE not RMW	tCCD_L_WR2	Min	10	ns	

NOTE 1: tAA_min (READ to data out) = RL_min +tCK2DQO_min
NOTE 2: "RU" means Round-Up
NOTE 3: If DDR6 operates below 4800Mbps, tCCD_L and tRRD_L are calculated as Max(BL/2, 5ns)
NOTE 4: Write Recovery time = tWTP - WL - RU(tCCD_L/tCK) + nACU

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10. AC and DC Operating Conditions

10.1. DC Operating Conditions

Table 219 — DC Operating Conditions

Symbol	Parameter	Low Freq Voltage Spec Freq:DC to 2 MHz				Notes
		Min.	Typ.	Max.	Unit	
VDD	Device Supply Voltage	0.994 (-3%)	1.025	1.087 (+6%)	V	1,2,3
VDDQ	Supply Voltage for I/O	0.582 (-3%)	0.6	0.636 (+6%)	V	1,2,3
VPP	Core Power Voltage	1.746 (-3%)	1.8	1.908 (+6%)	V	1,2,3

NOTE 1: This includes all voltage noise from DC to 2 MHz at the DRAM package ball
NOTE 2: For DIMM Z(f), JEDEC recommends to refer to the DIMM Z(f) specification
NOTE 3: For solder down component Z(f), JEDEC recommends to refer to each DRAM memory vendor's Z(f) guidance

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10.2. Absolute Maximum DC Ratings

Stresses greater than those listed may cause permanent damage to the device. This is a stress rating only, and functional operation of the device at these or any other conditions above those indicated in the operational sections of this standard is not implied. Exposure to absolute maximum rating conditions for extended periods may affect reliability.

Table 220 — Absolute Maximum DC Ratings

Parameter	Symbol	Min	Max	Unit	Note
VPP Supply Voltage relative to VSS	VPP	−0.4	2.1	V	1
VDD Supply Voltage relative to VSS	VDD	−0.4	1.2	V	1
VDDQ Supply Voltage relative to VSS	VDDQ	−0.4	1.2	V	1
Voltage on Any Signal Ball relative to VSS	V_{IN}, V_{OUT}	−0.4	1.2	V	
Voltage on ZQ (Reference) Ball relative to VSS	V_{ZQ}	−0.4	1.2	V	
Storage Temperature	TSTG	−55	100	°C	2

NOTE 1: See section x.x relationships between power supplies.
NOTE 2: Storage Temperature is the case surface temperature on the center/top side of the DDR6 device. For the measurement conditions, please refer to JESDxx-xx.

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10.3. Component Operating Temperature Range

Table 221 — DC Operating Temperature Range

Parameter	Symbol	Temperature Range (Unit: °C)		Grade	Notes
		Min	Max		
Normal Operating Temperature	Toper_normal	0	85	NT	1,2,4
Extended Operating Temperature	Toper_extended	0	95	XT	1,2,3,4
Extended Operating Temperature 2	Toper_extended 2	-5	95	XT2	1,2,3,4

NOTE 1: All operating temperature symbols, ranges, acronyms are referred from JESD402-1x.
NOTE 2: Operating Temperature is the case surface temperature on the center/top side of the DRAM. For the measurement conditions, please refer to JESD51-2x standard.
NOTE 3: All DDR6 SDRAMs are required to operate in NT / XT temperature ranges. Some applications require operation of DDR6 in Extended Operating Temperature range (XT2: -5°C~95°C).
NOTE 4: When operating above 85°C, the host shall provide appropriate Refresh mode controls associated with the increased temperature range. The full description of these settings is defined in section TBD.

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10.4. Electrostatic Discharge Sensitivity Characteristics

Table 222 — Electrostatic Discharge Sensitivity Characteristics

Parameter	Symbol	Min	Max	Unit	Notes
Human body model (HBM)	ESD_{HBM}	500	-	V	1,2
Charged-device model (CDM)	ESD_{CDM0}	200	-	V	1,3,4
	ESD_{CDM1}	TBD	-	V	1,3,4

NOTE 1: State-of-the-art basic ESD control measures have to be in place when handling devices
NOTE 2: Refer to ESDA / JEDEC Joint Standard JS-001 for measurement procedures
NOTE 3: Refer to ESDA / JEDEC Joint Standard JS-002 f for measurement procedures
NOTE 4: Transition from CDM0 to CDM1 parameter is intended to be speed bin dependent; however, the implementation details are TBD.

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11. AC and DC Input/Output Operating Conditions

11.1. Bit Error Rate

Unballoted.

This section provides an overview of the bit error rate (BER) and the desired statistical level of confidence.

$$n = \left(\frac{1}{BER}\right)\left[-\ln(1 - SLC) + \ln\left(\sum_{k=0}^{N}\frac{(n \times BER)^k}{k!}\right)\right]$$

Figure 199 — General Equation

NOTE: Where:

n = number of bits in a trial

SLC = statistical level of confidence
- BER = bit error rate
- k = intermediate number of specific errors found in trial
- N = number of errors recorded during trial

If no errors are assumed in a given test period, the second term drops out and the equation becomes:

$$n = \left(\frac{1}{BER}\right)\left[-\ln(1 - SLC)\right]$$

JEDEC recommends testing to 99.5% confidence levels; however, one may choose a number that is viable for their own manufacturing levels. To determine how many bits of data should be sent (again, assuming zero errors, or N = 0), using BER = $1E^{-9}$and confidence level SLC = 99.5%, the result is n = (1/BER)(-ln(1 - 0.995) = 5.298×10^9.

Table 223 — Estimated Number of Transmitted Bits (*n*) – Confidence Level of 70–99.5%

Number of Errors	*n* = -ln(1 - SLC)/BER							
	99.5%	99%	95%	90%	85%	80%	75%	70%
0	5.298/BER	4.61/BER	2.99/BER	2.3/BER	1.90/BER	1.61/BER	1.39/BER	1.20/BER

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11.1.1. Minimum BER Requirements

The following table specifies the average UI and BER requirements over which certain receiver and transmitter timing and voltage specifications need to be validated assuming a 99.5% confidence level at BER = $1E^{-9}$.

Table 224 — Minimum BER Requirements for Rx/Tx Timing and Voltage Tests (3200–8800)

Parameter	Symbol	3200–8800			Unit	Notes
		Min	Nom	Max		
Average UI	UI_{AVG}	0.999 * nom	1.000/f	1.001 * nom	ps	1
Number of UI (min)	$N_{Min_UI_Validation}$	5.3×10^9	–	–	UI	2
Bit error rate	BER_{Lane}	–	–	$1E^{-16}$	Events	3,4,5,6

NOTE 1: Average UI size; "f" is data rate.
NOTE 2: Number of UI over which certain Rx/Tx timing and voltage specifications need to be validated assuming a 99.5% confidence level at BER=$1E^{-9}$.
NOTE 3: This is a system parameter. It is the raw BER for every lane before any logical PHY or link layer-based correction. It may not be possible to have a validation methodology for this parameter for a standalone transmitter or standalone receiver; therefore, this parameter has to be validated in selected systems using a suitable methodology as deemed by the platform.
NOTE 4: This is a raw BER-per-lane before any correction. This parameter is primarily used to determine electrical margins during electrical analysis and measurements that are located between two interconnected devices.
NOTE 5: This is the minimum BER requirements for testing timing and voltage parameters listed in Input Clock Jitter, Rx DQS & DQ Voltage Sensitivity, Rx DQS Jitter Sensitivity, Rx DQ Stressed Eye, Tx DQS Jitter, Tx DQ Jitter, and Tx DQ Stressed eye height / eye width specifications.
NOTE 6: The BER for DDR5 during normal operation is $1E^{-16}$. For validation purposes, the BER used is $1E^{-9}$.

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11.2. Unit Interval and Jitter Definition

11.2.1. Introduction

This section describes the Unit Interval (UI) and UI Jitter definitions associated with the Jitter parameters specified in Tx DQS2DQ Stressed Edge Jitter and Input Clock Jitter specifications.

11.2.2. Unit Interval (UI)

The definition of Unit Interval (UI) is a minimum time interval between condition changes of a signal. DDR-based signals are referenced to the differential crossing point of CK_t and CK_c. 2UI=1tCK for DDR-based signals (for example, DQ, DQS).

The UI definitions shown in Figure 1and Figure 2 are for DDR-based signals. The times at which the differential crossing points of the clock occur are defined at t1, t2, …, tn-1, tn. The UI at index "n" is defined as an arbitrary time in steady state, where n=0 is chosen as the starting crossing point.

$$UI_n = t_n - t_{n-1}$$

Figure 200 — UI Definition in Terms of Adjacent Edge Timings

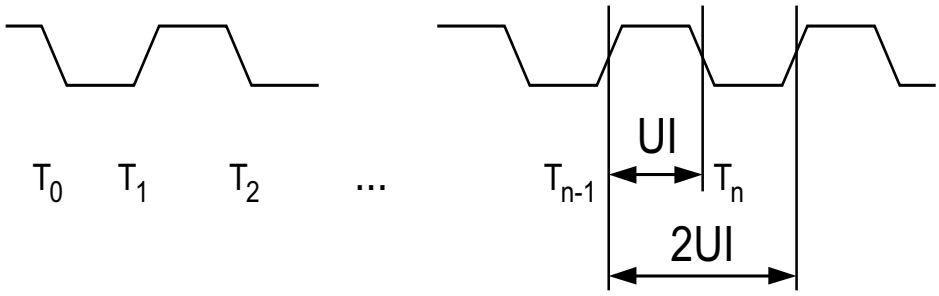


Figure 201 — UI Definition Using Clock Waveforms

11.2.3. UI Jitter Definition

If a number of UI edges are computed or measured at times t1, t2, …, tn-1, tn,..., tK, where K is the maximum number of samples, then the UI jitter at any instance "n" is defined in Figure3, where T = the ideal UI size.

$$UI(jit)_n = (t_n - t_{n-1}) - T, n = 1, 2, 3, ..., K$$

Figure 202 — UI Jitter for "nth" UI Definition (in terms of ideal UI)

In a large sample with random Gaussian-like jitter (therefore very close to symmetric distribution), the average of all UI sizes usually turns out to be very close to the ideal UI size.

The equation described in Figure 3 assumes starting from an instant steady state, where n=0 is chosen as the starting point.1 UI = one bit, which means 2 UI = one full cycle or time period of the forwarded strobe. Example: For 6.4 GT/s signaling, the forwarded strobe frequency is 3.2 GHz, or 1 UI = 156.25 ps.

Deterministic jitter is analyzed in terms of the peak-to-peak value and in terms of specific frequency components present in the jitter, isolating the causes for each frequency. Random jitter is unbounded and analyzed in terms of statistical distribution to convert to a bit error rate (BER) for the link.

11.2.4. UI-UI Jitter Definition

UI-UI (read as "UI to UI") jitter is defined to be the jitter between two consecutive UI as shown in Figure 4.

$$\Delta UI_n = UI_n - UI_{n-1}, n = 2, 3, ..., K$$

Figure 203 — UI-UI Jitter Definitions

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11.2.5. Accumulated Jitter (Over "N" UI)

Accumulated jitter is defined as the jitter accumulated over any consecutive "N" UI as shown in Figure 5 .

$$T_{acc}^{N} = \sum_{p=m}^{m+N-1} (UI_p - \overline{UI}) \quad m = 1, 2, 3, ..., K–N$$

Figure 204 — Definition of Accumulated Jitter (over "N" UI)

where $\overline{UI}$ is defined in the equation shown in Figure 6.

$$\overline{UI} = \frac{\sum_{p=1}^{K} UI_p}{K} \quad p = 1, 2, 3, ..., N, ...,K$$

Figure 205 — Defnition of $\overline{UI}$

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11.3. Differential Input Cross Point Voltage (V_{IX})

11.3.1. Differential Input Clock (CK_t, CK_c) Cross Point Voltage (VIX)

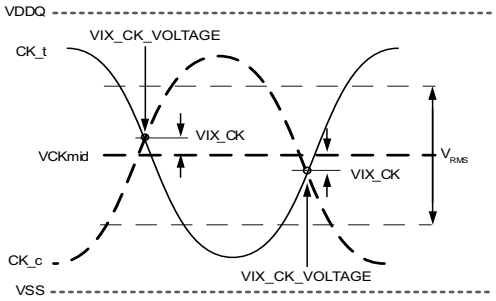


Figure 206 — VIX Definition (CK)

Table 225 — Crosspoint Voltage (VIX) for Differential Input Clock

Parameter	Symbol	DDR6 6400--12800		DDR6 14400-19200		Unit	Notes
		Min	Max	Min	Max		
Clock differential input crosspoint voltage ratio	VIX_CK_Ratio	-	TBD	-	TBD	%	1,2,3
Clock differential input crosspoint voltage	VIX_CK_Voltage	-	0.4	-	0.4	V	

NOTE 1: The VIX_CK voltage is referenced to VCKmid(mean) = (CK_t voltage + CK_c voltage) /2, where the mean is over 8 UI.
NOTE 2: VIX_CK_Ratio = (|VIX_CK| / |VRMS|)*100%, where VRMS = RMS(CK_t voltage - CK_c voltage)
NOTE 3: Only applies when both CK_t and CK_c are transitioning

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11.3.2. Differential Strobe (DQS_t, DQS_c) Cross Point Voltage (VIX)

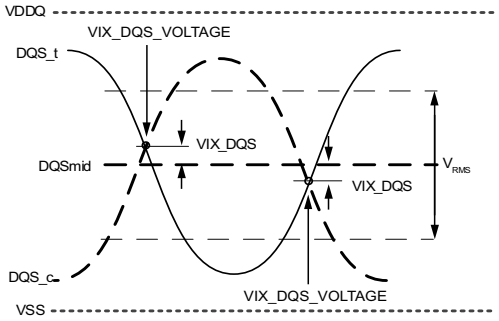


Figure 207 — VIX Definition (DQS)

Table 226 — Crosspoint Voltage (VIX) for Differential DQS

Parameter	Symbol	DDR6 6400--12800		DDR6 14400--19200		Unit	Notes
		Min	Max	Min	Max		
DQS differential input crosspoint voltage ratio	VIX_DQS_Ratio	-	TBD	-	TBD	%	1,2,3
DQS differential input crosspoint voltage	VIX_DQS_Voltage	-	0.4	-	0.4	V	

NOTE 1: The VIX_DQS voltage is referenced to VDQSmid(mean) = (DQS_t voltage + DQS_c voltage) /2, where the mean is over 8 UI
NOTE 2: VIX_DQS_Ratio = (|VIX_DQS| / |VRMS|)*100%, where VRMS = RMS(DQS_t voltage - DQS_c voltage)
NOTE 3: Only applies when both DQS_t and DQS_c are transitioning(including preamble)

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11.4. Read Timing tLZ (RDQS)

Unballoted.

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12. AC and DC Input Measurement Levels

12.1. CA Rx Voltage and Timings

To be balloted in Q4'26.

The command, address and control (including chip select) input receiver compliance mask for voltage and timing is shown in the figure below. All CA signals apply the same compliance mask and the CS signal applies its own compliance mask independently. All signals applied to the compliance mask operate in single data rate mode.

The CA input receiver mask (Rx Mask) for voltage and timing is applied across all CA[13:0] pins and the CS_n pin. For the remainder of this section, the reference to "CA" should be applied to the device CA[13:0] pins as well as the CS_n pin. The Rx Mask defines the area the input signal must not encroach for the device input receiver to successfully capture a valid input signal; it is not the valid data-eye.

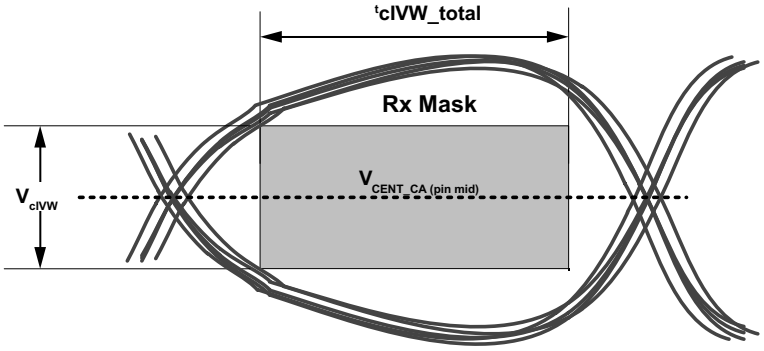


Figure 208 — CA/CS_n Rx Mask

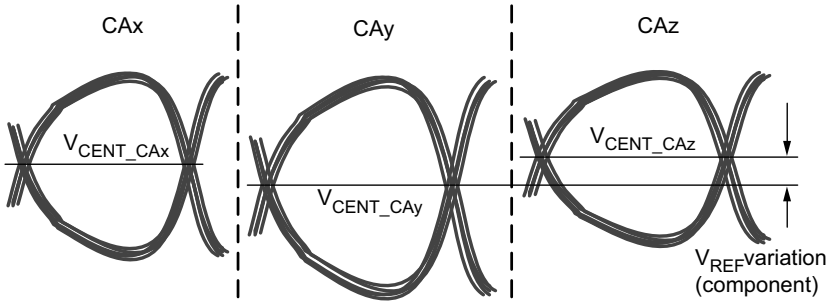


Figure 209 — Across-Pin V_{REFCA} Voltage Variation

$V_{cent_CA(pin\ mid)}$ is defined as the midpoint between the largest V_{cent_CA} voltage level and the smallest V_{cent_CA} voltage level across all CA and CS_n pins for a given device component.

Each CA V_{cent} level is defined by the center (widest opening) of the cumulative data input eye as depicted in the previous figure (Across-Pin V_{REFCA} Voltage Variation). This clarifies any device component level variation must be

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accounted for within the CA Rx Mask. The component-level V_{REF} is set by the system to account for R_{ON} and ODT settings.

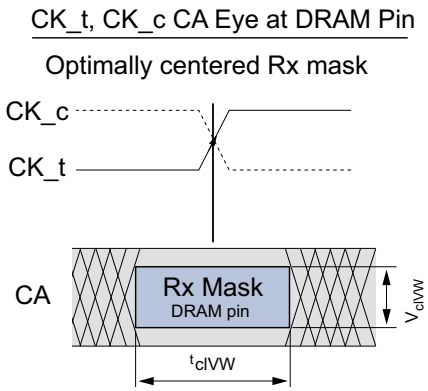


Figure 210 — CA/CS_n Timings at the Device Pins

^{t}cIVW is not necessarily center-aligned on CK_t/CK_c crossing at the DRAM pin, but is assumed to be center-aligned at the DRAM latch.

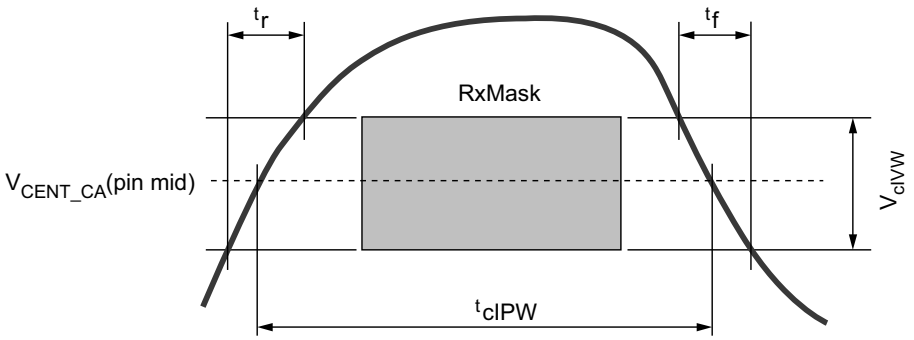


Figure 211 — CA T_{clPW} and S_{RIN_clVW} Definition (For Each Input Pulse)

NOTE: $S_{RIN_clVW} = V_{clVW_Total}/(^{t}r$ or $^{t}f)$, signal must be monotonic within ^{t}r and ^{t}f range.

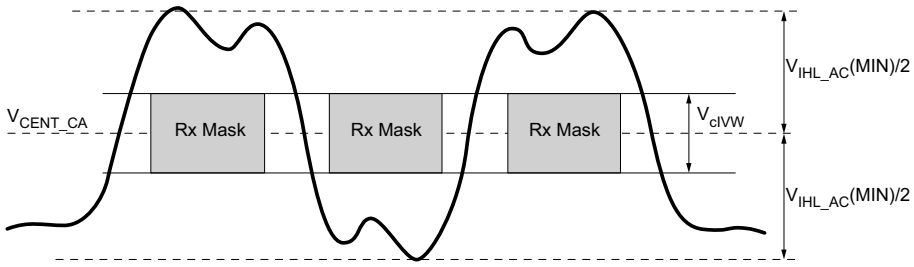


Figure 212 — CA V_{IHL_AC} Definition (For Each Input Pulse)

Table 227 — DDR5 3200-4800 Command/Address, CS Parametric Values

NOTE: CA Rx Mask voltage and timing parameters at the pin, including voltage and temperature drift.

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NOTE: Rx Mask voltage $V_{cIVW\ total(MIN)}$ must be centered around $V_{cent_CA(pin\ mid)}$.
NOTE: Rx differential CA to CK jitter total timing window at the V_{cIVW} voltage levels.
NOTE: Defined over the CA internal V_{REF} range. The Rx Mask at the pin must be within the internal V_{REFCA} (or V_{REFCS}) range, irrespective of the input signal common mode.
NOTE: CA only minimum input pulse width defined at the $V_{cent_CA(pin\ mid)}$.
NOTE: Input slew rate over $V_{cIVW\ Mask}$ centered at $V_{cent_CA(pin\ mid)}$.
NOTE: V_{IHL_AC} does not have to be met when no transitions are occurring.

Symbol	Parameter	3200		3600		4000		4400		4800		Unit	Notes
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
V_{cIVW}	Rx Mask voltage - p-p	140	-	140	-	140	-	130	-	130	-	mV	1,2,4
$^{t}cIVW$	Rx timing window	0.2	-	0.2	-	0.2	-	0.2	-	0.2	-	^{t}CK	1,2,3,4
V_{IHL_AC}	CA input pulse amplitude	160		160		160		150		150		mV	7
$^{t}cIPW$	CA input pulse width	0.58		0.58		0.58		0.58		0.58		^{t}CK	5
S_{RIN_cIVW}	Input slew rate over V_{cIVW}	1	7	1	7	1	7	1	7	1	7	V/ns	6

Table 228 — DDR5 5200-6400 Command/Address, CS Parametric Values

NOTE: CA Rx Mask voltage and timing parameters at the pin, including voltage and temperature drift.
NOTE: Rx Mask voltage $V_{cIVW\ total(MIN)}$ must be centered around $V_{cent_CA(pin\ mid)}$.
NOTE: Rx differential CA to CK jitter total timing window at the V_{cIVW} voltage levels.
NOTE: Defined over the CA internal V_{REF} range. The Rx Mask at the pin must be within the internal V_{REFCA} (or V_{REFCS}) range, irrespective of the input signal common mode.
NOTE: CA only minimum input pulse width defined at the $V_{cent_CA(pin\ mid)}$.
NOTE: Input slew rate over $V_{cIVW\ Mask}$ centered at $V_{cent_CA(pin\ mid)}$.
NOTE: V_{IHL_AC} does not have to be met when no transitions are occurring.

Symbol	Parameter	x4 and x8 Devices				x16 Devices				Unit	Notes
		5200-5600		6000-6400		5200-5600		6000-6400			
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
V_{cIVW}	Rx Mask voltage - p-p	100	-	85	-	110	-	95	-	mV	1,2,4
$^{t}cIVW$	Rx timing window	0.2	-	0.2	-	0.2	-	0.2	-	^{t}CK	1,2,3,4
V_{IHL_AC}	CA input pulse amplitude	115		105		125	-	115	-	mV	7

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Symbol	Parameter	x4 and x8 Devices				x16 Devices				Unit	Notes
		5200-5600		6000-6400		5200-5600		6000-6400			
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
$^{t}cIPW$	CA input pulse width	0.58		0.58		0.58		0.58		^{t}CK	5
S_{RIN_cIVW}	Input slew rate over V_{cIVW}	1	7	1	7	1	7	1	7	V/ns	6

Table 229 — DDR5 6800-8000 Command/Address, CS Parametric Values

NOTE: CA Rx Mask voltage and timing parameters at the pin, including voltage and temperature drift.

NOTE: Rx Mask voltage $V_{cIVW\ total(MIN)}$ must be centered around $V_{cent_CA(pin\ mid)}$.

NOTE: Rx differential CA to CK jitter total timing window at the V_{cIVW} voltage levels.

NOTE: Defined over the CA internal V_{REF} range. The Rx Mask at the pin must be within the internal V_{REFCA} (or V_{REFCS}) range, irrespective of the input signal common mode.

NOTE: CA only minimum input pulse width defined at the $V_{cent_CA(pin\ mid)}$.

NOTE: Input slew rate over $V_{cIVW\ Mask}$ centered at $V_{cent_CA(pin\ mid)}$.

NOTE: V_{IHL_AC} does not have to be met when no transitions are occurring.

Symbol	Parameter	x4 and x8 Devices				x16 Devices				Unit	Notes
		6800-7200		7600-8000		6800-7200		7600-8000			
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
V_{cIVW}	Rx Mask voltage - p-p	80	-	75	-	90	-	85	-	mV	1,2,4
$^{t}cIVW$	Rx timing window	0.2	-	0.2	-	0.2	-	0.2	-	^{t}CK	1,2,3,4
V_{IHL_AC}	CA input pulse amplitude	100		95		110	-	105	-	mV	7
$^{t}cIPW$	CA input pulse width	0.58		0.58		0.58		0.58		^{t}CK	5
S_{RIN_cIVW}	Input slew rate over V_{cIVW}	1	7	1	7	1	7	1	7	V/ns	6

Table 230 — DDR5 8400-8800 Command/Address, CS Parametric Values

NOTE: CA Rx Mask voltage and timing parameters at the pin, including voltage and temperature drift.

NOTE: Rx Mask voltage $V_{cIVW\ total(MIN)}$ must be centered around $V_{cent_CA(pin\ mid)}$.

NOTE: Rx differential CA to CK jitter total timing window at the V_{cIVW} voltage levels.

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NOTE: Defined over the CA internal V_{REF} range. The Rx Mask at the pin must be within the internal V_{REFCA} (or V_{REFCS}) range, irrespective of the input signal common mode.

NOTE: CA only minimum input pulse width defined at the $V_{cent_CA(pin\ mid)}$.

NOTE: Input slew rate over $V_{cIVW\ Mask}$ centered at $V_{cent_CA(pin\ mid)}$.

NOTE: V_{IHL_AC} does not have to be met when no transitions are occurring.

Symbol	Parameter	x4 and x8 Devices		x16 Devices		Unit	Notes
		8400-8800		8400-8800			
		MIN	MAX	MIN	MAX		
V_{cIVW}	Rx Mask voltage - p-p	70	-	80	-	mV	1,2,4
$^{t}cIVW$	Rx timing window	0.2	-	0.2	-	^{t}CK	1,2,3,4
V_{IHL_AC}	CA input pulse amplitude	90	-	100	-	mV	7
$^{t}cIPW$	CA input pulse width	0.58		0.58	0.58	^{t}CK	5
S_{RIN_cIVW}	Input slew rate over V_{cIVW}	1	7	1	7	V/ns	6

PROPOSED

12.2. Differential Input Slew Rate Definition for Clock

To be balloted in Q4'26.

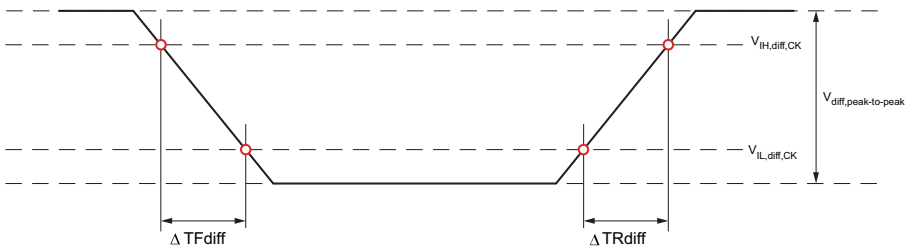


Figure 213 — Differential Input Slew Rate Definition for Clock (CK_t, CK_c)

Table 231 — Differential Input Slew Rate Definition for Clock (CK_t, CK_c)

Parameter	Measured		Defined By
	From	To	
Differential input slew rate for rising edge (CK_t - CK_c)	$V_{IL,diffCK}$	$V_{IH,diffCK}$	$\lvert(V_{IL,diffCK} - V_{IH,diffCK})\rvert$ / delta TR_{diff}
Differential input slew rate for falling edge (CK_t - CK_c)	$V_{IH,diffCK}$	$V_{IL,diffCK}$	$(V_{IH,diffCK} - V_{IL,diffCK})$ / delta TF_{diff}

Table 232 — Differential Input Slew Rate for Clock (CK_t, CK_c) — 3200-8000

Parameter	Symbol	3200-4800		5200-8800		Unit
		MIN	MAX	MIN	MAX	
Differential input slew rate for CK_t - CK_c	S_{RIdiff_CK}	2	14	2	30	V/ns

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12.3. Differential Input Clock Voltage Sensitivity

12.3.1. Differential Input Clock Voltage Sensitivity

The differential input clock voltage sensitivity test provides the methodology for testing the receiver's sensitivity to clock by varying input voltage in the absence of Inter-Symbol Interference (ISI), jitter (Rj, Dj, DCD) and crosstalk noise. This specifies the Rx voltage sensitivity requirement. The system input swing to the DRAM must be larger than the DRAM Rx at the specified BER

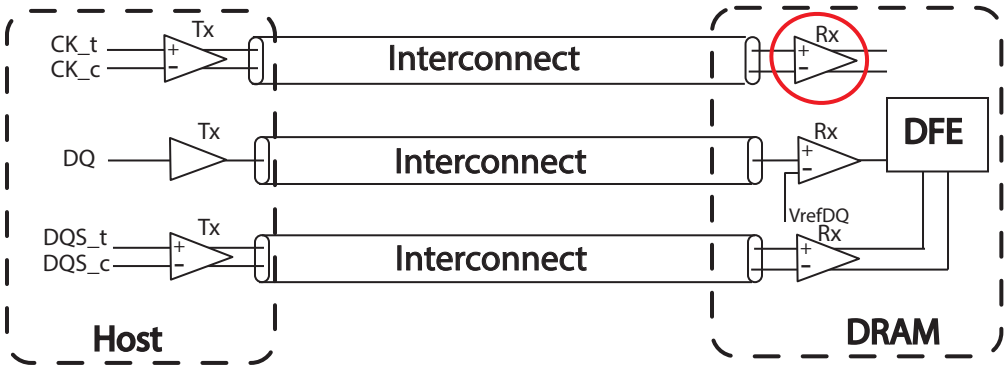


Figure 214 — Example of DDR6 Memory Interconnect

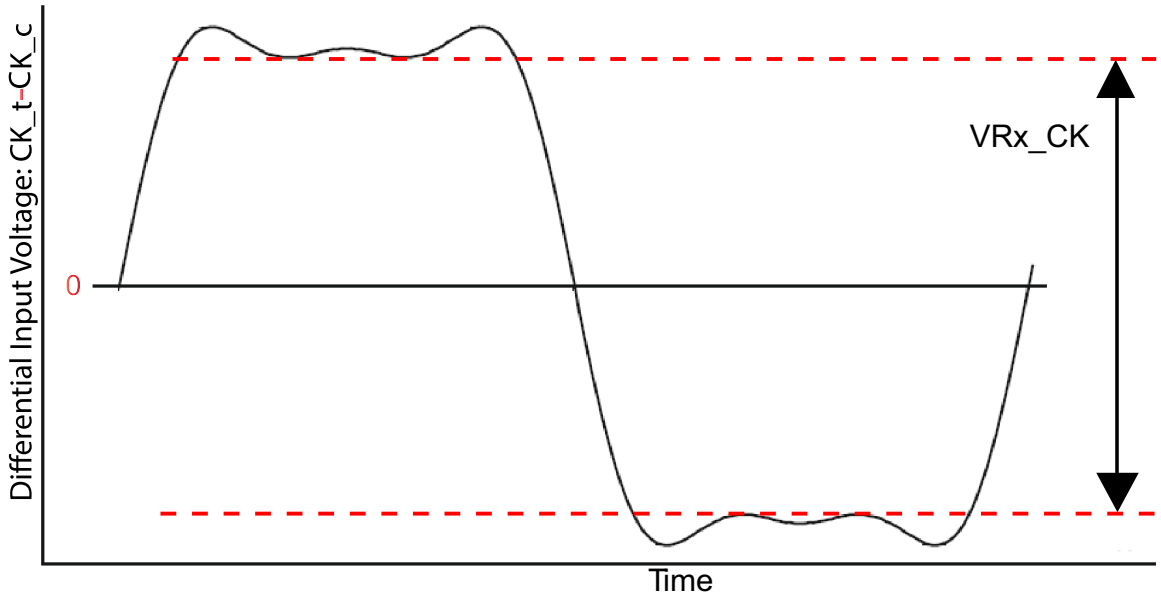


Figure 215 — VRx_CK

Table 233 — Differential Input Clock Voltage Sensitivity Parameter for DDR6 6400 to 19200

PARAMETER	SYMBOL	DDR6 6400--12800		DDR6 14400--19200		UNIT	NOTES
		MIN	MAX				
Input Clock Voltage (differential pp)	VRx_CK	100(*120)	-	-	TBD	mV	1,2,3

NOTE 1: Refer to the minimum BER requirements for DDR6
NOTE 2: The validation methodology for this parameter will be covered in future ballot(s)
NOTE 3: *indicate that is supported if the CK buffer is present on the module

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12.4. Input Clock Jitter Specification

To be balloted in Q4'26.

The clock is being driven to the device either by the RCD for L/RDIMM modules, or by the host for U/SODIMM modules.

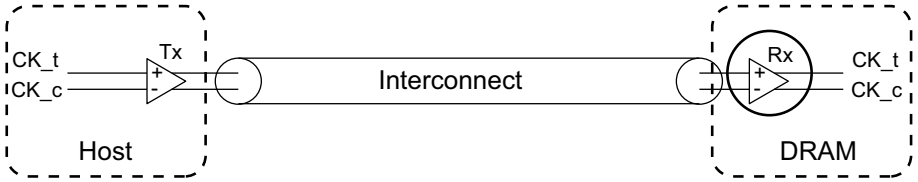


Figure 216 — HOST Driving Clock Signals to the device

12.4.1. Specifications for Input Clock Jitter

The Random Jitter (Rj) specified is a random jitter meeting a Gaussian distribution. The Deterministic Jitter (Dj) specified is bounded. Input clock violating the min/max jitter values may result in malfunction of the device.

NOTE: For the following tables: BUJ = Bounded Uncorrelated Jitter; DCD = Duty Cycle Distortion; Dj = Deterministic Jitter; Rj = Random Jitter; Tj = Total jitter; pp = Peak-to-Peak

Table 234 — Input Clock Jitter Specifications for 3200 to 4800

NOTE: f0 = Data Rate/2, example: if data rate is 3200MT/s, then f0 = 1600

NOTE: Rise and fall time slopes (V/ns) are measured between +100 mV and -100 mV of the differential output of reference clock

NOTE: On-die noise is similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of the cross-talk is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being a device component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining Tx lanes send patterns to the corresponding Rx receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specifications is met and another violated in which case the signaling analysis should be run to determine link feasibility.

NOTE: Duty Cycle Error defined as absolute difference between average value of all UI with that of average of odd UI, which in magnitude would equal absolute difference between average of all UI and average of all even UI.

NOTE: Rj RMS value of 1-UI jitter without BUJ, but on-die system-like noise present. This extraction is to be done after software correction of DCD

NOTE: Dj pp value of 1-UI jitter (after software assisted DCC). Without BUJ, but on-die system like noise present. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE: Rj RMS value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for $1 < N < 4$. This extraction is to be done after software correction of DCD

NOTE: Dj pp value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for $1 < N < 4$. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE: Rj RMS value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for $3 < N < 31$. This extraction is to be done after software correction of DCD

NOTE: Dj pp value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for $3 < N < 31$. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE: The validation methodology for these parameters will be covered in future ballots

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NOTE: If the clock meets total jitter Tj at BER of $1E^{-16}$, then meeting the individual Rj and Dj components of the spec can be considered optional. Tj is defined as Dj + 16.2*Rj for BER of $1E^{-16}$

Parameter	Symbol	3200		3600		4000		4400		4800		Unit	Notes
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Reference clock frequency	^{t}CK	0.9999 * f0	1.0001 * f0	0.9999 * f0	1.0001 * f0	0.9999 * f0	1.0001 * f0	0.9999 * f0	1.0001 * f0	0.9999 * f0	1.0001 * f0	MHz	1,11
Duty Cycle Error	$^{t}CK_Duty_UI_Error_Error$	-	0.05	-	0.05	-	0.05	-	0.05		0.05	UI	1,4,11
Rj RMS value of 1-UI Jitter	$^{t}CK_1UI_Rj__Error$	-	0.0037	-	0.0037	-	0.0037	-	0.0037	-	0.0037	UI (RMS)	3,5,11
Dj pp value of 1-UI Jitter	$^{t}CK_1UI_Dj_NoBUJ$	-	0.03	-	0.03	-	0.03	-	0.030	-	0.030	UI	3,6,11
Tj value of 1- UI Jitter	$^{t}CK_1UI_Tj_NoBUJ$	-	0.09	-	0.09	-	0.09	-	0.090	-	0.090	UI	3,6,11
Rj RMS value of N-UI Jitter, where N = 2,3	$^{t}CK_NUI_Rj_NoBUJ$,where N = 2,3	-	0.0040	-	0.0040	-	0.0040	-	0.0040	-	0.0040	UI (RMS)	3,7,11
Dj pp value of N-UI Jitter, where N = 2,3	$^{t}CK_NUI_Dj_NoBUJ$,where N = 2,3	-	0.074	-	0.074	-	0.074	-	0.074	-	0.074	UI	3,7,11
Tj value of N- UI Jitter, where N = 2,3	$^{t}CK_NUI_Tj_NoBUJ$,where N = 2,3	-	0.140	-	0.140	-	0.140	-	0.140	-	0.140	UI	3,8,11
Rj RMS value of N-UI Jitter, where N = 4,5,6,...,30	$^{t}CK_NUI_Rj_NoBUJ$,where N = 4,5,6,...,30	-	-	-	-	-	-	-	-	-	-	UI (RMS)	3,9,11,12
Dj pp value of N-UI Jitter, N = 4,5,6,...,30	$^{t}CK_NUI_Dj_NoBUJ$,where N = 4,5,6,...,30	-	-	-	-	-	-	-	-	-	-	UI	3,10,11,12

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Parameter	Symbol	3200		3600		4000		4400		4800		Unit	Notes
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Tj value of N- UI Jitter, N = 4,5,6,...,30	^{t}CK_NUI_Tj_NoBUJ ,where N = 4,5,6,...,30	-	-	-	-	-	-	-	-	-	-	UI	3,10, 11,12

Table 235 — Input Clock Jitter Specifications for 5200 to 6400

NOTE: f0 = Data Rate/2, example: if data rate is 3200MT/s, then f0 = 1600

NOTE: Rise and fall time slopes (V/ns) are measured between +100 mV and -100 mV of the differential output of reference clock

NOTE: On-die noise is similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of the cross-talk is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being a device component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining Tx lanes send patterns to the corresponding Rx receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specifications is met and another violated in which case the signaling analysis should be run to determine link feasibility.

NOTE: Duty Cycle Error defined as absolute difference between average value of all UI with that of average of odd UI, which in magnitude would equal absolute difference between average of all UI and average of all even UI.

NOTE: Rj RMS value of 1-UI jitter without BUJ, but on-die system-like noise present. This extraction is to be done after software correction of DCD

NOTE: Dj pp value of 1-UI jitter (after software assisted DCC). Without BUJ, but on-die system like noise present. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE: Rj RMS value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 1 < N < 6. This extraction is to be done after software correction of DCD

NOTE: Dj pp value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 1 < N < 6. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE: Rj RMS value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 5 < N < 31. This extraction is to be done after software correction of DCD

NOTE: Dj pp value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 5 < N < 31. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE: The validation methodology for these parameters will be covered in future ballots

NOTE: If the clock meets total jitter Tj at BER of $1E^{-16}$, then meeting the individual Rj and Dj components of the spec can be considered optional. Tj is defined as Dj + 16.2*Rj for BER of $1E^{-16}$

Parameter	Symbol	5200		5600		6000		6400		Unit	Notes
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Reference clock frequency	^{t}CK	0.9999*f0	1.0001*f0	0.9999*f0	1.0001*f0	0.9999*f0	1.0001*f0	0.9999*f0	1.0001*f0	MHz	1,11
Duty Cycle Error	^{t}CK_Duty_UI_Error	-	0.05	-	0.05	-	0.05		0.05	UI	1,4,11
Rj RMS value of 1-UI Jitter	^{t}CK_1UI_Rj_NoBUJ	-	0.0037	-	0.0037	-	0.0037		0.0037	UI (RMS)	3,5,11

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Parameter	Symbol	5200		5600		6000		6400		Unit	Notes
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Dj pp value of 1-UI Jitter	tCK_1UI_Dj_NoBUJ	-	0.030	-	0.030	-	0.030		0.030	UI	3,6,11
Tj value of 1-UI Jitter	tCK_1UI_Tj_NoBUJ	-	0.090	-	0.090	-	0.090		0.090	UI	3,6,11
Rj RMS value of N-UI Jitter, where N = 2,3,4,5	tCK_NUI_Rj_NoBUJ,where N = 2,3,4,5	-	0.0040	-	0.0040	-	0.0040	-	0.0040	UI (RMS)	3,7,11
Dj pp value of N-UI Jitter, where N = 2,3,4,5	tCK_NUI_Dj_NoBUJ,where N = 2,3,4,5	-	0.074	-	0.074	-	0.074	-	0.074	UI	3,7,11
Tj value of N-UI Jitter, where N = 2,3,4,5	tCK_NUI_Tj_NoBUJ,where N = 2,3,4,5	-	0.140	-	0.140	-	0.140	-	0.140	UI	3,8,11
Rj RMS value of N-UI Jitter, where N = 6,...,30	tCK_NUI_Rj_NoBUJ,where N = 6,...,30	-	-	-	-	-	-	-	-	UI (RMS)	3,9,11,12
Dj pp value of N-UI Jitter, N = 6,...,30	tCK_NUI_Dj_NoBUJ,where N = 6,...,30	-	-	-	-	-	-	-	-	UI	3,10,11,12
Tj value of N-UI Jitter, N = 6,...,30	tCK_NUI_Tj_NoBUJ,where N = 6,...,30	-	-	-	-	-	-	-	-	UI	3,10,11,12

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Table 236 — Input Clock Jitter Specifications for 6800 to 8800

NOTE: f0 = Data Rate/2, example: if data rate is 3200MT/s, then f0 = 1600

NOTE: Rise and fall time slopes (V/ns) are measured between +100 mV and -100 mV of the differential output of reference clock

NOTE: On-die noise is similar to that occurring with all the transmitter and receiver lanes toggling need to be stimulated. When there is no socket in transmitter measurement setup, in many cases, the contribution of the cross-talk is not significant or can be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being a device component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining Tx lanes send patterns to the corresponding Rx receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specifications is met and another violated in which case the signaling analysis should be run to determine link feasibility.

NOTE: Duty Cycle Error defined as absolute difference between average value of all UI with that of average of odd UI, which in magnitude would equal absolute difference between average of all UI and average of all even UI.

NOTE: Rj RMS value of 1-UI jitter without BUJ, but on-die system-like noise present. This extraction is to be done after software correction of DCD

NOTE: Dj pp value of 1-UI jitter (after software assisted DCC). Without BUJ, but on-die system like noise present. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE: Rj RMS value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 1 < N < 6. This extraction is to be done after software correction of DCD

NOTE: Dj pp value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 1 < N < 6. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE: Rj RMS value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 5 < N < 31. This extraction is to be done after software correction of DCD

NOTE: Dj pp value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 5 < N < 31. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD

NOTE: The validation methodology for these parameters will be covered in future ballots

NOTE: If the clock meets total jitter Tj at BER of $1E^{-16}$, then meeting the individual Rj and Dj components of the spec can be considered optional. Tj is defined as Dj + 16.2*Rj for BER of $1E^{-16}$

Parameter	Symbol	6800, 8000		7200, 8400		7600, 8800		Unit	Notes
		MIN	MAX	MIN	MAX	MIN	MAX		
Reference clock frequency	^{t}CK	0.9999 *f0	1.0001* f0	0.9999 * f0	1.0001* f0	0.9999 *f0	1.0001* f0	MHz	1,11
Duty Cycle Error	^{t}CK_Duty_UI_Error_Error	-	0.05	-	0.05	-	0.05	UI	1,4,11
Rj RMS value of 1-UI Jitter	^{t}CK_1UI_Rj__NoBUJ	-	0.003145	-	0.003145	-	0.003145	UI (RMS)	3,5,11
Dj pp value of 1-UI Jitter	^{t}CK_1UI_Dj_NoBUJ	-	0.0255	-	0.0255	-	0.0255	UI	3,6,11
Tj value of 1- UI Jitter	^{t}CK_1UI_Tj_NoBUJ	-	0.0765	-	0.0765	-	0.0765	UI	3,6,11
Rj RMS value of N-UI Jitter, where N = 2,3,4,5	^{t}CK_NUI_Rj_NoBUJ,where N = 2,3,4,5	-	0.0034	-	0.0034	-	0.0034	UI (RMS)	3,7,11
Dj pp value of N-UI Jitter, where N = 2,3,4,5	^{t}CK_NUI_Dj_NoBUJ,where N = 2,3,4,5	-	0.0629	-	0.0629	-	0.0629	UI	3,7,11
Tj value of N- UI Jitter, where N = 2,3,4,5	^{t}CK_NUI_Tj_NoBUJ,where N = 2,3,4,5	-	0.119	-	0.119	-	0.119	UI	3,8,11
Rj RMS value of N-UI Jitter, where N = 6,...,30	^{t}CK_NUI_Rj_NoBUJ,where N = 6,...,30	-	-	-	-	-	-	UI (RMS)	3,9,11, 12
Dj pp value of N-UI Jitter, N = 6,...,30	^{t}CK_NUI_Dj_NoBUJ,where N = 6,...,30	-	-	-	-	-	-	UI	3,10,11,12
Tj value of N- UI Jitter, N = 6,...,30	^{t}CK_NUI_Tj_NoBUJ,where N = 6,...,30	-	-	-	-	-	-	UI	3,10,11,12

PROPOSED

12.5. Differential Input Slew Rate for DQS_t, DQS_c

To be balloted in Q4'26.

Input slew rate for differential signals DQS_t, DQS_c are defined and measured as shown below.

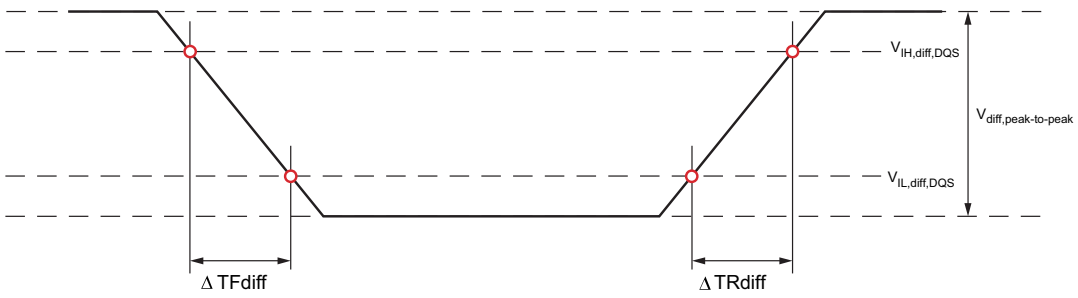


Figure 217 — Differential Input Slew Rate for DQS_t, DQS_c

Table 237 — Differential Input Slew Rate Definition for DQS_t, DQS_c

Parameter	Measured		Defined By
	From	To	
Differential input slew rate for rising edge (DQS_t - DQS_c)	$V_{IL,diff}DQS$	$V_{IH,diff}DQS$	$\lvert(V_{IL,diff}DQS - V_{IH,diff}DQS)\rvert$ / $deltaTR_{diff}$
Differential input slew rate for falling edge (DQS_t - DQS_c)	$V_{IH,diff}DQS$	$V_{IL,diff}DQS$	$(V_{IH,diff}DQS - V_{IL,diff}DQS)$ / $deltaTF_{diff}$

Table 238 — Differential Input Slew Rate for DQS_t, DQS_c — 3200-8800

NOTE: Applies only when both DQS_t and DQS_c are transitioning.

Parameter	Symbol	3200-4800		5200-8800		Unit	Note
		MIN	MAX	MIN	MAX		
Differential input slew rate for rising edge (DQS_t - DQS_c)	S_{RIdiff_DQS}	1.5	30	2.0	30	V/ns	1

PROPOSED

12.6. Rx DQS Voltage Sensitivity

To be balloted in Q4'26.

The receiver strobe input voltage sensitivity test provides the methodology for testing the receiver's sensitivity to varying input voltage in the absence of inter-symbol interference (ISI), jitter (Rj, Dj, DCD) and crosstalk noise.

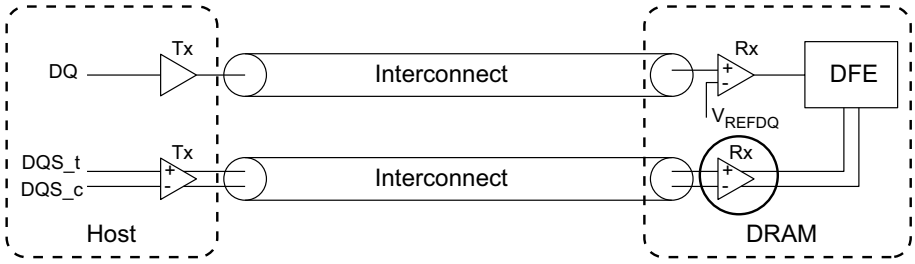


Figure 218 — Example of DDR5 Memory Interconnect

12.6.1. Receiver Strobe Voltage Sensitivity Parameters

The table below shows how input differential (DQS_t, DQS_c) V_{Rx_DQS} is defined and measured. The receiver must pass the minimum BER requirements for the device. These parameters are tested on the CTC2 card with neither additive gain nor Rx equalization set.

Table 239 — Rx DQS Input Voltage Sensitivity Parameter — 3200-4800

NOTE: Refer to the minimum BER requirements for DDR5.
NOTE: Test using clock-like pattern of repeating three 1s and three 0s

Parameter	Symbol	3200		3600		4000		4400-4800		Unit	Note
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Minimum DQS Rx input voltage sensitivity (differential)	V_{Rx_DQS}	-	130	-	115	-	105	-	100	mV	1, 2

Table 240 — Rx DQS Input Voltage Sensitivity Parameter — 5200-6400

NOTE: Refer to the minimum BER requirements for DDR5.
NOTE: Test using clock-like pattern of repeating three 1s and three 0s

Parameter	Symbol	5200-5600		6000-6400		Unit	Note
		MIN	MAX	MIN	MAX		
Minimum DQS Rx input voltage sensitivity (differential)	V_{Rx_DQS}	-	90	-	83	mV	1, 2

Table 241 — Rx DQS Input Voltage Sensitivity Parameter — 6800-7600

NOTE: Refer to the minimum BER requirements for DDR5.

PROPOSED

NOTE: Test using clock-like pattern of repeating three 1s and three 0s

Parameter	Symbol	6800-7200		7600		Unit	Note
		MIN	MAX	MIN	MAX		
Minimum DQS Rx input voltage sensitivity (differential)	V_{Rx_DQS}	-	80	-	77.5	mV	1, 2

Table 242 — Rx DQS Input Voltage Sensitivity Parameter — 8000-8800

NOTE: Refer to the minimum BER requirements for DDR5.

NOTE: Test using clock-like pattern of repeating three 1s and three 0s

Parameter	Symbol	8000		8400-800		Unit	Note
		MIN	MAX	MIN	MAX		
Minimum DQS Rx input voltage sensitivity (differential)	V_{Rx_DQS}	-	77.5	-	75	mV	1, 2

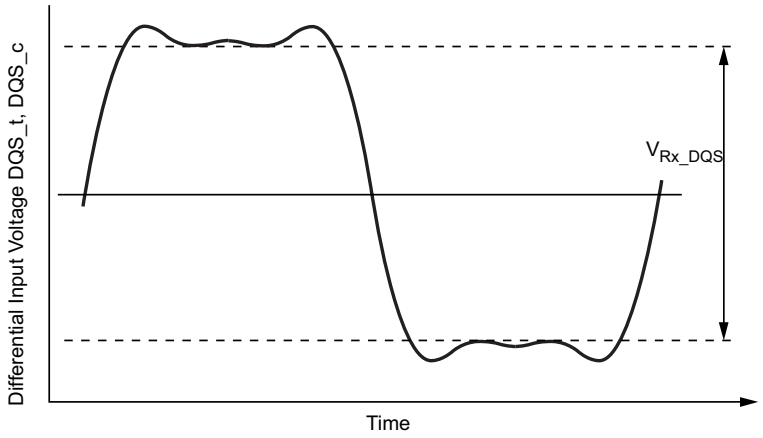


Figure 219 — V_{Rx_DQS}

PROPOSED

12.7. Rx DQS Jitter Sensitivity

To be balloted in Q4'26.

The receiver strobe jitter sensitivity test provides the methodology for testing the receiver's strobe sensitivity to an applied duty cycle distortion (DCD) and/or random jitter (Rj) at the forwarded strobe input without adding jitter, noise and ISI to the data. The receiver must pass the appropriate BER rate when no crosstalk or ISI is applied and it must pass through the combination of applied DCD and Rj.

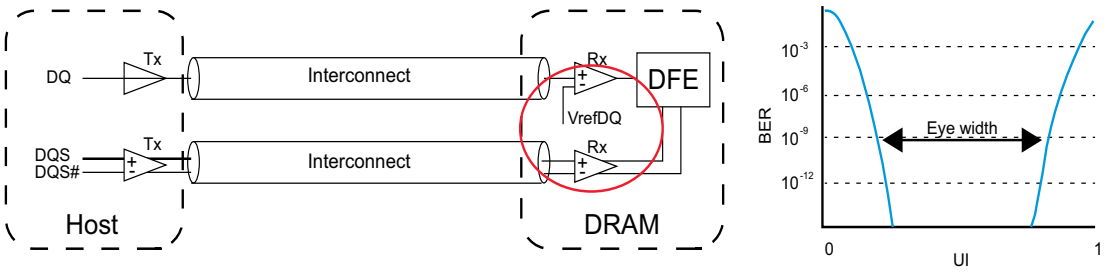


Figure 220 — Rx Forwarded Strobes for Jitter Sensitivity Testing

12.7.1. Specifications

The following tables provide Rx DQS jitter sensitivity specifications for the receivers when operating at various transfer rates. These parameters are tested on the CTC2 card without Rx equalization set. Additive DFE Gain Bias can be set.

Also note:

- BER = bit error rate; DCD = duty cycle distortion; Rj = random jitter
- Validation methodology will be defined in future ballots. 2UI is defined ast 1^{t}CK for the parameters.
- All measurements at BER=$1E^{-9}$.
- The test should be done after the DQS and DQ voltage sensitivity tests are complete and passing.
- Voltage swings and slew rates for strobe and DQ signals can be set as long as they meet specifications. The DQS and DQ input voltage swing and/or slew rate can be adjusted, without exceeding the specifications, for this test.

Table 243 — Rx DQS Jitter Sensitivity — 3200-4800

NOTE: Each of Δ^{t}Rx_DQ_tMargin_DQS_DCD, $^{\Delta t}$Rx_DQ_tMargin_DQS_Rj, and $^{\Delta t}$Rx_DQ_tMargin_DQS_DCD_Rj can be relaxed by up to 5% if ^{t}Rx_DQ_tMargin exceeds the spec by 5% or more.

NOTE: DQ timing width: timing width for any data lane using repetitive patterns (check note 3 for the pattern) measured at BER=$1E^{-9}$.

NOTE: Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD injection in forwarded strobe DQS compared to ^{t}Rx_DQ_tMargin, measured at BER=$1E^{-9}$. The magnitude of DCD is specified in Test Conditions for Rx DQS Jitter Sensitivity Testing. Test using clock-like pattern of repeating three 1s and three 0s.

NOTE: Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with only Rj injection in forwarded strobe DQS measured at BER=$1E^{-9}$, compared to ^{t}Rx_DQ_tMargin. The magnitude of Rj is specified in Test Conditions for Rx DQS Jitter Sensitivity Testing.

NOTE: Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD and Rj injection in forwarded strobe DQS measured at BER=$1E^{-9}$, compared to ^{t}Rx_DQ_tMargin. The magnitudes of DCD and Rj are specified in Test Conditions for Rx DQS Jitter Sensitivity Testing.

PROPOSED

PROPOSED

NOTE: Delay of any data lane relative to the strobe lane, as measured at the end of Tx+Channel. This parameter is a collective sum of effects of data clock mismatches in Tx and on the medium connecting Tx and Rx.
NOTE: DQ to DQ offset is skew between DQs within a nibble (x4) or a byte (x8, x16) at the device balls.
NOTE: This parameter is relative to the per-pin trained DQS2DQ value, and only applies after per-pin DQS2DQ training is complete.

Parameter	Symbol	3200		3600		4000		4400		4800		Unit	Note
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
DQ timing width	^{t}RX_DQ_tMargin	0.900	-	0.875	-	0.825	-	0.825	-	0.825	-	UI	1,2
Degradation of timing width compared to ^{t}RX_DQ_tMargin with DCD injection in DQS	Δ^{t}RX_DQ_tMargin_DQS_DCD	-	0.060	-	0.060	-	0.060	-	0.060	-	0.060	UI	3
Degradation of timing width compared to ^{t}RX_DQ_tMargin with Rj injection in DQS	Δ^{t}RX_DQ_tMargin_DQS_Rj	-	0.090	-	0.090	-	0.090	-	0.090	-	0.090	UI	4
Degradation of timing width compared to ^{t}RX_DQ_tMargin with DCD and Rj injection in DQS	Δ^{t}RX_DQ_tMargin_DQS_DCD_Rj	-	0.150	-	0.150	-	0.150	-	0.150	-	0.150	UI	1,5
Delay of any data lane relative to DQS_t/DQS_c crossing	^{t}Rx_DQS2DQ	114	937	114	833	114	750	114	738	114	729	ps	6
Delay of any data lane relative to any other data lane	^{t}Rx_DQ2DQ	-	50	-	50	-	50	-	50	-	50	ps	6,7,8

Table 244 — Rx DQS Jitter Sensitivity — 5200-6400

NOTE: Each of $^{\Delta t}$Rx_DQ_tMargin_DQS_DCD, $^{\Delta t}$Rx_DQ_tMargin_DQS_Rj, and $^{\Delta t}$Rx_DQ_tMargin_DQS_DCD_Rj can be relaxed by up to 5% if ^{t}Rx_DQ_tMargin exceeds the spec by 5% or more.

PROPOSED

NOTE: DQ timing width: timing width for any data lane using repetitive patterns (check note 3 for the pattern) measured at BER=E-9.

NOTE: Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD injection in forwarded strobe DQS compared to ^{t}Rx_DQ_tMargin, measured at BER=E-9. The magnitude of DCD is specified in Test Conditions for Rx DQS Jitter Sensitivity Testing. Test using clock-like pattern of repeating three 1s and three 0s.

NOTE: Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with only Rj injection in forwarded strobe DQS measured at BER=E-9, compared to ^{t}Rx_DQ_tMargin. The magnitude of Rj is specified in Test Conditions for Rx DQS Jitter Sensitivity Testing.

NOTE: Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD and Rj injection in forwarded strobe DQS measured at BER=E-9, compared to ^{t}Rx_DQ_tMargin. The magnitudes of DCD and Rj are specified in Test Conditions for Rx DQS Jitter Sensitivity Testing.

NOTE: Delay of any data lane relative to the strobe lane, as measured at the end of Tx+Channel. This parameter is a collective sum of effects of data clock mismatches in Tx and on the medium connecting Tx and Rx.

NOTE: DQ to DQ offset is skew between DQs within a nibble (x4) or a byte (x8, x16) at the device balls.

NOTE: This parameter is relative to the per-pin trained DQS2DQ value, and only applies after per-pin DQS2DQ training is complete.

Parameter	Symbol	5200		5600		6000		6400		Unit	Note
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
DQ timing width	^{t}RX_DQ_tMargin	0.835	-	0.835	-	0.845	-	0.845	-	UI	1,2
Degradation of timing width compared to ^{t}RX_DQ_tMargin with DCD injection in DQS	$^{\Delta t}$RX_DQ_tMargin_DQS_DCD	-	0.06	-	0.06	-	0.06	-	0.06	UI	3
Degradation of timing width compared to ^{t}RX_DQ_tMargin with Rj injection in DQS	$^{\Delta t}$RX_DQ_tMargin_DQS_Rj	-	0.09	-	0.09	-	0.09	-	0.09	UI	4
Degradation of timing width compared to ^{t}RX_DQ_tMargin with DCD and Rj injection in DQS	$^{\Delta t}$RX_DQ_tMargin_DQS_DCD_Rj	-	0.15	-	0.15	-	0.15	-	0.15	UI	1,5
Delay of any data lane relative to DQS_t/DQS_c crossing	^{t}Rx_DQS2DQ	114	721	114	714	114	708	114	703	ps	6

Parameter	Symbol	5200		5600		6000		6400		Unit	Note
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Delay of any data lane relative to any other data lane	^{t}Rx_DQ2DQ	-	50	-	50	-	50	-	50	ps	6,7,8

Table 245 — Rx DQS Jitter Sensitivity — 6800-7600

NOTE: Each of $^{\Delta t}$Rx_DQ_tMargin_DQS_DCD, $^{\Delta t}$Rx_DQ_tMargin_DQS_Rj, and $^{\Delta t}$Rx_DQ_tMargin_DQS_DCD_Rj can be relaxed by up to 5% if ^{t}Rx_DQ_tMargin exceeds the spec by 5% or more.

NOTE: DQ timing width: timing width for any data lane using repetitive patterns (check note 3 for the pattern) measured at BER=E-9.

NOTE: Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD injection in forwarded strobe DQS compared to ^{t}Rx_DQ_tMargin, measured at BER=E-9. The magnitude of DCD is specified in Test Conditions for Rx DQS Jitter Sensitivity Testing. Test using clock-like pattern of repeating three 1s and three 0s.

NOTE: Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with only Rj injection in forwarded strobe DQS measured at BER=E-9, compared to ^{t}Rx_DQ_tMargin. The magnitude of Rj is specified in Test Conditions for Rx DQS Jitter Sensitivity Testing.

NOTE: Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD and Rj injection in forwarded strobe DQS measured at BER=E-9, compared to ^{t}Rx_DQ_tMargin. The magnitudes of DCD and Rj are specified in Test Conditions for Rx DQS Jitter Sensitivity Testing.

NOTE: Delay of any data lane relative to the strobe lane, as measured at the end of Tx+Channel. This parameter is a collective sum of effects of data clock mismatches in Tx and on the medium connecting Tx and Rx.

NOTE: DQ to DQ offset is skew between DQs within a nibble (x4) or a byte (x8, x16) at the device balls.

NOTE: This parameter is relative to the per-pin trained DQS2DQ value, and only applies after per-pin DQS2DQ training is complete.

Parameter	Symbol	6800		7200		7600		Unit	Note
		MIN	MAX	MIN	MAX	MIN	MAX		
DQ timing width	^{t}RX_DQ_tMargin	0.845	-	0.845	-	0.845	-	UI	1,2
Degradation of timing width compared to ^{t}RX_DQ_tMargin with DCD injection in DQS	$^{\Delta t}$RX_DQ_tMargin_DQS_DCD	-	0.06	-	0.06	-	0.06	UI	3
Degradation of timing width compared to ^{t}RX_DQ_tMargin with Rj injection in DQS	$^{\Delta t}$RX_DQ_tMargin_DQS_Rj	-	0.09	-	0.09	-	0.09	UI	4
Degradation of timing width compared to ^{t}RX_DQ_tMargin with DCD and Rj injection in DQS	$^{\Delta t}$RX_DQ_^{t}Margin_DQS_DCD_Rj	-	0.15	-	0.15	-	0.15	UI	1,5

PROPOSED

Parameter	Symbol	6800		7200		7600		Unit	Note
		MIN	MAX	MIN	MAX	MIN	MAX		
Delay of any data lane relative to DQS_t/DQS_c crossing	^{t}Rx_DQS2DQ	114	703	114	703	114	703	ps	6
Delay of any data lane relative to any other data lane	^{t}Rx_DQ2DQ	-	50	-	50	-	50	ps	6,7,8

Table 246 — Rx DQS Input Voltage Sensitivity Parameter — 8000-8800

NOTE: Each of Δ^{t}Rx_DQ_tMargin_DQS_DCD, Δ^{t}Rx_DQ_tMargin_DQS_Rj, and Δ^{t}Rx_DQ_tMargin_DQS_DCD_Rj can be relaxed by up to 5% if ^{t}Rx_DQ_tMargin exceeds the spec by 5% or more.

NOTE: DQ timing width: timing width for any data lane using repetitive patterns (check note 3 for the pattern) measured at BER=E-9.

NOTE: Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD injection in forwarded strobe DQS compared to ^{t}Rx_DQ_tMargin, measured at BER=E-9. The magnitude of DCD is specified in Test Conditions for Rx DQS Jitter Sensitivity Testing. Test using clock-like pattern of repeating three 1s and three 0s.

NOTE: Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with only Rj injection in forwarded strobe DQS measured at BER=E-9, compared to ^{t}Rx_DQ_tMargin. The magnitude of Rj is specified in Test Conditions for Rx DQS Jitter Sensitivity Testing.

NOTE: Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD and Rj injection in forwarded strobe DQS measured at BER=E-9, compared to ^{t}Rx_DQ_tMargin. The magnitudes of DCD and Rj are specified in Test Conditions for Rx DQS Jitter Sensitivity Testing.

NOTE: Delay of any data lane relative to the strobe lane, as measured at the end of Tx+Channel. This parameter is a collective sum of effects of data clock mismatches in Tx and on the medium connecting Tx and Rx.

NOTE: DQ to DQ offset is skew between DQs within a nibble (x4) or a byte (x8, x16) at the device balls.

NOTE: This parameter is relative to the per-pin trained DQS2DQ value, and only applies after per-pin DQS2DQ training is complete.

Parameter	Symbol	8000		8400		8800		Unit	Note
		MIN	MAX	MIN	MAX	MIN	MAX		
DQ timing width	^{t}RX_DQ_tMargin	0.845	-	0.845	-	0.845	-	UI	1,2
Degradation of timing width compared to ^{t}RX_DQ_tMargin with DCD injection in DQS	$^{\Delta t}$RX_DQ_tMargin_DQS_DCD	-	0.06	-	0.06	-	0.06	UI	3
Degradation of timing width compared to ^{t}RX_DQ_tMargin with Rj injection in DQS	$^{\Delta t}$RX_DQ_tMargin_DQS_Rj	-	0.09	-	0.09	-	0.09	UI	4

PROPOSED

Parameter	Symbol	8000		8400		8800		Unit	Note
		MIN	MAX	MIN	MAX	MIN	MAX		
Degradation of timing width compared to ^{t}RX_DQ_tMargin with DCD and Rj injection in DQS	$^{\Delta t}$RX_DQ_tMargin_DQS_DCD_Rj	-	0.15	-	0.15	-	0.15	UI	1,5
Delay of any data lane relative to DQS_t/DQS_c crossing	^{t}Rx_DQS2DQ	114	703	114	703	114	703	ps	6
Delay of any data lane relative to any other data lane	^{t}Rx_DQ2DQ	-	50	-	50	-	50	ps	6,7,8

12.7.2. Test Conditions

The following table lists the amount of duty cycle distortion (DCD) and/or random jitter (Rj) that must be applied to the forwarded strobe when measuring the Rx strobe jitter sensitivity parameters specified in the previous tables.

Table 247 — Test Conditions for Rx DQS Jitter Sensitivity Testing — 3200-8800

NOTE: While imposing these specifications, the strobe lane is stressed, but the data input is kept large amplitude and no jitter or ISI injection. The specified voltages are at the Rx input pin. The DQS and DQ input voltage swing and/or slew rate can be adjusted, without exceeding the specifications, for this test.

NOTE: The jitter response of the forwarded strobe channel depends on the input voltage, primarily due to bandwidth limitations of the clock receiver. For this revision, no separate specification of jitter as a function of input amplitude is specified; instead, the response characterization is done at the specified clock amplitude only. The specified voltages are at the Rx input pin.

NOTE: Various DCD values should be tested, complying within the maximum limits.

NOTE: Various Rj values should be tested, complying within the maximum limits.

NOTE: Various combinations of DCD and Rj values should be tested, complying within the maximum limits. The maximum timing margin degradation as a result of the injected jitter is specified in a separate table.

NOTE: Although the device has bursty traffic, current available BERTs that can be used for this test do not support burst traffic patterns. A continuous strobe and continuous DQ are used for this parameter. The clock-like pattern repeating three 1s and three 0s is used for this test.

NOTE: Duty cycle distortion (in UI DCD) as applied to the input forwarded DQS from BERT (UI).

NOTE: RMS value of Rj (specified as edge jitter) applied to the input forwarded DQS from BERT (values of the edge jitter RMS values specified as % of UI).

NOTE: Duty cycle distortion (specified as UI DCD) and rms values of Rj (specified as edge jitter) applied to the input forwarded DQS from BERT (values of the edge jitter RMS values specified as % of UI).

NOTE: Voltage swing and slew rates can be set for strobe and DQ signals as long as they meet the specification. The DQS and DQ input voltage swing and/or slew rate can be adjusted, without exceeding the specifications, for this test.

Parameter	Symbol	3200-8800		Unit	Note
		MIN	MAX		
Applied DCD to the DQS	^{t}RX_DQS_DCD	-	0.045	UI	1,2,3,6,7,10
Applied to Rj RMS to the DQS	^{t}RX_DQS_Rj	-	0.0075	UI (RMS)	1,2,4,6,8,10

PROPOSED

Parameter	Symbol	3200-8800		Unit	Note
		MIN	MAX		
Applied DCD and Rj RMS to the DQS	^{t}RX_DQS_DCD_Rj	-	0.045 UI DCD + 0.0075 UI Rj RMS	UI	1,2,5,6,7,9,10

PROPOSED

12.8. Rx DQ Voltage Sensitivity

To be balloted in Q4'26.

The receiver data input voltage sensitivity test provides the methodology for testing the receiver's sensitivity to varying input voltage in the absence of inter-symbol interference (ISI), jitter (Rj, Dj, DCD) and crosstalk noise.

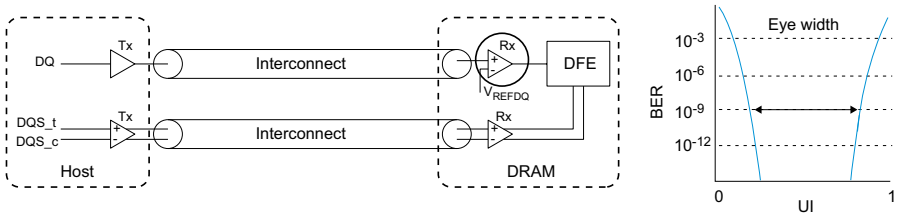


Figure 221 — Example of DDR5 Memory Interconnect

12.8.1. Receiver DQ Input Voltage Sensitivity Parameters

Input single-ended V_{Rx_DQ} is defined and measured as shown below. The receiver must pass the minimum BER requirements for DDR5. These parameters are tested on the CTC2 card with neither additive gain nor Rx Equalization set.

Table 248 — Rx DQ Input Voltage Sensitivity Parameter for DDR5 3200-4800

NOTE: Refer to the minimum BER requirements for DDR5.
NOTE: The validation methodology for this parameter will be covered in future ballot(s).
NOTE: Test using clock-like pattern of repeating three 1s and three 0s

Parameter	Symbol	3200		3600		4000		4400-4800		Unit	Note
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Minimum DQ Rx input voltage sensitivity applied around V_{REF}	V_{Rx_DQ}	-	85	-	75	-	70	-	65	mV	1, 2, 3

Table 249 — Rx DQ Input Voltage Sensitivity Parameter for DDR5 5200-6400

NOTE: Refer to the minimum BER requirements for DDR5.
NOTE: The validation methodology for this parameter will be covered in future ballot(s).
NOTE: Test using clock-like pattern of repeating three 1s and three 0s

Parameter	Symbol	5200-5600		6000-6400		Unit	Note
		MIN	MAX	MIN	MAX		
Minimum DQ Rx input voltage sensitivity applied around V_{REF}	V_{Rx_DQ}	-	60	-	55	mV	1, 2, 3

Table 250 — Rx DQ Input Voltage Sensitivity Parameter for DDR5 6800-7600

NOTE: Refer to the minimum BER requirements for DDR5.

PROPOSED

NOTE: The validation methodology for this parameter will be covered in future ballot(s).
NOTE: Test using clock-like pattern of repeating three 1s and three 0s

Parameter	Symbol	6800-7600		Unit	Note
		MIN	MAX		
Minimum DQ Rx input voltage sensitivity applied around V_{REF}	V_{Rx_DQ}	-	50	mV	1, 2, 3

Table 251 — Rx DQ Input Voltage Sensitivity Parameter for DDR5 8000-8800

NOTE: Refer to the minimum BER requirements for DDR5.
NOTE: The validation methodology for this parameter will be covered in future ballot(s).
NOTE: Test using clock-like pattern of repeating three 1s and three 0s

Parameter	Symbol	8000		8400-8800		Unit	Note
		MIN	MAX	MIN	MAX		
Minimum DQ Rx input voltage sensitivity applied around V_{REF}	V_{Rx_DQ}	-	50	-	47.5	mV	1, 2, 3

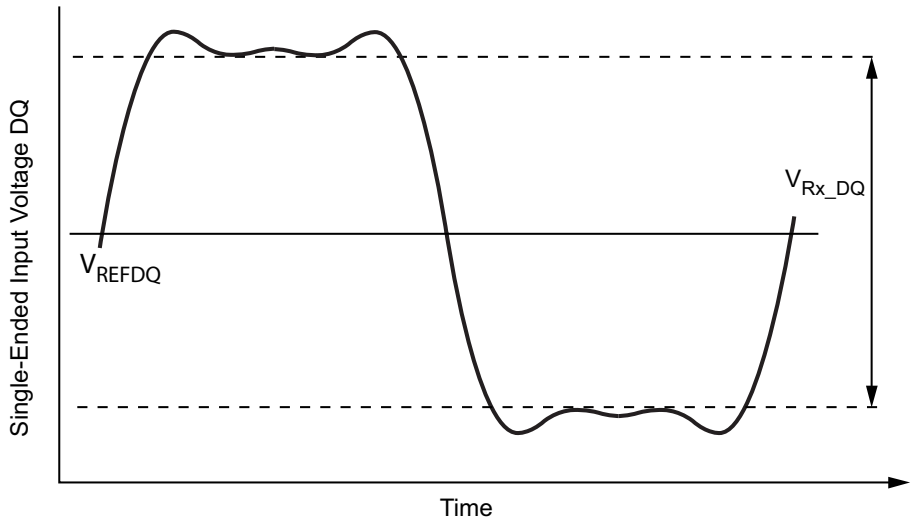


Figure 222 — V_{Rx_DQ}

PROPOSED

12.9. Rx Stressed Eye

To be balloted in Q4'26.

The stressed eye tests provide a methodology for creating the appropriate stress for the device's receiver with the combination of ISI (both loss and reflective), jitter (Rj, Dj, DCD), and crosstalk noise. The receiver must pass the appropriate BER rate when the equivalent stressed eye is applied through the combination of ISI, jitter and crosstalk.

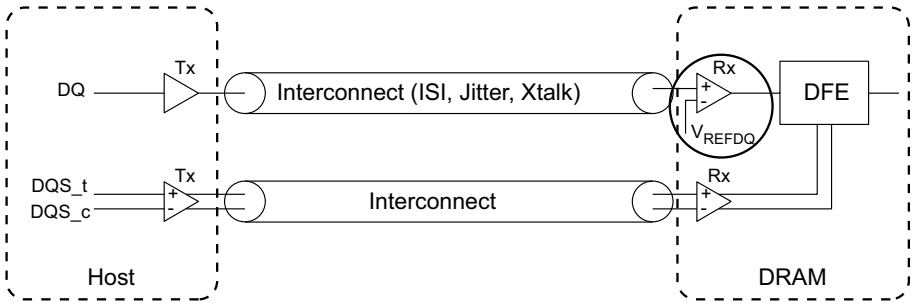


Figure 223 — Example: RX Stressed Test Setup in the Presence of ISI, Jitter and Crosstalk

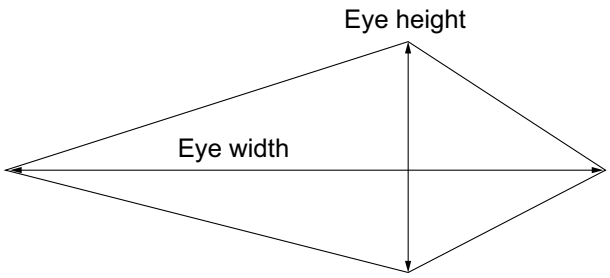


Figure 224 — Example of Rx Stressed Eye Height and Eye Width

12.9.1. Parameters

In the following tables, BER = bit error rate, DCD = duty cycle distortion, Rj = random jitter, Sj = sinusoidal jitter, p-p = peak to peak.

Table 252 — Test Conditions for Rx Stressed Eye Tests for DDR5 3200-4800

NOTE: Must meet minimum BER of IE^{-16} or better requirements with the stressed eye at the slicer of the receiver (after equalization is applied in the summer). The eye shape is verified by measuring to BER $1E^{-9}$ and extrapolating to BER $1E^{-16}$.
NOTE: These parameters are applied on the defined golden reference channel with parameters TBD.
NOTE: DFE tap 1-4 bias settings that give the best eye margin are used and referr to the values in the DFE Tap Coefficients MIN/MAX Ranges table. DFE tap range limits apply: sum of absolute values of Tap-2, Tap-3, and Tap-4 should be less than 60mV (|Tap-2| + |Tap-3| + |Tap-4| < 60mV) after the tap multiplier is applied.
NOTE: Evaluated with no DC supply voltage drift.
NOTE: Evaluated with no temperature drift.
NOTE: Supply voltage noise limited according to DC bandwidth specifications (see DC Operating Conditions).
NOTE: The stressed eye is assumed to have a diamond shape.

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NOTE: For this test, the following can be adjusted as needed without exceeding the specifications, including the limits described in note 1: V_{REFDQ}, DFE gain bias step, and DFE Taps 1, 2, 3, 4 bias step.

NOTE: The stressed eye is defined as centered on the DQS_t/DQS_c crossing during the calibration. Measurement includes an optimal set of DQS_t/DQS_c location, V_{REFDQ}, and DFE solution to give the best eye margin.

NOTE: The Rx stressed eye specification applies at 2933 and faster data rates.

NOTE: EH/EW are measured at the slicer of the receiver.

Parameter	Symbol	3200		3600		4000		4400		4800		Unit	Note
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX		
Eye height of stressed eye for golden reference channel 1	RxEH_Stressed_Eye_Golden_Ref_Channel_1	95	-	85	-	80	-	75	-	70	-	mV	1,2,3,4,5,6,7,8,9
Eye width of stressed eye golden reference channel 1	RxEW_Stressed_Eye_Golden_Ref_Channel_1	0.25	-	0.25	-	0.25	-	0.25	-	0.25	-	UI	1,2,3,4,5,6,7,8,9
Vswing stress to meet above data eye	Vswing_Stressed_Eye_Golden_Ref_Channel_1	-	600	-	600	-	600	-	600	-	600	mV	1,2
Injected sinusoidal jitter at 200 MHz to meet above data eye	Sj_Stressed_Eye_Golden_Ref_Channel_1	0	0.45	0	0.45	0	0.45	0	0.45	0	0.45	UI p-p	1,2
Injected random wide band (10 MHz-1 GHz) jitter to meet above data eye	Rj_Stressed_Eye_Golden_Ref_Channel_1	0	0.04	0	0.04	0	0.04	0	0.04	0	0.04	UI RMS	1,2
Injected voltage noise as PRBS23, or injected voltage noise at 2.1 GHz	Vnoise_Stressed_Eye_Golden_Ref_Channel_1	0	125	0	125	0	125	0	125	0	125	mV p-p	1,2
Golden reference channel 1 characteristics as measured at TBD	Golden_Ref_Channel_1_Characteristics	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	3

PROPOSED

Table 253 — Test Conditions for Rx Stressed Eye Tests for DDR5 5200-6400

NOTE: Must meet minimum BER of IE^{-16} or better requirements with the stressed eye at the slicer of the receiver (after equalization is applied in the summer). The eye shape is verified by measuring to BER $1E^{-9}$ and extrapolating to BER $1E^{-16}$.

NOTE: These parameters are applied on the defined golden reference channel with parameters TBD.

NOTE: DFE tap 1-4 bias settings that give the best eye margin are used and refer to the values in the DFE Tap Coefficients MIN/MAX Ranges table. DFE tap range limits apply: sum of absolute values of Tap-2, Tap-3, and Tap-4 must be less than 60mV (|Tap-2| + |Tap-3| + |Tap-4| < 60mV) after the tap multiplier is applied.

NOTE: Evaluated with no DC supply voltage drift.

NOTE: Evaluated with no temperature drift.

NOTE: Supply voltage noise limited according to DC bandwidth specifications (see DC Operating Conditions).

NOTE: The stressed eye is assumed to have a diamond shape.

NOTE: For this test, the following can be adjusted as needed without exceeding the specifications, including the limits described in note 1: V_{REFDQ}, DFE gain bias step, and DFE Taps 1, 2, 3, 4 bias step.

NOTE: The stressed eye is defined as centered on the DQS_t/DQS_c crossing during the calibration. Measurement includes an optimal set of DQS_t/DQS_c location, V_{REFDQ}, and DFE solution to give the best eye margin.

NOTE: The Rx stressed eye specification applies at 2933 and faster data rates.

NOTE: EH/EW are measured at the slicer of the receiver.

Parameter	Symbol	5200		5600		6000-6400		Unit	Note
		MIN	MAX	MIN	MAX	MIN	MAX		
Eye height of stressed eye for golden reference channel 1	RxEH_Stressed_Eye_Golden_Ref_Channel_1	65	-	60	-	57.5	-	mV	1,2,3,4,5,6,7,8,9
Eye width of stressed eye golden reference channel 1	RxEW_Stressed_Eye_Golden_Ref_Channel_1	0.235	-	0.235	-	0.230	-	UI	1,2,3,4,5,6,7,8,9
Vswing stress to meet above data eye	Vswing_Stressed_Eye_Golden_Ref_Channel_1	-	600	-	600	-	600	mV	1,2
Injected sinusoidal jitter at 200 MHz to meet above data eye	Sj_Stressed_Eye_Golden_Ref_Channel_1	-	0.45	-	0.45	-	0.45	UI p-p	1,2
Injected random wide band (10 MHz-1 GHz) jitter to meet above data eye	Rj_Stressed_Eye_Golden_Ref_Channel_1	-	0.04	-	0.04	-	0.04	UI RMS	1,2
Injected voltage noise as PRBS23, or injected voltage noise at 2.1 GHz	Vnoise_Stressed_Eye_Golden_Ref_Channel_1	-	125	-	125	-	125	mV p-p	1,2
Golden reference channel 1 characteristics as measured at TBD	Golden_Ref_Channel_1_Characteristics	TBD	TBD	TBD	TBD	TBD	TBD	TBD	3

PROPOSED

Table 254 — Test Conditions for Rx Stressed Eye Tests for DDR5 6800-7600

NOTE: Must meet minimum BER of IE^{-16} or better requirements with the stressed eye at the slicer of the receiver (after equalization is applied in the summer). The eye shape is verified by measuring to BER $1E^{-9}$ and extrapolating to BER $1E^{-16}$.

NOTE: These parameters are applied on the defined golden reference channel with parameters TBD.

NOTE: DFE tap 1-4 bias settings that give the best eye margin are used and refer to the values in the DFE Tap Coefficients MIN/MAX Ranges table. DFE tap range limits apply: sum of absolute values of Tap-2, Tap-3, and Tap-4 must be less than 60mV (|Tap-2| + |Tap-3| + |Tap-4| < 60mV) after the tap multiplier is applied.

NOTE: Evaluated with no DC supply voltage drift.

NOTE: Evaluated with no temperature drift.

NOTE: Supply voltage noise limited according to DC bandwidth specifications (see DC Operating Conditions).

NOTE: The stressed eye is assumed to have a diamond shape.

NOTE: For this test, the following can be adjusted as needed without exceeding the specifications, including the limits described in note 1: V_{REFDQ}, DFE gain bias step, and DFE Taps 1, 2, 3, 4 bias step.

NOTE: The stressed eye is defined as centered on the DQS_t/DQS_c crossing during the calibration. Measurement includes an optimal set of DQS_t/DQS_c location, V_{REFDQ} and DFE solution to give the best eye margin..

NOTE: The Rx stressed eye specification applies at 2933 and faster data rates.

NOTE: EH/EW are measured at the slicer of the receiver.

Parameter	Symbol	6800		7200		7600		Unit	Note
		MIN	MAX	MIN	MAX	MIN	MAX		
Eye height of stressed eye for golden reference channel 1	RxEH_Stressed_Eye_Golden_Ref_Channel_1	55	-	55	-	54	-	mV	1,2,3,4,5,6,7,8,9
Eye width of stressed eye golden reference channel 1	RxEW_Stressed_Eye_Golden_Ref_Channel_1	0.23	-	0.23	-	0.23	-	UI	2,3,4,5,6,7,8
Vswing stress to meet above data eye	Vswing_Stressed_Eye_Golden_Ref_Channel_1	-	600	-	600	-	600	mV	1,2
Injected sinusoidal jitter at 200 MHz to meet above data eye	Sj_Stressed_Eye_Golden_Ref_Channel_1	0	0.45	0	0.45	0	0.45	UI p-p	1,2
Injected random wide band (10 MHz-1 GHz) jitter to meet above data eye	Rj_Stressed_Eye_Golden_Ref_Channel_1	0	0.45	0	0.45	0	0.04	UI RMS	1,2
Injected voltage noise as PRBS23, or injected voltage noise at 2.1 GHz	Vnoise_Stressed_Eye_Golden_Ref_Channel_1	0	125	0	125	0	125	mV p-p	1,2
Golden reference channel 1 characteristics as measured at TBD	Golden_Ref_Channel_1_Characteristics	TBD	TBD	TBD	TBD	TBD	TBD	TBD	3

PROPOSED

Table 255 — Test Conditions for Rx Stressed Eye Tests for DDR5 8000-8800

NOTE: Must meet minimum BER of IE^{-16} or better requirements with the stressed eye at the slicer of the receiver (after equalization is applied in the summer). The eye shape is verified by measuring to BER $1E^{-9}$ and extrapolating to BER $1E^{-16}$.

NOTE: These parameters are applied on the defined golden reference channel with parameters TBD.

NOTE: DFE tap 1-4 bias settings that give the best eye margin are used and refer to the values in the DFE Tap Coefficients MIN/MAX Ranges table. DFE tap range limits apply: sum of absolute values of Tap-2, Tap-3, and Tap-4 must be less than 60mV (|Tap-2| + |Tap-3| + |Tap-4| < 60mV) after the tap multiplier is applied.

NOTE: Evaluated with no DC supply voltage drift.

NOTE: Evaluated with no temperature drift.

NOTE: Supply voltage noise limited according to DC bandwidth specifications (see DC Operating Conditions).

NOTE: The stressed eye is assumed to have a diamond shape.

NOTE: For this test, the following can be adjusted as needed without exceeding the specifications, including the limits described in note 1: V_{REFDQ}, DFE gain bias step, and DFE Taps 1, 2, 3, 4 bias step.

NOTE: The stressed eye is defined as centered on the DQS_t/DQS_c crossing during the calibration. Measurement includes an optimal set of DQS_t/DQS_c location, V_{REFDQ} and DFE solution to give the best eye margin..

NOTE: The Rx stressed eye specification applies at 2933 and faster data rates.

NOTE: EH/EW are measured at the slicer of the receiver.

Parameter	Symbol	8000		8400		8800		Unit	Note
		MIN	MAX	MIN	MAX	MIN	MAX		
Eye height of stressed eye for golden reference channel 1	RxEH_Stressed_Eye_Golden_Ref_Channel_1	54	-	53	-	53	-	mV	1,2,3,4,5,6,7,8,9
Eye width of stressed eye golden reference channel 1	RxEW_Stressed_Eye_Golden_Ref_Channel_1	0.23	-	0.23	-	0.23	-	UI	2,3,4,5,6,7,8
Vswing stress to meet above data eye	Vswing_Stressed_Eye_Golden_Ref_Channel_1	-	600	-	600	-	600	mV	1,2
Injected sinusoidal jitter at 200 MHz to meet above data eye	Sj_Stressed_Eye_Golden_Ref_Channel_1	0	0.45	0	0.45	0	0.45	UI p-p	1,2
Injected random wide band (10 MHz-1 GHz) jitter to meet above data eye	Rj_Stressed_Eye_Golden_Ref_Channel_1	0	0.04	0	0.04	0	0.04	UI RMS	1,2
Injected voltage noise as PRBS23, or injected voltage noise at 2.1 GHz	Vnoise_Stressed_Eye_Golden_Ref_Channel_1	0	125	0	125	0	125	mV p-p	1,2
Golden reference channel 1 characteristics as measured at TBD	Golden_Ref_Channel_1_Characteristics	TBD	TBD	TBD	TBD	TBD	TBD	TBD	3

PROPOSED

13. AC and DC Output Levels and Timing

13.1. DDR6 Ron Pullup/Pulldown DC Characteristic

DDR6 Ron Pullup/Pulldown DC characteristic table as follows shall be based on ZQ=120ohm and ZQ Calibrated at 0.5*VDDQ Voltage.

Table 256 — Pulldown/Pullup Driver DC Characteristic

ZQ=120ohm, entire operation temperature range, after proper ZQ calibration

Ronpd,nom	Vout	Min	Nom	Max	Unit	Note
Ron34pd	Vout=0.5*VDDQ	0.9	1	1.1	RZQ/3.5	1,2,3
Ron34pu	Vout=0.5*VDDQ	0.7	1	1.3	RZQ/3.5	1,2,3
Ron40pd	Vout=0.5*VDDQ	0.9	1	1.1	RZQ/3	1,2,3
Ron40pu	Vout=0.5*VDDQ	0.7	1	1.3	RZQ/3	1,2,3
Ron48pd	Vout=0.5*VDDQ	0.9	1	1.1	RZQ/2.5	1,2,3
Ron48pu	Vout=0.5*VDDQ	0.7	1	1.3	RZQ/2.5	1,2,3
Ron60pd	Vout=0.5*VDDQ	0.9	1	1.1	RZQ/2	1,2,3
Ron60pu	Vout=0.5*VDDQ	0.7	1	1.3	RZQ/2	1,2,3
Ron80pd	Vout=0.5*VDDQ	0.9	1	1.1	RZQ/1.5	1,2,3
Ron80pu	Vout=0.5*VDDQ	0.7	1	1.3	RZQ/1.5	1,2,3
Ron120pd	Vout=0.5*VDDQ	0.9	1	1.1	RZQ/1	1,2,3
Ron120pu	Vout=0.5*VDDQ	0.7	1	1.3	RZQ/1	1,2,3
Ron240pd	Vout=0.5*VDDQ	0.9	1	1.1	RZQ/0.5	1,2,3
Ron240pu	Vout=0.5*VDDQ	0.7	1	1.3	RZQ/0.5	1,2,3
Mismatch DQ-DQ within byte variation pull-up, MMPudd	Vout=0.5*VDDQ			10	%	1,2,3,4
Mismatch DQ-DQ within byte variation pull-dn, MMPddd	Vout=0.5*VDDQ			10	%	1,2,3,4

NOTE 1: The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see “ZQ Calibration Commands” section
NOTE 2: Pull-up and pull-dn output driver impedances are required to be calibrated at 0.5 * VDDQ
NOTE 3: The tolerance limits are specified under the typ voltage condition that VDDQ=0.6v, VDD=1.025v
NOTE 4: RON variance range ratio to RON Nominal value in a given component, including DQS_t and DQS_c

$$MM_{Pu,dd} = \frac{R_{ON,Pu,Max} - R_{ON,Pu,Min}}{R_{ON,NOM}} \times 100$$

$$MM_{Pd,dd} = \frac{R_{ON,Pd,Max} - R_{ON,Pd,Min}}{R_{ON,NOM}} \times 100$$

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13.2. Alert Output

13.2.1. Alert Signal

Alert is an output pin which signals fault status defined.

13.2.2. Alert Output

The Alert output shall be inactive when the DRAM has not detected a fault condition. The Alert output will be Hi-Z when inactive and the Host shall provide termination to VSS. The Alert output shall be active when the DRAM has detected a covered fault condition. The Alert output will drive high, to VDDQ, when active and the Host shall provide termination to VSS.

The DRAM pull-up driver strength is fixed at RZQ/3.5. The Host shall set the pull-down impedance(s), i.e., External Pulldown, to derive the desired VOH Level.

13.2.3. Alert Pull-Up Driver Characteristics

Alert output pin driver characteristics are defined by the following table.

Table 257 — Alert Output Driver Pull-up Impedance – RonPU_Alert

RonPU_Alert	Vout	Min	Nom	Max
34 Ohm	0.5 * VDDQ	0.2	1	1.3

NOTE 1: All values are without ZQ calibration. {Fixed ZQcode applied to avoid hazard on Alert pin.} **NOTE 2:** Vout assumes typical VDDQ and VDD; external pulldown with matching impedance.
NOTE 3: VOH level is dependent on Host on pull-down resistance, i.e., External Pulldown. The Signal integrity of the channel between the DRAM and the Host should be confirmed during simulation and design effort.

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13.2.4. DRAM Alert Output Driver with Alert Equivalent Circuit

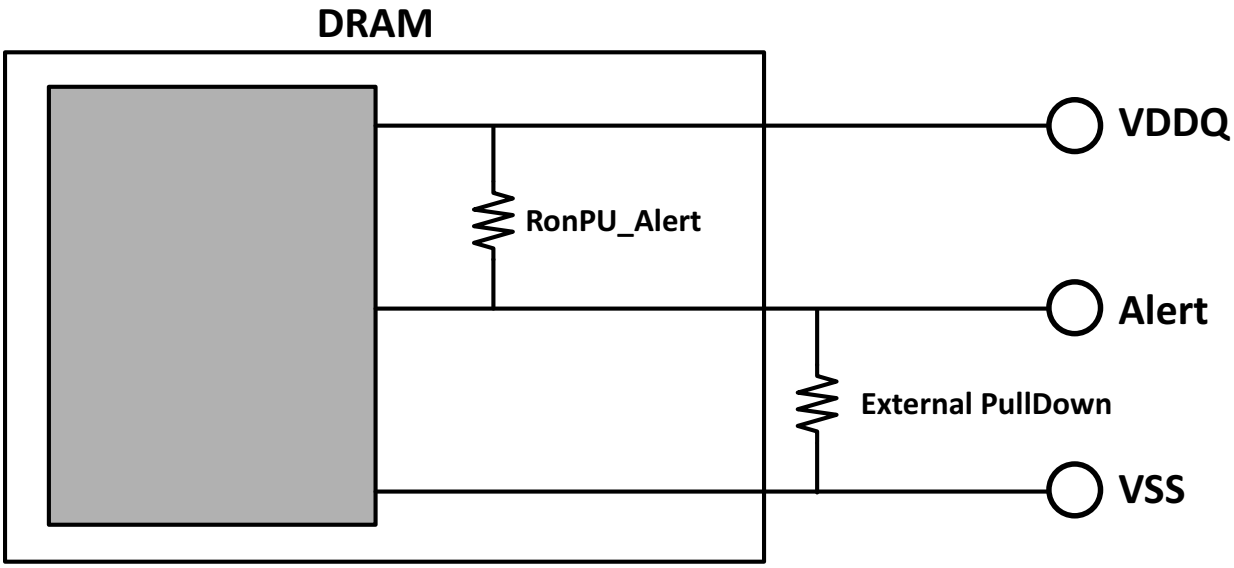


Figure 225 — Alert Output Driver with Alert Equivalent Circuit

13.2.5. VOH calculations

Alert VOH at Host can be determined by:

- VOH = (VDDQ * External Pulldown) / (RonPU_Alert + External Pulldown)

PROPOSED

13.3. Output Slew Rate

13.3.1. Single-Ended Output Levels - V_{OL}/V_{OH}

Table 258 — Single-Ended Output Levels

Parameter	Symbol	Value	Unit	Notes
Single-ended high measurement level for output slew rate	V_{OH}	0.75 x V_{pk-pk}	V	1
Single-ended low measurement level for output slew rate	V_{OL}	0.25 x V_{pk-pk}	V	1
NOTE: V_{pk-pk} is the mean high voltage minus the mean low voltage over 8UI samples.				

13.3.2. Single-Ended Output Slew Rate

With the reference load for timing measurements, output slew rate for falling and rising edges is defined and measured between V_{OL} and V_{OH} for single-ended signals, as described in the following tables and figure.

Table 259 — Single-Ended Output Slew Rate Definition

Description	Measured		Defined by	Note
	From	To		
Single-ended output slew rate for rising edge	V_{OL}	V_{OH}	[V_{OH} - V_{OL}] / delta TRse	1
Single-ended output slew rate for failing edge	V_{OH}	V_{OL}	[V_{OL} - V_{OH}] / delta TFse	1
NOTE: Output slew rate is verified by design and characterization, and may not be subject to production test.				

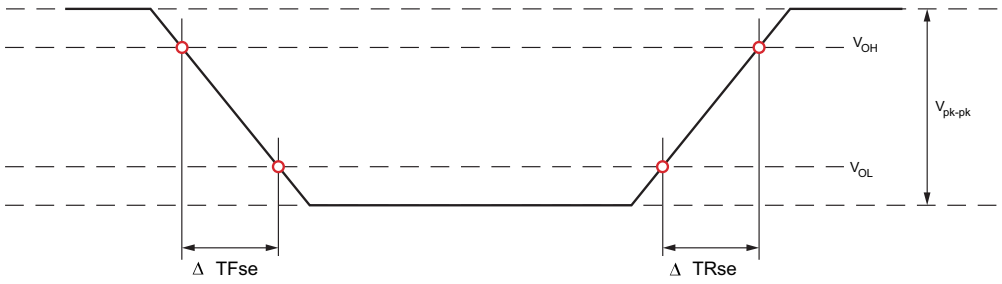


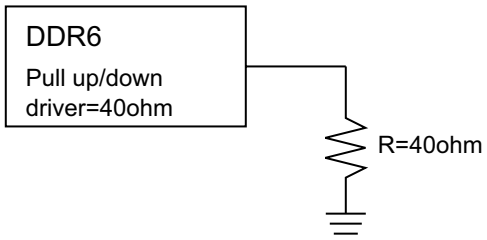
Figure 226 — Single-Ended Output Slew Rate Definition

PROPOSED

Table 260 — Single-Ended Output Slew Rate

Parameter	Symbol	DDR6-6400 to DDR6-12800		>DDR6-12800		Unit	Note
		MIN	MAX	MIN	MAX		
Single-ended output slew rate	SRQse	8	24	TBD	TBD	V/ns	1

NOTE: These values are guaranteed by design, and may not be subject to production test. Assumption: Pullup/Pulldown driver Ron=40ohm; reference load 40ohm.



13.3.3. Differential Output Levels - V_{OLdiff}/V_{OHdiff}

Table 261 — Differential Output Levels

Parameter	Symbol	Value	Unit	Notes
Differential high measurement level for output slew rate	V_{OHdiff}	0.75 x $V_{diffpk-pk}$	V	1
Differential low measurement level for output slew rate	V_{OLdiff}	0.25 x $V_{diffpk-pk}$	V	1

NOTE: $V_{diffpk-pk}$ is the mean high voltage minus the mean low voltage over 8UI samples.

13.3.4. Differential Output Slew Rate

With the reference load for timing measurements, output slew rate for falling and rising edges is defined and measured between V_{OLdiff} and V_{OHdiff} for differential signals, as described in the following tables and figure.

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Table 262 — Differential Output Slew Rate Definition

Description	Measured		Defined by	Note
	From	To		
Differential output slew rate for rising edge	V_{OLdiff}	V_{OHdiff}	[V_{OHdiff} - V_{OLdiff}] / delta TRdiff	1
Differential output slew rate for failing edge	V_{OHdiff}	V_{OLdiff}	[V_{OLdiff} - V_{OHdiff}] / delta TFdiff	1
NOTE: Output slew rate is verified by design and characterization, and may not be subject to production test.				

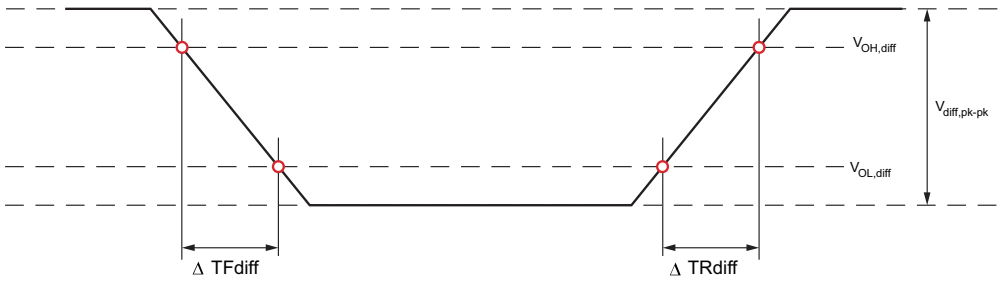
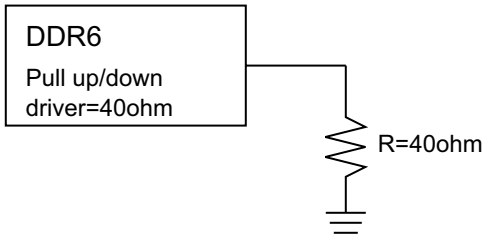


Figure 227 — Differential Output Slew Rate Definition

Table 263 — Differential Output Slew Rate

Parameter	Symbol	DDR6-6400 to DDR6-12800		>DDR6-12800		Unit	Note
		MIN	MAX	MIN	MAX		
Differential output slew rate	SRQdiff	16	48	TBD	TBD	V/ns	1

NOTE: These values are guaranteed by design. Assumption: Pullup/Pulldown driver Ron=40ohm; reference load 40ohm.



PROPOSED

13.4. Tx DQS_to_DQ Stressed Edge Jitter

13.4.1. Introduction

Tx DQS-to-DQ Stress Edge provides the methodology for testing the DRAM transmitter's timing characteristics under various DQS to DQ skew conditions. This test measures how each DQ edge aligns to its associated strobe when stressed and reports Random Jitter (Rj), Deterministic Jitter (Dj), and Total Jitter (Tj). The applied DQS-to-DQ skew conditions are determined by the Host's responsibility and capability to adjust advanced DQS edges through the Read DQS Offset Timing (mode TBD).

The total Tx DQS_to_DQ Stressed Edge jitter, Tj, is defined as Dj+Q*Rj, where Q isTBD 14.07 calculated at BER of 1E-12. The Deterministic Jitter (Dj) as specified is bounded. The Random Jitter (Rj) as specified meets a Gaussian distribution.

13.4.2. TX DQS_to_DQ Stressed Edge Jitter Parameters

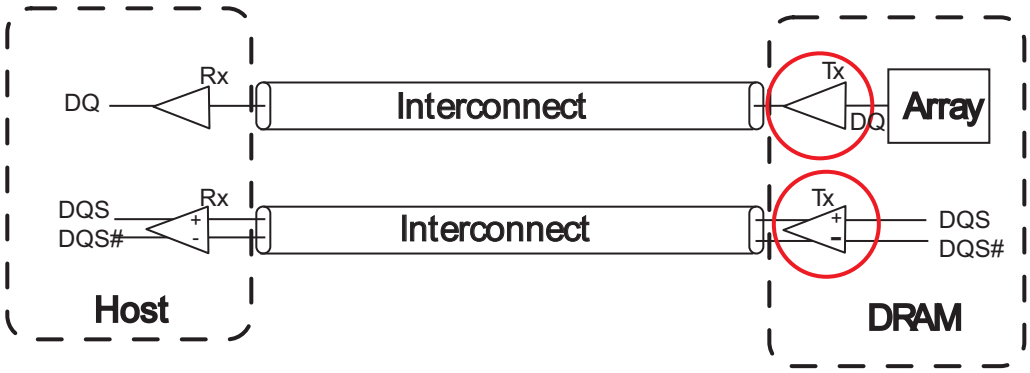


Figure 228 — Tx DQS_to_DQ Stressed Edge Jitter

Table 264 — DDR6 Tx DQS_to_DQ Stressed Edge 6400-9600 (Sheet 1 of 2)

Parameter	Symbol	DDR6 6400-9600		Unit	Notes
		Min	Max		
Tj specified at transmitter with a skew between DQ and DQS of NUI, where 0<=N<7	tTx_DQS2DQ_NUI_Tj	-	0.25	UI	1,2,3,4,7
Dj_dd specified at the transmitter with a skew between DQ and DQS of NUI, where 0<=N<7	tTx_DQS2DQ_NUI_Dj	-	0.15	UI	1,2,3,4,7
Rj RMS specified at the transmitter with a skew between DQ and DQS of NUI, where 0<=N<7	tTx_DQS2DQ_NUI_Rj	-	0.007~0.018	UI(RMS)	1,2,5,7

Table 264 — DDR6 Tx DQS_to_DQ Stressed Edge 6400-9600 (Continued) (Sheet 2

Parameter	Symbol	DDR6 6400-9600		Unit	Notes
		Min	Max		
Delay of any data lane relative to strobe lane	tTx_DQS2DQ	TBD	TBD	ps	6,7

NOTE 1: Rj (tTx_DQS2DQ_NUI_Rj) is informative parameters only. Range of Rj is defined to have zero to maximum allowed for tTx_DQS2DQ_NUI_Dj
NOTE 2: The 'NUI' in the specification symbol refers to the number of UI skew between DQ and DQS. N is to be replaced to represent this value of Nth UI skew for validation
NOTE 3: The duty cycle of the DQ pins must be adjusted to 50% using the Global and Per-Pin Duty Cycle Adjuster feature. If 50% duty cycle cannot be achieved, any residual Duty Cycle Distortion (DCD) shall be included in the reported Dj value
NOTE 4: The Tj and Dj parameters use the continuous PRBS8 LFSR training pattern which are sent out on all DQ lanes from the DRAM device (method TBD). The target IO lane runs PRBS pattern A while all other IO lanes run PRBS pattern B. This includes the effects of VOX and SSO
NOTE 5: The Rj parameter uses the continuous clock pattern which is sent out on the target IO lane (method TBD)**NOTE 6:** Delay of any data lane relative to strobe lane at TX output
NOTE 7: This parameter is defined at DRAM die pad

Table 265 — DDR6 Tx DQS_to_DQ Stressed Edge 10400-12800 (Sheet 1 of 2)

Parameter	Symbol	DDR6 10400-12800		Unit	Notes
		Min	Max		
Tj specified at transmitter with a skew between DQ and DQS of NUI, where 0<=N<7	tTx_DQS2DQ_NUI_Tj	-	0.25	UI	1,2,3,4,7
Dj_dd specified at the transmitter with a skew between DQ and DQS of NUI, where 0<=N<7	tTx_DQS2DQ_NUI_Dj	-	0.15	UI	1,2,3,4,7
Rj RMS specified at the transmitter with a skew between DQ and DQS of NUI, where 0<=N<7	tTx_DQS2DQ_NUI_Rj	-	0.007~0.018	UI(RMS)	1,2,5,7
Delay of any data lane relative to strobe lane	tTx_DQS2DQ	TBD	TBD	ps	6,7

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Table 265 — DDR6 Tx DQS_to_DQ Stressed Edge 10400-12800 (Continued) (Sheet

Parameter	Symbol	DDR6 10400-12800		Unit	Notes
		Min	Max		
NOTE 1: Rj (tTx_DQS2DQ_NUI_Rj) is informative parameters only. Range of Rj is defined to have zero to maximum allowed for tTx_DQS2DQ_NUI_Dj **NOTE 2:** The 'NUI' in the specification symbol refers to the number of UI skew between DQ and DQS. N is to be replaced to represent this value of Nth UI skew for validation **NOTE 3:** The duty cycle of the DQ pins must be adjusted to 50% using the Global and Per-Pin Duty Cycle Adjuster. If 50% duty cycle cannot be achieved, any residual Duty Cycle Distortion (DCD) shall be included in the reported Dj value **NOTE 4:** The Tj and Dj parameters use the continuous PRBS8 LFSR training pattern which are sent out on all DQ lanes from the DRAM device (method TBD). The target IO lane runs PRBS pattern A while all other IO lanes run PRBS pattern B. This includes the effects of VOX and SSO **NOTE 5:** The Rj parameter uses the continuous clock pattern which is sent out on the target IO lane (method TBD) **NOTE 6:** Delay of any data lane relative to strobe lane at TX output **NOTE 7:** This parameter is defined at DRAM die pad					

Table 266 — DDR6 Tx DQS_to_DQ Stressed Edge 14400-16400 (Sheet 1 of 2)

Parameter	Symbol	DDR6 14400		DDR6 16400		Unit	Notes
		Min	Max	Min	Max		
Tj specified at transmitter with a skew between DQ and DQS of NUI, where 0<=N<7	tTx_DQS2DQ_NUI_Tj	-	TBD	-	TBD	UI	1,2,3,4,7
Dj_dd specified at the transmitter with a skew between DQ and DQS of NUI, where 0<=N<7	tTx_DQS2DQ_NUI_Dj	-	TBD	-	TBD	UI	1,2,3,4,7
Rj RMS specified at the transmitter with a skew between DQ and DQS of NUI, where 0<=N<7	tTx_DQS2DQ_NUI_Rj	-	TBD	-	TBD	UI(RMS)	1,2,5,7
Delay of any data lane relative to strobe lane	tTx_DQS2DQ	TBD	TBD	TBD	TBD	ps	6,7

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Table 266 — DDR6 Tx DQS_to_DQ Stressed Edge 14400-16400 (Continued) (Sheet

Parameter	Symbol	DDR6 14400		DDR6 16400		Unit	Notes
		Min	Max	Min	Max		

NOTE 1: Rj (tTx_DQS2DQ_NUI_Rj) is informative parameters only. Range of Rj is defined to have zero to maximum allowed for tTx_DQS2DQ_NUI_Dj
NOTE 2: The 'NUI' in the specification symbol refers to the number of UI skew between DQ and DQS. N is to be replaced to represent this value of Nth UI skew for validation
NOTE 3: The duty cycle of the DQ pins must be adjusted to 50% using the Global and Per-Pin Duty Cycle Adjuster feature. If 50% duty cycle cannot be achieved, any residual Duty Cycle Distortion (DCD) shall be included in the reported Dj value
NOTE 4: The Tj and Dj parameters use the continuous PRBS8 LFSR training pattern which are sent out on all DQ lanes from the DRAM device (method TBD). The target IO lane runs PRBS pattern A while all other IO lanes run PRBS pattern B. This includes the effects of VOX and SSO
NOTE 5: The Rj parameter uses the continuous clock pattern which is sent out on the target IO lane (method TBD)
NOTE 6: Delay of any data lane relative to strobe lane at TX output
NOTE 7: This parameter is defined at DRAM die pad

Table 267 — DDR6 Tx DQS_to_DQ Stressed Edge 19200 (Sheet 1 of 2)

Parameter	Symbol	DDR6 19200		Unit	Notes
		Min	Max		
Tj specified at transmitter with a skew between DQ and DQS of NUI, where 0<=N<7	tTx_DQS2DQ_NUI_Tj	-	TBD	UI	1,2,3,4,7
Dj_dd specified at the transmitter with a skew between DQ and DQS of NUI, where 0<=N<7	tTx_DQS2DQ_NUI_Dj	-	TBD	UI	1,2,3,4,7
Rj RMS specified at the transmitter with a skew between DQ and DQS of NUI, where 0<=N<7	tTx_DQS2DQ_NUI_Rj	-	TBD	UI(RMS)	1,2,5,7
Delay of any data lane relative to strobe lane	tTx_DQS2DQ	TBD	TBD	ps	6,7

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Table 267 — DDR6 Tx DQS_to_DQ Stressed Edge 19200 (Continued) (Sheet 2 of 2)

Parameter	Symbol	DDR6 19200		Unit	Notes
		Min	Max		
NOTE 1: Rj (tTx_DQS2DQ_NUI_Rj) is informative parameters only. Range of Rj is defined to have zero to maximum allowed for tTx_DQS2DQ_NUI_Dj **NOTE 2:** The 'NUI' in the specification symbol refers to the number of UI skew between DQ and DQS. N is to be replaced to represent this value of Nth UI skew for validation **NOTE 3:** The duty cycle of the DQ pins must be adjusted to 50% using the Global and Per-Pin Duty Cycle Adjuster feature. If 50% duty cycle cannot be achieved, any residual Duty Cycle Distortion (DCD) shall be included in the reported Dj value **NOTE 4:** The Tj and Dj parameters use the continuous PRBS8 LFSR training pattern which are sent out on all DQ lanes from the DRAM device (method TBD). The target IO lane runs PRBS pattern A while all other IO lanes run PRBS pattern B. This includes the effects of VOX and SSO **NOTE 5:** The Rj parameter uses the continuous clock pattern which is sent out on the target IO lane (method TBD) **NOTE 6:** Delay of any data lane relative to strobe lane at TX output **NOTE 7:** This parameter is defined at DRAM die pad					

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13.5. Tx DQ Stressed Eye

To be balloted in Q4'26.

Tx DQ stressed eye height and eye width must meet minimum specification values at BER=E-9 and confidence level 99.5%. Tx DQ Stressed Eye shows the DQS to DQ skew for both eye width and height. To support different host receiver (Rx) designs, it is the responsibility of the host to ensure the advanced DQS edges are adjusted accordingly via the Read DQS Offset Timing mode register settings (MR40:OP[2:0]).

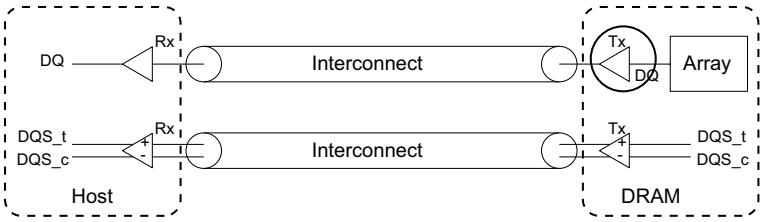


Figure 229 — Tx DQ Stressed Eye

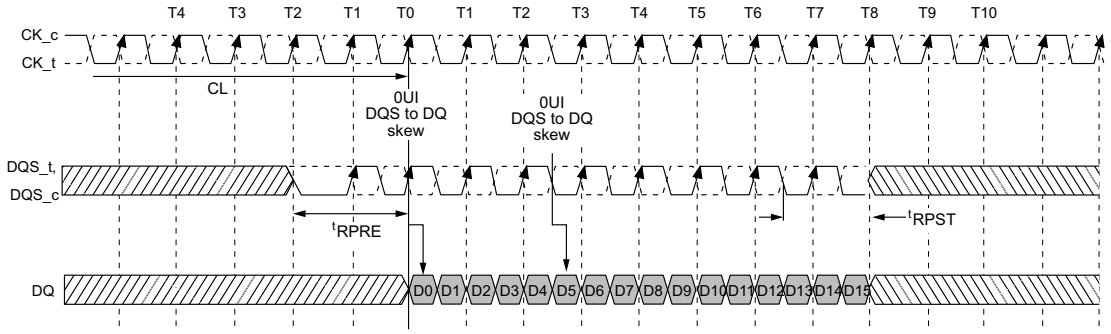


Figure 230 — Read Burst Example for Pin DQx Depicting Bit 0 and 5 Relative to the DQS Edge for 0 UI Skew

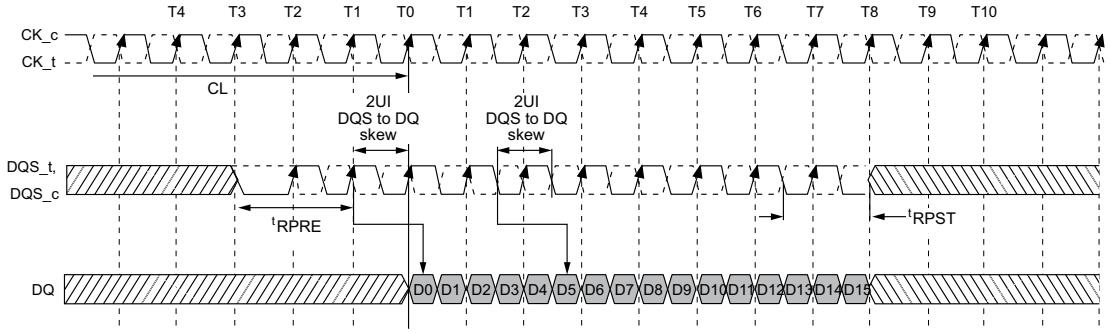


Figure 231 — Read Burst Example for Pin DQx Depicting Bit 0 and 5 Relative to the DQS Edge for 2 UI Skew with Read DQS Offset Timing Set to 1 Clock (2UI)

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13.5.1. Tx DQ Stressed Eye Parameters

The following notes apply to all tables:

- EH = Eye Height; EW = Eye Width; BER = Bit Error Rate, SES = Stressed Eye Skew
- Minimum BER E^{-9} and Confidence Level of 99.5% per pin.
- Refer to the minimum Bit Error Rate (BER) requirements for DDR5.
- The validation methodology for these parameters will be covered in the future.
- Mismatch is defined as DQS to DQ mismatch in UI increments.
- The DQ pins' duty cycle must be adjusted as close to 50% as possible using the Duty Cycle Adjuster feature prior to running the Tx DQS jtter test.
- The mode registers for the Duty Cycle Adjuster are MR43 and MR44. The mode registers for the per-pin DCA of DQS are MR103-MR110. The mode registers for the per-pin DCA of DQLx are MR(133+8x) and MR(134+8x), where $0 \leq x \leq 7$. The mode registers for the per-pin DCA of DQUy are MR(197+8y) and MR(198+8y), where $0 \leq y \leq 7$.
- Spread spectrum clocking (SSC) must be disabled while running these tests.
- These parameters are tested using the continuous PRBS8 LFSR training pattern, which is sent out on all DQ lanes off the device without the need for sending out continuous MRR commands. The MR25 OP[3] is set to 1 to enable this feature.
- These parameters are tested on the CTC2 card only.
- Matched DQS to DQ would require the DQs to be adjusted by 0.5UI to place it in the center of the DQ eye. 1UI mismatch would require the DQS to be adjusted 1.5UI. Generally, for XUI mismatch the DQ must be adjusted XUI + 0.5UI to be placed in the center of the eye.

Table 268 — Tx DQ Stressed Eye Parameters for DDR5 3200–4800

Parameter	Symbol	3200-4800		Unit	Note
		MIN	MAX		
Eye width specified at the transmitter with a skew between DQ and DQS of 1UI	TXEW_DQ_SES_1UI	0.72	–	UI	
Eye width specified at the transmitter with a skew between DQ and DQS of 2UI	TXEW_DQ_SES_2UI	0.72	–	UI	
Eye width specified at the transmitter with a skew between DQ and DQS of 3UI	TXEW_DQ_SES_3UI	0.72	–	UI	

Table 269 — Tx DQ Stressed Eye Parameters for DDR5 5200–6400

NOTE: The number of UI's accumulated will depend on the speed of the link. For higher speeds, higher UI accumulation may be specified. For lower speeds, N=4,5 UI may not be applicable.

Parameter	Symbol	5200-5600		6000-6400		Unit	Note
		MIN	MAX	MIN	MAX		
Eye width specified at the transmitter with a skew between DQ and DQS of 1UI	TXEW_DQ_SES_1UI	0.74	-	0.75	–	UI	
Eye width specified at the transmitter with a skew between DQ and DQS of 2UI	TXEW_DQ_SES_2UI	0.74	-	0.75	–	UI	

Parameter	Symbol	5200-5600		6000-6400		Unit	Note
		MIN	MAX	MIN	MAX		
Eye width specified at the transmitter with a skew between DQ and DQS of 3UI	TXEW_DQ_SES_3UI	0.74	-	0.75	–	UI	
Eye width specified at the transmitter with a skew between DQ and DQS of 4UI	TXEW_DQ_SES_4UI	0.74	-	0.75	–	UI	1
Eye width specified at the transmitter with a skew between DQ and DQS of 5UI	TXEW_DQ_SES_5UI	-	-	0.75	–	UI	1

Table 270 — Tx DQ Stressed Eye Parameters for DDR5 6800–8800

NOTE: The number of UI's accumulated will depend on the speed of the link. For higher speeds, higher UI accumulation may be specified. For lower speeds, N=4,5 UI may not be applicable.

Parameter	Symbol	6800-8800		Unit	Note
		MIN	MAX		
Eye width specified at the transmitter with a skew between DQ and DQS of 1UI	TXEW_DQ_SES_1UI	0.76	–	UI	
Eye width specified at the transmitter with a skew between DQ and DQS of 2UI	TXEW_DQ_SES_2UI	0.76	–	UI	
Eye width specified at the transmitter with a skew between DQ and DQS of 3UI	TXEW_DQ_SES_3UI	0.76	–	UI	
Eye width specified at the transmitter with a skew between DQ and DQS of 4UI	TXEW_DQ_SES_4UI	0.76	–	UI	1
Eye width specified at the transmitter with a skew between DQ and DQS of 5UI	TXEW_DQ_SES_5UI	0.76	–	UI	1

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14. DDR6 Silicon Pad IO Capacitance

Table 271 — Silicon Pad IO Capacitance DDR6-6400 to DDR6-12800

Symbol	Parameter	DDR6 6400-9600		DDR6 10400-12800		Unit	NOTE
		min	max	min	max		
C_{IO}	Input/output capacitance(DQ, DQS_t, DQS_c)	0.2	0.55	0.2	0.45	pF	1,2,10
C_{DIO}	Input/output capacitance delta(DQ)	-0.1	0.1	-0.1	0.1	pF	1,2,8
C_{DDQS}	Input/output capacitance delta (DQS_t and DQS_c)		0.04		0.04	pF	1,2,4
C_{CK}	Input capacitance (CK_t and CK_c)	0.15	0.4	0.15	0.375	pF	1,2,10
C_{DCK}	Input capacitance delta (CK_t and CK_c)		0.05		0.05	pF	1,2,3
C_{I}	Input capacitance(CS & CA pins only)	0.15	0.4	0.15	0.375	pF	1,2,5,10
C_{DI_CS}	Input capacitance delta(CS pin only)	-0.1	0.1	-0.1	0.1	pF	1,2,6
C_{DI_CA}	Input capacitance delta(CA pins only)	-0.1	0.1	-0.1	0.1	pF	1,2,7
C_{ALERT}	Output capacitance of ALERT	0.1	1.0	0.1	1.0	pF	1,2,10
C_{ZQ}	Input/output capacitance of ZQ	-	5	-	5	pF	1,2,9,10
C_{Reset_n}	Input capacitance of Reset_n	1.0	1.0	1.0	1.0	pF	1,2,10

NOTE 1: This parameter is not subject to production test. This parameter is measured by using vendor specific measurement methodology
NOTE 2: This parameter applies to die device, including IO capacitance, RDL if needed (does not include package capacitance such as bond wire)
NOTE 3: Absolute value CI(CK_t)-CI(CK_c)
NOTE 4: Absolute value of CIO(DQS_t)-CIO(DQS_c)
NOTE 5: CI applies to CS and CA
NOTE 6: CDI_CS = CI(CS)-0.5*(CI(CK_t)+CI(CK_c))
NOTE 7: CDI_CA = CI(CA)-0.5*(CI(CK_t)+CI(CK_c))
NOTE 8: CDIO = CIO(DQ)-Avg(CIO(DQ))
NOTE 9: Maximum external load capacitance on ZQ pin: TBD. The ZQ functionality/accuracy with the max capacitive load is characterized
NOTE 10: Contact memory vendor for typical value of silicon pad IO capacitance

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Table 272 — Silicon Pad IO Capacitance DDR6-14400 to DDR6-19200

Symbol	Parameter	DDR6 14400		DDR6 16400		DDR6 19200		Unit	NOTE
		min	max	min	max	min	max		
C_{IO}	Input/output capacitance(DQ, DQS_t, DQS_c)	TBD	TBD	TBD	TBD	TBD	TBD	pF	1,2,10
C_{DIO}	Input/output capacitance delta(DQ)	TBD	TBD	TBD	TBD	TBD	TBD	pF	1,2,8
C_{DDQS}	Input/output capacitance delta (DQS_t and DQS_c)		TBD		TBD		TBD	pF	1,2,4
C_{CK}	Input capacitance (CK_t and CK_c)	TBD	TBD	TBD	TBD	TBD	TBD	pF	1,2,10
C_{DCK}	Input capacitance delta (CK_t and CK_c)		TBD		TBD		TBD	pF	1,2,3
C_{I}	Input capacitance(CS & CA pins only)	TBD	TBD	TBD	TBD	TBD	TBD	pF	1,2,5,10
C_{DI_CS}	Input capacitance delta(CS pin only)	TBD	TBD	TBD	TBD	TBD	TBD	pF	1,2,6
C_{DI_CA}	Input capacitance delta(CA pins only)	TBD	TBD	TBD	TBD	TBD	TBD	pF	1,2,7
C_{ALERT}	Output capacitance of ALERT	TBD	TBD	TBD	TBD	TBD	TBD	pF	1,2,10
C_{ZQ}	Input/output capacitance of ZQ	-	TBD	-	TBD	-	TBD	pF	1,2,9,10
C_{Reset_n}	Input capacitance of Reset_n	TBD	TBD	TBD	TBD	TBD	TBD	pF	1,2,10

NOTE 1: This parameter is not subject to production test. This parameter is measured by using vendor specific measurement methodology
NOTE 2: This parameter applies to die device, including IO capacitance, RDL if needed (does not include package capacitance such as bond wire)
NOTE 3: Absolute value CI(CK_t)-CI(CK_c)
NOTE 4: Absolute value of CIO(DQS_t)-CIO(DQS_c)
NOTE 5: CI applies to CS and CA
NOTE 6: CDI_CS = CI(CS)-0.5*(CI(CK_t)+CI(CK_c))
NOTE 7: CDI_CA = CI(CA)-0.5*(CI(CK_t)+CI(CK_c))
NOTE 8: CDIO = CIO(DQ)-Avg(CIO(DQ))
NOTE 9: Maximum external load capacitance on ZQ pin: TBD. The ZQ functionality/accuracy with the max capacitive load is characterized
NOTE 10: Contact memory vendor for typical value of silicon pad IO capacitance

15. IDD Specification Parameters and Test Conditions

Unballoted.

15.1. IDD Specifications

Unballoted.

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16. Electrical Characteristics and AC Timing

Unballoted.

16.1. Reference Load for AC Timing and Output Slew Rate

The figure below represents the effective reference load of 50 ohms used in defining the relevant AC timing parameters of the device as well as output slew rate measurements.

R_{ON} nominal of DQ, DQS_t, and DQS_c drivers uses 34 ohms to specify the relevant AC timing parameter values of the device.

- The maximum DC high level of output signal = $1.0 \times V_{DDQ}$.
- The minimum DC low level of output signal = $\{34/(34 + 50)\} \times V_{DDQ} = 0.4 \times V_{DDQ}$
- The nominal reference level of an output signal can be approximated by the following:
  - The center of maximum DC high and minimum DC low
-
- $= \{(1 + 0.4)/2\} \times V_{DDQ} = 0.7 \times V_{DDQ}$

The actual reference level of output signal might vary with driver R_{ON} and reference load tolerances. Thus, the actual reference level or midpoint of an output signal is at the widest part of the output signal's eye. Prior to measuring AC parameters, the reference level of the verification tool should be set to an appropriate level.

It is not intended as a precise representation of any particular system environment or a depiction of the actual load presented by a production tester. System designers should use IBIS or other simulation tools to correlate the timing reference load to a system environment. Manufacturers correlate to their production test conditions, generally one or more coaxial transmission lines terminated at the tester electronics.

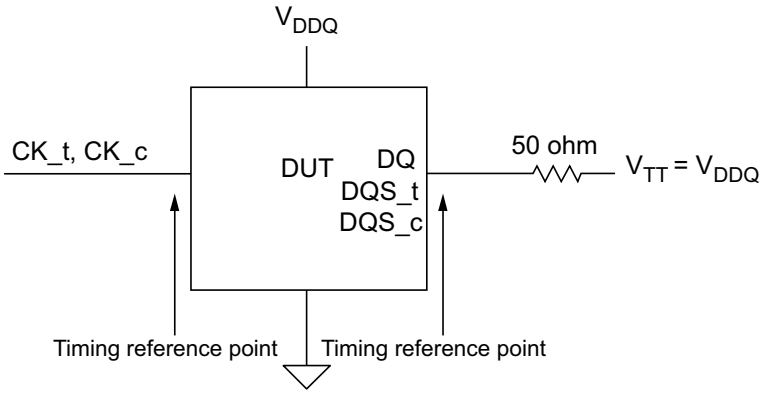


Figure 232 — Reference Load for AC Timing and Output Slew Rate

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17. Revision History

17.1. Rev0.x R0 - 09/2026

- Ballots included: 31
  - RB26475-DDR6 CRC_R12
  - RB26476-DDR6 Command Ignore_R4
  - RB26477-DDR6 Power-up and Initialize sequence R6.1
  - RB26478-DDR6 Frequency MR set R5.1
  - RB26479-DDR6 CA Parity_R14
  - RB26480-DDR6 UI and Jitter Definition
  - RB26481-DDR6 Self Refresh R4.1
  - RB26502-DDR6 BL24 SDRAM Core Architecture
  - RB26503-DDR6 Soft Post Package Repair (sPPR) R5
  - RB26504-DDR6 Command Truth Table R3
  - RB26505-DDR6 Speed Grades R0.1
  - RB26506-DDR6 Mode Register R2
  - RB26507-DDR6 Per Row Activation Counting (PRAC) R2
  - RB26508-CA Training Mode R2
  - RB26509-CS Training Mode R3
  - RB26510-DDR6 Scratch Pad MR
  - RB26511-DDR6 Power-down mode R4.2.1
  - RB26512-DDR6 DQS and CLK Interval Oscillator R2.5
  - RB26513-DDR6 DRAM Ballout and Definition R6.2
  - RB26514-DDR6 Command Timing Constraint R5.4
  - RB26515-DDR6 Mode Register Read (MRR) R5.7
  - RB26516-DDR6 Mode Register Write (MRW) R5.6
  - RB26517-DDR6 Read Preamble Training Mode R1.1
  - RB26518-DDR6 ZQ Calibration R3.4
  - RB26519-DDR6 Write Training Mode R3.1
  - RB26520-DDR6 Host Receiver Training (HRT) R1.31
  - RB26521-DDR6 DQS2DQ and CK2DQ AC parameters R0.2
  - RB26522-DDR6 Per-pin VREFCA Training R5.3
  - RB26523-DDR6 ACTIVATE COMMAND
  - RB26524-DDR6 Wakeup CS R2
  - RB26526-DDR6 Base Die Buffering DRAM Package R2

17.2. Rev0.7 R1.2 - 07/2026

- Editorial updates:
  - Moved TG423_1^20260629^1905.68^SKhynix^DDR6_Alert_Output_Driver_Characteristic_Ballot_R2 from Chapter 7 to Chapter 13
  - TG423_1^20260707^1897.72^SKhynix^DDR6_Write DCA_ballot_R3
  - TG423_1^20260629^1897.80^Micron^DDR6_Silicon_Pad_IO_Capacitance_Ballot_R3
  - TG423_1^20260708^1897.71C^Samsung^DDR6_DQ_RX_Offset_Calibration_Ballot_R4.3
  - TG423_1^20260707^xxxx.yy^Samsung^DDR6_Interamble_Ballot_Feedback
  - TG423_1^20260709^1868.54B^Samsung^DDR6_Temperature_Sensor_Specification_Ballot_R3.6

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- TG423_1^20260708^1897.75B^Micron^DDR6_Hard_Post_Package_Repair_(hPPR)_Ballot_R4
- TG423_1^20260713^1905.70^SKhynix^DDR6_DFE_ballot_R1.4
- TG423_1^20260713^1905.67^SKhynix^DDR6_VREFCS_ballot
- TG423_1^20260715^1905.70^SKhynix^DDR6_DFE_ballot_R1.5
- TG423_1^20260720^1905.33B^Micron^DDR6_Functional_Description_Ballot_R2.1
- TG423_1^20260720^1897.77C^Micron^DDR6_Memory_Built-in_Self_Test_(MBIST)_and_Repair_(mPPR)_Ballot_R4.1

17.3. Rev0.7 R1.1 - 07/2026

- Editorial updates:
  - TG423_1^20260629^1905.68^SKhynix^DDR6_Alert_Output_Driver_Characteristic_Ballot_R2
  - TG423_1^20260629^1905_75^SKhynix^DDR6_Read_DCA_ballot_R2
  - TG423_1^20260622^1905.43^Micron^DDR6_VRx_CK_Ballot_R1
  - TG423_1^20260622^1905.04^Micron^DDR6_TX_DQS-to-DQ_Stressed_Edge_Ballot_R3
  - TG423_1^20260629^1905.33B^Micron^DDR6_Functional_Description_Ballot_R2
    - Includes additional editorial update to remove 1p12 from "with an additional Bank Group term in the 1p6 and 1p12 configurations"
  - TG423_1^20260629^1897.78B^Micron^DDR6_MCME_Command_Ballot_R2
  - TG423_1^20260629^1905.74A^Micron^DDR6_PDA_Mode_Ballot_R1
  - TG423_1^20260629^1905.26B^Samsung^DDR6_Read_Write_Latency_Ballot_R3.2
  - RB26340-Proposed DDR6 Interambles Update- R5
    - Added figures 11-13, write2write with 4/8/12 clock gap

17.4. Rev0.7 R1 - 06/2026

- Ballots included: 53
  - RB26291-Proposed CK Sync Operation R0.1
  - RB26292-DDR6 CA Parity
  - RB26293-DDR6 CRC
  - RB26294-DDR6 Clocking R1.2
  - RB26295-DDR6 Alert Output Driver Characteristic
  - RB26296-DDR6 Command Timing Constraint R5.1
  - RB26297-DDR6 Command Ignore
  - RB26298-DDR6 Read Write Latency R3.1
  - RB26299-Proposed DDR6 Per DRAM Addressability (PDA)
  - RB26300-Proposed DDR6 VRx_CK Ballot
  - RB26301-Proposed DDR6 Silicon Pad IO Capacitance R2
  - RB26302-DDR6 Read Write ODT Timing R2.1
  - RB26303-DDR6 Temperature Sensor Specification R3.2
  - RB26304-Proposed DDR6 Read DQS and DQs Duty Cycle Adjuster (DCA) R1
  - RB26305-Proposed DDR6 Scope
  - RB26306-Proposed DDR6 Functional Description
  - RB26307-Proposed DDR6 MCME Command R1
  - RB26308-Proposed DDR6 MRUPDATE Mode
  - RB26309-Proposed DDR6 Self Refresh R3
  - RB26310-Proposed DDR6 DQ Preamble R2

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  - RB26311-Proposed DDR6 Read Preamble Training Mode R0
  - RB26312-Proposed DDR6 Write Preamble and Postamble
  - RB26313-DDR6 ACTIVATE COMMAND
  - RB26314-Proposed DDR6 CA Bus 2N Mode
  - RB26315-Proposed DDR6 Mode Register Write (MRW) R5.4
  - RB26316-Proposed DDR6 Mode Register Read (MRR) R5.4
  - RB26317-Proposed DDR6 Write DQS Duty Cycle Adjuster (DCA)
  - RB26318-DDR6_VREFCS
  - RB26319-DDR6 DQ RX Offset Calibration R4.2
  - RB26320-Proposed DDR6 TX DQS_to_DQ Stressed Edge R2
  - RB26321-Proposed DDR6 DQ RX DFE (Decision Feedback Equalization)
  - RB26322-DDR6 Per-pin VREFCA Training R5.1
  - RB26323-Proposed DDR6 Per Row Activation Counting (PRAC) R1
  - RB26324-Proposed DDR6 ZQ Calibration R3.1
  - RB26338-DDR6 Read Preambles, Postambles, Pre-Shift R3.1
  - RB26339-Proposed CS Training Mode R2
  - RB26340-Proposed DDR6 Interambles Update- R5
  - RB26341-Proposed DDR6 DRAM Ballout R5.1
  - RB26342-Proposed CA Training Mode R1
  - RB26343-Proposed DDR6 Write Training Mode R2.1
  - RB26344-Proposed DDR6 Host Receiver Training (HRT) R1.1
  - RB26345-DDR6 WRITE COMMAND
  - RB26346-Proposed DDR6 Mode Register R1
  - RB26347-DDR6 Power-down mode R3.1
  - RB26348-Proposed DDR6 DQS and CLK Interval Oscillator
  - RB26349-DDR6 READ COMMAND
  - RB26371-DDR6 Power-up and Initialize sequence R5.1
  - RB26372-DDR6 Rx CK DQS CA CS CTLE R5.3.1
  - RB26373-Proposed DDR6 Soft Post Package Repair (sPPR) R4.3.1
  - RB26374-Proposed DDR6 Hard Post Package Repair (hPPR) R3.3.1
  - RB26375-Proposed DDR6 Memory Built-in Self Test (MBIST) and Repair (mPPR) R3.1.1
  - RB26376-DDR6 BL24 SDRAM Core Architecture
  - RB26377-Proposed Update to DDR6 ESD Specification R4

- Ballots with missing or incomplete figures
  - RB26310-Proposed DDR6 DQ Preamble R2
  - RB26340-Proposed DDR6 Interambles Update- R5

17.5. Rev0.7 R0.1 – 04/2026

- Updated to include Rev0.7 R0 ballot source that was missing by mistake
  - RB26051-DDR6 DRAM Signal Ballot R1
  - RB26082-DDR6 RX CS CA CK DQS CTLE Ballot R3
  - RB26079-DDR6 Interambles

- Included editorially passed content updates within the TG
  - hPPR from Q2'26
  - sPPR from Q2'26

  - Refresh from Q1'26

- Other updates
  - DC Operating Conditions (removed directional content showing typical VDDQ=0.3V for buffered modules that was superseded)

17.6. Rev0.7 R0 – 03/2026

- Draft Rev0.7 R0 - Ballot source from Q1'26.
- Ballots included: 42
  - RB26022-DDR6 ESD Specification Proposal
  - RB26023-DDR6 Functional Description Ballot
  - RB26024-DDR6 Mode Register Write R5.1
  - RB26025-DDR6 Mode Register Read R5.1
  - RB26026-DDR6 ZQCAL Ballot Draft R2.1
  - RB26027-DDR6 DQ RX Offset Training Ballot R3
  - RB26028-DDR6 Read Write Latency Ballot R2
  - RB26029-DDR6 Write Preamble and Postamble Ballot R1
  - RB26030-Read Preambles Postambles Pre-Shift Proposal Ballot R2
  - RB26031-DDR6 Write Leveling Ballot R2
  - RB26032-DDR6 Clocking Ballot R0
  - RB26033-DDR6 Absolute Maximum DC Ratings Proposal Ballot R2
  - RB26034-DDR6 DQ Swizzle Discovery Mode R1
  - RB26036-DDR6 PRAC Ballot R0
  - RB26037-DDR6 Vix CK
  - RB26038-DDR6 Vix DQS
  - RB26039-DDR6 Command Ignore R1
  - RB26040-DDR6 Addressing Tables Ballot R3
  - RB26041-DDR6 Command Truth Table Ballot R2
  - RB26048-DDR6 Scope Ballot
  - RB26050-DDR6 Per-Pin VrefCA Training Ballot R4
  - ~~RB26051-DDR6 DRAM Signal Ballot R1~~
  - RB26078-DDR6 AC Timing Parameter Ballot R7
  - ~~RB26079-DDR6 Interambles~~
  - RB26080-DDR6 PerPin VREFDQ Training v0
  - RB26081-DDR6 Input Clock Frequency Change v0
  - ~~RB26082-DDR6 RX CS CA CK DQS CTLE Ballot R3~~
  - RB26083-DDR6 Simplified State Diagram Ballot R2
  - RB26084-DDR6 Data Packet Ballot
  - RB26085-DDR6 Temperature Sensor Specification Ballot R2
  - RB26102-DDR6 Tx DQS-to-DQ Stressed Edge Ballot R1
  - RB26103-DDR6 Frequency MR Ballot R4
  - RB26104-DDR6 RD WR ODT Timing Ballot R1
  - RB26105-DDR6 Self Refresh Ballot R2
  - RB26106-DDR6 Host Receiver Drift Training Mode Ballot R1
  - RB26107-DDR6 Mode Register Tables
  - RB26108-DDR6 Write DQ Training Mode Ballot R1.1

PROPOSED

  - RB26109-DDR6 WRITE Command
  - RB26110-DDR6 Command Timing Constraint Ballot R4
  - RB26111-DDR6 READ COMMAND R1.2
  - RB26112-DDR6 CS Training Mode (CSTM) Ballot R1
  - RB26113-DDR6 DC Operating Temperature Range Ballot R0

- Other changes
  - Removed LP6 reference material

17.7. Rev0.5 – 12/2025

- Initial version - Rev0.5 ballot source + directional ballots.
- Ballots included: 42
  - RB24455-DDR6 On-Die ECC Scheme - Directional
  - RB24456-DDR6 ECC Transparency - Directional
  - RB24567-DDR6 Bounded Fault Ballot
  - RB25049-DDR6 Stacking Directional Ballot
  - RB25132-DDR6_Hybrid_Clocking_Directional Ballot
  - RB25140-DDR6 Metadata Ballot
  - RB25259-Proposed DDR6 Command Truth Table
  - RB25260-DDR6 AC Timing Parameter
  - RB25261-DDR6 DRAM Tx DQ Jitter
  - RB25262-Proposed DDR6 Refresh Operation
  - RB25263-Proposed DDR6 ACTIVATE COMMAND
  - RB25265-DDR6 DC Operating Condition
  - RB25267-Proposed DDR6 Ron_ODT_ZQ_Calibration_Voltage Ballot_R1
  - RB25268-DDR6 Metadata Ballot
  - RB25269-Proposed DDR6 DQ RX DFE Step Size Directional Ballot
  - RB25428-DDR6 ACTIVATE COMMAND
  - RB25429-DDR6 MRUPDATE Mode
  - RB25430-DDR6 Mode Register Write (MRW)
  - RB25436-DDR6 Hard Post Package Repair (hPPR)
  - RB25432-DDR6 Memory Built-in Self Test (MBIST) and Repair (mPPR)
  - RB25433-DDR6 Addressing Tables
  - RB25434-DDR6 Refresh Operation
  - RB25435-DDR6 Pad Definition and Description
  - RB25437-DDR6 Soft Post Package Repair (sPPR)
  - RB25438-DDR6 CSTM Operation
  - RB25440-DDR6 CATM Operation
  - RB25442-DDR6 MCME Command
  - RB25443-DDR6 Vref Level Directional Ballot R1
  - RB25444-DDR6 Write Leveling R1
  - RB25446-DDR6 Per-pin VREFCA R2
  - RB25448-DDR6 AC Timing Parameter R5a
  - RB25449-DDR6 CA Parity
  - RB25450-DDR6 Write DQ Training Mode
  - RB25451-DDR6 Burst Length and Order

PROPOSED

- RB25452-DDR6 ZQ Calibration
- RB25453-DDR6 Output Slew Rate Ballot R3
- RB25454-DDR6 PRECHARGE COMMAND
- RB25455-DDR6 DRAM Package Size Directional Ballot R0
- RB25456-DDR6 CRC
- RB25457-DDR6 Speed Bins - Option A1
- RB25458-DDR6 DQ RX Offset Calibration
- RB25461-DDR6 Mode Register Read (MRR)

PROPOSED